

lyche-numerical-linear-algebra — formula report

4638 inline formulas (0 shown: the ones that did not render), 1056 display equations, 7 tables, 58 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-253 EQ0706 [conf 0.001]	253	■ 0.001 ● C +16	$\begin{array}{rl} \boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} &= \boldsymbol{h}^2 \boldsymbol{F} \\ \boldsymbol{V} \boldsymbol{V} = \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} & \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} + \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} = \boldsymbol{h}^2 \boldsymbol{F} \\ \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}^2 + \boldsymbol{S}^2 \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} &= \boldsymbol{h}^2 \boldsymbol{S} \boldsymbol{F} \boldsymbol{S} = \boldsymbol{h}^2 \boldsymbol{G} \\ \boldsymbol{T} \boldsymbol{S} = \boldsymbol{S} \boldsymbol{D} & \boldsymbol{S}^2 \boldsymbol{D} \boldsymbol{X} \boldsymbol{S}^2 + \boldsymbol{S}^2 \boldsymbol{X} \boldsymbol{S}^2 \boldsymbol{D} = \boldsymbol{h}^2 \boldsymbol{G} \\ \boldsymbol{S}^2 = \boldsymbol{I} / (2 \boldsymbol{h}) & \boldsymbol{D} \boldsymbol{X} + \boldsymbol{X} \boldsymbol{D} = 4 \boldsymbol{h}^4 \boldsymbol{G}. \end{array}$	$\begin{array}{rl} \boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} &= \boldsymbol{h}^2 \boldsymbol{F} \\ \boldsymbol{V} \boldsymbol{V} = \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} & \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} + \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} = \boldsymbol{h}^2 \boldsymbol{F} \\ \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}^2 + \boldsymbol{S}^2 \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} &= \boldsymbol{h}^2 \boldsymbol{S} \boldsymbol{F} \boldsymbol{S} = \boldsymbol{h}^2 \boldsymbol{G} \\ \boldsymbol{T} \boldsymbol{S} = \boldsymbol{S} \boldsymbol{D} & \boldsymbol{S}^2 \boldsymbol{D} \boldsymbol{X} \boldsymbol{S}^2 + \boldsymbol{S}^2 \boldsymbol{X} \boldsymbol{S}^2 \boldsymbol{D} = \boldsymbol{h}^2 \boldsymbol{G} \\ \boldsymbol{S}^2 = \boldsymbol{I} / (2 \boldsymbol{h}) & \boldsymbol{D} \boldsymbol{X} + \boldsymbol{X} \boldsymbol{D} = 4 \boldsymbol{h}^4 \boldsymbol{G}. \end{array}$	$\begin{array}{rl} \boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} &= \boldsymbol{h}^2 \boldsymbol{F} \\ \boldsymbol{V} \boldsymbol{V} = \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} & \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} + \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} = \boldsymbol{h}^2 \boldsymbol{F} \\ \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}^2 + \boldsymbol{S}^2 \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} &= \boldsymbol{h}^2 \boldsymbol{S} \boldsymbol{F} \boldsymbol{S} = \boldsymbol{h}^2 \boldsymbol{G} \\ \boldsymbol{T} \boldsymbol{S} = \boldsymbol{S} \boldsymbol{D} & \boldsymbol{S}^2 \boldsymbol{D} \boldsymbol{X} \boldsymbol{S}^2 + \boldsymbol{S}^2 \boldsymbol{X} \boldsymbol{S}^2 \boldsymbol{D} = \boldsymbol{h}^2 \boldsymbol{G} \\ \boldsymbol{S}^2 = \boldsymbol{I} / (2 \boldsymbol{h}) & \boldsymbol{D} \boldsymbol{X} + \boldsymbol{X} \boldsymbol{D} = 4 \boldsymbol{h}^4 \boldsymbol{G}. \end{array}$
lyche-numerical-linear-algebra-258 EQ0729 [conf 0.002]	258	■ 0.002 ● C +19	$\begin{array}{l} \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p,q} = \omega_{2m}^{j(2k)} = \omega_m^{jk}, \\ \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p+m,q} = \omega_{2m}^{(j+m)(2k)} = \omega_m^{(j+m)k}, \\ \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p,q+m} = \omega_{2m}^{j(2k+1)} = \omega_{2m}^j \omega_m^{jk}, \\ \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p+m,q+m} = \omega_{2m}^{j+m} \left(2k + 1 \right) = \omega_{2m}^{j+m} \omega_m^{j+m} k \\ (j + m \end{array}$	$\begin{array}{l} \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p,q} = \omega_{2m}^{j(2k)} = \omega_m^{jk}, \\ \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p+m,q} = \omega_{2m}^{(j+m)(2k)} = \omega_m^{(j+m)k}, \\ \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p,q+m} = \omega_{2m}^{j(2k+1)} = \omega_{2m}^j \omega_m^{jk}, \\ \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p+m,q+m} = \omega_{2m}^{(j+m)(2k+1)} = \omega_{2m}^{j+m} \omega_m^{(j+m)k} = -\omega_{2m}^j \omega_m^{jk}. \end{array}$	$\begin{array}{l} \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p,q} = \omega_{2m}^{j(2k)} = \omega_m^{jk}, \\ \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p+m,q} = \omega_{2m}^{(j+m)(2k)} = \omega_m^{(j+m)k}, \\ \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p,q+m} = \omega_{2m}^{j(2k+1)} = \omega_{2m}^j \omega_m^{jk}, \\ \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p+m,q+m} = \omega_{2m}^{(j+m)(2k+1)} = \omega_{2m}^{j+m} \omega_m^{(j+m)k} = -\omega_{2m}^j \omega_m^{jk}. \end{array}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra E00326 (5.18)	133	■0.118	$\begin{aligned} & \boldsymbol{Q}_1 := \left[\right. \\ & \begin{array}{lllll} \boldsymbol{q}_1, & \ldots, & \boldsymbol{q}_n, & \end{array} \\ & \left. \boldsymbol{R}_1 := \right. \\ & \boldsymbol{q}_j := \frac{\boldsymbol{v}_j}{\ \boldsymbol{v}_j\ _2}, \quad j = 1, \dots, n \text{ and} \\ & \left[\begin{array}{ccccc} \ \boldsymbol{v}_1\ _2 & \boldsymbol{a}_2^T \boldsymbol{q}_1 & \boldsymbol{a}_3^T \boldsymbol{q}_1 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_1 \\ 0 & \ \boldsymbol{v}_2\ _2 & \boldsymbol{a}_3^T \boldsymbol{q}_2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_2 \\ & 0 & \ \boldsymbol{v}_3\ _2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_3 \\ & & \ddots & \ddots & \vdots \\ & & & \ddots & \ \boldsymbol{v}_{n-1}\ _2 \\ & & & & 0 \end{array} \right] \end{aligned}$	$\boldsymbol{Q}_1 := [\boldsymbol{q}_1, \dots, \boldsymbol{q}_n], \quad \boldsymbol{q}_j := \frac{\boldsymbol{v}_j}{\ \boldsymbol{v}_j\ _2}, \quad j = 1, \dots, n \text{ and}$ $\boldsymbol{R}_1 := \begin{bmatrix} \ \boldsymbol{v}_1\ _2 & \boldsymbol{a}_2^T \boldsymbol{q}_1 & \boldsymbol{a}_3^T \boldsymbol{q}_1 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_1 \\ 0 & \ \boldsymbol{v}_2\ _2 & \boldsymbol{a}_3^T \boldsymbol{q}_2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_2 \\ & 0 & \ \boldsymbol{v}_3\ _2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_3 \\ & & \ddots & \ddots & \vdots \\ & & & \ddots & \ \boldsymbol{v}_{n-1}\ _2 \\ & & & & 0 \end{bmatrix}.$	
lyche-numerical-linear-algebra E00029	031	■0.138 S +0	$\begin{array}{c} \boldsymbol{a}_{11} x_1 + \boldsymbol{a}_{12} x_2 + \cdots + \boldsymbol{a}_{1n} x_n = b_1 \\ \boldsymbol{a}_{21} x_1 + \boldsymbol{a}_{22} x_2 + \cdots + \boldsymbol{a}_{2n} x_n = b_2 \\ \vdots \\ \boldsymbol{a}_{m1} x_1 + \boldsymbol{a}_{m2} x_2 + \cdots + \boldsymbol{a}_{mn} x_n = b_m \end{array}$	$\begin{array}{rcl} \boldsymbol{a}_{11}x_1 + \boldsymbol{a}_{12}x_2 + \cdots + \boldsymbol{a}_{1n}x_n & = & b_1 \\ \boldsymbol{a}_{21}x_1 + \boldsymbol{a}_{22}x_2 + \cdots + \boldsymbol{a}_{2n}x_n & = & b_2 \\ \vdots & & \vdots \\ \boldsymbol{a}_{m1}x_1 + \boldsymbol{a}_{m2}x_2 + \cdots + \boldsymbol{a}_{mn}x_n & = & b_m \end{array}$	$\begin{array}{l} \boldsymbol{a}_{11}x_1 + \boldsymbol{a}_{12}x_2 + \cdots + \boldsymbol{a}_{1n}x_n = b_1 \\ \boldsymbol{a}_{21}x_1 + \boldsymbol{a}_{22}x_2 + \cdots + \boldsymbol{a}_{2n}x_n = b_2 \\ \vdots \\ \boldsymbol{a}_{m1}x_1 + \boldsymbol{a}_{m2}x_2 + \cdots + \boldsymbol{a}_{mn}x_n = b_m \end{array}$

Conf.

MathPix confidence:

■ ≥0.9

■ 0.5–0.9

■ <0.5

— no value

LaTeX source

MathPix’s reading, unchanged;

Rendered

the same reading, with a display environment (align/algnat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual

render vs scan (inkdrill):

● C component

● W weak

● S stable

● N noise

● K clean

● not measured

[refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-229 EQ0630	229	■0.167 ●	$\begin{aligned} & \left[\begin{array}{cc} \boldsymbol{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \boldsymbol{0} \end{array} \right] \left[\begin{array}{c} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{array} \right] = \left[\begin{array}{c} \boldsymbol{A} \boldsymbol{v}_i \\ \boldsymbol{A}^* \boldsymbol{u}_i \end{array} \right] = \left[\begin{array}{c} \alpha_i \boldsymbol{u}_i \\ \alpha_i \boldsymbol{v}_i \end{array} \right] = \alpha_i \left[\begin{array}{c} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{array} \right], \quad i = 1, \dots, r, \\ & \left[\begin{array}{cc} \boldsymbol{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \boldsymbol{0} \end{array} \right] \left[\begin{array}{c} \boldsymbol{u}_i \\ -\boldsymbol{v}_i \end{array} \right] = \left[\begin{array}{c} -\boldsymbol{A} \boldsymbol{v}_i \\ \boldsymbol{A}^* \boldsymbol{u}_i \end{array} \right] = \left[\begin{array}{c} -\alpha_i \boldsymbol{u}_i \\ \alpha_i \boldsymbol{v}_i \end{array} \right] = -\alpha_i \left[\begin{array}{c} \boldsymbol{u}_i \\ -\boldsymbol{v}_i \end{array} \right], \quad i = 1, \dots, r, \\ & \left[\begin{array}{cc} \boldsymbol{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \boldsymbol{0} \end{array} \right] \left[\begin{array}{c} \boldsymbol{u}_i \\ \boldsymbol{0} \end{array} \right] = \left[\begin{array}{c} \boldsymbol{0} \\ \boldsymbol{A}^* \boldsymbol{u}_i \end{array} \right] = \left[\begin{array}{c} \boldsymbol{0} \\ \boldsymbol{0} \end{array} \right] = 0 \left[\begin{array}{c} \boldsymbol{u}_i \\ \boldsymbol{0} \end{array} \right], \quad i = r + 1, \dots, m, \\ & \left[\begin{array}{cc} \boldsymbol{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \boldsymbol{0} \end{array} \right] \left[\begin{array}{c} \boldsymbol{0} \\ \boldsymbol{v}_i \end{array} \right] = \left[\begin{array}{c} \boldsymbol{A} \boldsymbol{v}_i \\ \boldsymbol{0} \end{array} \right] = \left[\begin{array}{c} \boldsymbol{0} \\ \boldsymbol{0} \end{array} \right] = 0 \left[\begin{array}{c} \boldsymbol{0} \\ \boldsymbol{v}_i \end{array} \right], \quad i = r + 1, \dots, n \end{aligned}$	$\begin{aligned} & \left[\begin{array}{cc} \boldsymbol{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \boldsymbol{0} \end{array} \right] \left[\begin{array}{c} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{array} \right] = \left[\begin{array}{c} \boldsymbol{A} \boldsymbol{v}_i \\ \boldsymbol{A}^* \boldsymbol{u}_i \end{array} \right] = \left[\begin{array}{c} \alpha_i \boldsymbol{u}_i \\ \alpha_i \boldsymbol{v}_i \end{array} \right] = \alpha_i \left[\begin{array}{c} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{array} \right], \quad i = 1, \dots, r, \\ & \left[\begin{array}{cc} \boldsymbol{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \boldsymbol{0} \end{array} \right] \left[\begin{array}{c} \boldsymbol{u}_i \\ -\boldsymbol{v}_i \end{array} \right] = \left[\begin{array}{c} -\boldsymbol{A} \boldsymbol{v}_i \\ \boldsymbol{A}^* \boldsymbol{u}_i \end{array} \right] = \left[\begin{array}{c} -\alpha_i \boldsymbol{u}_i \\ \alpha_i \boldsymbol{v}_i \end{array} \right] = -\alpha_i \left[\begin{array}{c} \boldsymbol{u}_i \\ -\boldsymbol{v}_i \end{array} \right], \quad i = 1, \dots, r, \\ & \left[\begin{array}{cc} \boldsymbol{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \boldsymbol{0} \end{array} \right] \left[\begin{array}{c} \boldsymbol{u}_i \\ \boldsymbol{0} \end{array} \right] = \left[\begin{array}{c} \boldsymbol{0} \\ \boldsymbol{A}^* \boldsymbol{u}_i \end{array} \right] = \left[\begin{array}{c} \boldsymbol{0} \\ \boldsymbol{0} \end{array} \right] = 0 \left[\begin{array}{c} \boldsymbol{u}_i \\ \boldsymbol{0} \end{array} \right], \quad i = r + 1, \dots, m, \\ & \left[\begin{array}{cc} \boldsymbol{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \boldsymbol{0} \end{array} \right] \left[\begin{array}{c} \boldsymbol{0} \\ \boldsymbol{v}_i \end{array} \right] = \left[\begin{array}{c} \boldsymbol{A} \boldsymbol{v}_i \\ \boldsymbol{0} \end{array} \right] = \left[\begin{array}{c} \boldsymbol{0} \\ \boldsymbol{0} \end{array} \right] = 0 \left[\begin{array}{c} \boldsymbol{0} \\ \boldsymbol{v}_i \end{array} \right], \quad i = r + 1, \dots, n. \end{aligned}$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0369 (6.7)	150	0.298 • N +1	$\boldsymbol{J}:=\operatorname{diag}\left(\boldsymbol{U}_{\{1\}},\boldsymbol{U}_{\{2\}}\right)=\left[\begin{array}{cccccc} 2 & 1 & 0 & & & \\ 0 & 2 & 1 & & & \\ 0 & 0 & 2 & 1 & & \\ & & 0 & 1 & & \\ & & 0 & 2 & & \\ & & & & 0 & \\ & & & & 0 & 1 \end{array}\right] \in \mathbb{R}^{8 \times 8}.$	$\boldsymbol{J}:=\operatorname{diag}(\boldsymbol{U}_1,\boldsymbol{U}_2)=\left[\begin{array}{cccc} 2 & 1 & 0 & \\ 0 & 2 & 1 & \\ 0 & 0 & 2 & \\ & & 2 & 1 \\ & & & 3 & 1 \\ & & & 0 & 3 \end{array}\right] \in \mathbb{R}^{8 \times 8}.$	
lyche-numerical-linear-algebra- EQ0375	152	0.305 •	$\begin{aligned} &\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U}=\boldsymbol{W}^*\boldsymbol{V}^*\boldsymbol{A}\boldsymbol{V}\boldsymbol{W}=\left[\begin{array}{c c} 1 & \boldsymbol{0}^* \\ \hline \boldsymbol{0} & \boldsymbol{W}_1^* \end{array}\right]\left[\begin{array}{c c} \lambda_1 & \boldsymbol{x}^* \\ \hline \boldsymbol{0} & \boldsymbol{M} \end{array}\right]\left[\begin{array}{c c} 1 & \boldsymbol{0}^* \\ \hline \boldsymbol{0} & \boldsymbol{W}_1 \end{array}\right] \\ &=\left[\begin{array}{c c} \lambda_1 & \boldsymbol{x}^*\boldsymbol{W}_1 \\ \hline \boldsymbol{0} & \boldsymbol{W}_1^*\boldsymbol{M}\boldsymbol{W}_1 \end{array}\right] \end{aligned}$	$\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U}=\boldsymbol{W}^*(\boldsymbol{V}^*\boldsymbol{A}\boldsymbol{V})\boldsymbol{W}=\left[\begin{array}{c c} 1 & \boldsymbol{0}^* \\ \hline \boldsymbol{0} & \boldsymbol{W}_1^* \end{array}\right]\left[\begin{array}{c c} \lambda_1 & \boldsymbol{x}^* \\ \hline \boldsymbol{0} & \boldsymbol{M} \end{array}\right]\left[\begin{array}{c c} 1 & \boldsymbol{0}^* \\ \hline \boldsymbol{0} & \boldsymbol{W}_1 \end{array}\right]$ $=\left[\begin{array}{c c} \lambda_1 & \boldsymbol{x}^*\boldsymbol{W}_1 \\ \hline \boldsymbol{0} & \boldsymbol{W}_1^*\boldsymbol{M}\boldsymbol{W}_1 \end{array}\right]$	
lyche-numerical-linear-algebra- EQ0366 (6.4)	149	0.315 • C -4	$\boldsymbol{J}_{\{m\}}(\lambda):=\left[\begin{array}{cccccc} \lambda & 1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & 1 & \cdots & 0 & 0 \\ 0 & 0 & \lambda & \cdots & 0 & 0 \\ \vdots & & & \ddots & & \\ 0 & 0 & 0 & \cdots & \lambda & \lambda \\ 0 & 0 & 0 & \cdots & 0 & \lambda \end{array}\right]=\lambda\boldsymbol{I}_m+\boldsymbol{E}_m,\quad \boldsymbol{E}_m:=\left[\begin{array}{cccccc} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & & & \ddots & & \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{array}\right].$	$\boldsymbol{J}_m(\lambda):=\left[\begin{array}{cccccc} \lambda & 1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & 1 & \cdots & 0 & 0 \\ 0 & 0 & \lambda & \cdots & 0 & 0 \\ \vdots & & & \ddots & & \\ 0 & 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & 0 & \cdots & 0 & \lambda \end{array}\right]=\lambda\boldsymbol{I}_m+\boldsymbol{E}_m,\quad \boldsymbol{E}_m:=\left[\begin{array}{cccccc} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & & & \ddots & & \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{array}\right].$	
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ08579	215	0.322 W +1	$\begin{aligned} & \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x} = \left[\begin{array}{ccc} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{array} \right] \left[\begin{array}{c} 1 \\ \vdots \\ 1 \\ m \end{array} \right] \left[\begin{array}{c} x_1 \\ x_2 \end{array} \right] = \left[\begin{array}{cc} m & \sum t_k \\ \sum t_k & \sum t_k^2 \end{array} \right] \left[\begin{array}{c} x_1 \\ x_2 \end{array} \right], \\ & = \left[\begin{array}{ccc} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{array} \right] \left[\begin{array}{c} y_1 \\ \vdots \\ y_m \end{array} \right] = \left[\begin{array}{c} \sum y_k \\ \sum t_k y_k \end{array} \right] = \boldsymbol{A}^* \boldsymbol{b}, \end{aligned}$	$\begin{aligned} \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x} &= \begin{bmatrix} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 1 \\ m \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} m & \sum t_k \\ \sum t_k & \sum t_k^2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \\ &= \begin{bmatrix} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} = \begin{bmatrix} \sum y_k \\ \sum t_k y_k \end{bmatrix} = \boldsymbol{A}^* \boldsymbol{b}, \end{aligned}$	
lyche-numerical-linear-algebra EQ0835 (5.19)	135	0.329	$\left[\begin{array}{l} \boldsymbol{B}(i,:) \\ \boldsymbol{B}(j,:) \end{array} \right] = \left[\begin{array}{cc} c & s \\ -s & c \end{array} \right] \left[\begin{array}{c} \boldsymbol{A}(i,:) \\ \boldsymbol{A}(j,:) \end{array} \right], [\boldsymbol{C}(:,i) \boldsymbol{C}(:,j)] = [\boldsymbol{A}(:,i) \boldsymbol{A}(:,j)] \left[\begin{array}{cc} c & s \\ -s & c \end{array} \right].$	$\left[\begin{array}{c} \boldsymbol{B}(i,:) \\ \boldsymbol{B}(j,:) \end{array} \right] = \left[\begin{array}{cc} c & s \\ -s & c \end{array} \right] \left[\begin{array}{c} \boldsymbol{A}(i,:) \\ \boldsymbol{A}(j,:) \end{array} \right], [\boldsymbol{C}(:,i) \boldsymbol{C}(:,j)] = [\boldsymbol{A}(:,i) \boldsymbol{A}(:,j)] \left[\begin{array}{cc} c & s \\ -s & c \end{array} \right].$	
lyche-numerical-linear-algebra EQ0917	312	0.371 W +2	$\frac{\ \boldsymbol{x} - \boldsymbol{x}_k\ _{\boldsymbol{A}}}{\ \boldsymbol{x} - \boldsymbol{x}_0\ _{\boldsymbol{A}}} \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k \quad \text{for } k \geq 0$	$\frac{\ \boldsymbol{x} - \boldsymbol{x}_k\ _{\boldsymbol{A}}}{\ \boldsymbol{x} - \boldsymbol{x}_0\ _{\boldsymbol{A}}} \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k, \quad \text{for } k \geq 0,$	
lyche-numerical-linear-algebra EQ0547	207	0.373 N +0	$c_{i,k+1} = \begin{cases} 0 & \text{if } \sigma_i = \lambda_{k+1}, \\ c_{i,k} \left(1 - \frac{\sigma_i}{\lambda_{k+1}} \right) & \text{otherwise} \end{cases}$	$c_{i,k+1} = \begin{cases} 0 & \text{if } \sigma_i = \lambda_{k+1}, \\ c_{i,k} \left(1 - \frac{\sigma_i}{\lambda_{k+1}} \right) & \text{otherwise} \end{cases}$	
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0855	296	0.386 W +1	$\begin{array}{lll} \boldsymbol{t}_0=\left[\begin{array}{c} 3 \\ -3/2 \end{array} \right], & \alpha_0=1/2, & \boldsymbol{x}_1=\left[\begin{array}{c} -1/4 \\ -1/2 \end{array} \right], & \boldsymbol{r}_1=\left[\begin{array}{c} 0 \\ 3/4 \end{array} \right], & \beta_0=1/4, \\ \boldsymbol{p}_1=\left[\begin{array}{c} 3/8 \\ 3/4 \end{array} \right], & \boldsymbol{t}_1=\left[\begin{array}{c} 0 \\ 9/8 \end{array} \right], & \alpha_1=2/3, & \boldsymbol{x}_2=\mathbf{0}, & \boldsymbol{r}_2=\mathbf{0}. \end{array}$	$\begin{array}{l} \boldsymbol{t}_0=\left[\begin{array}{c} 3 \\ -3/2 \end{array} \right], \quad \alpha_0=1/2, \quad \boldsymbol{x}_1=\left[\begin{array}{c} -1/4 \\ -1/2 \end{array} \right], \quad \boldsymbol{r}_1=\left[\begin{array}{c} 0 \\ 3/4 \end{array} \right], \quad \beta_0=1/4, \\ \boldsymbol{p}_1=\left[\begin{array}{c} 3/8 \\ 3/4 \end{array} \right], \quad \boldsymbol{t}_1=\left[\begin{array}{c} 0 \\ 9/8 \end{array} \right], \quad \alpha_1=2/3, \quad \boldsymbol{x}_2=\mathbf{0}, \quad \boldsymbol{r}_2=\mathbf{0}. \end{array}$	
lyche-numerical-linear-algebra EQ0670 (10.10)	241	0.390 W +2	$\begin{aligned} a_{ii} &= 2d, i=1, \ldots, n, \\ a_{i,i+1} &= a_{i+1,i} = a, \quad i=1, \ldots, n-1, \quad i \neq m, 2m, \ldots, (m-1)m, \\ a_{i,i+m} &= a_{i+m,i} = a, \quad i=1, \ldots, n-m, \\ a_{ij} &= 0, \text{ otherwise} \end{aligned}$	$\begin{array}{l} a_{ii} = 2d, i = 1, \ldots, n, \\ a_{i,i+1} = a_{i+1,i} = a, \quad i = 1, \ldots, n-1, \quad i \neq m, 2m, \ldots, (m-1)m, \\ a_{i,i+m} = a_{i+m,i} = a, \quad i = 1, \ldots, n-m, \\ a_{ij} = 0, \text{ otherwise} \end{array}$	$\begin{array}{l} a_{ii} = 2d, i = 1, \ldots, n, \\ a_{i,i+1} = a_{i+1,i} = a, \quad i = 1, \ldots, n-1, \quad i \neq m, 2m, \ldots, (m-1)m, \\ a_{i,i+m} = a_{i+m,i} = a, \quad i = 1, \ldots, n-m, \\ a_{ij} = 0, \text{ otherwise}, \end{array}$
lyche-numerical-linear-algebra EQ0763 (12.14)	271	0.402 N +0	$\boldsymbol{G}_{\omega}=\left[\begin{array}{ll} l & 0 \\ 0 & 1 \end{array} \right]-\left[\begin{array}{cc} \omega/2 & 0 \\ \omega^2/4 & \omega/2 \end{array} \right]\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array} \right]=\left[\begin{array}{cc} 1-\omega & \omega/2 \\ \omega(1-\omega)/2 & 1-\omega+\omega^2/4 \end{array} \right].$	$\boldsymbol{G}_{\omega}=\left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right]-\left[\begin{array}{cc} \omega/2 & 0 \\ \omega^2/4 & \omega/2 \end{array} \right]\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array} \right]=\left[\begin{array}{cc} 1-\omega & \omega/2 \\ \omega(1-\omega)/2 & 1-\omega+\omega^2/4 \end{array} \right].$	$\boldsymbol{G}_{\omega}=\left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right]-\left[\begin{array}{cc} \omega/2 & 0 \\ \omega^2/4 & \omega/2 \end{array} \right]\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array} \right]=\left[\begin{array}{cc} 1-\omega & \omega/2 \\ \omega(1-\omega)/2 & 1-\omega+\omega^2/4 \end{array} \right].$
lyche-numerical-linear-algebra EQ0498	194	0.420 C +4	$\begin{aligned} & \& x_{\{1\}}+\left(1\cdot10^{\{-16\}}\right) x_{\{2\}}=20 \\ & \& x_{\{1\}}+\left(1\cdot10^{\{-15\}}\right) \end{aligned}$	$\begin{array}{l} x_1 + (1 - 10^{-16}) x_2 = 20 \\ x_1 + (1 - 10^{-15} \end{array}$	$\begin{array}{l} x_1 + \qquad \qquad \qquad x_2 = 20 \\ x_1 + (1 - 10^{-16})x_2 = 20 - 10^{-15} \end{array}$
lyche-numerical-linear-algebra EQ0262	107	0.421	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \boldsymbol{z}^* \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} \boldsymbol{z} = \boldsymbol{z}^* \text{diag}(\lambda_1, \ldots, \lambda_n) \boldsymbol{z} = \sum_{j=1}^n \lambda_j z_j ^2 \geq 0.$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \boldsymbol{z}^* \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} \boldsymbol{z} = \boldsymbol{z}^* \text{diag}(\lambda_1, \ldots, \lambda_n) \boldsymbol{z} = \sum_{j=1}^n \lambda_j z_j ^2 \geq 0.$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \boldsymbol{z}^* \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} \boldsymbol{z} = \boldsymbol{z}^* \text{diag}(\lambda_1, \ldots, \lambda_n) \boldsymbol{z} = \sum_{j=1}^n \lambda_j z_j ^2 \geq 0.$
lyche-numerical-linear-algebra EQ0404	161	0.448 C +10	$\left[\begin{array}{llllllll} 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & \end{array} \right] \in \mathbb{R}^{8,8}?$	$\left[\begin{array}{cccccccc} 2 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 3 & \end{array} \right] \in \mathbb{R}^{8,8}?$	$\left[\begin{array}{cccccccc} 2 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 3 & \end{array} \right] \in \mathbb{R}^{8,8}?$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0914 (13.48)	311	0.453 W +0	$\begin{aligned} & \& \boldsymbol{y}_{k+1}= \\ & \boldsymbol{y}_{k+1}=\boldsymbol{y}_k+\alpha_k\boldsymbol{q}_k, \quad \boldsymbol{q}_k, \quad \quad \\ & \alpha_k=z_k^Tz_k/\boldsymbol{q}_k^T(\boldsymbol{C}\boldsymbol{A}\boldsymbol{C}^T)\boldsymbol{q}_k, \\ & \boldsymbol{z}_{k+1}=\boldsymbol{z}_k-\alpha_k(\boldsymbol{C}\boldsymbol{A}\boldsymbol{C}^T)\boldsymbol{q}_k, \\ & \boldsymbol{q}_{k+1}=\boldsymbol{z}_{k+1}+\beta_k\boldsymbol{q}_k, \quad \beta_k=z_{k+1}^Tz_{k+1}/z_k^Tz_k. \end{aligned}$	$\boldsymbol{y}_{k+1} = \boldsymbol{y}_k + \alpha_k \boldsymbol{q}_k, \quad \alpha_k = z_k^T z_k / \boldsymbol{q}_k^T (\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T) \boldsymbol{q}_k, \\ \boldsymbol{z}_{k+1} = \boldsymbol{z}_k - \alpha_k (\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T) \boldsymbol{q}_k, \\ \boldsymbol{q}_{k+1} = \boldsymbol{z}_{k+1} + \beta_k \boldsymbol{q}_k, \quad \beta_k = z_{k+1}^T z_{k+1} / z_k^T z_k.$	
lyche-numerical-linear-algebra-EQ0245	100	0.458 N -1	$\left[\begin{array}{ll} a & \bar{b} \\ b & d \end{array}\right] = \left[\begin{array}{ll} 1 & 0 \\ l_1 & 1 \end{array}\right] \left[\begin{array}{ll} d_1 & 0 \\ 0 & d_2 \end{array}\right] \left[\begin{array}{ll} 1 & \bar{l}_1 \\ 0 & 1 \end{array}\right] = \left[\begin{array}{ll} d_1 & d_1 \bar{l}_1 \\ d_1 l_1 & d_1 l_1 ^2 + d_2 \end{array}\right]$	$\begin{bmatrix} a & \bar{b} \\ b & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ l_1 & 1 \end{bmatrix} \begin{bmatrix} d_1 & 0 \\ 0 & d_2 \end{bmatrix} \begin{bmatrix} 1 & \bar{l}_1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} d_1 & d_1 \bar{l}_1 \\ d_1 l_1 & d_1 l_1 ^2 + d_2 \end{bmatrix}$	
lyche-numerical-linear-algebra-EQ0370	150	0.466 N +1	$\begin{aligned} & \& \boldsymbol{s}_1=2 \\ & \boldsymbol{s}_1, \quad \quad \boldsymbol{A} \\ & \boldsymbol{s}_2=\boldsymbol{s}_1+2 \\ & \boldsymbol{s}_2, \quad \quad \boldsymbol{A} \\ & \boldsymbol{s}_3=\boldsymbol{s}_2+2 \\ & \boldsymbol{s}_3, \quad \& \boldsymbol{A} \\ & \boldsymbol{s}_4=2 \boldsymbol{s}_4, \quad \quad \boldsymbol{A} \\ & \boldsymbol{s}_5=\boldsymbol{s}_4+2 \\ & \boldsymbol{s}_5, \quad \& \boldsymbol{A} \\ & \boldsymbol{s}_6=2 \boldsymbol{s}_6, \quad \& \\ & \boldsymbol{s}_7=3 \boldsymbol{s}_7, \quad \boldsymbol{A} \\ & \boldsymbol{s}_8=\boldsymbol{s}_7+3 \boldsymbol{s}_8. \end{aligned}$	$\boldsymbol{A} \boldsymbol{s}_1 = 2 \boldsymbol{s}_1, \quad \boldsymbol{A} \boldsymbol{s}_2 = \boldsymbol{s}_1 + 2 \boldsymbol{s}_2, \quad \boldsymbol{A} \boldsymbol{s}_3 = \boldsymbol{s}_2 + 2 \boldsymbol{s}_3, \\ \boldsymbol{A} \boldsymbol{s}_4 = 2 \boldsymbol{s}_4, \quad \boldsymbol{A} \boldsymbol{s}_5 = \boldsymbol{s}_4 + 2 \boldsymbol{s}_5 \\ \boldsymbol{A} \boldsymbol{s}_6 = 2 \boldsymbol{s}_6, \\ \boldsymbol{A} \boldsymbol{s}_7 = 3 \boldsymbol{s}_7, \quad \boldsymbol{A} \boldsymbol{s}_8 = \boldsymbol{s}_7 + 3 \boldsymbol{s}_8.$	

Conf.

MathPix confidence:

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0396	158	0.467	$\begin{aligned} \beta_k &\leq \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{B}}(\mathbf{x}) \leq \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{A}}(\mathbf{x}) + \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{E}}(\mathbf{x}) \\ &\leq \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{A}}(\mathbf{x}) + \max_{\substack{\mathbf{x} \in \mathbb{C}^n \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{E}}(\mathbf{x}) = \alpha_k + \varepsilon_1, \end{aligned}$	$\beta_k \leq \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{B}}(\mathbf{x}) \leq \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{A}}(\mathbf{x}) + \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{E}}(\mathbf{x})$ $\leq \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{A}}(\mathbf{x}) + \max_{\substack{\mathbf{x} \in \mathbb{C}^n \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{E}}(\mathbf{x}) = \alpha_k + \varepsilon_1,$	$\beta_k \leq \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{B}}(\mathbf{x}) \leq \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{A}}(\mathbf{x}) + \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{E}}(\mathbf{x})$ $\leq \max_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{A}}(\mathbf{x}) + \max_{\substack{\mathbf{x} \in \mathbb{C}^n \\ \mathbf{x} \neq \mathbf{0}}} R_{\mathbf{E}}(\mathbf{x}) = \alpha_k + \varepsilon_1,$
lyche-numerical-linear-algebra- EQ0308	125	0.469 N +0	$\mathbf{P}\mathbf{x} = \mathbf{x} - \frac{\mathbf{v}^* \mathbf{x}}{\mathbf{v}^* \mathbf{v}} \mathbf{v} \stackrel{(5.13)}{=} \mathbf{x} - \frac{1}{2} \mathbf{v} = \frac{1}{2}(\mathbf{x} + \mathbf{y}).$	$\mathbf{P}\mathbf{x} = \mathbf{x} - \frac{\mathbf{v}^* \mathbf{x}}{\mathbf{v}^* \mathbf{v}} \mathbf{v} \stackrel{(5.13)}{=} \mathbf{x} - \frac{1}{2} \mathbf{v} = \frac{1}{2}(\mathbf{x} + \mathbf{y}).$	$\mathbf{P}\mathbf{x} = \mathbf{x} - \frac{\mathbf{v}^* \mathbf{x}}{\mathbf{v}^* \mathbf{v}} \mathbf{v} \stackrel{(5.13)}{=} \mathbf{x} - \frac{1}{2} \mathbf{v} = \frac{1}{2}(\mathbf{x} + \mathbf{y}).$
lyche-numerical-linear-algebra- EQ0600 (0)	221	0.474 N +0	$\left[\begin{array}{lll} 2 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{array} \right] \left[\begin{array}{l} x_1 \\ x_2 \\ x_3 \end{array} \right] = \frac{1}{2} \left[\begin{array}{llll} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ -1 & 1 & -1 & 1 \end{array} \right] \times \left[\begin{array}{l} 1 \\ 1 \\ 1 \\ 1 \end{array} \right] = \left[\begin{array}{l} 2 \\ 0 \\ 0 \end{array} \right],$	$\left[\begin{array}{lll} 2 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{array} \right] \left[\begin{array}{l} x_1 \\ x_2 \\ x_3 \end{array} \right] = \frac{1}{2} \left[\begin{array}{llll} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ -1 & 1 & -1 & 1 \end{array} \right] \times \left[\begin{array}{l} 1 \\ 1 \\ 1 \\ 1 \end{array} \right] = \left[\begin{array}{l} 2 \\ 0 \\ 0 \end{array} \right],$	$\left[\begin{array}{lll} 2 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{array} \right] \left[\begin{array}{l} x_1 \\ x_2 \\ x_3 \end{array} \right] = \frac{1}{2} \left[\begin{array}{llll} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ -1 & 1 & -1 & 1 \end{array} \right] \times \left[\begin{array}{l} 1 \\ 1 \\ 1 \\ 1 \end{array} \right] = \left[\begin{array}{l} 2 \\ 0 \\ 0 \end{array} \right],$
lyche-numerical-linear-algebra- EQ0028 (1.11)	031	0.479 N +0	$ x _2 := \left(\sum_{j=1}^n x_j ^2 \right)^{1/2} = \sqrt{x^* x}.$	$\ x\ _2 := \left(\sum_{j=1}^n x_j ^2 \right)^{1/2} = \sqrt{x^* x}.$	$\ \mathbf{x}\ _2 := \left(\sum_{j=1}^n x_j ^2 \right)^{1/2} = \sqrt{\mathbf{x}^* \mathbf{x}}.$
lyche-numerical-linear-algebra- EQ0241 (3.33)	097	0.492 C +5	$\mathbf{B} := \left[\begin{array}{cc} (1 + \mathbf{u}_1^T \mathbf{B}_{2,2} \boldsymbol{\ell}_1) / u_{1,1} & -\mathbf{u}_1^T \mathbf{B}_{2,2} / u_{1,1} \\ -\mathbf{u}_{2,2}^T \mathbf{B}_{2,2} \boldsymbol{\ell}_1 & \mathbf{B}_{2,2} \end{array} \right].$	$\mathbf{B} := \left[\begin{array}{cc} (1 + \mathbf{u}_1^T \mathbf{B}_{2,2} \boldsymbol{\ell}_1) / u_{1,1} & -\mathbf{u}_1^T \mathbf{B}_{2,2} / u_{1,1} \\ -\mathbf{B}_{2,2} \boldsymbol{\ell}_1 & \mathbf{B}_{2,2} \end{array} \right].$	$\mathbf{B} := \left[\begin{array}{cc} (1 + \mathbf{u}_1^T \mathbf{B}_{2,2} \boldsymbol{\ell}_1) / u_{1,1} & -\mathbf{u}_1^T \mathbf{B}_{2,2} / u_{1,1} \\ -\mathbf{B}_{2,2} \boldsymbol{\ell}_1 & \mathbf{B}_{2,2} \end{array} \right].$
lyche-numerical-linear-algebra- EQ0757 (12.11)	270	0.498 C +6	$\mathbf{A}_{\mathbf{L}} := \left[\begin{array}{ccc} 0 & & \\ -a_{2,1} & 0 & \\ \vdots & \ddots & \ddots \\ -a_{n,1} & \cdots & -a_{n,n-1} \end{array} \right], \quad \mathbf{A}_{\mathbf{R}} := \left[\begin{array}{ccc} 0 & -a_{1,2} & \cdots \\ \ddots & \ddots & \vdots \\ & 0 & -a_{n-1,n} \\ & & 0 \end{array} \right].$	$\mathbf{A}_{\mathbf{L}} := \left[\begin{array}{ccc} 0 & & \\ -a_{2,1} & 0 & \\ \vdots & \ddots & \ddots \\ -a_{n,1} & \cdots & -a_{n,n-1} & 0 \end{array} \right], \quad \mathbf{A}_{\mathbf{R}} := \left[\begin{array}{ccc} 0 & -a_{1,2} & \cdots \\ \ddots & \ddots & \vdots \\ & 0 & -a_{n-1,n} \\ & & 0 \end{array} \right].$	$\mathbf{A}_{\mathbf{L}} := \left[\begin{array}{ccc} 0 & & \\ -a_{2,1} & 0 & \\ \vdots & \ddots & \ddots \\ -a_{n,1} & \cdots & -a_{n,n-1} & 0 \end{array} \right], \quad \mathbf{A}_{\mathbf{R}} := \left[\begin{array}{ccc} 0 & -a_{1,2} & \cdots \\ \ddots & \ddots & \vdots \\ & 0 & -a_{n-1,n} \\ & & 0 \end{array} \right].$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0138	164	0.503	$\left[\begin{array}{ll} A_{11} & A_{12} \\ A_{21} & A_{22} \end{array}\right] \left[\begin{array}{ll} B_{11} & B_{12} \\ B_{21} & B_{22} \end{array}\right] = \left[\begin{array}{ll} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{array}\right],$	$\left[\begin{array}{ll} A_{11} & A_{12} \\ A_{21} & A_{22} \end{array}\right] \left[\begin{array}{ll} B_{11} & B_{12} \\ B_{21} & B_{22} \end{array}\right] = \left[\begin{array}{ll} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{array}\right],$	$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{bmatrix},$
lyche-numerical-linear-algebra-EQ0465 (8.1)	186	0.507 N +0	$\begin{aligned} & \ \mathbf{x}\ _p := \left(\sum_{j=1}^n x_j ^p \right)^{1/p}, \\ & \ \mathbf{x}\ _\infty := \max_{1 \leq j \leq n} x_j . \end{aligned}$	$\ \mathbf{x}\ _p := \left(\sum_{j=1}^n x_j ^p \right)^{1/p},$ $\ \mathbf{x}\ _\infty := \max_{1 \leq j \leq n} x_j .$	$\ \mathbf{x}\ _p := \left(\sum_{j=1}^n x_j ^p \right)^{1/p},$ $\ \mathbf{x}\ _\infty := \max_{1 \leq j \leq n} x_j .$
lyche-numerical-linear-algebra-EQ0486	191	0.508 N -1	$\ x\ _p := \left(\sum_{j=1}^n x_j ^p \right)^{1/p}, p \geq 1, \quad \ x\ _\infty := \max_{1 \leq j \leq n} x_j .$	$\ x\ _p := \left(\sum_{j=1}^n x_j ^p \right)^{1/p}, p \geq 1, \quad \ x\ _\infty := \max_{1 \leq j \leq n} x_j .$	$\ \mathbf{x}\ _p := \left(\sum_{j=1}^n x_j ^p \right)^{1/p}, p \geq 1, \quad \ \mathbf{x}\ _\infty := \max_{1 \leq j \leq n} x_j .$
lyche-numerical-linear-algebra-EQ0766	272	0.512 N +0	$x_{k+1}(2) = \frac{1}{2} \left(\frac{1}{2} x_k(2) + \frac{1}{2} \right) + \frac{1}{2} = \frac{1}{4} x_k(2) + \frac{3}{4}.$	$x_{k+1}(2) = \frac{1}{2} \left(\frac{1}{2} x_k(2) + \frac{1}{2} \right) + \frac{1}{2} = \frac{1}{4} x_k(2) + \frac{3}{4}.$	$\mathbf{x}_{k+1}(2) = \frac{1}{2} \left(\frac{1}{2} \mathbf{x}_k(2) + \frac{1}{2} \right) + \frac{1}{2} = \frac{1}{4} \mathbf{x}_k(2) + \frac{3}{4}.$
lyche-numerical-linear-algebra-EQ1017	345	0.514 K +0	$z_0 = \frac{1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2.$	$z_0 = \frac{1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2.$	$\mathbf{z}_0 = \frac{1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2.$
lyche-numerical-linear-algebra-EQ0798	280	0.516 W +1	$\begin{aligned} & \ \mathbf{x}_k - \mathbf{x}\ = \ \mathbf{G}(\mathbf{x}_{k-1} - \mathbf{x})\ \leq \ \mathbf{G}\ \ \mathbf{x}_{k-1} - \mathbf{x}\ \\ & = \ \mathbf{G}\ \ \mathbf{x}_{k-1} - \mathbf{x}_k + \mathbf{x}_k - \mathbf{x}\ \leq \ \mathbf{G}\ (\ \mathbf{x}_{k-1} - \mathbf{x}_k\ + \ \mathbf{x}_k - \mathbf{x}\) \end{aligned}$	$\ \mathbf{x}_k - \mathbf{x}\ = \ \mathbf{G}(\mathbf{x}_{k-1} - \mathbf{x})\ \leq \ \mathbf{G}\ \ \mathbf{x}_{k-1} - \mathbf{x}\ $ $= \ \mathbf{G}\ \ \mathbf{x}_{k-1} - \mathbf{x}_k + \mathbf{x}_k - \mathbf{x}\ \leq \ \mathbf{G}\ (\ \mathbf{x}_{k-1} - \mathbf{x}_k\ + \ \mathbf{x}_k - \mathbf{x}\)$	$\ \mathbf{x}_k - \mathbf{x}\ = \ \mathbf{G}(\mathbf{x}_{k-1} - \mathbf{x})\ \leq \ \mathbf{G}\ \ \mathbf{x}_{k-1} - \mathbf{x}\ $ $= \ \mathbf{G}\ \ \mathbf{x}_{k-1} - \mathbf{x}_k + \mathbf{x}_k - \mathbf{x}\ \leq \ \mathbf{G}\ (\ \mathbf{x}_{k-1} - \mathbf{x}_k\ + \ \mathbf{x}_k - \mathbf{x}\).$
lyche-numerical-linear-algebra-EQ0837 (13.3)	291	0.516 N +0	$Q(\mathbf{y}) := \frac{1}{2} \mathbf{y}^T \mathbf{A} \mathbf{y} - \mathbf{b}^T \mathbf{y} + c.$	$Q(\mathbf{y}) := \frac{1}{2} \mathbf{y}^T \mathbf{A} \mathbf{y} - \mathbf{b}^T \mathbf{y} + c.$	$Q(\mathbf{y}) := \frac{1}{2} \mathbf{y}^T \mathbf{A} \mathbf{y} - \mathbf{b}^T \mathbf{y} + c.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0854 (13.18)	296	0.521	$\begin{aligned} \boldsymbol{t}_k &= \boldsymbol{A} \boldsymbol{p}_k, \\ \boldsymbol{p}_{k+1} &= \boldsymbol{r}_k - \alpha_k \boldsymbol{p}_k, \\ \boldsymbol{r}_{k+1} &= \boldsymbol{r}_k - \alpha_k \boldsymbol{t}_k, \\ \beta_k &= (\boldsymbol{r}_{k+1}^T \boldsymbol{r}_{k+1}) / (\boldsymbol{r}_k^T \boldsymbol{r}_k), \\ \boldsymbol{p}_{k+1} &:= \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k. \end{aligned}$	$\begin{aligned} \boldsymbol{t}_k &= \boldsymbol{A} \boldsymbol{p}_k, \\ \alpha_k &= (\boldsymbol{r}_k^T \boldsymbol{r}_k) / (\boldsymbol{p}_k^T \boldsymbol{t}_k), \\ \boldsymbol{x}_{k+1} &= \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k, \\ \boldsymbol{r}_{k+1} &= \boldsymbol{r}_k - \alpha_k \boldsymbol{t}_k, \\ \beta_k &= (\boldsymbol{r}_{k+1}^T \boldsymbol{r}_{k+1}) / (\boldsymbol{r}_k^T \boldsymbol{r}_k), \\ \boldsymbol{p}_{k+1} &:= \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k. \end{aligned}$	$\begin{aligned} \boldsymbol{t}_k &= \boldsymbol{A} \boldsymbol{p}_k, \\ \alpha_k &= (\boldsymbol{r}_k^T \boldsymbol{r}_k) / (\boldsymbol{p}_k^T \boldsymbol{t}_k), \\ \boldsymbol{x}_{k+1} &= \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k, \\ \boldsymbol{r}_{k+1} &= \boldsymbol{r}_k - \alpha_k \boldsymbol{p}_k, \\ \boldsymbol{r}_{k+1} &= \boldsymbol{r}_k - \alpha_k \boldsymbol{t}_k, \\ \beta_k &= (\boldsymbol{r}_{k+1}^T \boldsymbol{r}_{k+1}) / (\boldsymbol{r}_k^T \boldsymbol{r}_k), \\ \boldsymbol{p}_{k+1} &:= \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k. \end{aligned}$
lyche-numerical-linear-algebra-EQ0329	134	0.523 ● N -2	$\boldsymbol{A} = \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{bmatrix}.$	$\boldsymbol{A} = \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{bmatrix}.$	$\boldsymbol{A} = \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{bmatrix}.$
lyche-numerical-linear-algebra-EQ0862	300	0.523 ● W -1	$\kappa_P = \frac{\lambda_{\max}}{\lambda_{\min}} = \frac{\cos^2(\pi h/2)}{\sin^2(\pi h/2)} \text{ and } \sqrt{\kappa_P} = \frac{\cos(\pi h/2)}{\sin(\pi h/2)} \approx \frac{2}{\pi h} \approx \frac{2}{\pi} \sqrt{n}.$	$\kappa_P = \frac{\lambda_{\max}}{\lambda_{\min}} = \frac{\cos^2(\pi h/2)}{\sin^2(\pi h/2)} \text{ and } \sqrt{\kappa_P} = \frac{\cos(\pi h/2)}{\sin(\pi h/2)} \approx \frac{2}{\pi h} \approx \frac{2}{\pi} \sqrt{n}.$	$\kappa_P = \frac{\lambda_{\max}}{\lambda_{\min}} = \frac{\cos^2(\pi h/2)}{\sin^2(\pi h/2)} \text{ and } \sqrt{\kappa_P} = \frac{\cos(\pi h/2)}{\sin(\pi h/2)} \approx \frac{2}{\pi h} \approx \frac{2}{\pi} \sqrt{n}.$
lyche-numerical-linear-algebra-EQ0504 (8.26)	197	0.531 ● W +0	$K_2(\boldsymbol{A}) = \begin{cases} \lambda_1 / \lambda_n, & \text{if } \boldsymbol{A} \text{ is positive definite,} \\ \lambda_1 / \lambda_n , & \text{if } \boldsymbol{A} \text{ is normal,} \\ \sigma_1 / \sigma_n, & \text{in general.} \end{cases}$	$K_2(\boldsymbol{A}) = \begin{cases} \lambda_1 / \lambda_n, & \text{if } \boldsymbol{A} \text{ is positive definite,} \\ \lambda_1 / \lambda_n , & \text{if } \boldsymbol{A} \text{ is normal,} \\ \sigma_1 / \sigma_n, & \text{in general.} \end{cases}$	$K_2(\boldsymbol{A}) = \begin{cases} \lambda_1 / \lambda_n, & \text{if } \boldsymbol{A} \text{ is positive definite,} \\ \lambda_1 / \lambda_n , & \text{if } \boldsymbol{A} \text{ is normal,} \\ \sigma_1 / \sigma_n, & \text{in general.} \end{cases}$
lyche-numerical-linear-algebra-EQ0830 (12.46)	287	0.531 ● N +0	$\boldsymbol{A} = \boldsymbol{M} - \boldsymbol{N}, \quad \boldsymbol{M} = (\boldsymbol{I} - \boldsymbol{L})(\boldsymbol{I} - \boldsymbol{L}^T), \quad \boldsymbol{N} = \boldsymbol{L} \boldsymbol{L}^T.$	$\boldsymbol{A} = \boldsymbol{M} - \boldsymbol{N}, \quad \boldsymbol{M} = (\boldsymbol{I} - \boldsymbol{L})(\boldsymbol{I} - \boldsymbol{L}^T), \quad \boldsymbol{N} = \boldsymbol{L} \boldsymbol{L}^T.$	$\boldsymbol{A} = \boldsymbol{M} - \boldsymbol{N}, \quad \boldsymbol{M} = (\boldsymbol{I} - \boldsymbol{L})(\boldsymbol{I} - \boldsymbol{L}^T), \quad \boldsymbol{N} = \boldsymbol{L} \boldsymbol{L}^T.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ00998	054	0.532 C +3	$\begin{aligned} & \texttt{\textbackslash begin{aligned}} \ \& \texttt{u_{1}=d_{1}}, \ \texttt{\quad l_{1}=a_{1} \ / \ u_{1}}, \\ & \texttt{\quad u_{2}=d_{2}-l_{1} \ c_{1}}, \ \texttt{\quad l_{2}=a_{2} \ / \ u_{2}}, \\ & \texttt{\quad u_{3}=d_{3}-l_{2} \ c_{2}} \ \& \ \texttt{z_{1}=b_{1}}, \ \texttt{\quad} \\ & \texttt{z_{2}=b_{2}-l_{1} \ z_{1}}, \ \texttt{\quad} \texttt{z_{3}=b_{3}-l_{2} \ z_{2}} \\ & \texttt{\quad} \& \ \texttt{x_{3}=z_{3} \ / \ u_{3}}, \ \texttt{\quad} \texttt{x_{2}=\left(z_{2}-c_{2} \ x_{3}} \ / \ u_{2}}, \\ & \texttt{\quad} \texttt{x_{1}=\left(z_{1}-c_{1} \ x_{2}} \ / \ u_{1}} \ \texttt{\end{aligned}}$	$\begin{aligned} u_1 &= d_1, \quad l_1 = a_1/u_1, \quad u_2 = d_2 - l_1 c_1, \quad l_2 = a_2/u_2, \quad u_3 = d_3 - l_2 c_2 \\ z_1 &= b_1, \quad z_2 = b_2 - l_1 z_1, \quad z_3 = b_3 - l_2 z_2 \\ x_3 &= z_3/u_3, \quad x_2 = (z_2 - c_2 x_3)/u_2, \quad x_1 = (z_1 - c_1 x_2)/u_1 \end{aligned}$	$\begin{aligned} u_1 &= d_1, \quad l_1 = a_1/u_1, \quad u_2 = d_2 - l_1 c_1, \quad l_2 = a_2/u_2, \quad u_3 = d_3 - l_2 c_2, \\ z_1 &= b_1, \quad z_2 = b_2 - l_1 z_1, \quad z_3 = b_3 - l_2 z_2, \\ x_3 &= z_3/u_3, \quad x_2 = (z_2 - c_2 x_3)/u_2, \quad x_1 = (z_1 - c_1 x_2)/u_1. \end{aligned}$
lyche-numerical-linear-algebra-EQ0836 (13.2)	291	0.546	$\left \left(\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y} \right) \boldsymbol{A} \right ^2 \leq \left(\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} \right) \left(\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y} \right), \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n.$	$ \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y} ^2 \leq (\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x})(\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}), \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n.$	
lyche-numerical-linear-algebra-EQ0868	301	0.546 N +0	$\left \left \boldsymbol{x} - \boldsymbol{x}_0 - \boldsymbol{w} \right _{\boldsymbol{A}} - \left \boldsymbol{x} - \boldsymbol{x}_k + \boldsymbol{u} \right _{\boldsymbol{A}} \right \geq \left \boldsymbol{x} - \boldsymbol{x}_k \right _{\boldsymbol{A}},$	$\ \boldsymbol{x} - \boldsymbol{x}_0 - \boldsymbol{w}\ _{\boldsymbol{A}} = \ \boldsymbol{x} - \boldsymbol{x}_k + \boldsymbol{u}\ _{\boldsymbol{A}} \geq \ \boldsymbol{x} - \boldsymbol{x}_k\ _{\boldsymbol{A}},$	
lyche-numerical-linear-algebra-EQ0918	312	0.547 W +0	$\frac{\left \left \boldsymbol{y} - \boldsymbol{y}_k \right _{\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T} - \left \left \boldsymbol{y} - \boldsymbol{y}_0 \right _{\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T} \right \right }{\left \left \boldsymbol{y} - \boldsymbol{y}_0 \right _{\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T} \right } \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k, \quad \text{for } k \geq 0,$	$\frac{\ \boldsymbol{y} - \boldsymbol{y}_k\ _{\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T}}{\ \boldsymbol{y} - \boldsymbol{y}_0\ _{\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T}} \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k, \quad \text{for } k \geq 0,$	
lyche-numerical-linear-algebra-EQ0386	156	0.548 N +0	$\begin{aligned} \boldsymbol{R} &:= \left[\begin{array}{ccc} \boldsymbol{D}_1 & \boldsymbol{R}_{1,2} & \boldsymbol{R}_{1,3} \\ \boldsymbol{0} & \boldsymbol{D}_2 & \boldsymbol{R}_{2,3} \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{D}_3 \end{array} \right], \boldsymbol{D}_1 := \left[\begin{array}{cc} 2 & 1 \\ -1 & 2 \end{array} \right], \boldsymbol{D}_2 := [1], \boldsymbol{D}_3 := \left[\begin{array}{cc} 3 & 2 \\ -1 & 1 \end{array} \right]. \\ \boldsymbol{D}_1 &:= \left[\begin{array}{cc} 2 & 1 \\ -1 & 2 \end{array} \right], \boldsymbol{D}_2 := [1], \boldsymbol{D}_3 := \left[\begin{array}{cc} 3 & 2 \\ -1 & 1 \end{array} \right]. \end{aligned}$	$\boldsymbol{R} := \left[\begin{array}{ccc} \boldsymbol{D}_1 & \boldsymbol{R}_{1,2} & \boldsymbol{R}_{1,3} \\ \boldsymbol{0} & \boldsymbol{D}_2 & \boldsymbol{R}_{2,3} \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{D}_3 \end{array} \right], \boldsymbol{D}_1 := \left[\begin{array}{cc} 2 & 1 \\ -1 & 2 \end{array} \right], \boldsymbol{D}_2 := [1], \boldsymbol{D}_3 := \left[\begin{array}{cc} 3 & 2 \\ -1 & 1 \end{array} \right].$	
lyche-numerical-linear-algebra-EQ0185	079	0.548 W +0	$l_{ik}^{(k)} a_{kj}^{(k)} = a_{ij}^{(k)} - a_{ij}^{(k+1)} \text{ for } k < \min(i, j), \text{ and } l_{ij}^{(k)} a_{jj}^{(j)} = a_{ij}^{(j)} \text{ for } i > j.$	$l_{ik}^{(k)} a_{kj}^{(k)} = a_{ij}^{(k)} - a_{ij}^{(k+1)} \text{ for } k < \min(i, j), \text{ and } l_{ij}^{(k)} a_{jj}^{(j)} = a_{ij}^{(j)} \text{ for } i > j.$	
lyche-numerical-linear-algebra-EQ0313	126	0.549 N +1	$\begin{aligned} \boldsymbol{H} \boldsymbol{x} &= \boldsymbol{x} - (\boldsymbol{u}^* \boldsymbol{x}) \boldsymbol{u} = \rho \ \boldsymbol{x}\ _2 (\boldsymbol{z} - (\boldsymbol{u}^* \boldsymbol{z}) \boldsymbol{u}) = \rho \ \boldsymbol{x}\ _2 \left(\boldsymbol{z} - \frac{(\boldsymbol{z}^* + \boldsymbol{e}_1^*) \boldsymbol{z}}{1 + z_1} (\boldsymbol{z} + \boldsymbol{e}_1) \right) \\ &= \rho \ \boldsymbol{x}\ _2 (\boldsymbol{z} - (\boldsymbol{z} + \boldsymbol{e}_1)) = -\rho \ \boldsymbol{x}\ _2 \boldsymbol{e}_1 = \boldsymbol{a} \boldsymbol{e}_1. \end{aligned}$	$\begin{aligned} \boldsymbol{H} \boldsymbol{x} &= \boldsymbol{x} - (\boldsymbol{u}^* \boldsymbol{x}) \boldsymbol{u} = \rho \ \boldsymbol{x}\ _2 (\boldsymbol{z} - (\boldsymbol{u}^* \boldsymbol{z}) \boldsymbol{u}) = \rho \ \boldsymbol{x}\ _2 \left(\boldsymbol{z} - \frac{(\boldsymbol{z}^* + \boldsymbol{e}_1^*) \boldsymbol{z}}{1 + z_1} (\boldsymbol{z} + \boldsymbol{e}_1) \right) \\ &= \rho \ \boldsymbol{x}\ _2 (\boldsymbol{z} - (\boldsymbol{z} + \boldsymbol{e}_1)) = -\rho \ \boldsymbol{x}\ _2 \boldsymbol{e}_1 = \boldsymbol{a} \boldsymbol{e}_1. \end{aligned}$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0129 (89)	063	0.550 ● N +0	$\boldsymbol{A}_{11}=[1], \quad \boldsymbol{A}_{12}=[2,3],$ $\boldsymbol{A}_{21}=\left[\begin{array}{l} 4 \\ 7 \end{array}\right], \quad \text{and } \boldsymbol{A}_{22}=\left[\begin{array}{ll} 5 & 6 \\ 8 & 9 \end{array}\right],$	$\boldsymbol{A}_{11}=[1], \quad \boldsymbol{A}_{12}=[2,3], \quad \boldsymbol{A}_{21}=\left[\begin{array}{l} 4 \\ 7 \end{array}\right], \quad \text{and } \boldsymbol{A}_{22}=\left[\begin{array}{ll} 5 & 6 \\ 8 & 9 \end{array}\right],$	$\boldsymbol{A}_{11}=\begin{bmatrix} 1 \end{bmatrix}, \; \boldsymbol{A}_{12}=\begin{bmatrix} 2, 3 \end{bmatrix}, \; \boldsymbol{A}_{21}=\begin{bmatrix} 4 \\ 7 \end{bmatrix}, \; \text{and } \boldsymbol{A}_{22}=\begin{bmatrix} 5 & 6 \\ 8 & 9 \end{bmatrix},$
lyche-numerical-linear-algebra-EQ0117 (2.27)	060	0.551 ●	$\boldsymbol{T}=\boldsymbol{T}_m:=\operatorname{tridiag}_m(-1,2,-1)=\left[\begin{array}{ccccccc} & & & & & & \\ & & & & & & \\ & & & & & & \\ & & & & & & \\ & & & & & & \\ & & & & & & \\ & & & & & & \end{array}\right] \in \mathbb{R}^{m \times m}.$	$\boldsymbol{T}=\boldsymbol{T}_m:=\operatorname{tridiag}_m(-1,2,-1)=\left[\begin{array}{ccccccc} 2 & -1 & 0 & & & & \\ -1 & 2 & -1 & & & & \\ 0 & \ddots & \ddots & \ddots & & & \\ & & 0 & & & & \\ & & & -1 & 2 & -1 & \\ & & & 0 & -1 & 2 & \end{array}\right] \in \mathbb{R}^{m \times m}.$	$\boldsymbol{T}=\boldsymbol{T}_m:=\operatorname{tridiag}_m(-1,2,-1)=\left[\begin{array}{ccccccc} 2 & -1 & 0 & & & & \\ -1 & 2 & -1 & & & & \\ 0 & \ddots & \ddots & \ddots & & & \\ & & 0 & & & & \\ & & & -1 & 2 & -1 & \\ & & & 0 & -1 & 2 & \end{array}\right] \in \mathbb{R}^{m \times m}.$
lyche-numerical-linear-algebra-EQ0577	214	0.552 ● N +0	$\boldsymbol{A} \boldsymbol{x}=\left[\begin{array}{c} p\left(t_1\right) \\ \vdots \\ p\left(t_m\right) \end{array}\right]=\left[\begin{array}{cc} 1 & t_1 \\ \vdots & \\ 1 & t_m \end{array}\right]\left[\begin{array}{c} x_1 \\ x_2 \end{array}\right]=\left[\begin{array}{c} y_1 \\ \vdots \\ y_m \end{array}\right]=\boldsymbol{b}.$	$\boldsymbol{A} \boldsymbol{x}=\left[\begin{array}{c} p\left(t_1\right) \\ \vdots \\ p\left(t_m\right) \end{array}\right]=\left[\begin{array}{cc} 1 & t_1 \\ \vdots & \\ 1 & t_m \end{array}\right]\left[\begin{array}{c} x_1 \\ x_2 \end{array}\right]=\left[\begin{array}{c} y_1 \\ \vdots \\ y_m \end{array}\right]=\boldsymbol{b}.$	$\boldsymbol{A} \boldsymbol{x}=\left[\begin{array}{c} p\left(t_1\right) \\ \vdots \\ p\left(t_m\right) \end{array}\right]=\left[\begin{array}{cc} 1 & t_1 \\ \vdots & \\ 1 & t_m \end{array}\right]\left[\begin{array}{c} x_1 \\ x_2 \end{array}\right]=\left[\begin{array}{c} y_1 \\ \vdots \\ y_m \end{array}\right]=\boldsymbol{b}.$
lyche-numerical-linear-algebra-EQ0569	211	0.554 ● N +0	$\boldsymbol{A} \hat{\boldsymbol{s}}=\boldsymbol{b}+\boldsymbol{e}.$	$\boldsymbol{A} \hat{\boldsymbol{s}}=\boldsymbol{b}+\boldsymbol{e}.$	$\boldsymbol{A} \hat{\boldsymbol{s}}=\boldsymbol{b}+\boldsymbol{e}.$
lyche-numerical-linear-algebra-EQ0336	136	0.557 ● W +0	$\boldsymbol{A}=\left[\begin{array}{cccc} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right] \xrightarrow{P_{12}}\left[\begin{array}{cccc} r_{11} & r_{12} & r_{13} & r_{14} \\ \mathbf{0} & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right] \xrightarrow{P_{23}}\left[\begin{array}{cccc} r_{11} & r_{12} & r_{13} & r_{14} \\ 0 & r_{22} & r_{23} & r_{24} \\ 0 & \mathbf{0} & x & x \\ 0 & 0 & x & x \end{array}\right] \xrightarrow{P_{34}}\left[\begin{array}{cccc} r_{11} & r_{12} & r_{13} & r_{14} \\ 0 & r_{22} & r_{23} & r_{24} \\ 0 & 0 & r_{33} & r_{34} \\ 0 & 0 & \mathbf{0} & r_{44} \end{array}\right].$	$\boldsymbol{A}=\left[\begin{array}{cccc} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right] \xrightarrow{P_{12}}\left[\begin{array}{cccc} r_{11} & r_{12} & r_{13} & r_{14} \\ \mathbf{0} & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right] \xrightarrow{P_{23}}\left[\begin{array}{cccc} r_{11} & r_{12} & r_{13} & r_{14} \\ 0 & r_{22} & r_{23} & r_{24} \\ 0 & \mathbf{0} & x & x \\ 0 & 0 & x & x \end{array}\right] \xrightarrow{P_{34}}\left[\begin{array}{cccc} r_{11} & r_{12} & r_{13} & r_{14} \\ 0 & r_{22} & r_{23} & r_{24} \\ 0 & 0 & r_{33} & r_{34} \\ 0 & 0 & \mathbf{0} & r_{44} \end{array}\right].$	$\boldsymbol{A}=\left[\begin{array}{cccc} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right] \xrightarrow{P_{12}}\left[\begin{array}{cccc} r_{11} & r_{12} & r_{13} & r_{14} \\ \mathbf{0} & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right] \xrightarrow{P_{23}}\left[\begin{array}{cccc} r_{11} & r_{12} & r_{13} & r_{14} \\ 0 & r_{22} & r_{23} & r_{24} \\ 0 & \mathbf{0} & x & x \\ 0 & 0 & x & x \end{array}\right] \xrightarrow{P_{34}}\left[\begin{array}{cccc} r_{11} & r_{12} & r_{13} & r_{14} \\ 0 & r_{22} & r_{23} & r_{24} \\ 0 & 0 & r_{33} & r_{34} \\ 0 & 0 & \mathbf{0} & r_{44} \end{array}\right].$
lyche-numerical-linear-algebra-EQ1015	344	0.561 ● N +0	$\boldsymbol{A}=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right], \quad \boldsymbol{z}_0:=\left[\begin{array}{c} 1 \\ 0 \end{array}\right].$	$\boldsymbol{A}=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right], \quad \boldsymbol{z}_0:=\left[\begin{array}{c} 1 \\ 0 \end{array}\right].$	$\boldsymbol{A}=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right], \quad \boldsymbol{z}_0:=\left[\begin{array}{c} 1 \\ 0 \end{array}\right].$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0919	312	0.563 W +0	$\begin{aligned} \left\ \boldsymbol{y}-\boldsymbol{y}_k\right\ _{\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T}^2 &=\left(\boldsymbol{y}-\boldsymbol{y}_k\right)^T\left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T\right)\left(\boldsymbol{y}-\boldsymbol{y}_k\right) \\ &=\left(\boldsymbol{C}^T\left(\boldsymbol{y}-\boldsymbol{y}_k\right)\right)^T \boldsymbol{A}\left(\boldsymbol{C}^T\left(\boldsymbol{y}-\boldsymbol{y}_k\right)\right)=\left\ \boldsymbol{x}-\boldsymbol{x}_k\right\ _{\boldsymbol{A}}^2, \end{aligned}$	$\left\ \boldsymbol{y}-\boldsymbol{y}_k\right\ _{\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T}^2=\left(\boldsymbol{y}-\boldsymbol{y}_k\right)^T\left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T\right)\left(\boldsymbol{y}-\boldsymbol{y}_k\right) \\ =\left(\boldsymbol{C}^T\left(\boldsymbol{y}-\boldsymbol{y}_k\right)\right)^T \boldsymbol{A}\left(\boldsymbol{C}^T\left(\boldsymbol{y}-\boldsymbol{y}_k\right)\right)=\left\ \boldsymbol{x}-\boldsymbol{x}_k\right\ _{\boldsymbol{A}}^2,$	
lyche-numerical-linear-algebra-EQ0132	063	0.564	$\boldsymbol{A} \boldsymbol{B}=\left[\begin{array}{c} \boldsymbol{a}_{1:}^T \\ \boldsymbol{a}_{2:}^T \\ \vdots \\ \boldsymbol{a}_{m:}^T \end{array}\right] \boldsymbol{B}=\left[\begin{array}{c} \boldsymbol{a}_{1:}^T \boldsymbol{B} \\ \boldsymbol{a}_{2:}^T \boldsymbol{B} \\ \vdots \\ \boldsymbol{a}_{m:}^T \boldsymbol{B} \end{array}\right],$	$\boldsymbol{A} \boldsymbol{B}=\left[\begin{array}{c} \boldsymbol{a}_{1:}^T \\ \boldsymbol{a}_{2:}^T \\ \vdots \\ \boldsymbol{a}_{m:}^T \end{array}\right] \boldsymbol{B}=\left[\begin{array}{c} \boldsymbol{a}_{1:}^T \boldsymbol{B} \\ \boldsymbol{a}_{2:}^T \boldsymbol{B} \\ \vdots \\ \boldsymbol{a}_{m:}^T \boldsymbol{B} \end{array}\right],$	
lyche-numerical-linear-algebra-EQ0425 (7.4)	171	0.564 S -1	$\begin{aligned} \boldsymbol{U} &=\left[\boldsymbol{U}_1, \boldsymbol{U}_2\right] \in \mathbb{C}^{m \times m}, \quad \boldsymbol{U}_1:=\left[\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r\right], \quad \boldsymbol{U}_2:=\left[\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_m\right], \\ \boldsymbol{V} &=\left[\boldsymbol{V}_1, \boldsymbol{V}_2\right] \in \mathbb{C}^{n \times n}, \quad \boldsymbol{V}_1:=\left[\boldsymbol{v}_1, \ldots, \boldsymbol{v}_r\right], \quad \boldsymbol{V}_2:=\left[\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_n\right], \\ \boldsymbol{\Sigma} &=\left[\begin{array}{cc} \boldsymbol{\Sigma}_1 & \boldsymbol{0}_{r, n-r} \\ \boldsymbol{0}_{m-r, r} & \boldsymbol{0}_{m-r, n-r} \end{array}\right] \in \mathbb{R}^{m \times n}, \text { where } \boldsymbol{\Sigma}_1:=\text {diag}\left(\sigma_1, \ldots, \sigma_r\right) . \end{aligned}$	$\begin{aligned} \boldsymbol{U} &=\left[\boldsymbol{U}_1, \boldsymbol{U}_2\right] \in \mathbb{C}^{m \times m}, \quad \boldsymbol{U}_1:=\left[\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r\right], \quad \boldsymbol{U}_2:=\left[\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_m\right], \\ \boldsymbol{V} &=\left[\boldsymbol{V}_1, \boldsymbol{V}_2\right] \in \mathbb{C}^{n \times n}, \quad \boldsymbol{V}_1:=\left[\boldsymbol{v}_1, \ldots, \boldsymbol{v}_r\right], \quad \boldsymbol{V}_2:=\left[\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_n\right], \\ \boldsymbol{\Sigma} &=\left[\begin{array}{cc} \boldsymbol{\Sigma}_1 & \boldsymbol{0}_{r, n-r} \\ \boldsymbol{0}_{m-r, r} & \boldsymbol{0}_{m-r, n-r} \end{array}\right] \in \mathbb{R}^{m \times n}, \text { where } \boldsymbol{\Sigma}_1:=\text {diag}\left(\sigma_1, \ldots, \sigma_r\right) . \end{aligned}$	
lyche-numerical-linear-algebra-EQ1016	345	0.571 N +0	$\begin{aligned} \boldsymbol{z}_1 &=\boldsymbol{A} \boldsymbol{z}_0=\left[\begin{array}{c} 2 \\ -1 \end{array}\right], \quad \boldsymbol{z}_2=\boldsymbol{A} \boldsymbol{z}_1=\left[\begin{array}{c} 5 \\ -4 \end{array}\right], \cdots, \boldsymbol{z}_k=\frac{1}{2}\left[\begin{array}{c} 1+3^k \\ 1-3^k \end{array}\right], \cdots . \end{aligned}$	$\boldsymbol{z}_1=\boldsymbol{A} \boldsymbol{z}_0=\left[\begin{array}{c} 2 \\ -1 \end{array}\right], \quad \boldsymbol{z}_2=\boldsymbol{A} \boldsymbol{z}_1=\left[\begin{array}{c} 5 \\ -4 \end{array}\right], \cdots, \boldsymbol{z}_k=\frac{1}{2}\left[\begin{array}{c} 1+3^k \\ 1-3^k \end{array}\right], \cdots .$	
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-2019-09-09 EQ0834	289	■0.578 ●N +1	$\boldsymbol{A}:=\left[\begin{array}{ccccc}\lambda&0&\cdots&0&0\\0&\lambda&a&\cdots&0&0\\0&0&\lambda&\cdots&0&0\\ \vdots & & & & \vdots \\0&0&0&\cdots&\lambda&a\\0&0&0&\cdots&0&\lambda\end{array}\right]\in\mathbb{R}^{n\times n}.$	$\boldsymbol{A}:=\begin{bmatrix}\lambda&a&0&\cdots&0&0\\0&\lambda&a&\cdots&0&0\\0&0&\lambda&\cdots&0&0\\ \vdots & & & & \vdots \\0&0&0&\cdots&\lambda&a\\0&0&0&\cdots&0&\lambda\end{bmatrix}\in\mathbb{R}^{n\times n}.$	
lyche-numerical-linear-algebra-2019-09-09 EQ0831 (12.47)	287	■0.581 ●N +0	$\lambda=\frac{\boldsymbol{x}^T\boldsymbol{N}\boldsymbol{x}}{\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x}+\boldsymbol{x}^T\boldsymbol{N}\boldsymbol{x}}.$	$\lambda=\frac{\boldsymbol{x}^T\boldsymbol{N}\boldsymbol{x}}{\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x}+\boldsymbol{x}^T\boldsymbol{N}\boldsymbol{x}}.$	$\lambda=\frac{\boldsymbol{x}^T\boldsymbol{N}\boldsymbol{x}}{\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x}+\boldsymbol{x}^T\boldsymbol{N}\boldsymbol{x}}.$
lyche-numerical-linear-algebra-2019-09-09 EQ0420	169	■0.585 ●K +0	$\lambda^m\pi_{\boldsymbol{A}^*\boldsymbol{A}}(\lambda)=\lambda^n\pi_{\boldsymbol{A}\boldsymbol{A}^*}(\lambda),\quad\lambda\in\mathbb{C},$	$\lambda^m\pi_{\boldsymbol{A}^*\boldsymbol{A}}(\lambda)=\lambda^n\pi_{\boldsymbol{A}\boldsymbol{A}^*}(\lambda),\quad\lambda\in\mathbb{C},$	$\lambda^m\pi_{\boldsymbol{A}^*\boldsymbol{A}}(\lambda)=\lambda^n\pi_{\boldsymbol{A}\boldsymbol{A}^*}(\lambda),\quad\lambda\in\mathbb{C},$
lyche-numerical-linear-algebra-2019-09-09 EQ0709	254	■0.587 ●	$\begin{aligned}&\boldsymbol{X}=\boldsymbol{h}^4\boldsymbol{G}/\boldsymbol{M},\text{ where }\boldsymbol{M}:=\left[\begin{array}{c}\sigma_1\\\vdots\\\sigma_m\end{array}\right][1,\ldots,1]+\left[\begin{array}{c}1\\\vdots\\i\end{array}\right][\sigma_1\ldots\sigma_m],\\&\boldsymbol{S}=\sin\left(\pi h\left[\begin{array}{c}1\\2\\\vdots\\m\end{array}\right][12\ldots m]\right),\quad\boldsymbol{\sigma}=\sin\left(\frac{\pi h}{2}\left[\begin{array}{c}1\\2\\\vdots\\m\end{array}\right]\right)\wedge2.\end{aligned}$	$\boldsymbol{X}=\boldsymbol{h}^4\boldsymbol{G}/\boldsymbol{M},\text{ where }\boldsymbol{M}:=\left[\begin{array}{c}\sigma_1\\\vdots\\\sigma_m\end{array}\right][1,\ldots,1]+\left[\begin{array}{c}1\\\vdots\\i\end{array}\right][\sigma_1\ldots\sigma_m],$ $\boldsymbol{S}=\sin\left(\pi h\left[\begin{array}{c}1\\2\\\vdots\\m\end{array}\right][12\ldots m]\right),\quad\boldsymbol{\sigma}=\sin\left(\frac{\pi h}{2}\left[\begin{array}{c}1\\2\\\vdots\\m\end{array}\right]\right)\wedge2.$	$\boldsymbol{X}=\boldsymbol{h}^4\boldsymbol{G}/\boldsymbol{M},\text{ where }\boldsymbol{M}:=\left[\begin{array}{c}\sigma_1\\\vdots\\\sigma_m\end{array}\right][1,\ldots,1]+\left[\begin{array}{c}1\\\vdots\\i\end{array}\right][\sigma_1\ldots\sigma_m],$ $\boldsymbol{S}=\sin\left(\pi h\left[\begin{array}{c}1\\2\\\vdots\\m\end{array}\right][1\ 2\ \ldots\ m]\right),\quad\boldsymbol{\sigma}=\sin\left(\frac{\pi h}{2}\left[\begin{array}{c}1\\2\\\vdots\\m\end{array}\right]\right)\wedge2.$
lyche-numerical-linear-algebra-2019-09-09 EQ0644	232	■0.587 ●W +0	$\begin{array}{l}\boldsymbol{A}\boldsymbol{B}\boldsymbol{A}=\boldsymbol{A}\quad(1)\quad\boldsymbol{A}\boldsymbol{C}\boldsymbol{A}=\boldsymbol{A},\\ \boldsymbol{B}\boldsymbol{A}\boldsymbol{B}=\boldsymbol{B}\quad(2)\quad\boldsymbol{C}\boldsymbol{A}\boldsymbol{C}=\boldsymbol{C},\\ (\boldsymbol{A}\boldsymbol{B})^*=\boldsymbol{A}\boldsymbol{B}\quad(3)\quad(\boldsymbol{A}\boldsymbol{C})^*=\boldsymbol{A}\boldsymbol{C},\\ (\boldsymbol{B}\boldsymbol{A})^*=\boldsymbol{B}\boldsymbol{A}\quad(4)\quad(\boldsymbol{C}\boldsymbol{A})^*=\boldsymbol{C}\boldsymbol{A}.\end{array}$	$\begin{array}{l}\boldsymbol{A}\boldsymbol{B}\boldsymbol{A}=\boldsymbol{A}\quad(1)\quad\boldsymbol{A}\boldsymbol{C}\boldsymbol{A}=\boldsymbol{A},\\ \boldsymbol{B}\boldsymbol{A}\boldsymbol{B}=\boldsymbol{B}\quad(2)\quad\boldsymbol{C}\boldsymbol{A}\boldsymbol{C}=\boldsymbol{C},\\ (\boldsymbol{A}\boldsymbol{B})^*=\boldsymbol{A}\boldsymbol{B}\quad(3)\quad(\boldsymbol{A}\boldsymbol{C})^*=\boldsymbol{A}\boldsymbol{C},\\ (\boldsymbol{B}\boldsymbol{A})^*=\boldsymbol{B}\boldsymbol{A}\quad(4)\quad(\boldsymbol{C}\boldsymbol{A})^*=\boldsymbol{C}\boldsymbol{A}.\end{array}$	$\begin{array}{l}\boldsymbol{A}\boldsymbol{B}\boldsymbol{A}=\boldsymbol{A}\quad(1)\quad\boldsymbol{A}\boldsymbol{C}\boldsymbol{A}=\boldsymbol{A},\\ \boldsymbol{B}\boldsymbol{A}\boldsymbol{B}=\boldsymbol{B}\quad(2)\quad\boldsymbol{C}\boldsymbol{A}\boldsymbol{C}=\boldsymbol{C},\\ (\boldsymbol{A}\boldsymbol{B})^*=\boldsymbol{A}\boldsymbol{B}\quad(3)\quad(\boldsymbol{A}\boldsymbol{C})^*=\boldsymbol{A}\boldsymbol{C},\\ (\boldsymbol{B}\boldsymbol{A})^*=\boldsymbol{B}\boldsymbol{A}\quad(4)\quad(\boldsymbol{C}\boldsymbol{A})^*=\boldsymbol{C}\boldsymbol{A}.\end{array}$
lyche-numerical-linear-algebra-2019-09-09 EQ0171	076	■0.588 ●W +0	$\begin{aligned}&a_{11}^{(1)}x_1+a_{12}^{(1)}x_2+a_{13}^{(1)}x_3=b_1^{(1)},\text{I}\\&a_{22}^{(2)}x_2+a_{23}^{(2)}x_3=b_2^{(2)},\text{II}'\\&a_{32}^{(2)}x_2+a_{33}^{(2)}x_3=b_3^{(2)},\text{III}',\end{aligned}$	$\begin{aligned}&a_{11}^{(1)}x_1+a_{12}^{(1)}x_2+a_{13}^{(1)}x_3=b_1^{(1)},\text{I}\\&a_{22}^{(2)}x_2+a_{23}^{(2)}x_3=b_2^{(2)},\text{II}'\\&a_{32}^{(2)}x_2+a_{33}^{(2)}x_3=b_3^{(2)},\text{III}',\end{aligned}$	$\begin{aligned}&a_{11}^{(1)}x_1+a_{12}^{(1)}x_2+a_{13}^{(1)}x_3=b_1^{(1)},\quad\text{I}\\&a_{22}^{(2)}x_2+a_{23}^{(2)}x_3=b_2^{(2)},\quad\text{II}'\\&a_{32}^{(2)}x_2+a_{33}^{(2)}x_3=b_3^{(2)},\quad\text{III}',\end{aligned}$
Conf. MathPix confidence: ■ ■ ■ — ■≥ 0.9 ■0.5-0.9 ■< 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● ● ● ● ● ● ●C component ●W weak ●S stable ●N noise ●K clean ●not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0467 (8.4)	186	0.589 N -1	$\lim_{p \rightarrow \infty} \ \boldsymbol{x}\ _p = \ \boldsymbol{x}\ _\infty$ for all $\boldsymbol{x} \in \mathbb{C}^n$.		$\lim_{p \rightarrow \infty} \ \boldsymbol{x}\ _p = \ \boldsymbol{x}\ _\infty$ for all $\boldsymbol{x} \in \mathbb{C}^n$.
lyche-numerical-linear-algebra- EQ0793	279	0.592 K +0	$k = -\frac{s \log(10)}{\log(1-\eta)} \approx \frac{s \log(10)}{\eta} = \tilde{k}$.		$k = -\frac{s \log(10)}{\log(1-\eta)} \approx \frac{s \log(10)}{\eta} = \tilde{k}$.
lyche-numerical-linear-algebra- EQ1023 (15.6)	346	0.600 W +0	$\begin{array}{ll} \text{ (i) } & \boldsymbol{y}_k = \boldsymbol{A} \boldsymbol{x}_{k-1} \\ \text{ (ii) } & \boldsymbol{x}_k = \boldsymbol{y}_k / \ \boldsymbol{y}_k\ . \end{array}$		$\begin{array}{ll} \text{ (i) } & \boldsymbol{y}_k = \boldsymbol{A} \boldsymbol{x}_{k-1} \\ \text{ (ii) } & \boldsymbol{x}_k = \boldsymbol{y}_k / \ \boldsymbol{y}_k\ . \end{array}$
lyche-numerical-linear-algebra- EQ0935	317	0.600 N +0	$\lambda = \frac{x^T \boldsymbol{A} \boldsymbol{x}}{x^T \boldsymbol{A}_p \boldsymbol{x}}$.		$\lambda = \frac{x^T \boldsymbol{A} \boldsymbol{x}}{x^T \boldsymbol{A}_p \boldsymbol{x}}$.
lyche-numerical-linear-algebra- EQ0901	308	0.602	$\begin{aligned} & \left\ \boldsymbol{\epsilon}_k \right\ _{\boldsymbol{A}}^2 = \boldsymbol{\epsilon}_k^T \boldsymbol{A} \boldsymbol{\epsilon}_k = \boldsymbol{r}_k^T \boldsymbol{A}^{-1} \boldsymbol{r}_k, \\ & \left\ \boldsymbol{\epsilon}_{k+1} \right\ _{\boldsymbol{A}}^2 = (\boldsymbol{\epsilon}_k - \alpha_k \boldsymbol{r}_k)^T \boldsymbol{A} (\boldsymbol{\epsilon}_k - \alpha_k \boldsymbol{r}_k) \\ & = \boldsymbol{\epsilon}_k^T \boldsymbol{A} \boldsymbol{\epsilon}_k - 2\alpha_k \boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{\epsilon}_k + \alpha_k^2 \boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k = \left\ \boldsymbol{\epsilon}_k \right\ _{\boldsymbol{A}}^2 - \frac{(\boldsymbol{r}_k^T \boldsymbol{r}_k)^2}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k}. \end{aligned}$		$\begin{aligned} \left\ \boldsymbol{\epsilon}_k \right\ _{\boldsymbol{A}}^2 &= \boldsymbol{\epsilon}_k^T \boldsymbol{A} \boldsymbol{\epsilon}_k = \boldsymbol{r}_k^T \boldsymbol{A}^{-1} \boldsymbol{r}_k, \\ \left\ \boldsymbol{\epsilon}_{k+1} \right\ _{\boldsymbol{A}}^2 &= (\boldsymbol{\epsilon}_k - \alpha_k \boldsymbol{r}_k)^T \boldsymbol{A} (\boldsymbol{\epsilon}_k - \alpha_k \boldsymbol{r}_k) \\ &= \boldsymbol{\epsilon}_k^T \boldsymbol{A} \boldsymbol{\epsilon}_k - 2\alpha_k \boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{\epsilon}_k + \alpha_k^2 \boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k = \left\ \boldsymbol{\epsilon}_k \right\ _{\boldsymbol{A}}^2 - \frac{(\boldsymbol{r}_k^T \boldsymbol{r}_k)^2}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k}. \end{aligned}$
lyche-numerical-linear-algebra- EQ0892 (13.37)	307	0.606 W +0	$a := \frac{\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}}{\boldsymbol{y}^T \boldsymbol{y}} = \sum_{i=1}^n t_i \lambda_i, \quad b := \frac{\boldsymbol{y}^T \boldsymbol{A}^{-1} \boldsymbol{y}}{\boldsymbol{y}^T \boldsymbol{y}} = \sum_{i=1}^n \frac{t_i}{\lambda_i},$		$a := \frac{\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}}{\boldsymbol{y}^T \boldsymbol{y}} = \sum_{i=1}^n t_i \lambda_i, \quad b := \frac{\boldsymbol{y}^T \boldsymbol{A}^{-1} \boldsymbol{y}}{\boldsymbol{y}^T \boldsymbol{y}} = \sum_{i=1}^n \frac{t_i}{\lambda_i},$
lyche-numerical-linear-algebra- EQ1029	348	0.608 K +0	$\boldsymbol{A}_1 := \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \boldsymbol{A}_2 := \begin{bmatrix} 1.7 & -0.4 \\ 0.15 & 2.2 \end{bmatrix}, \quad \text{and } \boldsymbol{A}_3 = \begin{bmatrix} 1 & 2 \\ -3 & 4 \end{bmatrix}.$		$\boldsymbol{A}_1 := \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \boldsymbol{A}_2 := \begin{bmatrix} 1.7 & -0.4 \\ 0.15 & 2.2 \end{bmatrix}, \quad \text{and } \boldsymbol{A}_3 = \begin{bmatrix} 1 & 2 \\ -3 & 4 \end{bmatrix}.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0379	154	0.611 S +0	$\begin{aligned} \boldsymbol{A} \boldsymbol{A}^* &= \left(\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^* \right) \boldsymbol{A} \boldsymbol{A}^* = \left(\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^* \right) \boldsymbol{A} \boldsymbol{A}^* \\ \boldsymbol{A}^* \boldsymbol{A} &= \left(\boldsymbol{U} \boldsymbol{B}^* \boldsymbol{U}^* \right) \left(\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^* \right) = \boldsymbol{U} \boldsymbol{B} \boldsymbol{B}^* \boldsymbol{U}^* \end{aligned}$	$\boldsymbol{A} \boldsymbol{A}^* = (\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^*) (\boldsymbol{U} \boldsymbol{B}^* \boldsymbol{U}^*) = \boldsymbol{U} \boldsymbol{B} \boldsymbol{B}^* \boldsymbol{U}^* \text{ and}$ $\boldsymbol{A}^* \boldsymbol{A} = (\boldsymbol{U} \boldsymbol{B}^* \boldsymbol{U}^*) (\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^*) = \boldsymbol{U} \boldsymbol{B}^* \boldsymbol{B} \boldsymbol{U}^*.$	$\boldsymbol{A} \boldsymbol{A}^* = (\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^*) (\boldsymbol{U} \boldsymbol{B}^* \boldsymbol{U}^*) = \boldsymbol{U} \boldsymbol{B} \boldsymbol{B}^* \boldsymbol{U}^* \text{ and}$ $\boldsymbol{A}^* \boldsymbol{A} = (\boldsymbol{U} \boldsymbol{B}^* \boldsymbol{U}^*) (\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^*) = \boldsymbol{U} \boldsymbol{B}^* \boldsymbol{B} \boldsymbol{U}^*.$
lyche-numerical-linear-algebra-EQ1031 (15.7)	349	0.611 W +0	$\begin{array}{ll} \text{(i)} & (\boldsymbol{A} - s \boldsymbol{I}) \boldsymbol{y}_k = \boldsymbol{x}_{k-1} \\ \text{(ii)} & \boldsymbol{x}_k = \boldsymbol{y}_k / \ \boldsymbol{y}_k\ . \end{array}$	$\begin{array}{ll} \text{(i)} & (\boldsymbol{A} - s \boldsymbol{I}) \boldsymbol{y}_k = \boldsymbol{x}_{k-1} \\ \text{(ii)} & \boldsymbol{x}_k = \boldsymbol{y}_k / \ \boldsymbol{y}_k\ . \end{array}$	$\text{(i)} \quad (\boldsymbol{A} - s \boldsymbol{I}) \boldsymbol{y}_k = \boldsymbol{x}_{k-1}$ $\text{(ii)} \quad \boldsymbol{x}_k = \boldsymbol{y}_k / \ \boldsymbol{y}_k\ .$
lyche-numerical-linear-algebra-EQ1011	341	0.614	$\boldsymbol{A} := \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \boldsymbol{E} := \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \epsilon & 0 & 0 & 0 \end{bmatrix}, \quad \boldsymbol{B} := \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \epsilon & 0 & 0 & 0 \end{bmatrix}.$	$\boldsymbol{A} := \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \boldsymbol{E} := \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \epsilon & 0 & 0 & 0 \end{bmatrix}, \quad \boldsymbol{B} := \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \epsilon & 0 & 0 & 0 \end{bmatrix}.$	$\boldsymbol{A} := \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \boldsymbol{E} := \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \epsilon & 0 & 0 & 0 \end{bmatrix}, \quad \boldsymbol{B} := \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \epsilon & 0 & 0 & 0 \end{bmatrix}.$
lyche-numerical-linear-algebra-EQ0011	024	0.618 N +0	$\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{j}) := \boldsymbol{A} \begin{pmatrix} i_1 & i_2 & \cdots & i_r \\ j_1 & j_2 & \cdots & j_c \end{pmatrix} = \begin{bmatrix} a_{i_1, j_1} & a_{i_1, j_2} & \cdots & a_{i_1, j_c} \\ a_{i_2, j_1} & a_{i_2, j_2} & \cdots & a_{i_2, j_c} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_r, j_1} & a_{i_r, j_2} & \cdots & a_{i_r, j_c} \end{bmatrix}.$	$\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{j}) := \boldsymbol{A} \begin{pmatrix} i_1 & i_2 & \cdots & i_r \\ j_1 & j_2 & \cdots & j_c \end{pmatrix} = \begin{bmatrix} a_{i_1, j_1} & a_{i_1, j_2} & \cdots & a_{i_1, j_c} \\ a_{i_2, j_1} & a_{i_2, j_2} & \cdots & a_{i_2, j_c} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_r, j_1} & a_{i_r, j_2} & \cdots & a_{i_r, j_c} \end{bmatrix}.$	$\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{j}) := \boldsymbol{A} \begin{pmatrix} i_1 & i_2 & \cdots & i_r \\ j_1 & j_2 & \cdots & j_c \end{pmatrix} = \begin{bmatrix} a_{i_1, j_1} & a_{i_1, j_2} & \cdots & a_{i_1, j_c} \\ a_{i_2, j_1} & a_{i_2, j_2} & \cdots & a_{i_2, j_c} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_r, j_1} & a_{i_r, j_2} & \cdots & a_{i_r, j_c} \end{bmatrix}.$
lyche-numerical-linear-algebra-EQ0249 (4.4)	102	0.629 S +0	$\boldsymbol{a}_n = \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{l}_n, \quad \boldsymbol{a}_{nn} = \boldsymbol{l}_n^* \boldsymbol{D}_{n-1} \boldsymbol{l}_n + d_{nn}.$	$\boldsymbol{a}_n = \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{l}_n, \quad \boldsymbol{a}_{nn} = \boldsymbol{l}_n^* \boldsymbol{D}_{n-1} \boldsymbol{l}_n + d_{nn}.$	$\boldsymbol{a}_n = \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{l}_n, \quad \boldsymbol{a}_{nn} = \boldsymbol{l}_n^* \boldsymbol{D}_{n-1} \boldsymbol{l}_n + d_{nn}.$
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-2020-080849	295	■ 0.636 ● W +0	$\begin{aligned} & \boldsymbol{r}_{k+1}^T \boldsymbol{r}_j \stackrel{(13.13)}{=} (\boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k)^T \boldsymbol{r}_j \\ & \stackrel{(13.11)}{=} \boldsymbol{r}_k^T \boldsymbol{r}_j - \alpha_k \boldsymbol{p}_k^T \boldsymbol{A} \left(\boldsymbol{p}_j + \sum_{i=0}^{j-1} \left(\frac{\boldsymbol{r}_j^T \boldsymbol{A} \boldsymbol{p}_i}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i} \right) \boldsymbol{p}_i \right) \\ & \stackrel{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_i = 0}{=} \boldsymbol{r}_k^T \boldsymbol{r}_j - \alpha_k \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_j = 0, \quad j = 0, 1, \dots, k. \end{aligned}$	$\boldsymbol{r}_{k+1}^T \boldsymbol{r}_j \stackrel{(13.13)}{=} (\boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k)^T \boldsymbol{r}_j$ $\stackrel{(13.11)}{=} \boldsymbol{r}_k^T \boldsymbol{r}_j - \alpha_k \boldsymbol{p}_k^T \boldsymbol{A} \left(\boldsymbol{p}_j + \sum_{i=0}^{j-1} \left(\frac{\boldsymbol{r}_j^T \boldsymbol{A} \boldsymbol{p}_i}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i} \right) \boldsymbol{p}_i \right)$ $\stackrel{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_i = 0}{=} \boldsymbol{r}_k^T \boldsymbol{r}_j - \alpha_k \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_j = 0, \quad j = 0, 1, \dots, k.$	$\boldsymbol{r}_{k+1}^T \boldsymbol{r}_j \stackrel{(13.13)}{=} (\boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k)^T \boldsymbol{r}_j$ $\stackrel{(13.11)}{=} \boldsymbol{r}_k^T \boldsymbol{r}_j - \alpha_k \boldsymbol{p}_k^T \boldsymbol{A} \left(\boldsymbol{p}_j + \sum_{i=0}^{j-1} \left(\frac{\boldsymbol{r}_j^T \boldsymbol{A} \boldsymbol{p}_i}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i} \right) \boldsymbol{p}_i \right)$ $\stackrel{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_i = 0}{=} \boldsymbol{r}_k^T \boldsymbol{r}_j - \alpha_k \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_j = 0, \quad j = 0, 1, \dots, k.$
lyche-numerical-linear-algebra-2020-080592 (9.4)	218	■ 0.637 ● K +0	$\mathbb{C}^m = \mathcal{R}(\boldsymbol{A}) \stackrel{\perp}{\oplus} \mathcal{N}(\boldsymbol{A}^*).$	$\mathbb{C}^m = \mathcal{R}(\boldsymbol{A}) \stackrel{\perp}{\oplus} \mathcal{N}(\boldsymbol{A}^*).$	$\mathbb{C}^m = \mathcal{R}(\boldsymbol{A}) \stackrel{\perp}{\oplus} \mathcal{N}(\boldsymbol{A}^*).$
lyche-numerical-linear-algebra-2020-080553 (8.42)	208	■ 0.638 ● N +1	$\operatorname{cond}_1(\boldsymbol{T}) = \operatorname{cond}_\infty(\boldsymbol{T}) = \frac{1}{2} \begin{cases} h^{-2}, & m \text{ odd} \\ h^{-2} - 1, & m \text{ even} \end{cases}$	$\operatorname{cond}_1(\boldsymbol{T}) = \operatorname{cond}_\infty(\boldsymbol{T}) = \frac{1}{2} \begin{cases} h^{-2}, & m \text{ odd} \\ h^{-2} - 1, & m \text{ even} \end{cases}$	$\operatorname{cond}_1(\boldsymbol{T}) = \operatorname{cond}_\infty(\boldsymbol{T}) = \frac{1}{2} \begin{cases} h^{-2}, & m \text{ odd} \\ h^{-2} - 1, & m \text{ even} \end{cases}$
lyche-numerical-linear-algebra-2020-0801032	349	■ 0.642 ●	$\begin{aligned} & \text{(i) } \boldsymbol{A} - s_{k-1} \boldsymbol{I} \boldsymbol{y}_k = \boldsymbol{x}_{k-1}, \\ & \text{(ii) } \boldsymbol{x}_k = \boldsymbol{y}_k / \ \boldsymbol{y}_k\ , \\ & \text{(iii) } \boldsymbol{s}_k = \boldsymbol{x}_k^* \boldsymbol{A} \boldsymbol{x}_k, \\ & \text{(iv) } \boldsymbol{r}_k = \boldsymbol{A} \boldsymbol{x}_k - s_k \boldsymbol{x}_k. \end{aligned}$	$\begin{aligned} & \text{(i) } (\boldsymbol{A} - s_{k-1} \boldsymbol{I}) \boldsymbol{y}_k = \boldsymbol{x}_{k-1}, \\ & \text{(ii) } \boldsymbol{x}_k = \boldsymbol{y}_k / \ \boldsymbol{y}_k\ , \\ & \text{(iii) } \boldsymbol{s}_k = \boldsymbol{x}_k^* \boldsymbol{A} \boldsymbol{x}_k, \\ & \text{(iv) } \boldsymbol{r}_k = \boldsymbol{A} \boldsymbol{x}_k - s_k \boldsymbol{x}_k. \end{aligned}$	$\begin{aligned} & \text{(i) } (\boldsymbol{A} - s_{k-1} \boldsymbol{I}) \boldsymbol{y}_k = \boldsymbol{x}_{k-1}, \\ & \text{(ii) } \boldsymbol{x}_k = \boldsymbol{y}_k / \ \boldsymbol{y}_k\ , \\ & \text{(iii) } \boldsymbol{s}_k = \boldsymbol{x}_k^* \boldsymbol{A} \boldsymbol{x}_k, \\ & \text{(iv) } \boldsymbol{r}_k = \boldsymbol{A} \boldsymbol{x}_k - s_k \boldsymbol{x}_k. \end{aligned}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-2019-09-06-098	250	■ 0.645 ● W +0	$\begin{aligned} \boldsymbol{A} &= \left[\begin{array}{rrr rrr rrr} 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 4 & 0 & 0 & -1 & 0 & 0 & 0 \\ \hline -1 & 0 & 0 & 4 & -1 & 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & -1 & 4 & -1 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & -1 & 4 & 0 & 0 & -1 \\ \hline 0 & 0 & 0 & -1 & 0 & 0 & 4 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 \end{array} \right] \\ &= \begin{bmatrix} \boldsymbol{T} + 2\boldsymbol{I} & -\boldsymbol{I} & \boldsymbol{0} \\ -\boldsymbol{I} & \boldsymbol{T} + 2\boldsymbol{I} & -\boldsymbol{I} \\ \boldsymbol{0} & -\boldsymbol{I} & \boldsymbol{T} + 2\boldsymbol{I} \end{bmatrix}, \end{aligned}$	$\boldsymbol{A} = \left[\begin{array}{rrr rrr rrr} 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 4 & 0 & 0 & -1 & 0 & 0 & 0 \\ \hline -1 & 0 & 0 & 4 & -1 & 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & -1 & 4 & -1 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & -1 & 4 & 0 & 0 & -1 \\ \hline 0 & 0 & 0 & -1 & 0 & 0 & 4 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 \end{array} \right] \\ = \begin{bmatrix} \boldsymbol{T} + 2\boldsymbol{I} & -\boldsymbol{I} & \boldsymbol{0} \\ -\boldsymbol{I} & \boldsymbol{T} + 2\boldsymbol{I} & -\boldsymbol{I} \\ \boldsymbol{0} & -\boldsymbol{I} & \boldsymbol{T} + 2\boldsymbol{I} \end{bmatrix},$	
lyche-numerical-linear-algebra-2019-09-06-041	232	■ 0.645 ● N +0	$\boldsymbol{r} = \boldsymbol{r}(x) = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}.$	$\boldsymbol{r} = \boldsymbol{r}(x) = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}.$	$\boldsymbol{r} = \boldsymbol{r}(x) = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}.$
lyche-numerical-linear-algebra-2019-09-06-081 (9.7)	222	■ 0.650 ● N -1	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^* = [\boldsymbol{U}_1, \boldsymbol{U}_2] \begin{bmatrix} \boldsymbol{\Sigma}_1 & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{V}_1^* \\ \boldsymbol{V}_2^* \end{bmatrix} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*, \quad \boldsymbol{\Sigma}_1 = \text{diag}(\sigma_1, \dots, \sigma_r),$	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^* = [\boldsymbol{U}_1, \boldsymbol{U}_2] \begin{bmatrix} \boldsymbol{\Sigma}_1 & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{V}_1^* \\ \boldsymbol{V}_2^* \end{bmatrix} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*, \quad \boldsymbol{\Sigma}_1 = \text{diag}(\sigma_1, \dots, \sigma_r),$	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^* = [\boldsymbol{U}_1, \boldsymbol{U}_2] \begin{bmatrix} \boldsymbol{\Sigma}_1 & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{V}_1^* \\ \boldsymbol{V}_2^* \end{bmatrix} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*, \quad \boldsymbol{\Sigma}_1 = \text{diag}(\sigma_1, \dots, \sigma_r),$
lyche-numerical-linear-algebra-2019-09-06-081 (3.11)	085	■ 0.651 ● N -1	$\boldsymbol{p} := \boldsymbol{p}_n, \text{ where } \boldsymbol{p}_1 := [1, 2, \dots, n]^T, \text{ and } \boldsymbol{p}_{k+1} := \boldsymbol{I}_{r_k, k} \boldsymbol{p}_k \text{ for } k = 1, \dots, n-1.$	$\boldsymbol{p} := \boldsymbol{p}_n, \text{ where } \boldsymbol{p}_1 := [1, 2, \dots, n]^T, \text{ and } \boldsymbol{p}_{k+1} := \boldsymbol{I}_{r_k, k} \boldsymbol{p}_k \text{ for } k = 1, \dots, n-1.$	$\boldsymbol{p} := \boldsymbol{p}_n, \text{ where } \boldsymbol{p}_1 := [1, 2, \dots, n]^T, \text{ and } \boldsymbol{p}_{k+1} := \boldsymbol{I}_{r_k, k} \boldsymbol{p}_k \text{ for } k = 1, \dots, n-1.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0947	320	0.654	$\begin{aligned} & \left\{ \begin{array}{l} \boldsymbol{x}_0, \\ \boldsymbol{x}_1, \boldsymbol{x}_2, \\ \boldsymbol{x}_3 \end{array} \right\} = \left\{ \begin{array}{l} \boldsymbol{x}_0, \\ \boldsymbol{x}_1, \boldsymbol{x}_2, \\ \boldsymbol{x}_3 \end{array} \right\} \\ & \left\{ \begin{array}{l} \boldsymbol{r}_0, \\ \boldsymbol{r}_1, \boldsymbol{r}_2, \boldsymbol{r}_3 \end{array} \right\} = \left\{ \begin{array}{l} \boldsymbol{r}_0, \\ \boldsymbol{r}_1, \boldsymbol{r}_2, \boldsymbol{r}_3 \end{array} \right\} \end{aligned}$	$\begin{aligned} [\boldsymbol{x}_0, \boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3] &= \begin{bmatrix} 0 & 2 & 8/3 & 3 \\ 0 & 0 & 4/3 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \\ [\boldsymbol{r}_0, \boldsymbol{r}_1, \boldsymbol{r}_2, \boldsymbol{r}_3] &= \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 4/3 & 0 \end{bmatrix}, \end{aligned}$	$\begin{aligned} [\boldsymbol{x}_0, \boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3] &= \begin{bmatrix} 0 & 2 & 8/3 & 3 \\ 0 & 0 & 4/3 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \\ [\boldsymbol{r}_0, \boldsymbol{r}_1, \boldsymbol{r}_2, \boldsymbol{r}_3] &= \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 4/3 & 0 \end{bmatrix}, \end{aligned}$
lyche-numerical-linear-algebra- EQ0415	166	0.666	$R(\boldsymbol{x} - t\boldsymbol{y}) = R(\boldsymbol{x}) - 2t(\boldsymbol{A}\boldsymbol{x} - R(\boldsymbol{x})\boldsymbol{x})^T \boldsymbol{y} + \mathcal{O}(t^2),$	$R(\boldsymbol{x} - t\boldsymbol{y}) = R(\boldsymbol{x}) - 2t(\boldsymbol{A}\boldsymbol{x} - R(\boldsymbol{x})\boldsymbol{x})^T \boldsymbol{y} + \mathcal{O}(t^2),$	$R(\boldsymbol{x} - t\boldsymbol{y}) = R(\boldsymbol{x}) - 2t(\boldsymbol{A}\boldsymbol{x} - R(\boldsymbol{x})\boldsymbol{x})^T \boldsymbol{y} + \mathcal{O}(t^2),$
lyche-numerical-linear-algebra- EQ0602	222	0.667	$\begin{aligned} & \left\{ \begin{array}{l} \boldsymbol{U}_1, \dots, \boldsymbol{U}_r, \\ \boldsymbol{U}_2, \dots, \boldsymbol{U}_m, \\ \boldsymbol{V}_1, \dots, \boldsymbol{V}_r, \\ \boldsymbol{V}_2, \dots, \boldsymbol{V}_n \end{array} \right\} = \left\{ \begin{array}{l} \boldsymbol{U}_1, \dots, \boldsymbol{U}_r, \\ \boldsymbol{U}_2, \dots, \boldsymbol{U}_m, \\ \boldsymbol{V}_1, \dots, \boldsymbol{V}_r, \\ \boldsymbol{V}_2, \dots, \boldsymbol{V}_n \end{array} \right\} \\ & \left\{ \begin{array}{l} \boldsymbol{U}_1^* \boldsymbol{U}_1 = \boldsymbol{I}, \boldsymbol{U}_2^* \boldsymbol{U}_2 = \boldsymbol{I}, \\ \boldsymbol{V}_1^* \boldsymbol{V}_1 = \boldsymbol{I}, \boldsymbol{V}_2^* \boldsymbol{V}_2 = \boldsymbol{I} \end{array} \right\} \end{aligned}$	$\begin{aligned} \boldsymbol{U}_1 &= [\boldsymbol{u}_1, \dots, \boldsymbol{u}_r], \quad \boldsymbol{U}_2 = [\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m], \quad \boldsymbol{V}_1^* \boldsymbol{V}_1 = \boldsymbol{I}, \quad \boldsymbol{V}_2^* \boldsymbol{V}_2 = \boldsymbol{I}, \\ \boldsymbol{V}_1 &= [\boldsymbol{v}_1, \dots, \boldsymbol{v}_r], \quad \boldsymbol{V}_2 = [\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n], \end{aligned}$	$\begin{aligned} \boldsymbol{U}_1 &= [\boldsymbol{u}_1, \dots, \boldsymbol{u}_r], \quad \boldsymbol{U}_2 = [\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m], \quad \boldsymbol{U}_1^* \boldsymbol{U}_1 = \boldsymbol{I}, \quad \boldsymbol{U}_2^* \boldsymbol{U}_2 = \boldsymbol{I}, \\ \boldsymbol{V}_1 &= [\boldsymbol{v}_1, \dots, \boldsymbol{v}_r], \quad \boldsymbol{V}_2 = [\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n], \quad \boldsymbol{V}_1^* \boldsymbol{V}_1 = \boldsymbol{I}, \quad \boldsymbol{V}_2^* \boldsymbol{V}_2 = \boldsymbol{I} \end{aligned}$
lyche-numerical-linear-algebra- EQ0751	267	0.670	$\begin{aligned} & \left\{ \begin{array}{l} \boldsymbol{J} : \boldsymbol{v}_{k+1}(i, j) = \left(\boldsymbol{v}_k(i-1, j) + \boldsymbol{v}_k(i, j-1) + \boldsymbol{v}_k(i+1, j) + \boldsymbol{v}_k(i, j+1) \right. \\ \left. + \boldsymbol{e}(i, j) \right) / 4 \\ \boldsymbol{G} \boldsymbol{S} : \boldsymbol{v}_{k+1}(i, j) = \left(\boldsymbol{v}_{k+1}(i-1, j) + \boldsymbol{v}_{k+1}(i, j-1) + \boldsymbol{v}_{k+1}(i+1, j) + \boldsymbol{v}_{k+1}(i, j+1) \right. \\ \left. + \boldsymbol{e}(i, j) \right) / 4 \end{array} \right\} \end{aligned}$	$\begin{aligned} \boldsymbol{J} : \boldsymbol{v}_{k+1}(i, j) &= (\boldsymbol{v}_k(i-1, j) + \boldsymbol{v}_k(i, j-1) + \boldsymbol{v}_k(i+1, j) + \boldsymbol{v}_k(i, j+1) \\ &+ \boldsymbol{e}(i, j)) / 4 \\ \boldsymbol{G} \boldsymbol{S} : \boldsymbol{v}_{k+1}(i, j) &= (\boldsymbol{v}_{k+1}(i-1, j) + \boldsymbol{v}_{k+1}(i, j-1) + \boldsymbol{v}_{k+1}(i+1, j) + \boldsymbol{v}_{k+1}(i, j+1) \\ &+ \boldsymbol{e}(i, j)) / 4 \end{aligned}$	$\begin{aligned} \boldsymbol{J} : \boldsymbol{v}_{k+1}(i, j) &= (\boldsymbol{v}_k(i-1, j) + \boldsymbol{v}_k(i, j-1) + \boldsymbol{v}_k(i+1, j) + \boldsymbol{v}_k(i, j+1) \\ &+ \boldsymbol{e}(i, j)) / 4 \\ \boldsymbol{G} \boldsymbol{S} : \boldsymbol{v}_{k+1}(i, j) &= (\boldsymbol{v}_{k+1}(i-1, j) + \boldsymbol{v}_{k+1}(i, j-1) + \boldsymbol{v}_{k+1}(i+1, j) + \boldsymbol{v}_{k+1}(i, j+1) \\ &+ \boldsymbol{e}(i, j)) / 4 \end{aligned}$
lyche-numerical-linear-algebra- EQ0190 (3.8)	080	0.672	$\boldsymbol{x}_k = \left(\boldsymbol{b}_k - \sum_{j=l_k}^{k-1} \boldsymbol{a}_{k,j} \boldsymbol{x}_j \right) / \boldsymbol{a}_{kk}, \quad k = 1, 2, \dots, n.$	$\boldsymbol{x}_k = \left(\boldsymbol{b}_k - \sum_{j=l_k}^{k-1} \boldsymbol{a}_{k,j} \boldsymbol{x}_j \right) / \boldsymbol{a}_{kk}, \quad k = 1, 2, \dots, n.$	$\boldsymbol{x}_k = \left(\boldsymbol{b}_k - \sum_{j=l_k}^{k-1} \boldsymbol{a}_{k,j} \boldsymbol{x}_j \right) / \boldsymbol{a}_{kk}, \quad k = 1, 2, \dots, n.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0719	257	0.674	$\begin{aligned} \left(\boldsymbol{F}_{2m+2}\boldsymbol{z}\right)_{k+1} &= \sum_{j=0}^{2m-1} \omega^{jk} z_{j+1} = \sum_{j=1}^m x_j \omega^{jk} - \sum_{j=1}^m x_j \omega^{(2m+2-j)k} \\ &= \sum_{j=1}^m x_j \left(e^{-\pi i j k / (m+1)} - e^{\pi i j k / (m+1)} \right) \\ &= -2i \sum_{j=1}^m x_j \sin \left(\frac{jk\pi}{m+1} \right) = -2i (\boldsymbol{S}_m \boldsymbol{x})_k. \end{aligned}$	$\begin{aligned} (\boldsymbol{F}_{2m+2}\boldsymbol{z})_{k+1} &= \sum_{j=0}^{2m-1} \omega^{jk} z_{j+1} = \sum_{j=1}^m x_j \omega^{jk} - \sum_{j=1}^m x_j \omega^{(2m+2-j)k} \\ &= \sum_{j=1}^m x_j \left(e^{-\pi i j k / (m+1)} - e^{\pi i j k / (m+1)} \right) \\ &= -2i \sum_{j=1}^m x_j \sin \left(\frac{jk\pi}{m+1} \right) = -2i (\boldsymbol{S}_m \boldsymbol{x})_k. \end{aligned}$	$\begin{aligned} (\boldsymbol{F}_{2m+2}\boldsymbol{z})_{k+1} &= \sum_{j=0}^{2m-1} \omega^{jk} z_{j+1} = \sum_{j=1}^m x_j \omega^{jk} - \sum_{j=1}^m x_j \omega^{(2m+2-j)k} \\ &= \sum_{j=1}^m x_j \left(e^{-\pi i j k / (m+1)} - e^{\pi i j k / (m+1)} \right) \\ &= -2i \sum_{j=1}^m x_j \sin \left(\frac{jk\pi}{m+1} \right) = -2i (\boldsymbol{S}_m \boldsymbol{x})_k. \end{aligned}$
lyche-numerical-linear-algebra EQ0599	221	0.676 N -1	$\boldsymbol{A}=\left[\begin{array}{ccc} 1 & 3 & 1 \\ 1 & 3 & 7 \\ 1 & -1 & -4 \\ 1 & -1 & 2 \end{array}\right] \text{ and } \boldsymbol{b}=\left[\begin{array}{l} 1 \\ 1 \\ 1 \\ 1 \end{array}\right].$	$\boldsymbol{A}=\left[\begin{array}{ccc} 1 & 3 & 1 \\ 1 & 3 & 7 \\ 1 & -1 & -4 \\ 1 & -1 & 2 \end{array}\right] \text{ and } \boldsymbol{b}=\left[\begin{array}{l} 1 \\ 1 \\ 1 \\ 1 \end{array}\right].$	$\boldsymbol{A}=\left[\begin{array}{ccc} 1 & 3 & 1 \\ 1 & 3 & 7 \\ 1 & -1 & -4 \\ 1 & -1 & 2 \end{array}\right] \text{ and } \boldsymbol{b}=\left[\begin{array}{l} 1 \\ 1 \\ 1 \\ 1 \end{array}\right].$
lyche-numerical-linear-algebra EQ0726 (11.7)	258	0.678 W +0	$\boldsymbol{F}_{2m}\boldsymbol{P}_{2m}=\left[\begin{array}{c c} \boldsymbol{F}_m & \boldsymbol{D}_m\boldsymbol{F}_m \\ \hline \boldsymbol{F}_m & -\boldsymbol{D}_m\boldsymbol{F}_m \end{array}\right],$	$\boldsymbol{F}_{2m}\boldsymbol{P}_{2m}=\left[\begin{array}{c c} \boldsymbol{F}_m & \boldsymbol{D}_m\boldsymbol{F}_m \\ \hline \boldsymbol{F}_m & -\boldsymbol{D}_m\boldsymbol{F}_m \end{array}\right],$	$\boldsymbol{F}_{2m}\boldsymbol{P}_{2m}=\left[\begin{array}{c c} \boldsymbol{F}_m & \boldsymbol{D}_m\boldsymbol{F}_m \\ \hline \boldsymbol{F}_m & -\boldsymbol{D}_m\boldsymbol{F}_m \end{array}\right],$
lyche-numerical-linear-algebra EQ0513	200	0.679 N +1	$\begin{aligned} f(x) &= \frac{x_2-x}{x_2-x_1} f(x_1) + \frac{x-x_1}{x_2-x_1} f(x_2) + (x-x_1)(x-x_2) f''(c)/2 \\ &= (1-\lambda) f(x_1) + \lambda f(x_2) + (x_2-x_1)^2 \lambda (\lambda-1) f''(c)/2, \quad \lambda := \frac{x-x_1}{x_2-x_1}. \end{aligned}$	$\begin{aligned} f(x) &= \frac{x_2-x}{x_2-x_1} f(x_1) + \frac{x-x_1}{x_2-x_1} f(x_2) + (x-x_1)(x-x_2) f''(c)/2 \\ &= (1-\lambda) f(x_1) + \lambda f(x_2) + (x_2-x_1)^2 \lambda (\lambda-1) f''(c)/2, \quad \lambda := \frac{x-x_1}{x_2-x_1}. \end{aligned}$	$\begin{aligned} f(x) &= \frac{x_2-x}{x_2-x_1} f(x_1) + \frac{x-x_1}{x_2-x_1} f(x_2) + (x-x_1)(x-x_2) f''(c)/2 \\ &= (1-\lambda) f(x_1) + \lambda f(x_2) + (x_2-x_1)^2 \lambda (\lambda-1) f''(c)/2, \quad \lambda := \frac{x-x_1}{x_2-x_1}. \end{aligned}$
lyche-numerical-linear-algebra EQ0607	222	0.679 S +1	$\begin{aligned} \boldsymbol{A}\boldsymbol{A}^\dagger\boldsymbol{A} &= \boldsymbol{U}_1\boldsymbol{\Sigma}_1\boldsymbol{V}_1^*\boldsymbol{V}_1\boldsymbol{V}_1^* = \boldsymbol{U}_1\boldsymbol{\Sigma}_1\boldsymbol{V}_1^* = \boldsymbol{A} \\ \boldsymbol{A}^\dagger\boldsymbol{A}\boldsymbol{A}^\dagger &= \boldsymbol{V}_1\boldsymbol{\Sigma}_1^{-1}\boldsymbol{U}_1^*\boldsymbol{U}_1\boldsymbol{U}_1^* = \boldsymbol{V}_1\boldsymbol{\Sigma}^{-1}\boldsymbol{U}_1^* = \boldsymbol{A}^\dagger \end{aligned}$	$\begin{aligned} \boldsymbol{A}\boldsymbol{A}^\dagger\boldsymbol{A} &= \boldsymbol{U}_1\boldsymbol{\Sigma}_1\boldsymbol{V}_1^*\boldsymbol{V}_1\boldsymbol{V}_1^* = \boldsymbol{U}_1\boldsymbol{\Sigma}_1\boldsymbol{V}_1^* = \boldsymbol{A} \\ \boldsymbol{A}^\dagger\boldsymbol{A}\boldsymbol{A}^\dagger &= \boldsymbol{V}_1\boldsymbol{\Sigma}_1^{-1}\boldsymbol{U}_1^*\boldsymbol{U}_1\boldsymbol{U}_1^* = \boldsymbol{V}_1\boldsymbol{\Sigma}^{-1}\boldsymbol{U}_1^* = \boldsymbol{A}^\dagger \end{aligned}$	$\begin{aligned} \boldsymbol{A}\boldsymbol{A}^\dagger\boldsymbol{A} &= \boldsymbol{U}_1\boldsymbol{\Sigma}_1\boldsymbol{V}_1^*\boldsymbol{V}_1\boldsymbol{V}_1^* = \boldsymbol{U}_1\boldsymbol{\Sigma}_1\boldsymbol{V}_1^* = \boldsymbol{A} \\ \boldsymbol{A}^\dagger\boldsymbol{A}\boldsymbol{A}^\dagger &= \boldsymbol{V}_1\boldsymbol{\Sigma}_1^{-1}\boldsymbol{U}_1^*\boldsymbol{U}_1\boldsymbol{U}_1^* = \boldsymbol{V}_1\boldsymbol{\Sigma}^{-1}\boldsymbol{U}_1^* = \boldsymbol{A}^\dagger. \end{aligned}$
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environmont (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- E00038	035	■0.682 ●W +1	$\begin{aligned} & \left \begin{array}{ccc} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{array} \right = a_{11}a_{22}a_{33} - a_{11}a_{32}a_{23} - a_{21}a_{12}a_{33} \\ & + a_{21}a_{32}a_{13} + a_{31}a_{12}a_{23} - a_{31}a_{22}a_{13} \end{aligned}$	$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33} - a_{11}a_{32}a_{23} - a_{21}a_{12}a_{33} + a_{21}a_{32}a_{13} + a_{31}a_{12}a_{23} - a_{31}a_{22}a_{13}.$	
lyche-numerical-linear-algebra- E00383	155	■0.685 ●	$R(x)=R_{\{A\}}(x):=\frac{x^{*}A}{x^{*}x}x$	$R(x)=R_A(x):=\frac{x^*Ax}{x^*x}$	$R(\boldsymbol{x})=R_A(\boldsymbol{x}):=\frac{\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}}{\boldsymbol{x}^*\boldsymbol{x}}$
lyche-numerical-linear-algebra- E00578	214	■0.686 ●N +0	$c_{\{1\}}\left[\begin{array}{c} 1 \\ \vdots \\ t_1 \\ \vdots \\ t_m \\ 1 \end{array}\right]+c_{\{2\}}\left[\begin{array}{c} t_1 \\ \vdots \\ t_m \end{array}\right]=\left[\begin{array}{c} 0 \\ \vdots \\ 0 \end{array}\right]\Rightarrow\left[\begin{array}{cc} 1 & t_i \\ 1 & t_j \end{array}\right]\left[\begin{array}{c} c_1 \\ c_2 \end{array}\right]=\left[\begin{array}{c} 0 \\ 0 \end{array}\right]\Rightarrow c_1=c_2=0.$	$c_1\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}+c_2\begin{bmatrix} t_1 \\ \vdots \\ t_m \end{bmatrix}=\begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}\Rightarrow\left[\begin{array}{cc} 1 & t_i \\ 1 & t_j \end{array}\right]\begin{bmatrix} c_1 \\ c_2 \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \end{bmatrix}\Rightarrow c_1=c_2=0.$	$c_1\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}+c_2\begin{bmatrix} t_1 \\ \vdots \\ t_m \end{bmatrix}=\begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}\Rightarrow\left[\begin{array}{cc} 1 & t_i \\ 1 & t_j \end{array}\right]\begin{bmatrix} c_1 \\ c_2 \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \end{bmatrix}\Rightarrow c_1=c_2=0.$
lyche-numerical-linear-algebra- E00535	203	■0.693 ●N +0	$a_{\{n\}}\langle\boldsymbol{x},\boldsymbol{y}\rangle=\left\langle a_{\{n\}}\boldsymbol{x},\boldsymbol{y}\right\rangle\stackrel{(8.36)}{=}\frac{1}{4}\left(\left\ a_{\{n\}}\boldsymbol{x}+\boldsymbol{y}\right\ ^2-\left\ a_{\{n\}}\boldsymbol{x}-\boldsymbol{y}\right\ ^2\right).$	$a_n\langle\boldsymbol{x},\boldsymbol{y}\rangle=\langle a_n\boldsymbol{x},\boldsymbol{y}\rangle\stackrel{(8.36)}{=}\frac{1}{4}\left(\ a_n\boldsymbol{x}+\boldsymbol{y}\ ^2-\ a_n\boldsymbol{x}-\boldsymbol{y}\ ^2\right).$	$a_n\langle\boldsymbol{x},\boldsymbol{y}\rangle=\langle a_n\boldsymbol{x},\boldsymbol{y}\rangle\stackrel{(8.36)}{=}\frac{1}{4}\left(\ a_n\boldsymbol{x}+\boldsymbol{y}\ ^2-\ a_n\boldsymbol{x}-\boldsymbol{y}\ ^2\right).$
lyche-numerical-linear-algebra- EQ1026	347	■0.696 ●N +0	$\left[\begin{array}{c} \boldsymbol{u}^* \\ \boldsymbol{U}^* \end{array}\right](\boldsymbol{A}\boldsymbol{u}-\lambda\boldsymbol{u})=\left[\begin{array}{c} \boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u}-\lambda\boldsymbol{u}^*\boldsymbol{u} \\ \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{u}-\lambda\boldsymbol{U}^*\boldsymbol{u} \end{array}\right]=\left[\begin{array}{c} \boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u}-\lambda \\ \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{u} \end{array}\right].$	$\begin{bmatrix} \boldsymbol{u}^* \\ \boldsymbol{U}^* \end{bmatrix}(\boldsymbol{A}\boldsymbol{u}-\lambda\boldsymbol{u})=\begin{bmatrix} \boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u}-\lambda\boldsymbol{u}^*\boldsymbol{u} \\ \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{u}-\lambda\boldsymbol{U}^*\boldsymbol{u} \end{bmatrix}=\begin{bmatrix} \boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u}-\lambda \\ \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{u} \end{bmatrix}.$	$\begin{bmatrix} \boldsymbol{u}^* \\ \boldsymbol{U}^* \end{bmatrix}(\boldsymbol{A}\boldsymbol{u}-\lambda\boldsymbol{u})=\begin{bmatrix} \boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u}-\lambda\boldsymbol{u}^*\boldsymbol{u} \\ \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{u}-\lambda\boldsymbol{U}^*\boldsymbol{u} \end{bmatrix}=\begin{bmatrix} \boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u}-\lambda \\ \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{u} \end{bmatrix}.$
lyche-numerical-linear-algebra- E00410 (6.19)	164	■0.702 ●N +0	$\mu_{\{\boldsymbol{A}\}}(\lambda):=\prod_{i=1}^k\left(\lambda-\lambda_i\right)^{m_i}\text{ where }\left\{ \begin{array}{l} m_i:=\max_{1\leq j\leq g_i}m_{i,j}, \\ m_{\{i\}}:=\max_{1\leq j\leq g_{\{i\}}}m_{\{i,j\}} \end{array} \right.$	$\mu_A(\lambda):=\prod_{i=1}^k(\lambda_i-\lambda)^{m_i}\text{ where }m_i:=\max_{1\leq j\leq g_i}m_{i,j},$	$\mu_A(\lambda):=\prod_{i=1}^k(\lambda_i-\lambda)^{m_i}\text{ where }m_i:=\max_{1\leq j\leq g_i}m_{i,j},$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-251 EQ0699 (11.1)	251	0.706 W -2	$\left[\begin{array}{cccc} \boldsymbol{D}_1 & \boldsymbol{C}_1 & & \\ \boldsymbol{A}_1 & \boldsymbol{D}_2 & \boldsymbol{C}_2 & \\ & \ddots & \ddots & \ddots \\ & & \boldsymbol{A}_{m-2} & \boldsymbol{D}_{m-1} \\ & & \boldsymbol{A}_{m-1} & \boldsymbol{D}_m \end{array}\right] = \left[\begin{array}{cccc} \boldsymbol{I} & & & \\ \boldsymbol{L}_1 & \boldsymbol{I} & & \\ & \ddots & \ddots & \\ & & \boldsymbol{L}_{m-1} & \boldsymbol{I} \end{array}\right] \left[\begin{array}{cccc} \boldsymbol{U}_1 & \boldsymbol{C}_1 & & \\ & \ddots & \ddots & \\ & & \boldsymbol{U}_{m-1} & \boldsymbol{C}_{m-1} \\ & & & \boldsymbol{U}_m \end{array}\right].$	$\left[\begin{array}{cccc} \boldsymbol{D}_1 & \boldsymbol{C}_1 & & \\ \boldsymbol{A}_1 & \boldsymbol{D}_2 & \boldsymbol{C}_2 & \\ & \ddots & \ddots & \ddots \\ & & \boldsymbol{A}_{m-2} & \boldsymbol{D}_{m-1} \\ & & \boldsymbol{A}_{m-1} & \boldsymbol{D}_m \end{array}\right] = \left[\begin{array}{cccc} \boldsymbol{I} & & & \\ \boldsymbol{L}_1 & \boldsymbol{I} & & \\ & \ddots & \ddots & \\ & & \boldsymbol{L}_{m-1} & \boldsymbol{I} \end{array}\right] \left[\begin{array}{cccc} \boldsymbol{U}_1 & \boldsymbol{C}_1 & & \\ & \ddots & \ddots & \\ & & \boldsymbol{U}_{m-1} & \boldsymbol{C}_{m-1} \\ & & & \boldsymbol{U}_m \end{array}\right].$	
lyche-numerical-linear-algebra-256 EQ0717	256	0.709 N +0	$(\boldsymbol{S})_{\boldsymbol{x}}_{\boldsymbol{k}} = \frac{i}{2} \left(\boldsymbol{F}_{2m+2} \boldsymbol{z} \right)_{k+1}, \quad k = 1, \dots, m.$	$(\boldsymbol{S} \boldsymbol{x})_k = \frac{i}{2} (\boldsymbol{F}_{2m+2} \boldsymbol{z})_{k+1}, \quad k = 1, \dots, m.$	$(\boldsymbol{S} \boldsymbol{x})_k = \frac{i}{2} (\boldsymbol{F}_{2m+2} \boldsymbol{z})_{k+1}, \quad k = 1, \dots, m.$
lyche-numerical-linear-algebra-218 EQ0593	218	0.712	$\ A \, x - b\ _2^2 = \ (A \, x - b_1) - b_2\ _2^2 = \ A \, x - b_1\ _2^2 + \ b_2\ _2^2 \geq \ b_2\ _2^2$	$\ Ax - b\ _2^2 = \ (Ax - b_1) - b_2\ _2^2 = \ Ax - b_1\ _2^2 + \ b_2\ _2^2 \geq \ b_2\ _2^2$	$\ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2^2 = \ (\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1) - \boldsymbol{b}_2\ _2^2 = \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1\ _2^2 + \ \boldsymbol{b}_2\ _2^2 \geq \ \boldsymbol{b}_2\ _2^2$
lyche-numerical-linear-algebra-203 EQ0530 (8.38)	203	0.713 W +1	$\begin{aligned} & 4 \angle x, z \stackrel{(8.36)}{=} \ x+z\ ^2 - \ x-z\ ^2 + \ y+z\ ^2 - \ y-z\ ^2 \\ & = \left\ \left(z + \frac{x+y}{2} \right) + \frac{x-y}{2} \right\ ^2 - \left\ \left(z - \frac{x+y}{2} \right) + \frac{y-x}{2} \right\ ^2 \\ & + \left\ \left(z + \frac{x+y}{2} \right) - \frac{x-y}{2} \right\ ^2 - \left\ \left(z - \frac{x+y}{2} \right) - \frac{y-x}{2} \right\ ^2 \\ & \stackrel{(8.35)}{=} 2 \left\ z + \frac{x+y}{2} \right\ ^2 + 2 \left\ \frac{x-y}{2} \right\ ^2 - 2 \left\ z - \frac{x+y}{2} \right\ ^2 - 2 \left\ \frac{y-x}{2} \right\ ^2 \\ & \stackrel{(8.36)}{=} 8 \left\langle \frac{x+y}{2}, z \right\rangle \end{aligned}$	$\begin{aligned} & 4 \langle x, z \rangle + 4 \langle y, z \rangle \stackrel{(8.36)}{=} \ x+z\ ^2 - \ x-z\ ^2 + \ y+z\ ^2 - \ y-z\ ^2 \\ & = \left\ \left(z + \frac{x+y}{2} \right) + \frac{x-y}{2} \right\ ^2 - \left\ \left(z - \frac{x+y}{2} \right) + \frac{y-x}{2} \right\ ^2 \\ & + \left\ \left(z + \frac{x+y}{2} \right) - \frac{x-y}{2} \right\ ^2 - \left\ \left(z - \frac{x+y}{2} \right) - \frac{y-x}{2} \right\ ^2 \\ & \stackrel{(8.35)}{=} 2 \left\ z + \frac{x+y}{2} \right\ ^2 + 2 \left\ \frac{x-y}{2} \right\ ^2 - 2 \left\ z - \frac{x+y}{2} \right\ ^2 - 2 \left\ \frac{y-x}{2} \right\ ^2 \\ & \stackrel{(8.36)}{=} 8 \left\langle \frac{x+y}{2}, z \right\rangle \end{aligned}$	$\begin{aligned} & 4 \langle \boldsymbol{x}, \boldsymbol{z} \rangle + 4 \langle \boldsymbol{y}, \boldsymbol{z} \rangle \stackrel{(8.36)}{=} \ \boldsymbol{x} + \boldsymbol{z}\ ^2 - \ \boldsymbol{x} - \boldsymbol{z}\ ^2 + \ \boldsymbol{y} + \boldsymbol{z}\ ^2 - \ \boldsymbol{y} - \boldsymbol{z}\ ^2 \\ & = \left\ \left(\boldsymbol{z} + \frac{\boldsymbol{x} + \boldsymbol{y}}{2} \right) + \frac{\boldsymbol{x} - \boldsymbol{y}}{2} \right\ ^2 - \left\ \left(\boldsymbol{z} - \frac{\boldsymbol{x} + \boldsymbol{y}}{2} \right) + \frac{\boldsymbol{y} - \boldsymbol{x}}{2} \right\ ^2 \\ & + \left\ \left(\boldsymbol{z} + \frac{\boldsymbol{x} + \boldsymbol{y}}{2} \right) - \frac{\boldsymbol{x} - \boldsymbol{y}}{2} \right\ ^2 - \left\ \left(\boldsymbol{z} - \frac{\boldsymbol{x} + \boldsymbol{y}}{2} \right) - \frac{\boldsymbol{y} - \boldsymbol{x}}{2} \right\ ^2 \\ & \stackrel{(8.35)}{=} 2 \ \boldsymbol{z} + \frac{\boldsymbol{x} + \boldsymbol{y}}{2}\ ^2 + 2 \ \frac{\boldsymbol{x} - \boldsymbol{y}}{2}\ ^2 - 2 \ \boldsymbol{z} - \frac{\boldsymbol{x} + \boldsymbol{y}}{2}\ ^2 - 2 \ \frac{\boldsymbol{y} - \boldsymbol{x}}{2}\ ^2 \\ & \stackrel{(8.36)}{=} 8 \left\langle \frac{\boldsymbol{x} + \boldsymbol{y}}{2}, \boldsymbol{z} \right\rangle, \end{aligned}$
lyche-numerical-linear-algebra-226 EQ0615	226	0.721 K +0	$\boldsymbol{A}^{\dagger} \boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, (\boldsymbol{A}^* \boldsymbol{A})^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}, \boldsymbol{A}^{\dagger} = (\boldsymbol{A}^* \boldsymbol{A})^{-1} \boldsymbol{A}^* = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$	$\boldsymbol{A}^* \boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, (\boldsymbol{A}^* \boldsymbol{A})^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}, \boldsymbol{A}^{\dagger} = (\boldsymbol{A}^* \boldsymbol{A})^{-1} \boldsymbol{A}^* = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$	$\boldsymbol{A}^* \boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, (\boldsymbol{A}^* \boldsymbol{A})^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}, \boldsymbol{A}^{\dagger} = (\boldsymbol{A}^* \boldsymbol{A})^{-1} \boldsymbol{A}^* = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0460 (2)	182	0.722 C +6	$\boldsymbol{A}:=\left[\begin{array}{rrr} 0 & 3 & 3 \\ 4 & 1 & -1 \\ 4 & 1 & -1 \\ 0 & 3 & 3 \end{array}\right]=\left[\begin{array}{rrrr} \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{array}\right]\left[\begin{array}{rrr} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right]\left[\begin{array}{rrr} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right]$	$\boldsymbol{A}:=\left[\begin{array}{rrr} 0 & 3 & 3 \\ 4 & 1 & -1 \\ 4 & 1 & -1 \\ 0 & 3 & 3 \end{array}\right]=\left[\begin{array}{rrrr} \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{array}\right]\left[\begin{array}{rrr} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right]\left[\begin{array}{rrr} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right]$	
lyche-numerical-linear-algebra-EQ0736	261	0.724 N -1	$\boldsymbol{T} \boldsymbol{X}+\boldsymbol{X} \boldsymbol{D}=\boldsymbol{C}, \text { where } \boldsymbol{X}=\boldsymbol{V} \boldsymbol{S}, \text { and } \boldsymbol{C}=\boldsymbol{h}^2 \boldsymbol{F} \boldsymbol{S} .$	$\boldsymbol{T} \boldsymbol{X}+\boldsymbol{X} \boldsymbol{D}=\boldsymbol{C}, \text { where } \boldsymbol{X}=\boldsymbol{V} \boldsymbol{S}, \text { and } \boldsymbol{C}=\boldsymbol{h}^2 \boldsymbol{F} \boldsymbol{S} .$	
lyche-numerical-linear-algebra-EQ0668	241	0.727	$\boldsymbol{A}=\left[\begin{array}{rrrrrrrr} 4 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 4 & -1 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & 4 & 0 & 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 4 & -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & -1 & 4 & -1 & 0 & -1 \\ 0 & 0 & -1 & 0 & -1 & 4 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 4 & -1 \\ 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 \end{array}\right] .$	$\boldsymbol{A}=\left[\begin{array}{rrrrrrrr} 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 \\ -1 & 4 & -1 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & 4 & 0 & 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 4 & -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & -1 & 4 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 & -1 & 4 & 0 & -1 \\ 0 & 0 & 0 & -1 & 0 & 0 & 4 & -1 \\ 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 \end{array}\right] .$	
lyche-numerical-linear-algebra-EQ0976 (14.1)	330	0.729 N +0	$ \mu-\lambda \leq K\left E\right _{2}^{1 / n}, \quad K=\left(\left\ A\right\ _{2}+\left\ A+E\right\ _{2}\right)^{1-1 / n} .$	$ \mu-\lambda \leq K\left\ E\right\ _{2}^{1 / n}, \quad K=\left(\left\ A\right\ _{2}+\left\ A+E\right\ _{2}\right)^{1-1 / n} .$	

Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0676 (10.13)	243	0.737 W +2	$\begin{aligned} & (\lambda \boldsymbol{A}) \otimes (\mu \boldsymbol{B}) = \lambda \mu (\boldsymbol{A} \otimes \boldsymbol{B}) \\ & (\boldsymbol{A}_1 + \boldsymbol{A}_2) \otimes \boldsymbol{B} = \boldsymbol{A}_1 \otimes \boldsymbol{B} + \boldsymbol{A}_2 \otimes \boldsymbol{B}, \\ & \boldsymbol{A} \otimes (\boldsymbol{B}_1 + \boldsymbol{B}_2) = \boldsymbol{A} \otimes \boldsymbol{B}_1 + \boldsymbol{A} \otimes \boldsymbol{B}_2, \\ & (\boldsymbol{A} \otimes \boldsymbol{B}) \otimes \boldsymbol{C} = \boldsymbol{A} \otimes (\boldsymbol{B} \otimes \boldsymbol{C}). \end{aligned}$	$(\lambda \boldsymbol{A}) \otimes (\mu \boldsymbol{B}) = \lambda \mu (\boldsymbol{A} \otimes \boldsymbol{B})$ $(\boldsymbol{A}_1 + \boldsymbol{A}_2) \otimes \boldsymbol{B} = \boldsymbol{A}_1 \otimes \boldsymbol{B} + \boldsymbol{A}_2 \otimes \boldsymbol{B},$ $\boldsymbol{A} \otimes (\boldsymbol{B}_1 + \boldsymbol{B}_2) = \boldsymbol{A} \otimes \boldsymbol{B}_1 + \boldsymbol{A} \otimes \boldsymbol{B}_2$ $(\boldsymbol{A} \otimes \boldsymbol{B}) \otimes \boldsymbol{C} = \boldsymbol{A} \otimes (\boldsymbol{B} \otimes \boldsymbol{C}).$	$(\lambda \boldsymbol{A}) \otimes (\mu \boldsymbol{B}) = \lambda \mu (\boldsymbol{A} \otimes \boldsymbol{B}),$ $(\boldsymbol{A}_1 + \boldsymbol{A}_2) \otimes \boldsymbol{B} = \boldsymbol{A}_1 \otimes \boldsymbol{B} + \boldsymbol{A}_2 \otimes \boldsymbol{B},$ $\boldsymbol{A} \otimes (\boldsymbol{B}_1 + \boldsymbol{B}_2) = \boldsymbol{A} \otimes \boldsymbol{B}_1 + \boldsymbol{A} \otimes \boldsymbol{B}_2,$ $(\boldsymbol{A} \otimes \boldsymbol{B}) \otimes \boldsymbol{C} = \boldsymbol{A} \otimes (\boldsymbol{B} \otimes \boldsymbol{C}).$
lyche-numerical-linear-algebra EQ0466 (8.3)	186	0.737 N +0	$\sum_{j=1}^n x_j y_j \leq \ \boldsymbol{x} \ _p \ \boldsymbol{y} \ _q, \quad \frac{1}{p} + \frac{1}{q} = 1, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n.$	$\sum_{j=1}^n x_j y_j \leq \ \boldsymbol{x} \ _p \ \boldsymbol{y} \ _q, \quad \frac{1}{p} + \frac{1}{q} = 1, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n.$	$\sum_{j=1}^n x_j y_j \leq \ \boldsymbol{x} \ _p \ \boldsymbol{y} \ _q, \quad \frac{1}{p} + \frac{1}{q} = 1, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n.$
lyche-numerical-linear-algebra EQ0897	308	0.743 W +0	$\frac{\left(\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y} \right) \left(\boldsymbol{y}^T \boldsymbol{A}^{-1} \boldsymbol{y} \right)}{\left(\boldsymbol{y}^T \boldsymbol{y} \right)^2} = ab \leq \frac{(M+m)^2}{4Mm},$	$\frac{(\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y})(\boldsymbol{y}^T \boldsymbol{A}^{-1} \boldsymbol{y})}{(\boldsymbol{y}^T \boldsymbol{y})^2} = ab \leq \frac{(M+m)^2}{4Mm},$	$\frac{(\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y})(\boldsymbol{y}^T \boldsymbol{A}^{-1} \boldsymbol{y})}{(\boldsymbol{y}^T \boldsymbol{y})^2} = ab \leq \frac{(M+m)^2}{4Mm},$
lyche-numerical-linear-algebra EQ0582	216	0.744 W +0	$\begin{array}{c c c c c} \boldsymbol{u} & \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_m \\ \hline & y_1 & y_2 & \cdots & y_m \end{array}.$	$\frac{\boldsymbol{u} \mid \boldsymbol{u}_1 \mid \boldsymbol{u}_2 \mid \cdots \mid \boldsymbol{u}_m}{y \mid y_1 \mid y_2 \mid \cdots \mid y_m}.$	$\frac{\boldsymbol{u} \mid \boldsymbol{u}_1 \mid \boldsymbol{u}_2 \mid \cdots \mid \boldsymbol{u}_m}{y \mid y_1 \mid y_2 \mid \cdots \mid y_m}.$
lyche-numerical-linear-algebra EQ0204	085	0.749	$\begin{aligned} & \text{for } i = k+1 : n \\ & \text{for } j = k : n \\ & a_{p_i,k}^{(k)} = a_{p_i,k}^{(k)} / a_{p_k,k}^{(k)} \\ & a_{p_i,j}^{(k+1)} = a_{p_i,j}^{(k)} - a_{p_i,k}^{(k)} a_{p_k,j}^{(k)} \end{aligned}$	$\text{for } i = k+1 : n$ $a_{p_i,k}^{(k)} = a_{p_i,k}^{(k)} / a_{p_k,k}^{(k)}$ $\text{for } j = k : n$ $a_{p_i,j}^{(k+1)} = a_{p_i,j}^{(k)} - a_{p_i,k}^{(k)} a_{p_k,j}^{(k)}$	$\text{for } i = k+1 : n$ $a_{p_i,k}^{(k)} = a_{p_i,k}^{(k)} / a_{p_k,k}^{(k)}$ $\text{for } j = k : n$ $a_{p_i,j}^{(k+1)} = a_{p_i,j}^{(k)} - a_{p_i,k}^{(k)} a_{p_k,j}^{(k)}$
lyche-numerical-linear-algebra EQ0142	065	0.749 K +0	$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \mathbf{0} & \boldsymbol{A}_{22} \end{bmatrix}$	$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \mathbf{0} & \boldsymbol{A}_{22} \end{bmatrix}$	$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \mathbf{0} & \boldsymbol{A}_{22} \end{bmatrix}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-253 EQ0707	253	0.749 N +0	$\begin{aligned} &(\boldsymbol{D})_{j,k}=\sum_{\ell=1}^m d_{j,\ell} x_{\ell,k} \\ &x_{\ell,k}=\sum_{j=1}^m \lambda_{j,k} d_{\ell,j} \end{aligned}$	$(\boldsymbol{D}\boldsymbol{X}+\boldsymbol{X}\boldsymbol{D})_{j,k}=\sum_{\ell=1}^m d_{j,\ell} x_{\ell,k}+\sum_{\ell=1}^m x_{j,\ell} d_{\ell,k}=\lambda_j x_{j,k}+\lambda_k x_{j,k}$	$(\boldsymbol{D}\boldsymbol{X}+\boldsymbol{X}\boldsymbol{D})_{j,k}=\sum_{\ell=1}^m d_{j,\ell} x_{\ell,k}+\sum_{\ell=1}^m x_{j,\ell} d_{\ell,k}=\lambda_j x_{j,k}+\lambda_k x_{j,k}$
lyche-numerical-linear-algebra-322 EQ0959 (13.68)	322	0.749 N +0	$\omega_k:=\begin{cases} 1, & \text{if } k=0 \\ \left(1+\omega_{k-1}^{-1}\rho_k/\rho_{k-1}\right)^{-1}, & \text{otherwise.} \end{cases}$	$\omega_k:=\begin{cases} 1, & \text{if } k=0 \\ \left(1+\omega_{k-1}^{-1}\rho_k/\rho_{k-1}\right)^{-1}, & \text{otherwise.} \end{cases}$	$\omega_k:=\begin{cases} 1, & \text{if } k=0 \\ \left(1+\omega_{k-1}^{-1}\rho_k/\rho_{k-1}\right)^{-1}, & \text{otherwise.} \end{cases}$
lyche-numerical-linear-algebra-213 EQ0575	213	0.751 W +1	$\begin{aligned} &x_1=1 \\ &\boldsymbol{A}=\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad \boldsymbol{x}=[x_1], \quad \boldsymbol{b}=\begin{bmatrix} 1 \\ 1 \end{bmatrix}. \end{aligned}$	$x_1=1, \quad \boldsymbol{A}=\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad \boldsymbol{x}=[x_1], \quad \boldsymbol{b}=\begin{bmatrix} 1 \\ 1 \end{bmatrix}.$	$x_1=1, \quad \boldsymbol{A}=\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad \boldsymbol{x}=[x_1], \quad \boldsymbol{b}=\begin{bmatrix} 1 \\ 1 \end{bmatrix}.$
lyche-numerical-linear-algebra-234 EQ0649	234	0.756 K +0	$U^*AV=\begin{bmatrix} \Sigma_1 & 0 \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$	$U^*AV=\begin{bmatrix} \Sigma_1 & 0 \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$	$U^*AV=\begin{bmatrix} \Sigma_1 & 0 \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$
lyche-numerical-linear-algebra-240 EQ0667 (10.8)	240	0.758 W -2	$\begin{aligned} &a_{ii}=4, i=1,\dots,n, \\ &a_{i+1,i}=a_{i,i+1}=-1, i=1,\dots,n-1, \\ &a_{i+m,i}=a_{i,i+m}=-1, i=1,\dots,n-m, \\ &a_{ij}=0, \text{ otherwise.} \end{aligned}$	$\begin{aligned} &a_{ii}=4, i=1,\dots,n, \\ &a_{i+1,i}=a_{i,i+1}=-1, i=1,\dots,n-1, \\ &a_{i+m,i}=a_{i,i+m}=-1, i=1,\dots,n-m, \\ &a_{ij}=0, \text{ otherwise.} \end{aligned}$	$\begin{aligned} &a_{ii}=4, \quad i=1,\dots,n, \\ &a_{i+1,i}=a_{i,i+1}=-1, \quad i=1,\dots,n-1, \quad i\neq m, 2m,\dots,(m-1)m, \\ &a_{i+m,i}=a_{i,i+m}=-1, \quad i=1,\dots,n-m, \\ &a_{ij}=0, \quad \text{otherwise.} \end{aligned}$
lyche-numerical-linear-algebra-345 EQ1018	345	0.760 K +0	$z_k=c_1\lambda_1^k v_1+c_2\lambda_2^k v_2=c_13^k v_1+c_21^k v_2.$	$z_k=c_1\lambda_1^k v_1+c_2\lambda_2^k v_2=c_13^k v_1+c_21^k v_2.$	$\boldsymbol{z}_k=c_1\lambda_1^k \boldsymbol{v}_1+c_2\lambda_2^k \boldsymbol{v}_2=c_13^k \boldsymbol{v}_1+c_21^k \boldsymbol{v}_2.$
lyche-numerical-linear-algebra-163 EQ0408	163	0.761 K +0	$\boldsymbol{A}=\frac{1}{6}\begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$	$\boldsymbol{A}=\frac{1}{6}\begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$	$\boldsymbol{A}=\frac{1}{6}\begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$
lyche-numerical-linear-algebra-234 EQ0651	234	0.763 N +0	$\begin{aligned} &\text{(1) } \boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}, \quad \text{(2) } \boldsymbol{A}^*\boldsymbol{y}=\mathbf{0}, \boldsymbol{y}^*\boldsymbol{b}\neq 0. \end{aligned}$	$\text{(1) } \boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}, \quad \text{(2) } \boldsymbol{A}^*\boldsymbol{y}=\mathbf{0}, \boldsymbol{y}^*\boldsymbol{b}\neq 0.$	$\text{(1) } \boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}, \quad \text{(2) } \boldsymbol{A}^*\boldsymbol{y}=\mathbf{0}, \boldsymbol{y}^*\boldsymbol{b}\neq 0.$
lyche-numerical-linear-algebra-077 EQ0176	077	0.770 W +0	$\text{for } i=k+1:n \quad l_{ik}^{(k)}=a_{ik}^{(k)}/a_{kk}^{(k)}$	$\text{for } i=k+1:n \quad l_{ik}^{(k)}=a_{ik}^{(k)}/a_{kk}^{(k)}$	$\text{for } i=k+1:n \quad l_{ik}^{(k)}=a_{ik}^{(k)}/a_{kk}^{(k)}$
lyche-numerical-linear-algebra-273 EQ0772 (12.19)	273	0.772	$\begin{aligned} &\rho(\boldsymbol{I}-\alpha\boldsymbol{A})>\rho(\boldsymbol{I}-\alpha_o\boldsymbol{A})=\frac{\kappa-1}{\kappa+1}, \quad \alpha\in\mathbb{R}\setminus\{\alpha_o\}, \\ &\kappa:=\frac{\lambda_1}{\lambda_n}, \quad \alpha_o:=\frac{2}{\lambda_1+\lambda_n}. \end{aligned}$	$\rho(\boldsymbol{I}-\alpha\boldsymbol{A})>\rho(\boldsymbol{I}-\alpha_o\boldsymbol{A})=\frac{\kappa-1}{\kappa+1}, \quad \alpha\in\mathbb{R}\setminus\{\alpha_o\},$ $\kappa:=\frac{\lambda_1}{\lambda_n}, \quad \alpha_o:=\frac{2}{\lambda_1+\lambda_n}.$	$\rho(\boldsymbol{I}-\alpha\boldsymbol{A})>\rho(\boldsymbol{I}-\alpha_o\boldsymbol{A})=\frac{\kappa-1}{\kappa+1}, \quad \alpha\in\mathbb{R}\setminus\{\alpha_o\},$ $\kappa:=\frac{\lambda_1}{\lambda_n}, \quad \alpha_o:=\frac{2}{\lambda_1+\lambda_n}.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0915 (13.47)	311	0.779 K +0	$\begin{aligned} \boldsymbol{x}_{\{k\}} &:= \boldsymbol{C}^{\mathrm{T}} \\ \boldsymbol{y}_{\{k\}}, \quad &\boldsymbol{p}_{\{k\}} := \\ \boldsymbol{C}^{\mathrm{T}} \boldsymbol{q}_{\{k\}}, \quad &\boldsymbol{q}_{\{k\}}, \quad \\ \boldsymbol{s}_{\{k\}} &:= \boldsymbol{C}^{\mathrm{T}} \boldsymbol{z}_{\{k\}}, \quad \\ \boldsymbol{r}_{\{k\}} &:= \boldsymbol{C}^{-1} \boldsymbol{z}_{\{k\}} \end{aligned}$	$\boldsymbol{x}_k := \boldsymbol{C}^T \boldsymbol{y}_k, \quad \boldsymbol{p}_k := \boldsymbol{C}^T \boldsymbol{q}_k, \quad \boldsymbol{s}_k := \boldsymbol{C}^T \boldsymbol{z}_k, \quad \boldsymbol{r}_k := \boldsymbol{C}^{-1} \boldsymbol{z}_k$	$\boldsymbol{x}_k := \boldsymbol{C}^T \boldsymbol{y}_k, \quad \boldsymbol{p}_k := \boldsymbol{C}^T \boldsymbol{q}_k, \quad \boldsymbol{s}_k := \boldsymbol{C}^T \boldsymbol{z}_k, \quad \boldsymbol{r}_k := \boldsymbol{C}^{-1} \boldsymbol{z}_k$
lyche-numerical-linear-algebra-EQ0610	223	0.780 N +0	$\begin{aligned} \boldsymbol{v} &= \boldsymbol{U} \boldsymbol{U}^* \boldsymbol{v} = \left[\boldsymbol{U}_1, \boldsymbol{U}_2 \left[\begin{array}{l} \boldsymbol{U}_1^* \\ \boldsymbol{U}_2^* \end{array} \right] \right] \boldsymbol{v} = \boldsymbol{s} + \boldsymbol{t}, \\ \boldsymbol{v} &= \boldsymbol{U} \boldsymbol{U}^* \boldsymbol{v} = \left[\boldsymbol{U}_1, \boldsymbol{U}_2 \left[\begin{array}{l} \boldsymbol{U}_1^* \\ \boldsymbol{U}_2^* \end{array} \right] \right] \boldsymbol{v} = \boldsymbol{s} + \boldsymbol{t}, \end{aligned}$	$\boldsymbol{v} = \boldsymbol{U} \boldsymbol{U}^* \boldsymbol{v} = \left[\boldsymbol{U}_1, \boldsymbol{U}_2 \left[\begin{array}{l} \boldsymbol{U}_1^* \\ \boldsymbol{U}_2^* \end{array} \right] \right] \boldsymbol{v} = \boldsymbol{s} + \boldsymbol{t},$	$\boldsymbol{v} = \boldsymbol{U} \boldsymbol{U}^* \boldsymbol{v} = [\boldsymbol{U}_1, \boldsymbol{U}_2 \left[\begin{array}{l} \boldsymbol{U}_1^* \\ \boldsymbol{U}_2^* \end{array} \right] \boldsymbol{v} = \boldsymbol{s} + \boldsymbol{t},$
lyche-numerical-linear-algebra-EQ0203 (3.13)	085	0.781 W +0	$\begin{aligned} &\begin{aligned} \boldsymbol{p} &= [\boldsymbol{1}, \ldots, \boldsymbol{n}]^T; \\ \text{for } k &= 1 : n - 1 \\ \text{choose } r_k &\geq k \text{ so that } a_{p_{r_k}, k}^{(k)} \neq 0. \\ \boldsymbol{p} &= I_{r_k, k} \boldsymbol{p} \end{aligned} \end{aligned}$	$\begin{aligned} \boldsymbol{p} &= [\boldsymbol{1}, \ldots, \boldsymbol{n}]^T; \\ \text{for } k &= 1 : n - 1 \\ \text{choose } r_k &\geq k \text{ so that } a_{p_{r_k}, k}^{(k)} \neq 0. \\ \boldsymbol{p} &= I_{r_k, k} \boldsymbol{p} \end{aligned}$	$\begin{aligned} \boldsymbol{p} &= [\boldsymbol{1}, \ldots, \boldsymbol{n}]^T; \\ \text{for } k &= 1 : n - 1 \\ \text{choose } r_k &\geq k \text{ so that } a_{p_{r_k}, k}^{(k)} \neq 0. \\ \boldsymbol{p} &= I_{r_k, k} \boldsymbol{p} \end{aligned}$
lyche-numerical-linear-algebra-EQ0852 (13.15)	295	0.782 W +0	$\begin{aligned} &\begin{aligned} \text{For } k &= 0, 1, 2, \ldots \\ \boldsymbol{x}_{k+1} &:= \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k, \quad \alpha_k := \frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k}, \\ \boldsymbol{r}_{k+1} &:= \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k, \end{aligned} \end{aligned}$	$\begin{aligned} \text{For } k &= 0, 1, 2, \ldots \\ \boldsymbol{x}_{k+1} &:= \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k, \quad \alpha_k := \frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k}, \\ \boldsymbol{r}_{k+1} &:= \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k, \end{aligned}$	$\begin{aligned} \text{For } k &= 0, 1, 2, \ldots \\ \boldsymbol{x}_{k+1} &:= \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k, \quad \alpha_k := \frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k}, \\ \boldsymbol{r}_{k+1} &:= \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k, \end{aligned}$
lyche-numerical-linear-algebra-EQ0531	203	0.783 N +0	$\begin{aligned} \langle \boldsymbol{x}, \boldsymbol{z} \rangle + \langle \boldsymbol{y}, \boldsymbol{z} \rangle &= 2 \left\langle \frac{\boldsymbol{x} + \boldsymbol{y}}{2}, \boldsymbol{z} \right\rangle, \quad \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}. \\ \langle \boldsymbol{x}, \boldsymbol{z} \rangle + \langle \boldsymbol{y}, \boldsymbol{z} \rangle &= 2 \left\langle \frac{\boldsymbol{x} + \boldsymbol{y}}{2}, \boldsymbol{z} \right\rangle, \quad \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}. \end{aligned}$	$\langle \boldsymbol{x}, \boldsymbol{z} \rangle + \langle \boldsymbol{y}, \boldsymbol{z} \rangle = 2 \left\langle \frac{\boldsymbol{x} + \boldsymbol{y}}{2}, \boldsymbol{z} \right\rangle, \quad \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}.$	$\langle \boldsymbol{x}, \boldsymbol{z} \rangle + \langle \boldsymbol{y}, \boldsymbol{z} \rangle = 2 \langle \frac{\boldsymbol{x} + \boldsymbol{y}}{2}, \boldsymbol{z} \rangle, \quad \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}.$
lyche-numerical-linear-algebra-EQ0009 (1.1)	023	0.784 N +0	$\begin{aligned} \delta_{ij} &:= \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases} \\ \delta_{ij} &:= \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases} \end{aligned}$	$\delta_{ij} := \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$	$\delta_{ij} := \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{otherwise,} \end{cases}$
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-E00108 (2.21)	057	0.788	$\begin{array}{l} \text{\textbf{T}} \text{\textbf{v}} = \left[\begin{array}{cccc} 2 & -1 & 0 & \\ -1 & 2 & -1 & \\ 0 & \ddots & \ddots & \ddots \\ & & -1 & 2-1 \\ & & 0-1 & 2 \end{array} \right] \left[\begin{array}{c} v_1 \\ v_2 \\ \vdots \\ v_{m-1} \\ v_m \end{array} \right] = h^2 \left[\begin{array}{c} f(h) \\ f(2h) \\ \vdots \\ f((m-1)h) \\ f(mh) \end{array} \right] =: \text{\textbf{b}}. \end{array}$	$T\boldsymbol{v} = \begin{bmatrix} 2 & -1 & 0 & & \\ -1 & 2 & -1 & & \\ 0 & \ddots & \ddots & \ddots & \\ & & -1 & 2-1 & \\ & & 0-1 & 2 & \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_{m-1} \\ v_m \end{bmatrix} = h^2 \begin{bmatrix} f(h) \\ f(2h) \\ \vdots \\ f((m-1)h) \\ f(mh) \end{bmatrix} =: \boldsymbol{b}.$	$T\boldsymbol{v} = \begin{bmatrix} 2 & -1 & 0 & & \\ -1 & 2 & -1 & & \\ 0 & \ddots & \ddots & \ddots & \\ & & -1 & 2-1 & \\ & & 0-1 & 2 & \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_{m-1} \\ v_m \end{bmatrix} = h^2 \begin{bmatrix} f(h) \\ f(2h) \\ \vdots \\ f((m-1)h) \\ f(mh) \end{bmatrix} =: \boldsymbol{b}.$
lyche-numerical-linear-algebra-E00401	160	0.791	$\text{\textbf{V}}^{\text{\textbf{A}}}\text{\textbf{V}}^{\text{\textbf{V}}} \text{\textbf{A}} \text{\textbf{V}} = \left[\begin{array}{c} \lambda \\ \mathbf{0} \end{array} \right] \text{\textbf{M}}^{\text{\textbf{z}}},$	$\boldsymbol{V}^* \boldsymbol{A} \boldsymbol{V} = \left[\begin{array}{c c} \lambda & z^* \\ \hline \mathbf{0} & \boldsymbol{M} \end{array} \right],$	$\boldsymbol{V}^* \boldsymbol{A} \boldsymbol{V} = \left[\begin{array}{c c} \lambda & \boldsymbol{z}^* \\ \hline \mathbf{0} & \boldsymbol{M} \end{array} \right],$
lyche-numerical-linear-algebra-E00590	218	0.792	$\begin{array}{l} \frac{1}{m} b_{i,j} = \frac{1}{m} \sum_{k=1}^m t_k^{i+j-2} = \frac{1}{m} \sum_{k=1}^m \left(\frac{k-1}{m-1} \right)^{i+j-2} \\ \approx \int_0^1 x^{i+j-2} dx = \frac{1}{i+j-1} = h_{i,j}. \end{array}$	$\frac{1}{m} b_{i,j} = \frac{1}{m} \sum_{k=1}^m t_k^{i+j-2} = \frac{1}{m} \sum_{k=1}^m \left(\frac{k-1}{m-1} \right)^{i+j-2} \\ \approx \int_0^1 x^{i+j-2} dx = \frac{1}{i+j-1} = h_{i,j}$	$\frac{1}{m} b_{i,j} = \frac{1}{m} \sum_{k=1}^m t_k^{i+j-2} = \frac{1}{m} \sum_{k=1}^m \left(\frac{k-1}{m-1} \right)^{i+j-2} \\ \approx \int_0^1 x^{i+j-2} dx = \frac{1}{i+j-1} = h_{i,j}.$
lyche-numerical-linear-algebra-E00096 (0)	054	0.793	$\begin{array}{l} \left[\begin{array}{ccc} d_1 & c_1 & 0 \\ a_1 & d_2 & c_2 \\ 0 & a_2 & d_3 \end{array} \right] = \left[\begin{array}{ccc} 1 & 0 & 0 \\ l_1 & 1 & 0 \\ 0 & l_2 & 1 \end{array} \right] \left[\begin{array}{ccc} u_1 & c_1 & 0 \\ 0 & u_2 & c_2 \\ 0 & 0 & u_3 \end{array} \right] = \left[\begin{array}{ccc} u_1 & c_1 & 0 \\ l_1 u_1 & l_1 c_1 + u_2 & c_2 \\ 0 & l_2 u_2 & l_2 c_2 + u_3 \end{array} \right], \end{array}$	$\begin{bmatrix} d_1 & c_1 & 0 \\ a_1 & d_2 & c_2 \\ 0 & a_2 & d_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_1 & 1 & 0 \\ 0 & l_2 & 1 \end{bmatrix} \begin{bmatrix} u_1 & c_1 & 0 \\ 0 & u_2 & c_2 \\ 0 & 0 & u_3 \end{bmatrix} = \begin{bmatrix} u_1 & c_1 & 0 \\ l_1 u_1 & l_1 c_1 + u_2 & c_2 \\ 0 & l_2 u_2 & l_2 c_2 + u_3 \end{bmatrix},$	$\begin{bmatrix} d_1 & c_1 & 0 \\ a_1 & d_2 & c_2 \\ 0 & a_2 & d_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_1 & 1 & 0 \\ 0 & l_2 & 1 \end{bmatrix} \begin{bmatrix} u_1 & c_1 & 0 \\ 0 & u_2 & c_2 \\ 0 & 0 & u_3 \end{bmatrix} = \begin{bmatrix} u_1 & c_1 & 0 \\ l_1 u_1 & l_1 c_1 + u_2 & c_2 \\ 0 & l_2 u_2 & l_2 c_2 + u_3 \end{bmatrix},$
lyche-numerical-linear-algebra-E00752 (12.7)	268	0.800	$\begin{array}{l} \text{\textbf{S}} \text{\textbf{O}} \text{\textbf{R}}: \text{\textbf{v}}_{k+1}(i, j) = \omega \left(\text{\textbf{v}}_{k+1}(i-1, j) + \text{\textbf{v}}_{k+1}(i, j-1) + \text{\textbf{v}}_k(i+1, j) \right. \\ \left. + \text{\textbf{v}}_k(i, j+1) + \text{\textbf{e}}(i, j) \right) / 4 + (1-\omega) \text{\textbf{v}}_k(i, j). \end{array}$	$SOR: \boldsymbol{v}_{k+1}(i, j) = \omega (\boldsymbol{v}_{k+1}(i-1, j) + \boldsymbol{v}_{k+1}(i, j-1) + \boldsymbol{v}_k(i+1, j) + \boldsymbol{v}_k(i, j+1) + \boldsymbol{e}(i, j)) / 4 + (1-\omega) \boldsymbol{v}_k(i, j).$	$SOR: \boldsymbol{v}_{k+1}(i, j) = \omega (\boldsymbol{v}_{k+1}(i-1, j) + \boldsymbol{v}_{k+1}(i, j-1) + \boldsymbol{v}_k(i+1, j) + \boldsymbol{v}_k(i, j+1) + \boldsymbol{e}(i, j)) / 4 + (1-\omega) \boldsymbol{v}_k(i, j).$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0403	161	■0.801 ● N +0	$\boldsymbol{V}^* \boldsymbol{A}^* \boldsymbol{V} \boldsymbol{u} = \left[\begin{array}{c c} \bar{\lambda} & \boldsymbol{0}^* \\ \hline \boldsymbol{z} & \boldsymbol{M}^* \end{array} \right] \left[\begin{array}{c} 0 \\ \boldsymbol{v} \end{array} \right] = \left[\begin{array}{c} 0 \\ \boldsymbol{M}^* \boldsymbol{v} \end{array} \right] = \bar{\lambda} \left[\begin{array}{c} 0 \\ \boldsymbol{v} \end{array} \right]$	$\boldsymbol{V}^* \boldsymbol{A}^* \boldsymbol{V} \boldsymbol{u} = \left[\begin{array}{c c} \bar{\lambda} & \boldsymbol{0}^* \\ \hline \boldsymbol{z} & \boldsymbol{M}^* \end{array} \right] \left[\begin{array}{c} 0 \\ \boldsymbol{v} \end{array} \right] = \left[\begin{array}{c} 0 \\ \boldsymbol{M}^* \boldsymbol{v} \end{array} \right] = \bar{\lambda} \left[\begin{array}{c} 0 \\ \boldsymbol{v} \end{array} \right]$	
lyche-numerical-linear-algebra-EQ0588	217	■0.812 ●	$\boldsymbol{B}_3 \boldsymbol{x} := \left[\begin{array}{ccc} m & \sum t_k & \sum t_k^2 \\ \sum t_k & \sum t_k^2 & \sum t_k^3 \\ \sum t_k^2 & \sum t_k^3 & \sum t_k^4 \end{array} \right] \left[\begin{array}{c} x_1 \\ x_2 \\ x_3 \end{array} \right] = \left[\begin{array}{c} \sum y_k \\ \sum t_k y_k \\ \sum t_k^2 y_k \end{array} \right].$	$\boldsymbol{B}_3 \boldsymbol{x} := \left[\begin{array}{ccc} m & \sum t_k & \sum t_k^2 \\ \sum t_k & \sum t_k^2 & \sum t_k^3 \\ \sum t_k^2 & \sum t_k^3 & \sum t_k^4 \end{array} \right] \left[\begin{array}{c} x_1 \\ x_2 \\ x_3 \end{array} \right] = \left[\begin{array}{c} \sum y_k \\ \sum t_k y_k \\ \sum t_k^2 y_k \end{array} \right].$	$\boldsymbol{B}_3 \boldsymbol{x} := \left[\begin{array}{ccc} m & \sum t_k & \sum t_k^2 \\ \sum t_k & \sum t_k^2 & \sum t_k^3 \\ \sum t_k^2 & \sum t_k^3 & \sum t_k^4 \end{array} \right] \left[\begin{array}{c} x_1 \\ x_2 \\ x_3 \end{array} \right] = \left[\begin{array}{c} \sum y_k \\ \sum t_k y_k \\ \sum t_k^2 y_k \end{array} \right].$
lyche-numerical-linear-algebra-EQ0529 (8.37)	203	■0.818 ● W +1	$\begin{gathered} \langle \boldsymbol{x}, \boldsymbol{z} \rangle, \boldsymbol{z} \in \mathcal{V} \\ \langle \boldsymbol{x}, \boldsymbol{y} \rangle, \boldsymbol{y} \in \mathcal{V} \\ \langle \boldsymbol{x}, \boldsymbol{z} \rangle + \langle \boldsymbol{y}, \boldsymbol{z} \rangle = \langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{z} \rangle, \quad \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V} \\ \langle \boldsymbol{a} \boldsymbol{x}, \boldsymbol{y} \rangle = a \langle \boldsymbol{x}, \boldsymbol{y} \rangle, \quad a \in \mathbb{R}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}. \end{gathered}$	$\langle \boldsymbol{x}, \boldsymbol{z} \rangle + \langle \boldsymbol{y}, \boldsymbol{z} \rangle = \langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{z} \rangle, \quad \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}$ $\langle \boldsymbol{a} \boldsymbol{x}, \boldsymbol{y} \rangle = a \langle \boldsymbol{x}, \boldsymbol{y} \rangle, \quad a \in \mathbb{R}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}.$	$\langle \boldsymbol{x}, \boldsymbol{z} \rangle + \langle \boldsymbol{y}, \boldsymbol{z} \rangle = \langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{z} \rangle, \quad \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V},$ $\langle \boldsymbol{a} \boldsymbol{x}, \boldsymbol{y} \rangle = a \langle \boldsymbol{x}, \boldsymbol{y} \rangle, \quad a \in \mathbb{R}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}.$
lyche-numerical-linear-algebra-EQ0089 (2.11)	051	■0.822 ● C -6	$\left[\begin{array}{ccccc} 4 & 1 & & & \\ 1 & 4 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & 4 & 1 \\ & & & 1 & 4 \end{array} \right] \left[\begin{array}{c} \mu_2 \\ \mu_3 \\ \vdots \\ \mu_{n-1} \\ \mu_n \end{array} \right] = \frac{6}{h^2} \left[\begin{array}{c} \delta^2 y_2 - \mu_1 \\ \delta^2 y_3 \\ \vdots \\ \delta^2 y_{n-1} \\ \delta^2 y_n - \mu_{n+1} \end{array} \right], \delta^2 y_i := y_{i+1} - 2y_i + y_{i-1}.$	$\left[\begin{array}{ccc} 4 & 1 & \\ 1 & 4 & 1 \\ & \ddots & \ddots & \ddots \\ & & 1 & 4 & 1 \\ & & & 1 & 4 \end{array} \right] \left[\begin{array}{c} \mu_2 \\ \mu_3 \\ \vdots \\ \mu_{n-1} \\ \mu_n \end{array} \right] = \frac{6}{h^2} \left[\begin{array}{c} \delta^2 y_2 - \mu_1 \\ \delta^2 y_3 \\ \vdots \\ \delta^2 y_{n-1} \\ \delta^2 y_n - \mu_{n+1} \end{array} \right], \delta^2 y_i := y_{i+1} - 2y_i + y_{i-1}.$	$\left[\begin{array}{ccc} 4 & 1 & \\ 1 & 4 & 1 \\ & \ddots & \ddots & \ddots \\ & & 1 & 4 & 1 \\ & & & 1 & 4 \end{array} \right] \left[\begin{array}{c} \mu_2 \\ \mu_3 \\ \vdots \\ \mu_{n-1} \\ \mu_n \end{array} \right] = \frac{6}{h^2} \left[\begin{array}{c} \delta^2 y_2 - \mu_1 \\ \delta^2 y_3 \\ \vdots \\ \delta^2 y_{n-1} \\ \delta^2 y_n - \mu_{n+1} \end{array} \right], \delta^2 y_i := y_{i+1} - 2y_i + y_{i-1}.$
lyche-numerical-linear-algebra-EQ0563 (8.44)	210	■0.822 ● K +0	$\boldsymbol{A} \boldsymbol{s} = \boldsymbol{b},$	$\boldsymbol{A} \boldsymbol{s} = \boldsymbol{b},$	$\boldsymbol{A} \boldsymbol{s} = \boldsymbol{b},$
lyche-numerical-linear-algebra-EQ0354	142	■0.825 ● N +0	$\left[\begin{array}{c} \boldsymbol{L}^* \\ \boldsymbol{z}^* \end{array} \right] = \boldsymbol{Q} \left[\begin{array}{c} \boldsymbol{R} \\ 0 \end{array} \right],$	$\left[\begin{array}{c} \boldsymbol{L}^* \\ \boldsymbol{z}^* \end{array} \right] = \boldsymbol{Q} \left[\begin{array}{c} \boldsymbol{R} \\ 0 \end{array} \right],$	$\left[\begin{array}{c} \boldsymbol{L}^* \\ \boldsymbol{z}^* \end{array} \right] = \boldsymbol{Q} \left[\begin{array}{c} \boldsymbol{R} \\ 0 \end{array} \right],$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0040 (1.14)	036	■ 0.826 ● N +0	$\begin{aligned} \operatorname{det}(\boldsymbol{A}) &= \sum_{j=1}^n (-1)^{i+j} a_{ij} \operatorname{det}(\boldsymbol{A}_{ij}) \text{ for } i = 1, \dots, n, \text{ row} \\ \operatorname{det}(\boldsymbol{A}) &= \sum_{i=1}^n (-1)^{i+j} a_{ij} \operatorname{det}(\boldsymbol{A}_{ij}) \text{ for } j = 1, \dots, n, \text{ column.} \end{aligned}$	$\det(\boldsymbol{A}) = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det(\boldsymbol{A}_{ij}) \text{ for } i = 1, \dots, n, \text{ row}$ $\det(\boldsymbol{A}) = \sum_{i=1}^n (-1)^{i+j} a_{ij} \det(\boldsymbol{A}_{ij}) \text{ for } j = 1, \dots, n, \text{ column.}$	$\det(\boldsymbol{A}) = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det(\boldsymbol{A}_{ij}) \text{ for } i = 1, \dots, n, \text{ row}$ $\det(\boldsymbol{A}) = \sum_{i=1}^n (-1)^{i+j} a_{ij} \det(\boldsymbol{A}_{ij}) \text{ for } j = 1, \dots, n, \text{ column.}$
lyche-numerical-linear-algebra- EQ0338	136	■ 0.827 ● N +0	$\left \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y} \right ^2 \leq \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}$	$ \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y} ^2 \leq \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}$	$ \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y} ^2 \leq \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}$
lyche-numerical-linear-algebra- EQ1004	338	■ 0.827 ●	$\boldsymbol{y}^* \boldsymbol{A} \boldsymbol{y} = \boldsymbol{y}^* \boldsymbol{E}^* \boldsymbol{B} \boldsymbol{E} \boldsymbol{y} = \boldsymbol{z}^* \boldsymbol{B} \boldsymbol{z} = \sum_{i=m+1}^n \mu_i z_i ^2 \leq 0,$	$\boldsymbol{y}^* \boldsymbol{A} \boldsymbol{y} = \boldsymbol{y}^* \boldsymbol{E}^* \boldsymbol{B} \boldsymbol{E} \boldsymbol{y} = \boldsymbol{z}^* \boldsymbol{B} \boldsymbol{z} = \sum_{i=m+1}^n \mu_i z_i ^2 \leq 0,$	$\boldsymbol{y}^* \boldsymbol{A} \boldsymbol{y} = \boldsymbol{y}^* \boldsymbol{E}^* \boldsymbol{B} \boldsymbol{E} \boldsymbol{y} = \boldsymbol{z}^* \boldsymbol{B} \boldsymbol{z} = \sum_{i=m+1}^n \mu_i z_i ^2 \leq 0,$
lyche-numerical-linear-algebra- EQ0298	123	■ 0.828 ● W +0	$\langle \boldsymbol{x}, \overline{\boldsymbol{A} \boldsymbol{y}} \rangle = \overline{\langle \boldsymbol{x}, \boldsymbol{A} \boldsymbol{y} \rangle} = \overline{\boldsymbol{y}^* \boldsymbol{A}^* \boldsymbol{x}} = \boldsymbol{y}^* \boldsymbol{A}^T \boldsymbol{x} = \boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x} = \langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{y} \rangle.$	$\langle \boldsymbol{x}, \overline{\boldsymbol{A} \boldsymbol{y}} \rangle = \overline{\langle \boldsymbol{x}, \boldsymbol{A} \boldsymbol{y} \rangle} = \boldsymbol{y}^* \overline{\boldsymbol{A}^*} \boldsymbol{x} = \boldsymbol{y}^* \boldsymbol{A}^T \boldsymbol{x} = \boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x} = \langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{y} \rangle.$	$\langle \boldsymbol{x}, \overline{\boldsymbol{A} \boldsymbol{y}} \rangle = \overline{\langle \boldsymbol{x}, \boldsymbol{A} \boldsymbol{y} \rangle} = \boldsymbol{y}^* \overline{\boldsymbol{A}^*} \boldsymbol{x} = \boldsymbol{y}^* \boldsymbol{A}^T \boldsymbol{x} = \boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x} = \langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{y} \rangle.$
lyche-numerical-linear-algebra- EQ0263	108	■ 0.830 ● N +0	$\boldsymbol{A} = \boldsymbol{B} \boldsymbol{B}^*, \quad \boldsymbol{B} = \begin{bmatrix} 1 & 0 \\ 1/3 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{3} & 0 \\ 0 & \sqrt{8/3} \end{bmatrix}.$	$\boldsymbol{A} = \boldsymbol{B} \boldsymbol{B}^*, \quad \boldsymbol{B} = \begin{bmatrix} 1 & 0 \\ 1/3 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{3} & 0 \\ 0 & \sqrt{8/3} \end{bmatrix}.$	$\boldsymbol{A} = \boldsymbol{B} \boldsymbol{B}^*, \quad \boldsymbol{B} = \begin{bmatrix} 1 & 0 \\ 1/3 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{3} & 0 \\ 0 & \sqrt{8/3} \end{bmatrix}.$
lyche-numerical-linear-algebra- EQ0267	109	■ 0.831 ● N +0	$\boldsymbol{L} \boldsymbol{L}^* = \begin{bmatrix} \beta & \mathbf{0} \\ \boldsymbol{v}/\beta & \boldsymbol{L}_1 \end{bmatrix} \begin{bmatrix} \beta & \boldsymbol{v}^*/\beta \\ \mathbf{0} & \boldsymbol{L}_1^* \end{bmatrix} = \begin{bmatrix} \alpha & \boldsymbol{v}^* \\ \boldsymbol{v} & \boldsymbol{B} \end{bmatrix} = \boldsymbol{A}$	$\boldsymbol{L} \boldsymbol{L}^* = \begin{bmatrix} \beta & \mathbf{0} \\ \boldsymbol{v}/\beta & \boldsymbol{L}_1 \end{bmatrix} \begin{bmatrix} \beta & \boldsymbol{v}^*/\beta \\ \mathbf{0} & \boldsymbol{L}_1^* \end{bmatrix} = \begin{bmatrix} \alpha & \boldsymbol{v}^* \\ \boldsymbol{v} & \boldsymbol{B} \end{bmatrix} = \boldsymbol{A}$	$\boldsymbol{L} \boldsymbol{L}^* = \begin{bmatrix} \beta & \mathbf{0} \\ \boldsymbol{v}/\beta & \boldsymbol{L}_1 \end{bmatrix} \begin{bmatrix} \beta & \boldsymbol{v}^*/\beta \\ \mathbf{0} & \boldsymbol{L}_1^* \end{bmatrix} = \begin{bmatrix} \alpha & \boldsymbol{v}^* \\ \boldsymbol{v} & \boldsymbol{B} \end{bmatrix} = \boldsymbol{A}$
lyche-numerical-linear-algebra- EQ0189 (3.7)	080	■ 0.831 ● N +0	$x_{\{k\}} = \left(b_{\{k\}} - \sum_{j=1}^{k-1} a_{k,j} x_j \right) / a_{kk}, \quad k=1,2, \ldots, n.$	$x_k = \left(b_k - \sum_{j=1}^{k-1} a_{k,j} x_j \right) / a_{kk}, \quad k = 1, 2, \dots, n.$	$x_k = \left(b_k - \sum_{j=1}^{k-1} a_{k,j} x_j \right) / a_{kk}, \quad k = 1, 2, \dots, n.$
lyche-numerical-linear-algebra- EQ0122 (2.30)	061	■ 0.833 ● W +0	$\begin{aligned} \boldsymbol{s}_j &= [\sin(j\pi h), \sin(2j\pi h), \dots, \sin(mj\pi h)]^T, \\ \lambda_j &= d + 2a \cos(j\pi h). \end{aligned}$	$\boldsymbol{s}_j = [\sin(j\pi h), \sin(2j\pi h), \dots, \sin(mj\pi h)]^T,$ $\lambda_j = d + 2a \cos(j\pi h).$	$\boldsymbol{s}_j = [\sin(j\pi h), \sin(2j\pi h), \dots, \sin(mj\pi h)]^T,$ $\lambda_j = d + 2a \cos(j\pi h).$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0950 (13.64)	321	0.838 N +0	$\begin{aligned} & \langle \boldsymbol{B} \boldsymbol{x}, \boldsymbol{x} \rangle = 0, \quad \boldsymbol{x} \in \mathbb{R}^n, \\ & \langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{x} \rangle = \langle \boldsymbol{x}, \boldsymbol{x} \rangle = \ \boldsymbol{x}\ _2^2. \end{aligned}$	$\langle \boldsymbol{B} \boldsymbol{x}, \boldsymbol{x} \rangle = 0, \quad \boldsymbol{x} \in \mathbb{R}^n, \\ \langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{x} \rangle = \langle \boldsymbol{x}, \boldsymbol{x} \rangle = \ \boldsymbol{x}\ _2^2.$	$\langle \boldsymbol{B} \boldsymbol{x}, \boldsymbol{x} \rangle = 0, \quad \boldsymbol{x} \in \mathbb{R}^n, \\ \langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{x} \rangle = \langle \boldsymbol{x}, \boldsymbol{x} \rangle = \ \boldsymbol{x}\ _2^2.$
lyche-numerical-linear-algebra-EQ0321 (5.16)	130	0.839 N +0	$\int_0^n 4(m-k)(n+r-k) dk = 2m(n+r)^2 - \frac{2}{3}(n+r)^3.$	$\int_0^n 4(m-k)(n+r-k) dk = 2m(n+r)^2 - \frac{2}{3}(n+r)^3.$	$\int_0^n 4(m-k)(n+r-k) dk = 2m(n+r)^2 - \frac{2}{3}(n+r)^3.$
lyche-numerical-linear-algebra-EQ0693 (10.23)	248	0.839 C +9	$\begin{aligned} & \Delta^2 u(s, t) := \Delta(\Delta u(s, t)) = f(s, t) \quad (s, t) \in \Omega, \\ & u(s, t) = 0, \quad \Delta u(s, t) = 0 \quad (s, t) \in \partial \Omega. \end{aligned}$	$\Delta^2 u(s, t) := \Delta(\Delta u(s, t)) = f(s, t) \quad (s, t) \in \Omega, \\ u(s, t) = 0, \quad \Delta u(s, t) = 0 \quad (s, t) \in \partial \Omega.$	$\Delta^2 u(s, t) := \Delta(\Delta u(s, t)) = f(s, t) \quad (s, t) \in \Omega, \\ u(s, t) = 0, \quad \Delta u(s, t) = 0 \quad (s, t) \in \partial \Omega.$
lyche-numerical-linear-algebra-EQ0828	287	0.840 K +0	$\boldsymbol{x}_k(1) = \boldsymbol{x}_k(2) = 1 - a^k, \quad k \geq 0,$	$\boldsymbol{x}_k(1) = \boldsymbol{x}_k(2) = 1 - a^k, \quad k \geq 0,$	$\boldsymbol{x}_k(1) = \boldsymbol{x}_k(2) = 1 - a^k, \quad k \geq 0,$
lyche-numerical-linear-algebra-EQ0282	118	0.842 W +0	$\boldsymbol{z} := \boldsymbol{x} - a \boldsymbol{y}, \quad a := \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\langle \boldsymbol{y}, \boldsymbol{y} \rangle}.$	$\boldsymbol{z} := \boldsymbol{x} - a \boldsymbol{y}, \quad a := \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\langle \boldsymbol{y}, \boldsymbol{y} \rangle}.$	$\boldsymbol{z} := \boldsymbol{x} - a \boldsymbol{y}, \quad a := \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\langle \boldsymbol{y}, \boldsymbol{y} \rangle}.$
lyche-numerical-linear-algebra-EQ0624	228	0.842	$\begin{aligned} & \alpha_j \leq \max_{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ (\boldsymbol{B} + (\boldsymbol{A} - \boldsymbol{B}))\boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2} \leq \max_{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{B}\boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2} + \max_{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ (\boldsymbol{A} - \boldsymbol{B})\boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2} \\ & \leq \beta_j + \ \boldsymbol{A} - \boldsymbol{B}\ _2. \end{aligned}$	$\alpha_j \leq \max_{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ (\boldsymbol{B} + (\boldsymbol{A} - \boldsymbol{B}))\boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2} \leq \max_{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{B}\boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2} + \max_{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ (\boldsymbol{A} - \boldsymbol{B})\boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2} \\ \leq \beta_j + \ \boldsymbol{A} - \boldsymbol{B}\ _2.$	$\alpha_j \leq \max_{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ (\boldsymbol{B} + (\boldsymbol{A} - \boldsymbol{B}))\boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2} \leq \max_{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{B}\boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2} + \max_{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ (\boldsymbol{A} - \boldsymbol{B})\boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2} \\ \leq \beta_j + \ \boldsymbol{A} - \boldsymbol{B}\ _2.$
lyche-numerical-linear-algebra-EQ0049	038	0.844 W +1	$\begin{aligned} & \operatorname{det}(\boldsymbol{A} - \lambda \boldsymbol{I}) = \left(a_{11} - \lambda \right) \begin{vmatrix} a_{22} - \lambda & a_{23} \\ a_{32} & a_{33} - \lambda \end{vmatrix} - a_{21} \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} - \lambda \end{vmatrix} \\ & + a_{31} \begin{vmatrix} a_{12} & a_{13} \\ a_{22} - \lambda & a_{23} \end{vmatrix} = (a_{11} - \lambda)(a_{22} - \lambda)(a_{33} - \lambda) + r(\lambda) \end{aligned}$	$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = (a_{11} - \lambda) \begin{vmatrix} a_{22} - \lambda & a_{23} \\ a_{32} & a_{33} - \lambda \end{vmatrix} - a_{21} \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} - \lambda \end{vmatrix} \\ + a_{31} \begin{vmatrix} a_{12} & a_{13} \\ a_{22} - \lambda & a_{23} \end{vmatrix} = (a_{11} - \lambda)(a_{22} - \lambda)(a_{33} - \lambda) + r(\lambda)$	$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = (a_{11} - \lambda) \begin{vmatrix} a_{22} - \lambda & a_{23} \\ a_{32} & a_{33} - \lambda \end{vmatrix} - a_{21} \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} - \lambda \end{vmatrix} \\ + a_{31} \begin{vmatrix} a_{12} & a_{13} \\ a_{22} - \lambda & a_{23} \end{vmatrix} = (a_{11} - \lambda)(a_{22} - \lambda)(a_{33} - \lambda) + r(\lambda)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0851 (13.14)	295	■0.846 ●W +0	$\backslash beta_{\{k\}} := - \frac{\backslash boldsymbol{r}_{\{k+1\}}^{\wedge T} \backslash boldsymbol{A} \backslash boldsymbol{p}_{\{k\}}}{\backslash boldsymbol{p}_{\{k\}}^T \backslash boldsymbol{A} \backslash boldsymbol{p}_{\{k\}}} \stackrel{\text{(13.13)}}{=} \frac{\backslash boldsymbol{r}_{\{k+1\}}^{\wedge T} \backslash boldsymbol{A} \backslash boldsymbol{p}_{\{k\}}}{\alpha_k \backslash boldsymbol{p}_{\{k\}}^T \backslash boldsymbol{A} \backslash boldsymbol{p}_{\{k\}}} \stackrel{\text{(13.12)}}{=} \frac{\backslash boldsymbol{r}_{\{k+1\}}^T \backslash boldsymbol{r}_{k+1}}{\backslash boldsymbol{r}_k^T \backslash boldsymbol{r}_k}.$	$\beta_k := - \frac{\boldsymbol{r}_{k+1}^T \boldsymbol{A} \boldsymbol{p}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k} \stackrel{(13.13)}{=} \frac{\boldsymbol{r}_{k+1}^T (\boldsymbol{r}_{k+1} - \boldsymbol{r}_k)}{\alpha_k \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k} \stackrel{(13.12)}{=} \frac{\boldsymbol{r}_{k+1}^T \boldsymbol{r}_{k+1}}{\boldsymbol{r}_k^T \boldsymbol{r}_k}.$	$\beta_k := - \frac{\boldsymbol{r}_{k+1}^T \boldsymbol{A} \boldsymbol{p}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k} \stackrel{(13.13)}{=} \frac{\boldsymbol{r}_{k+1}^T (\boldsymbol{r}_{k+1} - \boldsymbol{r}_k)}{\alpha_k \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k} \stackrel{(13.12)}{=} \frac{\boldsymbol{r}_{k+1}^T \boldsymbol{r}_{k+1}}{\boldsymbol{r}_k^T \boldsymbol{r}_k}.$
lyche-numerical-linear-algebra-EQ0837	156	■0.846 ●N +1	$\backslash pi_{\{D_1\}}(\backslash lambda) = \backslash pi_{\{D_3\}}(\backslash lambda) = \backslash lambda^2 - 4 \backslash lambda + 5, \quad \backslash quad \backslash pi_{\{D_2\}}(\backslash lambda) = \backslash lambda - 1,$	$\pi_{D_1}(\lambda) = \pi_{D_3}(\lambda) = \lambda^2 - 4\lambda + 5, \quad \pi_{D_2}(\lambda) = \lambda - 1,$	$\pi_{D_1}(\lambda) = \pi_{D_3}(\lambda) = \lambda^2 - 4\lambda + 5, \quad \pi_{D_2}(\lambda) = \lambda - 1,$
lyche-numerical-linear-algebra-EQ0650	234	■0.847 ●N +1	$\backslash left[\backslash begin{array}{cc} \backslash Sigma_{\{1\}} \& 0 \backslash \ 0 \& 0 \backslash end{array} \backslash right] y = c.$	$\begin{bmatrix} \Sigma_1 & 0 \\ 0 & 0 \end{bmatrix} y = c.$	$\begin{bmatrix} \boldsymbol{\Sigma}_1 & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} \end{bmatrix} \boldsymbol{y} = \boldsymbol{c}.$
lyche-numerical-linear-algebra-EQ0034	033	■0.848 ●K +0	$C = C \quad I = C(A \ B) = (C \ A) \quad B = I \quad B = B$	$C = CI = C(AB) = (CA)B = IB = B$	$\boldsymbol{C} = \boldsymbol{C} \boldsymbol{I} = \boldsymbol{C}(\boldsymbol{A} \boldsymbol{B}) = (\boldsymbol{C} \boldsymbol{A}) \boldsymbol{B} = \boldsymbol{I} \boldsymbol{B} = \boldsymbol{B}$
lyche-numerical-linear-algebra-EQ0084 (2.6)	050	■0.850 ●K +0	$p_{\{i\}} \backslash left(x_{\{i\}} \backslash right) = y_{\{i\}}, \quad \backslash quad p_{\{i\}} \backslash left(x_{\{i+1\}} \backslash right) = y_{\{i+1\}}, \quad \backslash quad p_{\{i\}}^{\backslash prime \backslash prime} \backslash left(x_{\{i\}} \backslash right) = \backslash mu_{\{i\}}, \quad \backslash quad p_{\{i\}}^{\backslash prime \backslash prime} \backslash left(x_{\{i+1\}} \backslash right) = \backslash mu_{\{i+1\}}.$	$p_i(x_i) = y_i, \quad p_i(x_{i+1}) = y_{i+1}, \quad p_i''(x_i) = \mu_i, \quad p_i''(x_{i+1}) = \mu_{i+1}.$	$p_i(x_i) = y_i, \quad p_i(x_{i+1}) = y_{i+1}, \quad p_i''(x_i) = \mu_i, \quad p_i''(x_{i+1}) = \mu_{i+1}.$
lyche-numerical-linear-algebra-EQ0409	163	■0.850 ●K +0	$\backslash boldsymbol{A} = \backslash left[\backslash begin{array}{cc} a \& 1 \backslash \ 0 \& a \backslash end{array} \backslash right]$	$\boldsymbol{A} = \begin{bmatrix} a & 1 \\ 0 & a \end{bmatrix}$	$\boldsymbol{A} = \begin{bmatrix} a & 1 \\ 0 & a \end{bmatrix}$
lyche-numerical-linear-algebra-EQ0144	065	■0.853 ●	$\backslash boldsymbol{B} \backslash boldsymbol{A} = \backslash left[\backslash begin{array}{ll} \backslash boldsymbol{B}_{\{11\}} \& \backslash boldsymbol{B}_{\{12\}} \backslash \backslash \backslash boldsymbol{B}_{\{21\}} \& \backslash boldsymbol{B}_{\{22\}} \backslash end{array} \backslash right] \backslash left[\backslash begin{array}{cc} \backslash boldsymbol{A}_{\{11\}} \& \backslash boldsymbol{A}_{\{12\}} \backslash \backslash \backslash mathbf{0} \& \backslash boldsymbol{A}_{\{22\}} \backslash end{array} \backslash right] = \backslash left[\backslash begin{array}{cc} \backslash boldsymbol{I} \& \backslash mathbf{0} \backslash \backslash \backslash mathbf{0} \& \backslash boldsymbol{I} \backslash end{array} \backslash right] = \backslash boldsymbol{I}$	$\boldsymbol{B} \boldsymbol{A} = \begin{bmatrix} \boldsymbol{B}_{11} & \boldsymbol{B}_{12} \\ \boldsymbol{B}_{21} & \boldsymbol{B}_{22} \end{bmatrix} \begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{0} & \boldsymbol{A}_{22} \end{bmatrix} = \begin{bmatrix} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{bmatrix} = \boldsymbol{I}$	$\boldsymbol{B} \boldsymbol{A} = \begin{bmatrix} \boldsymbol{B}_{11} & \boldsymbol{B}_{12} \\ \boldsymbol{B}_{21} & \boldsymbol{B}_{22} \end{bmatrix} \begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{0} & \boldsymbol{A}_{22} \end{bmatrix} = \begin{bmatrix} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{bmatrix} = \boldsymbol{I}$
lyche-numerical-linear-algebra-EQ0264 (4.8)	109	■0.854 ●N +0	$\backslash boldsymbol{A} = \backslash left[\backslash begin{array}{ll} \backslash alpha \& \backslash boldsymbol{v}^{\wedge *} \backslash \backslash \backslash boldsymbol{v} \& \backslash boldsymbol{B} \backslash end{array} \backslash right], \quad \backslash quad \backslash alpha \backslash in \backslash mathbb{C}, \quad \backslash quad \backslash boldsymbol{v} \backslash in \backslash mathbb{C}^{\{n-1\}}, \quad \backslash quad \backslash boldsymbol{B} \backslash in \backslash mathbb{C}^{\{n-1\}} \backslash times (n-1).$	$\boldsymbol{A} = \begin{bmatrix} \alpha & \boldsymbol{v}^* \\ \boldsymbol{v} & \boldsymbol{B} \end{bmatrix}, \quad \alpha \in \mathbb{C}, \quad \boldsymbol{v} \in \mathbb{C}^{n-1}, \quad \boldsymbol{B} \in \mathbb{C}^{(n-1) \times (n-1)}.$	$\boldsymbol{A} = \begin{bmatrix} \alpha & \boldsymbol{v}^* \\ \boldsymbol{v} & \boldsymbol{B} \end{bmatrix}, \quad \alpha \in \mathbb{C}, \quad \boldsymbol{v} \in \mathbb{C}^{n-1}, \quad \boldsymbol{B} \in \mathbb{C}^{(n-1) \times (n-1)}.$
lyche-numerical-linear-algebra-EQ0392	157	■0.857 ●N +0	$\backslash max_{\{ \backslash substack{x \backslash in \backslash mathcal{S}} \backslash \backslash \backslash boldsymbol{x} \backslash neq \backslash mathbf{0} \}} R(\backslash boldsymbol{x}) \backslash geq R(\backslash boldsymbol{y}) = \backslash sum_{\{j=1\}^{\wedge n}} \backslash lambda_{\{j\}} \backslash left c_{\{j\}} \backslash right ^{\wedge 2} = \backslash sum_{\{j=1\}^{\wedge k}} \backslash lambda_{\{j\}} \backslash left c_{\{j\}} \backslash right ^{\wedge 2} \backslash geq \backslash sum_{\{j=1\}^{\wedge k}} \backslash lambda_{\{k\}} \backslash left c_{\{j\}} \backslash right ^{\wedge 2} = \backslash lambda_{\{k\}},$	$\max_{\substack{x \in S \\ x \neq \mathbf{0}}} R(\boldsymbol{x}) \geq R(\boldsymbol{y}) = \sum_{j=1}^n \lambda_j c_j ^2 = \sum_{j=1}^k \lambda_j c_j ^2 \geq \sum_{j=1}^k \lambda_k c_j ^2 = \lambda_k,$	$\max_{\substack{x \in S \\ x \neq \mathbf{0}}} R(\boldsymbol{x}) \geq R(\boldsymbol{y}) = \sum_{j=1}^n \lambda_j c_j ^2 = \sum_{j=1}^k \lambda_j c_j ^2 \geq \sum_{j=1}^k \lambda_k c_j ^2 = \lambda_k,$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0032	332	■ 0.859 ● N +0	$\boldsymbol{B} \boldsymbol{z} = \left[\begin{array}{l} \boldsymbol{A} \\ \boldsymbol{b} \end{array} \right] \left[\begin{array}{l} \tilde{\boldsymbol{z}} \\ z_{n+1} \end{array} \right] = \boldsymbol{A} \tilde{\boldsymbol{z}} + z_{n+1} \boldsymbol{b} = \boldsymbol{0}.$	$\boldsymbol{B} \boldsymbol{z} = \left[\begin{array}{l} \boldsymbol{A} \\ \boldsymbol{b} \end{array} \right] \left[\begin{array}{l} \tilde{\boldsymbol{z}} \\ z_{n+1} \end{array} \right] = \boldsymbol{A} \tilde{\boldsymbol{z}} + z_{n+1} \boldsymbol{b} = \boldsymbol{0}.$	$\boldsymbol{B} \boldsymbol{z} = \left[\begin{array}{l} \boldsymbol{A} \\ \boldsymbol{b} \end{array} \right] \left[\begin{array}{l} \tilde{\boldsymbol{z}} \\ z_{n+1} \end{array} \right] = \boldsymbol{A} \tilde{\boldsymbol{z}} + z_{n+1} \boldsymbol{b} = \boldsymbol{0}.$
lyche-numerical-linear-algebra- EQ0091 (13.36)	307	■ 0.859 ● W +0	$1 \leq \frac{\left(\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y} \right) \left(\boldsymbol{y}^T \boldsymbol{A}^{-1} \boldsymbol{y} \right)}{\left(\boldsymbol{y}^T \boldsymbol{y} \right)^2} \leq \frac{(M+m)^2}{4Mm} \quad \boldsymbol{y} \neq \boldsymbol{0}, \boldsymbol{y} \in \mathbb{R}^n,$	$1 \leq \frac{(\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y})(\boldsymbol{y}^T \boldsymbol{A}^{-1} \boldsymbol{y})}{(\boldsymbol{y}^T \boldsymbol{y})^2} \leq \frac{(M+m)^2}{4Mm} \quad \boldsymbol{y} \neq \boldsymbol{0}, \boldsymbol{y} \in \mathbb{R}^n,$	$1 \leq \frac{(\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y})(\boldsymbol{y}^T \boldsymbol{A}^{-1} \boldsymbol{y})}{(\boldsymbol{y}^T \boldsymbol{y})^2} \leq \frac{(M+m)^2}{4Mm} \quad \boldsymbol{y} \neq \boldsymbol{0}, \boldsymbol{y} \in \mathbb{R}^n,$
lyche-numerical-linear-algebra- EQ0033	333	■ 0.860 ● S +0	$\boldsymbol{A} \boldsymbol{x} = -\boldsymbol{A} \left(\frac{\tilde{\boldsymbol{z}}}{z_{n+1}} \right) = -\frac{1}{z_{n+1}} \boldsymbol{A} \tilde{\boldsymbol{z}} = -\frac{1}{z_{n+1}} (-z_{n+1} \boldsymbol{b}) = \boldsymbol{b},$	$\boldsymbol{A} \boldsymbol{x} = -\boldsymbol{A} \left(\frac{\tilde{\boldsymbol{z}}}{z_{n+1}} \right) = -\frac{1}{z_{n+1}} \boldsymbol{A} \tilde{\boldsymbol{z}} = -\frac{1}{z_{n+1}} (-z_{n+1} \boldsymbol{b}) = \boldsymbol{b},$	$\boldsymbol{A} \boldsymbol{x} = -\boldsymbol{A} \left(\frac{\tilde{\boldsymbol{z}}}{z_{n+1}} \right) = -\frac{1}{z_{n+1}} \boldsymbol{A} \tilde{\boldsymbol{z}} = -\frac{1}{z_{n+1}} (-z_{n+1} \boldsymbol{b}) = \boldsymbol{b},$
lyche-numerical-linear-algebra- EQ0451	180	■ 0.869 ● S +0	$\boldsymbol{p}_i = \left[\begin{array}{l} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{array} \right], \quad \boldsymbol{q}_i = \left[\begin{array}{l} \boldsymbol{u}_i \\ -\boldsymbol{v}_i \end{array} \right], \quad \boldsymbol{r}_j = \left[\begin{array}{l} \boldsymbol{u}_j \\ \boldsymbol{0} \end{array} \right], \text{ for } i = 1, \dots, n, j = n + 1, \dots, m.$	$\boldsymbol{p}_i = \left[\begin{array}{l} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{array} \right], \quad \boldsymbol{q}_i = \left[\begin{array}{l} \boldsymbol{u}_i \\ -\boldsymbol{v}_i \end{array} \right], \quad \boldsymbol{r}_j = \left[\begin{array}{l} \boldsymbol{u}_j \\ \boldsymbol{0} \end{array} \right], \text{ for } i = 1, \dots, n, j = n + 1, \dots, m.$	$\boldsymbol{p}_i = \left[\begin{array}{l} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{array} \right], \quad \boldsymbol{q}_i = \left[\begin{array}{l} \boldsymbol{u}_i \\ -\boldsymbol{v}_i \end{array} \right], \quad \boldsymbol{r}_j = \left[\begin{array}{l} \boldsymbol{u}_j \\ \boldsymbol{0} \end{array} \right], \text{ for } i = 1, \dots, n, j = n + 1, \dots, m.$
lyche-numerical-linear-algebra- EQ0035 (13.1)	290	■ 0.869 ● W +0	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle_{\boldsymbol{A}} := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y}, \quad \ \boldsymbol{y}\ _{\boldsymbol{A}} := \sqrt{\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n.$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle_{\boldsymbol{A}} := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y}, \quad \ \boldsymbol{y}\ _{\boldsymbol{A}} := \sqrt{\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n.$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle_{\boldsymbol{A}} := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y}, \quad \ \boldsymbol{y}\ _{\boldsymbol{A}} := \sqrt{\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0910 (13.46)	310	0.869	$\begin{aligned} & \left \left \epsilon_k \right _2^2 - \left \epsilon_{k+1} \right _2^2 \right = \alpha_k \left(2 \mathbf{p}_k^T \epsilon_{k+1} + \alpha_k \mathbf{p}_k^T \mathbf{p}_k \right) \\ & \stackrel{(13.41)}{=} \alpha_k \left(2 \sum_{i=k+1}^m \alpha_i \mathbf{p}_i^T \mathbf{p}_k + \alpha_k \mathbf{p}_k^T \mathbf{p}_k \right) = \left(\sum_{i=k}^m + \sum_{i=k+1}^m \right) \alpha_k \alpha_i \mathbf{p}_i^T \mathbf{p}_k \\ & \stackrel{(13.43)}{=} \left(\sum_{i=k}^m + \sum_{i=k+1}^m \right) \frac{\mathbf{r}_k^T \mathbf{r}_k}{\mathbf{p}_k^T \mathbf{A} \mathbf{p}_k} \frac{\mathbf{r}_i^T \mathbf{r}_i}{\mathbf{p}_i^T \mathbf{A} \mathbf{p}_i} \frac{\mathbf{r}_i^T \mathbf{r}_i}{\mathbf{r}_k^T \mathbf{r}_k} \mathbf{p}_k^T \mathbf{p}_k \\ & \stackrel{(13.42)}{=} \frac{\ \mathbf{p}_k\ _2^2}{\ \mathbf{p}_k\ _A^2} \left(\ \epsilon_k\ _A^2 + \ \epsilon_{k+1}\ _A^2 \right). \end{aligned}$	$\begin{aligned} & \left \left \epsilon_k \right _2^2 - \left \epsilon_{k+1} \right _2^2 \right = \alpha_k \left(2 \mathbf{p}_k^T \epsilon_{k+1} + \alpha_k \mathbf{p}_k^T \mathbf{p}_k \right) \\ & \stackrel{(13.41)}{=} \alpha_k \left(2 \sum_{i=k+1}^m \alpha_i \mathbf{p}_i^T \mathbf{p}_k + \alpha_k \mathbf{p}_k^T \mathbf{p}_k \right) = \left(\sum_{i=k}^m + \sum_{i=k+1}^m \right) \alpha_k \alpha_i \mathbf{p}_i^T \mathbf{p}_k \\ & \stackrel{(13.43)}{=} \left(\sum_{i=k}^m + \sum_{i=k+1}^m \right) \frac{\mathbf{r}_k^T \mathbf{r}_k}{\mathbf{p}_k^T \mathbf{A} \mathbf{p}_k} \frac{\mathbf{r}_i^T \mathbf{r}_i}{\mathbf{p}_i^T \mathbf{A} \mathbf{p}_i} \frac{\mathbf{r}_i^T \mathbf{r}_i}{\mathbf{r}_k^T \mathbf{r}_k} \mathbf{p}_k^T \mathbf{p}_k \\ & \stackrel{(13.42)}{=} \frac{\ \mathbf{p}_k\ _2^2}{\ \mathbf{p}_k\ _A^2} \left(\ \epsilon_k\ _A^2 + \ \epsilon_{k+1}\ _A^2 \right). \end{aligned}$	$\ \epsilon_k\ _2^2 - \ \epsilon_{k+1}\ _2^2 = \alpha_k (2 \mathbf{p}_k^T \epsilon_{k+1} + \alpha_k \mathbf{p}_k^T \mathbf{p}_k)$ $\stackrel{(13.41)}{=} \alpha_k \left(2 \sum_{i=k+1}^m \alpha_i \mathbf{p}_i^T \mathbf{p}_k + \alpha_k \mathbf{p}_k^T \mathbf{p}_k \right) = \left(\sum_{i=k}^m + \sum_{i=k+1}^m \right) \alpha_k \alpha_i \mathbf{p}_i^T \mathbf{p}_k$ $\stackrel{(13.43)}{=} \left(\sum_{i=k}^m + \sum_{i=k+1}^m \right) \frac{\mathbf{r}_k^T \mathbf{r}_k}{\mathbf{p}_k^T \mathbf{A} \mathbf{p}_k} \frac{\mathbf{r}_i^T \mathbf{r}_i}{\mathbf{p}_i^T \mathbf{A} \mathbf{p}_i} \frac{\mathbf{r}_i^T \mathbf{r}_i}{\mathbf{r}_k^T \mathbf{r}_k} \mathbf{p}_k^T \mathbf{p}_k$ $\stackrel{(13.42)}{=} \frac{\ \mathbf{p}_k\ _2^2}{\ \mathbf{p}_k\ _A^2} \left(\ \epsilon_k\ _A^2 + \ \epsilon_{k+1}\ _A^2 \right).$
lyche-numerical-linear-algebra- EQ1014 (15.1)	344	0.870 K +0	$\begin{aligned} \mathbf{z}_k &:= \mathbf{A}^k \mathbf{z}_0 = \mathbf{A} \mathbf{z}_{k-1}, \quad k = 1, 2, \dots \\ \mathbf{z}_0 &:= \mathbf{A}^0 \mathbf{z}_0 = \mathbf{A} \mathbf{z}_{-1}, \quad k = 1, 2, \dots \end{aligned}$	$\mathbf{z}_k := \mathbf{A}^k \mathbf{z}_0 = \mathbf{A} \mathbf{z}_{k-1}, \quad k = 1, 2, \dots$	$\mathbf{z}_k := \mathbf{A}^k \mathbf{z}_0 = \mathbf{A} \mathbf{z}_{k-1}, \quad k = 1, 2, \dots$
lyche-numerical-linear-algebra- EQ0591	218	0.877 N +0	$\begin{aligned} E(\mathbf{x}) &:= \ \mathbf{A} \mathbf{x} - \mathbf{b}\ _2^2 = (\mathbf{A} \mathbf{x} - \mathbf{b})^* (\mathbf{A} \mathbf{x} - \mathbf{b}) = \mathbf{x}^* \mathbf{A}^* \mathbf{A} \mathbf{x} - 2 \mathbf{x}^* \mathbf{A}^* \mathbf{b} + \mathbf{b}^* \mathbf{b}. \end{aligned}$	$E(\mathbf{x}) := \ \mathbf{A} \mathbf{x} - \mathbf{b}\ _2^2 = (\mathbf{A} \mathbf{x} - \mathbf{b})^* (\mathbf{A} \mathbf{x} - \mathbf{b}) = \mathbf{x}^* \mathbf{A}^* \mathbf{A} \mathbf{x} - 2 \mathbf{x}^* \mathbf{A}^* \mathbf{b} + \mathbf{b}^* \mathbf{b}.$	$E(\mathbf{x}) := \ \mathbf{A} \mathbf{x} - \mathbf{b}\ _2^2 = (\mathbf{A} \mathbf{x} - \mathbf{b})^* (\mathbf{A} \mathbf{x} - \mathbf{b}) = \mathbf{x}^* \mathbf{A}^* \mathbf{A} \mathbf{x} - 2 \mathbf{x}^* \mathbf{A}^* \mathbf{b} + \mathbf{b}^* \mathbf{b}.$
lyche-numerical-linear-algebra- EQ0181	078	0.880 N +0	$\begin{aligned} & [1], [5], [9], \left[\begin{array}{ll} 1 & 2 \\ 4 & 5 \end{array} \right], \left[\begin{array}{ll} 1 & 3 \\ 7 & 9 \end{array} \right], \left[\begin{array}{ll} 5 & 6 \\ 8 & 9 \end{array} \right], \mathbf{A}. \end{aligned}$	$[1], [5], [9], \left[\begin{array}{ll} 1 & 2 \\ 4 & 5 \end{array} \right], \left[\begin{array}{ll} 1 & 3 \\ 7 & 9 \end{array} \right], \left[\begin{array}{ll} 5 & 6 \\ 8 & 9 \end{array} \right], \mathbf{A}.$	$[1], [5], [9], \left[\begin{array}{ll} 1 & 2 \\ 4 & 5 \end{array} \right], \left[\begin{array}{ll} 1 & 3 \\ 7 & 9 \end{array} \right], \left[\begin{array}{ll} 5 & 6 \\ 8 & 9 \end{array} \right], \mathbf{A}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0323	132	■0.881 ●	$\boldsymbol{A}=\left[\begin{array}{rrr} 1 & 3 & 1 \\ 3 & 7 & 1 \\ 1 & -1 & -4 \end{array}\right]=\frac{1}{2}\left[\begin{array}{rrrr} 1 & 1 & -1 & -1 \\ 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \end{array}\right]\times\left[\begin{array}{rrr} 2 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{array}\right]=\boldsymbol{QR}.$	$\boldsymbol{A}=\left[\begin{array}{rrr} 1 & 3 & 1 \\ 1 & 3 & 7 \\ 1 & -1 & -4 \\ 1 & -1 & 2 \end{array}\right]=\frac{1}{2}\left[\begin{array}{rrrr} 1 & 1 & -1 & -1 \\ 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{array}\right]\times\left[\begin{array}{rrr} 2 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \\ 0 & 0 & 0 \end{array}\right]=\boldsymbol{QR}.$	
lyche-numerical-linear-algebra- EQ0314	127	■0.884 ●N +1	$\boldsymbol{H}\boldsymbol{x}=\left[\begin{array}{c} \boldsymbol{y} \\ a\boldsymbol{e}_1 \end{array}\right], \text{ where } \boldsymbol{H}:=\boldsymbol{I}-\boldsymbol{u}\boldsymbol{u}^*=\left[\begin{array}{rr} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{array}\right]-\left[\begin{array}{r} \boldsymbol{0} \\ \hat{\boldsymbol{u}} \end{array}\right]\left[\begin{array}{rr} \boldsymbol{0} & \hat{\boldsymbol{u}}^* \end{array}\right]=\left[\begin{array}{rr} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \hat{\boldsymbol{H}} \end{array}\right],$	$\boldsymbol{H}\boldsymbol{x}=\left[\begin{array}{r} \boldsymbol{y} \\ a\boldsymbol{e}_1 \end{array}\right], \text{ where } \boldsymbol{H}:=\boldsymbol{I}-\boldsymbol{u}\boldsymbol{u}^*=\left[\begin{array}{rr} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{array}\right]-\left[\begin{array}{r} \boldsymbol{0} \\ \hat{\boldsymbol{u}} \end{array}\right]\left[\begin{array}{rr} \boldsymbol{0} & \hat{\boldsymbol{u}}^* \end{array}\right]=\left[\begin{array}{rr} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \hat{\boldsymbol{H}} \end{array}\right],$	
lyche-numerical-linear-algebra- EQ0367 (6.5)	150	■0.884 ●N +0	$\boldsymbol{J}:=\boldsymbol{S}^{-1}\boldsymbol{A}\boldsymbol{S}=\text{diag}(\boldsymbol{U}_1,\dots,\boldsymbol{U}_k), \text{ with } \boldsymbol{U}_i\in\mathbb{C}^{a_i\times a_i},$	$\boldsymbol{J}:=\boldsymbol{S}^{-1}\boldsymbol{A}\boldsymbol{S}=\text{diag}(\boldsymbol{U}_1,\dots,\boldsymbol{U}_k), \text{ with } \boldsymbol{U}_i\in\mathbb{C}^{a_i\times a_i},$	
lyche-numerical-linear-algebra- EQ0594 (9.5)	219	■0.885 ●N +0	$\left\{\boldsymbol{x}\in\mathbb{C}^n:\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}_1\right\}.$	$\{\boldsymbol{x}\in\mathbb{C}^n:\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}_1\}.$	
lyche-numerical-linear-algebra- EQ0064	042	■0.887 ●N +0	$\boldsymbol{A}=\left[\begin{array}{rrr} 2 & -6 & 3 \\ 3 & -2 & -6 \\ 6 & 3 & 2 \end{array}\right],$	$\boldsymbol{A}=\left[\begin{array}{rrr} 2 & -6 & 3 \\ 3 & -2 & -6 \\ 6 & 3 & 2 \end{array}\right],$	
lyche-numerical-linear-algebra- EQ0977	330	■0.892 ●N +0	$\begin{gathered} \left\ \left(\mu\boldsymbol{I}-\boldsymbol{A}\right)\boldsymbol{u}_1\right\ _2=\left\ \left(\boldsymbol{A}+\boldsymbol{E}\right)\boldsymbol{u}_1-\boldsymbol{A}\boldsymbol{u}_1\right\ _2=\left\ \boldsymbol{E}\boldsymbol{u}_1\right\ _2\leq\left\ \boldsymbol{E}\right\ _2, \\ \prod_{j=2}^n\left\ \left(\mu\boldsymbol{I}-\boldsymbol{A}\right)\boldsymbol{u}_j\right\ _2\leq\prod_{j=2}^n\left(\left \mu\right +\left\ \boldsymbol{A}\boldsymbol{u}_j\right\ _2\right)\leq\left(\left\ \boldsymbol{A}+\boldsymbol{E}\right\ _2+\left\ \boldsymbol{A}\right\ _2\right)^{n-1}. \end{gathered}$	$\left\ \left(\mu\boldsymbol{I}-\boldsymbol{A}\right)\boldsymbol{u}_1\right\ _2=\left\ \left(\boldsymbol{A}+\boldsymbol{E}\right)\boldsymbol{u}_1-\boldsymbol{A}\boldsymbol{u}_1\right\ _2=\left\ \boldsymbol{E}\boldsymbol{u}_1\right\ _2\leq\left\ \boldsymbol{E}\right\ _2, \\ \prod_{j=2}^n\left\ \left(\mu\boldsymbol{I}-\boldsymbol{A}\right)\boldsymbol{u}_j\right\ _2\leq\prod_{j=2}^n\left(\left \mu\right +\left\ \boldsymbol{A}\boldsymbol{u}_j\right\ _2\right)\leq\left(\left\ \boldsymbol{A}+\boldsymbol{E}\right\ _2+\left\ \boldsymbol{A}\right\ _2\right)^{n-1}.$	
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 —no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0147	066	0.896 N +0	$\left[\begin{array}{cc} \boldsymbol{A}_{11}^{-1} & -\boldsymbol{A}_{11}^{-1}\boldsymbol{A}_{12}\boldsymbol{A}_{22}^{-1} \\ \boldsymbol{0} & \boldsymbol{A}_{22}^{-1} \end{array}\right] \left[\begin{array}{cc} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{0} & \boldsymbol{A}_{22} \end{array}\right] = \left[\begin{array}{cc} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{array}\right] = \boldsymbol{I}$	$\left[\begin{array}{cc} \boldsymbol{A}_{11}^{-1} & -\boldsymbol{A}_{11}^{-1}\boldsymbol{A}_{12}\boldsymbol{A}_{22}^{-1} \\ \boldsymbol{0} & \boldsymbol{A}_{22}^{-1} \end{array}\right] \left[\begin{array}{cc} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{0} & \boldsymbol{A}_{22} \end{array}\right] = \left[\begin{array}{cc} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{array}\right] = \boldsymbol{I}$	
lyche-numerical-linear-algebra-EQ0825	286	0.896	$\rho(\boldsymbol{G}_J)=\sqrt{\left \mathrm{a}_{21}\mathrm{a}_{12}/\mathrm{a}_{11}\mathrm{a}_{22}\right }.$	$\rho(\boldsymbol{G}_J)=\sqrt{\left \mathrm{a}_{21}\mathrm{a}_{12}/\mathrm{a}_{11}\mathrm{a}_{22}\right }.$	
lyche-numerical-linear-algebra-EQ0989	333	0.899 N +0	$\boldsymbol{A}_k=\left[\begin{array}{cc} \boldsymbol{B}_k & \boldsymbol{C}_k \\ \boldsymbol{D}_k & \boldsymbol{E}_k \end{array}\right].$	$\boldsymbol{A}_k=\left[\begin{array}{cc} \boldsymbol{B}_k & \boldsymbol{C}_k \\ \boldsymbol{D}_k & \boldsymbol{E}_k \end{array}\right].$	
lyche-numerical-linear-algebra-EQ0024 (1.9)	029	0.902 N +0	$\boldsymbol{s}_j=\sum_{i=1}^ma_{ij}\boldsymbol{v}_i\text{ for }j=1,\ldots,n.$	$\boldsymbol{s}_j=\sum_{i=1}^ma_{ij}\boldsymbol{v}_i\text{ for }j=1,\ldots,n.$	
lyche-numerical-linear-algebra-EQ0614	225	0.902 K +0	$\boldsymbol{A}=\left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{array}\right],\quad \boldsymbol{b}=\left[\begin{array}{c} 10^{-4} \\ 0 \\ 1 \end{array}\right],\quad \boldsymbol{e}=\left[\begin{array}{c} 10^{-6} \\ 0 \\ 0 \end{array}\right].$	$\boldsymbol{A}=\left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{array}\right],\quad \boldsymbol{b}=\left[\begin{array}{c} 10^{-4} \\ 0 \\ 1 \end{array}\right],\quad \boldsymbol{e}=\left[\begin{array}{c} 10^{-6} \\ 0 \\ 0 \end{array}\right].$	
lyche-numerical-linear-algebra-EQ0168	074	0.904 N +0	$\boldsymbol{A}:=\left[\begin{array}{cc} \lambda & \boldsymbol{a}^T \\ \boldsymbol{0} & \boldsymbol{A}_1 \end{array}\right],\quad \boldsymbol{B}:=\left[\begin{array}{cc} 1 & \boldsymbol{0}^T \\ \boldsymbol{0} & \boldsymbol{B}_1 \end{array}\right],\quad \boldsymbol{C}:=\left[\begin{array}{cc} 1 & \boldsymbol{0}^T \\ \boldsymbol{0} & \boldsymbol{C}_1 \end{array}\right],$	$\boldsymbol{A}:=\left[\begin{array}{cc} \lambda & \boldsymbol{a}^T \\ \boldsymbol{0} & \boldsymbol{A}_1 \end{array}\right],\quad \boldsymbol{B}:=\left[\begin{array}{cc} 1 & \boldsymbol{0}^T \\ \boldsymbol{0} & \boldsymbol{B}_1 \end{array}\right],\quad \boldsymbol{C}:=\left[\begin{array}{cc} 1 & \boldsymbol{0}^T \\ \boldsymbol{0} & \boldsymbol{C}_1 \end{array}\right],$	
lyche-numerical-linear-algebra-EQ0925 (13.55)	314	0.907 W -1	$\begin{aligned} & \left(d_1v\right)_{j,k}:=c_{j-\frac{1}{2},k}\left(v_{j,k}-v_{j-1,k}\right)-c_{j+\frac{1}{2},k}\left(v_{j+1,k}-v_{j,k}\right) \approx -h^2\frac{\partial}{\partial x}\left(c\frac{\partial u}{\partial x}\right)_{j,k}, \\ & \left(d_2v\right)_{j,k}:=c_{j,k-\frac{1}{2}}\left(v_{j,k}-v_{j,k-1}\right)-c_{j,k+\frac{1}{2}}\left(v_{j,k+1}-v_{j,k}\right) \approx -h^2\frac{\partial}{\partial y}\left(c\frac{\partial u}{\partial y}\right)_{j,k}, \end{aligned}$	$\begin{aligned} & \left(d_1v\right)_{j,k}:=c_{j-\frac{1}{2},k}\left(v_{j,k}-v_{j-1,k}\right)-c_{j+\frac{1}{2},k}\left(v_{j+1,k}-v_{j,k}\right) \approx -h^2\frac{\partial}{\partial x}\left(c\frac{\partial u}{\partial x}\right)_{j,k}, \\ & \left(d_2v\right)_{j,k}:=c_{j,k-\frac{1}{2}}\left(v_{j,k}-v_{j,k-1}\right)-c_{j,k+\frac{1}{2}}\left(v_{j,k+1}-v_{j,k}\right) \approx -h^2\frac{\partial}{\partial y}\left(c\frac{\partial u}{\partial y}\right)_{j,k}, \end{aligned}$	
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0714 (11.6)	256	■ 0.908 ● W +0	$\boldsymbol{F}_4=\left[\begin{array}{cccc} 1 & 1 & 1 & 1 \\ 1 & \omega_4 & \omega_4^2 & \omega_4^3 \\ 1 & \omega_4^2 & \omega_4^4 & \omega_4^6 \\ 1 & \omega_4^3 & \omega_4^6 & \omega_4^9 \end{array}\right]=\left[\begin{array}{cccc} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{array}\right].$		
lyche-numerical-linear-algebra-EQ0488 (8.18)	191	■ 0.909 ●	$\begin{aligned} & \& \boldsymbol{A}\boldsymbol{A}^\top := \sum_{j=1}^m a_{k,j} ^2, \quad (\text{max column sum}) \\ & \boldsymbol{A}\boldsymbol{A}^\top := \sigma_1^2, \quad (\text{largest singular value of } \boldsymbol{A}) \\ & \boldsymbol{A}\boldsymbol{A}^\top = \max_{1 \leq k \leq m} \ \boldsymbol{e}_k^\top \boldsymbol{A}\ _1 = \max_{1 \leq k \leq m} \sum_{j=1}^n a_{k,j} . \quad (\text{max row sum}) \end{aligned}$	$\ \boldsymbol{A}\ _1 := \max_{1 \leq j \leq n} \ \boldsymbol{A} \boldsymbol{e}_j\ _1 = \max_{1 \leq j \leq n} \sum_{k=1}^m a_{k,j} , \quad (\text{max column sum})$ $\ \boldsymbol{A}\ _2 := \sigma_1, \quad (\text{largest singular value of } \boldsymbol{A})$ $\ \boldsymbol{A}\ _\infty = \max_{1 \leq k \leq m} \ \boldsymbol{e}_k^\top \boldsymbol{A}\ _1 = \max_{1 \leq k \leq m} \sum_{j=1}^n a_{k,j} . \quad (\text{max row sum})$	$\ \boldsymbol{A}\ _1 := \max_{1 \leq j \leq n} \ \boldsymbol{A} \boldsymbol{e}_j\ _1 = \max_{1 \leq j \leq n} \sum_{k=1}^m a_{k,j} , \quad (\text{max column sum})$ $\ \boldsymbol{A}\ _2 := \sigma_1, \quad (\text{largest singular value of } \boldsymbol{A})$ $\ \boldsymbol{A}\ _\infty = \max_{1 \leq k \leq m} \ \boldsymbol{e}_k^\top \boldsymbol{A}\ _1 = \max_{1 \leq k \leq m} \sum_{j=1}^n a_{k,j} . \quad (\text{max row sum})$
lyche-numerical-linear-algebra-EQ0303	124	■ 0.910 ● N +0	$\boldsymbol{H}=\left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array}\right]-\left[\begin{array}{c} u_1 \\ u_2 \end{array}\right]\left[\begin{array}{cc} u_1 & u_2 \end{array}\right]=\left[\begin{array}{cc} 1-u_1^2 & -u_1 u_2 \\ -u_2 u_1 & 1-u_2^2 \end{array}\right].$		
lyche-numerical-linear-algebra-EQ0252	103	■ 0.911 ● W +0	$\begin{aligned} & z^* \boldsymbol{A} z = (\boldsymbol{x} - i \boldsymbol{y})^T \boldsymbol{A} (\boldsymbol{x} + i \boldsymbol{y}) = \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} - i \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{x} + i \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y} - i^2 \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y} \\ & = \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} + \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}, \end{aligned}$	$\boldsymbol{z}^* \boldsymbol{A} \boldsymbol{z} = (\boldsymbol{x} - i \boldsymbol{y})^T \boldsymbol{A} (\boldsymbol{x} + i \boldsymbol{y}) = \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} - i \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{x} + i \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y} - i^2 \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y} = \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} + \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y},$	
lyche-numerical-linear-algebra-EQ0170	075	■ 0.914 ● W +0	$\begin{aligned} & \begin{aligned} & \boldsymbol{a}_{11}^{(1)} x_1 + \boldsymbol{a}_{12}^{(1)} x_2 + \boldsymbol{a}_{13}^{(1)} x_3 = \boldsymbol{b}_1^{(1)}, \quad \text{I} \\ & \boldsymbol{a}_{21}^{(1)} x_1 + \boldsymbol{a}_{22}^{(1)} x_2 + \boldsymbol{a}_{23}^{(1)} x_3 = \boldsymbol{b}_2^{(1)}, \quad \text{II} \\ & \boldsymbol{a}_{31}^{(1)} x_1 + \boldsymbol{a}_{32}^{(1)} x_2 + \boldsymbol{a}_{33}^{(1)} x_3 = \boldsymbol{b}_3^{(1)}. \quad \text{III.} \end{aligned} \end{aligned}$	$\begin{aligned} & \boldsymbol{a}_{11}^{(1)} x_1 + \boldsymbol{a}_{12}^{(1)} x_2 + \boldsymbol{a}_{13}^{(1)} x_3 = \boldsymbol{b}_1^{(1)}, \quad \text{I} \\ & \boldsymbol{a}_{21}^{(1)} x_1 + \boldsymbol{a}_{22}^{(1)} x_2 + \boldsymbol{a}_{23}^{(1)} x_3 = \boldsymbol{b}_2^{(1)}, \quad \text{II} \\ & \boldsymbol{a}_{31}^{(1)} x_1 + \boldsymbol{a}_{32}^{(1)} x_2 + \boldsymbol{a}_{33}^{(1)} x_3 = \boldsymbol{b}_3^{(1)}. \quad \text{III.} \end{aligned}$	
lyche-numerical-linear-algebra-EQ0494	194	■ 0.915 ● K +0	$\left\ \boldsymbol{A}^* \right\ _F = \left\ \boldsymbol{A} \right\ _F \text{ and } \left\ \boldsymbol{A}^* \right\ _2 = \left\ \boldsymbol{A} \right\ _2.$	$\ \boldsymbol{A}^*\ _F = \ \boldsymbol{A}\ _F \text{ and } \ \boldsymbol{A}^*\ _2 = \ \boldsymbol{A}\ _2.$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0485	191	0.917 C +34	$\begin{aligned} & \ \boldsymbol{A}\boldsymbol{B}\ = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B}\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B}\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{B}\boldsymbol{x}\ } \frac{\ \boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } \\ & \leq \max_{\boldsymbol{y} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{y}\ }{\ \boldsymbol{y}\ } \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \ \boldsymbol{A}\ \ \boldsymbol{B}\ \end{aligned}$	$\ \boldsymbol{A}\boldsymbol{B}\ = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B}\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B}\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{B}\boldsymbol{x}\ } \frac{\ \boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } \\ \leq \max_{\boldsymbol{y} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{y}\ }{\ \boldsymbol{y}\ } \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \ \boldsymbol{A}\ \ \boldsymbol{B}\ .$	
lyche-numerical-linear-algebra-EQ1022 (15.5)	346	0.918	$\lim_{k \rightarrow \infty} \frac{z_k}{\lambda_1^k} = c_1 \boldsymbol{v}_1,$	$\lim_{k \rightarrow \infty} \frac{z_k}{\lambda_1^k} = c_1 \boldsymbol{v}_1,$	$\lim_{k \rightarrow \infty} \frac{z_k}{\lambda_1^k} = c_1 \boldsymbol{v}_1,$
lyche-numerical-linear-algebra-EQ0580	215	0.924 N +0	$\begin{array}{c c c c c} t & 1.0 & 2.0 & 3.0 & 4.0 \\ \hline y & 3.1 & 1.8 & 1.0 & 0.1 \end{array}$	$\begin{array}{c c c c c} t & 1.0 & 2.0 & 3.0 & 4.0 \\ \hline y & 3.1 & 1.8 & 1.0 & 0.1 \end{array}$	$\begin{array}{c c c c c} t & 1.0 & 2.0 & 3.0 & 4.0 \\ \hline y & 3.1 & 1.8 & 1.0 & 0.1 \end{array}$
lyche-numerical-linear-algebra-EQ0912 (13.45)	310	0.931 W +0	$\begin{aligned} & \boldsymbol{A} \boldsymbol{x} = \boldsymbol{b} \quad \boldsymbol{B} \boldsymbol{A} \boldsymbol{x} = \boldsymbol{B} \boldsymbol{b} \quad \left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T \right) \boldsymbol{y} = \boldsymbol{C} \boldsymbol{b}, \quad \boldsymbol{x} = \boldsymbol{C}^T \boldsymbol{y}. \end{aligned}$	$\begin{aligned} & \boldsymbol{A} \boldsymbol{x} = \boldsymbol{b} \\ & \boldsymbol{B} \boldsymbol{A} \boldsymbol{x} = \boldsymbol{B} \boldsymbol{b} \\ & \left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T \right) \boldsymbol{y} = \boldsymbol{C} \boldsymbol{b}, \boldsymbol{x} = \boldsymbol{C}^T \boldsymbol{y}. \end{aligned}$	$\begin{aligned} & \boldsymbol{A} \boldsymbol{x} = \boldsymbol{b} \\ & \boldsymbol{B} \boldsymbol{A} \boldsymbol{x} = \boldsymbol{B} \boldsymbol{b} \\ & \left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T \right) \boldsymbol{y} = \boldsymbol{C} \boldsymbol{b}, \text{ \& } \boldsymbol{x} = \boldsymbol{C}^T \boldsymbol{y}. \end{aligned}$
lyche-numerical-linear-algebra-EQ0229 (3.20)	092	0.933 N +0	$\begin{aligned} & \boldsymbol{A} = \begin{bmatrix} \boldsymbol{L}_{21} & \boldsymbol{I} & \cdots & \boldsymbol{L}_{m,m-1} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{U}_{11} & \cdots & \boldsymbol{U}_{1m} \\ \boldsymbol{U}_{22} & \cdots & \boldsymbol{U}_{2m} \\ \cdots & \cdots & \cdots \\ \boldsymbol{U}_{mm} \end{bmatrix} \\ & \boldsymbol{L}_{21} & \boldsymbol{I} & \cdots & \boldsymbol{L}_{m,m-1} & \boldsymbol{I} \\ & \vdots & & \ddots & & \\ & \boldsymbol{L}_{m1} & \cdots & \boldsymbol{L}_{m,m-1} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{U}_{11} & \cdots & \boldsymbol{U}_{1m} \\ \boldsymbol{U}_{22} & \cdots & \boldsymbol{U}_{2m} \\ \cdots & \cdots & \cdots \\ \boldsymbol{U}_{mm} \end{bmatrix} \end{aligned}$	$\boldsymbol{A} = \boldsymbol{L} \boldsymbol{U} = \begin{bmatrix} \boldsymbol{I} & & & \\ \boldsymbol{L}_{21} & \boldsymbol{I} & & \\ \vdots & & \ddots & \\ \boldsymbol{L}_{m1} & \cdots & \boldsymbol{L}_{m,m-1} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{U}_{11} & \cdots & \boldsymbol{U}_{1m} \\ \boldsymbol{U}_{22} & \cdots & \boldsymbol{U}_{2m} \\ \cdots & \cdots & \cdots \\ \boldsymbol{U}_{mm} \end{bmatrix}$	$\boldsymbol{A} = \boldsymbol{L} \boldsymbol{U} = \begin{bmatrix} \boldsymbol{I} & & & \\ \boldsymbol{L}_{21} & \boldsymbol{I} & & \\ \vdots & & \ddots & \\ \boldsymbol{L}_{m1} & \cdots & \boldsymbol{L}_{m,m-1} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{U}_{11} & \cdots & \boldsymbol{U}_{1m} \\ \boldsymbol{U}_{22} & \cdots & \boldsymbol{U}_{2m} \\ \cdots & \cdots & \cdots \\ \boldsymbol{U}_{mm} \end{bmatrix}$
Conf. MathPix confidence: 0.917 0.918 0.924 0.931 0.933 0.5-0.9 0.5-0.9 0.5-0.9 0.5-0.9 0.5-0.9 C +34 N +0 W +0 N +0 N +0 no value no value no value no value no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C +34 N +0 W +0 N +0 N +0 no value no value no value no value no value [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-200253	103	0.934 W +1	$\nabla f(\boldsymbol{x})=\left[\begin{array}{c}\frac{\partial f(\boldsymbol{x})}{\partial x_1}\\\vdots\\\frac{\partial f(\boldsymbol{x})}{\partial x_n}\end{array}\right]\in\mathbb{R}^n,\quad Hf(\boldsymbol{x})=\left[\begin{array}{ccc}\frac{\partial^2 f(\boldsymbol{x})}{\partial x_1\partial x_1}&\cdots&\frac{\partial^2 f(\boldsymbol{x})}{\partial x_1\partial x_n}\\\vdots&&\vdots\\\frac{\partial^2 f(\boldsymbol{x})}{\partial x_n\partial x_1}&\cdots&\frac{\partial^2 f(\boldsymbol{x})}{\partial x_n\partial x_n}\end{array}\right]\in\mathbb{R}^{n\times n}.$	$\nabla f(\boldsymbol{x})=\left[\begin{array}{c}\frac{\partial f(\boldsymbol{x})}{\partial x_1}\\\vdots\\\frac{\partial f(\boldsymbol{x})}{\partial x_n}\end{array}\right]\in\mathbb{R}^n,\quad Hf(\boldsymbol{x})=\left[\begin{array}{ccc}\frac{\partial^2 f(\boldsymbol{x})}{\partial x_1\partial x_1}&\cdots&\frac{\partial^2 f(\boldsymbol{x})}{\partial x_1\partial x_n}\\\vdots&&\vdots\\\frac{\partial^2 f(\boldsymbol{x})}{\partial x_n\partial x_1}&\cdots&\frac{\partial^2 f(\boldsymbol{x})}{\partial x_n\partial x_n}\end{array}\right]\in\mathbb{R}^{n\times n}.$	$\nabla f(\boldsymbol{x})=\left[\begin{array}{c}\frac{\partial f(\boldsymbol{x})}{\partial x_1}\\\vdots\\\frac{\partial f(\boldsymbol{x})}{\partial x_n}\end{array}\right]\in\mathbb{R}^n,\quad Hf(\boldsymbol{x})=\left[\begin{array}{ccc}\frac{\partial^2 f(\boldsymbol{x})}{\partial x_1\partial x_1}&\cdots&\frac{\partial^2 f(\boldsymbol{x})}{\partial x_1\partial x_n}\\\vdots&&\vdots\\\frac{\partial^2 f(\boldsymbol{x})}{\partial x_n\partial x_1}&\cdots&\frac{\partial^2 f(\boldsymbol{x})}{\partial x_n\partial x_n}\end{array}\right]\in\mathbb{R}^{n\times n}.$
lyche-numerical-linear-algebra-200659 (10.3)	239	0.934 W +0	$\begin{aligned} -\Delta_h v_{j,k} &= f_{j,k}, & (jh,kh) \in \Omega_h, \\ v_{j,k} &= 0, & (jh,kh) \in \partial\Omega_h, \end{aligned}$	$-\Delta_h v_{j,k} = f_{j,k}, \quad (jh,kh) \in \Omega_h, \\ v_{j,k} = 0, \quad (jh,kh) \in \partial\Omega_h,$	$-\Delta_h v_{j,k} = f_{j,k}, \quad (jh,kh) \in \Omega_h, \\ v_{j,k} = 0, \quad (jh,kh) \in \partial\Omega_h,$
lyche-numerical-linear-algebra-200873	303	0.936 N -1	$\left\ \boldsymbol{x}-\boldsymbol{x}_k\right\ _{\boldsymbol{A}}^2 \leq\left\ \boldsymbol{x}-\boldsymbol{x}_0-\boldsymbol{w}\right\ _{\boldsymbol{A}}^2 \leq \max _{a \leq x \leq b} Q(x) ^2 \sum_{j=1}^n \frac{\sigma_j^2}{\lambda_j}=\max _{a \leq x \leq b} Q(x) ^2\left\ \boldsymbol{x}-\boldsymbol{x}_0\right\ _{\boldsymbol{A}}^2,$	$\left\ \boldsymbol{x}-\boldsymbol{x}_k\right\ _{\boldsymbol{A}}^2 \leq\left\ \boldsymbol{x}-\boldsymbol{x}_0-\boldsymbol{w}\right\ _{\boldsymbol{A}}^2 \leq \max _{a \leq x \leq b} Q(x) ^2 \sum_{j=1}^n \frac{\sigma_j^2}{\lambda_j}=\max _{a \leq x \leq b} Q(x) ^2\left\ \boldsymbol{x}-\boldsymbol{x}_0\right\ _{\boldsymbol{A}}^2,$	$\left\ \boldsymbol{x}-\boldsymbol{x}_k\right\ _{\boldsymbol{A}}^2 \leq\left\ \boldsymbol{x}-\boldsymbol{x}_0-\boldsymbol{w}\right\ _{\boldsymbol{A}}^2 \leq \max _{a \leq x \leq b} Q(x) ^2 \sum_{j=1}^n \frac{\sigma_j^2}{\lambda_j}=\max _{a \leq x \leq b} Q(x) ^2\left\ \boldsymbol{x}-\boldsymbol{x}_0\right\ _{\boldsymbol{A}}^2,$
lyche-numerical-linear-algebra-201052	359	0.936	$\begin{aligned} &\nabla f(x,y)=\left[\begin{array}{c}2x-y\\-x+2y\end{array}\right],\quad \nabla^T\boldsymbol{g}(x,y)=\left[\begin{array}{cc}2x-y&-x+2y\\1&-1\end{array}\right],\\ \boldsymbol{H}f(x,y)&=\left[\begin{array}{cc}\frac{\partial^2 f}{\partial x^2}&\frac{\partial^2 f}{\partial x\partial y}\\\frac{\partial^2 f}{\partial y\partial x}&\frac{\partial^2 f}{\partial y^2}\end{array}\right]=\left[\begin{array}{cc}2&-1\\-1&2\end{array}\right]. \end{aligned}$	$\nabla f(x,y)=\left[\begin{array}{c}2x-y\\-x+2y\end{array}\right],\quad \nabla^T\boldsymbol{g}(x,y)=\left[\begin{array}{cc}2x-y&-x+2y\\1&-1\end{array}\right], \\ \boldsymbol{H}f(x,y)=\left[\begin{array}{cc}\frac{\partial^2 f}{\partial x^2}&\frac{\partial^2 f}{\partial x\partial y}\\\frac{\partial^2 f}{\partial y\partial x}&\frac{\partial^2 f}{\partial y^2}\end{array}\right]=\left[\begin{array}{cc}2&-1\\-1&2\end{array}\right].$	$\nabla f(x,y)=\left[\begin{array}{c}2x-y\\-x+2y\end{array}\right],\quad \nabla^T\boldsymbol{g}(x,y)=\left[\begin{array}{cc}2x-y&-x+2y\\1&-1\end{array}\right], \\ \boldsymbol{H}f(x,y)=\left[\begin{array}{cc}\frac{\partial^2 f}{\partial x^2}&\frac{\partial^2 f}{\partial x\partial y}\\\frac{\partial^2 f}{\partial y\partial x}&\frac{\partial^2 f}{\partial y^2}\end{array}\right]=\left[\begin{array}{cc}2&-1\\-1&2\end{array}\right].$

Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-verified EQ0782	276	0.938 W +0	$\begin{aligned} & (\boldsymbol{A} \boldsymbol{x})^* \boldsymbol{y} + \boldsymbol{y}^* (\lambda \boldsymbol{A} \boldsymbol{x}) = \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_R) \boldsymbol{y} + \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} + \boldsymbol{A}_R - \boldsymbol{D}) \boldsymbol{y} \\ & = \boldsymbol{y}^* (2\omega^{-1} - 1) \boldsymbol{D} \boldsymbol{y} = \omega^{-1} (2 - \omega) 1 - \lambda ^2 \boldsymbol{x}^* \boldsymbol{D} \boldsymbol{x}. \end{aligned}$	$(\boldsymbol{A} \boldsymbol{x})^* \boldsymbol{y} + \boldsymbol{y}^* (\lambda \boldsymbol{A} \boldsymbol{x}) = \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_R) \boldsymbol{y} + \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} + \boldsymbol{A}_R - \boldsymbol{D}) \boldsymbol{y} \\ = \boldsymbol{y}^* (2\omega^{-1} - 1) \boldsymbol{D} \boldsymbol{y} = \omega^{-1} (2 - \omega) 1 - \lambda ^2 \boldsymbol{x}^* \boldsymbol{D} \boldsymbol{x}.$	$(\boldsymbol{A} \boldsymbol{x})^* \boldsymbol{y} + \boldsymbol{y}^* (\lambda \boldsymbol{A} \boldsymbol{x}) = \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_R) \boldsymbol{y} + \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} + \boldsymbol{A}_R - \boldsymbol{D}) \boldsymbol{y} \\ = \boldsymbol{y}^* (2\omega^{-1} - 1) \boldsymbol{D} \boldsymbol{y} = \omega^{-1} (2 - \omega) 1 - \lambda ^2 \boldsymbol{x}^* \boldsymbol{D} \boldsymbol{x}.$
lyche-numerical-linear-algebra-verified EQ0330	134	0.938 K +0	$\boldsymbol{P} := \left[\begin{array}{cc} c & s \\ -s & c \end{array} \right], \text{ where } c^2 + s^2 = 1.$	$\boldsymbol{P} := \left[\begin{array}{cc} c & s \\ -s & c \end{array} \right], \text{ where } c^2 + s^2 = 1.$	$\boldsymbol{P} := \left[\begin{array}{cc} c & s \\ -s & c \end{array} \right], \text{ where } c^2 + s^2 = 1.$
lyche-numerical-linear-algebra-verified EQ0770 (12.17)	273	0.939 N -1	$\rho(\boldsymbol{A}) := \max_{\lambda \in \sigma(\boldsymbol{A})} \lambda .$	$\rho(\boldsymbol{A}) := \max_{\lambda \in \sigma(\boldsymbol{A})} \lambda .$	$\rho(\boldsymbol{A}) := \max_{\lambda \in \sigma(\boldsymbol{A})} \lambda .$
lyche-numerical-linear-algebra-verified EQ0482	190	0.940 N +0	$\max_{\boldsymbol{x} \neq 0} \frac{\ \boldsymbol{A} \boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{x} \neq 0} \left\ \boldsymbol{A} \left(\frac{\boldsymbol{x}}{\ \boldsymbol{x}\ } \right) \right\ = \max_{\ \boldsymbol{y}\ =1} \ \boldsymbol{A} \boldsymbol{y}\ .$	$\max_{\boldsymbol{x} \neq 0} \frac{\ \boldsymbol{A} \boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{x} \neq 0} \left\ \boldsymbol{A} \left(\frac{\boldsymbol{x}}{\ \boldsymbol{x}\ } \right) \right\ = \max_{\ \boldsymbol{y}\ =1} \ \boldsymbol{A} \boldsymbol{y}\ .$	$\max_{\boldsymbol{x} \neq 0} \frac{\ \boldsymbol{A} \boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{x} \neq 0} \left\ \boldsymbol{A} \left(\frac{\boldsymbol{x}}{\ \boldsymbol{x}\ } \right) \right\ = \max_{\ \boldsymbol{y}\ =1} \ \boldsymbol{A} \boldsymbol{y}\ .$
lyche-numerical-linear-algebra-verified EQ0604 (9.9)	222	0.940 N +0	$\boldsymbol{A}^{\dagger} = \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^*.$	$\boldsymbol{A}^{\dagger} = \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^*.$	$\boldsymbol{A}^{\dagger} = \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^*.$
lyche-numerical-linear-algebra-verified EQ0806 (12.35)	282	0.941 K +0	$\ (I - B)^{-1}\ \leq \frac{1}{1 - \ B\ }.$	$\ (I - B)^{-1}\ \leq \frac{1}{1 - \ B\ }.$	$\ (I - B)^{-1}\ \leq \frac{1}{1 - \ B\ }.$
lyche-numerical-linear-algebra-verified EQ1012	342	0.942 N +0	$\boldsymbol{A}_n = \begin{bmatrix} 10 & 1 & 0 & \cdots & 0 \\ 1 & 10 & 1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & 1 & 10 & 1 \\ 0 & \cdots & 0 & 1 & 10 \end{bmatrix} \in \mathbb{R}^{n \times n}.$	$\boldsymbol{A}_n = \begin{bmatrix} 10 & 1 & 0 & \cdots & 0 \\ 1 & 10 & 1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & 1 & 10 & 1 \\ 0 & \cdots & 0 & 1 & 10 \end{bmatrix} \in \mathbb{R}^{n \times n}.$	$\boldsymbol{A}_n = \begin{bmatrix} 10 & 1 & 0 & \cdots & 0 \\ 1 & 10 & 1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & 1 & 10 & 1 \\ 0 & \cdots & 0 & 1 & 10 \end{bmatrix} \in \mathbb{R}^{n \times n}.$
Conf. MathPix confidence: 0.938					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-076 EQ0174	076	0.942	$\begin{aligned} \boldsymbol{L} \boldsymbol{U} &:= \left[\begin{array}{ccc} 1 & 0 & 0 \\ l_{21}^{(1)} & 1 & 0 \\ l_{31}^{(1)} & l_{32}^{(2)} & 1 \end{array} \right] \left[\begin{array}{ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ 0 & a_{22}^{(2)} & a_{23}^{(2)} \\ 0 & 0 & a_{33}^{(3)} \end{array} \right] \\ &= \left[\begin{array}{ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ l_{21}^{(1)} a_{11}^{(1)} & l_{21}^{(1)} a_{12}^{(1)} + a_{22}^{(2)} & l_{21}^{(1)} a_{13}^{(1)} + a_{23}^{(2)} \\ l_{31}^{(1)} a_{11}^{(1)} & l_{31}^{(1)} a_{12}^{(1)} + l_{32}^{(2)} a_{22}^{(2)} & l_{31}^{(1)} a_{13}^{(1)} + l_{32}^{(2)} a_{23}^{(2)} + a_{33}^{(3)} \end{array} \right] = \left[\begin{array}{ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ a_{21}^{(1)} & a_{22}^{(1)} & a_{23}^{(1)} \\ a_{31}^{(1)} & a_{32}^{(1)} & a_{33}^{(1)} \end{array} \right] = \boldsymbol{A}. \end{aligned}$	$\boldsymbol{L} \boldsymbol{U} := \left[\begin{array}{ccc} 1 & 0 & 0 \\ l_{21}^{(1)} & 1 & 0 \\ l_{31}^{(1)} & l_{32}^{(2)} & 1 \end{array} \right] \left[\begin{array}{ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ 0 & a_{22}^{(2)} & a_{23}^{(2)} \\ 0 & 0 & a_{33}^{(3)} \end{array} \right]$ $= \left[\begin{array}{ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ l_{21}^{(1)} a_{11}^{(1)} & l_{21}^{(1)} a_{12}^{(1)} + a_{22}^{(2)} & l_{21}^{(1)} a_{13}^{(1)} + a_{23}^{(2)} \\ l_{31}^{(1)} a_{11}^{(1)} & l_{31}^{(1)} a_{12}^{(1)} + l_{32}^{(2)} a_{22}^{(2)} & l_{31}^{(1)} a_{13}^{(1)} + l_{32}^{(2)} a_{23}^{(2)} + a_{33}^{(3)} \end{array} \right] = \left[\begin{array}{ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ a_{21}^{(1)} & a_{22}^{(1)} & a_{23}^{(1)} \\ a_{31}^{(1)} & a_{32}^{(1)} & a_{33}^{(1)} \end{array} \right] = \boldsymbol{A}.$	
lyche-numerical-linear-algebra-229 EQ0628	229	0.943 N -1	$\boldsymbol{C} := \left[\begin{array}{cc} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \mathbf{0} \end{array} \right] \text{ and } \boldsymbol{D} := \left[\begin{array}{cc} \mathbf{0} & \boldsymbol{B} \\ \boldsymbol{B}^* & \mathbf{0} \end{array} \right] \in \mathbb{C}^{(m+n) \times (m+n)}.$	$\boldsymbol{C} := \left[\begin{array}{cc} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \mathbf{0} \end{array} \right] \text{ and } \boldsymbol{D} := \left[\begin{array}{cc} \mathbf{0} & \boldsymbol{B} \\ \boldsymbol{B}^* & \mathbf{0} \end{array} \right] \in \mathbb{C}^{(m+n) \times (m+n)}.$	
lyche-numerical-linear-algebra-359 EQ1054	359	0.944 W +1	$\begin{aligned} g(0) &= f(\boldsymbol{x}) \quad g(1) = f(\boldsymbol{x} + \boldsymbol{h}), \\ g'(t) &= \sum_{i=1}^n h_i \frac{\partial f(\boldsymbol{x} + t\boldsymbol{h})}{\partial x_i} = \boldsymbol{h}^T \nabla f(\boldsymbol{x} + t\boldsymbol{h}), \\ g''(t) &= \sum_{i=1}^n \sum_{j=1}^n h_i h_j \frac{\partial^2 f(\boldsymbol{x} + t\boldsymbol{h})}{\partial x_i \partial x_j} = \boldsymbol{h}^T \nabla \nabla^T f(\boldsymbol{x} + t\boldsymbol{h}) \boldsymbol{h}. \end{aligned}$	$g(0) = f(\boldsymbol{x}) \quad g(1) = f(\boldsymbol{x} + \boldsymbol{h}),$ $g'(t) = \sum_{i=1}^n h_i \frac{\partial f(\boldsymbol{x} + t\boldsymbol{h})}{\partial x_i} = \boldsymbol{h}^T \nabla f(\boldsymbol{x} + t\boldsymbol{h}),$ $g''(t) = \sum_{i=1}^n \sum_{j=1}^n h_i h_j \frac{\partial^2 f(\boldsymbol{x} + t\boldsymbol{h})}{\partial x_i \partial x_j} = \boldsymbol{h}^T \nabla \nabla^T f(\boldsymbol{x} + t\boldsymbol{h}) \boldsymbol{h}.$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value					
LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					
[refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0373 (6.8)	152	■0.945 ● W -1	$\boldsymbol{V}^{\boldsymbol{A}}=\left[\begin{array}{c c}\lambda_1&\boldsymbol{x}^*\\\hline\mathbf{0}&\boldsymbol{M}\end{array}\right],\text{ for some }\boldsymbol{M}\in\mathbb{C}^{k\times k}\text{ and }\boldsymbol{x}\in\mathbb{C}^k.$	$\boldsymbol{V}^*\boldsymbol{A}\boldsymbol{V}=\left[\begin{array}{c c}\lambda_1&\boldsymbol{x}^*\\\hline\mathbf{0}&\boldsymbol{M}\end{array}\right],\text{ for some }\boldsymbol{M}\in\mathbb{C}^{k\times k}\text{ and }\boldsymbol{x}\in\mathbb{C}^k.$	
lyche-numerical-linear-algebra-EQ0533	203	■0.946 ●	$m^2\left\langle\frac{n}{m}\boldsymbol{x},\boldsymbol{y}\right\rangle\stackrel{(8.39)}{=}m\langle n\boldsymbol{x},\boldsymbol{y}\rangle\stackrel{(8.39)}{=}mn\langle\boldsymbol{x},\boldsymbol{y}\rangle,$	$m^2\left\langle\frac{n}{m}\boldsymbol{x},\boldsymbol{y}\right\rangle\stackrel{(8.39)}{=}m\langle n\boldsymbol{x},\boldsymbol{y}\rangle\stackrel{(8.39)}{=}mn\langle\boldsymbol{x},\boldsymbol{y}\rangle,$	
lyche-numerical-linear-algebra-EQ0686 (10.19)	246	■0.946 ● N +0	$\left(\boldsymbol{s}_j\otimes\boldsymbol{s}_k\right)^T\left(\boldsymbol{s}_p\otimes\boldsymbol{s}_q\right)=\frac{1}{4h^2}\delta_{j,p}\delta_{k,q},\quad j,k,p,q=1,\dots,m,$	$\left(\boldsymbol{s}_j\otimes\boldsymbol{s}_k\right)^T\left(\boldsymbol{s}_p\otimes\boldsymbol{s}_q\right)=\frac{1}{4h^2}\delta_{j,p}\delta_{k,q},\quad j,k,p,q=1,\dots,m,$	
lyche-numerical-linear-algebra-EQ0324	132	■0.946 ● N +0	$\boldsymbol{A}=\frac{1}{2}\left[\begin{array}{ccc}1&1&-1\\1&1&1\\1&-1&-1\\1&-1&1\end{array}\right]\times\left[\begin{array}{ccc}2&2&3\\0&4&5\\0&0&6\end{array}\right]=\boldsymbol{Q}_1\boldsymbol{R}_1.$	$\boldsymbol{A}=\frac{1}{2}\left[\begin{array}{ccc}1&1&-1\\1&1&1\\1&-1&-1\\1&-1&1\end{array}\right]\times\left[\begin{array}{ccc}2&2&3\\0&4&5\\0&0&6\end{array}\right]=\boldsymbol{Q}_1\boldsymbol{R}_1.$	
lyche-numerical-linear-algebra-EQ0331	135	■0.947 ● N +0	$\boldsymbol{x}=\left[\begin{array}{l}x_1\\x_2\end{array}\right]\neq\mathbf{0},\quad c:=\frac{x_1}{r},\quad s:=\frac{x_2}{r},\quad r:=\ \boldsymbol{x}\ _2.$	$\boldsymbol{x}=\left[\begin{array}{l}x_1\\x_2\end{array}\right]\neq\mathbf{0},\quad c:=\frac{x_1}{r},\quad s:=\frac{x_2}{r},\quad r:=\ \boldsymbol{x}\ _2.$	
lyche-numerical-linear-algebra-EQ1000	337	■0.947 ● N +0	$y_0<y_1<z_1<y_2<z_2<\cdots<z_{k-1}<y_k<y_{k+1}.$	$y_0<y_1<z_1<y_2<z_2<\cdots<z_{k-1}<y_k<y_{k+1}.$	
lyche-numerical-linear-algebra-EQ0999	337	■0.950 ● N +0	$p_3(-M)<0,\quad p_3(y_1)>0,\quad p_3(y_2)<0,\quad p_3(+M)>0.$	$p_3(-M)<0,\quad p_3(y_1)>0,\quad p_3(y_2)<0,\quad p_3(+M)>0.$	
lyche-numerical-linear-algebra-EQ0948 (13.61)	321	■0.950 ● W +0	$\begin{aligned}&\left[\boldsymbol{A}\boldsymbol{p}_0,\boldsymbol{A}\boldsymbol{p}_1,\boldsymbol{A}\boldsymbol{p}_2\right]=\left[\begin{array}{ccc}8&0&0\\-4&3&0\\0&-2&16/9\end{array}\right],\\&\left[\boldsymbol{p}_0,\boldsymbol{p}_1,\boldsymbol{p}_2,\boldsymbol{p}_3\right]=\left[\begin{array}{cccc}4&1&4/9&0\\0&2&8/9&0\\0&0&12/9&0\end{array}\right],\end{aligned}$	$\begin{aligned}&\left[\boldsymbol{A}\boldsymbol{p}_0,\boldsymbol{A}\boldsymbol{p}_1,\boldsymbol{A}\boldsymbol{p}_2\right]=\left[\begin{array}{ccc}8&0&0\\-4&3&0\\0&-2&16/9\end{array}\right],\\&\left[\boldsymbol{p}_0,\boldsymbol{p}_1,\boldsymbol{p}_2,\boldsymbol{p}_3\right]=\left[\begin{array}{cccc}4&1&4/9&0\\0&2&8/9&0\\0&0&12/9&0\end{array}\right],\end{aligned}$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0058	041	0.951 ● N +1	$\left[\begin{array}{ll} a & b \\ c & d \end{array}\right]^{-1}=\alpha\left[\begin{array}{cc} d & -b \\ -c & a \end{array}\right], \quad \alpha=\frac{1}{ad-bc}$	$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \alpha \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}, \quad \alpha = \frac{1}{ad - bc},$	
lyche-numerical-linear-algebra-EQ0220	089	0.951 ● N +0	$\left[\begin{array}{ll} a & b \\ c & d \end{array}\right]=\left[\begin{array}{ll} 1 & 0 \\ l_1 & 1 \end{array}\right]\left[\begin{array}{ll} u_1 & u_2 \\ 0 & u_3 \end{array}\right]=\left[\begin{array}{ll} u_1 & u_2 \\ u_1l_1 & u_2l_1+u_3 \end{array}\right]$	$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ l_1 & 1 \end{bmatrix} \begin{bmatrix} u_1 & u_2 \\ 0 & u_3 \end{bmatrix} = \begin{bmatrix} u_1 & u_2 \\ u_1l_1 & u_2l_1 + u_3 \end{bmatrix}$	
lyche-numerical-linear-algebra-EQ0110	058	0.951 ●	$\boldsymbol{A}_{_1}=\left[\begin{array}{lll} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{array}\right], \quad \boldsymbol{A}_{_2}=\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{array}\right], \quad \boldsymbol{A}_{_3}=\left[\begin{array}{lll} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{array}\right].$	$\boldsymbol{A}_1 = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad \boldsymbol{A}_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{A}_3 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}.$	
lyche-numerical-linear-algebra-EQ0278	117	0.953 ● N -2	$\angle\boldsymbol{x}, \boldsymbol{y}\rangle=\angle\boldsymbol{y}, \boldsymbol{x}\rangle \quad \text{ (symmetry). }$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{y}, \boldsymbol{x} \rangle \quad (\text{symmetry}).$	
lyche-numerical-linear-algebra-EQ0570	211	0.953 ● K +0	$\frac{ \hat{\boldsymbol{s}}-\boldsymbol{s} _{\infty}}{\ \boldsymbol{s}\ _{\infty}} \quad \text{ and } \quad \frac{\ \hat{\boldsymbol{s}}-\boldsymbol{s}\ _1}{\ \boldsymbol{s}\ _1}.$	$\frac{\ \hat{\boldsymbol{s}} - \boldsymbol{s}\ _{\infty}}{\ \boldsymbol{s}\ _{\infty}} \quad \text{ and } \quad \frac{\ \hat{\boldsymbol{s}} - \boldsymbol{s}\ _1}{\ \boldsymbol{s}\ _1}.$	
lyche-numerical-linear-algebra-EQ0620 (9.16)	227	0.953 ● C -5	$\sigma_k=\min_{\operatorname{dim}(\mathcal{S})=n-k+1}\max_{\substack{\boldsymbol{x}\in\mathcal{S} \\ \boldsymbol{x}\neq\boldsymbol{0}}}\frac{\ \boldsymbol{Ax}\ _2}{\ \boldsymbol{x}\ _2}=\max_{\operatorname{dim}(\mathcal{S})=k}\min_{\substack{\boldsymbol{x}\in\mathcal{S} \\ \boldsymbol{x}\neq\boldsymbol{0}}}\frac{\ \boldsymbol{Ax}\ _2}{\ \boldsymbol{x}\ _2}.$	$\sigma_k = \min_{\dim(S)=n-k+1} \max_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \boldsymbol{0}}} \frac{\ \boldsymbol{Ax}\ _2}{\ \boldsymbol{x}\ _2} = \max_{\dim(S)=k} \min_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \boldsymbol{0}}} \frac{\ \boldsymbol{Ax}\ _2}{\ \boldsymbol{x}\ _2}.$	
lyche-numerical-linear-algebra-EQ0722	257	0.954 ● N +0	$\boldsymbol{P}_{_4}=\left[\begin{array}{llll} \boldsymbol{e}_{_1} & \boldsymbol{e}_{_3} & \boldsymbol{e}_{_2} & \end{array}\right]=\left[\begin{array}{llll} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] \quad \boldsymbol{F}_4\boldsymbol{P}_{_4}=\left[\begin{array}{ll ll} 1 & 1 & 1 & 1 \\ 1 & -1 & -i & i \\ 1 & 1 & -1 & -1 \\ 1 & -1 & i & -i \end{array}\right],$	$\boldsymbol{P}_4 = [\boldsymbol{e}_1 \ \boldsymbol{e}_3 \ \boldsymbol{e}_2 \ \boldsymbol{e}_4] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \boldsymbol{F}_4 \boldsymbol{P}_4 = \left[\begin{array}{ll ll} 1 & 1 & 1 & 1 \\ 1 & -1 & -i & i \\ 1 & 1 & -1 & -1 \\ 1 & -1 & i & -i \end{array} \right],$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0057	040	0.969	$\begin{array}{ll} \mathbf{P}_1=(\mathbf{A}+\mathbf{D})(\mathbf{E}+\mathbf{H}), & \\ \mathbf{P}_2=(\mathbf{C}+\mathbf{D})\mathbf{E}, & \mathbf{P}_5=(\mathbf{A}+\mathbf{B})\mathbf{H}, \\ \mathbf{P}_3=\mathbf{A}(\mathbf{F}-\mathbf{H}), & \mathbf{P}_6=(\mathbf{C}-\mathbf{A})(\mathbf{E}+\mathbf{F}), \\ \mathbf{P}_4=\mathbf{D}(\mathbf{G}-\mathbf{E}), & \mathbf{P}_7=(\mathbf{B}-\mathbf{D})(\mathbf{G}+\mathbf{H}), \\ \mathbf{W}=\mathbf{P}_1+\mathbf{P}_4-\mathbf{P}_5+\mathbf{P}_7, & \mathbf{X}=\mathbf{P}_3+\mathbf{P}_5, \\ \mathbf{Y}=\mathbf{P}_2+\mathbf{P}_4, & \mathbf{Z}=\mathbf{P}_1+\mathbf{P}_3-\mathbf{P}_2+\mathbf{P}_6. \end{array}$		
lyche-numerical-linear-algebra-EQ0224 (3.16)	090	0.969 W +0	$\left[\begin{array}{cc} \mathbf{A}_{[k]} & \mathbf{B}_k \\ \mathbf{C}_k & \mathbf{F}_k \end{array}\right] = \left[\begin{array}{cc} \mathbf{L}_{[k]} & \mathbf{0} \\ \mathbf{M}_k & \mathbf{N}_k \end{array}\right] \left[\begin{array}{cc} \mathbf{U}_{[k]} & \mathbf{S}_k \\ \mathbf{0} & \mathbf{T}_k \end{array}\right] = \left[\begin{array}{cc} \mathbf{L}_{[k]}\mathbf{U}_{[k]} & \mathbf{L}_{[k]}\mathbf{S}_k \\ \mathbf{M}_k\mathbf{U}_{[k]} & \mathbf{M}_k\mathbf{S}_k + \mathbf{N}_k\mathbf{T}_k \end{array}\right],$	$\left[\begin{array}{cc} \mathbf{A}_{[k]} & \mathbf{B}_k \\ \mathbf{C}_k & \mathbf{F}_k \end{array}\right] = \left[\begin{array}{cc} \mathbf{L}_{[k]} & \mathbf{0} \\ \mathbf{M}_k & \mathbf{N}_k \end{array}\right] \left[\begin{array}{cc} \mathbf{U}_{[k]} & \mathbf{S}_k \\ \mathbf{0} & \mathbf{T}_k \end{array}\right] = \left[\begin{array}{cc} \mathbf{L}_{[k]}\mathbf{U}_{[k]} & \mathbf{L}_{[k]}\mathbf{S}_k \\ \mathbf{M}_k\mathbf{U}_{[k]} & \mathbf{M}_k\mathbf{S}_k + \mathbf{N}_k\mathbf{T}_k \end{array}\right],$	
lyche-numerical-linear-algebra-EQ0106	057	0.970 C +7	$\begin{aligned} u^{\prime\prime}(x) &= \lim_{h \rightarrow 0} \frac{u'(x+\frac{h}{2}) - u'(x-\frac{h}{2})}{h} = \lim_{h \rightarrow 0} \frac{\frac{u(x+h)-u(x)}{h} - \frac{u(x)-u(x-h)}{h}}{h} \\ &= \lim_{h \rightarrow 0} \frac{u(x+h) - 2u(x) + u(x-h)}{h^2}. \end{aligned}$	$u''(x) = \lim_{h \rightarrow 0} \frac{u'(x+\frac{h}{2}) - u'(x-\frac{h}{2})}{h} = \lim_{h \rightarrow 0} \frac{\frac{u(x+h)-u(x)}{h} - \frac{u(x)-u(x-h)}{h}}{h} = \lim_{h \rightarrow 0} \frac{u(x+h) - 2u(x) + u(x-h)}{h^2}.$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0680	245	0.970	$\begin{aligned} & (\boldsymbol{A} \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F}) \\ & \Leftrightarrow \begin{bmatrix} \boldsymbol{A} b_{11} & \cdots & \boldsymbol{A} b_{1s} \\ \vdots & & \vdots \\ \boldsymbol{A} b_{s1} & \cdots & \boldsymbol{A} b_{ss} \end{bmatrix} \begin{bmatrix} \boldsymbol{v}_1 \\ \vdots \\ \boldsymbol{v}_s \end{bmatrix} = \begin{bmatrix} \boldsymbol{f}_1 \\ \vdots \\ \boldsymbol{f}_s \end{bmatrix} \\ & \Leftrightarrow \boldsymbol{A} \left[\sum_j b_{1j} \boldsymbol{v}_j, \dots, \sum_j b_{sj} \boldsymbol{v}_j \right] = [\boldsymbol{f}_1, \dots, \boldsymbol{f}_s] \\ & \Leftrightarrow \boldsymbol{A} [\boldsymbol{V} \boldsymbol{b}_1, \dots, \boldsymbol{V} \boldsymbol{b}_s] = \boldsymbol{F} \quad \Leftrightarrow \quad \boldsymbol{A} \boldsymbol{V} \boldsymbol{B}^T = \boldsymbol{F}. \end{aligned}$	$(\boldsymbol{A} \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F})$ $\Leftrightarrow \begin{bmatrix} \boldsymbol{A} b_{11} & \cdots & \boldsymbol{A} b_{1s} \\ \vdots & & \vdots \\ \boldsymbol{A} b_{s1} & \cdots & \boldsymbol{A} b_{ss} \end{bmatrix} \begin{bmatrix} \boldsymbol{v}_1 \\ \vdots \\ \boldsymbol{v}_s \end{bmatrix} = \begin{bmatrix} \boldsymbol{f}_1 \\ \vdots \\ \boldsymbol{f}_s \end{bmatrix}$ $\Leftrightarrow \boldsymbol{A} \left[\sum_j b_{1j} \boldsymbol{v}_j, \dots, \sum_j b_{sj} \boldsymbol{v}_j \right] = [\boldsymbol{f}_1, \dots, \boldsymbol{f}_s]$ $\Leftrightarrow \boldsymbol{A} [\boldsymbol{V} \boldsymbol{b}_1, \dots, \boldsymbol{V} \boldsymbol{b}_s] = \boldsymbol{F} \quad \Leftrightarrow \quad \boldsymbol{A} \boldsymbol{V} \boldsymbol{B}^T = \boldsymbol{F}.$	$(\boldsymbol{A} \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F})$ $\Leftrightarrow \begin{bmatrix} \boldsymbol{A} b_{11} & \cdots & \boldsymbol{A} b_{1s} \\ \vdots & & \vdots \\ \boldsymbol{A} b_{s1} & \cdots & \boldsymbol{A} b_{ss} \end{bmatrix} \begin{bmatrix} \boldsymbol{v}_1 \\ \vdots \\ \boldsymbol{v}_s \end{bmatrix} = \begin{bmatrix} \boldsymbol{f}_1 \\ \vdots \\ \boldsymbol{f}_s \end{bmatrix}$ $\Leftrightarrow \boldsymbol{A} \left[\sum_j b_{1j} \boldsymbol{v}_j, \dots, \sum_j b_{sj} \boldsymbol{v}_j \right] = [\boldsymbol{f}_1, \dots, \boldsymbol{f}_s]$ $\Leftrightarrow \boldsymbol{A} [\boldsymbol{V} \boldsymbol{b}_1, \dots, \boldsymbol{V} \boldsymbol{b}_s] = \boldsymbol{F} \quad \Leftrightarrow \quad \boldsymbol{A} \boldsymbol{V} \boldsymbol{B}^T = \boldsymbol{F}.$
lyche-numerical-linear-algebra-EQ0161	071	0.971 ● N +0	$a_{\{j\}} = -1 - \frac{h}{2} r(x_{\{j\}}), \quad c_{\{j\}} = -1 + \frac{h}{2} r(x_{\{j\}}), \quad d_{\{j\}} = 2 + h^2 q(x_{\{j\}})$	$a_j = -1 - \frac{h}{2} r(x_j), \quad c_j = -1 + \frac{h}{2} r(x_j), \quad d_j = 2 + h^2 q(x_j),$	$a_j = -1 - \frac{h}{2} r(x_j), \quad c_j = -1 + \frac{h}{2} r(x_j), \quad d_j = 2 + h^2 q(x_j),$
lyche-numerical-linear-algebra-EQ0312 (5.14)	126	0.971 ● N +0	$\boldsymbol{u} := \frac{\boldsymbol{z} + \boldsymbol{e}_1}{\sqrt{1 + z_1}}, \quad \text{where } \boldsymbol{z} := \bar{\rho} \boldsymbol{x} / \ \boldsymbol{x}\ _2.$	$\boldsymbol{u} := \frac{\boldsymbol{z} + \boldsymbol{e}_1}{\sqrt{1 + z_1}}, \quad \text{where } \boldsymbol{z} := \bar{\rho} \boldsymbol{x} / \ \boldsymbol{x}\ _2.$	$\boldsymbol{u} := \frac{\boldsymbol{z} + \boldsymbol{e}_1}{\sqrt{1 + z_1}}, \quad \text{where } \boldsymbol{z} := \bar{\rho} \boldsymbol{x} / \ \boldsymbol{x}\ _2.$
lyche-numerical-linear-algebra-EQ1043	353	0.971 ● C +78	$\begin{aligned} & \begin{bmatrix} x & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{P_{12}^*} \begin{bmatrix} x & x & x & x \\ \mathbf{x} & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{P_{23}^*} \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & \mathbf{x} & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{P_{34}^*} \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & \mathbf{x} & x \end{bmatrix} \\ & \text{Post multiplication in a QR step} \end{aligned}$	(not rendered)	$\boldsymbol{A} = \begin{bmatrix} x & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{P_{12}^*} \begin{bmatrix} x & x & x & x \\ \mathbf{x} & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{P_{23}^*} \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & \mathbf{x} & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{P_{34}^*} \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & \mathbf{x} & x \end{bmatrix}.$

Conf.

MathPix confidence:

≥0.9

0.5-0.9

<0.5

— no value

LaTeX source

MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual

render vs scan (inkdrill):

● C component

● W weak

● S stable

● N noise

● K clean

● not measured

[refined: *basis*]

beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0407	162	0.971	$\boldsymbol{B}=\left[\begin{array}{ccccc} 0&0&\cdots&0&\cdots \\ 0&-q_0&\cdots&0&-q_1 \\ 0&-q_1&\cdots&0&1 \\ \vdots&\vdots&\ddots&\vdots&\vdots \\ 0&-q_2&\cdots&\vdots&\vdots \\ \vdots&\vdots&\ddots&\vdots&\vdots \\ 0&\vdots&\cdots&1&-q_{n-1} \end{array}\right]$	$\boldsymbol{B}=\begin{bmatrix} 0&0&\cdots&0&-q_0 \\ 1&0&\cdots&0&-q_1 \\ 0&1&\cdots&0&-q_2 \\ \vdots&\vdots&\ddots&\vdots&\vdots \\ 0&0&\cdots&1&-q_{n-1} \end{bmatrix}.$	$\boldsymbol{B}=\begin{bmatrix} 0&0&\cdots&0&-q_0 \\ 1&0&\cdots&0&-q_1 \\ 0&1&\cdots&0&-q_2 \\ \vdots&\vdots&\ddots&\vdots&\vdots \\ 0&0&\cdots&1&-q_{n-1} \end{bmatrix}.$
lyche-numerical-linear-algebra- EQ0289 (5.8)	120	0.971 W +0	$\boldsymbol{v}_1:=\boldsymbol{s}_1, \quad \boldsymbol{v}_j:=\boldsymbol{s}_j-\sum_{i=1}^{j-1}\frac{\langle \boldsymbol{s}_j,\boldsymbol{v}_i\rangle}{\langle \boldsymbol{v}_i,\boldsymbol{v}_i\rangle}\boldsymbol{v}_i, \quad j=2,\ldots,k.$	$\boldsymbol{v}_1:=\boldsymbol{s}_1, \quad \boldsymbol{v}_j:=\boldsymbol{s}_j-\sum_{i=1}^{j-1}\frac{\langle \boldsymbol{s}_j,\boldsymbol{v}_i\rangle}{\langle \boldsymbol{v}_i,\boldsymbol{v}_i\rangle}\boldsymbol{v}_i, \quad j=2,\ldots,k.$	$\boldsymbol{v}_1:=\boldsymbol{s}_1, \quad \boldsymbol{v}_j:=\boldsymbol{s}_j-\sum_{i=1}^{j-1}\frac{\langle \boldsymbol{s}_j,\boldsymbol{v}_i\rangle}{\langle \boldsymbol{v}_i,\boldsymbol{v}_i\rangle}\boldsymbol{v}_i, \quad j=2,\ldots,k.$
lyche-numerical-linear-algebra- EQ0783	276	0.973 K +0	$(\boldsymbol{A}\boldsymbol{A}^*\boldsymbol{x})^*\boldsymbol{y}+\boldsymbol{y}^*(\boldsymbol{A}\boldsymbol{A}^*\boldsymbol{x})=(1-\lambda)\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{A}^*\boldsymbol{x}+\lambda(1-\bar{\lambda})\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{A}^*\boldsymbol{x}=(1- \lambda ^2)\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{A}^*\boldsymbol{x},$	$(\boldsymbol{A}\boldsymbol{x})^*\boldsymbol{y}+\boldsymbol{y}^*(\lambda\boldsymbol{A}\boldsymbol{x})=(1-\lambda)\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}+\lambda(1-\bar{\lambda})\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}=(1- \lambda ^2)\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x},$	$(\boldsymbol{A}\boldsymbol{x})^*\boldsymbol{y}+\boldsymbol{y}^*(\lambda\boldsymbol{A}\boldsymbol{x})=(1-\lambda)\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}+\lambda(1-\bar{\lambda})\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}=(1- \lambda ^2)\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x},$
lyche-numerical-linear-algebra- EQ0677 (10.14)	243	0.974 K +0	$(\boldsymbol{A}\otimes\boldsymbol{B})(\boldsymbol{C}\otimes\boldsymbol{D})=(\boldsymbol{A}\boldsymbol{C})\otimes(\boldsymbol{B}\boldsymbol{D}).$	$(\boldsymbol{A}\otimes\boldsymbol{B})(\boldsymbol{C}\otimes\boldsymbol{D})=(\boldsymbol{A}\boldsymbol{C})\otimes(\boldsymbol{B}\boldsymbol{D}).$	$(\boldsymbol{A}\otimes\boldsymbol{B})(\boldsymbol{C}\otimes\boldsymbol{D})=(\boldsymbol{A}\boldsymbol{C})\otimes(\boldsymbol{B}\boldsymbol{D}).$
lyche-numerical-linear-algebra- EQ0527	202	0.975 K +0	$\ \boldsymbol{x}+\boldsymbol{y}\ ^2+\ \boldsymbol{x}-\boldsymbol{y}\ ^2=\langle\boldsymbol{x}+\boldsymbol{y},\boldsymbol{x}+\boldsymbol{y}\rangle+\langle\boldsymbol{x}-\boldsymbol{y},\boldsymbol{x}-\boldsymbol{y}\rangle=2\ \boldsymbol{x}\ ^2+2\ \boldsymbol{y}\ ^2$	$\ \boldsymbol{x}+\boldsymbol{y}\ ^2+\ \boldsymbol{x}-\boldsymbol{y}\ ^2=\langle\boldsymbol{x}+\boldsymbol{y},\boldsymbol{x}+\boldsymbol{y}\rangle+\langle\boldsymbol{x}-\boldsymbol{y},\boldsymbol{x}-\boldsymbol{y}\rangle=2\ \boldsymbol{x}\ ^2+2\ \boldsymbol{y}\ ^2$	$\ \boldsymbol{x}+\boldsymbol{y}\ ^2+\ \boldsymbol{x}-\boldsymbol{y}\ ^2=\langle\boldsymbol{x}+\boldsymbol{y},\boldsymbol{x}+\boldsymbol{y}\rangle+\langle\boldsymbol{x}-\boldsymbol{y},\boldsymbol{x}-\boldsymbol{y}\rangle=2\ \boldsymbol{x}\ ^2+2\ \boldsymbol{y}\ ^2$
lyche-numerical-linear-algebra- EQ0994 (14.6)	336	0.975 W +1	$\boldsymbol{A}=\left[\begin{array}{ccccc} d_1&c_1&&& \\ c_1&d_2&c_2&& \\ &\ddots&\ddots&\ddots& \\ &&c_{n-2}&d_{n-1}&c_{n-1} \\ &&&c_{n-1}&d_n \end{array}\right]$	$\boldsymbol{A}=\begin{bmatrix} d_1&c_1&&& \\ c_1&d_2&c_2&& \\ &\ddots&\ddots&\ddots& \\ &&c_{n-2}&d_{n-1}&c_{n-1} \\ &&&c_{n-1}&d_n \end{bmatrix}$	$\boldsymbol{A}=\begin{bmatrix} d_1&c_1&&& \\ c_1&d_2&c_2&& \\ &\ddots&\ddots&\ddots& \\ &&c_{n-2}&d_{n-1}&c_{n-1} \\ &&&c_{n-1}&d_n \end{bmatrix}.$
lyche-numerical-linear-algebra- EQ0081 (2.4)	047	0.975 W +0	$g(x):=\begin{cases} p_1(x), & \text{if } x_1\leq x<x_2, \\ p_2(x), & \text{if } x_2\leq x<x_3, \\ \vdots \\ p_{n-1}(x), & \text{if } x_{n-1}\leq x<x_n, \\ p_n(x), & \text{if } x_n\leq x\leq x_{n+1}, \end{cases}$	$g(x):=\begin{cases} p_1(x), & \text{if } x_1\leq x<x_2, \\ p_2(x), & \text{if } x_2\leq x<x_3, \\ \vdots \\ p_{n-1}(x), & \text{if } x_{n-1}\leq x<x_n, \\ p_n(x), & \text{if } x_n\leq x\leq x_{n+1}, \end{cases}$	$g(x):=\begin{cases} p_1(x), & \text{if } x_1\leq x<x_2, \\ p_2(x), & \text{if } x_2\leq x<x_3, \\ \vdots \\ p_{n-1}(x), & \text{if } x_{n-1}\leq x<x_n, \\ p_n(x), & \text{if } x_n\leq x\leq x_{n+1}, \end{cases}$
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean • not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0998	336	0.976 N +0	$\begin{aligned} p_3(x) &= \left(x - d_3\right) p_2(x) - c_2^2(x) \\ p_1(x) &= \left(x - d_3\right) \left(x - y_1\right) \left(x - y_2\right) - c_2^2(x - d_1) \\ &\quad \left(x - y_2\right) - c_2^2(x) \left(x - d_1\right) \end{aligned}$	$p_3(x) = (x - d_3) p_2(x) - c_2^2 p_1(x) = (x - d_3) (x - y_1) (x - y_2) - c_2^2 (x - d_1).$	$p_3(x) = (x - d_3) p_2(x) - c_2^2 p_1(x) = (x - d_3) (x - y_1) (x - y_2) - c_2^2 (x - d_1).$
lyche-numerical-linear-algebra- EQ0731	259	0.976	$\begin{aligned} &\begin{aligned} &\mathbf{F}_{2m} \mathbf{y} = \mathbf{F}_{2m} \mathbf{P}_{2m} \mathbf{P}_{2m}^T \mathbf{y} = \mathbf{F}_{2m} \mathbf{P}_{2m} \mathbf{w} \\ &= \left[\begin{array}{c c} \mathbf{F}_m & \mathbf{D}_m \mathbf{F}_m \\ \hline \mathbf{F}_m & -\mathbf{D}_m \mathbf{F}_m \end{array} \right] \begin{bmatrix} \mathbf{w}_1 \\ \mathbf{w}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{q}_1 + \mathbf{q}_2 \\ \mathbf{q}_1 - \mathbf{q}_2 \end{bmatrix}, \end{aligned} \\ &\begin{aligned} &\mathbf{F}_{2m} \mathbf{y} = \mathbf{F}_{2m} \mathbf{P}_{2m} \mathbf{P}_{2m}^T \mathbf{y} = \mathbf{F}_{2m} \mathbf{P}_{2m} \mathbf{w} \\ &= \left[\begin{array}{c c} \mathbf{F}_m & \mathbf{D}_m \mathbf{F}_m \\ \hline \mathbf{F}_m & -\mathbf{D}_m \mathbf{F}_m \end{array} \right] \begin{bmatrix} \mathbf{w}_1 \\ \mathbf{w}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{q}_1 + \mathbf{q}_2 \\ \mathbf{q}_1 - \mathbf{q}_2 \end{bmatrix}, \end{aligned} \end{aligned}$	$\mathbf{F}_{2m} \mathbf{y} = \mathbf{F}_{2m} \mathbf{P}_{2m} \mathbf{P}_{2m}^T \mathbf{y} = \mathbf{F}_{2m} \mathbf{P}_{2m} \mathbf{w}$ $= \left[\begin{array}{c c} \mathbf{F}_m & \mathbf{D}_m \mathbf{F}_m \\ \hline \mathbf{F}_m & -\mathbf{D}_m \mathbf{F}_m \end{array} \right] \begin{bmatrix} \mathbf{w}_1 \\ \mathbf{w}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{q}_1 + \mathbf{q}_2 \\ \mathbf{q}_1 - \mathbf{q}_2 \end{bmatrix},$	$\mathbf{F}_{2m} \mathbf{y} = \mathbf{F}_{2m} \mathbf{P}_{2m} \mathbf{P}_{2m}^T \mathbf{y} = \mathbf{F}_{2m} \mathbf{P}_{2m} \mathbf{w}$ $= \left[\begin{array}{c c} \mathbf{F}_m & \mathbf{D}_m \mathbf{F}_m \\ \hline \mathbf{F}_m & -\mathbf{D}_m \mathbf{F}_m \end{array} \right] \begin{bmatrix} \mathbf{w}_1 \\ \mathbf{w}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{q}_1 + \mathbf{q}_2 \\ \mathbf{q}_1 - \mathbf{q}_2 \end{bmatrix},$
lyche-numerical-linear-algebra- EQ0046	037	0.976 C -6	$\left[\begin{array}{ccc} c_{i_1, j_1} & \cdots & c_{i_1, j_r} \\ \vdots & & \vdots \\ c_{i_r, j_1} & \cdots & c_{i_r, j_r} \end{array} \right] = \sum_k \left[\begin{array}{ccc} a_{i_1, k_1} & \cdots & a_{i_1, k_r} \\ \vdots & & \vdots \\ a_{i_r, k_1} & \cdots & a_{i_r, k_r} \end{array} \right] \left[\begin{array}{ccc} b_{k_1, j_1} & \cdots & b_{k_1, j_r} \\ \vdots & & \vdots \\ b_{k_r, j_1} & \cdots & b_{k_r, j_r} \end{array} \right],$	$\left[\begin{array}{ccc} c_{i_1, j_1} & \cdots & c_{i_1, j_r} \\ \vdots & & \vdots \\ c_{i_r, j_1} & \cdots & c_{i_r, j_r} \end{array} \right] = \sum_k \left[\begin{array}{ccc} a_{i_1, k_1} & \cdots & a_{i_1, k_r} \\ \vdots & & \vdots \\ a_{i_r, k_1} & \cdots & a_{i_r, k_r} \end{array} \right] \left[\begin{array}{ccc} b_{k_1, j_1} & \cdots & b_{k_1, j_r} \\ \vdots & & \vdots \\ b_{k_r, j_1} & \cdots & b_{k_r, j_r} \end{array} \right],$	$\left[\begin{array}{ccc} c_{i_1, j_1} & \cdots & c_{i_1, j_r} \\ \vdots & & \vdots \\ c_{i_r, j_1} & \cdots & c_{i_r, j_r} \end{array} \right] = \sum_k \left[\begin{array}{ccc} a_{i_1, k_1} & \cdots & a_{i_1, k_r} \\ \vdots & & \vdots \\ a_{i_r, k_1} & \cdots & a_{i_r, k_r} \end{array} \right] \left[\begin{array}{ccc} b_{k_1, j_1} & \cdots & b_{k_1, j_r} \\ \vdots & & \vdots \\ b_{k_r, j_1} & \cdots & b_{k_r, j_r} \end{array} \right],$
lyche-numerical-linear-algebra- EQ0470 (8.6)	187	0.977 N +0	$\begin{aligned} &\ \mathbf{x}\ _{p'} \leq \ \mathbf{x}\ _p \leq n^{1/p-1/p'} \ \mathbf{x}\ _{p'}, \quad \mathbf{x} \in \mathbb{C}^n, \quad 1 \leq p \leq p' \leq \infty. \\ &\ \mathbf{x}\ _{p'} \leq \ \mathbf{x}\ _p \leq n^{1/p-1/p'} \ \mathbf{x}\ _{p'}, \quad \mathbf{x} \in \mathbb{C}^n, \quad 1 \leq p \leq p' \leq \infty. \end{aligned}$	$\ \mathbf{x}\ _{p'} \leq \ \mathbf{x}\ _p \leq n^{1/p-1/p'} \ \mathbf{x}\ _{p'}, \quad \mathbf{x} \in \mathbb{C}^n, \quad 1 \leq p \leq p' \leq \infty.$	$\ \mathbf{x}\ _{p'} \leq \ \mathbf{x}\ _p \leq n^{1/p-1/p'} \ \mathbf{x}\ _{p'}, \quad \mathbf{x} \in \mathbb{C}^n, \quad 1 \leq p \leq p' \leq \infty.$
lyche-numerical-linear-algebra- EQ0969	327	0.977 N +1	$\begin{aligned} &\mathbf{R}_i = \{z \in \mathbb{C} : z - a_{ii} \leq r_i\}, \quad r_i := \sum_{\substack{j=1 \\ j \neq i}}^n a_{ij} , \\ &\mathbf{C}_j = \{z \in \mathbb{C} : z - a_{jj} \leq c_j\}, \quad c_j := \sum_{\substack{i=1 \\ i \neq j}}^n a_{ij} . \end{aligned}$	$\mathbf{R}_i = \{z \in \mathbb{C} : z - a_{ii} \leq r_i\}, \quad r_i := \sum_{\substack{j=1 \\ j \neq i}}^n a_{ij} ,$ $\mathbf{C}_j = \{z \in \mathbb{C} : z - a_{jj} \leq c_j\}, \quad c_j := \sum_{\substack{i=1 \\ i \neq j}}^n a_{ij} .$	$\mathbf{R}_i = \{z \in \mathbb{C} : z - a_{ii} \leq r_i\}, \quad r_i := \sum_{\substack{j=1 \\ j \neq i}}^n a_{ij} ,$ $\mathbf{C}_j = \{z \in \mathbb{C} : z - a_{jj} \leq c_j\}, \quad c_j := \sum_{\substack{i=1 \\ i \neq j}}^n a_{ij} .$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0342	137	0.977 S +0	$\boldsymbol{A}=\left[\begin{array}{ll} 1&2\\ 1&0\\ 1&0 \end{array}\right], \quad \boldsymbol{Q}=\frac{1}{2}\left[\begin{array}{cccc} 1&1&1&1\\ 1&1&-1&-1\\ 1&-1&-1&1\\ 1&-1&1&-1 \end{array}\right], \quad \boldsymbol{R}=\left[\begin{array}{cc} 2&2\\ 0&2\\ 0&0 \end{array}\right].$	$\boldsymbol{A}=\left[\begin{array}{cc} 1&2\\ 1&2\\ 1&0 \end{array}\right], \quad \boldsymbol{Q}=\frac{1}{2}\left[\begin{array}{cccc} 1&1&1&1\\ 1&1&-1&-1\\ 1&-1&-1&1\\ 1&-1&1&-1 \end{array}\right], \quad \boldsymbol{R}=\left[\begin{array}{cc} 2&2\\ 0&2\\ 0&0 \end{array}\right].$	
lyche-numerical-linear-algebra-EQ0418 (7.2)	168	0.977	$\boldsymbol{A}=\frac{1}{15}\left[\begin{array}{ccc} 14&2&4\\ 4&22&16\\ 16&13&2 \end{array}\right]=\frac{1}{3}\left[\begin{array}{ccc} 1&2&2\\ 2&-2&1\\ 2&1&-2 \end{array}\right]\left[\begin{array}{cc} 2&0\\ 0&1\\ 0&0 \end{array}\right]\frac{1}{5}\left[\begin{array}{cc} 3&4\\ 4&-3 \end{array}\right]=\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^*.$	$\boldsymbol{A}=\frac{1}{15}\left[\begin{array}{cc} 14&2\\ 4&22\\ 16&13 \end{array}\right]=\frac{1}{3}\left[\begin{array}{ccc} 1&2&2\\ 2&-2&1\\ 2&1&-2 \end{array}\right]\left[\begin{array}{cc} 2&0\\ 0&1\\ 0&0 \end{array}\right]\frac{1}{5}\left[\begin{array}{cc} 3&4\\ 4&-3 \end{array}\right]=\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^*.$	
lyche-numerical-linear-algebra-EQ0536	204	0.978 N +0	$\begin{aligned} (-a)\angle\boldsymbol{x}, \boldsymbol{y}\angle &= \angle(-a)\boldsymbol{x}, \boldsymbol{y}\angle \\ &\stackrel{(8.36)}{=} \frac{1}{4}\left(\ -a\boldsymbol{x} + \boldsymbol{y}\ ^2 - \ -a\boldsymbol{x} - \boldsymbol{y}\ ^2\right) = -\langle a\boldsymbol{x}, \boldsymbol{y}\rangle, \end{aligned}$	$(-a)\langle \boldsymbol{x}, \boldsymbol{y}\rangle = \langle (-a)\boldsymbol{x}, \boldsymbol{y}\rangle \stackrel{(8.36)}{=} \frac{1}{4}\left(\ -a\boldsymbol{x} + \boldsymbol{y}\ ^2 - \ -a\boldsymbol{x} - \boldsymbol{y}\ ^2\right) = -\langle a\boldsymbol{x}, \boldsymbol{y}\rangle,$	
lyche-numerical-linear-algebra-EQ0767	272	0.979 N +0	$\boldsymbol{x}_{k+1}=\left[\begin{array}{l} \boldsymbol{x}_{k+1}(1)\\ \boldsymbol{x}_{k+1}(2) \end{array}\right]=\left[\begin{array}{cc} 0&1/2\\ 0&1/4 \end{array}\right]\left[\begin{array}{l} \boldsymbol{x}_k(1)\\ \boldsymbol{x}_k(2) \end{array}\right]+\left[\begin{array}{c} 1/2\\ 3/4 \end{array}\right]=\boldsymbol{G}_1\boldsymbol{x}_k+\boldsymbol{c}.$	$\boldsymbol{x}_{k+1}=\left[\begin{array}{l} \boldsymbol{x}_{k+1}(1)\\ \boldsymbol{x}_{k+1}(2) \end{array}\right]=\left[\begin{array}{cc} 0&1/2\\ 0&1/4 \end{array}\right]\left[\begin{array}{l} \boldsymbol{x}_k(1)\\ \boldsymbol{x}_k(2) \end{array}\right]+\left[\begin{array}{c} 1/2\\ 3/4 \end{array}\right]=\boldsymbol{G}_1\boldsymbol{x}_k+\boldsymbol{c}.$	
lyche-numerical-linear-algebra-EQ0125	061	0.979 W +1	$\begin{aligned} \boldsymbol{s}_{\boldsymbol{j}}^T\boldsymbol{s}_{\boldsymbol{j}} &= \sum_{k=1}^m \sin^2(kj\pi h) = \sum_{k=0}^m \sin^2(kj\pi h) = \frac{1}{2} \sum_{k=0}^m (1 - \cos(2kj\pi h)) \\ &= \frac{m+1}{2} - \frac{1}{2} \sum_{k=0}^m \cos(2kj\pi h) = \frac{m+1}{2}, \end{aligned}$	$\begin{aligned} \boldsymbol{s}_{\boldsymbol{j}}^T\boldsymbol{s}_{\boldsymbol{j}} &= \sum_{k=1}^m \sin^2(kj\pi h) = \sum_{k=0}^m \sin^2(kj\pi h) = \frac{1}{2} \sum_{k=0}^m (1 - \cos(2kj\pi h)) \\ &= \frac{m+1}{2} - \frac{1}{2} \sum_{k=0}^m \cos(2kj\pi h) = \frac{m+1}{2}, \end{aligned}$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0172	076	0.980 W -2	$\begin{aligned} & a_{11}^{(1)} x_1 + a_{12}^{(1)} x_2 + a_{13}^{(1)} x_3 = b_1^{(1)}, \\ & a_{22}^{(2)} x_2 + a_{23}^{(2)} x_3 = b_2^{(2)}, \\ & a_{33}^{(3)} x_3 = b_3^{(3)}, \end{aligned}$	$\begin{aligned} & a_{11}^{(1)} x_1 + a_{12}^{(1)} x_2 + a_{13}^{(1)} x_3 = b_1^{(1)}, \\ & a_{22}^{(2)} x_2 + a_{23}^{(2)} x_3 = b_2^{(2)}, \\ & a_{33}^{(3)} x_3 = b_3^{(3)}, \end{aligned}$	$\begin{aligned} & a_{11}^{(1)} x_1 + a_{12}^{(1)} x_2 + a_{13}^{(1)} x_3 = b_1^{(1)}, \\ & a_{22}^{(2)} x_2 + a_{23}^{(2)} x_3 = b_2^{(2)}, \\ & a_{33}^{(3)} x_3 = b_3^{(3)}, \end{aligned}$
lyche-numerical-linear-algebra EQ0703	253	0.981 W +0	$\begin{aligned} & \boldsymbol{s}_j = [\sin(j\pi h), \sin(2j\pi h), \dots, \sin(mj\pi h)]^T, \\ & \lambda_j = 2 - 2\cos(j\pi h) = 4\sin^2(j\pi h/2), \quad h = 1/(m+1), \\ & \boldsymbol{s}_j^T \boldsymbol{s}_k = \delta_{j,k}/(2h) \text{ for all } j, k. \end{aligned}$	$\begin{aligned} & \boldsymbol{s}_j = [\sin(j\pi h), \sin(2j\pi h), \dots, \sin(mj\pi h)]^T, \\ & \lambda_j = 2 - 2\cos(j\pi h) = 4\sin^2(j\pi h/2), \quad h = 1/(m+1), \\ & \boldsymbol{s}_j^T \boldsymbol{s}_k = \delta_{j,k}/(2h) \text{ for all } j, k. \end{aligned}$	$\begin{aligned} & \boldsymbol{s}_j = [\sin(j\pi h), \sin(2j\pi h), \dots, \sin(mj\pi h)]^T, \\ & \lambda_j = 2 - 2\cos(j\pi h) = 4\sin^2(j\pi h/2), \quad h = 1/(m+1), \\ & \boldsymbol{s}_j^T \boldsymbol{s}_k = \delta_{j,k}/(2h) \text{ for all } j, k. \end{aligned}$
lyche-numerical-linear-algebra EQ0674	243	0.981	$\boldsymbol{T}_1 \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}_1 = \begin{bmatrix} \boldsymbol{T}_1 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{T}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{T}_1 \end{bmatrix} + \begin{bmatrix} d\boldsymbol{I} & a\boldsymbol{I} & \mathbf{0} \\ a\boldsymbol{I} & d\boldsymbol{I} & a\boldsymbol{I} \\ \mathbf{0} & a\boldsymbol{I} & d\boldsymbol{I} \end{bmatrix} = \boldsymbol{T}_2$	$\boldsymbol{T}_1 \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}_1 = \begin{bmatrix} \boldsymbol{T}_1 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{T}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{T}_1 \end{bmatrix} + \begin{bmatrix} d\boldsymbol{I} & a\boldsymbol{I} & \mathbf{0} \\ a\boldsymbol{I} & d\boldsymbol{I} & a\boldsymbol{I} \\ \mathbf{0} & a\boldsymbol{I} & d\boldsymbol{I} \end{bmatrix} = \boldsymbol{T}_2$	$\boldsymbol{T}_1 \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}_1 = \begin{bmatrix} \boldsymbol{T}_1 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{T}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{T}_1 \end{bmatrix} + \begin{bmatrix} d\boldsymbol{I} & a\boldsymbol{I} & \mathbf{0} \\ a\boldsymbol{I} & d\boldsymbol{I} & a\boldsymbol{I} \\ \mathbf{0} & a\boldsymbol{I} & d\boldsymbol{I} \end{bmatrix} = \boldsymbol{T}_2$
lyche-numerical-linear-algebra EQ0613 (9.13)	225	0.981 N +0	$\frac{1}{K(\boldsymbol{A})} \frac{\ \boldsymbol{e}_1\ }{\ \boldsymbol{b}_1\ } \leq \frac{\ \boldsymbol{y} - \boldsymbol{x}\ }{\ \boldsymbol{x}\ } \leq K(\boldsymbol{A}) \frac{\ \boldsymbol{e}_1\ }{\ \boldsymbol{b}_1\ }, \quad K(\boldsymbol{A}) = \ \boldsymbol{A}\ \ \boldsymbol{A}^\dagger\ .$	$\frac{1}{K(\boldsymbol{A})} \frac{\ \boldsymbol{e}_1\ }{\ \boldsymbol{b}_1\ } \leq \frac{\ \boldsymbol{y} - \boldsymbol{x}\ }{\ \boldsymbol{x}\ } \leq K(\boldsymbol{A}) \frac{\ \boldsymbol{e}_1\ }{\ \boldsymbol{b}_1\ }, \quad K(\boldsymbol{A}) = \ \boldsymbol{A}\ \ \boldsymbol{A}^\dagger\ .$	$\frac{1}{K(\boldsymbol{A})} \frac{\ \boldsymbol{e}_1\ }{\ \boldsymbol{b}_1\ } \leq \frac{\ \boldsymbol{y} - \boldsymbol{x}\ }{\ \boldsymbol{x}\ } \leq K(\boldsymbol{A}) \frac{\ \boldsymbol{e}_1\ }{\ \boldsymbol{b}_1\ }, \quad K(\boldsymbol{A}) = \ \boldsymbol{A}\ \ \boldsymbol{A}^\dagger\ .$
lyche-numerical-linear-algebra EQ0127	062	0.981 N +2	$\lambda \boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x} = (\boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{y})^* = (\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{y})^* = (\mu \boldsymbol{x}^* \boldsymbol{y})^* = \mu \boldsymbol{y}^* \boldsymbol{x},$	$\lambda \boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x} = (\boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{y})^* = (\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{y})^* = (\mu \boldsymbol{x}^* \boldsymbol{y})^* = \mu \boldsymbol{y}^* \boldsymbol{x},$	$\lambda \boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x} = (\boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{y})^* = (\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{y})^* = (\mu \boldsymbol{x}^* \boldsymbol{y})^* = \mu \boldsymbol{y}^* \boldsymbol{x},$
lyche-numerical-linear-algebra EQ1020 (15.3)	345	0.981 N +0	$\boldsymbol{z}_k = c_1 \lambda_1^k \boldsymbol{v}_1 + c_2 \lambda_2^k \boldsymbol{v}_2 + \dots + c_n \lambda_n^k \boldsymbol{v}_n, \quad k = 0, 1, 2, \dots$	$\boldsymbol{z}_k = c_1 \lambda_1^k \boldsymbol{v}_1 + c_2 \lambda_2^k \boldsymbol{v}_2 + \dots + c_n \lambda_n^k \boldsymbol{v}_n, \quad k = 0, 1, 2, \dots$	$\boldsymbol{z}_k = c_1 \lambda_1^k \boldsymbol{v}_1 + c_2 \lambda_2^k \boldsymbol{v}_2 + \dots + c_n \lambda_n^k \boldsymbol{v}_n, \quad k = 0, 1, 2, \dots$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ1021 (15.4)	345	0.982 W +1	$\frac{z_{\{k\}}}{\lambda_{\{1\}}^k}=c_{\{1\}}\boldsymbol{v}_{\{1\}}+c_{\{2\}}\left(\frac{\lambda_{\{2\}}}{\lambda_{\{1\}}}\right)^k\boldsymbol{v}_{\{2\}}+\cdots+c_{\{n\}}\left(\frac{\lambda_{\{n\}}}{\lambda_{\{1\}}}\right)^k\boldsymbol{v}_{\{n\}},\quad k=0,1,2,\ldots$	$\frac{z_k}{\lambda_1^k}=c_1\boldsymbol{v}_1+c_2\left(\frac{\lambda_2}{\lambda_1}\right)^k\boldsymbol{v}_2+\cdots+c_n\left(\frac{\lambda_n}{\lambda_1}\right)^k\boldsymbol{v}_n,\quad k=0,1,2,\ldots$	$\frac{z_k}{\lambda_1^k}=c_1\boldsymbol{v}_1+c_2\left(\frac{\lambda_2}{\lambda_1}\right)^k\boldsymbol{v}_2+\cdots+c_n\left(\frac{\lambda_n}{\lambda_1}\right)^k\boldsymbol{v}_n,\quad k=0,1,2,\ldots$
lyche-numerical-linear-algebra- EQ0497	194	0.982 N +0	$\left x_{\{i\}}\right \leq\left y_{\{i\}}\right ,\,i=1,\,\ldots,\,\ldots,\,n\,\Longrightarrow\ \boldsymbol{x}\ \leq\ \boldsymbol{y}\ ,\,\text{for all }\boldsymbol{x},\boldsymbol{y}\in\mathbb{C}^n.$	$ x_i \leq y_i ,\,i=1,\ldots,n\implies\ \boldsymbol{x}\ \leq\ \boldsymbol{y}\ ,\text{ for all }\boldsymbol{x},\boldsymbol{y}\in\mathbb{C}^n.$	$ x_i \leq y_i ,\,i=1,\ldots,n\implies\ \boldsymbol{x}\ \leq\ \boldsymbol{y}\ ,\text{ for all }\boldsymbol{x},\boldsymbol{y}\in\mathbb{C}^n.$
lyche-numerical-linear-algebra- EQ0537	204	0.983 N +0	$\ \boldsymbol{x}\ _{\boldsymbol{A}}\leq\ \boldsymbol{y}\ _{\boldsymbol{A}}\implies\ \boldsymbol{x}\ _2\leq\sqrt{\frac{\lambda_1}{\lambda_n}}\ \boldsymbol{y}\ _2,$	$\ \boldsymbol{x}\ _{\boldsymbol{A}}\leq\ \boldsymbol{y}\ _{\boldsymbol{A}}\implies\ \boldsymbol{x}\ _2\leq\sqrt{\frac{\lambda_1}{\lambda_n}}\ \boldsymbol{y}\ _2,$	$\ \boldsymbol{x}\ _{\boldsymbol{A}}\leq\ \boldsymbol{y}\ _{\boldsymbol{A}}\implies\ \boldsymbol{x}\ _2\leq\sqrt{\frac{\lambda_1}{\lambda_n}}\ \boldsymbol{y}\ _2,$
lyche-numerical-linear-algebra- EQ0573	212	0.984 N +0	$\ \boldsymbol{x}\ _p\leq\ \boldsymbol{x}\ _q\leq n^{1/q-1/p}\ \boldsymbol{x}\ _p,\quad\boldsymbol{x}\in\mathbb{C}^n.$	$\ \boldsymbol{x}\ _p\leq\ \boldsymbol{x}\ _q\leq n^{1/q-1/p}\ \boldsymbol{x}\ _p,\quad\boldsymbol{x}\in\mathbb{C}^n.$	$\ \boldsymbol{x}\ _p\leq\ \boldsymbol{x}\ _q\leq n^{1/q-1/p}\ \boldsymbol{x}\ _p,\quad\boldsymbol{x}\in\mathbb{C}^n.$
lyche-numerical-linear-algebra- EQ0184 (3.4)	079	0.984	$\boldsymbol{L}=\left[\begin{array}{cccc} 1&&&\\ l_{21}^{(1)}&1&&\\ \vdots&&\ddots&\\ l_{n1}^{(1)}&l_{n2}^{(2)}&\cdots&1\end{array}\right],\quad\boldsymbol{U}=\left[\begin{array}{ccc} a_{11}^{(1)}&\cdots&a_{1n}^{(1)}\\ &\ddots&\vdots\\ &&a_{nn}^{(n)}\end{array}\right],$	$\boldsymbol{L}=\left[\begin{array}{ccc} 1&&\\ l_{21}^{(1)}&1&\\ \vdots&&\ddots\\ l_{n1}^{(1)}&l_{n2}^{(2)}&\cdots&1\end{array}\right],\quad\boldsymbol{U}=\left[\begin{array}{ccc} a_{11}^{(1)}&\cdots&a_{1n}^{(1)}\\ &\ddots&\vdots\\ &&a_{nn}^{(n)}\end{array}\right],$	$\boldsymbol{L}=\left[\begin{array}{ccc} 1&&\\ l_{21}^{(1)}&1&\\ \vdots&&\ddots\\ l_{n1}^{(1)}&l_{n2}^{(2)}&\cdots&1\end{array}\right],\quad\boldsymbol{U}=\left[\begin{array}{ccc} a_{11}^{(1)}&\cdots&a_{1n}^{(1)}\\ &\ddots&\vdots\\ &&a_{nn}^{(n)}\end{array}\right],$
lyche-numerical-linear-algebra- EQ0019 (1.7)	029	0.984 N +0	$\operatorname{dim}(\mathcal{S}+\mathcal{T})=\operatorname{dim}(\mathcal{S})+\operatorname{dim}(\mathcal{T})-\operatorname{dim}(\mathcal{S}\cap\mathcal{T}).$	$\dim(\mathcal{S}+\mathcal{T})=\dim(\mathcal{S})+\dim(\mathcal{T})-\dim(\mathcal{S}\cap\mathcal{T}).$	$\dim(\mathcal{S}+\mathcal{T})=\dim(\mathcal{S})+\dim(\mathcal{T})-\dim(\mathcal{S}\cap\mathcal{T}).$
lyche-numerical-linear-algebra- EQ0502 (8.23)	196	0.984 N +0	$\frac{1}{K(\boldsymbol{A})}\frac{\ \boldsymbol{r}(\boldsymbol{y})\ }{\ \boldsymbol{b}\ }\leq\frac{\ \boldsymbol{y}-\boldsymbol{x}\ }{\ \boldsymbol{x}\ }\leq K(\boldsymbol{A})\frac{\ \boldsymbol{r}(\boldsymbol{y})\ }{\ \boldsymbol{b}\ }.$	$\frac{1}{K(\boldsymbol{A})}\frac{\ \boldsymbol{r}(\boldsymbol{y})\ }{\ \boldsymbol{b}\ }\leq\frac{\ \boldsymbol{y}-\boldsymbol{x}\ }{\ \boldsymbol{x}\ }\leq K(\boldsymbol{A})\frac{\ \boldsymbol{r}(\boldsymbol{y})\ }{\ \boldsymbol{b}\ }.$	$\frac{1}{K(\boldsymbol{A})}\frac{\ \boldsymbol{r}(\boldsymbol{y})\ }{\ \boldsymbol{b}\ }\leq\frac{\ \boldsymbol{y}-\boldsymbol{x}\ }{\ \boldsymbol{x}\ }\leq K(\boldsymbol{A})\frac{\ \boldsymbol{r}(\boldsymbol{y})\ }{\ \boldsymbol{b}\ }.$
lyche-numerical-linear-algebra- EQ0740	262	0.984 W +1	$x_{\{j\},\,k}=\frac{h^4g_{j,k}}{\sigma_j+\sigma_k-\frac{2}{3}\sigma_j\sigma_k},\,\text{where }\sigma_j=\sin^2((j\pi h)/2)\,\text{for }j,k=1,2,\ldots,m.$	$x_{j,k}=\frac{h^4g_{j,k}}{\sigma_j+\sigma_k-\frac{2}{3}\sigma_j\sigma_k},\,\text{where }\sigma_j=\sin^2((j\pi h)/2)\,\text{for }j,k=1,2,\ldots,m.$	$x_{j,k}=\frac{h^4g_{j,k}}{\sigma_j+\sigma_k-\frac{2}{3}\sigma_j\sigma_k},\,\text{where }\sigma_j=\sin^2((j\pi h)/2)\,\text{for }j,k=1,2,\ldots,m.$
lyche-numerical-linear-algebra- EQ0114	059	0.985 N +0	$u^{\prime\prime}(j\,h)\approx\frac{u((j+1)\,h)-2u(j\,h)+u((j-1)\,h)}{h^2},\quad j=1,\,\ldots,\,m,$	$u''(jh)\approx\frac{u((j+1)h)-2u(jh)+u((j-1)h)}{h^2},\quad j=1,\ldots,m,$	$u''(jh)\approx\frac{u((j+1)h)-2u(jh)+u((j-1)h)}{h^2},\quad j=1,\ldots,m,$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-2021-090564	210	0.985 N +0	A:=\left[\begin{array}{cccccc} 2 & \mu_1 & 0 & \cdots & 0 & \lambda_1 \\ 0 & \lambda_2 & 2 & \mu_2 & \ddots & 0 \\ 0 & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & \ddots & \ddots & \ddots & \ddots & 0 \\ \mu_n & 0 & \cdots & 0 & \lambda_n & 2 \end{array}\right],	$A := \begin{bmatrix} 2 & \mu_1 & 0 & \cdots & 0 & \lambda_1 \\ \lambda_2 & 2 & \mu_2 & \ddots & & 0 \\ 0 & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & & \ddots & \lambda_{n-1} & 2 & \mu_{n-1} \\ \mu_n & 0 & \cdots & 0 & \lambda_n & 2 \end{bmatrix},$	
lyche-numerical-linear-algebra-2021-090701 (11.3)	252	0.985 N +0	\begin{aligned} \boldsymbol{y}_{\{1\}} &= \boldsymbol{b}_{\{1\}}, \\ \quad \boldsymbol{y}_{\{k\}} &= \boldsymbol{b}_{\{k\}} - \\ \boldsymbol{y}_{\{k-1\}} \boldsymbol{L}_{\{k-1\}} & \boldsymbol{y}_{\{k-1\}}, \\ \quad k=2,3,\ldots,m, \quad \boldsymbol{x}_{\{m\}} &= \boldsymbol{U}_{\{m\}}^{-1} \boldsymbol{y}_{\{m\}}, \\ \quad \boldsymbol{x}_{\{k\}} &= \boldsymbol{U}_{\{k\}}^{-1} \left(\boldsymbol{y}_{\{k\}} - \boldsymbol{C}_{\{k\}} \boldsymbol{x}_{\{k+1\}} \right. \\ \left. \boldsymbol{x}_{\{k+1\}} \right), \quad k=m-1,\ldots,2,1. \end{aligned}	$\boldsymbol{y}_1 = \boldsymbol{b}_1, \quad \boldsymbol{y}_k = \boldsymbol{b}_k - \boldsymbol{L}_{k-1} \boldsymbol{y}_{k-1}, \quad k = 2, 3, \dots, m,$ $\boldsymbol{x}_m = \boldsymbol{U}_m^{-1} \boldsymbol{y}_m, \quad \boldsymbol{x}_k = \boldsymbol{U}_k^{-1} (\boldsymbol{y}_k - \boldsymbol{C}_k \boldsymbol{x}_{k+1}), \quad k = m - 1, \dots, 2, 1.$	$\boldsymbol{y}_1 = \boldsymbol{b}_1, \quad \boldsymbol{y}_k = \boldsymbol{b}_k - \boldsymbol{L}_{k-1} \boldsymbol{y}_{k-1}, \quad k = 2, 3, \dots, m,$ $\boldsymbol{x}_m = \boldsymbol{U}_m^{-1} \boldsymbol{y}_m, \quad \boldsymbol{x}_k = \boldsymbol{U}_k^{-1} (\boldsymbol{y}_k - \boldsymbol{C}_k \boldsymbol{x}_{k+1}), \quad k = m - 1, \dots, 2, 1.$
lyche-numerical-linear-algebra-2021-090848 (13.12)	294	0.985	\begin{aligned} \boldsymbol{p}_{\{j\}} &:= \boldsymbol{r}_{\{j\}} - \sum_{i=0}^{j-1} \left(\frac{\boldsymbol{r}_{\{j\}}^T \boldsymbol{A} \boldsymbol{p}_{\{i\}}}{\boldsymbol{p}_{\{i\}}^T \boldsymbol{A} \boldsymbol{p}_{\{i\}}} \right) \boldsymbol{p}_{\{i\}}, \\ \boldsymbol{x}_{\{j+1\}} &:= \boldsymbol{x}_{\{j\}} + \alpha_{\{j\}} \boldsymbol{p}_{\{j\}} \quad \alpha_{\{j\}} := \frac{\boldsymbol{r}_{\{j\}}^T \boldsymbol{r}_{\{j\}}}{\boldsymbol{p}_{\{j\}}^T \boldsymbol{A} \boldsymbol{p}_{\{j\}}}, \\ \boldsymbol{r}_{\{j+1\}} &= \boldsymbol{r}_{\{j\}} - \alpha_{\{j\}} \boldsymbol{A} \boldsymbol{p}_{\{j\}}. \end{aligned}	$\boldsymbol{p}_j := \boldsymbol{r}_j - \sum_{i=0}^{j-1} \left(\frac{\boldsymbol{r}_j^T \boldsymbol{A} \boldsymbol{p}_i}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i} \right) \boldsymbol{p}_i,$ $\boldsymbol{x}_{j+1} := \boldsymbol{x}_j + \alpha_j \boldsymbol{p}_j \quad \alpha_j := \frac{\boldsymbol{r}_j^T \boldsymbol{r}_j}{\boldsymbol{p}_j^T \boldsymbol{A} \boldsymbol{p}_j},$ $\boldsymbol{r}_{j+1} = \boldsymbol{r}_j - \alpha_j \boldsymbol{A} \boldsymbol{p}_j.$	$\boldsymbol{p}_j := \boldsymbol{r}_j - \sum_{i=0}^{j-1} \left(\frac{\boldsymbol{r}_j^T \boldsymbol{A} \boldsymbol{p}_i}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i} \right) \boldsymbol{p}_i,$ $\boldsymbol{x}_{j+1} := \boldsymbol{x}_j + \alpha_j \boldsymbol{p}_j \quad \alpha_j := \frac{\boldsymbol{r}_j^T \boldsymbol{r}_j}{\boldsymbol{p}_j^T \boldsymbol{A} \boldsymbol{p}_j},$ $\boldsymbol{r}_{j+1} = \boldsymbol{r}_j - \alpha_j \boldsymbol{A} \boldsymbol{p}_j.$
lyche-numerical-linear-algebra-2021-090210	087	0.985 N +0	\begin{array}{r} 10^{-4} x_1 + 2 x_2 = 4 \\ x_1 + x_2 = 3 \end{array}	$10^{-4} x_1 + 2 x_2 = 4$ $x_1 + x_2 = 3$	$10^{-4} x_1 + 2 x_2 = 4$ $x_1 + x_2 = 3$
lyche-numerical-linear-algebra-2021-090429	172	0.985 K +0	\boldsymbol{A}=\left[\begin{array}{ll} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{array}\right].	$\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}.$	$\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}.$
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0145	065	0.985 K +0	$\begin{aligned} \boldsymbol{B}_{\{11\}} \boldsymbol{A}_{\{11\}} &= \boldsymbol{I}, \\ \boldsymbol{B}_{\{21\}} \boldsymbol{A}_{\{11\}} &= \mathbf{0}, \\ \boldsymbol{B}_{\{21\}} \boldsymbol{A}_{\{12\}} + \boldsymbol{B}_{\{22\}} \boldsymbol{A}_{\{22\}} &= \boldsymbol{I}, \\ \boldsymbol{B}_{\{11\}} \boldsymbol{A}_{\{12\}} + \boldsymbol{B}_{\{12\}} \boldsymbol{A}_{\{22\}} &= \mathbf{0} . \end{aligned}$	$\boldsymbol{B}_{11} \boldsymbol{A}_{11} = \boldsymbol{I}, \boldsymbol{B}_{21} \boldsymbol{A}_{11} = \mathbf{0}, \boldsymbol{B}_{21} \boldsymbol{A}_{12} + \boldsymbol{B}_{22} \boldsymbol{A}_{22} = \boldsymbol{I}, \boldsymbol{B}_{11} \boldsymbol{A}_{12} + \boldsymbol{B}_{12} \boldsymbol{A}_{22} = \mathbf{0}.$	$\boldsymbol{B}_{11} \boldsymbol{A}_{11} = \boldsymbol{I}, \boldsymbol{B}_{21} \boldsymbol{A}_{11} = \mathbf{0}, \boldsymbol{B}_{21} \boldsymbol{A}_{12} + \boldsymbol{B}_{22} \boldsymbol{A}_{22} = \boldsymbol{I}, \boldsymbol{B}_{11} \boldsymbol{A}_{12} + \boldsymbol{B}_{12} \boldsymbol{A}_{22} = \mathbf{0}.$
lyche-numerical-linear-algebra- EQ0622 (9.17)	227	0.985 W -1	$\begin{aligned} \sigma_1 &= \max_{\substack{\boldsymbol{x} \in \mathbb{C}^n \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2}, \quad \sigma_n = \min_{\substack{\boldsymbol{x} \in \mathbb{C}^n \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2}. \end{aligned}$	$\sigma_1 = \max_{\substack{\boldsymbol{x} \in \mathbb{C}^n \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2}, \quad \sigma_n = \min_{\substack{\boldsymbol{x} \in \mathbb{C}^n \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2}.$	$\sigma_1 = \max_{\substack{\boldsymbol{x} \in \mathbb{C}^n \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2}, \quad \sigma_n = \min_{\substack{\boldsymbol{x} \in \mathbb{C}^n \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2}.$
lyche-numerical-linear-algebra- EQ0272	113	0.986 K +0	$\boldsymbol{A} = \begin{pmatrix} 1 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & -1 & 2 & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{pmatrix}.$	$\boldsymbol{A} = \begin{pmatrix} 1 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & -1 & 2 & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{pmatrix}.$	$\boldsymbol{A} = \begin{pmatrix} 1 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & -1 & 2 & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{pmatrix}.$
lyche-numerical-linear-algebra- EQ0268	110	0.986	$\boldsymbol{L} \boldsymbol{L}^* = \begin{bmatrix} 0 & \mathbf{0}^* \\ 0 & \boldsymbol{L}_1 \end{bmatrix} \begin{bmatrix} 0 & \mathbf{0}^* \\ 0 & \boldsymbol{L}_1^* \end{bmatrix} = \begin{bmatrix} 0 & \mathbf{0}^* \\ 0 & \boldsymbol{B} \end{bmatrix} = \boldsymbol{A}.$	$\boldsymbol{L} \boldsymbol{L}^* = \begin{bmatrix} 0 & \mathbf{0}^* \\ 0 & \boldsymbol{L}_1 \end{bmatrix} \begin{bmatrix} 0 & \mathbf{0}^* \\ 0 & \boldsymbol{L}_1^* \end{bmatrix} = \begin{bmatrix} 0 & \mathbf{0}^* \\ 0 & \boldsymbol{B} \end{bmatrix} = \boldsymbol{A}.$	$\boldsymbol{L} \boldsymbol{L}^* = \begin{bmatrix} 0 & \mathbf{0}^* \\ 0 & \boldsymbol{L}_1 \end{bmatrix} \begin{bmatrix} 0 & \mathbf{0}^* \\ 0 & \boldsymbol{L}_1^* \end{bmatrix} = \begin{bmatrix} 0 & \mathbf{0}^* \\ 0 & \boldsymbol{B} \end{bmatrix} = \boldsymbol{A}.$
lyche-numerical-linear-algebra- EQ0205 (3.14)	086	0.986 W +0	$\begin{aligned} \boldsymbol{L} &= \begin{bmatrix} 1 & & \\ a_{p_2,1}^{(1)} & 1 & \\ \vdots & & \ddots \\ a_{p_n,1}^{(1)} & a_{p_n,2}^{(2)} & \cdots 1 \end{bmatrix}, \quad \boldsymbol{U} = \begin{bmatrix} a_{p_1,1}^{(1)} & \cdots & a_{p_1,n}^{(1)} \\ & \ddots & \vdots \\ & & a_{p_n,n}^{(n)} \end{bmatrix}. \\ \boldsymbol{U} &= \begin{bmatrix} a_{p_1,1}^{(1)} & \cdots & a_{p_1,n}^{(1)} \\ & \ddots & \vdots \\ & & a_{p_n,n}^{(n)} \end{bmatrix}. \end{aligned}$	$\boldsymbol{L} = \begin{bmatrix} 1 & & \\ a_{p_2,1}^{(1)} & 1 & \\ \vdots & & \ddots \\ a_{p_n,1}^{(1)} & a_{p_n,2}^{(2)} & \cdots 1 \end{bmatrix}, \quad \boldsymbol{U} = \begin{bmatrix} a_{p_1,1}^{(1)} & \cdots & a_{p_1,n}^{(1)} \\ & \ddots & \vdots \\ & & a_{p_n,n}^{(n)} \end{bmatrix}.$	$\boldsymbol{L} = \begin{bmatrix} 1 & & \\ a_{p_2,1}^{(1)} & 1 & \\ \vdots & & \ddots \\ a_{p_n,1}^{(1)} & a_{p_n,2}^{(2)} & \cdots 1 \end{bmatrix}, \quad \boldsymbol{U} = \begin{bmatrix} a_{p_1,1}^{(1)} & \cdots & a_{p_1,n}^{(1)} \\ & \ddots & \vdots \\ & & a_{p_n,n}^{(n)} \end{bmatrix}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0995	336	0.986 N -1	$\boldsymbol{A}_1=\left[\begin{array}{ccccc} d_1 & c_1 & c_2 & & \\ c_1 & d_2 & & & \\ & \ddots & \ddots & \ddots & \\ & & c_{i-2} & d_{i-1} & c_{i-1} \\ & & & c_{i-1} & d_i \end{array}\right] \text{ and } \boldsymbol{A}_2=\left[\begin{array}{ccccc} d_{i+1} & c_{i+1} & c_{i+2} & & \\ c_{i+1} & d_{i+2} & & & \\ & \ddots & \ddots & \ddots & \\ & & c_{n-2} & d_{n-1} & c_{n-1} \\ & & & c_{n-1} & d_n \end{array}\right].$	$\boldsymbol{A}_1=\left[\begin{array}{ccc} d_1 & c_1 & \\ c_1 & d_2 & c_2 \\ & \ddots & \ddots & \ddots \\ & & c_{i-2} & d_{i-1} & c_{i-1} \\ & & & c_{i-1} & d_i \end{array}\right] \text{ and } \boldsymbol{A}_2=\left[\begin{array}{ccc} d_{i+1} & c_{i+1} & \\ c_{i+1} & d_{i+2} & c_{i+2} \\ & \ddots & \ddots & \ddots \\ & & c_{n-2} & d_{n-1} & c_{n-1} \\ & & & c_{n-1} & d_n \end{array}\right].$	
lyche-numerical-linear-algebra- EQ0300	124	0.987 N +0	$\langle Ux,Uy\rangle=(Uy)^*(Ux)=y^*U^*Ux=y^*x=\langle x,y\rangle.$ $x=y^{\wedge\{*\}}\quad x=\langle x,y\rangle.$	$\langle Ux,Uy\rangle=(Uy)^*(Ux)=y^*U^*Ux=y^*x=\langle x,y\rangle.$	
lyche-numerical-linear-algebra- EQ0426 (7.5)	172	0.987 N +0	$\boldsymbol{A}=\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^*=\boldsymbol{U}_1\boldsymbol{\Sigma}_1\boldsymbol{V}_1^*.$	$\boldsymbol{A}=\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^*=\boldsymbol{U}_1\boldsymbol{\Sigma}_1\boldsymbol{V}_1^*.$	
lyche-numerical-linear-algebra- EQ0397 (6.15)	159	0.987 U +1	$\sum_{j=1}^n\left \mu_{i_j}-\lambda_j\right ^2\leq\sum_{i=1}^n\sum_{j=1}^n\left a_{ij}-b_{ij}\right ^2.$	$\sum_{j=1}^n\left \mu_{i_j}-\lambda_j\right ^2\leq\sum_{i=1}^n\sum_{j=1}^n\left a_{ij}-b_{ij}\right ^2.$	
lyche-numerical-linear-algebra- EQ0511	199	0.987 N +2	$\ \boldsymbol{x}\ _p:=\left(\sum_{j=1}^n\left x_j\right ^p\right)^{1/p},\quad\ \boldsymbol{x}\ _\infty:=\max_{1\leq j\leq n}\left x_j\right .$	$\ \boldsymbol{x}\ _p:=\left(\sum_{j=1}^n\left x_j\right ^p\right)^{1/p},\quad\ \boldsymbol{x}\ _\infty:=\max_{1\leq j\leq n}\left x_j\right .$	
lyche-numerical-linear-algebra- EQ0139	064	0.987	$\boldsymbol{A}\boldsymbol{B}=\left[\left[\begin{array}{cc} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{array}\right]\left[\begin{array}{c} \boldsymbol{B}_{11} \\ \boldsymbol{B}_{21} \end{array}\right],\left[\begin{array}{cc} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{array}\right]\left[\begin{array}{c} \boldsymbol{B}_{12} \\ \boldsymbol{B}_{22} \end{array}\right]\right].$	$\boldsymbol{A}\boldsymbol{B}=\left[\left[\begin{array}{cc} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{array}\right]\left[\begin{array}{c} \boldsymbol{B}_{11} \\ \boldsymbol{B}_{21} \end{array}\right],\left[\begin{array}{cc} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{array}\right]\left[\begin{array}{c} \boldsymbol{B}_{12} \\ \boldsymbol{B}_{22} \end{array}\right]\right].$	
Conf. MathPix confidence: 0.986 0.5-0.9 <0.5 no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured refined: basis beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0164 (2.38)	073	■0.987 ●C -4	$\boldsymbol{L}=\left[\begin{array}{ccccc} 1 & 0 & \cdots & \cdots & 0 \\ & -\frac{1}{2} & 1 & \ddots & \vdots \\ 0 & -\frac{2}{3} & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & -\frac{m-1}{m} & 1 \end{array}\right], \boldsymbol{U}=\left[\begin{array}{ccccc} 2 & -1 & 0 & \cdots & 0 \\ 0 & \frac{3}{2} & -1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & \frac{m}{m-1} & -1 & \frac{m+1}{m} \\ 0 & \cdots & \cdots & 0 & \frac{m+1}{m} \end{array}\right].$	$\boldsymbol{L}=\left[\begin{array}{ccccc} 1 & 0 & \cdots & \cdots & 0 \\ -\frac{1}{2} & 1 & \ddots & & \vdots \\ 0 & -\frac{2}{3} & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & -\frac{m-1}{m} & 1 \end{array}\right], \boldsymbol{U}=\left[\begin{array}{ccccc} 2 & -1 & 0 & \cdots & 0 \\ 0 & \frac{3}{2} & -1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ \vdots & & \ddots & \frac{m}{m-1} & -1 \\ 0 & \cdots & \cdots & 0 & \frac{m+1}{m} \end{array}\right].$	
lyche-numerical-linear-algebra EQ0572	212	■0.988 ●K +0	$S_p=\left\{\boldsymbol{x}\in\mathbb{R}^n:\ \boldsymbol{x}\ _p=1\right\}$	$S_p=\{\boldsymbol{x}\in\mathbb{R}^n:\ \boldsymbol{x}\ _p=1\}$	$S_p=\{\boldsymbol{x}\in\mathbb{R}^n:\ \boldsymbol{x}\ _p=1\}$
lyche-numerical-linear-algebra EQ0949 (13.62)	321	■0.988 ●W +0	$\begin{aligned} \angle\boldsymbol{x}, \boldsymbol{y} &= \angle\boldsymbol{y}, \boldsymbol{x}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n, \\ \angle\boldsymbol{Cx}, \boldsymbol{y} &= \angle\boldsymbol{y}, \boldsymbol{C}^T\boldsymbol{x} \stackrel{(13.61)}{=} \angle\boldsymbol{C}^T\boldsymbol{x}, \boldsymbol{y}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n, \boldsymbol{C} \in \mathbb{R}^{n\times n}. \end{aligned}$	$\langle\boldsymbol{x},\boldsymbol{y}\rangle=\langle\boldsymbol{y},\boldsymbol{x}\rangle,\quad\boldsymbol{x},\boldsymbol{y}\in\mathbb{R}^n,$ $\langle\boldsymbol{C}\boldsymbol{x},\boldsymbol{y}\rangle=\langle\boldsymbol{y},\boldsymbol{C}^T\boldsymbol{x}\rangle\stackrel{(13.61)}{=}\langle\boldsymbol{C}^T\boldsymbol{x},\boldsymbol{y}\rangle,\quad\boldsymbol{x},\boldsymbol{y}\in\mathbb{R}^n,\boldsymbol{C}\in\mathbb{R}^{n\times n}.$	$\langle\boldsymbol{x},\boldsymbol{y}\rangle=\langle\boldsymbol{y},\boldsymbol{x}\rangle,\quad\boldsymbol{x},\boldsymbol{y}\in\mathbb{R}^n,$ $\langle\boldsymbol{C}\boldsymbol{x},\boldsymbol{y}\rangle=\langle\boldsymbol{y},\boldsymbol{C}^T\boldsymbol{x}\rangle\stackrel{(13.61)}{=}\langle\boldsymbol{C}^T\boldsymbol{x},\boldsymbol{y}\rangle,\quad\boldsymbol{x},\boldsymbol{y}\in\mathbb{R}^n,\boldsymbol{C}\in\mathbb{R}^{n\times n}.$
lyche-numerical-linear-algebra EQ0166	073	■0.988 ●N +0	$\boldsymbol{A}\boldsymbol{B}^T=\boldsymbol{a}_{:1}\boldsymbol{b}_{:1}^T+\boldsymbol{a}_{:2}\boldsymbol{b}_{:2}^T+\cdots+\boldsymbol{a}_{:n}\boldsymbol{b}_{:n}^T.$	$\boldsymbol{AB}^T=\boldsymbol{a}_{:1}\boldsymbol{b}_{:1}^T+\boldsymbol{a}_{:2}\boldsymbol{b}_{:2}^T+\cdots+\boldsymbol{a}_{:n}\boldsymbol{b}_{:n}^T.$	$\boldsymbol{AB}^T=\boldsymbol{a}_{:1}\boldsymbol{b}_{:1}^T+\boldsymbol{a}_{:2}\boldsymbol{b}_{:2}^T+\cdots+\boldsymbol{a}_{:n}\boldsymbol{b}_{:n}^T.$
lyche-numerical-linear-algebra EQ0589	217	■0.989 ●W +2	$\boldsymbol{H}_3=\left[\begin{array}{ccc} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{array}\right].$	$\boldsymbol{H}_3=\left[\begin{array}{ccc} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{array}\right].$	$\boldsymbol{H}_3=\left[\begin{array}{ccc} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{array}\right].$
lyche-numerical-linear-algebra EQ0911 (13.44)	310	■0.989 ●	$\boldsymbol{B}\boldsymbol{A}\boldsymbol{x}=\boldsymbol{B}\boldsymbol{b}\Leftrightarrow\boldsymbol{C}^T\left(\boldsymbol{C}\boldsymbol{A}\boldsymbol{C}^T\right)\boldsymbol{C}^{-T}\boldsymbol{x}=\boldsymbol{C}^T\boldsymbol{C}\boldsymbol{b}\Leftrightarrow\left(\boldsymbol{C}\boldsymbol{A}\boldsymbol{C}^T\right)\boldsymbol{y}=\boldsymbol{C}\boldsymbol{b},$	$\boldsymbol{BAx=Bb}\Leftrightarrow\boldsymbol{C}^T\left(\boldsymbol{CAC}^T\right)\boldsymbol{C}^{-T}\boldsymbol{x}=\boldsymbol{C}^T\boldsymbol{Cb}\Leftrightarrow\left(\boldsymbol{CAC}^T\right)\boldsymbol{y}=\boldsymbol{Cb},$	$\boldsymbol{BAx=Bb}\Leftrightarrow\boldsymbol{C}^T\left(\boldsymbol{CAC}^T\right)\boldsymbol{C}^{-T}\boldsymbol{x}=\boldsymbol{C}^T\boldsymbol{Cb}\Leftrightarrow\left(\boldsymbol{CAC}^T\right)\boldsymbol{y}=\boldsymbol{Cb},$
lyche-numerical-linear-algebra EQ0500	195	■0.989 ●K +0	$\ \boldsymbol{AB}\ \leq\ \boldsymbol{A}\ \ \boldsymbol{B}\ \text{ and }\ \boldsymbol{Ax}\ \leq\ \boldsymbol{A}\ \ \boldsymbol{x}\ .$	$\ \boldsymbol{AB}\ \leq\ \boldsymbol{A}\ \ \boldsymbol{B}\ \text{ and }\ \boldsymbol{Ax}\ \leq\ \boldsymbol{A}\ \ \boldsymbol{x}\ .$	$\ \boldsymbol{AB}\ \leq\ \boldsymbol{A}\ \ \boldsymbol{B}\ \text{ and }\ \boldsymbol{Ax}\ \leq\ \boldsymbol{A}\ \ \boldsymbol{x}\ .$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 —no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0137	064	■ 0.989 ● W +1	$\begin{aligned} & (\boldsymbol{A} \boldsymbol{B})_{ij} = \sum_{k=1}^p a_{ik} b_{kj} = \sum_{k=1}^s a_{ik} b_{kj} + \sum_{k=s+1}^p a_{ik} b_{kj} \\ & = (\boldsymbol{A}_1 \boldsymbol{B}_1)_{ij} + (\boldsymbol{A}_2 \boldsymbol{B}_2)_{ij} = (\boldsymbol{A}_1 \boldsymbol{B}_1 + \boldsymbol{A}_2 \boldsymbol{B}_2)_{ij}. \end{aligned}$	$(\boldsymbol{A} \boldsymbol{B})_{ij} = \sum_{k=1}^p a_{ik} b_{kj} = \sum_{k=1}^s a_{ik} b_{kj} + \sum_{k=s+1}^p a_{ik} b_{kj} \\ = (\boldsymbol{A}_1 \boldsymbol{B}_1)_{ij} + (\boldsymbol{A}_2 \boldsymbol{B}_2)_{ij} = (\boldsymbol{A}_1 \boldsymbol{B}_1 + \boldsymbol{A}_2 \boldsymbol{B}_2)_{ij}.$	$(\boldsymbol{A} \boldsymbol{B})_{ij} = \sum_{k=1}^p a_{ik} b_{kj} = \sum_{k=1}^s a_{ik} b_{kj} + \sum_{k=s+1}^p a_{ik} b_{kj} \\ = (\boldsymbol{A}_1 \boldsymbol{B}_1)_{ij} + (\boldsymbol{A}_2 \boldsymbol{B}_2)_{ij} = (\boldsymbol{A}_1 \boldsymbol{B}_1 + \boldsymbol{A}_2 \boldsymbol{B}_2)_{ij}.$
lyche-numerical-linear-algebra- EQ0436 (7.7)	174	■ 0.989 ● S +0	$\begin{aligned} & \boldsymbol{A} \boldsymbol{v}_i = \sigma_i \boldsymbol{u}_i, i = 1, \dots, r, \quad \boldsymbol{A} \boldsymbol{v}_i = 0, i = r + 1, \dots, n, \\ & \boldsymbol{A}^* \boldsymbol{u}_i = \sigma_i \boldsymbol{v}_i, i = 1, \dots, r, \quad \boldsymbol{A}^* \boldsymbol{u}_i = 0, i = r + 1, \dots, m. \end{aligned}$	$\boldsymbol{A} \boldsymbol{v}_i = \sigma_i \boldsymbol{u}_i, i = 1, \dots, r, \quad \boldsymbol{A} \boldsymbol{v}_i = 0, i = r + 1, \dots, n, \\ \boldsymbol{A}^* \boldsymbol{u}_i = \sigma_i \boldsymbol{v}_i, i = 1, \dots, r, \quad \boldsymbol{A}^* \boldsymbol{u}_i = 0, i = r + 1, \dots, m.$	$\boldsymbol{A} \boldsymbol{v}_i = \sigma_i \boldsymbol{u}_i, i = 1, \dots, r, \quad \boldsymbol{A} \boldsymbol{v}_i = 0, i = r + 1, \dots, n, \\ \boldsymbol{A}^* \boldsymbol{u}_i = \sigma_i \boldsymbol{v}_i, i = 1, \dots, r, \quad \boldsymbol{A}^* \boldsymbol{u}_i = 0, i = r + 1, \dots, m.$
lyche-numerical-linear-algebra- EQ0001	021	■ 0.990 ● K +0	$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix},$	$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix},$	$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix},$
lyche-numerical-linear-algebra- EQ0060	041	■ 0.990 ● K +0	$(\boldsymbol{A} + \boldsymbol{B} \boldsymbol{C}^T)^{-1} = \boldsymbol{A}^{-1} - \boldsymbol{A}^{-1} \boldsymbol{B} (\boldsymbol{I} + \boldsymbol{C}^T \boldsymbol{A}^{-1} \boldsymbol{B})^{-1} \boldsymbol{C}^T \boldsymbol{A}^{-1}.$	$(\boldsymbol{A} + \boldsymbol{B} \boldsymbol{C}^T)^{-1} = \boldsymbol{A}^{-1} - \boldsymbol{A}^{-1} \boldsymbol{B} (\boldsymbol{I} + \boldsymbol{C}^T \boldsymbol{A}^{-1} \boldsymbol{B})^{-1} \boldsymbol{C}^T \boldsymbol{A}^{-1}.$	$(\boldsymbol{A} + \boldsymbol{B} \boldsymbol{C}^T)^{-1} = \boldsymbol{A}^{-1} - \boldsymbol{A}^{-1} \boldsymbol{B} (\boldsymbol{I} + \boldsymbol{C}^T \boldsymbol{A}^{-1} \boldsymbol{B})^{-1} \boldsymbol{C}^T \boldsymbol{A}^{-1}.$
lyche-numerical-linear-algebra- EQ0231 (3.21)	092	■ 0.990 ●	$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_{\{m-1\}} & \boldsymbol{B} \\ \boldsymbol{C}^T & \boldsymbol{A}_{mm} \end{bmatrix}$	$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_{\{m-1\}} & \boldsymbol{B} \\ \boldsymbol{C}^T & \boldsymbol{A}_{mm} \end{bmatrix} = \begin{bmatrix} \boldsymbol{L}_{\{m-1\}} & \boldsymbol{0} \\ \boldsymbol{C}^T \boldsymbol{U}_{\{m-1\}}^{-1} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{U}_{\{m-1\}} & \boldsymbol{L}_{\{m-1\}}^{-1} \boldsymbol{B} \\ 0 & \boldsymbol{A}_{mm} - \boldsymbol{C}^T \boldsymbol{U}_{\{m-1\}}^{-1} \boldsymbol{L}_{\{m-1\}}^{-1} \boldsymbol{B} \end{bmatrix},$	$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_{\{m-1\}} & \boldsymbol{B} \\ \boldsymbol{C}^T & \boldsymbol{A}_{mm} \end{bmatrix} \\ = \begin{bmatrix} \boldsymbol{L}_{\{m-1\}} & \boldsymbol{0} \\ \boldsymbol{C}^T \boldsymbol{U}_{\{m-1\}}^{-1} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{U}_{\{m-1\}} & \boldsymbol{L}_{\{m-1\}}^{-1} \boldsymbol{B} \\ 0 & \boldsymbol{A}_{mm} - \boldsymbol{C}^T \boldsymbol{U}_{\{m-1\}}^{-1} \boldsymbol{L}_{\{m-1\}}^{-1} \boldsymbol{B} \end{bmatrix},$
lyche-numerical-linear-algebra- EQ0293 (5.9)	122	■ 0.990 ● N +1	$\langle \boldsymbol{s}_0, \boldsymbol{s} \rangle = \langle \boldsymbol{v}, \boldsymbol{s} \rangle, \quad \text{for all } \boldsymbol{s} \in \mathcal{S}, \quad \langle \boldsymbol{t}_0, \boldsymbol{t} \rangle = \langle \boldsymbol{v}, \boldsymbol{t} \rangle, \quad \text{for all } \boldsymbol{t} \in \mathcal{T}.$	$\langle \boldsymbol{s}_0, \boldsymbol{s} \rangle = \langle \boldsymbol{v}, \boldsymbol{s} \rangle, \quad \text{for all } \boldsymbol{s} \in \mathcal{S}, \quad \langle \boldsymbol{t}_0, \boldsymbol{t} \rangle = \langle \boldsymbol{v}, \boldsymbol{t} \rangle, \quad \text{for all } \boldsymbol{t} \in \mathcal{T}.$	$\langle \boldsymbol{s}_0, \boldsymbol{s} \rangle = \langle \boldsymbol{v}, \boldsymbol{s} \rangle, \quad \text{for all } \boldsymbol{s} \in \mathcal{S}, \quad \langle \boldsymbol{t}_0, \boldsymbol{t} \rangle = \langle \boldsymbol{v}, \boldsymbol{t} \rangle, \quad \text{for all } \boldsymbol{t} \in \mathcal{T}.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-206 EQ0543	206	0.990 N +0	$\left \left \boldsymbol{A}^{-1}\right \right _2=\max_{\boldsymbol{x}\neq\mathbf{0}}\frac{\left \left \boldsymbol{x}\right \right _2}{\left \left \boldsymbol{A}\boldsymbol{x}\right \right _2}.$	$\left\ \boldsymbol{A}^{-1}\right\ _2=\max_{\boldsymbol{x}\neq\mathbf{0}}\frac{\left\ \boldsymbol{x}\right\ _2}{\left\ \boldsymbol{A}\boldsymbol{x}\right\ _2}.$	$\left\ \boldsymbol{A}^{-1}\right\ _2=\max_{\boldsymbol{x}\neq\mathbf{0}}\frac{\left\ \boldsymbol{x}\right\ _2}{\left\ \boldsymbol{A}\boldsymbol{x}\right\ _2}.$
lyche-numerical-linear-algebra-313 EQ0920 (13.52)	313	0.990 W +0	$\begin{aligned} &-\frac{\partial}{\partial x}\left(c(x,y)\frac{\partial u}{\partial x}\right)-\frac{\partial}{\partial y}\left(c(x,y)\frac{\partial u}{\partial y}\right)=f(x,y),\quad (x,y)\in\Omega=(0,1)^2, \\ &u(x,y)=0,\quad (x,y)\in\partial\Omega. \end{aligned}$	$-\frac{\partial}{\partial x}\left(c(x,y)\frac{\partial u}{\partial x}\right)-\frac{\partial}{\partial y}\left(c(x,y)\frac{\partial u}{\partial y}\right)=f(x,y),\quad (x,y)\in\Omega=(0,1)^2, \\ u(x,y)=0,\quad (x,y)\in\partial\Omega.$	$-\frac{\partial}{\partial x}\left(c(x,y)\frac{\partial u}{\partial x}\right)-\frac{\partial}{\partial y}\left(c(x,y)\frac{\partial u}{\partial y}\right)=f(x,y),\quad (x,y)\in\Omega=(0,1)^2, \\ u(x,y)=0,\quad (x,y)\in\partial\Omega.$
lyche-numerical-linear-algebra-244 EQ0679	244	0.990 N +0	$((\boldsymbol{A}\otimes\boldsymbol{B})(\boldsymbol{C}\otimes\boldsymbol{D}))_{i,j}=\boldsymbol{A}\boldsymbol{C}\sum_{k=1}^tb_{i,k}d_{k,j}=(\boldsymbol{A}\boldsymbol{C})(\boldsymbol{B}\boldsymbol{D})_{i,j}=((\boldsymbol{A}\boldsymbol{C})\otimes(\boldsymbol{B}\boldsymbol{D}))_{i,j}.$	$((\boldsymbol{A}\otimes\boldsymbol{B})(\boldsymbol{C}\otimes\boldsymbol{D}))_{i,j}=\boldsymbol{A}\boldsymbol{C}\sum_{k=1}^tb_{i,k}d_{k,j}=(\boldsymbol{A}\boldsymbol{C})(\boldsymbol{B}\boldsymbol{D})_{i,j}=((\boldsymbol{A}\boldsymbol{C})\otimes(\boldsymbol{B}\boldsymbol{D}))_{i,j}.$	$((\boldsymbol{A}\otimes\boldsymbol{B})(\boldsymbol{C}\otimes\boldsymbol{D}))_{i,j}=\boldsymbol{A}\boldsymbol{C}\sum_{k=1}^tb_{i,k}d_{k,j}=(\boldsymbol{A}\boldsymbol{C})(\boldsymbol{B}\boldsymbol{D})_{i,j}=((\boldsymbol{A}\boldsymbol{C})\otimes(\boldsymbol{B}\boldsymbol{D}))_{i,j}.$
lyche-numerical-linear-algebra-036 EQ0042	036	0.991 N +1	$\begin{aligned} &\left \begin{array}{ccc} 1 & x & y \\ 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \end{array}\right =\left \begin{array}{ccc} 1 & x & y \\ 0 & x_1-x & y_1-y \\ 0 & x_2-x_1 & y_2-y_1 \end{array}\right \\ &=\left \begin{array}{cc} x_1-x & y_1-y \\ x_2-x_1 & y_2-y_1 \end{array}\right =(x_1-x)(y_2-y_1)-(y_1-y)(x_2-x_1). \end{aligned}$	$\left \begin{array}{ccc} 1 & x & y \\ 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \end{array}\right =\left \begin{array}{ccc} 1 & x & y \\ 0 & x_1-x & y_1-y \\ 0 & x_2-x_1 & y_2-y_1 \end{array}\right \\ =\left \begin{array}{cc} x_1-x & y_1-y \\ x_2-x_1 & y_2-y_1 \end{array}\right =(x_1-x)(y_2-y_1)-(y_1-y)(x_2-x_1).$	$\left \begin{array}{ccc} 1 & x & y \\ 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \end{array}\right =\left \begin{array}{ccc} 1 & x & y \\ 0 & x_1-x & y_1-y \\ 0 & x_2-x_1 & y_2-y_1 \end{array}\right \\ =\left \begin{array}{cc} x_1-x & y_1-y \\ x_2-x_1 & y_2-y_1 \end{array}\right =(x_1-x)(y_2-y_1)-(y_1-y)(x_2-x_1).$
lyche-numerical-linear-algebra-087 EQ0213	087	0.991 K +0	$\mathrm{fl}\left(x_2\right)=2,\quad \mathrm{fl}\left(x_1\right)=0.$	$\mathrm{fl}\left(x_2\right)=2,\quad \mathrm{fl}\left(x_1\right)=0.$	$\mathrm{fl}\left(x_2\right)=2,\quad \mathrm{fl}\left(x_1\right)=0.$
lyche-numerical-linear-algebra-094 EQ0234 (3.25)	094	0.991	$\begin{aligned} &1+2+\cdots+m=\frac{1}{2}m(m+1), \\ &1^2+2^2+\cdots+m^2=\frac{1}{3}m(m+\frac{1}{2})(m+1), \\ &1+3+5+\cdots+2m-1=m^2, \\ &1*2+2*3+3*4+\cdots+(m-1)m=\frac{1}{3}(m-1)m(m+1). \end{aligned}$	$\begin{aligned} &1+2+\cdots+m=\frac{1}{2}m(m+1), \\ &1^2+2^2+\cdots+m^2=\frac{1}{3}m\left(m+\frac{1}{2}\right)(m+1), \\ &1+3+5+\cdots+2m-1=m^2, \\ &1*2+2*3+3*4+\cdots+(m-1)m=\frac{1}{3}(m-1)m(m+1). \end{aligned}$	$\begin{aligned} &1+2+\cdots+m=\frac{1}{2}m(m+1), \\ &1^2+2^2+\cdots+m^2=\frac{1}{3}m(m+\frac{1}{2})(m+1), \\ &1+3+5+\cdots+2m-1=m^2, \\ &1*2+2*3+3*4+\cdots+(m-1)m=\frac{1}{3}(m-1)m(m+1). \end{aligned}$
lyche-numerical-linear-algebra-186 EQ0468 (8.2)	186	0.991 N +0	$\left \left \boldsymbol{x}\right \right _p:=\left \left \boldsymbol{x}\right \right _{\infty}^{\frac{1}{p}}.$	$\left\ \boldsymbol{x}\right\ _p:=\left\ \boldsymbol{x}\right\ _{\infty}^{\left(\sum_{j=1}^n\left(\frac{\left x_j\right }{\left\ \boldsymbol{x}\right\ _{\infty}}\right)^p\right)^{1/p}}.$	$\left\ \boldsymbol{x}\right\ _p:=\left\ \boldsymbol{x}\right\ _{\infty}^{\left(\sum_{j=1}^n\left(\frac{\left x_j\right }{\left\ \boldsymbol{x}\right\ _{\infty}}\right)^p\right)^{1/p}}.$

Conf. MathPix confidence:

■

≥0.9

■

0.5–0.9

■

<0.5

—

no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ
Residual render vs scan (inkdrill):

●

C component

●

W weak

●

S stable

●

N noise

●

K clean

●

not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0584 (9.1)	216	0.991 W +0	$\boldsymbol{A} \boldsymbol{x} = \left[\begin{array}{c} p \left(t_1 \right) \\\vdots \\ p \left(t_m \right) \end{array} \right] \\\vdots \\ \boldsymbol{p} \left(t_m \right) \end{array} \right] = \left[\begin{array}{ccc} \phi_1 \left(t_1 \right) & \cdots & \phi_n \left(t_1 \right) \\\vdots & & \vdots \\ \phi_1 \left(t_m \right) & \cdots & \phi_n \left(t_m \right) \end{array} \right] \left[\begin{array}{c} x_1 \\\vdots \\ x_n \end{array} \right] = \left[\begin{array}{c} y_1 \\\vdots \\ y_m \end{array} \right] =: \boldsymbol{b}.$	$\boldsymbol{Ax} = \begin{bmatrix} p(t_1) \\ \vdots \\ p(t_m) \end{bmatrix} = \begin{bmatrix} \phi_1(t_1) & \cdots & \phi_n(t_1) \\ \vdots & & \vdots \\ \phi_1(t_m) & \cdots & \phi_n(t_m) \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} =: \boldsymbol{b}.$	
lyche-numerical-linear-algebra EQ0725	258	0.991 K +0	$\boldsymbol{F}_4 \boldsymbol{P}_4 = \left[\begin{array}{c c} \boldsymbol{F}_2 & \boldsymbol{D}_2 \boldsymbol{F}_2 \\\hline \boldsymbol{F}_2 & -\boldsymbol{D}_2 \boldsymbol{F}_2 \end{array} \right].$	$\boldsymbol{F}_4 \boldsymbol{P}_4 = \left[\begin{array}{c c} \boldsymbol{F}_2 & \boldsymbol{D}_2 \boldsymbol{F}_2 \\\hline \boldsymbol{F}_2 & -\boldsymbol{D}_2 \boldsymbol{F}_2 \end{array} \right].$	
lyche-numerical-linear-algebra EQ0169	074	0.991 N +0	$\boldsymbol{C} \boldsymbol{A} \boldsymbol{B} = \left[\begin{array}{cc} \lambda & \boldsymbol{a}^T \boldsymbol{B}_1 \\ \boldsymbol{0} & \boldsymbol{C}_1 \boldsymbol{A}_1 \boldsymbol{B}_1 \end{array} \right].$	$\boldsymbol{CAB} = \left[\begin{array}{cc} \lambda & \boldsymbol{a}^T \boldsymbol{B}_1 \\ \boldsymbol{0} & \boldsymbol{C}_1 \boldsymbol{A}_1 \boldsymbol{B}_1 \end{array} \right].$	
lyche-numerical-linear-algebra EQ0047 (1.15)	037	0.991 K +0	$\operatorname{det}(\boldsymbol{C}(\boldsymbol{i}, \boldsymbol{j})) = \sum_{\boldsymbol{k}} \operatorname{det}(\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{k})) \operatorname{det}(\boldsymbol{B}(\boldsymbol{k}, \boldsymbol{j})),$	$\det\bigl(\boldsymbol{C}(\boldsymbol{i}, \boldsymbol{j})\bigr) = \sum_k \det\bigl(\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{k})\bigr) \det\bigl(\boldsymbol{B}(\boldsymbol{k}, \boldsymbol{j})\bigr),$	
lyche-numerical-linear-algebra EQ0422	170	0.992 N +0	$\begin{aligned} \boldsymbol{U} \boldsymbol{\Sigma} &= \left[\begin{array}{c} \boldsymbol{U} \left[\begin{array}{c} \sigma_1 \\ \vdots \\ \sigma_r \\ 0 \\ \vdots \\ 0 \end{array} \right] \\ \vdots \\ \boldsymbol{U} \left[\begin{array}{c} \sigma_1 \\ \vdots \\ \sigma_r \\ 0 \\ \vdots \\ 0 \end{array} \right] \end{array} \right] \\ &= \left[\begin{array}{c} \boldsymbol{U} \left[\begin{array}{c} \sigma_1 \\ \vdots \\ \sigma_r \\ 0 \\ \vdots \\ 0 \end{array} \right] \\ \vdots \\ \boldsymbol{U} \left[\begin{array}{c} \sigma_1 \\ \vdots \\ \sigma_r \\ 0 \\ \vdots \\ 0 \end{array} \right] \end{array} \right] \\ &= \left[\begin{array}{c} \boldsymbol{U} \left[\begin{array}{c} \sigma_1 \\ \vdots \\ \sigma_r \\ 0 \\ \vdots \\ 0 \end{array} \right] \\ \vdots \\ \boldsymbol{U} \left[\begin{array}{c} \sigma_1 \\ \vdots \\ \sigma_r \\ 0 \\ \vdots \\ 0 \end{array} \right] \end{array} \right] \end{aligned}$	$\begin{aligned} \boldsymbol{U} \boldsymbol{\Sigma} &= \boldsymbol{U} [\sigma_1 \boldsymbol{e}_1, \dots, \sigma_r \boldsymbol{e}_r, 0, \dots, 0] \\ &= [\sigma_1 \boldsymbol{u}_1, \dots, \sigma_r \boldsymbol{u}_r, 0, \dots, 0] \\ &= [\boldsymbol{A} \boldsymbol{v}_1, \dots, \boldsymbol{A} \boldsymbol{v}_n]. \end{aligned}$	
lyche-numerical-linear-algebra EQ0702	252	0.992	$\boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = h^2 \boldsymbol{F} \quad \text{with} \quad h = 1/(m+1),$	$\boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = h^2 \boldsymbol{F} \quad \text{with} \quad h = 1/(m+1),$	
lyche-numerical-linear-algebra EQ0857 (13.19)	299	0.992 N +0	$\frac{\ \boldsymbol{x} - \boldsymbol{x}_k\ _{\boldsymbol{A}}}{\ \boldsymbol{x} - \boldsymbol{x}_0\ _{\boldsymbol{A}}} \leq \left(\frac{\kappa - 1}{\kappa + 1} \right)^k < e^{-\frac{2}{\kappa} k}, \quad k > 0,$	$\frac{\ \boldsymbol{x} - \boldsymbol{x}_k\ _{\boldsymbol{A}}}{\ \boldsymbol{x} - \boldsymbol{x}_0\ _{\boldsymbol{A}}} \leq \left(\frac{\kappa - 1}{\kappa + 1} \right)^k < e^{-\frac{2}{\kappa} k}, \quad k > 0,$	
Conf. MathPix confidence: 0.991 0.5–0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0687 (4h)	246	0.992 N +0	$\left(\boldsymbol{s}_{\textit{j}}\otimes\boldsymbol{s}_{\textit{k}}\right)^{\textit{T}}\left(\boldsymbol{s}_{\textit{p}}\otimes\boldsymbol{s}_{\textit{q}}\right)=\left(\boldsymbol{s}_{\textit{j}}^{\textit{T}}\otimes\boldsymbol{s}_{\textit{k}}^{\textit{T}}\right)\left(\boldsymbol{s}_{\textit{p}}\otimes\boldsymbol{s}_{\textit{q}}\right)=\left(\boldsymbol{s}_{\textit{j}}^{\textit{T}}\boldsymbol{s}_{\textit{p}}\right)\otimes\left(\boldsymbol{s}_{\textit{k}}^{\textit{T}}\boldsymbol{s}_{\textit{q}}\right)=\frac{1}{4h^2}\delta_{j,p}\delta_{k,q}$	$\left(s_j\otimes s_k\right)^T\left(s_p\otimes s_q\right)=\left(s_j^T\otimes s_k^T\right)\left(s_p\otimes s_q\right)=\left(s_j^Ts_p\right)\otimes\left(s_k^Ts_q\right)=\frac{1}{4h^2}\delta_{j,p}\delta_{k,q}$	
lyche-numerical-linear-algebra-EQ0775 (12.20)	274	0.992 N +0	$\left \left \boldsymbol{x}_{\textit{k}}-\boldsymbol{x}\right \right _2\leq\left(\frac{\kappa-1}{\kappa+1}\right)^k\left\ \boldsymbol{x}_0-\boldsymbol{x}\right\ _2,\quad k=0,1,2,\ldots$	$\ \boldsymbol{x}_k-\boldsymbol{x}\ _2\leq\left(\frac{\kappa-1}{\kappa+1}\right)^k\ \boldsymbol{x}_0-\boldsymbol{x}\ _2,\quad k=0,1,2,\ldots$	
lyche-numerical-linear-algebra-EQ0904	309	0.992 N +0	$\boldsymbol{\epsilon}_{\textit{j}}=\boldsymbol{x}_{\textit{m}+1}-\boldsymbol{x}_{\textit{j}}=\boldsymbol{x}_{\textit{m}}-\boldsymbol{x}_{\textit{j}}+\alpha_{\textit{m}}\boldsymbol{p}_{\textit{m}}=\boldsymbol{x}_{\textit{m}-1}-\boldsymbol{x}_{\textit{j}}+\alpha_{\textit{m}-1}\boldsymbol{p}_{\textit{m}-1}+\alpha_{\textit{m}}\boldsymbol{p}_{\textit{m}}=\ldots$	$\boldsymbol{\epsilon}_{\textit{j}}=\boldsymbol{x}_{\textit{m}+1}-\boldsymbol{x}_{\textit{j}}=\boldsymbol{x}_{\textit{m}}-\boldsymbol{x}_{\textit{j}}+\alpha_{\textit{m}}\boldsymbol{p}_{\textit{m}}=\boldsymbol{x}_{\textit{m}-1}-\boldsymbol{x}_{\textit{j}}+\alpha_{\textit{m}-1}\boldsymbol{p}_{\textit{m}-1}+\alpha_{\textit{m}}\boldsymbol{p}_{\textit{m}}=\ldots$	
lyche-numerical-linear-algebra-EQ0462	183	0.992 K +0	$\boldsymbol{A}=\left[\begin{array}{cc}3&1\\2&3\\-1&5\end{array}\right].$	$\boldsymbol{A}=\left[\begin{array}{cc}3&1\\2&3\\-1&5\end{array}\right].$	
lyche-numerical-linear-algebra-EQ0015 (1.3)	026	0.993 N +0	$c_{\textit{1}}\boldsymbol{x}_{\textit{1}}+\cdots+c_{\textit{n}}\boldsymbol{x}_{\textit{n}}=\boldsymbol{\mathbf{0}}\quad\Longrightarrow\quad c_{\textit{1}}=\cdots=c_{\textit{n}}=0.$	$c_1\boldsymbol{x}_1+\cdots+c_n\boldsymbol{x}_n=\boldsymbol{0}\quad\Longrightarrow\quad c_1=\cdots=c_n=0.$	
lyche-numerical-linear-algebra-EQ0248 (4.3)	101	0.993 W +0	$\begin{aligned}\boldsymbol{A}&=\left[\begin{array}{cc}\boldsymbol{L}_{\textit{n}-1}&\boldsymbol{0}\\\boldsymbol{l}_n^*&1\end{array}\right]\left[\begin{array}{cc}\boldsymbol{D}_{\textit{n}-1}&\boldsymbol{0}\\0&d_{nn}\end{array}\right]\left[\begin{array}{cc}\boldsymbol{L}_{\textit{n}-1}^*&l_n\\\boldsymbol{0}^*&1\end{array}\right] \\&=\left[\begin{array}{cc}\boldsymbol{L}_{\textit{n}-1}\boldsymbol{D}_{\textit{n}-1}\boldsymbol{L}_{\textit{n}-1}^*&\boldsymbol{L}_{\textit{n}-1}\boldsymbol{D}_{\textit{n}-1}l_n\\\boldsymbol{l}_n^*\boldsymbol{D}_{\textit{n}-1}\boldsymbol{L}_{\textit{n}-1}^*&\boldsymbol{l}_n^*\boldsymbol{D}_{\textit{n}-1}l_n+d_{nn}\end{array}\right]\end{aligned}$	$\begin{aligned}\boldsymbol{A}&=\left[\begin{array}{cc}\boldsymbol{A}_{[\textit{n}-1]}&\boldsymbol{a}_n\\\boldsymbol{a}_n^*&a_{nn}\end{array}\right]=\left[\begin{array}{cc}\boldsymbol{L}_{\textit{n}-1}&\boldsymbol{0}\\\boldsymbol{l}_n^*&1\end{array}\right]\left[\begin{array}{cc}\boldsymbol{D}_{\textit{n}-1}&\boldsymbol{0}\\0&d_{nn}\end{array}\right]\left[\begin{array}{cc}\boldsymbol{L}_{\textit{n}-1}^*&l_n\\\boldsymbol{0}^*&1\end{array}\right] \\&=\left[\begin{array}{cc}\boldsymbol{L}_{\textit{n}-1}\boldsymbol{D}_{\textit{n}-1}\boldsymbol{L}_{\textit{n}-1}^*&\boldsymbol{L}_{\textit{n}-1}\boldsymbol{D}_{\textit{n}-1}l_n\\\boldsymbol{l}_n^*\boldsymbol{D}_{\textit{n}-1}\boldsymbol{L}_{\textit{n}-1}^*&\boldsymbol{l}_n^*\boldsymbol{D}_{\textit{n}-1}l_n+d_{nn}\end{array}\right]\end{aligned}$	
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0583	216	0.993	$\boldsymbol{A} \, \boldsymbol{x} = \left[\begin{array}{c} \boldsymbol{u}_1^* \\ \boldsymbol{u}_2^* \\ \vdots \\ \boldsymbol{u}_m^* \end{array} \right] \boldsymbol{x} = \left[\begin{array}{c} y_1 \\ y_2 \\ \vdots \\ y_m \end{array} \right] = \boldsymbol{b}.$	$\boldsymbol{A} \boldsymbol{x} = \left[\begin{array}{c} \boldsymbol{u}_1^* \\ \boldsymbol{u}_2^* \\ \vdots \\ \boldsymbol{u}_m^* \end{array} \right] \boldsymbol{x} = \left[\begin{array}{c} y_1 \\ y_2 \\ \vdots \\ y_m \end{array} \right] = \boldsymbol{b}.$	
lyche-numerical-linear-algebra-EQ0432	173	0.993 ● N +0	$\boldsymbol{\Sigma} = \left[\begin{array}{cc} \boldsymbol{\Sigma}_1 & 0 \\ 0 & 0 \\ 0 & 0 \end{array} \right], \quad \boldsymbol{\Sigma}_1 = [2], \quad \boldsymbol{V} = \frac{1}{\sqrt{2}} \left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array} \right].$	$\boldsymbol{\Sigma} = \left[\begin{array}{cc} \boldsymbol{\Sigma}_1 & 0 \\ 0 & 0 \\ 0 & 0 \end{array} \right], \quad \boldsymbol{\Sigma}_1 = [2], \quad \boldsymbol{V} = \frac{1}{\sqrt{2}} \left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array} \right].$	
lyche-numerical-linear-algebra-EQ0769	273	0.993 ● N +0	$\left \left \boldsymbol{x}_k - \boldsymbol{x} \right \right = \left \left \boldsymbol{G}^k (\boldsymbol{x}_0 - \boldsymbol{x}) \right \right \leq \left \left \boldsymbol{G}^k \right \right \left \left \boldsymbol{x}_0 - \boldsymbol{x} \right \right \leq \left \left \boldsymbol{G} \right \right ^k \left \left \boldsymbol{x}_0 - \boldsymbol{x} \right \right \rightarrow 0, \quad k \rightarrow \infty.$	$\left\ \boldsymbol{x}_k - \boldsymbol{x} \right\ = \left\ \boldsymbol{G}^k (\boldsymbol{x}_0 - \boldsymbol{x}) \right\ \leq \left\ \boldsymbol{G}^k \right\ \left\ \boldsymbol{x}_0 - \boldsymbol{x} \right\ \leq \left\ \boldsymbol{G} \right\ ^k \left\ \boldsymbol{x}_0 - \boldsymbol{x} \right\ \rightarrow 0, \quad k \rightarrow \infty.$	
lyche-numerical-linear-algebra-EQ0863	300	0.993 ● N -1	$\mathbb{W}_k = \text{span}(\boldsymbol{r}_0, \boldsymbol{A} \boldsymbol{r}_0, \boldsymbol{A}^2 \boldsymbol{r}_0, \dots, \boldsymbol{A}^{k-1} \boldsymbol{r}_0), \quad k = 1, 2, 3, \dots.$	$\mathbb{W}_k = \text{span}(\boldsymbol{r}_0, \boldsymbol{A} \boldsymbol{r}_0, \boldsymbol{A}^2 \boldsymbol{r}_0, \dots, \boldsymbol{A}^{k-1} \boldsymbol{r}_0), \quad k = 1, 2, 3, \dots.$	
lyche-numerical-linear-algebra-EQ0279	117	0.993 ● N +0	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{x}^T \overline{\boldsymbol{y}} = \sum_{j=1}^n x_j \overline{y_j}.$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{x}^T \overline{\boldsymbol{y}} = \sum_{j=1}^n x_j \overline{y_j}.$	
lyche-numerical-linear-algebra-EQ0414	165	0.994 ● N +1	$R(\boldsymbol{x}) := \frac{\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^T \boldsymbol{x}}.$	$R(\boldsymbol{x}) := \frac{\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^T \boldsymbol{x}}.$	
lyche-numerical-linear-algebra-EQ0802	281	0.994 ● W +0	$\left\ \boldsymbol{A} \right\ = \left\ \boldsymbol{D}_t \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} \boldsymbol{D}_t^{-1} \right\ _1 = \left\ \boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1} \right\ _1 = \max_{1 \leq j \leq n} \sum_{i=1}^n \left (\boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1})_{ij} \right $ $\leq \max_{1 \leq j \leq n} (\lambda_j + \epsilon) = \rho(\boldsymbol{A}) + \epsilon.$	$\left\ \boldsymbol{A} \right\ = \left\ \boldsymbol{D}_t \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} \boldsymbol{D}_t^{-1} \right\ _1 = \left\ \boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1} \right\ _1 = \max_{1 \leq j \leq n} \sum_{i=1}^n \left (\boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1})_{ij} \right $ $\leq \max_{1 \leq j \leq n} (\lambda_j + \epsilon) = \rho(\boldsymbol{A}) + \epsilon.$	
Conf. MathPix confidence: 0.993 0.5–0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0175 (3.1)	077	0.994	$\boldsymbol{A}^{(k)}=\left[\begin{array}{cccccc} a_{1,1}^{(1)} & \cdots & a_{1,k-1}^{(1)} & a_{1,k}^{(1)} & \cdots & a_{1,j}^{(1)} & \cdots & a_{1,n}^{(1)} \\ & \ddots & \vdots & \vdots & & \vdots & & \vdots \\ & & a_{k-1,k-1}^{(k-1)} & a_{k-1,k}^{(k-1)} & \cdots & a_{k-1,j}^{(k-1)} & \cdots & a_{k-1,n}^{(k-1)} \\ \hline & & & a_{k,k}^{(k)} & \cdots & a_{k,j}^{(k)} & \cdots & a_{k,n}^{(k)} \\ & & & \vdots & & \vdots & & \vdots \\ & & & a_{i,k}^{(k)} & \cdots & a_{i,j}^{(k)} & \cdots & a_{i,n}^{(k)} \\ & & & \vdots & & \vdots & & \vdots \\ & & & a_{n,k}^{(k)} & \cdots & a_{n,j}^{(k)} & \cdots & a_{n,n}^{(k)} \end{array}\right].$		
lyche-numerical-linear-algebra- EQ0833	288	0.994 K +0	$A:=\left[\begin{array}{lll} 3 & 0 & 1 \\ 0 & 7 & 2 \\ 1 & 2 & 4 \end{array}\right]$	$A:=\left[\begin{array}{lll} 3 & 0 & 1 \\ 0 & 7 & 2 \\ 1 & 2 & 4 \end{array}\right]$	$A:=\left[\begin{array}{lll} 3 & 0 & 1 \\ 0 & 7 & 2 \\ 1 & 2 & 4 \end{array}\right]$
lyche-numerical-linear-algebra- EQ0992	335	0.994 K +0	$\boldsymbol{Q}_{n-1}=\boldsymbol{I} \text{ and } \boldsymbol{Q}_k=\boldsymbol{H}_k \boldsymbol{Q}_{k+1} \text{ for } k=n-2, n-3, \ldots, 1.$	$\boldsymbol{Q}_{n-1}=\boldsymbol{I} \text{ and } \boldsymbol{Q}_k=\boldsymbol{H}_k \boldsymbol{Q}_{k+1} \text{ for } k=n-2, n-3, \ldots, 1.$	$\boldsymbol{Q}_{n-1}=\boldsymbol{I} \text{ and } \boldsymbol{Q}_k=\boldsymbol{H}_k \boldsymbol{Q}_{k+1} \text{ for } k=n-2, n-3, \ldots, 1.$
lyche-numerical-linear-algebra- EQ1019 (15.2)	345	0.994 W +0	$\begin{aligned} & \text{(i) } \lambda_1 > \lambda_2 \geq \lambda_3 \geq \cdots \geq \lambda_n , \\ & \text{(ii) } z_0^T v_1 \neq 0 \end{aligned}$	$\begin{aligned} & \text{(i) } \lambda_1 > \lambda_2 \geq \lambda_3 \geq \cdots \geq \lambda_n , \\ & \text{(ii) } z_0^T v_1 \neq 0 \end{aligned}$	$\begin{aligned} & \text{(i) } \lambda_1 > \lambda_2 \geq \lambda_3 \geq \cdots \geq \lambda_n , \\ & \text{(ii) } z_0^T v_1 \neq 0 \end{aligned}$
lyche-numerical-linear-algebra- EQ0124	061	0.995 W +0	$\begin{aligned} & \left(\boldsymbol{T}_1 \boldsymbol{s}_j \right)_k = \sum_{l=1}^m t_{k,l} \sin(lj\pi h) \\ & = a[\sin((k-1)j\pi h) + \sin((k+1)j\pi h)] + d \sin(kj\pi h) \\ & = 2a \cos(j\pi h) \sin(kj\pi h) + d \sin(kj\pi h) = \lambda_j s_{k,j}. \end{aligned}$	$\begin{aligned} & \left(\boldsymbol{T}_1 \boldsymbol{s}_j \right)_k = \sum_{l=1}^m t_{k,l} \sin(lj\pi h) \\ & = a[\sin((k-1)j\pi h) + \sin((k+1)j\pi h)] + d \sin(kj\pi h) \\ & = 2a \cos(j\pi h) \sin(kj\pi h) + d \sin(kj\pi h) = \lambda_j s_{k,j}. \end{aligned}$	$\begin{aligned} & \left(\boldsymbol{T}_1 \boldsymbol{s}_j \right)_k = \sum_{l=1}^m t_{k,l} \sin(lj\pi h) \\ & = a[\sin((k-1)j\pi h) + \sin((k+1)j\pi h)] + d \sin(kj\pi h) \\ & = 2a \cos(j\pi h) \sin(kj\pi h) + d \sin(kj\pi h) = \lambda_j s_{k,j}. \end{aligned}$
lyche-numerical-linear-algebra- EQ0384 (6.9)	155	0.995 N +0	$R_{\boldsymbol{A}}(\boldsymbol{x})=\frac{\sum_{i=1}^n \lambda_i \left c_i \right ^2}{\sum_{j=1}^n \left c_j \right ^2}, \quad \boldsymbol{x} \neq \mathbf{0}, \quad \boldsymbol{x}=\sum_{j=1}^n c_j \boldsymbol{u}_j.$	$R_{\boldsymbol{A}}(\boldsymbol{x})=\frac{\sum_{i=1}^n \lambda_i \left c_i \right ^2}{\sum_{j=1}^n \left c_j \right ^2}, \quad \boldsymbol{x} \neq \mathbf{0}, \quad \boldsymbol{x}=\sum_{j=1}^n c_j \boldsymbol{u}_j.$	$R_{\boldsymbol{A}}(\boldsymbol{x})=\frac{\sum_{i=1}^n \lambda_i \left c_i \right ^2}{\sum_{j=1}^n \left c_j \right ^2}, \quad \boldsymbol{x} \neq \mathbf{0}, \quad \boldsymbol{x}=\sum_{j=1}^n c_j \boldsymbol{u}_j.$

Conf. MathPix confidence:

≥0.9

0.5–0.9

<0.5

— no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ
Residual render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

61

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0632	229	0.995	$\begin{gathered} \lambda_1 \geq \cdots \geq \lambda_{m+n} = \alpha_1 \geq \cdots \geq \alpha_t \geq 0 = \cdots = 0 \geq -\alpha_t \geq \cdots \geq -\alpha_1, \\ \mu_1 \geq \cdots \geq \mu_{m+n} = \beta_1 \geq \cdots \geq \beta_t \geq 0 = \cdots = 0 \geq -\beta_t \geq \cdots \geq -\beta_1. \end{gathered}$	$\lambda_1 \geq \cdots \geq \lambda_{m+n} = \alpha_1 \geq \cdots \geq \alpha_t \geq 0 = \cdots = 0 \geq -\alpha_t \geq \cdots \geq -\alpha_1, \\ \mu_1 \geq \cdots \geq \mu_{m+n} = \beta_1 \geq \cdots \geq \beta_t \geq 0 = \cdots = 0 \geq -\beta_t \geq \cdots \geq -\beta_1.$	$\lambda_1 \geq \cdots \geq \lambda_{m+n} = \alpha_1 \geq \cdots \geq \alpha_t \geq 0 = \cdots = 0 \geq -\alpha_t \geq \cdots \geq -\alpha_1, \\ \mu_1 \geq \cdots \geq \mu_{m+n} = \beta_1 \geq \cdots \geq \beta_t \geq 0 = \cdots = 0 \geq -\beta_t \geq \cdots \geq -\beta_1.$
lyche-numerical-linear-algebra- EQ0647	233	0.995 K +0	$\boldsymbol{A}^{\dagger} = \frac{1}{\alpha} \boldsymbol{A}^*, \quad \alpha = \ \boldsymbol{u}\ _2^2 \ \boldsymbol{v}\ _2^2.$	$\boldsymbol{A}^{\dagger} = \frac{1}{\alpha} \boldsymbol{A}^*, \quad \alpha = \ \boldsymbol{u}\ _2^2 \ \boldsymbol{v}\ _2^2.$	$\boldsymbol{A}^{\dagger} = \frac{1}{\alpha} \boldsymbol{A}^*, \quad \alpha = \ \boldsymbol{u}\ _2^2 \ \boldsymbol{v}\ _2^2.$
lyche-numerical-linear-algebra- EQ0881 (13.31)	304	0.995 N +0	$x_{n+1} - 2tx_n + x_{n-1} = 0 \text{ for } n \geq 1, \text{ with } x_0 = 1, x_1 = t.$	$x_{n+1} - 2tx_n + x_{n-1} = 0 \text{ for } n \geq 1, \text{ with } x_0 = 1, x_1 = t.$	$x_{n+1} - 2tx_n + x_{n-1} = 0 \text{ for } n \geq 1, \text{ with } x_0 = 1, x_1 = t.$
lyche-numerical-linear-algebra- EQ0301	124	0.995 N +0	$\left(\boldsymbol{U}^* \boldsymbol{U} \right)_{i,j} = \boldsymbol{e}_i^* \boldsymbol{U}^* \boldsymbol{U} \boldsymbol{e}_j = (\boldsymbol{U} \boldsymbol{e}_i)^* (\boldsymbol{U} \boldsymbol{e}_j) = \langle \boldsymbol{U} \boldsymbol{e}_j, \boldsymbol{U} \boldsymbol{e}_i \rangle = \langle \boldsymbol{e}_j, \boldsymbol{e}_i \rangle = \boldsymbol{e}_i^* \boldsymbol{e}_j,$	$(\boldsymbol{U}^* \boldsymbol{U})_{i,j} = \boldsymbol{e}_i^* \boldsymbol{U}^* \boldsymbol{U} \boldsymbol{e}_j = (\boldsymbol{U} \boldsymbol{e}_i)^* (\boldsymbol{U} \boldsymbol{e}_j) = \langle \boldsymbol{U} \boldsymbol{e}_j, \boldsymbol{U} \boldsymbol{e}_i \rangle = \langle \boldsymbol{e}_j, \boldsymbol{e}_i \rangle = \boldsymbol{e}_i^* \boldsymbol{e}_j,$	$(\boldsymbol{U}^* \boldsymbol{U})_{i,j} = \boldsymbol{e}_i^* \boldsymbol{U}^* \boldsymbol{U} \boldsymbol{e}_j = (\boldsymbol{U} \boldsymbol{e}_i)^* (\boldsymbol{U} \boldsymbol{e}_j) = \langle \boldsymbol{U} \boldsymbol{e}_j, \boldsymbol{U} \boldsymbol{e}_i \rangle = \langle \boldsymbol{e}_j, \boldsymbol{e}_i \rangle = \boldsymbol{e}_i^* \boldsymbol{e}_j,$
lyche-numerical-linear-algebra- EQ0311	126	0.995 S -1	$\boldsymbol{H} \boldsymbol{x} = \boldsymbol{a} \boldsymbol{e}_1, \quad \boldsymbol{a} = -\rho \ \boldsymbol{x}\ _2, \quad \rho := \begin{cases} x_1/ x_1 , & \text{if } x_1 \neq 0, \\ 1, & \text{otherwise.} \end{cases}$	$\boldsymbol{H} \boldsymbol{x} = \boldsymbol{a} \boldsymbol{e}_1, \quad \boldsymbol{a} = -\rho \ \boldsymbol{x}\ _2, \quad \rho := \begin{cases} x_1/ x_1 , & \text{if } x_1 \neq 0, \\ 1, & \text{otherwise.} \end{cases}$	$\boldsymbol{H} \boldsymbol{x} = \boldsymbol{a} \boldsymbol{e}_1, \quad \boldsymbol{a} = -\rho \ \boldsymbol{x}\ _2, \quad \rho := \begin{cases} x_1/ x_1 , & \text{if } x_1 \neq 0, \\ 1, & \text{otherwise.} \end{cases}$
lyche-numerical-linear-algebra- EQ0013	025	0.995 N +0	$(\boldsymbol{f} + \boldsymbol{g})(t) := \boldsymbol{f}(t) + \boldsymbol{g}(t), \quad (c\boldsymbol{f})(t) := c\boldsymbol{f}(t), \quad t \in \mathcal{D}, \quad c \in \mathbb{R}.$	$(\boldsymbol{f} + \boldsymbol{g})(t) := \boldsymbol{f}(t) + \boldsymbol{g}(t), \quad (c\boldsymbol{f})(t) := c\boldsymbol{f}(t), \quad t \in \mathcal{D}, \quad c \in \mathbb{R}.$	$(\boldsymbol{f} + \boldsymbol{g})(t) := \boldsymbol{f}(t) + \boldsymbol{g}(t), \quad (c\boldsymbol{f})(t) := c\boldsymbol{f}(t), \quad t \in \mathcal{D}, \quad c \in \mathbb{R}.$
lyche-numerical-linear-algebra- EQ0061 (1.20)	041	0.995 K +0	$\lambda := \boldsymbol{a}^T \boldsymbol{C} \boldsymbol{e}_i \neq 0,$	$\lambda := \boldsymbol{a}^T \boldsymbol{C} \boldsymbol{e}_i \neq 0,$	$\lambda := \boldsymbol{a}^T \boldsymbol{C} \boldsymbol{e}_i \neq 0,$
lyche-numerical-linear-algebra- EQ0216	088	0.995 N +2	$x_2 = \frac{3.9997}{1.9999} \approx 2, \quad x_1 = 3 - x_2 \approx 1.$	$x_2 = \frac{3.9997}{1.9999} \approx 2, \quad x_1 = 3 - x_2 \approx 1.$	$x_2 = \frac{3.9997}{1.9999} \approx 2, \quad x_1 = 3 - x_2 \approx 1.$
lyche-numerical-linear-algebra- EQ0284	118	0.995 W +0	$\begin{aligned} \ \boldsymbol{x}\ ^2 &= \langle \boldsymbol{x}, \boldsymbol{x} \rangle = \langle \boldsymbol{z} + \boldsymbol{a} \boldsymbol{y}, \boldsymbol{z} + \boldsymbol{a} \boldsymbol{y} \rangle \\ &\stackrel{(5.4)}{=} \langle \boldsymbol{z}, \boldsymbol{z} \rangle + \langle \boldsymbol{a} \boldsymbol{y}, \boldsymbol{a} \boldsymbol{y} \rangle \stackrel{(5.1)}{=} \ \boldsymbol{z}\ ^2 + \boldsymbol{a} ^2 \ \boldsymbol{y}\ ^2 \\ &\geq \boldsymbol{a} ^2 \ \boldsymbol{y}\ ^2 = \frac{ \langle \boldsymbol{x}, \boldsymbol{y} \rangle ^2}{\ \boldsymbol{y}\ ^2}. \end{aligned}$	$\ \boldsymbol{x}\ ^2 = \langle \boldsymbol{x}, \boldsymbol{x} \rangle = \langle \boldsymbol{z} + \boldsymbol{a} \boldsymbol{y}, \boldsymbol{z} + \boldsymbol{a} \boldsymbol{y} \rangle \\ \stackrel{(5.4)}{=} \langle \boldsymbol{z}, \boldsymbol{z} \rangle + \langle \boldsymbol{a} \boldsymbol{y}, \boldsymbol{a} \boldsymbol{y} \rangle \stackrel{(5.1)}{=} \ \boldsymbol{z}\ ^2 + \boldsymbol{a} ^2 \ \boldsymbol{y}\ ^2 \\ \geq \boldsymbol{a} ^2 \ \boldsymbol{y}\ ^2 = \frac{ \langle \boldsymbol{x}, \boldsymbol{y} \rangle ^2}{\ \boldsymbol{y}\ ^2}.$	$\ \boldsymbol{x}\ ^2 = \langle \boldsymbol{x}, \boldsymbol{x} \rangle = \langle \boldsymbol{z} + \boldsymbol{a} \boldsymbol{y}, \boldsymbol{z} + \boldsymbol{a} \boldsymbol{y} \rangle \\ \stackrel{(5.4)}{=} \langle \boldsymbol{z}, \boldsymbol{z} \rangle + \langle \boldsymbol{a} \boldsymbol{y}, \boldsymbol{a} \boldsymbol{y} \rangle \stackrel{(5.1)}{=} \ \boldsymbol{z}\ ^2 + \boldsymbol{a} ^2 \ \boldsymbol{y}\ ^2 \\ \geq \boldsymbol{a} ^2 \ \boldsymbol{y}\ ^2 = \frac{ \langle \boldsymbol{x}, \boldsymbol{y} \rangle ^2}{\ \boldsymbol{y}\ ^2}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0374	152	0.995 N -1	$\boldsymbol{W}=\left[\begin{array}{c c} 1 & \\ \hline \mathbf{0}^*& \boldsymbol{W}_1\end{array}\right] \text{ and } \boldsymbol{U}=\boldsymbol{V}\boldsymbol{W}.$	$\boldsymbol{W}=\left[\begin{array}{c c} 1 & \mathbf{0}^* \\ \hline \mathbf{0} & \boldsymbol{W}_1\end{array}\right] \text{ and } \boldsymbol{U}=\boldsymbol{V}\boldsymbol{W}.$	$\boldsymbol{W}=\left[\begin{array}{c c} 1 & \mathbf{0}^* \\ \hline \mathbf{0} & \boldsymbol{W}_1\end{array}\right] \text{ and } \boldsymbol{U}=\boldsymbol{V}\boldsymbol{W}.$
lyche-numerical-linear-algebra-EQ0858 (13.20)	299	0.995	$\frac{\left\ \boldsymbol{x}-\boldsymbol{x}_k\right\ _{\boldsymbol{A}}}{\left\ \boldsymbol{x}-\boldsymbol{x}_0\right\ _{\boldsymbol{A}}} \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k < 2 e^{-\frac{2}{\sqrt{\kappa}} k}, \quad k \geq 0 .$	$\frac{\left\ \boldsymbol{x}-\boldsymbol{x}_k\right\ _{\boldsymbol{A}}}{\left\ \boldsymbol{x}-\boldsymbol{x}_0\right\ _{\boldsymbol{A}}} \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k < 2 e^{-\frac{2}{\sqrt{\kappa}} k}, \quad k \geq 0 .$	$\frac{\left\ \boldsymbol{x}-\boldsymbol{x}_k\right\ _{\boldsymbol{A}}}{\left\ \boldsymbol{x}-\boldsymbol{x}_0\right\ _{\boldsymbol{A}}} \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k < 2 e^{-\frac{2}{\sqrt{\kappa}} k}, \quad k \geq 0 .$
lyche-numerical-linear-algebra-EQ0459	182	0.995 K +0	$\begin{aligned} & x_1-x_3=4, \\ & x_1+x_2+x_3=12 . \end{aligned}$	$x_1-x_3=4, \\ x_1+x_2+x_3=12.$	$x_1-x_3=4, \\ x_1+x_2+x_3=12.$
lyche-numerical-linear-algebra-EQ0982	331	0.995 N -1	$1=\left\ \boldsymbol{x}\right\ _p=\left\ \boldsymbol{X}\boldsymbol{D}_1^{-1}\boldsymbol{X}^{-1}\boldsymbol{r}\right\ _p\leq\left\ \boldsymbol{D}_1^{-1}\right\ _pK_p(\boldsymbol{X})\ \boldsymbol{r}\ _p=\frac{K_p(\boldsymbol{X})\ \boldsymbol{r}\ _p}{\min _j\left \lambda_j-\mu\right }.$	$1=\ \boldsymbol{x}\ _p=\ \boldsymbol{X}\boldsymbol{D}_1^{-1}\boldsymbol{X}^{-1}\boldsymbol{r}\ _p\leq\ \boldsymbol{D}_1^{-1}\ _pK_p(\boldsymbol{X})\ \boldsymbol{r}\ _p=\frac{K_p(\boldsymbol{X})\ \boldsymbol{r}\ _p}{\min _j\left \lambda_j-\mu\right }.$	$1=\ \boldsymbol{x}\ _p=\ \boldsymbol{X}\boldsymbol{D}_1^{-1}\boldsymbol{X}^{-1}\boldsymbol{r}\ _p\leq\ \boldsymbol{D}_1^{-1}\ _pK_p(\boldsymbol{X})\ \boldsymbol{r}\ _p=\frac{K_p(\boldsymbol{X})\ \boldsymbol{r}\ _p}{\min _j\left \lambda_j-\mu\right }.$
lyche-numerical-linear-algebra-EQ0951 (13.63)	321	0.995 N -1	$\left\ \boldsymbol{A}^{-1}\right\ _2=\max _{\boldsymbol{x} \neq \mathbf{0}} \frac{\left\ \boldsymbol{x}\right\ _2}{\left\ \boldsymbol{A} \boldsymbol{x}\right\ _2},$	$\left\ \boldsymbol{A}^{-1}\right\ _2=\max _{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{x}\ _2}{\ \boldsymbol{A} \boldsymbol{x}\ _2},$	$\left\ \boldsymbol{A}^{-1}\right\ _2=\max _{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{x}\ _2}{\ \boldsymbol{A} \boldsymbol{x}\ _2},$
lyche-numerical-linear-algebra-EQ0532 (8.39)	203	0.995 N +0	$\langle n \boldsymbol{x}, \boldsymbol{y}\rangle=\langle(n-1) \boldsymbol{x}+\boldsymbol{x}, \boldsymbol{y}\rangle \stackrel{(8.37)}{=} \langle(n-1) \boldsymbol{x}, \boldsymbol{y}\rangle+\langle \boldsymbol{x}, \boldsymbol{y}\rangle=n\langle \boldsymbol{x}, \boldsymbol{y}\rangle .$	$\langle n \boldsymbol{x}, \boldsymbol{y}\rangle=\langle(n-1) \boldsymbol{x}+\boldsymbol{x}, \boldsymbol{y}\rangle \stackrel{(8.37)}{=} \langle(n-1) \boldsymbol{x}, \boldsymbol{y}\rangle+\langle \boldsymbol{x}, \boldsymbol{y}\rangle=n\langle \boldsymbol{x}, \boldsymbol{y}\rangle .$	$\langle n \boldsymbol{x}, \boldsymbol{y}\rangle=\langle(n-1) \boldsymbol{x}+\boldsymbol{x}, \boldsymbol{y}\rangle \stackrel{(8.37)}{=} \langle(n-1) \boldsymbol{x}, \boldsymbol{y}\rangle+\langle \boldsymbol{x}, \boldsymbol{y}\rangle=n\langle \boldsymbol{x}, \boldsymbol{y}\rangle .$
lyche-numerical-linear-algebra-EQ0648	233	0.995 K +0	$\lambda_i^{\dagger}=\left\{\begin{array}{cc} 1 / \lambda_i, & \lambda_i \neq 0 \\ 0 & \lambda_i=0 . \end{array}\right.$	$\lambda_i^{\dagger}=\left\{\begin{array}{cc} 1 / \lambda_i, & \lambda_i \neq 0 \\ 0 & \lambda_i=0 . \end{array}\right.$	$\lambda_i^{\dagger}=\left\{\begin{array}{cc} 1 / \lambda_i, & \lambda_i \neq 0 \\ 0 & \lambda_i=0 . \end{array}\right.$
lyche-numerical-linear-algebra-EQ0975	330	0.995 N +0	$\boldsymbol{A}_1:=\left[\begin{array}{cccccc} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{array}\right], \quad \boldsymbol{E}:=\left[\begin{array}{cccccc} 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \epsilon & 0 & 0 & \cdots & 0 & 0 \end{array}\right]=\epsilon \boldsymbol{e}_n \boldsymbol{e}_1^T .$	$\boldsymbol{A}_1:=\left[\begin{array}{cccccc} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{array}\right], \quad \boldsymbol{E}:=\left[\begin{array}{cccccc} 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \epsilon & 0 & 0 & \cdots & 0 & 0 \end{array}\right]=\epsilon \boldsymbol{e}_n \boldsymbol{e}_1^T .$	
Conf. MathPix confidence: 0.995 0.5-0.9 <0.5 no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0926 (13.56)	314	0.996 W +1	$\mathbf{L}_h \mathbf{v} = \frac{1}{h^2} \begin{bmatrix} (dv)_{1,1} & \dots & (dv)_{1,m} \\ \vdots & & \vdots \\ (dv)_{m,1} & \dots & (dv)_{m,m} \end{bmatrix}, \quad \mathbf{F} := \begin{bmatrix} f_{1,1} & \dots & f_{1,m} \\ \vdots & & \vdots \\ f_{m,1} & \dots & f_{m,m} \end{bmatrix}.$	$L_h v = \mathbf{F}, \quad L_h v := \frac{1}{h^2} \begin{bmatrix} (dv)_{1,1} & \dots & (dv)_{1,m} \\ \vdots & & \vdots \\ (dv)_{m,1} & \dots & (dv)_{m,m} \end{bmatrix}, \quad \mathbf{F} := \begin{bmatrix} f_{1,1} & \dots & f_{1,m} \\ \vdots & & \vdots \\ f_{m,1} & \dots & f_{m,m} \end{bmatrix}.$	$L_h v = \mathbf{F}, \quad L_h v := \frac{1}{h^2} \begin{bmatrix} (dv)_{1,1} & \dots & (dv)_{1,m} \\ \vdots & & \vdots \\ (dv)_{m,1} & \dots & (dv)_{m,m} \end{bmatrix}, \quad \mathbf{F} := \begin{bmatrix} f_{1,1} & \dots & f_{1,m} \\ \vdots & & \vdots \\ f_{m,1} & \dots & f_{m,m} \end{bmatrix}.$
lyche-numerical-linear-algebra- EQ1024	346	0.996	$\lim_{k \rightarrow \infty} \left(\frac{ \lambda_1 }{\lambda_1} \right)^k \mathbf{x}_k = \frac{c_1}{ c_1 } \frac{\mathbf{v}_1}{\ \mathbf{v}_1\ }.$	$\lim_{k \rightarrow \infty} \left(\frac{ \lambda_1 }{\lambda_1} \right)^k \mathbf{x}_k = \frac{c_1}{ c_1 } \frac{\mathbf{v}_1}{\ \mathbf{v}_1\ }.$	$\lim_{k \rightarrow \infty} \left(\frac{ \lambda_1 }{\lambda_1} \right)^k \mathbf{x}_k = \frac{c_1}{ c_1 } \frac{\mathbf{v}_1}{\ \mathbf{v}_1\ }.$
lyche-numerical-linear-algebra- EQ0031	032	0.996 W +0	$\mathbf{A} \mathbf{x} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} = \mathbf{b}.$	$\mathbf{A} \mathbf{x} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} = \mathbf{b}.$	$\mathbf{A} \mathbf{x} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} = \mathbf{b}.$
lyche-numerical-linear-algebra- EQ0207	086	0.996 C +3	$\begin{aligned} (\mathbf{L} \mathbf{U})_{ij} &= \sum_{k=1}^n l_{i,k} u_{kj} = \sum_{k=1}^{i-1} a_{pi,k}^{(k)} a_{pk,j}^{(k)} + a_{pi,j}^{(i)} \\ &= \sum_{k=1}^{i-1} \left(a_{pi,j}^{(k)} - a_{pi,j}^{(k+1)} \right) + a_{pi,j}^{(i)} = a_{pi,j}^{(1)} = a_{pi,j} = (\mathbf{P}^T \mathbf{A})_{ij}, \end{aligned}$	$(\mathbf{L} \mathbf{U})_{ij} = \sum_{k=1}^n l_{i,k} u_{kj} = \sum_{k=1}^{i-1} a_{pi,k}^{(k)} a_{pk,j}^{(k)} + a_{pi,j}^{(i)} = \sum_{k=1}^{i-1} \left(a_{pi,j}^{(k)} - a_{pi,j}^{(k+1)} \right) + a_{pi,j}^{(i)} = a_{pi,j}^{(1)} = a_{pi,j} = (\mathbf{P}^T \mathbf{A})_{ij},$	$(\mathbf{L} \mathbf{U})_{ij} = \sum_{k=1}^n l_{i,k} u_{kj} = \sum_{k=1}^{i-1} a_{pi,k}^{(k)} a_{pk,j}^{(k)} + a_{pi,j}^{(i)} = \sum_{k=1}^{i-1} \left(a_{pi,j}^{(k)} - a_{pi,j}^{(k+1)} \right) + a_{pi,j}^{(i)} = a_{pi,j}^{(1)} = a_{pi,j} = (\mathbf{P}^T \mathbf{A})_{ij},$
lyche-numerical-linear-algebra- EQ0739	261	0.996 N +0	$\mathbf{D} \mathbf{X} + \mathbf{X} \mathbf{D} - \frac{1}{6} \mathbf{D} \mathbf{X} \mathbf{D} = 4h^4 \mathbf{G}, \text{ where } \mathbf{G} = S \mu \mathbf{F} S,$	$\mathbf{D} \mathbf{X} + \mathbf{X} \mathbf{D} - \frac{1}{6} \mathbf{D} \mathbf{X} \mathbf{D} = 4h^4 \mathbf{G}, \text{ where } \mathbf{G} = S \mu \mathbf{F} S,$	$\mathbf{D} \mathbf{X} + \mathbf{X} \mathbf{D} - \frac{1}{6} \mathbf{D} \mathbf{X} \mathbf{D} = 4h^4 \mathbf{G}, \text{ where } \mathbf{G} = S \mu \mathbf{F} S,$
lyche-numerical-linear-algebra- EQ0787 (12.27)	277	0.996 N +0	$\rho(\mathbf{G}_\omega) = \begin{cases} \frac{1}{4} \left(\omega \beta + \sqrt{(\omega \beta)^2 - 4(\omega - 1)} \right)^2, & \text{for } 0 < \omega \leq \omega^*, \\ \omega - 1, & \text{for } \omega^* < \omega < 2, \end{cases}$	$\rho(\mathbf{G}_\omega) = \begin{cases} \frac{1}{4} \left(\omega \beta + \sqrt{(\omega \beta)^2 - 4(\omega - 1)} \right)^2, & \text{for } 0 < \omega \leq \omega^*, \\ \omega - 1, & \text{for } \omega^* < \omega < 2, \end{cases}$	$\rho(\mathbf{G}_\omega) = \begin{cases} \frac{1}{4} \left(\omega \beta + \sqrt{(\omega \beta)^2 - 4(\omega - 1)} \right)^2, & \text{for } 0 < \omega \leq \omega^*, \\ \omega - 1, & \text{for } \omega^* < \omega < 2, \end{cases}$
lyche-numerical-linear-algebra- EQ0819	285	0.996 K +0	$d(\omega) := (\omega \mu)^2 - 4(\omega - 1).$	$d(\omega) := (\omega \mu)^2 - 4(\omega - 1).$	$d(\omega) := (\omega \mu)^2 - 4(\omega - 1).$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra E08850 (13.16)	295	■0.996 ● N +0	$\boldsymbol{r}_{j}^T \boldsymbol{A} \boldsymbol{p}_{i}^{\stackrel{(13.13)}{=}} \boldsymbol{r}_{j}^T \left(\frac{\boldsymbol{r}_i - \boldsymbol{r}_{i+1}}{\alpha_i} \right) = 0, \quad i = 0, 1, \dots, j - 2.$		$\boldsymbol{r}_j^T \boldsymbol{A} \boldsymbol{p}_i \stackrel{(13.13)}{=} \boldsymbol{r}_j^T \left(\frac{\boldsymbol{r}_i - \boldsymbol{r}_{i+1}}{\alpha_i} \right) = 0, \quad i = 0, 1, \dots, j - 2.$
lyche-numerical-linear-algebra E09991	334	■0.996 ●	$\begin{aligned} \boldsymbol{A}_{k+1} &= \\ \boldsymbol{H}_k \boldsymbol{A}_k \boldsymbol{H}_k &= \left[\begin{array}{cc} \boldsymbol{I}_k & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{V}_k \end{array} \right] \left[\begin{array}{cc} \boldsymbol{B}_k & \boldsymbol{C}_k \\ \boldsymbol{D}_k & \boldsymbol{E}_k \end{array} \right] \left[\begin{array}{cc} \boldsymbol{I}_k & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{V}_k \end{array} \right] \\ &= \left[\begin{array}{cc} \boldsymbol{B}_k & \boldsymbol{C}_k \boldsymbol{V}_k \\ \boldsymbol{V}_k \boldsymbol{D}_k & \boldsymbol{V}_k \boldsymbol{E}_k \boldsymbol{V}_k \end{array} \right]. \end{aligned}$		$\begin{aligned} \boldsymbol{A}_{k+1} &= \boldsymbol{H}_k \boldsymbol{A}_k \boldsymbol{H}_k = \left[\begin{array}{cc} \boldsymbol{I}_k & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{V}_k \end{array} \right] \left[\begin{array}{cc} \boldsymbol{B}_k & \boldsymbol{C}_k \\ \boldsymbol{D}_k & \boldsymbol{E}_k \end{array} \right] \left[\begin{array}{cc} \boldsymbol{I}_k & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{V}_k \end{array} \right] \\ &= \left[\begin{array}{cc} \boldsymbol{B}_k & \boldsymbol{C}_k \boldsymbol{V}_k \\ \boldsymbol{V}_k \boldsymbol{D}_k & \boldsymbol{V}_k \boldsymbol{E}_k \boldsymbol{V}_k \end{array} \right]. \end{aligned}$
lyche-numerical-linear-algebra E08454 (7.14)	181	■0.997 ● K +0	$\boldsymbol{X}_{k+1} = \frac{1}{2} (\boldsymbol{X}_k + \boldsymbol{X}_k^{-T}), k = 0, 1, 2, \dots \text{ with } \boldsymbol{X}_0 = \boldsymbol{A},$		$\boldsymbol{X}_{k+1} = \frac{1}{2} (\boldsymbol{X}_k + \boldsymbol{X}_k^{-T}), k = 0, 1, 2, \dots \text{ with } \boldsymbol{X}_0 = \boldsymbol{A},$
lyche-numerical-linear-algebra E00135	064	■0.997 ● K +0	$\left[\begin{array}{l} A_1 \\ A_2 \end{array} \right] B = \left[\begin{array}{l} A_1 B \\ A_2 B \end{array} \right].$		$\left[\begin{array}{l} A_1 \\ A_2 \end{array} \right] B = \left[\begin{array}{l} A_1 B \\ A_2 B \end{array} \right].$
lyche-numerical-linear-algebra E09269 (4.11)	110	■0.997 ● W +2	$\begin{aligned} &\text{if } A(1,1) > 0 \quad \& \quad A(1,1) = \sqrt{A(1,1)} \\ &A(2:n,1) = A(2:n,1)/A(1,1) \\ &\text{for } j = 2:n \\ &A(j:n,j) = A(j:n,j) - A(j,1) * A(j:n,1) \end{aligned}$	$\begin{aligned} &\text{if } A(1,1) > 0 \\ &A(1,1) = \sqrt{A(1,1)} \\ &A(2:n,1) = A(2:n,1)/A(1,1) \\ &\text{for } j = 2:n \\ &A(j:n,j) = A(j:n,j) - A(j,1) * A(j:n,1) \end{aligned}$	if $A(1,1) > 0$ $A(1,1) = \sqrt{A(1,1)}$ $A(2:n,1) = A(2:n,1)/A(1,1)$ for $j = 2:n$ $A(j:n,j) = A(j:n,j) - A(j,1) * A(j:n,1)$
lyche-numerical-linear-algebra E09431	173	■0.997 ● N +0	$\boldsymbol{B} \left[\begin{array}{l} 1 \\ 1 \end{array} \right] = 4 \left[\begin{array}{l} 1 \\ 1 \end{array} \right], \quad \boldsymbol{B} \left[\begin{array}{l} 1 \\ -1 \end{array} \right] = 0 \left[\begin{array}{l} 1 \\ -1 \end{array} \right],$		$\boldsymbol{B} \left[\begin{array}{l} 1 \\ 1 \end{array} \right] = 4 \left[\begin{array}{l} 1 \\ 1 \end{array} \right], \quad \boldsymbol{B} \left[\begin{array}{l} 1 \\ -1 \end{array} \right] = 0 \left[\begin{array}{l} 1 \\ -1 \end{array} \right],$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-alignat-EQ0656 (10.1)	238	0.997 N +0	$\begin{gathered} -\Delta u := -\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = f \text{ on } \Omega, \\ u := 0 \text{ on } \partial\Omega. \end{gathered}$	$-\Delta u := -\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = f \text{ on } \Omega, \\ u := 0 \text{ on } \partial\Omega.$	$-\Delta u := -\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = f \text{ on } \Omega, \\ u := 0 \text{ on } \partial\Omega.$
lyche-numerical-linear-alignat-EQ0666 (10.7)	240	0.997 N +0	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}, \quad \boldsymbol{A} \in \mathbb{R}^{n \times n}, \quad \boldsymbol{b} \in \mathbb{R}^n, \quad n = m^2,$	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}, \quad \boldsymbol{A} \in \mathbb{R}^{n \times n}, \quad \boldsymbol{b} \in \mathbb{R}^n, \quad n = m^2,$	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}, \quad \boldsymbol{A} \in \mathbb{R}^{n \times n}, \quad \boldsymbol{b} \in \mathbb{R}^n, \quad n = m^2,$
lyche-numerical-linear-alignat-EQ0671 (10.11)	242	0.997	$\boldsymbol{T}_2 = \left[\begin{array}{rrr rrr rrr} 2d & a & 0 & a & 0 & 0 & 0 & 0 & 0 \\ a & 2d & a & 0 & a & 0 & 0 & 0 & 0 \\ 0 & a & 2d & 0 & 0 & a & 0 & 0 & 0 \\ \hline a & 0 & 0 & 2d & a & 0 & a & 0 & 0 \\ 0 & a & 0 & a & 2d & a & 0 & a & 0 \\ 0 & 0 & a & 0 & a & 2d & 0 & 0 & a \\ \hline 0 & 0 & 0 & a & 0 & 0 & 2d & a & 0 \\ 0 & 0 & 0 & 0 & a & 0 & a & 2d & a \\ 0 & 0 & 0 & 0 & 0 & a & 0 & a & 2d \end{array} \right].$	$\boldsymbol{T}_2 = \left[\begin{array}{rrr rrr rrr} 2d & a & 0 & a & 0 & 0 & 0 & 0 & 0 \\ a & 2d & a & 0 & a & 0 & 0 & 0 & 0 \\ 0 & a & 2d & 0 & 0 & a & 0 & 0 & 0 \\ \hline a & 0 & 0 & 2d & a & 0 & a & 0 & 0 \\ 0 & a & 0 & a & 2d & a & 0 & a & 0 \\ 0 & 0 & a & 0 & a & 2d & 0 & 0 & a \\ \hline 0 & 0 & 0 & a & 0 & 0 & 2d & a & 0 \\ 0 & 0 & 0 & 0 & a & 0 & a & 2d & a \\ 0 & 0 & 0 & 0 & 0 & a & 0 & a & 2d \end{array} \right].$	$\boldsymbol{T}_2 = \left[\begin{array}{rrr rrr rrr} 2d & a & 0 & a & 0 & 0 & 0 & 0 & 0 \\ a & 2d & a & 0 & a & 0 & 0 & 0 & 0 \\ 0 & a & 2d & 0 & 0 & a & 0 & 0 & 0 \\ \hline a & 0 & 0 & 2d & a & 0 & a & 0 & 0 \\ 0 & a & 0 & a & 2d & a & 0 & a & 0 \\ 0 & 0 & a & 0 & a & 2d & 0 & 0 & a \\ \hline 0 & 0 & 0 & a & 0 & 0 & 2d & a & 0 \\ 0 & 0 & 0 & 0 & a & 0 & a & 2d & a \\ 0 & 0 & 0 & 0 & 0 & a & 0 & a & 2d \end{array} \right].$
lyche-numerical-linear-alignat-EQ1010 (14.10)	240	0.997 W +1	$a_{kj} = \begin{cases} 1, & j = k + 1, \\ 0, & \text{otherwise} \end{cases}, \quad e_{kj} = \begin{cases} \epsilon, & k = n, j = 1, \\ 0, & \text{otherwise} \end{cases},$	$a_{kj} = \begin{cases} 1, & j = k + 1, \\ 0, & \text{otherwise} \end{cases}, \quad e_{kj} = \begin{cases} \epsilon, & k = n, j = 1, \\ 0, & \text{otherwise} \end{cases},$	$a_{kj} = \begin{cases} 1, & j = k + 1, \\ 0, & \text{otherwise} \end{cases}, \quad e_{kj} = \begin{cases} \epsilon, & k = n, j = 1, \\ 0, & \text{otherwise} \end{cases},$
lyche-numerical-linear-alignat-EQ0294 (5.10)	122	0.997 N +0	$\boldsymbol{s}_0 = \sum_{i=1}^k \frac{\langle \boldsymbol{v}, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle} \boldsymbol{v}_i.$	$\boldsymbol{s}_0 = \sum_{i=1}^k \frac{\langle \boldsymbol{v}, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle} \boldsymbol{v}_i.$	$\boldsymbol{s}_0 = \sum_{i=1}^k \frac{\langle \boldsymbol{v}, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle} \boldsymbol{v}_i.$
lyche-numerical-linear-alignat-EQ0416	166	0.997 N +0	$\boldsymbol{B}_1 := \{ \boldsymbol{x} \in \mathbb{R}^n \mid \ \boldsymbol{x}\ _2 = 1 \}.$	$\boldsymbol{B}_1 := \{ \boldsymbol{x} \in \mathbb{R}^n \mid \ \boldsymbol{x}\ _2 = 1 \}.$	$\boldsymbol{B}_1 := \{ \boldsymbol{x} \in \mathbb{R}^n \mid \ \boldsymbol{x}\ _2 = 1 \}.$
lyche-numerical-linear-alignat-EQ0694 (10.24)	248	0.997 W +1	$\begin{array}{l} -\Delta v(s, t) = f(s, t) \\ -\Delta u(s, t) = v(s, t) \\ u(s, t) = v(s, t) = 0 \end{array} \quad \begin{array}{l} (s, t) \in \Omega \\ (s, t) \in \Omega \\ (s, t) \in \partial\Omega \end{array}$	$\begin{array}{l} -\Delta v(s, t) = f(s, t) \\ -\Delta u(s, t) = v(s, t) \\ u(s, t) = v(s, t) = 0 \end{array} \quad \begin{array}{l} (s, t) \in \Omega \\ (s, t) \in \Omega \\ (s, t) \in \partial\Omega \end{array}$	$\begin{array}{l} -\Delta v(s, t) = f(s, t) \\ -\Delta u(s, t) = v(s, t) \\ u(s, t) = v(s, t) = 0 \end{array} \quad \begin{array}{l} (s, t) \in \Omega \\ (s, t) \in \Omega \\ (s, t) \in \partial\Omega \end{array}$
lyche-numerical-linear-alignat-EQ0768 (12.16)	272	0.997 K +0	$\boldsymbol{x}_k - \boldsymbol{x} = \boldsymbol{G}^k (\boldsymbol{x}_0 - \boldsymbol{x}), \quad k = 0, 1, 2, \dots$	$\boldsymbol{x}_k - \boldsymbol{x} = \boldsymbol{G}^k (\boldsymbol{x}_0 - \boldsymbol{x}), \quad k = 0, 1, 2, \dots$	$\boldsymbol{x}_k - \boldsymbol{x} = \boldsymbol{G}^k (\boldsymbol{x}_0 - \boldsymbol{x}), \quad k = 0, 1, 2, \dots$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0815	284	0.997 K +0	$\boldsymbol{G}_J \boldsymbol{v} = (\boldsymbol{L} + \boldsymbol{R}) \boldsymbol{v} = \frac{\lambda + \omega - 1}{\omega \lambda^{1/2}} \boldsymbol{v} = \mu \boldsymbol{v}.$		$\boldsymbol{G}_J \boldsymbol{v} = (\boldsymbol{L} + \boldsymbol{R}) \boldsymbol{v} = \frac{\lambda + \omega - 1}{\omega \lambda^{1/2}} \boldsymbol{v} = \mu \boldsymbol{v}.$
lyche-numerical-linear-algebra-EQ0847 (13.11)	294	0.997 S +0	$\begin{aligned} & \boldsymbol{t}_{2k-2} = 3 * 4^{1-k} \begin{bmatrix} 1 \\ -1/2 \end{bmatrix}, \quad \boldsymbol{x}_{2k-1} = -4^{-k} \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \boldsymbol{r}_{2k-1} = 3 * 4^{-k} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ & \boldsymbol{t}_{2k-1} = 3 * 4^{-k} \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \quad \boldsymbol{x}_{2k} = -4^{-k} \begin{bmatrix} 1 \\ 1/2 \end{bmatrix}, \quad \boldsymbol{r}_{2k} = 3 * 4^{-k} \begin{bmatrix} 1/2 \\ 0 \end{bmatrix}. \end{aligned}$	$\boldsymbol{t}_{2k-2} = 3 * 4^{1-k} \begin{bmatrix} 1 \\ -1/2 \end{bmatrix}, \quad \boldsymbol{x}_{2k-1} = -4^{-k} \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \boldsymbol{r}_{2k-1} = 3 * 4^{-k} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $\boldsymbol{t}_{2k-1} = 3 * 4^{-k} \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \quad \boldsymbol{x}_{2k} = -4^{-k} \begin{bmatrix} 1 \\ 1/2 \end{bmatrix}, \quad \boldsymbol{r}_{2k} = 3 * 4^{-k} \begin{bmatrix} 1/2 \\ 0 \end{bmatrix}.$	$\boldsymbol{t}_{2k-2} = 3 * 4^{1-k} \begin{bmatrix} 1 \\ -1/2 \end{bmatrix}, \quad \boldsymbol{x}_{2k-1} = -4^{-k} \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \boldsymbol{r}_{2k-1} = 3 * 4^{-k} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $\boldsymbol{t}_{2k-1} = 3 * 4^{-k} \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \quad \boldsymbol{x}_{2k} = -4^{-k} \begin{bmatrix} 1 \\ 1/2 \end{bmatrix}, \quad \boldsymbol{r}_{2k} = 3 * 4^{-k} \begin{bmatrix} 1/2 \\ 0 \end{bmatrix}.$
lyche-numerical-linear-algebra-EQ0355 (5.22)	142	0.998	$P_{i_1, n+1} P_{i_2, n+1} \cdots P_{i_n, n+1} \begin{bmatrix} \boldsymbol{L}^* \\ \boldsymbol{z}^* \end{bmatrix} = \begin{bmatrix} \boldsymbol{R}' \\ 0 \end{bmatrix},$	$P_{i_1, n+1} P_{i_2, n+1} \cdots P_{i_n, n+1} \begin{bmatrix} \boldsymbol{L}^* \\ \boldsymbol{z}^* \end{bmatrix} = \begin{bmatrix} \boldsymbol{R}' \\ 0 \end{bmatrix},$	$P_{i_1, n+1} P_{i_2, n+1} \cdots P_{i_n, n+1} \begin{bmatrix} \boldsymbol{L}^* \\ \boldsymbol{z}^* \end{bmatrix} = \begin{bmatrix} \boldsymbol{R}' \\ 0 \end{bmatrix},$
lyche-numerical-linear-algebra-EQ0390 (6.11)	157	0.998 K +0	$\lambda_k \leq \max_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}),$	$\lambda_k \leq \max_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}),$	$\lambda_k \leq \max_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}),$
lyche-numerical-linear-algebra-EQ0448	179	0.998 N +0	$\boldsymbol{A} := \frac{1}{15} \begin{bmatrix} 14 & 4 & 16 \\ 2 & 22 & 13 \end{bmatrix} \in \mathbb{R}^{2 \times 3}.$	$\boldsymbol{A} := \frac{1}{15} \begin{bmatrix} 14 & 4 & 16 \\ 2 & 22 & 13 \end{bmatrix} \in \mathbb{R}^{2 \times 3}.$	$\boldsymbol{A} := \frac{1}{15} \begin{bmatrix} 14 & 4 & 16 \\ 2 & 22 & 13 \end{bmatrix} \in \mathbb{R}^{2 \times 3}.$
lyche-numerical-linear-algebra-EQ0765	272	0.998 N +1	$x_{k+1}(1) = \frac{1}{2} x_k(2) + \frac{1}{2}, \quad x_{k+1}(2) = \frac{1}{2} x_{k+1}(1) + \frac{1}{2}.$	$x_{k+1}(1) = \frac{1}{2} x_k(2) + \frac{1}{2}, \quad x_{k+1}(2) = \frac{1}{2} x_{k+1}(1) + \frac{1}{2}.$	$\boldsymbol{x}_{k+1}(1) = \frac{1}{2} \boldsymbol{x}_k(2) + \frac{1}{2}, \quad \boldsymbol{x}_{k+1}(2) = \frac{1}{2} \boldsymbol{x}_{k+1}(1) + \frac{1}{2}.$
lyche-numerical-linear-algebra-EQ0228 (3.19)	092	0.998 N +0	$\boldsymbol{A} := \begin{bmatrix} \boldsymbol{A}_{11} & \cdots & \boldsymbol{A}_{1m} \\ \vdots & & \vdots \\ \boldsymbol{A}_{m1} & \cdots & \boldsymbol{A}_{mm} \end{bmatrix},$	$\boldsymbol{A} := \begin{bmatrix} \boldsymbol{A}_{11} & \cdots & \boldsymbol{A}_{1m} \\ \vdots & & \vdots \\ \boldsymbol{A}_{m1} & \cdots & \boldsymbol{A}_{mm} \end{bmatrix},$	$\boldsymbol{A} := \begin{bmatrix} \boldsymbol{A}_{11} & \cdots & \boldsymbol{A}_{1m} \\ \vdots & & \vdots \\ \boldsymbol{A}_{m1} & \cdots & \boldsymbol{A}_{mm} \end{bmatrix},$
lyche-numerical-linear-algebra-EQ0603 (9.8)	222	0.998 N +0	$\boldsymbol{A} \boldsymbol{A}^\dagger \boldsymbol{A} = \boldsymbol{A}, \boldsymbol{A}^\dagger \boldsymbol{A} \boldsymbol{A}^\dagger = \boldsymbol{A}^\dagger, (\boldsymbol{A}^\dagger \boldsymbol{A})^* = \boldsymbol{A}^\dagger \boldsymbol{A}, (\boldsymbol{A} \boldsymbol{A}^\dagger)^* = \boldsymbol{A} \boldsymbol{A}^\dagger.$	$\boldsymbol{A} \boldsymbol{A}^\dagger \boldsymbol{A} = \boldsymbol{A}, \boldsymbol{A}^\dagger \boldsymbol{A} \boldsymbol{A}^\dagger = \boldsymbol{A}^\dagger, (\boldsymbol{A}^\dagger \boldsymbol{A})^* = \boldsymbol{A}^\dagger \boldsymbol{A}, (\boldsymbol{A} \boldsymbol{A}^\dagger)^* = \boldsymbol{A} \boldsymbol{A}^\dagger.$	$\boldsymbol{A} \boldsymbol{A}^\dagger \boldsymbol{A} = \boldsymbol{A}, \boldsymbol{A}^\dagger \boldsymbol{A} \boldsymbol{A}^\dagger = \boldsymbol{A}^\dagger, (\boldsymbol{A}^\dagger \boldsymbol{A})^* = \boldsymbol{A}^\dagger \boldsymbol{A}, (\boldsymbol{A} \boldsymbol{A}^\dagger)^* = \boldsymbol{A} \boldsymbol{A}^\dagger.$
Conf. MathPix confidence: 0.997 0.998 0.998 0.998 0.998 0.997 0.998 0.998 0.998 0.998 K +0 N +0 N +0 no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-E00691	247	0.998 N +0	$\boldsymbol{A}=\left[\begin{array}{rrrr} 2\,d & a & a & 0 \\ a & 2d & 0 & a \\ a & 0 & 2d & a \\ 0 & a & a & 2d \end{array}\right]\,.$	$\boldsymbol{A}=\left[\begin{array}{rrrr} 2d & a & a & 0 \\ a & 2d & 0 & a \\ a & 0 & 2d & a \\ 0 & a & a & 2d \end{array}\right]\,.$	$\boldsymbol{A}=\left[\begin{array}{rrrr} 2d & a & a & 0 \\ a & 2d & 0 & a \\ a & 0 & 2d & a \\ 0 & a & a & 2d \end{array}\right]\,.$
lyche-numerical-linear-algebra-E00981	331	0.998 K +0	$\boldsymbol{X}\,\boldsymbol{D}^{-1}\boldsymbol{X}^{-1}\boldsymbol{r}=\left(\boldsymbol{X}(\boldsymbol{D}-\mu\boldsymbol{I})\boldsymbol{X}^{-1}\right)^{-1}\boldsymbol{r}=(\boldsymbol{A}-\mu\boldsymbol{I})^{-1}(\boldsymbol{A}-\mu\boldsymbol{I})\boldsymbol{x}=\boldsymbol{x}.$	$\boldsymbol{X}\boldsymbol{D}_1^{-1}\boldsymbol{X}^{-1}\boldsymbol{r}=\left(\boldsymbol{X}(\boldsymbol{D}-\mu\boldsymbol{I})\boldsymbol{X}^{-1}\right)^{-1}\boldsymbol{r}=(\boldsymbol{A}-\mu\boldsymbol{I})^{-1}(\boldsymbol{A}-\mu\boldsymbol{I})\boldsymbol{x}=\boldsymbol{x}.$	$\boldsymbol{X}\boldsymbol{D}_1^{-1}\boldsymbol{X}^{-1}\boldsymbol{r}=\left(\boldsymbol{X}(\boldsymbol{D}-\mu\boldsymbol{I})\boldsymbol{X}^{-1}\right)^{-1}\boldsymbol{r}=(\boldsymbol{A}-\mu\boldsymbol{I})^{-1}(\boldsymbol{A}-\mu\boldsymbol{I})\boldsymbol{x}=\boldsymbol{x}.$
lyche-numerical-linear-algebra-E01035 (15.8)	351	0.998 W -1	$\begin{aligned} &\boldsymbol{A}_{\boldsymbol{k}}=\boldsymbol{A} \\ &\text{for }k=1,2,\ldots \\ &\boldsymbol{Q}_k\boldsymbol{R}_k=\boldsymbol{A}_k \quad (\text{QR factorization of } \boldsymbol{A}_k) \\ &\boldsymbol{A}_{k+1}=\boldsymbol{R}_k\boldsymbol{Q}_k. \end{aligned}$	$\boldsymbol{A}_1=\boldsymbol{A}$ for $k=1,2,\ldots$ $\boldsymbol{Q}_k\boldsymbol{R}_k=\boldsymbol{A}_k \quad (\text{QR factorization of } \boldsymbol{A}_k)$ $\boldsymbol{A}_{k+1}=\boldsymbol{R}_k\boldsymbol{Q}_k.$	$\boldsymbol{A}_1=\boldsymbol{A}$ for $k=1,2,\ldots$ $\boldsymbol{Q}_k\boldsymbol{R}_k=\boldsymbol{A}_k \quad (\text{QR factorization of } \boldsymbol{A}_k)$ $\boldsymbol{A}_{k+1}=\boldsymbol{R}_k\boldsymbol{Q}_k.$
lyche-numerical-linear-algebra-E00012	024	0.998	$\boldsymbol{A}\left(r_1:r_2,\,c_1:c_2\right):=\left[\begin{array}{cccc} a_{r_1,c_1} & a_{r_1,c_1+1} & \cdots & a_{r_1,c_2} \\ a_{r_1+1,c_1} & a_{r_1+1,c_1+1} & \cdots & a_{r_1+1,c_2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{r_2,c_1} & a_{r_2,c_1+1} & \cdots & a_{r_2,c_2} \end{array}\right]\,.$	$\boldsymbol{A}\left(r_1:r_2,\,c_1:c_2\right):=\left[\begin{array}{cccc} a_{r_1,c_1} & a_{r_1,c_1+1} & \cdots & a_{r_1,c_2} \\ a_{r_1+1,c_1} & a_{r_1+1,c_1+1} & \cdots & a_{r_1+1,c_2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{r_2,c_1} & a_{r_2,c_1+1} & \cdots & a_{r_2,c_2} \end{array}\right]\,.$	$\boldsymbol{A}(r_1:r_2,\,c_1:c_2):=\left[\begin{array}{cccc} a_{r_1,c_1} & a_{r_1,c_1+1} & \cdots & a_{r_1,c_2} \\ a_{r_1+1,c_1} & a_{r_1+1,c_1+1} & \cdots & a_{r_1+1,c_2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{r_2,c_1} & a_{r_2,c_1+1} & \cdots & a_{r_2,c_2} \end{array}\right]\,.$
lyche-numerical-linear-algebra-E00247 (4.2)	101	0.998 N +0	$\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{\left[\boldsymbol{k}\right]} & \boldsymbol{B}_{\boldsymbol{k}}^* \\ \boldsymbol{B}_{\boldsymbol{k}} & \boldsymbol{F}_{\boldsymbol{k}} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{L}_{\left[\boldsymbol{k}\right]} & \boldsymbol{0} \\ \boldsymbol{M}_{\boldsymbol{k}} & \boldsymbol{N}_{\boldsymbol{k}} \end{array}\right]\left[\begin{array}{cc} \boldsymbol{D}_{\left[\boldsymbol{k}\right]} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{E}_{\boldsymbol{k}} \end{array}\right]\left[\begin{array}{cc} \boldsymbol{L}_{\left[\boldsymbol{k}\right]}^* & \boldsymbol{M}_{\boldsymbol{k}}^* \\ \boldsymbol{0} & \boldsymbol{N}_{\boldsymbol{k}}^* \end{array}\right]=\boldsymbol{LDU},$	$\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{\left[\boldsymbol{k}\right]} & \boldsymbol{B}_{\boldsymbol{k}}^* \\ \boldsymbol{B}_{\boldsymbol{k}} & \boldsymbol{F}_{\boldsymbol{k}} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{L}_{\left[\boldsymbol{k}\right]} & \boldsymbol{0} \\ \boldsymbol{M}_{\boldsymbol{k}} & \boldsymbol{N}_{\boldsymbol{k}} \end{array}\right]\left[\begin{array}{cc} \boldsymbol{D}_{\left[\boldsymbol{k}\right]} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{E}_{\boldsymbol{k}} \end{array}\right]\left[\begin{array}{cc} \boldsymbol{L}_{\left[\boldsymbol{k}\right]}^* & \boldsymbol{M}_{\boldsymbol{k}}^* \\ \boldsymbol{0} & \boldsymbol{N}_{\boldsymbol{k}}^* \end{array}\right]=\boldsymbol{LDU},$	$\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{\left[\boldsymbol{k}\right]} & \boldsymbol{B}_{\boldsymbol{k}}^* \\ \boldsymbol{B}_{\boldsymbol{k}} & \boldsymbol{F}_{\boldsymbol{k}} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{L}_{\left[\boldsymbol{k}\right]} & \boldsymbol{0} \\ \boldsymbol{M}_{\boldsymbol{k}} & \boldsymbol{N}_{\boldsymbol{k}} \end{array}\right]\left[\begin{array}{cc} \boldsymbol{D}_{\left[\boldsymbol{k}\right]} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{E}_{\boldsymbol{k}} \end{array}\right]\left[\begin{array}{cc} \boldsymbol{L}_{\left[\boldsymbol{k}\right]}^* & \boldsymbol{M}_{\boldsymbol{k}}^* \\ \boldsymbol{0} & \boldsymbol{N}_{\boldsymbol{k}}^* \end{array}\right]=\boldsymbol{LDU},$
lyche-numerical-linear-algebra-E00306 (5.13)	125	0.998 N +0	$\boldsymbol{v}^*\boldsymbol{v}=(\boldsymbol{x}-\boldsymbol{y})^*(\boldsymbol{x}-\boldsymbol{y})=2\boldsymbol{x}^*\boldsymbol{x}-2\operatorname{Re}(\boldsymbol{y}^*\boldsymbol{x})=2\boldsymbol{v}^*\boldsymbol{x}.$	$\boldsymbol{v}^*\boldsymbol{v}=(\boldsymbol{x}-\boldsymbol{y})^*(\boldsymbol{x}-\boldsymbol{y})=2\boldsymbol{x}^*\boldsymbol{x}-2\operatorname{Re}(\boldsymbol{y}^*\boldsymbol{x})=2\boldsymbol{v}^*\boldsymbol{x}.$	$\boldsymbol{v}^*\boldsymbol{v}=(\boldsymbol{x}-\boldsymbol{y})^*(\boldsymbol{x}-\boldsymbol{y})=2\boldsymbol{x}^*\boldsymbol{x}-2\operatorname{Re}(\boldsymbol{y}^*\boldsymbol{x})=2\boldsymbol{v}^*\boldsymbol{x}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0320	129	0.998 W +0	$\boldsymbol{A}=\left[\begin{array}{lll} u_{11} & r_{12} & r_{13} \\ u_{21} & u_{22} & r_{23} \\ u_{31} & u_{32} & u_{33} \\ u_{41} & u_{42} & u_{43} \end{array}\right] \text{ or } \boldsymbol{A}=\left[\begin{array}{lll} r_{11} & r_{12} & r_{13} \\ u_{21} & r_{22} & r_{23} \\ u_{31} & u_{32} & r_{33} \\ u_{41} & u_{42} & u_{43} \end{array}\right].$	$\boldsymbol{A}=\left[\begin{array}{lll} u_{11} & r_{12} & r_{13} \\ u_{21} & u_{22} & r_{23} \\ u_{31} & u_{32} & u_{33} \\ u_{41} & u_{42} & u_{43} \end{array}\right] \text{ or } \boldsymbol{A}=\left[\begin{array}{lll} r_{11} & r_{12} & r_{13} \\ u_{21} & r_{22} & r_{23} \\ u_{31} & u_{32} & r_{33} \\ u_{41} & u_{42} & u_{43} \end{array}\right].$	$\boldsymbol{A}=\left[\begin{array}{lll} u_{11} & r_{12} & r_{13} \\ u_{21} & u_{22} & r_{23} \\ u_{31} & u_{32} & u_{33} \\ u_{41} & u_{42} & u_{43} \end{array}\right] \text{ or } \boldsymbol{A}=\left[\begin{array}{lll} r_{11} & r_{12} & r_{13} \\ u_{21} & r_{22} & r_{23} \\ u_{31} & u_{32} & r_{33} \\ u_{41} & u_{42} & u_{43} \end{array}\right].$
lyche-numerical-linear-algebra-EQ0438	175	0.998 W +0	$1=\ z\ _2^2=\left\ \boldsymbol{V}_1\boldsymbol{\Sigma}_1^{-1}\boldsymbol{y}\right\ _2^2=\left\ \boldsymbol{\Sigma}_1^{-1}\boldsymbol{y}\right\ _2^2=\frac{y_1^2}{\sigma_1^2}+\cdots+\frac{y_n^2}{\sigma_n^2}.$	$1=\ z\ _2^2=\left\ \boldsymbol{V}_1\boldsymbol{\Sigma}_1^{-1}\boldsymbol{y}\right\ _2^2=\left\ \boldsymbol{\Sigma}_1^{-1}\boldsymbol{y}\right\ _2^2=\frac{y_1^2}{\sigma_1^2}+\cdots+\frac{y_n^2}{\sigma_n^2}.$	$1=\ z\ _2^2=\left\ \boldsymbol{V}_1\boldsymbol{\Sigma}_1^{-1}\boldsymbol{y}\right\ _2^2=\left\ \boldsymbol{\Sigma}_1^{-1}\boldsymbol{y}\right\ _2^2=\frac{y_1^2}{\sigma_1^2}+\cdots+\frac{y_n^2}{\sigma_n^2}.$
lyche-numerical-linear-algebra-EQ0477 (8.10)	189	0.998 N +0	$\left\ \boldsymbol{A}^k\right\ \leq\left\ \boldsymbol{A}\right\ ^k \text { for } \boldsymbol{A} \in \mathbb{C}^{n \times n} \text { and } k \in \mathbb{N}.$	$\left\ \boldsymbol{A}^k\right\ \leq\left\ \boldsymbol{A}\right\ ^k \text { for } \boldsymbol{A} \in \mathbb{C}^{n \times n} \text { and } k \in \mathbb{N}.$	$\left\ \boldsymbol{A}^k\right\ \leq\left\ \boldsymbol{A}\right\ ^k \text { for } \boldsymbol{A} \in \mathbb{C}^{n \times n} \text { and } k \in \mathbb{N}.$
lyche-numerical-linear-algebra-EQ0487 (8.17)	191	0.998 N -1	$\ \boldsymbol{A}\ _p:=\max _{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ _p}{\ \boldsymbol{x}\ _p}=\max _{\ \boldsymbol{y}\ _p=1}\ \boldsymbol{A} \boldsymbol{y}\ _p.$	$\ \boldsymbol{A}\ _p:=\max _{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ _p}{\ \boldsymbol{x}\ _p}=\max _{\ \boldsymbol{y}\ _p=1}\ \boldsymbol{A} \boldsymbol{y}\ _p.$	$\ \boldsymbol{A}\ _p:=\max _{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ _p}{\ \boldsymbol{x}\ _p}=\max _{\ \boldsymbol{y}\ _p=1}\ \boldsymbol{A} \boldsymbol{y}\ _p.$
lyche-numerical-linear-algebra-EQ0517 (8.32)	201	0.998 N +0	$a_1^{\lambda_1} a_2^{\lambda_2} \cdots a_n^{\lambda_n} \leq \sum_{j=1}^n \lambda_j a_j,$	$a_1^{\lambda_1} a_2^{\lambda_2} \cdots a_n^{\lambda_n} \leq \sum_{j=1}^n \lambda_j a_j,$	$a_1^{\lambda_1} a_2^{\lambda_2} \cdots a_n^{\lambda_n} \leq \sum_{j=1}^n \lambda_j a_j,$
lyche-numerical-linear-algebra-EQ0544	206	0.998	$\boldsymbol{A}=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right].$	$\boldsymbol{A}=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right].$	$\boldsymbol{A}=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right].$
lyche-numerical-linear-algebra-EQ0997	336	0.998 N +0	$p_2(-M)>0, \quad p_2\left(d_1\right)<0, \quad p_2(+M)>0.$	$p_2(-M)>0, \quad p_2\left(d_1\right)<0, \quad p_2(+M)>0.$	$p_2(-M)>0, \quad p_2\left(d_1\right)<0, \quad p_2(+M)>0.$
lyche-numerical-linear-algebra-EQ1049	358	0.998 N +0	$D_i f(\boldsymbol{x}):=\frac{\partial f(\boldsymbol{x})}{\partial x_i}:=\lim _{h \rightarrow 0} \frac{f\left(\boldsymbol{x}+h \boldsymbol{e}_i\right)-f(\boldsymbol{x})}{h}, \quad \boldsymbol{x} \in \mathbb{R}^n,$	$D_i f(\boldsymbol{x}):=\frac{\partial f(\boldsymbol{x})}{\partial x_i}:=\lim _{h \rightarrow 0} \frac{f\left(\boldsymbol{x}+h \boldsymbol{e}_i\right)-f(\boldsymbol{x})}{h}, \quad \boldsymbol{x} \in \mathbb{R}^n,$	$D_i f(\boldsymbol{x}):=\frac{\partial f(\boldsymbol{x})}{\partial x_i}:=\lim _{h \rightarrow 0} \frac{f\left(\boldsymbol{x}+h \boldsymbol{e}_i\right)-f(\boldsymbol{x})}{h}, \quad \boldsymbol{x} \in \mathbb{R}^n,$
lyche-numerical-linear-algebra-EQ0027 (1.10)	031	0.998 N +0	$\langle \boldsymbol{x}, \boldsymbol{y}\rangle:=\boldsymbol{y}^* \boldsymbol{x}=\boldsymbol{x}^T \overline{\boldsymbol{y}}=\sum_{j=1}^n x_j \overline{y_j}.$	$\langle \boldsymbol{x}, \boldsymbol{y}\rangle:=\boldsymbol{y}^* \boldsymbol{x}=\boldsymbol{x}^T \overline{\boldsymbol{y}}=\sum_{j=1}^n x_j \overline{y_j}.$	$\langle \boldsymbol{x}, \boldsymbol{y}\rangle:=\boldsymbol{y}^* \boldsymbol{x}=\boldsymbol{x}^T \overline{\boldsymbol{y}}=\sum_{j=1}^n x_j \overline{y_j}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0318 (5.15)	128	■ 0.998 ● W +1	$\begin{aligned} \boldsymbol{A}_{\boldsymbol{k}} &= \left[\begin{array}{cccc} a_{1,1}^{(1)} & \cdots & a_{1,k-1}^{(1)} & \vdots \\ & \ddots & & \vdots \\ & & a_{k-1,k-1}^{(k-1)} & \vdots \\ \hline & & & a_{k,k}^{(k)} \cdots a_{k,j}^{(k)} \cdots a_{k,n}^{(k)} \\ & & & \vdots \\ & & & a_{i,k}^{(k)} \cdots a_{i,j}^{(k)} \cdots a_{i,n}^{(k)} \\ & & & \vdots \\ & & & a_{m,k}^{(k)} \cdots a_{m,j}^{(k)} \cdots a_{m,n}^{(k)} \end{array} \right] \\ &= \begin{bmatrix} \boldsymbol{B}_k & \boldsymbol{C}_k \\ \boldsymbol{0} & \boldsymbol{D}_k \end{bmatrix}. \end{aligned}$	$\boldsymbol{A}_k = \left[\begin{array}{cccc cccc} a_{1,1}^{(1)} & \cdots & a_{1,k-1}^{(1)} & & a_{1,k}^{(1)} & \cdots & a_{1,j}^{(1)} & \cdots & a_{1,n}^{(1)} \\ & & & & \vdots & & \vdots & & \vdots \\ & & & & \vdots & & \vdots & & \vdots \\ & & & & a_{k-1,k}^{(k-1)} & \cdots & a_{k-1,j}^{(k-1)} & \cdots & a_{k-1,n}^{(k-1)} \\ \hline & & & & a_{k,k}^{(k)} & \cdots & a_{k,j}^{(k)} & \cdots & a_{k,n}^{(k)} \\ & & & & \vdots & & \vdots & & \vdots \\ & & & & a_{i,k}^{(k)} & \cdots & a_{i,j}^{(k)} & \cdots & a_{i,n}^{(k)} \\ & & & & \vdots & & \vdots & & \vdots \\ & & & & a_{m,k}^{(k)} & \cdots & a_{m,j}^{(k)} & \cdots & a_{m,n}^{(k)} \end{array} \right] \\ &= \begin{bmatrix} \boldsymbol{B}_k & \boldsymbol{C}_k \\ \boldsymbol{0} & \boldsymbol{D}_k \end{bmatrix}.$	$\boldsymbol{A}_k = \left[\begin{array}{cccc cccc} a_{1,1}^{(1)} & \cdots & a_{1,k-1}^{(1)} & & a_{1,k}^{(1)} & \cdots & a_{1,j}^{(1)} & \cdots & a_{1,n}^{(1)} \\ & & & & \vdots & & \vdots & & \vdots \\ & & & & \vdots & & \vdots & & \vdots \\ & & & & a_{k-1,k}^{(k-1)} & \cdots & a_{k-1,j}^{(k-1)} & \cdots & a_{k-1,n}^{(k-1)} \\ \hline & & & & a_{k,k}^{(k)} & \cdots & a_{k,j}^{(k)} & \cdots & a_{k,n}^{(k)} \\ & & & & \vdots & & \vdots & & \vdots \\ & & & & a_{i,k}^{(k)} & \cdots & a_{i,j}^{(k)} & \cdots & a_{i,n}^{(k)} \\ & & & & \vdots & & \vdots & & \vdots \\ & & & & a_{m,k}^{(k)} & \cdots & a_{m,j}^{(k)} & \cdots & a_{m,n}^{(k)} \end{array} \right] \\ &= \begin{bmatrix} \boldsymbol{B}_k & \boldsymbol{C}_k \\ \boldsymbol{0} & \boldsymbol{D}_k \end{bmatrix}.$
lyche-numerical-linear-algebra- EQ0362 (6.1)	147	■ 0.998 ● N +0	$\pi_{AC}(\lambda) = \lambda^{m-n} \pi_{CA}(\lambda), \quad \lambda \in \mathbb{C}.$	$\pi_{AC}(\lambda) = \lambda^{m-n} \pi_{CA}(\lambda), \quad \lambda \in \mathbb{C}.$	$\pi_{AC}(\lambda) = \lambda^{m-n} \pi_{CA}(\lambda), \quad \lambda \in \mathbb{C}.$
lyche-numerical-linear-algebra- EQ0611	224	■ 0.998 ● N +0	$\boldsymbol{A} := \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix}, \boldsymbol{A}^\dagger = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \end{bmatrix} = \frac{1}{4} \boldsymbol{A}, \boldsymbol{A}^\dagger \boldsymbol{b} = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}.$	$\boldsymbol{A} := \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix}, \boldsymbol{A}^\dagger = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \end{bmatrix} = \frac{1}{4} \boldsymbol{A}, \boldsymbol{A}^\dagger \boldsymbol{b} = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}.$	$\boldsymbol{A} := \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix}, \boldsymbol{A}^\dagger = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \end{bmatrix} = \frac{1}{4} \boldsymbol{A}, \boldsymbol{A}^\dagger \boldsymbol{b} = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}.$
lyche-numerical-linear-algebra- EQ0872 (13.28)	303	■ 0.998 ●	$\frac{\ \boldsymbol{x} - \boldsymbol{x}_k\ _{\boldsymbol{A}}}{\ \boldsymbol{x} - \boldsymbol{x}_0\ _{\boldsymbol{A}}} \leq \min_{\substack{Q \in \Pi_k \\ Q(0)=1}} \max_{a \leq x \leq b} Q(x) ,$	$\frac{\ \boldsymbol{x} - \boldsymbol{x}_k\ _{\boldsymbol{A}}}{\ \boldsymbol{x} - \boldsymbol{x}_0\ _{\boldsymbol{A}}} \leq \min_{\substack{Q \in \Pi_k \\ Q(0)=1}} \max_{a \leq x \leq b} Q(x) ,$	$\frac{\ \boldsymbol{x} - \boldsymbol{x}_k\ _{\boldsymbol{A}}}{\ \boldsymbol{x} - \boldsymbol{x}_0\ _{\boldsymbol{A}}} \leq \min_{\substack{Q \in \Pi_k \\ Q(0)=1}} \max_{a \leq x \leq b} Q(x) ,$
lyche-numerical-linear-algebra- EQ0016 (1.4)	028	■ 0.999 ● N -1	$\mathcal{S} + \mathcal{T} := \{s + t : s \in \mathcal{S} \text{ and } t \in \mathcal{T}\}.$	$\mathcal{S} + \mathcal{T} := \{s + t : s \in \mathcal{S} \text{ and } t \in \mathcal{T}\}.$	$\mathcal{S} + \mathcal{T} := \{s + t : s \in \mathcal{S} \text{ and } t \in \mathcal{T}\}.$
lyche-numerical-linear-algebra- EQ0073	044	■ 0.999 ● N +0	$b_{j,k} = (\alpha_k + \beta_j) A_k (-\beta_j) B_j (-\alpha_k),$	$b_{j,k} = (\alpha_k + \beta_j) A_k (-\beta_j) B_j (-\alpha_k),$	$b_{j,k} = (\alpha_k + \beta_j) A_k (-\beta_j) B_j (-\alpha_k),$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-055 EQ0101 (2.18)	055	0.999 W +1	$\begin{aligned} \left u_{k+1}\right &= \left d_{k+1}-l_k c_k\right \geq\left d_{k+1}-\frac{\left a_k\right \left c_k\right }{\left u_k\right }\right \\ &\geq\left d_{k+1}\right -\left a_k\right =\sigma_{k+1}+\left c_{k+1}\right \end{aligned}$	$\left u_{k+1}\right =\left d_{k+1}-l_k c_k\right =\left d_{k+1}-\frac{a_k c_k}{u_k}\right \geq\left d_{k+1}\right -\frac{\left a_k\right \left c_k\right }{\left u_k\right } \geq\left d_{k+1}\right -\left a_k\right =\sigma_{k+1}+\left c_{k+1}\right .$	
lyche-numerical-linear-algebra-071 EQ0156	071	0.999 N +0	$\Delta^2 f(x):=\frac{f(x+h)-2 f(x)+f(x-h)}{h^2}, \quad h>0, \quad f:[x-h, x+h] \rightarrow \mathbb{R} .$	$\delta^2 f(x):=\frac{f(x+h)-2 f(x)+f(x-h)}{h^2}, \quad h>0, \quad f:[x-h, x+h] \rightarrow \mathbb{R} .$	
lyche-numerical-linear-algebra-102 EQ0250	102	0.999 N +1	$\left[\begin{array}{ll} 3 & 1 \\ 1 & 3 \end{array}\right]=\left[\begin{array}{cc} 1 & 0 \\ 1 / 3 & 1 \end{array}\right]\left[\begin{array}{cc} 3 & 0 \\ 0 & 8 / 3 \end{array}\right]\left[\begin{array}{cc} 1 & 1 / 3 \\ 0 & 1 \end{array}\right]$	$\left[\begin{array}{cc} 3 & 1 \\ 1 & 3 \end{array}\right]=\left[\begin{array}{cc} 1 & 0 \\ 1 / 3 & 1 \end{array}\right]\left[\begin{array}{cc} 3 & 0 \\ 0 & 8 / 3 \end{array}\right]\left[\begin{array}{cc} 1 & 1 / 3 \\ 0 & 1 \end{array}\right]$	
lyche-numerical-linear-algebra-123 EQ0296 (5.11)	123	0.999 N +0	$\left\ \boldsymbol{v}-\boldsymbol{s}_0\right\ <\left\ \boldsymbol{v}-\boldsymbol{s}\right\ , \text { for all } \boldsymbol{s} \in \mathcal{S}, \boldsymbol{s} \neq \boldsymbol{s}_0 .$	$\left\ \boldsymbol{v}-\boldsymbol{s}_0\right\ <\left\ \boldsymbol{v}-\boldsymbol{s}\right\ , \text { for all } \boldsymbol{s} \in \mathcal{S}, \boldsymbol{s} \neq \boldsymbol{s}_0 .$	
lyche-numerical-linear-algebra-147 EQ0361	147	0.999 W +0	$\begin{aligned} \pi_{\boldsymbol{B}}(\lambda) &= \det \left(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S} - \lambda \boldsymbol{I} \right) = \det \left(\boldsymbol{S}^{-1} (\boldsymbol{A} - \lambda \boldsymbol{I}) \boldsymbol{S} \right) \\ &= \det \left(\boldsymbol{S}^{-1} \right) \det (\boldsymbol{A} - \lambda \boldsymbol{I}) \det (\boldsymbol{S}) = \det \left(\boldsymbol{S}^{-1} \boldsymbol{S} \right) \det (\boldsymbol{A} - \lambda \boldsymbol{I}) = \pi_{\boldsymbol{A}}(\lambda), \end{aligned}$	$\begin{aligned} \pi_{\boldsymbol{B}}(\lambda) &= \det (\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S} - \lambda \boldsymbol{I}) = \det \left(\boldsymbol{S}^{-1} (\boldsymbol{A} - \lambda \boldsymbol{I}) \boldsymbol{S} \right) \\ &= \det (\boldsymbol{S}^{-1}) \det (\boldsymbol{A} - \lambda \boldsymbol{I}) \det (\boldsymbol{S}) = \det (\boldsymbol{S}^{-1} \boldsymbol{S}) \det (\boldsymbol{A} - \lambda \boldsymbol{I}) = \pi_{\boldsymbol{A}}(\lambda), \end{aligned}$	
lyche-numerical-linear-algebra-154 EQ0381	154	0.999 W -1	$e_{ii}=\sum_{k=1}^n \bar{b}_{k i} b_{k i}=\sum_{k=1}^i\left b_{k i}\right ^2 \text { and } f_{i i}=\sum_{k=1}^n b_{i k} \bar{b}_{i k}=\sum_{k=i}^n\left b_{i k}\right ^2 .$	$e_{ii}=\sum_{k=1}^n \bar{b}_{k i} b_{k i}=\sum_{k=1}^i\left b_{k i}\right ^2 \text { and } f_{i i}=\sum_{k=1}^n b_{i k} \bar{b}_{i k}=\sum_{k=i}^n\left b_{i k}\right ^2 .$	
lyche-numerical-linear-algebra-162 EQ0406	162	0.999	$\boldsymbol{A}=\left[\begin{array}{ccccc} -q_{n-1} & -q_{n-2} & \cdots & -q_1 & -q_0 \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{array}\right] .$	$\boldsymbol{A}=\left[\begin{array}{ccccc} -q_{n-1} & -q_{n-2} & \cdots & -q_1 & -q_0 \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{array}\right] .$	
Conf. MathPix confidence: 0.999 0.5-0.9 <0.5 no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0446 (7.10)	178	0.999 N +0	$\boldsymbol{A}:=\left[\begin{array}{ll}1&1\\1&1\end{array}\right]=\frac{1}{\sqrt{2}}\left[\begin{array}{ll}1&1\\1&-1\end{array}\right]\frac{1}{\sqrt{2}}\left[\begin{array}{cc}1&1\\1&-1\end{array}\right]=\boldsymbol{U}\boldsymbol{D}\boldsymbol{U}^T$	$\boldsymbol{A}:=\left[\begin{array}{ll}1&1\\1&1\end{array}\right]=\frac{1}{\sqrt{2}}\left[\begin{array}{ll}1&1\\1&-1\end{array}\right]\left[\begin{array}{ll}2&0\\0&0\end{array}\right]\frac{1}{\sqrt{2}}\left[\begin{array}{ll}1&1\\1&-1\end{array}\right]=\boldsymbol{U}\boldsymbol{D}\boldsymbol{U}^T$	$\boldsymbol{A}:=\left[\begin{array}{ll}1&1\\1&1\end{array}\right]=\frac{1}{\sqrt{2}}\left[\begin{array}{ll}1&1\\1&-1\end{array}\right]\left[\begin{array}{ll}2&0\\0&0\end{array}\right]\frac{1}{\sqrt{2}}\left[\begin{array}{ll}1&1\\1&-1\end{array}\right]=\boldsymbol{U}\boldsymbol{D}\boldsymbol{U}^T$
lyche-numerical-linear-algebra- EQ0471 (8.7)	187	0.999 N +0	$m\ \boldsymbol{x}\ '\leq\ \boldsymbol{x}\ \leq M\ \boldsymbol{x}\ '.$	$m\ \boldsymbol{x}\ '\leq\ \boldsymbol{x}\ \leq M\ \boldsymbol{x}\ '.$	$m\ \boldsymbol{x}\ '\leq\ \boldsymbol{x}\ \leq M\ \boldsymbol{x}\ '.$
lyche-numerical-linear-algebra- EQ0481 (8.14)	190	0.999 N +0	$\mathcal{S}:=\{x\in\mathbb{C}^n:\ \boldsymbol{x}\ =1\}$	$\mathcal{S}:=\{x\in\mathbb{C}^n:\ \boldsymbol{x}\ =1\}$	$\mathcal{S}:=\{x\in\mathbb{C}^n:\ \boldsymbol{x}\ =1\}$
lyche-numerical-linear-algebra- EQ0538	204	0.999 N +0	$\ \boldsymbol{x}\ _{\boldsymbol{A}}:=\sqrt{\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x}},\quad\boldsymbol{x}\in\mathbb{R}^n.$	$\ \boldsymbol{x}\ _{\boldsymbol{A}}:=\sqrt{\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x}},\quad\boldsymbol{x}\in\mathbb{R}^n.$	$\ \boldsymbol{x}\ _{\boldsymbol{A}}:=\sqrt{\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x}},\quad\boldsymbol{x}\in\mathbb{R}^n.$
lyche-numerical-linear-algebra- EQ0634	230	0.999 N +0	$\ \boldsymbol{C}-\boldsymbol{D}\ _F^2=\left\ \left[\begin{array}{cc}\boldsymbol{0}&\boldsymbol{A}-\boldsymbol{B}\\\boldsymbol{A}^*-\boldsymbol{B}^*&\boldsymbol{0}\end{array}\right]\right\ _F^2=\ \boldsymbol{B}-\boldsymbol{A}\ _F^2+\ (\boldsymbol{B}-\boldsymbol{A})^*\ _F^2=2\ \boldsymbol{B}-\boldsymbol{A}\ _F^2.$	$\ \boldsymbol{C}-\boldsymbol{D}\ _F^2=\left\ \left[\begin{array}{cc}\boldsymbol{0}&\boldsymbol{A}-\boldsymbol{B}\\\boldsymbol{A}^*-\boldsymbol{B}^*&\boldsymbol{0}\end{array}\right]\right\ _F^2=\ \boldsymbol{B}-\boldsymbol{A}\ _F^2+\ (\boldsymbol{B}-\boldsymbol{A})^*\ _F^2=2\ \boldsymbol{B}-\boldsymbol{A}\ _F^2.$	$\ \boldsymbol{C}-\boldsymbol{D}\ _F^2=\left\ \left[\begin{array}{cc}\boldsymbol{0}&\boldsymbol{A}-\boldsymbol{B}\\\boldsymbol{A}^*-\boldsymbol{B}^*&\boldsymbol{0}\end{array}\right]\right\ _F^2=\ \boldsymbol{B}-\boldsymbol{A}\ _F^2+\ (\boldsymbol{B}-\boldsymbol{A})^*\ _F^2=2\ \boldsymbol{B}-\boldsymbol{A}\ _F^2.$
lyche-numerical-linear-algebra- EQ0816	284	0.999 K +0	$\frac{1}{4}\left(\lambda^{1/2}w_{i-1,j}+\lambda^{1/2}w_{i,j-1}+\lambda^{-1/2}w_{i+1,j}+\lambda^{-1/2}w_{i,j+1}\right)=\mu w_{i,j}.$	$\frac{1}{4}\left(\lambda^{1/2}w_{i-1,j}+\lambda^{1/2}w_{i,j-1}+\lambda^{-1/2}w_{i+1,j}+\lambda^{-1/2}w_{i,j+1}\right)=\mu w_{i,j}.$	$\frac{1}{4}(\lambda^{1/2}w_{i-1,j}+\lambda^{1/2}w_{i,j-1}+\lambda^{-1/2}w_{i+1,j}+\lambda^{-1/2}w_{i,j+1})=\mu w_{i,j}.$
lyche-numerical-linear-algebra- EQ0861	300	0.999 N +0	$\kappa_A=\frac{5+4\cos(\pi h)}{5-4\cos(\pi h)}\leq 9,$	$\kappa_A=\frac{5+4\cos(\pi h)}{5-4\cos(\pi h)}\leq 9,$	$\kappa_A=\frac{5+4\cos(\pi h)}{5-4\cos(\pi h)}\leq 9,$
lyche-numerical-linear-algebra- EQ0954 (13.67)	322	0.999 N -1	$\left\ \boldsymbol{x}^*-\boldsymbol{x}\right\ _2\leq\ \boldsymbol{A}\ _2\min_{\boldsymbol{w}\in\mathcal{W}}\ \boldsymbol{x}^*-\boldsymbol{w}\ _2.$	$\ \boldsymbol{x}^*-\boldsymbol{x}\ _2\leq\ \boldsymbol{A}\ _2\min_{\boldsymbol{w}\in\mathcal{W}}\ \boldsymbol{x}^*-\boldsymbol{w}\ _2.$	$\ \boldsymbol{x}^*-\boldsymbol{x}\ _2\leq\ \boldsymbol{A}\ _2\min_{\boldsymbol{w}\in\mathcal{W}}\ \boldsymbol{x}^*-\boldsymbol{w}\ _2.$
lyche-numerical-linear-algebra- EQ0208	086	0.999	$\begin{aligned} &(\boldsymbol{L}\boldsymbol{U})_{ij}=\sum_{k=1}^n l_{ik}^{(k)}u_{kj}=\sum_{k=1}^{j-1}a_{pi,k}^{(k)}a_{pk,j}^{(k)}+a_{pi,j}^{(k)}a_{pj,j}^{(j)}\\ &=\sum_{k=1}^{j-1}\left(a_{pi,j}^{(k)}-a_{pi,j}^{(k+1)}\right)+a_{pi,j}^{(j)}=a_{pi,j}^{(1)}=a_{pi,j}=\left(\boldsymbol{P}^T\boldsymbol{A}\right)_{ij}. \end{aligned}$	$(\boldsymbol{L}\boldsymbol{U})_{ij}=\sum_{k=1}^n l_{ik}^{(k)}u_{kj}=\sum_{k=1}^{j-1}a_{pi,k}^{(k)}a_{pk,j}^{(k)}+a_{pi,j}^{(k)}a_{pj,j}^{(j)}\\ =\sum_{k=1}^{j-1}\left(a_{pi,j}^{(k)}-a_{pi,j}^{(k+1)}\right)+a_{pi,j}^{(j)}=a_{pi,j}^{(1)}=a_{pi,j}=\left(\boldsymbol{P}^T\boldsymbol{A}\right)_{ij}.$	$(\boldsymbol{L}\boldsymbol{U})_{ij}=\sum_{k=1}^n l_{ik}^{(k)}u_{kj}=\sum_{k=1}^{j-1}a_{pi,k}^{(k)}a_{pk,j}^{(k)}+a_{pi,j}^{(k)}a_{pj,j}^{(j)}\\ =\sum_{k=1}^{j-1}\left(a_{pi,j}^{(k)}-a_{pi,j}^{(k+1)}\right)+a_{pi,j}^{(j)}=a_{pi,j}^{(1)}=a_{pi,j}=\left(\boldsymbol{P}^T\boldsymbol{A}\right)_{ij}.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0265 (4.9)	109	0.999 W +0	$\begin{aligned} 0 &\leq \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \left[-\boldsymbol{y}^* \boldsymbol{v} / \alpha, \boldsymbol{y}^* \right] \begin{bmatrix} \alpha & \boldsymbol{v}^* \\ \boldsymbol{v} & \boldsymbol{B} \end{bmatrix} \begin{bmatrix} -\boldsymbol{v}^* \boldsymbol{y} / \alpha \\ \boldsymbol{y} \end{bmatrix} \\ &= [0, -(\boldsymbol{y}^* \boldsymbol{v}) \boldsymbol{v}^* / \alpha + \boldsymbol{y}^* \boldsymbol{B} \boldsymbol{y}] \begin{bmatrix} -\boldsymbol{v}^* \boldsymbol{y} / \alpha \\ \boldsymbol{y} \end{bmatrix} \\ &= -\boldsymbol{y}^* \boldsymbol{v} \boldsymbol{v}^* \boldsymbol{y} / \alpha + \boldsymbol{y}^* \boldsymbol{B} \boldsymbol{y} = \boldsymbol{y}^* \boldsymbol{C} \boldsymbol{y}. \end{aligned}$	$0 \leq \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = [-\boldsymbol{y}^* \boldsymbol{v} / \alpha, \boldsymbol{y}^*] \begin{bmatrix} \alpha & \boldsymbol{v}^* \\ \boldsymbol{v} & \boldsymbol{B} \end{bmatrix} \begin{bmatrix} -\boldsymbol{v}^* \boldsymbol{y} / \alpha \\ \boldsymbol{y} \end{bmatrix}$ $= [0, -(\boldsymbol{y}^* \boldsymbol{v}) \boldsymbol{v}^* / \alpha + \boldsymbol{y}^* \boldsymbol{B} \boldsymbol{y}] \begin{bmatrix} -\boldsymbol{v}^* \boldsymbol{y} / \alpha \\ \boldsymbol{y} \end{bmatrix}$ $= -\boldsymbol{y}^* \boldsymbol{v} \boldsymbol{v}^* \boldsymbol{y} / \alpha + \boldsymbol{y}^* \boldsymbol{B} \boldsymbol{y} = \boldsymbol{y}^* \boldsymbol{C} \boldsymbol{y}.$	$0 \leq \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = [-\boldsymbol{y}^* \boldsymbol{v} / \alpha, \boldsymbol{y}^*] \begin{bmatrix} \alpha & \boldsymbol{v}^* \\ \boldsymbol{v} & \boldsymbol{B} \end{bmatrix} \begin{bmatrix} -\boldsymbol{v}^* \boldsymbol{y} / \alpha \\ \boldsymbol{y} \end{bmatrix}$ $= [0, -(\boldsymbol{y}^* \boldsymbol{v}) \boldsymbol{v}^* / \alpha + \boldsymbol{y}^* \boldsymbol{B} \boldsymbol{y}] \begin{bmatrix} -\boldsymbol{v}^* \boldsymbol{y} / \alpha \\ \boldsymbol{y} \end{bmatrix}$ $= -\boldsymbol{y}^* \boldsymbol{v} \boldsymbol{v}^* \boldsymbol{y} / \alpha + \boldsymbol{y}^* \boldsymbol{B} \boldsymbol{y} = \boldsymbol{y}^* \boldsymbol{C} \boldsymbol{y}.$
lyche-numerical-linear-algebra- EQ0309	126	0.999 C -12	$\begin{gathered} \boldsymbol{H} := \boldsymbol{I} - \frac{2}{\boldsymbol{v}^T \boldsymbol{v}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{2}{4} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \\ \boldsymbol{P} := \boldsymbol{I} - \frac{\boldsymbol{v} \boldsymbol{v}^T}{\boldsymbol{v}^T \boldsymbol{v}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{1}{4} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \end{gathered}$	$\boldsymbol{H} := \boldsymbol{I} - \frac{2\boldsymbol{v}\boldsymbol{v}^T}{\boldsymbol{v}^T\boldsymbol{v}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{2}{4} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$ $\boldsymbol{P} := \boldsymbol{I} - \frac{\boldsymbol{v}\boldsymbol{v}^T}{\boldsymbol{v}^T\boldsymbol{v}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{1}{4} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$	$\boldsymbol{H} := \boldsymbol{I} - \frac{2\boldsymbol{v}\boldsymbol{v}^T}{\boldsymbol{v}^T\boldsymbol{v}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{2}{4} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$ $\boldsymbol{P} := \boldsymbol{I} - \frac{\boldsymbol{v}\boldsymbol{v}^T}{\boldsymbol{v}^T\boldsymbol{v}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{1}{4} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$
lyche-numerical-linear-algebra- EQ0645	232	0.999	$\begin{aligned} \boldsymbol{B} &= (\boldsymbol{B} \boldsymbol{A}) \boldsymbol{B} = (\boldsymbol{A}^*) \boldsymbol{B}^* \boldsymbol{B} = (\boldsymbol{A}^* \boldsymbol{C}^*) \boldsymbol{A}^* \boldsymbol{B}^* \boldsymbol{B} = \boldsymbol{C} \boldsymbol{A} (\boldsymbol{A}^* \boldsymbol{B}^*) \boldsymbol{B} \\ &= \boldsymbol{C} \boldsymbol{A} (\boldsymbol{B} \boldsymbol{A} \boldsymbol{B}) = (\boldsymbol{C}) \boldsymbol{A} \boldsymbol{B} = \boldsymbol{C} (\boldsymbol{A} \boldsymbol{C}) \boldsymbol{A} \boldsymbol{B} = \boldsymbol{C} \boldsymbol{C}^* \boldsymbol{A}^* (\boldsymbol{A} \boldsymbol{B}) \\ &= \boldsymbol{C} \boldsymbol{C}^* (\boldsymbol{A}^* \boldsymbol{B}^* \boldsymbol{A}^*) = \boldsymbol{C} (\boldsymbol{C}^* \boldsymbol{A}^*) = \boldsymbol{C} \boldsymbol{A} \boldsymbol{C} = \boldsymbol{C}. \end{aligned}$	$\boldsymbol{B} = (\boldsymbol{B} \boldsymbol{A}) \boldsymbol{B} = (\boldsymbol{A}^*) \boldsymbol{B}^* \boldsymbol{B} = (\boldsymbol{A}^* \boldsymbol{C}^*) \boldsymbol{A}^* \boldsymbol{B}^* \boldsymbol{B} = \boldsymbol{C} \boldsymbol{A} (\boldsymbol{A}^* \boldsymbol{B}^*) \boldsymbol{B}$ $= \boldsymbol{C} \boldsymbol{A} (\boldsymbol{B} \boldsymbol{A} \boldsymbol{B}) = (\boldsymbol{C}) \boldsymbol{A} \boldsymbol{B} = \boldsymbol{C} (\boldsymbol{A} \boldsymbol{C}) \boldsymbol{A} \boldsymbol{B} = \boldsymbol{C} \boldsymbol{C}^* \boldsymbol{A}^* (\boldsymbol{A} \boldsymbol{B})$ $= \boldsymbol{C} \boldsymbol{C}^* (\boldsymbol{A}^* \boldsymbol{B}^* \boldsymbol{A}^*) = \boldsymbol{C} (\boldsymbol{C}^* \boldsymbol{A}^*) = \boldsymbol{C} \boldsymbol{A} \boldsymbol{C} = \boldsymbol{C}.$	$\boldsymbol{B} = (\boldsymbol{B} \boldsymbol{A}) \boldsymbol{B} = (\boldsymbol{A}^*) \boldsymbol{B}^* \boldsymbol{B} = (\boldsymbol{A}^* \boldsymbol{C}^*) \boldsymbol{A}^* \boldsymbol{B}^* \boldsymbol{B} = \boldsymbol{C} \boldsymbol{A} (\boldsymbol{A}^* \boldsymbol{B}^*) \boldsymbol{B}$ $= \boldsymbol{C} \boldsymbol{A} (\boldsymbol{B} \boldsymbol{A} \boldsymbol{B}) = (\boldsymbol{C}) \boldsymbol{A} \boldsymbol{B} = \boldsymbol{C} (\boldsymbol{A} \boldsymbol{C}) \boldsymbol{A} \boldsymbol{B} = \boldsymbol{C} \boldsymbol{C}^* \boldsymbol{A}^* (\boldsymbol{A} \boldsymbol{B})$ $= \boldsymbol{C} \boldsymbol{C}^* (\boldsymbol{A}^* \boldsymbol{B}^* \boldsymbol{A}^*) = \boldsymbol{C} (\boldsymbol{C}^* \boldsymbol{A}^*) = \boldsymbol{C} \boldsymbol{A} \boldsymbol{C} = \boldsymbol{C}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0827	287	0.999 N +0	$\boldsymbol{x}:=\left[\begin{array}{l} x_1 \\ x_2 \end{array}\right]=\left[\begin{array}{cc} 0 & a \\ a & 0 \end{array}\right]\left[\begin{array}{l} x_1 \\ x_2 \end{array}\right]+\left[\begin{array}{l} 1-a \\ 1-a \end{array}\right]=:\boldsymbol{G}\boldsymbol{x}+\boldsymbol{c}.$	$\boldsymbol{x}:=\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}=\begin{bmatrix} 0 & a \\ a & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}+\begin{bmatrix} 1-a \\ 1-a \end{bmatrix}=:\boldsymbol{G}\boldsymbol{x}+\boldsymbol{c}.$	
lyche-numerical-linear-algebra-EQ0942	319	0.999 N +0	$\mathbb{W}_k:=\operatorname{span}\bigl(\boldsymbol{b},\boldsymbol{A}\boldsymbol{b},\dots,\boldsymbol{A}^{k-1}\boldsymbol{b}\bigr).$	$\mathbb{W}_k:=\operatorname{span}(\boldsymbol{b},\boldsymbol{A}\boldsymbol{b},\dots,\boldsymbol{A}^{k-1}\boldsymbol{b}).$	
lyche-numerical-linear-algebra-EQ0062 (1.21)	041	0.999 K +0	$\overline{\boldsymbol{C}}=\boldsymbol{C}\left(\boldsymbol{I}+\frac{1}{\lambda}\boldsymbol{e}_i\left(\boldsymbol{e}_i^T-\boldsymbol{a}^T\boldsymbol{C}\right)\right)$	$\overline{\boldsymbol{C}}=\boldsymbol{C}(\boldsymbol{I}+\frac{1}{\lambda}\boldsymbol{e}_i(\boldsymbol{e}_i^T-\boldsymbol{a}^T\boldsymbol{C}))$	
lyche-numerical-linear-algebra-EQ0202 (3.12)	085	0.999 N +0	$\begin{aligned} \boldsymbol{P}^T&=\boldsymbol{P}_{n-1}\cdots\boldsymbol{P}_1=\boldsymbol{I}(\boldsymbol{p},:),\quad \boldsymbol{P}=\boldsymbol{P}_1\boldsymbol{P}_2\cdots\boldsymbol{P}_{n-1}=\boldsymbol{I}(:,\boldsymbol{p}). \\ \boldsymbol{P}&=\boldsymbol{P}_1\boldsymbol{P}_2\cdots\boldsymbol{P}_{n-1}=\boldsymbol{I}(:,\boldsymbol{p}). \end{aligned}$	$\boldsymbol{P}^T=\boldsymbol{P}_{n-1}\cdots\boldsymbol{P}_1=\boldsymbol{I}(\boldsymbol{p},:),\quad \boldsymbol{P}=\boldsymbol{P}_1\boldsymbol{P}_2\cdots\boldsymbol{P}_{n-1}=\boldsymbol{I}(:,\boldsymbol{p}).$	
lyche-numerical-linear-algebra-EQ0254	104	0.999 W +0	$\begin{aligned} &\boldsymbol{x}^T\boldsymbol{T}\boldsymbol{x}=2\sum_{i=1}^nx_i^2-\sum_{i=1}^{n-1}x_ix_{i+1}-\sum_{i=2}^nx_{i-1}x_i \\ &=\sum_{i=1}^{n-1}x_i^2-2\sum_{i=1}^{n-1}x_ix_{i+1}+\sum_{i=1}^{n-1}x_{i+1}^2+x_1^2+x_n^2 \\ &=x_1^2+x_n^2+\sum_{i=1}^{n-1}(x_{i+1}-x_i)^2. \end{aligned}$	$\begin{aligned} \boldsymbol{x}^T\boldsymbol{T}\boldsymbol{x} &= 2\sum_{i=1}^n x_i^2 - \sum_{i=1}^{n-1} x_i x_{i+1} - \sum_{i=2}^n x_{i-1} x_i \\ &= \sum_{i=1}^{n-1} x_i^2 - 2\sum_{i=1}^{n-1} x_i x_{i+1} + \sum_{i=1}^{n-1} x_{i+1}^2 + x_1^2 + x_n^2 \\ &= x_1^2 + x_n^2 + \sum_{i=1}^{n-1} (x_{i+1} - x_i)^2. \end{aligned}$	
lyche-numerical-linear-algebra-EQ0286	119	0.999 N +0	$\ \boldsymbol{x}+\boldsymbol{y}\ ^2\leq\ \boldsymbol{x}\ ^2+2\ \boldsymbol{x}\ \ \boldsymbol{y}\ +\ \boldsymbol{y}\ ^2=(\ \boldsymbol{x}\ +\ \boldsymbol{y}\)^2.$	$\ \boldsymbol{x}+\boldsymbol{y}\ ^2\leq\ \boldsymbol{x}\ ^2+2\ \boldsymbol{x}\ \ \boldsymbol{y}\ +\ \boldsymbol{y}\ ^2=(\ \boldsymbol{x}\ +\ \boldsymbol{y}\)^2.$	
lyche-numerical-linear-algebra-EQ0310	126	0.999 W +0	$\mathcal{M}:=\left\{\boldsymbol{w}\in\mathbb{R}^3:\boldsymbol{w}^T\boldsymbol{v}=0\right\}=\left\{\left[\begin{array}{l} w_1 \\ w_2 \\ w_3 \end{array}\right]:2w_1=0\right\}$	$\mathcal{M}:=\{\boldsymbol{w}\in\mathbb{R}^3:\boldsymbol{w}^T\boldsymbol{v}=0\}=\left\{\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}:2w_1=0\right\}$	
lyche-numerical-linear-algebra-EQ0348	141	0.999	$\left[\begin{array}{ll} p_{ii} & p_{ij} \\ p_{ji} & p_{jj} \end{array}\right]=\left[\begin{array}{cc} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{array}\right],$	$\begin{bmatrix} p_{ii} & p_{ij} \\ p_{ji} & p_{jj} \end{bmatrix}=\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix},$	
Conf. MathPix confidence: 0.999 0.5–0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0412 (6.20)	165	0.999 N +0	$\boldsymbol{x}=\sum_{j=1}^n c_{\{j\}} \boldsymbol{u}_{\{j\}},$ $\quad \boldsymbol{A} \boldsymbol{x}=\sum_{j=1}^n c_{\{j\}} \lambda_{\{j\}} \boldsymbol{u}_{\{j\}},$ $\text{ where } c_{\{j\}}=\frac{\boldsymbol{u}_{\{j\}}^* \boldsymbol{x}}{\boldsymbol{u}_{\{j\}}^* \boldsymbol{u}_{\{j\}}}$	$\boldsymbol{x}=\sum_{j=1}^n c_j \boldsymbol{u}_j, \quad \boldsymbol{A} \boldsymbol{x}=\sum_{j=1}^n c_j \lambda_j \boldsymbol{u}_j, \text{ where } c_j=\frac{\boldsymbol{u}_j^* \boldsymbol{x}}{\boldsymbol{u}_j^* \boldsymbol{u}_j}.$	$\boldsymbol{x}=\sum_{j=1}^n c_j \boldsymbol{u}_j, \quad \boldsymbol{A} \boldsymbol{x}=\sum_{j=1}^n c_j \lambda_j \boldsymbol{u}_j, \text{ where } c_j=\frac{\boldsymbol{u}_j^* \boldsymbol{x}}{\boldsymbol{u}_j^* \boldsymbol{u}_j}.$
lyche-numerical-linear-algebra EQ0440	176	0.999 C -4	$\frac{\left(\frac{3}{5} x_1+\frac{4}{5} x_2\right)^2}{9}+\frac{\left(-\frac{4}{5} x_1+\frac{3}{5} x_2\right)^2}{1}=1,$	$\frac{\left(\frac{3}{5} x_1+\frac{4}{5} x_2\right)^2}{9}+\frac{\left(-\frac{4}{5} x_1+\frac{3}{5} x_2\right)^2}{1}=1,$	$\frac{\left(\frac{3}{5} x_1+\frac{4}{5} x_2\right)^2}{9}+\frac{\left(-\frac{4}{5} x_1+\frac{3}{5} x_2\right)^2}{1}=1,$
lyche-numerical-linear-algebra EQ0499	195	0.999 K +0	$x_{\{1\}}+\left(1+10^{-16}\right) x_{\{2\}}=20-10^{-15},$	$x_1+\left(1+10^{-16}\right) x_2=20-10^{-15},$	$x_1+\left(1+10^{-16}\right) x_2=20-10^{-15},$
lyche-numerical-linear-algebra EQ0510	198	0.999 K +0	$\left \boldsymbol{B}^{-1}\right \leq\left \boldsymbol{A}^{-1}\right +\left \boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{B}^{-1}\right \leq\left \boldsymbol{A}^{-1}\right +r\left \boldsymbol{B}^{-1}\right .$	$\left\ \boldsymbol{B}^{-1}\right\ \leq\left\ \boldsymbol{A}^{-1}\right\ +\left\ \boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{B}^{-1}\right\ \leq\left\ \boldsymbol{A}^{-1}\right\ +r\left\ \boldsymbol{B}^{-1}\right\ .$	$\left\ \boldsymbol{B}^{-1}\right\ \leq\left\ \boldsymbol{A}^{-1}\right\ +\left\ \boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{B}^{-1}\right\ \leq\left\ \boldsymbol{A}^{-1}\right\ +r\left\ \boldsymbol{B}^{-1}\right\ .$
lyche-numerical-linear-algebra EQ0559	209	0.999 N +0	$K\left(\boldsymbol{B}\right)^{-1} \frac{\left\ \boldsymbol{E}\right\ }{\left\ \boldsymbol{A}\right\ } \leq \frac{\left\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\ }{\left\ \boldsymbol{B}^{-1}\right\ }.$	$K(\boldsymbol{B})^{-1} \frac{\ \boldsymbol{E}\ }{\ \boldsymbol{A}\ } \leq \frac{\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\ }{\ \boldsymbol{B}^{-1}\ }.$	$K(\boldsymbol{B})^{-1} \frac{\ \boldsymbol{E}\ }{\ \boldsymbol{A}\ } \leq \frac{\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\ }{\ \boldsymbol{B}^{-1}\ }.$
lyche-numerical-linear-algebra EQ0617 (9.14)	226	0.999 N +0	$\frac{1}{4} 10^{-2} \leq \frac{\left\ \boldsymbol{y}-\boldsymbol{x}\right\ _{\infty}}{\left\ \boldsymbol{x}\right\ _{\infty}} \leq 4 \cdot 10^{-2}.$	$\frac{1}{4} 10^{-2} \leq \frac{\ \boldsymbol{y}-\boldsymbol{x}\ _{\infty}}{\ \boldsymbol{x}\ _{\infty}} \leq 4 \cdot 10^{-2}.$	$\frac{1}{4} 10^{-2} \leq \frac{\ \boldsymbol{y}-\boldsymbol{x}\ _{\infty}}{\ \boldsymbol{x}\ _{\infty}} \leq 4 \cdot 10^{-2}.$
lyche-numerical-linear-algebra EQ0661 (10.5)	239	0.999 N +0	$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}=\boldsymbol{h}^2 \boldsymbol{F}$ with $\boldsymbol{h}=1 /(m+1),$	$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}=\boldsymbol{h}^2 \boldsymbol{F} \quad \text { with } \quad \boldsymbol{h}=1 /(m+1),$	$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}=\boldsymbol{h}^2 \boldsymbol{F} \quad \text { with } \quad \boldsymbol{h}=1 /(m+1),$
lyche-numerical-linear-algebra EQ0723	258	0.999 K +0	$\boldsymbol{D}_{\{2\}}=\operatorname{diag}\left(1,\omega_{\{4\}}\right)=\left[\begin{array}{rr} 1 & 0 \\ 0 & -i \end{array}\right].$	$\boldsymbol{D}_2=\operatorname{diag}\left(1, \omega_4\right)=\left[\begin{array}{cc} 1 & 0 \\ 0 & -i \end{array}\right].$	$\boldsymbol{D}_2=\operatorname{diag}\left(1, \omega_4\right)=\left[\begin{array}{cc} 1 & 0 \\ 0 & -i \end{array}\right].$
lyche-numerical-linear-algebra EQ0846	293	0.999 N +0	$\begin{aligned} & \& \boldsymbol{t}_{\{0\}}=3 \left[\begin{array}{c} 1 \\ -1 / 2 \end{array}\right], \quad \alpha_{\{0\}}=\frac{1}{2}, \quad \boldsymbol{x}_{\{1\}}=-4^{-1}\left[\begin{array}{c} 1 \\ 2 \end{array}\right], \quad \boldsymbol{r}_{\{1\}}=3 * 4^{-1}\left[\begin{array}{c} 0 \\ 1 \end{array}\right] \\ & \& \boldsymbol{t}_{\{1\}}=3 * 4^{-1}\left[\begin{array}{c} -1 \\ 2 \end{array}\right], \quad \alpha_{\{1\}}=\frac{1}{2}, \quad \boldsymbol{x}_{\{2\}}=-4^{-1}\left[\begin{array}{c} 1 \\ 1 / 2 \end{array}\right], \quad \boldsymbol{r}_{\{2\}}=3 * 4^{-1}\left[\begin{array}{c} 1 / 2 \\ 0 \end{array}\right], \end{aligned}$	$\begin{aligned} & \& \boldsymbol{t}_{\{0\}}=3 \left[\begin{array}{c} 1 \\ -1 / 2 \end{array}\right], \quad \alpha_{\{0\}}=\frac{1}{2}, \quad \boldsymbol{x}_{\{1\}}=-4^{-1}\left[\begin{array}{c} 1 \\ 2 \end{array}\right], \quad \boldsymbol{r}_{\{1\}}=3 * 4^{-1}\left[\begin{array}{c} 0 \\ 1 \end{array}\right] \\ & \& \boldsymbol{t}_{\{1\}}=3 * 4^{-1}\left[\begin{array}{c} -1 \\ 2 \end{array}\right], \quad \alpha_{\{1\}}=\frac{1}{2}, \quad \boldsymbol{x}_{\{2\}}=-4^{-1}\left[\begin{array}{c} 1 \\ 1 / 2 \end{array}\right], \quad \boldsymbol{r}_{\{2\}}=3 * 4^{-1}\left[\begin{array}{c} 1 / 2 \\ 0 \end{array}\right], \end{aligned}$	$\begin{aligned} & \& \boldsymbol{t}_{\{0\}}=3 \left[\begin{array}{c} 1 \\ -1 / 2 \end{array}\right], \quad \alpha_{\{0\}}=\frac{1}{2}, \quad \boldsymbol{x}_{\{1\}}=-4^{-1}\left[\begin{array}{c} 1 \\ 2 \end{array}\right], \quad \boldsymbol{r}_{\{1\}}=3 * 4^{-1}\left[\begin{array}{c} 0 \\ 1 \end{array}\right] \\ & \& \boldsymbol{t}_{\{1\}}=3 * 4^{-1}\left[\begin{array}{c} -1 \\ 2 \end{array}\right], \quad \alpha_{\{1\}}=\frac{1}{2}, \quad \boldsymbol{x}_{\{2\}}=-4^{-1}\left[\begin{array}{c} 1 \\ 1 / 2 \end{array}\right], \quad \boldsymbol{r}_{\{2\}}=3 * 4^{-1}\left[\begin{array}{c} 1 / 2 \\ 0 \end{array}\right], \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-alignatEQ0899 (13.40)	308	■ 0.999	$\frac{\left \left \boldsymbol{\epsilon}_{k+1}\right \right _{\boldsymbol{A}}^2}{\left \left \boldsymbol{\epsilon}_k\right \right _{\boldsymbol{A}}^2}\leq\left(\frac{\kappa-1}{\kappa+1}\right)^2,\quad k=0,1,2,\ldots,$	$\frac{\left\ \boldsymbol{\epsilon}_{k+1}\right\ _{\boldsymbol{A}}^2}{\left\ \boldsymbol{\epsilon}_k\right\ _{\boldsymbol{A}}^2}\leq\left(\frac{\kappa-1}{\kappa+1}\right)^2,\quad k=0,1,2,\ldots,$	
lyche-numerical-linear-alignatEQ0937	317	■ 0.999 K +0	$c_{\{0\}}\left(x^T\boldsymbol{A}_{\{p\}}x\right)\leq x^T\boldsymbol{A}x\leq c_{\{1\}}\left(x^T\boldsymbol{A}_{\{p\}}x\right)$	$c_0\left(x^T\boldsymbol{A}_px\right)\leq x^T\boldsymbol{A}x\leq c_1\left(x^T\boldsymbol{A}_px\right)$	$c_0(x^T\boldsymbol{A}_px)\leq x^T\boldsymbol{A}x\leq c_1(x^T\boldsymbol{A}_px)$
lyche-numerical-linear-alignatEQ0945	320	■ 0.999 K +0	$\boldsymbol{A}\boldsymbol{p}_{k-1}\in\operatorname{span}\left(\boldsymbol{p}_{k-2},\boldsymbol{p}_{k-1},\boldsymbol{p}_k\right).$	$\boldsymbol{A}\boldsymbol{p}_{k-1}\in\operatorname{span}\left(\boldsymbol{p}_{k-2},\boldsymbol{p}_{k-1},\boldsymbol{p}_k\right).$	$\boldsymbol{A}\boldsymbol{p}_{k-1}\in\operatorname{span}(\boldsymbol{p}_{k-2},\boldsymbol{p}_{k-1},\boldsymbol{p}_k).$
lyche-numerical-linear-alignatEQ0960 (13.69)	323	■ 0.999 W +0	$\begin{aligned}&\&\boldsymbol{x}_{k+1}=\left(1-\omega_k\right)\boldsymbol{x}_{k-1}+\omega_k\boldsymbol{x}_{k-1}+\omega_k\boldsymbol{r}_k\\&\boldsymbol{r}_{k+1}=\left(1-\omega_k\right)\boldsymbol{r}_{k-1}+\omega_k\boldsymbol{B}\boldsymbol{r}_k,\end{aligned}$	$\begin{aligned}\boldsymbol{x}_{k+1}&=\left(1-\omega_k\right)\boldsymbol{x}_{k-1}+\omega_k\left(\boldsymbol{x}_k+\boldsymbol{r}_k\right),\\ \boldsymbol{r}_{k+1}&=\left(1-\omega_k\right)\boldsymbol{r}_{k-1}+\omega_k\boldsymbol{B}\boldsymbol{r}_k,\end{aligned}$	$\begin{aligned}\boldsymbol{x}_{k+1}&=\left(1-\omega_k\right)\boldsymbol{x}_{k-1}+\omega_k\left(\boldsymbol{x}_k+\boldsymbol{r}_k\right),\\ \boldsymbol{r}_{k+1}&=\left(1-\omega_k\right)\boldsymbol{r}_{k-1}+\omega_k\boldsymbol{B}\boldsymbol{r}_k,\end{aligned}$
lyche-numerical-linear-alignatEQ0986	332	■ 0.999 N +0	$\frac{1}{\lambda}\leq\left\ \boldsymbol{A}^{-1}\right\ _p=\frac{K_p(\boldsymbol{A})}{\left\ \boldsymbol{A}\right\ _p}$	$\frac{1}{\lambda}\leq\left\ \boldsymbol{A}^{-1}\right\ _p=\frac{K_p(\boldsymbol{A})}{\left\ \boldsymbol{A}\right\ _p}$	$\frac{1}{\lambda}\leq\left\ \boldsymbol{A}^{-1}\right\ _p=\frac{K_p(\boldsymbol{A})}{\left\ \boldsymbol{A}\right\ _p}$
lyche-numerical-linear-alignatEQ1025	346	■ 0.999 W +0	$\boldsymbol{x}_k=\frac{\boldsymbol{z}_k}{\left\ \boldsymbol{z}_k\right\ }=\frac{c_1\lambda_1^k}{\left c_1\lambda_1^k\right }\frac{\boldsymbol{v}_1+\frac{c_2}{c_1}\left(\frac{\lambda_2}{\lambda_1}\right)^k\boldsymbol{v}_2+\cdots+\frac{c_n}{c_1}\left(\frac{\lambda_n}{\lambda_1}\right)^k\boldsymbol{v}_n}{\left\ \boldsymbol{v}_1+\frac{c_2}{c_1}\left(\frac{\lambda_2}{\lambda_1}\right)^k\boldsymbol{v}_2+\cdots+\frac{c_n}{c_1}\left(\frac{\lambda_n}{\lambda_1}\right)^k\boldsymbol{v}_n\right\ },\quad k=0,1,2,\ldots,$	$\boldsymbol{x}_k=\frac{\boldsymbol{z}_k}{\left\ \boldsymbol{z}_k\right\ }=\frac{c_1\lambda_1^k}{\left c_1\lambda_1^k\right }\frac{\boldsymbol{v}_1+\frac{c_2}{c_1}\left(\frac{\lambda_2}{\lambda_1}\right)^k\boldsymbol{v}_2+\cdots+\frac{c_n}{c_1}\left(\frac{\lambda_n}{\lambda_1}\right)^k\boldsymbol{v}_n}{\left\ \boldsymbol{v}_1+\frac{c_2}{c_1}\left(\frac{\lambda_2}{\lambda_1}\right)^k\boldsymbol{v}_2+\cdots+\frac{c_n}{c_1}\left(\frac{\lambda_n}{\lambda_1}\right)^k\boldsymbol{v}_n\right\ },\quad k=0,1,2,\ldots,$	$\boldsymbol{x}_k=\frac{\boldsymbol{z}_k}{\left\ \boldsymbol{z}_k\right\ }=\frac{c_1\lambda_1^k}{\left c_1\lambda_1^k\right }\frac{\boldsymbol{v}_1+\frac{c_2}{c_1}\left(\frac{\lambda_2}{\lambda_1}\right)^k\boldsymbol{v}_2+\cdots+\frac{c_n}{c_1}\left(\frac{\lambda_n}{\lambda_1}\right)^k\boldsymbol{v}_n}{\left\ \boldsymbol{v}_1+\frac{c_2}{c_1}\left(\frac{\lambda_2}{\lambda_1}\right)^k\boldsymbol{v}_2+\cdots+\frac{c_n}{c_1}\left(\frac{\lambda_n}{\lambda_1}\right)^k\boldsymbol{v}_n\right\ },\quad k=0,1,2,\ldots,$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ1034	349	0.999 W +0	$\begin{aligned} & s_k = \frac{\boldsymbol{y}_k}{\boldsymbol{y}_k^* \boldsymbol{y}_k} = s_{k-1} + \frac{\boldsymbol{y}_k^* (\boldsymbol{A} - s_{k-1} \boldsymbol{I}) \boldsymbol{y}_k}{\boldsymbol{y}_k^* \boldsymbol{y}_k} = s_{k-1} + \frac{\boldsymbol{y}_k^* \boldsymbol{x}_{k-1}}{\boldsymbol{y}_k^* \boldsymbol{y}_k} = s_{k-1} + \rho_k, \\ & \boldsymbol{r}_k = \boldsymbol{A} \boldsymbol{x}_k - s_k \boldsymbol{x}_k = \frac{\boldsymbol{A} \boldsymbol{y}_k - (s_{k-1} + \rho_k) \boldsymbol{y}_k}{\ \boldsymbol{y}_k\ _2} = \frac{\boldsymbol{x}_{k-1} - \rho_k \boldsymbol{y}_k}{\ \boldsymbol{y}_k\ _2} = \boldsymbol{w}_k - \rho_k \boldsymbol{x}_k. \end{aligned}$	$s_k = \frac{\boldsymbol{y}_k^* \boldsymbol{A} \boldsymbol{y}_k}{\boldsymbol{y}_k^* \boldsymbol{y}_k} = s_{k-1} + \frac{\boldsymbol{y}_k^* (\boldsymbol{A} - s_{k-1} \boldsymbol{I}) \boldsymbol{y}_k}{\boldsymbol{y}_k^* \boldsymbol{y}_k} = s_{k-1} + \frac{\boldsymbol{y}_k^* \boldsymbol{x}_{k-1}}{\boldsymbol{y}_k^* \boldsymbol{y}_k} = s_{k-1} + \rho_k,$ $\boldsymbol{r}_k = \boldsymbol{A} \boldsymbol{x}_k - s_k \boldsymbol{x}_k = \frac{\boldsymbol{A} \boldsymbol{y}_k - (s_{k-1} + \rho_k) \boldsymbol{y}_k}{\ \boldsymbol{y}_k\ _2} = \frac{\boldsymbol{x}_{k-1} - \rho_k \boldsymbol{y}_k}{\ \boldsymbol{y}_k\ _2} = \boldsymbol{w}_k - \rho_k \boldsymbol{x}_k.$	$s_k = \frac{\boldsymbol{y}_k^* \boldsymbol{A} \boldsymbol{y}_k}{\boldsymbol{y}_k^* \boldsymbol{y}_k} = s_{k-1} + \frac{\boldsymbol{y}_k^* (\boldsymbol{A} - s_{k-1} \boldsymbol{I}) \boldsymbol{y}_k}{\boldsymbol{y}_k^* \boldsymbol{y}_k} = s_{k-1} + \frac{\boldsymbol{y}_k^* \boldsymbol{x}_{k-1}}{\boldsymbol{y}_k^* \boldsymbol{y}_k} = s_{k-1} + \rho_k,$ $\boldsymbol{r}_k = \boldsymbol{A} \boldsymbol{x}_k - s_k \boldsymbol{x}_k = \frac{\boldsymbol{A} \boldsymbol{y}_k - (s_{k-1} + \rho_k) \boldsymbol{y}_k}{\ \boldsymbol{y}_k\ _2} = \frac{\boldsymbol{x}_{k-1} - \rho_k \boldsymbol{y}_k}{\ \boldsymbol{y}_k\ _2} = \boldsymbol{w}_k - \rho_k \boldsymbol{x}_k.$
lyche-numerical-linear-algebra EQ0097	054	0.999	$\left[\begin{array}{lll} 1 & 0 & 0 \\ l_1 & 1 & 0 \\ 0 & l_2 & 1 \end{array}\right] \left[\begin{array}{l} z_1 \\ z_2 \\ z_3 \end{array}\right] = \left[\begin{array}{l} b_1 \\ b_2 \\ b_3 \end{array}\right], \quad \left[\begin{array}{ccc} u_1 & c_1 & 0 \\ 0 & u_2 & c_2 \\ 0 & 0 & u_3 \end{array}\right] \left[\begin{array}{l} x_1 \\ x_2 \\ x_3 \end{array}\right] = \left[\begin{array}{l} z_1 \\ z_2 \\ z_3 \end{array}\right].$	$\left[\begin{array}{ccc} 1 & 0 & 0 \\ l_1 & 1 & 0 \\ 0 & l_2 & 1 \end{array}\right] \left[\begin{array}{l} z_1 \\ z_2 \\ z_3 \end{array}\right] = \left[\begin{array}{l} b_1 \\ b_2 \\ b_3 \end{array}\right], \quad \left[\begin{array}{ccc} u_1 & c_1 & 0 \\ 0 & u_2 & c_2 \\ 0 & 0 & u_3 \end{array}\right] \left[\begin{array}{l} x_1 \\ x_2 \\ x_3 \end{array}\right] = \left[\begin{array}{l} z_1 \\ z_2 \\ z_3 \end{array}\right].$	$\left[\begin{array}{ccc} 1 & 0 & 0 \\ l_1 & 1 & 0 \\ 0 & l_2 & 1 \end{array}\right] \left[\begin{array}{l} z_1 \\ z_2 \\ z_3 \end{array}\right] = \left[\begin{array}{l} b_1 \\ b_2 \\ b_3 \end{array}\right], \quad \left[\begin{array}{ccc} u_1 & c_1 & 0 \\ 0 & u_2 & c_2 \\ 0 & 0 & u_3 \end{array}\right] \left[\begin{array}{l} x_1 \\ x_2 \\ x_3 \end{array}\right] = \left[\begin{array}{l} z_1 \\ z_2 \\ z_3 \end{array}\right].$
lyche-numerical-linear-algebra EQ0380	154	0.999 K +0	$\boldsymbol{B} \boldsymbol{B}^* = \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} \boldsymbol{U}^* \boldsymbol{A}^* \boldsymbol{U} = \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{A}^* \boldsymbol{U} = \boldsymbol{U}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{B}^* \boldsymbol{B}.$	$\boldsymbol{B} \boldsymbol{B}^* = \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} \boldsymbol{U}^* \boldsymbol{A}^* \boldsymbol{U} = \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{A}^* \boldsymbol{U} = \boldsymbol{U}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{B}^* \boldsymbol{B}.$	$\boldsymbol{B} \boldsymbol{B}^* = \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} \boldsymbol{U}^* \boldsymbol{A}^* \boldsymbol{U} = \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{A}^* \boldsymbol{U} = \boldsymbol{U}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{B}^* \boldsymbol{B}.$
lyche-numerical-linear-algebra EQ0394 (6.13)	158	0.999 W -2	$\lambda_k = \min_{\dim(S)=n-k+1} \max_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}) = \max_{\dim(S)=k} \min_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}), \quad k = 1, \dots, n.$	$\lambda_k = \min_{\dim(S)=n-k+1} \max_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}) = \max_{\dim(S)=k} \min_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}), \quad k = 1, \dots, n.$	$\lambda_k = \min_{\dim(S)=n-k+1} \max_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}) = \max_{\dim(S)=k} \min_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}), \quad k = 1, \dots, n.$
lyche-numerical-linear-algebra EQ0445	178	0.999 W +1	$\left\ \boldsymbol{A} - \boldsymbol{A}' \right\ _F = \min_{\substack{\boldsymbol{B} \in \mathbb{R}^{m \times n} \\ \text{rank}(\boldsymbol{B})=r}} \left\ \boldsymbol{A} - \boldsymbol{B} \right\ _F = \sqrt{\sigma_{r+1}^2 + \dots + \sigma_n^2}.$	$\left\ \boldsymbol{A} - \boldsymbol{A}' \right\ _F = \min_{\substack{\boldsymbol{B} \in \mathbb{R}^{m \times n} \\ \text{rank}(\boldsymbol{B})=r}} \left\ \boldsymbol{A} - \boldsymbol{B} \right\ _F = \sqrt{\sigma_{r+1}^2 + \dots + \sigma_n^2}.$	$\left\ \boldsymbol{A} - \boldsymbol{A}' \right\ _F = \min_{\substack{\boldsymbol{B} \in \mathbb{R}^{m \times n} \\ \text{rank}(\boldsymbol{B})=r}} \left\ \boldsymbol{A} - \boldsymbol{B} \right\ _F = \sqrt{\sigma_{r+1}^2 + \dots + \sigma_n^2}.$
Conf. MathPix confidence: 0.999 0.5-0.9 <0.5 no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-2027-EQ0621	227	0.999 W +0	$\frac{\ \boldsymbol{A} \boldsymbol{x}\ _2^2}{\ \boldsymbol{x}\ _2^2} = \frac{(\boldsymbol{A} \boldsymbol{x})^* (\boldsymbol{A} \boldsymbol{x})}{\boldsymbol{x}^* \boldsymbol{x}} = \frac{\boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}} = R_{\boldsymbol{A}^* \boldsymbol{A}}(\boldsymbol{x}),$	$\frac{\ \boldsymbol{A} \boldsymbol{x}\ _2^2}{\ \boldsymbol{x}\ _2^2} = \frac{(\boldsymbol{A} \boldsymbol{x})^* (\boldsymbol{A} \boldsymbol{x})}{\boldsymbol{x}^* \boldsymbol{x}} = \frac{\boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}} = R_{\boldsymbol{A}^* \boldsymbol{A}}(\boldsymbol{x}),$	$\frac{\ \boldsymbol{A} \boldsymbol{x}\ _2^2}{\ \boldsymbol{x}\ _2^2} = \frac{(\boldsymbol{A} \boldsymbol{x})^* (\boldsymbol{A} \boldsymbol{x})}{\boldsymbol{x}^* \boldsymbol{x}} = \frac{\boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}} = R_{\boldsymbol{A}^* \boldsymbol{A}}(\boldsymbol{x}),$
lyche-numerical-linear-algebra-2027-EQ0903	309	0.999 N +0	$\left\ \boldsymbol{\epsilon}_k \right\ _2 - \left\ \boldsymbol{\epsilon}_{k+1} \right\ _2 = \frac{\ \boldsymbol{p}_k\ _2^2}{\ \boldsymbol{p}_k\ _A^2} \left(\ \boldsymbol{\epsilon}_k\ _A^2 + \ \boldsymbol{\epsilon}_{k+1}\ _A^2 \right)$	$\ \boldsymbol{\epsilon}_k\ _2^2 - \ \boldsymbol{\epsilon}_{k+1}\ _2^2 = \frac{\ \boldsymbol{p}_k\ _2^2}{\ \boldsymbol{p}_k\ _A^2} \left(\ \boldsymbol{\epsilon}_k\ _A^2 + \ \boldsymbol{\epsilon}_{k+1}\ _A^2 \right)$	$\ \boldsymbol{\epsilon}_k\ _2^2 - \ \boldsymbol{\epsilon}_{k+1}\ _2^2 = \frac{\ \boldsymbol{p}_k\ _2^2}{\ \boldsymbol{p}_k\ _A^2} (\ \boldsymbol{\epsilon}_k\ _A^2 + \ \boldsymbol{\epsilon}_{k+1}\ _A^2)$
lyche-numerical-linear-algebra-2027-EQ0003	022	1.000 S +0	$\boldsymbol{A} = \left[\begin{array}{cccc} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{array} \right].$	$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$	$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$
lyche-numerical-linear-algebra-2027-EQ0008	023	1.000	$\boldsymbol{e}_1 := \left[\begin{array}{c} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{array} \right], \quad \boldsymbol{e}_2 := \left[\begin{array}{c} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{array} \right], \quad \boldsymbol{e}_3 := \left[\begin{array}{c} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{array} \right], \quad \dots, \quad \boldsymbol{e}_n := \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{array} \right],$	$\boldsymbol{e}_1 := \left[\begin{array}{c} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{array} \right], \quad \boldsymbol{e}_2 := \left[\begin{array}{c} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{array} \right], \quad \boldsymbol{e}_3 := \left[\begin{array}{c} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{array} \right], \quad \dots, \quad \boldsymbol{e}_n := \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{array} \right],$	$\boldsymbol{e}_1 := \left[\begin{array}{c} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{array} \right], \quad \boldsymbol{e}_2 := \left[\begin{array}{c} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{array} \right], \quad \boldsymbol{e}_3 := \left[\begin{array}{c} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{array} \right], \quad \dots, \quad \boldsymbol{e}_n := \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{array} \right],$
lyche-numerical-linear-algebra-2027-EQ0025	030	1.000 W +0	$\sum_{i=1}^m b_i \boldsymbol{v}_i = \boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j \stackrel{(1.9)}{=} \sum_{j=1}^n c_j \left(\sum_{i=1}^m a_{ij} \boldsymbol{v}_i \right) = \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} c_j \right) \boldsymbol{v}_i.$	$\sum_{i=1}^m b_i \boldsymbol{v}_i = \boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j \stackrel{(1.9)}{=} \sum_{j=1}^n c_j \left(\sum_{i=1}^m a_{ij} \boldsymbol{v}_i \right) = \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} c_j \right) \boldsymbol{v}_i.$	$\sum_{i=1}^m b_i \boldsymbol{v}_i = \boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j \stackrel{(1.9)}{=} \sum_{j=1}^n c_j \left(\sum_{i=1}^m a_{ij} \boldsymbol{v}_i \right) = \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} c_j \right) \boldsymbol{v}_i.$
lyche-numerical-linear-algebra-2027-EQ0053	039	1.000 N +0	$\pi_A(\lambda) = (-1)^n \lambda^n + c_{n-1} \lambda^{n-1} + \cdots + c_0,$	$\pi_A(\lambda) = (-1)^n \lambda^n + c_{n-1} \lambda^{n-1} + \cdots + c_0,$	$\pi_A(\lambda) = (-1)^n \lambda^n + c_{n-1} \lambda^{n-1} + \cdots + c_0,$
lyche-numerical-linear-algebra-2027-EQ0078 (2.2)	046	1.000 K +0	$g(\boldsymbol{x}_i) = \boldsymbol{y}_i, \text{ for } i=1, \ldots, n+1.$	$g(\boldsymbol{x}_i) = \boldsymbol{y}_i, \text{ for } i=1, \ldots, n+1.$	$g(\boldsymbol{x}_i) = \boldsymbol{y}_i, \text{ for } i=1, \ldots, n+1.$
lyche-numerical-linear-algebra-2027-EQ0087 (2.9)	051	1.000 N +0	$\mu_{i-1} + 4\mu_i + \mu_{i+1} = \frac{6}{h^2} (y_{i+1} - 2y_i + y_{i-1}), \quad i=2, \ldots, n,$	$\mu_{i-1} + 4\mu_i + \mu_{i+1} = \frac{6}{h^2} (y_{i+1} - 2y_i + y_{i-1}), \quad i=2, \ldots, n,$	$\mu_{i-1} + 4\mu_i + \mu_{i+1} = \frac{6}{h^2} (y_{i+1} - 2y_i + y_{i-1}), \quad i=2, \ldots, n,$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0240 (3.32)	1097	■ 1.000 ● K +0	$\boldsymbol{L}=\left[\begin{array}{cc} 1&0\\ \boldsymbol{\ell}_1&\boldsymbol{L}_{2,2}\end{array}\right],\quad \boldsymbol{U}=\left[\begin{array}{cc} u_{1,1}&\boldsymbol{u}_1^T\\ 0&\boldsymbol{U}_{2,2}\end{array}\right],$	$\boldsymbol{L}=\left[\begin{array}{cc} 1&0\\ \boldsymbol{\ell}_1&\boldsymbol{L}_{2,2}\end{array}\right],\quad \boldsymbol{U}=\left[\begin{array}{cc} u_{1,1}&\boldsymbol{u}_1^T\\ 0&\boldsymbol{U}_{2,2}\end{array}\right],$	$\boldsymbol{L}=\left[\begin{array}{cc} 1&0\\ \boldsymbol{\ell}_1&\boldsymbol{L}_{2,2}\end{array}\right],\quad \boldsymbol{U}=\left[\begin{array}{cc} u_{1,1}&\boldsymbol{u}_1^T\\ 0&\boldsymbol{U}_{2,2}\end{array}\right],$
lyche-numerical-linear-algebra-EQ0270	1113	■ 1.000 ● N +1	$\boldsymbol{A}[a]:=\left[\begin{array}{cc} 2&2-a\\ a&1\end{array}\right],\quad a\in\mathbb{R}$	$\boldsymbol{A}[a]:=\left[\begin{array}{cc} 2&2-a\\ a&1\end{array}\right],\quad a\in\mathbb{R}$	$\boldsymbol{A}[a]:=\left[\begin{array}{cc} 2&2-a\\ a&1\end{array}\right],\quad a\in\mathbb{R}$
lyche-numerical-linear-algebra-EQ0274	1114	■ 1.000 ● K +0	$\boldsymbol{A}_{+}=\left[\begin{array}{cc} \boldsymbol{A}&\boldsymbol{a}\\ \boldsymbol{a}^T&\alpha\end{array}\right],$	$\boldsymbol{A}_{+}=\left[\begin{array}{cc} \boldsymbol{A}&\boldsymbol{a}\\ \boldsymbol{a}^T&\alpha\end{array}\right],$	$\boldsymbol{A}_{+}=\left[\begin{array}{cc} \boldsymbol{A}&\boldsymbol{a}\\ \boldsymbol{a}^T&\alpha\end{array}\right],$
lyche-numerical-linear-algebra-EQ0283 (5.4)	1118	■ 1.000 ● N +0	$\angle a\boldsymbol{y},\boldsymbol{z},\boldsymbol{y}\angle+\angle\boldsymbol{z},\boldsymbol{y}\angle a\overline{\angle\boldsymbol{z},\boldsymbol{y}},\boldsymbol{y}\angle+\overline{\angle\boldsymbol{z},\boldsymbol{y}}\angle\boldsymbol{z},\boldsymbol{y}\angle=0.$	$\langle a\boldsymbol{y},\boldsymbol{z}\rangle+\langle\boldsymbol{z},a\boldsymbol{y}\rangle=a\overline{\langle\boldsymbol{z},\boldsymbol{y}\rangle}+\bar{a}\langle\boldsymbol{z},\boldsymbol{y}\rangle=0.$	$\langle a\boldsymbol{y},\boldsymbol{z}\rangle+\langle\boldsymbol{z},a\boldsymbol{y}\rangle=a\overline{\langle\boldsymbol{z},\boldsymbol{y}\rangle}+\bar{a}\langle\boldsymbol{z},\boldsymbol{y}\rangle=0.$
lyche-numerical-linear-algebra-EQ0328	1133	■ 1.000 ● N +0	$\boldsymbol{R}_{\{1\}}=\boldsymbol{R}=\left[\begin{array}{cc} \ \boldsymbol{v}_1\ _2&\boldsymbol{a}_2^T\boldsymbol{q}_1\\ 0&\ \boldsymbol{v}_2\ _2\end{array}\right]=\frac{1}{\sqrt{5}}\left[\begin{array}{cc} 5&-4\\ 0&3\end{array}\right]$	$\boldsymbol{R}_1=\boldsymbol{R}=\left[\begin{array}{cc} \ \boldsymbol{v}_1\ _2&\boldsymbol{a}_2^T\boldsymbol{q}_1\\ 0&\ \boldsymbol{v}_2\ _2\end{array}\right]=\frac{1}{\sqrt{5}}\left[\begin{array}{cc} 5&-4\\ 0&3\end{array}\right]$	$\boldsymbol{R}_1=\boldsymbol{R}=\left[\begin{array}{cc} \ \boldsymbol{v}_1\ _2&\boldsymbol{a}_2^T\boldsymbol{q}_1\\ 0&\ \boldsymbol{v}_2\ _2\end{array}\right]=\frac{1}{\sqrt{5}}\left[\begin{array}{cc} 5&-4\\ 0&3\end{array}\right]$
lyche-numerical-linear-algebra-EQ0341	1137	■ 1.000 ● K +0	$\boldsymbol{B}:=\left[\begin{array}{cccc} 0&1&0&0\\ 0&0&1&0\\ 0&0&0&1\\ \epsilon&0&0&0\end{array}\right],$	$\boldsymbol{B}:=\left[\begin{array}{cccc} 0&1&0&0\\ 0&0&1&0\\ 0&0&0&1\\ \epsilon&0&0&0\end{array}\right],$	$\boldsymbol{B}:=\left[\begin{array}{cccc} 0&1&0&0\\ 0&0&1&0\\ 0&0&0&1\\ \epsilon&0&0&0\end{array}\right],$
lyche-numerical-linear-algebra-EQ0345	1139	■ 1.000 ●	$\boldsymbol{V}_{\{r\}}:=\boldsymbol{I}_{\{r\}}-2\frac{\boldsymbol{v}_r\boldsymbol{v}_r^*}{\boldsymbol{v}_r^*\boldsymbol{v}_r}=\boldsymbol{I}_r-\boldsymbol{u}_r\boldsymbol{u}_r^*,\text{ with }\boldsymbol{u}_r:=\sqrt{2}\frac{\boldsymbol{v}_r}{\ \boldsymbol{v}_r\ _2}.$	$\boldsymbol{V}_r:=\boldsymbol{I}_r-2\frac{\boldsymbol{v}_r\boldsymbol{v}_r^*}{\boldsymbol{v}_r^*\boldsymbol{v}_r}=\boldsymbol{I}_r-\boldsymbol{u}_r\boldsymbol{u}_r^*,\text{ with }\boldsymbol{u}_r:=\sqrt{2}\frac{\boldsymbol{v}_r}{\ \boldsymbol{v}_r\ _2}.$	$\boldsymbol{V}_r:=\boldsymbol{I}_r-2\frac{\boldsymbol{v}_r\boldsymbol{v}_r^*}{\boldsymbol{v}_r^*\boldsymbol{v}_r}=\boldsymbol{I}_r-\boldsymbol{u}_r\boldsymbol{u}_r^*,\text{ with }\boldsymbol{u}_r:=\sqrt{2}\frac{\boldsymbol{v}_r}{\ \boldsymbol{v}_r\ _2}.$
lyche-numerical-linear-algebra-EQ0363	1147	■ 1.000 ● K +0	$E:=\left[\begin{array}{cc} A&C\\ C&0\end{array}\right],\quad F:=\left[\begin{array}{cc} 0&0\\ C&CA\end{array}\right],\quad S=\left[\begin{array}{cc} I&A\\ 0&I\end{array}\right].$	$E:=\left[\begin{array}{cc} A&C\\ C&0\end{array}\right],\quad F:=\left[\begin{array}{cc} 0&0\\ C&CA\end{array}\right],\quad S=\left[\begin{array}{cc} I&A\\ 0&I\end{array}\right].$	$E:=\left[\begin{array}{cc} A&C\\ C&0\end{array}\right],\quad F:=\left[\begin{array}{cc} 0&0\\ C&CA\end{array}\right],\quad S=\left[\begin{array}{cc} I&A\\ 0&I\end{array}\right].$
lyche-numerical-linear-algebra-EQ0364 (6.2)	1148	■ 1.000 ● K +0	$\pi_{\boldsymbol{A}}(\lambda):=\det(\boldsymbol{A}-\lambda\boldsymbol{I})=(\lambda_1-\lambda)^{a_1}\cdots(\lambda_k-\lambda)^{a_k},\quad \lambda_i\neq\lambda_j,i\neq j,\sum_{i=1}^ka_i=n.$	$\pi_{\boldsymbol{A}}(\lambda):=\det(\boldsymbol{A}-\lambda\boldsymbol{I})=(\lambda_1-\lambda)^{a_1}\cdots(\lambda_k-\lambda)^{a_k},\quad \lambda_i\neq\lambda_j,i\neq j,\sum_{i=1}^ka_i=n.$	$\pi_{\boldsymbol{A}}(\lambda):=\det(\boldsymbol{A}-\lambda\boldsymbol{I})=(\lambda_1-\lambda)^{a_1}\cdots(\lambda_k-\lambda)^{a_k},\quad \lambda_i\neq\lambda_j,i\neq j,\sum_{i=1}^ka_i=n.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0393 (6.12)	157	■ 1.000 ● N -1	$\lambda_{\mathbf{k}} \geq \min_{\mathbf{x} \in S} R(\mathbf{x})$	$\lambda_k \geq \min_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R(\mathbf{x}),$	$\lambda_k \geq \min_{\substack{\mathbf{x} \in S \\ \mathbf{x} \neq \mathbf{0}}} R(\mathbf{x}),$
lyche-numerical-linear-algebra- EQ0413	165	■ 1.000 ● W -1	$\lambda_{\min} = \min_{\mathbf{x} \neq \mathbf{0}} R(\mathbf{x}) = \min_{\ \mathbf{x}\ _2=1} R(\mathbf{x}),$	$\lambda_{\min} = \min_{\mathbf{x} \neq \mathbf{0}} R(\mathbf{x}) = \min_{\ \mathbf{x}\ _2=1} R(\mathbf{x}),$	$\lambda_{\min} = \min_{\mathbf{x} \neq \mathbf{0}} R(\mathbf{x}) = \min_{\ \mathbf{x}\ _2=1} R(\mathbf{x}),$
lyche-numerical-linear-algebra- EQ0427 (2)	172	■ 1.000 ● K +0	$\left[\begin{array}{ll} 1 & -1 \\ 1 & -1 \end{array} \right] = \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & 1 \\ 1 & -1 \end{array} \right] \left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \end{array} \right] \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & -1 \\ 1 & 1 \end{array} \right] = \frac{1}{\sqrt{2}} \left[\begin{array}{l} 1 \\ 1 \end{array} \right] [2] \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & -1 \end{array} \right].$	$\left[\begin{array}{ll} 1 & -1 \\ 1 & -1 \end{array} \right] = \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & 1 \\ 1 & -1 \end{array} \right] \left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \end{array} \right] \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & -1 \\ 1 & 1 \end{array} \right] = \frac{1}{\sqrt{2}} \left[\begin{array}{l} 1 \\ 1 \end{array} \right] [2] \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & -1 \end{array} \right].$	$\left[\begin{array}{ll} 1 & -1 \\ 1 & -1 \end{array} \right] = \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & 1 \\ 1 & -1 \end{array} \right] \left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \end{array} \right] \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & -1 \\ 1 & 1 \end{array} \right] = \frac{1}{\sqrt{2}} \left[\begin{array}{l} 1 \\ 1 \end{array} \right] [2] \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & -1 \end{array} \right].$
lyche-numerical-linear-algebra- EQ0435	173	■ 1.000 ● W -2	$\mathbf{U} := \left[\begin{array}{ccc} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{array} \right] \in \mathbb{R}^{3,3}, \quad \mathbf{\Sigma} := \left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \\ 0 & 0 \end{array} \right] \in \mathbb{R}^{3,2}, \quad \mathbf{V} := \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & 1 \\ 1 & -1 \end{array} \right] \in \mathbb{R}^{2,2}.$	$U := \left[\begin{array}{lll} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{array} \right] \in \mathbb{R}^{3,3}, \quad \Sigma := \left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \\ 0 & 0 \end{array} \right] \in \mathbb{R}^{3,2}, \quad V := \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & 1 \\ 1 & -1 \end{array} \right] \in \mathbb{R}^{2,2}.$	$U := \left[\begin{array}{lll} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{array} \right] \in \mathbb{R}^{3,3}, \quad \Sigma := \left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \\ 0 & 0 \end{array} \right] \in \mathbb{R}^{3,2}, \quad V := \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & 1 \\ 1 & -1 \end{array} \right] \in \mathbb{R}^{2,2}.$
lyche-numerical-linear-algebra- EQ0439	175	■ 1.000 ● K +0	$A := \frac{1}{25} \left[\begin{array}{ll} 11 & 48 \\ 48 & 39 \end{array} \right]$	$A := \frac{1}{25} \left[\begin{array}{ll} 11 & 48 \\ 48 & 39 \end{array} \right]$	$A := \frac{1}{25} \left[\begin{array}{ll} 11 & 48 \\ 48 & 39 \end{array} \right]$
lyche-numerical-linear-algebra- EQ0442	177	■ 1.000 ● N +0	$\ \mathbf{AB}\ _F^2 = \sum_{i=1}^m \sum_{j=1}^k \mathbf{a}_{i:}^* \mathbf{b}_{:j} ^2 \leq \sum_{i=1}^m \sum_{j=1}^k \ \mathbf{a}_{i:}\ _2^2 \ \mathbf{b}_{:j}\ _2^2 = \ \mathbf{A}\ _F^2 \ \mathbf{B}\ _F^2.$	$\ \mathbf{AB}\ _F^2 = \sum_{i=1}^m \sum_{j=1}^k \mathbf{a}_{i:}^* \mathbf{b}_{:j} ^2 \leq \sum_{i=1}^m \sum_{j=1}^k \ \mathbf{a}_{i:}\ _2^2 \ \mathbf{b}_{:j}\ _2^2 = \ \mathbf{A}\ _F^2 \ \mathbf{B}\ _F^2.$	$\ \mathbf{AB}\ _F^2 = \sum_{i=1}^m \sum_{j=1}^k \mathbf{a}_{i:}^* \mathbf{b}_{:j} ^2 \leq \sum_{i=1}^m \sum_{j=1}^k \ \mathbf{a}_{i:}\ _2^2 \ \mathbf{b}_{:j}\ _2^2 = \ \mathbf{A}\ _F^2 \ \mathbf{B}\ _F^2.$
lyche-numerical-linear-algebra- EQ0447 (7.11)	178	■ 1.000 ●	$\mathbf{A} := \left[\begin{array}{ll} 1 & -1 \\ 1 & -1 \end{array} \right] = \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & 1 \\ 1 & -1 \end{array} \right] \left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \end{array} \right] \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & -1 \\ 1 & 1 \end{array} \right] =: \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$	$\mathbf{A} := \left[\begin{array}{ll} 1 & -1 \\ 1 & -1 \end{array} \right] = \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & 1 \\ 1 & -1 \end{array} \right] \left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \end{array} \right] \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & -1 \\ 1 & 1 \end{array} \right] =: \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$	$\mathbf{A} := \left[\begin{array}{ll} 1 & -1 \\ 1 & -1 \end{array} \right] = \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & 1 \\ 1 & -1 \end{array} \right] \left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \end{array} \right] \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & -1 \\ 1 & 1 \end{array} \right] =: \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$
lyche-numerical-linear-algebra- EQ0457	181	■ 1.000 ● K +0	$\mathbf{A} = \left[\begin{array}{ll} 1 & 2 \\ 0 & 1 \\ -1 & 3 \end{array} \right].$	$\mathbf{A} = \left[\begin{array}{ll} 1 & 2 \\ 0 & 1 \\ -1 & 3 \end{array} \right].$	$\mathbf{A} = \left[\begin{array}{ll} 1 & 2 \\ 0 & 1 \\ -1 & 3 \end{array} \right].$
lyche-numerical-linear-algebra- EQ0463	183	■ 1.000 ● N +0	$\mathbf{T} = \left[\begin{array}{ll} 2 & -1 \\ -1 & 2 \end{array} \right].$	$\mathbf{T} = \left[\begin{array}{ll} 2 & -1 \\ -1 & 2 \end{array} \right].$	$\mathbf{T} = \left[\begin{array}{ll} 2 & -1 \\ -1 & 2 \end{array} \right].$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0484 (8.16)	190	■ 1.000 ● K +0	$\ \boldsymbol{A}\ =\left\ \boldsymbol{A}\right\ $ $\boldsymbol{x}^*\right\ \text{ for some } \boldsymbol{x}^* \in \mathcal{S}.$	$\ \boldsymbol{A}\ =\ \boldsymbol{A}\boldsymbol{x}^*\ \text{ for some } \boldsymbol{x}^* \in \mathcal{S}.$	$\ \boldsymbol{A}\ =\ \boldsymbol{A}\boldsymbol{x}^*\ \text{ for some } \boldsymbol{x}^* \in \mathcal{S}.$
lyche-numerical-linear-algebra- EQ0490	192	■ 1.000 ● N +1	$\left\ \boldsymbol{A}\right\ _{\infty}=\max_{1\leq k\leq m}\left \sum_{j=1}^na_{kj}x_j\right \leq\max_{1\leq k\leq m}\sum_{j=1}^n a_{kj} x_j \leq K_{\infty}.$	$\ \boldsymbol{A}\boldsymbol{x}\ _{\infty}=\max_{1\leq k\leq m}\left \sum_{j=1}^na_{kj}x_j\right \leq\max_{1\leq k\leq m}\sum_{j=1}^n a_{kj} x_j \leq K_{\infty}.$	$\ \boldsymbol{A}\boldsymbol{x}\ _{\infty}=\max_{1\leq k\leq m}\left \sum_{j=1}^na_{kj}x_j\right \leq\max_{1\leq k\leq m}\sum_{j=1}^n a_{kj} x_j \leq K_{\infty}.$
lyche-numerical-linear-algebra- EQ0493 (8.21)	193	■ 1.000 ● K +0	$\left\ \boldsymbol{A}\right\ _2^2\left\ \boldsymbol{v}\right\ _1=\sigma^2\left\ \boldsymbol{v}\right\ _1=\left\ \sigma^2\boldsymbol{v}\right\ _1=\left\ \boldsymbol{A}^*\boldsymbol{A}\boldsymbol{v}\right\ _1\leq\left\ \boldsymbol{A}^*\right\ _1\left\ \boldsymbol{A}\right\ _1\left\ \boldsymbol{v}\right\ _1.$	$\ \boldsymbol{A}\ _2^2\ \boldsymbol{v}\ _1=\sigma^2\ \boldsymbol{v}\ _1=\ \sigma^2\boldsymbol{v}\ _1=\ \boldsymbol{A}^*\boldsymbol{A}\boldsymbol{v}\ _1\leq\ \boldsymbol{A}^*\ _1\ \boldsymbol{A}\ _1\ \boldsymbol{v}\ _1.$	$\ \boldsymbol{A}\ _2^2\ \boldsymbol{v}\ _1=\sigma^2\ \boldsymbol{v}\ _1=\ \sigma^2\boldsymbol{v}\ _1=\ \boldsymbol{A}^*\boldsymbol{A}\boldsymbol{v}\ _1\leq\ \boldsymbol{A}^*\ _1\ \boldsymbol{A}\ _1\ \boldsymbol{v}\ _1.$
lyche-numerical-linear-algebra- EQ0505 (8.27)	198	■ 1.000 ● N +0	$\frac{\left\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\ }{\left\ \boldsymbol{B}^{-1}\right\ }\leq\left\ \boldsymbol{A}^{-1}\boldsymbol{E}\right\ \leq K(\boldsymbol{A})\frac{\left\ \boldsymbol{E}\right\ }{\left\ \boldsymbol{A}\right\ },$	$\frac{\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\ }{\ \boldsymbol{B}^{-1}\ }\leq\ \boldsymbol{A}^{-1}\boldsymbol{E}\ \leq K(\boldsymbol{A})\frac{\ \boldsymbol{E}\ }{\ \boldsymbol{A}\ },$	$\frac{\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\ }{\ \boldsymbol{B}^{-1}\ }\leq\ \boldsymbol{A}^{-1}\boldsymbol{E}\ \leq K(\boldsymbol{A})\frac{\ \boldsymbol{E}\ }{\ \boldsymbol{A}\ },$
lyche-numerical-linear-algebra- EQ0509	198	■ 1.000 ● N +0	$\left\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\ \leq\left\ \boldsymbol{A}^{-1}\boldsymbol{E}\right\ \left\ \boldsymbol{B}^{-1}\right\ \leq K(\boldsymbol{A})\frac{\left\ \boldsymbol{E}\right\ }{\left\ \boldsymbol{A}\right\ }\left\ \boldsymbol{B}^{-1}\right\ .$	$\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\ \leq\ \boldsymbol{A}^{-1}\boldsymbol{E}\ \ \boldsymbol{B}^{-1}\ \leq K(\boldsymbol{A})\frac{\ \boldsymbol{E}\ }{\ \boldsymbol{A}\ }\ \boldsymbol{B}^{-1}\ .$	$\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\ \leq\ \boldsymbol{A}^{-1}\boldsymbol{E}\ \ \boldsymbol{B}^{-1}\ \leq K(\boldsymbol{A})\frac{\ \boldsymbol{E}\ }{\ \boldsymbol{A}\ }\ \boldsymbol{B}^{-1}\ .$
lyche-numerical-linear-algebra- EQ0521 (8.34)	201	■ 1.000 ● N +1	$a^{\frac{1}{p}}b^{\frac{1}{q}}\leq\frac{1}{p}a+\frac{1}{q}b,\text{ where }\frac{1}{p}+\frac{1}{q}=1.$	$a^{\frac{1}{p}}b^{\frac{1}{q}}\leq\frac{1}{p}a+\frac{1}{q}b,\text{ where }\frac{1}{p}+\frac{1}{q}=1.$	$a^{\frac{1}{p}}b^{\frac{1}{q}}\leq\frac{1}{p}a+\frac{1}{q}b,\text{ where }\frac{1}{p}+\frac{1}{q}=1.$
lyche-numerical-linear-algebra- EQ0528 (8.36)	202	■ 1.000 ● N +0	$\langle\boldsymbol{x},\boldsymbol{y}\rangle:=\frac{1}{4}\left(\left\ \boldsymbol{x}+\boldsymbol{y}\right\ ^2-\left\ \boldsymbol{x}-\boldsymbol{y}\right\ ^2\right),\quad\boldsymbol{x},\boldsymbol{y}\in\mathcal{V}$	$\langle\boldsymbol{x},\boldsymbol{y}\rangle:=\frac{1}{4}\left(\ \boldsymbol{x}+\boldsymbol{y}\ ^2-\ \boldsymbol{x}-\boldsymbol{y}\ ^2\right),\quad\boldsymbol{x},\boldsymbol{y}\in\mathcal{V}$	$\langle\boldsymbol{x},\boldsymbol{y}\rangle:=\frac{1}{4}\left(\ \boldsymbol{x}+\boldsymbol{y}\ ^2-\ \boldsymbol{x}-\boldsymbol{y}\ ^2\right),\quad\boldsymbol{x},\boldsymbol{y}\in\mathcal{V}$
lyche-numerical-linear-algebra- EQ0551	208	■ 1.000 ●	$\boldsymbol{A}:=\left[\begin{array}{cc}1.1&1\\1&1\end{array}\right],\quad\boldsymbol{b}:=\left[\begin{array}{c}b_1\\b_2\end{array}\right]=\left[\begin{array}{c}2.1\\2.0\end{array}\right],\quad\boldsymbol{e}:=\left[\begin{array}{c}e_1\\e_2\end{array}\right],\quad\ \boldsymbol{e}\ _2=0.1.$	$\boldsymbol{A}:=\left[\begin{array}{cc}1.1&1\\1&1\end{array}\right],\quad\boldsymbol{b}:=\left[\begin{array}{c}b_1\\b_2\end{array}\right]=\left[\begin{array}{c}2.1\\2.0\end{array}\right],\quad\boldsymbol{e}:=\left[\begin{array}{c}e_1\\e_2\end{array}\right],\quad\ \boldsymbol{e}\ _2=0.1.$	$\boldsymbol{A}:=\left[\begin{array}{cc}1.1&1\\1&1\end{array}\right],\quad\boldsymbol{b}:=\left[\begin{array}{c}b_1\\b_2\end{array}\right]=\left[\begin{array}{c}2.1\\2.0\end{array}\right],\quad\boldsymbol{e}:=\left[\begin{array}{c}e_1\\e_2\end{array}\right],\quad\ \boldsymbol{e}\ _2=0.1.$
lyche-numerical-linear-algebra- EQ0598	220	■ 1.000 ● N +1	$\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{x}=\boldsymbol{A}^*\boldsymbol{b}\implies\boldsymbol{R}_1\boldsymbol{x}=\boldsymbol{c}_1,\quad\boldsymbol{c}_1:=\boldsymbol{Q}_1^*\boldsymbol{b}.$	$\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{x}=\boldsymbol{A}^*\boldsymbol{b}\implies\boldsymbol{R}_1\boldsymbol{x}=\boldsymbol{c}_1,\quad\boldsymbol{c}_1:=\boldsymbol{Q}_1^*\boldsymbol{b}.$	$\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{x}=\boldsymbol{A}^*\boldsymbol{b}\implies\boldsymbol{R}_1\boldsymbol{x}=\boldsymbol{c}_1,\quad\boldsymbol{c}_1:=\boldsymbol{Q}_1^*\boldsymbol{b}.$
lyche-numerical-linear-algebra- EQ0608 (9.11)	223	■ 1.000 ● K +0	$\boldsymbol{A}^{\dagger}=\left(\boldsymbol{A}^*\boldsymbol{A}\right)^{-1}\boldsymbol{A}^*.$	$\boldsymbol{A}^{\dagger}=\left(\boldsymbol{A}^*\boldsymbol{A}\right)^{-1}\boldsymbol{A}^*.$	$\boldsymbol{A}^{\dagger}=\left(\boldsymbol{A}^*\boldsymbol{A}\right)^{-1}\boldsymbol{A}^*.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-245 EQ0682 (10.15)	245	■ 1.000 ● N +0	$\lambda_j = d + 2 a \cos (j \pi h), \quad h := \frac{1}{m+1},$	$\lambda_j = d + 2a \cos(j\pi h), \quad h := \frac{1}{m+1},$	$\lambda_j = d + 2a \cos(j\pi h), \quad h := \frac{1}{m+1},$
lyche-numerical-linear-algebra-246 EQ0684 (10.17)	246	■ 1.000 ● N +0	$\boldsymbol{s}_{\boldsymbol{j}}^{\boldsymbol{T}} \boldsymbol{s}_{\boldsymbol{k}} = \frac{m+1}{2} \delta_{j,k} = \frac{1}{2h} \delta_{j,k}, \quad j,k = 1, \dots, m.$ $\delta_{\boldsymbol{j}, \boldsymbol{k}} = \frac{1}{2h} \delta_{j,k}, \quad j,k = 1, \dots, m.$	$\boldsymbol{s}_j^T \boldsymbol{s}_k = \frac{m+1}{2} \delta_{j,k} = \frac{1}{2h} \delta_{j,k}, \quad j,k = 1, \dots, m.$	$\boldsymbol{s}_j^T \boldsymbol{s}_k = \frac{m+1}{2} \delta_{j,k} = \frac{1}{2h} \delta_{j,k}, \quad j,k = 1, \dots, m.$
lyche-numerical-linear-algebra-248 EQ0692 (10.22)	248	■ 1.000 ● N +1	$\boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} - \frac{1}{6} \boldsymbol{T} \boldsymbol{V} \boldsymbol{T} = h^2 \mu \boldsymbol{F}$	$\boldsymbol{TV} + \boldsymbol{VT} - \frac{1}{6} \boldsymbol{TVT} = h^2 \mu \boldsymbol{F}$	$\boldsymbol{TV} + \boldsymbol{VT} - \frac{1}{6} \boldsymbol{TVT} = h^2 \mu \boldsymbol{F}.$
lyche-numerical-linear-algebra-248 EQ0695 (10.21)	248	■ 1.000 ● N +0	$\boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = h^2 \boldsymbol{F}, \quad \boldsymbol{T} \boldsymbol{U} + \boldsymbol{U} \boldsymbol{T} = h^2 \boldsymbol{V}.$	$\boldsymbol{TV} + \boldsymbol{VT} = h^2 \boldsymbol{F}, \quad \boldsymbol{TU} + \boldsymbol{UT} = h^2 \boldsymbol{V}.$	$\boldsymbol{TV} + \boldsymbol{VT} = h^2 \boldsymbol{F}, \quad \boldsymbol{TU} + \boldsymbol{UT} = h^2 \boldsymbol{V}.$
lyche-numerical-linear-algebra-257 EQ0721	257	■ 1.000 ● N +0	$\boldsymbol{A} \boldsymbol{P}_N = [\boldsymbol{a}_1, \boldsymbol{a}_3, \dots, \boldsymbol{a}_{N-1}, \boldsymbol{a}_2, \boldsymbol{a}_4, \dots, \boldsymbol{a}_N],$	$\boldsymbol{AP}_N = [\boldsymbol{a}_1, \boldsymbol{a}_3, \dots, \boldsymbol{a}_{N-1}, \boldsymbol{a}_2, \boldsymbol{a}_4, \dots, \boldsymbol{a}_N],$	$\boldsymbol{AP}_N = [\boldsymbol{a}_1, \boldsymbol{a}_3, \dots, \boldsymbol{a}_{N-1}, \boldsymbol{a}_2, \boldsymbol{a}_4, \dots, \boldsymbol{a}_N],$
lyche-numerical-linear-algebra-270 EQ0754 (12.9)	270	■ 1.000 ● N +0	$\boldsymbol{M} \boldsymbol{x}_{k+1} = (\boldsymbol{M} - \boldsymbol{A}) \boldsymbol{x}_k + \boldsymbol{b}$	$\boldsymbol{M} \boldsymbol{x}_{k+1} = (\boldsymbol{M} - \boldsymbol{A}) \boldsymbol{x}_k + \boldsymbol{b}$	$\boldsymbol{M} \boldsymbol{x}_{k+1} = (\boldsymbol{M} - \boldsymbol{A}) \boldsymbol{x}_k + \boldsymbol{b}$
lyche-numerical-linear-algebra-277 EQ0789 (12.29)	277	■ 1.000 ● K +0	$\rho(\boldsymbol{G}_{\omega}) > \rho(\boldsymbol{G}_{\omega^*}) \text{ for } \omega \in (0, 2) \setminus \{\omega^*\}.$	$\rho(\boldsymbol{G}_{\omega}) > \rho(\boldsymbol{G}_{\omega^*}) \text{ for } \omega \in (0, 2) \setminus \{\omega^*\}.$	$\rho(\boldsymbol{G}_{\omega}) > \rho(\boldsymbol{G}_{\omega^*}) \text{ for } \omega \in (0, 2) \setminus \{\omega^*\}.$
lyche-numerical-linear-algebra-278 EQ0791	278	■ 1.000 ● N +0	$\ \boldsymbol{x}_k - \boldsymbol{x}\ \leq \ \boldsymbol{G}^k\ \ \boldsymbol{x}_0 - \boldsymbol{x}\ \approx \rho(\boldsymbol{G})^k \ \boldsymbol{x}_0 - \boldsymbol{x}\ .$	$\ \boldsymbol{x}_k - \boldsymbol{x}\ \leq \ \boldsymbol{G}^k\ \ \boldsymbol{x}_0 - \boldsymbol{x}\ \approx \rho(\boldsymbol{G})^k \ \boldsymbol{x}_0 - \boldsymbol{x}\ .$	$\ \boldsymbol{x}_k - \boldsymbol{x}\ \leq \ \boldsymbol{G}^k\ \ \boldsymbol{x}_0 - \boldsymbol{x}\ \approx \rho(\boldsymbol{G})^k \ \boldsymbol{x}_0 - \boldsymbol{x}\ .$
lyche-numerical-linear-algebra-285 EQ0820	285	■ 1.000 ● K +0	$d'(\omega) = 2(\omega \mu^2 - 2) < 2(\omega - 2) < 0.$	$d'(\omega) = 2(\omega \mu^2 - 2) < 2(\omega - 2) < 0.$	$d'(\omega) = 2(\omega \mu^2 - 2) < 2(\omega - 2) < 0.$
lyche-numerical-linear-algebra-292 EQ0840 (13.6)	292	■ 1.000 ●	$\begin{aligned} \frac{\partial}{\partial y_i} Q(\boldsymbol{y}) &:= \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} (Q(\boldsymbol{y} + \varepsilon \boldsymbol{e}_i) - Q(\boldsymbol{y})) \\ &= \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} (-\varepsilon \boldsymbol{e}_i^T \boldsymbol{r}(\boldsymbol{y})) + \frac{1}{2} \varepsilon^2 \boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_i \Big) = -\boldsymbol{e}_i^T \boldsymbol{r}(\boldsymbol{y}), \quad i = 1, \dots, n, \end{aligned}$	$\begin{aligned} \frac{\partial}{\partial y_i} Q(\boldsymbol{y}) &:= \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} (Q(\boldsymbol{y} + \varepsilon \boldsymbol{e}_i) - Q(\boldsymbol{y})) \\ &= \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} (-\varepsilon \boldsymbol{e}_i^T \boldsymbol{r}(\boldsymbol{y})) + \frac{1}{2} \varepsilon^2 \boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_i \Big) = -\boldsymbol{e}_i^T \boldsymbol{r}(\boldsymbol{y}), \quad i = 1, \dots, n, \end{aligned}$	$\begin{aligned} \frac{\partial}{\partial y_i} Q(\boldsymbol{y}) &:= \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} (Q(\boldsymbol{y} + \varepsilon \boldsymbol{e}_i) - Q(\boldsymbol{y})) \\ &= \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \left(-\varepsilon \boldsymbol{e}_i^T \boldsymbol{r}(\boldsymbol{y})) + \frac{1}{2} \varepsilon^2 \boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_i \right) = -\boldsymbol{e}_i^T \boldsymbol{r}(\boldsymbol{y}), \quad i = 1, \dots, n, \end{aligned}$
lyche-numerical-linear-algebra-301 EQ0866 (13.23)	301	■ 1.000 ● N +0	$\boldsymbol{r}_k^T \boldsymbol{w} = \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{w} = 0, \quad \boldsymbol{w} \in \mathbb{W}_k.$	$\boldsymbol{r}_k^T \boldsymbol{w} = \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{w} = 0, \quad \boldsymbol{w} \in \mathbb{W}_k.$	$\boldsymbol{r}_k^T \boldsymbol{w} = \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{w} = 0, \quad \boldsymbol{w} \in \mathbb{W}_k.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0874 (13.29)	303	■ 1.000 ● N -1	$\frac{\left \left \boldsymbol{x}-\boldsymbol{x}_{k}\right \right _{\boldsymbol{A}}}{\left \left \boldsymbol{x}-\boldsymbol{x}_0\right \right _{\boldsymbol{A}}}\leq\min _{Q \in \Pi_k} \max _{\lambda_{\min } \leq x \leq \lambda_{\max }} Q(x) =\frac{2}{a^{-k}+a^k}, \quad a:=\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1},$	$\frac{\ \boldsymbol{x}-\boldsymbol{x}_k\ _{\boldsymbol{A}}}{\ \boldsymbol{x}-\boldsymbol{x}_0\ _{\boldsymbol{A}}} \leq \min_{Q \in \Pi_k} \max_{\lambda_{\min} \leq x \leq \lambda_{\max}} Q(x) = \frac{2}{a^{-k} + a^k}, \quad a := \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1},$	
lyche-numerical-linear-algebra- EQ0875 (13.30)	303	■ 1.000 ● K +0	$\frac{x-1}{x+1}<e^{-2 / x} \quad \text { for } \quad x>1$	$\frac{x-1}{x+1} < e^{-2/x} \quad \text{for} \quad x > 1$	$\frac{x-1}{x+1} < e^{-2/x} \quad \text{for} \quad x > 1$
lyche-numerical-linear-algebra- EQ0929 (13.58)	315	■ 1.000 ● N +0	$\langle \boldsymbol{V}, \boldsymbol{W} \rangle:=h^2 \sum_{j, k=1}^m v_{j, k} w_{j, k},$	$\langle \boldsymbol{V}, \boldsymbol{W} \rangle:=h^2 \sum_{j, k=1}^m v_{j, k} w_{j, k},$	$\langle \boldsymbol{V}, \boldsymbol{W} \rangle:=h^2 \sum_{j, k=1}^m v_{j, k} w_{j, k},$
lyche-numerical-linear-algebra- EQ0930 (13.59)	315	■ 1.000 ● W -1	$\begin{aligned} \left\langle L_h \boldsymbol{v}, \boldsymbol{W}\right\rangle &=\sum_{j=1}^m \sum_{k=0}^m c_{j, k+\frac{1}{2}}\left(v_{j, k+1}-v_{j, k}\right)\left(w_{j, k+1}-w_{j, k}\right) \\ &+\sum_{j=0}^m \sum_{k=1}^m c_{j+\frac{1}{2}, k}\left(v_{j+1, k}-v_{j, k}\right)\left(w_{j+1, k}-w_{j, k}\right) . \end{aligned}$	$\begin{aligned} \langle L_h \boldsymbol{v}, \boldsymbol{W} \rangle &= \sum_{j=1}^m \sum_{k=0}^m c_{j, k+\frac{1}{2}} (v_{j, k+1} - v_{j, k}) (w_{j, k+1} - w_{j, k}) \\ &+ \sum_{j=0}^m \sum_{k=1}^m c_{j+\frac{1}{2}, k} (v_{j+1, k} - v_{j, k}) (w_{j+1, k} - w_{j, k}). \end{aligned}$	$\begin{aligned} \left\langle L_h \boldsymbol{v}, \boldsymbol{W}\right\rangle &= \sum_{j=1}^m \sum_{k=0}^m c_{j, k+\frac{1}{2}} \left(v_{j, k+1} - v_{j, k}\right) \left(w_{j, k+1} - w_{j, k}\right) \\ &+ \sum_{j=0}^m \sum_{k=1}^m c_{j+\frac{1}{2}, k} \left(v_{j+1, k} - v_{j, k}\right) \left(w_{j+1, k} - w_{j, k}\right). \end{aligned}$
lyche-numerical-linear-algebra- EQ0946	320	■ 1.000 ● N -1	$\boldsymbol{A}=\left[\begin{array}{rrr} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{array}\right], \quad \text { and } \quad \boldsymbol{b}=\left[\begin{array}{l} 4 \\ 0 \\ 0 \end{array}\right] .$	$\boldsymbol{A}=\left[\begin{array}{rrr} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{array}\right], \quad \text{and} \quad \boldsymbol{b}=\left[\begin{array}{l} 4 \\ 0 \\ 0 \end{array}\right] .$	$\boldsymbol{A}=\left[\begin{array}{rrr} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{array}\right], \quad \text{and} \quad \boldsymbol{b}=\left[\begin{array}{l} 4 \\ 0 \\ 0 \end{array}\right] .$
lyche-numerical-linear-algebra- EQ0957	322	■ 1.000 ● N +0	$\boldsymbol{r}_{k}:=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_k, \quad \text { and } \quad \rho_k:=\left\ \boldsymbol{r}_k\right\ _2^2 .$	$\boldsymbol{r}_k := \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k, \text{ and } \rho_k := \ \boldsymbol{r}_k\ _2^2 .$	$\boldsymbol{r}_k := \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k, \text{ and } \rho_k := \ \boldsymbol{r}_k\ _2^2 .$
lyche-numerical-linear-algebra- EQ0967	326	■ 1.000 ●	$\begin{aligned} \boldsymbol{A} \boldsymbol{D} &=\operatorname{diag}\left(\boldsymbol{A}_{\{1\}}, \ldots, \boldsymbol{A}_{\{r\}}\right) \operatorname{diag}\left(\boldsymbol{X}_{\{1\}}, \ldots, \boldsymbol{X}_{\{r\}}\right) \\ &=\operatorname{diag}\left(\boldsymbol{A}_{\{1\}} \boldsymbol{X}_{\{1\}}, \ldots, \boldsymbol{A}_{\{r\}} \boldsymbol{X}_{\{r\}}\right) \\ &=\operatorname{diag}\left(\boldsymbol{X}_{\{1\}} \boldsymbol{D}_{\{1\}}, \ldots, \boldsymbol{X}_{\{r\}} \boldsymbol{D}_{\{r\}}\right)=\boldsymbol{X} \boldsymbol{D} . \end{aligned}$	$\begin{aligned} \boldsymbol{A} \boldsymbol{D} &= \operatorname{diag}\left(\boldsymbol{A}_1, \ldots, \boldsymbol{A}_r\right) \operatorname{diag}\left(\boldsymbol{X}_1, \ldots, \boldsymbol{X}_r\right)=\operatorname{diag}\left(\boldsymbol{A}_1 \boldsymbol{X}_1, \ldots, \boldsymbol{A}_r \boldsymbol{X}_r\right) \\ &= \operatorname{diag}\left(\boldsymbol{X}_1 \boldsymbol{D}_1, \ldots, \boldsymbol{X}_r \boldsymbol{D}_r\right)=\boldsymbol{X} \boldsymbol{D} . \end{aligned}$	$\begin{aligned} \boldsymbol{A} \boldsymbol{D} &= \operatorname{diag}\left(\boldsymbol{A}_1, \ldots, \boldsymbol{A}_r\right) \operatorname{diag}\left(\boldsymbol{X}_1, \ldots, \boldsymbol{X}_r\right)=\operatorname{diag}\left(\boldsymbol{A}_1 \boldsymbol{X}_1, \ldots, \boldsymbol{A}_r \boldsymbol{X}_r\right) \\ &= \operatorname{diag}\left(\boldsymbol{X}_1 \boldsymbol{D}_1, \ldots, \boldsymbol{X}_r \boldsymbol{D}_r\right)=\boldsymbol{X} \boldsymbol{D} . \end{aligned}$
lyche-numerical-linear-algebra- EQ1030	348	■ 1.000 ● N +0	$\mu_1(s)=\left(\lambda_1-s\right)^{-1}, \mu_2(s)=\left(\lambda_2-s\right)^{-1}, \ldots, \mu_n(s)=\left(\lambda_n-s\right)^{-1} .$	$\mu_1(s) = (\lambda_1 - s)^{-1}, \mu_2(s) = (\lambda_2 - s)^{-1}, \dots, \mu_n(s) = (\lambda_n - s)^{-1} .$	$\mu_1(s) = (\lambda_1 - s)^{-1}, \mu_2(s) = (\lambda_2 - s)^{-1}, \dots, \mu_n(s) = (\lambda_n - s)^{-1} .$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-022 EQ1041	352	1.000 N +0	$A_{14}=\left[\begin{array}{rrrr}r rr r\ 2.323\ \&\ 0.047223\ \&\ -0.39232\ \&\ -0.65056\ \\ \hline -2.1\ e-10\ \&\ 0.13029\ \&\ 0.36125\ \&\ 0.15946\ \\ -4.1\ e-10\ \&\ -0.58622\ \&\ 0.052576\ \&\ -0.25774\ \\ \hline 1.2\ e-14\ \&\ 3.3\ e-05\ \&\ -1.1\ e-05\ \&\ 0.22746\ \end{array}\right].$	$A_{14}=\left[\begin{array}{ccc c}2.323 & 0.047223 & -0.39232 & -0.65056 \\ -2.1e-10 & 0.13029 & 0.36125 & 0.15946 \\ -4.1e-10 & -0.58622 & 0.052576 & -0.25774 \\ \hline 1.2e-14 & 3.3e-05 & -1.1e-05 & 0.22746\end{array}\right].$	$A_{14}=\left[\begin{array}{ccc c}2.323 & 0.047223 & -0.39232 & -0.65056 \\ -2.1e-10 & 0.13029 & 0.36125 & 0.15946 \\ -4.1e-10 & -0.58622 & 0.052576 & -0.25774 \\ \hline 1.2e-14 & 3.3e-05 & -1.1e-05 & 0.22746\end{array}\right].$
lyche-numerical-linear-algebra-022 EQ1045	353	1.000 N +0	$\tilde{\boldsymbol{Q}}_{k-1}\tilde{\boldsymbol{R}}_k=\tilde{\boldsymbol{Q}}_{k-1}(\boldsymbol{Q}_k\boldsymbol{R}_k)\tilde{\boldsymbol{R}}_{k-1}=\tilde{\boldsymbol{Q}}_{k-1}\boldsymbol{A}_k\tilde{\boldsymbol{R}}_{k-1}=\left(\tilde{\boldsymbol{Q}}_{k-1}\tilde{\boldsymbol{Q}}_{k-1}^*\right)\boldsymbol{A}\tilde{\boldsymbol{Q}}_{k-1}\tilde{\boldsymbol{R}}_{k-1}=\boldsymbol{A}^k.$	$\tilde{\boldsymbol{Q}}_k\tilde{\boldsymbol{R}}_k=\tilde{\boldsymbol{Q}}_{k-1}(\boldsymbol{Q}_k\boldsymbol{R}_k)\tilde{\boldsymbol{R}}_{k-1}=\tilde{\boldsymbol{Q}}_{k-1}\boldsymbol{A}_k\tilde{\boldsymbol{R}}_{k-1}=(\tilde{\boldsymbol{Q}}_{k-1}\tilde{\boldsymbol{Q}}_{k-1}^*)\boldsymbol{A}\tilde{\boldsymbol{Q}}_{k-1}\tilde{\boldsymbol{R}}_{k-1}=\boldsymbol{A}^k.$	$\tilde{\boldsymbol{Q}}_k\tilde{\boldsymbol{R}}_k=\tilde{\boldsymbol{Q}}_{k-1}(\boldsymbol{Q}_k\boldsymbol{R}_k)\tilde{\boldsymbol{R}}_{k-1}=\tilde{\boldsymbol{Q}}_{k-1}\boldsymbol{A}_k\tilde{\boldsymbol{R}}_{k-1}=(\tilde{\boldsymbol{Q}}_{k-1}\tilde{\boldsymbol{Q}}_{k-1}^*)\boldsymbol{A}\tilde{\boldsymbol{Q}}_{k-1}\tilde{\boldsymbol{R}}_{k-1}=\boldsymbol{A}^k.$
lyche-numerical-linear-algebra-022 EQ1047	354	1.000 N +1	$\hat{\boldsymbol{A}}_k=\tilde{\boldsymbol{Q}}_{k-1}^*(\boldsymbol{A}+\boldsymbol{E})\tilde{\boldsymbol{Q}}_{k-1},\quad \boldsymbol{E}=\tilde{\boldsymbol{Q}}_{k-1}\left(a_{i+1,i}^{(k)}\boldsymbol{e}_{i+1}\boldsymbol{e}_i^T\right)\tilde{\boldsymbol{Q}}_{k-1}^*.$	$\hat{\boldsymbol{A}}_k=\tilde{\boldsymbol{Q}}_{k-1}^*(\boldsymbol{A}+\boldsymbol{E})\tilde{\boldsymbol{Q}}_{k-1},\quad \boldsymbol{E}=\tilde{\boldsymbol{Q}}_{k-1}\left(a_{i+1,i}^{(k)}\boldsymbol{e}_{i+1}\boldsymbol{e}_i^T\right)\tilde{\boldsymbol{Q}}_{k-1}^*.$	$\hat{\boldsymbol{A}}_k=\tilde{\boldsymbol{Q}}_{k-1}^*(\boldsymbol{A}+\boldsymbol{E})\tilde{\boldsymbol{Q}}_{k-1},\quad \boldsymbol{E}=\tilde{\boldsymbol{Q}}_{k-1}(a_{i+1,i}^{(k)}\boldsymbol{e}_{i+1}\boldsymbol{e}_i^T)\tilde{\boldsymbol{Q}}_{k-1}^*.$
lyche-numerical-linear-algebra-022 EQ0002	022	1.000 C -4	$\boldsymbol{x}+\boldsymbol{y}:=\left[\begin{array}{c}x_1+y_1\\\vdots\\x_n+y_n\end{array}\right],\quad \boldsymbol{a}\boldsymbol{x}:=\left[\begin{array}{c}ax_1\\\vdots\\ax_n\end{array}\right],\quad \boldsymbol{x},\boldsymbol{y}\in\mathbb{R}^n,\quad \boldsymbol{a}\in\mathbb{R}.$	$\boldsymbol{x}+\boldsymbol{y}:=\left[\begin{array}{c}x_1+y_1\\\vdots\\x_n+y_n\end{array}\right],\quad \boldsymbol{a}\boldsymbol{x}:=\left[\begin{array}{c}ax_1\\\vdots\\ax_n\end{array}\right],\quad \boldsymbol{x},\boldsymbol{y}\in\mathbb{R}^n,\quad \boldsymbol{a}\in\mathbb{R}.$	$\boldsymbol{x}+\boldsymbol{y}:=\left[\begin{array}{c}x_1+y_1\\\vdots\\x_n+y_n\end{array}\right],\quad \boldsymbol{a}\boldsymbol{x}:=\left[\begin{array}{c}ax_1\\\vdots\\ax_n\end{array}\right],\quad \boldsymbol{x},\boldsymbol{y}\in\mathbb{R}^n,\quad \boldsymbol{a}\in\mathbb{R}.$
lyche-numerical-linear-algebra-022 EQ0004	022	1.000 W +1	$\boldsymbol{a}_{:j}:=\left[\begin{array}{c}a_{1j}\\a_{2j}\\\vdots\\a_{mj}\end{array}\right],\quad \boldsymbol{a}_{i:}^T:=\left[a_{i1},a_{i2},\dots,a_{in}\right],\quad \boldsymbol{A}=\left[\boldsymbol{a}_{:1},\boldsymbol{a}_{:2},\dots,\boldsymbol{a}_{:n}\right]=\left[\begin{array}{c}a_{1:}^T\\a_{2:}^T\\\vdots\\a_{m:}^T\end{array}\right]$	$\boldsymbol{a}_{:j}:=\left[\begin{array}{c}a_{1j}\\a_{2j}\\\vdots\\a_{mj}\end{array}\right],\quad \boldsymbol{a}_{i:}^T:=\left[a_{i1},a_{i2},\dots,a_{in}\right],\quad \boldsymbol{A}=\left[\boldsymbol{a}_{:1},\boldsymbol{a}_{:2},\dots,\boldsymbol{a}_{:n}\right]=\left[\begin{array}{c}a_{1:}^T\\a_{2:}^T\\\vdots\\a_{m:}^T\end{array}\right]$	$\boldsymbol{a}_{:j}:=\left[\begin{array}{c}a_{1j}\\a_{2j}\\\vdots\\a_{mj}\end{array}\right],\quad \boldsymbol{a}_{i:}^T:=\left[a_{i1},a_{i2},\dots,a_{in}\right],\quad \boldsymbol{A}=\left[\boldsymbol{a}_{:1},\boldsymbol{a}_{:2},\dots,\boldsymbol{a}_{:n}\right]=\left[\begin{array}{c}a_{1:}^T\\a_{2:}^T\\\vdots\\a_{m:}^T\end{array}\right]$
lyche-numerical-linear-algebra-022 EQ0005	022	1.000	$e^x=e^{a+ib}:=e^a(\cos b+i\sin b).$	$e^x=e^{a+ib}:=e^a(\cos b+i\sin b).$	$e^x=e^{a+ib}:=e^a(\cos b+i\sin b).$
lyche-numerical-linear-algebra-022 EQ0006	022	1.000 N +0	$e^{i\pi/2}=i,\quad e^{i\pi}=-1,\quad e^{2i\pi}=1.$	$e^{i\pi/2}=i,\quad e^{i\pi}=-1,\quad e^{2i\pi}=1.$	$e^{i\pi/2}=i,\quad e^{i\pi}=-1,\quad e^{2i\pi}=1.$
lyche-numerical-linear-algebra-022 EQ0007	023	1.000 N +0	$x=a+ib=re^{i\theta},\quad r= x =\sqrt{a^2+b^2},\quad \cos\theta=\frac{a}{r},\quad \sin\theta=\frac{b}{r}.$	$x=a+ib=re^{i\theta},\quad r= x =\sqrt{a^2+b^2},\quad \cos\theta=\frac{a}{r},\quad \sin\theta=\frac{b}{r}.$	$x=a+ib=re^{i\theta},\quad r= x =\sqrt{a^2+b^2},\quad \cos\theta=\frac{a}{r},\quad \sin\theta=\frac{b}{r}.$
lyche-numerical-linear-algebra-022 EQ0014 (1.2)	025	1.000 N +1	$p(t):=a_0+a_1t+a_2t^2+\cdots+a_nt^n,$	$p(t):=a_0+a_1t+a_2t^2+\cdots+a_nt^n,$	$p(t):=a_0+a_1t+a_2t^2+\cdots+a_nt^n,$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-028 EQ0017 (1.5)	028	1.000 N +0	$\mathcal{S} \cap \mathcal{T} := \{\boldsymbol{x} : \boldsymbol{x} \in \mathcal{S} \text{ and } \boldsymbol{x} \in \mathcal{T}\}$	$\mathcal{S} \cap \mathcal{T} := \{\boldsymbol{x} : \boldsymbol{x} \in \mathcal{S} \text{ and } \boldsymbol{x} \in \mathcal{T}\}$	$\mathcal{S} \cap \mathcal{T} := \{\boldsymbol{x} : \boldsymbol{x} \in \mathcal{S} \text{ and } \boldsymbol{x} \in \mathcal{T}\}$
lyche-numerical-linear-algebra-028 EQ0018 (1.6)	028	1.000 N -1	$\mathcal{S} \cup \mathcal{T} := \{\boldsymbol{x} : \boldsymbol{x} \in \mathcal{S} \text{ or } \boldsymbol{x} \in \mathcal{T}\}$	$\mathcal{S} \cup \mathcal{T} := \{\boldsymbol{x} : \boldsymbol{x} \in \mathcal{S} \text{ or } \boldsymbol{x} \in \mathcal{T}\}$	$\mathcal{S} \cup \mathcal{T} := \{\boldsymbol{x} : \boldsymbol{x} \in \mathcal{S} \text{ or } \boldsymbol{x} \in \mathcal{T}\}$
lyche-numerical-linear-algebra-029 EQ0020 (1.8)	029	1.000 N -1	$\operatorname{dim}(\mathcal{S} \oplus \mathcal{T}) = \operatorname{dim}(\mathcal{S}) + \operatorname{dim}(\mathcal{T})$	$\dim(\mathcal{S} \oplus \mathcal{T}) = \dim(\mathcal{S}) + \dim(\mathcal{T})$	$\dim(\mathcal{S} \oplus \mathcal{T}) = \dim(\mathcal{S}) + \dim(\mathcal{T})$
lyche-numerical-linear-algebra-029 EQ0021	029	1.000 N +1	$\left\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{s}_1, \dots, \boldsymbol{s}_q, \boldsymbol{t}_1, \dots, \boldsymbol{t}_r\right\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{s}_1, \dots, \boldsymbol{s}_q, \boldsymbol{t}_1, \dots, \boldsymbol{t}_r\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{s}_1, \dots, \boldsymbol{s}_q, \boldsymbol{t}_1, \dots, \boldsymbol{t}_r\}$
lyche-numerical-linear-algebra-029 EQ0022	029	1.000 N +0	$\left\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{s}_1, \dots, \boldsymbol{s}_q\right\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{s}_1, \dots, \boldsymbol{s}_q\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{s}_1, \dots, \boldsymbol{s}_q\}$
lyche-numerical-linear-algebra-029 EQ0023	029	1.000 C -3	$\operatorname{dim}(\mathcal{S} + \mathcal{T}) = p + q + r = (p + q) + (p + r) - p = \operatorname{dim}(\mathcal{S}) + \operatorname{dim}(\mathcal{T}) - \operatorname{dim}(\mathcal{S} \cap \mathcal{T})$	$\dim(\mathcal{S} + \mathcal{T}) = p + q + r = (p + q) + (p + r) - p = \dim(\mathcal{S}) + \dim(\mathcal{T}) - \dim(\mathcal{S} \cap \mathcal{T})$	$\dim(\mathcal{S} + \mathcal{T}) = p + q + r = (p + q) + (p + r) - p = \dim(\mathcal{S}) + \dim(\mathcal{T}) - \dim(\mathcal{S} \cap \mathcal{T})$
lyche-numerical-linear-algebra-030 EQ0026	030	1.000 N +0	$\mathcal{R}(\boldsymbol{X}) := \{\boldsymbol{X}\boldsymbol{c} : \boldsymbol{c} \in \mathbb{R}^n\} = \operatorname{span}(\mathcal{X})$	$\mathcal{R}(\boldsymbol{X}) := \{\boldsymbol{X}\boldsymbol{c} : \boldsymbol{c} \in \mathbb{R}^n\} = \operatorname{span}(\mathcal{X})$	$\mathcal{R}(\boldsymbol{X}) := \{\boldsymbol{X}\boldsymbol{c} : \boldsymbol{c} \in \mathbb{R}^n\} = \operatorname{span}(\mathcal{X})$
lyche-numerical-linear-algebra-031 EQ0030	031	1.000 N +0	$x_1 \boldsymbol{a}_1 + x_2 \boldsymbol{a}_2 + \dots + x_n \boldsymbol{a}_n = \boldsymbol{b},$	$x_1 \boldsymbol{a}_1 + x_2 \boldsymbol{a}_2 + \dots + x_n \boldsymbol{a}_n = \boldsymbol{b},$	$x_1 \boldsymbol{a}_1 + x_2 \boldsymbol{a}_2 + \dots + x_n \boldsymbol{a}_n = \boldsymbol{b},$
lyche-numerical-linear-algebra-035 EQ0035 (1.12)	035	1.000 N +1	$\operatorname{det}(\boldsymbol{A}) = \sum_{\sigma \in S_n} \operatorname{sign}(\sigma) a_{\sigma(1),1} a_{\sigma(2),2} \cdots a_{\sigma(n),n}$	$\det(\boldsymbol{A}) = \sum_{\sigma \in S_n} \operatorname{sign}(\sigma) a_{\sigma(1),1} a_{\sigma(2),2} \cdots a_{\sigma(n),n}$	$\det(\boldsymbol{A}) = \sum_{\sigma \in S_n} \operatorname{sign}(\sigma) a_{\sigma(1),1} a_{\sigma(2),2} \cdots a_{\sigma(n),n}$
lyche-numerical-linear-algebra-035 EQ0036	035	1.000 N +0	$\left \begin{array}{cccc} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{array} \right $	$\left \begin{array}{cccc} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{array} \right $	$\left \begin{array}{cccc} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{array} \right $
lyche-numerical-linear-algebra-035 EQ0037	035	1.000 N +0	$\left \begin{array}{ll} a_{11} & a_{12} \\ a_{21} & a_{22} \end{array} \right = a_{11} a_{22} - a_{21} a_{12}$	$\left \begin{array}{ll} a_{11} & a_{12} \\ a_{21} & a_{22} \end{array} \right = a_{11} a_{22} - a_{21} a_{12}$	$\left \begin{array}{ll} a_{11} & a_{12} \\ a_{21} & a_{22} \end{array} \right = a_{11} a_{22} - a_{21} a_{12}$
lyche-numerical-linear-algebra-036 EQ0039 (1.13)	036	1.000 K +0	$\det(\boldsymbol{B}) = \det(\boldsymbol{A})$	$\det(\boldsymbol{B}) = \det(\boldsymbol{A})$	$\det(\boldsymbol{B}) = \det(\boldsymbol{A})$
lyche-numerical-linear-algebra-036 EQ0041	036	1.000	$\det(\boldsymbol{A}) := \left \begin{array}{ccc} 1 & x & y \\ 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \end{array} \right = 0$	$\det(\boldsymbol{A}) := \left \begin{array}{ccc} 1 & x & y \\ 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \end{array} \right = 0$	$\det(\boldsymbol{A}) := \left \begin{array}{ccc} 1 & x & y \\ 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \end{array} \right = 0$
lyche-numerical-linear-algebra-036 EQ0043	036	1.000 N +0	$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$	$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$	$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-037-EQ0044	037	■ 1.000 ● N +0	$x_{\{j\}}=\frac{\operatorname{det}\left(\boldsymbol{A}_{\{j\}}(\boldsymbol{b})\right)}{\operatorname{det}(\boldsymbol{A})}, \quad j=1,2,\ldots,n,$	$x_j = \frac{\det(\boldsymbol{A}_j(\boldsymbol{b}))}{\det(\boldsymbol{A})}, \quad j = 1, 2, \dots, n,$	$x_j = \frac{\det(\boldsymbol{A}_j(\boldsymbol{b}))}{\det(\boldsymbol{A})}, \quad j = 1, 2, \dots, n,$
lyche-numerical-linear-algebra-037-EQ0045	037	■ 1.000 ● K +0	$\boldsymbol{A}^{-1}=\frac{1}{\operatorname{det}(\boldsymbol{A})}\operatorname{adj}(\boldsymbol{A}),$	$\boldsymbol{A}^{-1} = \frac{1}{\det(\boldsymbol{A})} \operatorname{adj}(\boldsymbol{A}),$	$\boldsymbol{A}^{-1} = \frac{1}{\det(\boldsymbol{A})} \operatorname{adj}(\boldsymbol{A}),$
lyche-numerical-linear-algebra-038-EQ0048	038	■ 1.000 ● W +0	$\operatorname{det}(\boldsymbol{A}-\lambda\boldsymbol{I})=\left \begin{array}{ccc}a_{11}-\lambda&a_{12}&a_{13}\\a_{21}&a_{22}-\lambda&a_{23}\\a_{31}&a_{32}&a_{33}-\lambda\end{array}\right .$	$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = \begin{vmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ a_{21} & a_{22} - \lambda & a_{23} \\ a_{31} & a_{32} & a_{33} - \lambda \end{vmatrix}.$	$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = \begin{vmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ a_{21} & a_{22} - \lambda & a_{23} \\ a_{31} & a_{32} & a_{33} - \lambda \end{vmatrix}.$
lyche-numerical-linear-algebra-038-EQ0050 (1.16)	038	■ 1.000 ● N +0	$\operatorname{det}(\boldsymbol{A}-\lambda\boldsymbol{I})=\left(a_{11}-\lambda\right)\left(a_{22}-\lambda\right)\cdots\left(a_{nn}-\lambda\right)+r(\lambda),$	$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = (a_{11} - \lambda)(a_{22} - \lambda) \cdots (a_{nn} - \lambda) + r(\lambda),$	$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = (a_{11} - \lambda)(a_{22} - \lambda) \cdots (a_{nn} - \lambda) + r(\lambda),$
lyche-numerical-linear-algebra-039-EQ0051 (1.17)	039	■ 1.000 ● N +0	$\operatorname{trace}(\boldsymbol{A})=\lambda_1+\lambda_2+\cdots+\lambda_n, \quad \operatorname{det}(\boldsymbol{A})=\lambda_1\lambda_2\cdots\lambda_n,$	$\operatorname{trace}(\boldsymbol{A}) = \lambda_1 + \lambda_2 + \cdots + \lambda_n, \quad \det(\boldsymbol{A}) = \lambda_1 \lambda_2 \cdots \lambda_n,$	$\operatorname{trace}(\boldsymbol{A}) = \lambda_1 + \lambda_2 + \cdots + \lambda_n, \quad \det(\boldsymbol{A}) = \lambda_1 \lambda_2 \cdots \lambda_n,$
lyche-numerical-linear-algebra-039-EQ0052 (1.18)	039	■ 1.000 ● N -1	$\operatorname{trace}(\boldsymbol{A}):=a_{11}+a_{22}+\cdots+a_{nn}.$	$\operatorname{trace}(\boldsymbol{A}) := a_{11} + a_{22} + \cdots + a_{nn}.$	$\operatorname{trace}(\boldsymbol{A}) := a_{11} + a_{22} + \cdots + a_{nn}.$
lyche-numerical-linear-algebra-039-EQ0054	039	■ 1.000 ● N +0	$\pi_{\boldsymbol{A}}(\lambda)=\left(\lambda_1-\lambda\right)\cdots\left(\lambda_n-\lambda\right)=(-1)^n\lambda^n+d_{n-1}\lambda^{n-1}+\cdots+d_0,$	$\pi_{\boldsymbol{A}}(\lambda) = (\lambda_1 - \lambda) \cdots (\lambda_n - \lambda) = (-1)^n \lambda^n + d_{n-1} \lambda^{n-1} + \cdots + d_0,$	$\pi_{\boldsymbol{A}}(\lambda) = (\lambda_1 - \lambda) \cdots (\lambda_n - \lambda) = (-1)^n \lambda^n + d_{n-1} \lambda^{n-1} + \cdots + d_0,$
lyche-numerical-linear-algebra-039-EQ0055 (1.19)	039	■ 1.000 ● K +0	$\lambda^2-\operatorname{trace}(\boldsymbol{A})\lambda+\operatorname{det}(\boldsymbol{A})=0.$	$\lambda^2 - \operatorname{trace}(\boldsymbol{A})\lambda + \det(\boldsymbol{A}) = 0.$	$\lambda^2 - \operatorname{trace}(\boldsymbol{A})\lambda + \det(\boldsymbol{A}) = 0.$
lyche-numerical-linear-algebra-040-EQ0056	040	■ 1.000 ● K +0	$\left[\begin{array}{ll}W&X\\Y&Z\end{array}\right]=\left[\begin{array}{ll}A&B\\C&D\end{array}\right]\left[\begin{array}{ll}E&F\\G&H\end{array}\right],$	$\begin{bmatrix} W & X \\ Y & Z \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} E & F \\ G & H \end{bmatrix},$	$\begin{bmatrix} W & X \\ Y & Z \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} E & F \\ G & H \end{bmatrix},$
lyche-numerical-linear-algebra-041-EQ0059	041	■ 1.000 ● N +0	$A=\left[\begin{array}{cc}\cos\theta&-\sin\theta\\\sin\theta&\cos\theta\end{array}\right].$	$A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$	$A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$
lyche-numerical-linear-algebra-042-EQ0063	042	■ 1.000 ● K +0	$\left[\begin{array}{ll}1&2\\2&1\end{array}\right]\left[\begin{array}{l}x_1\\x_2\end{array}\right]=\left[\begin{array}{l}3\\6\end{array}\right]$	$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$
lyche-numerical-linear-algebra-042-EQ0065	042	■ 1.000 ● N +0	$\operatorname{adj}(\boldsymbol{A})=\left[\begin{array}{ccc}14&21&42\\-42&-14&21\\21&-42&14\end{array}\right].$	$\operatorname{adj}(\boldsymbol{A}) = \begin{bmatrix} 14 & 21 & 42 \\ -42 & -14 & 21 \\ 21 & -42 & 14 \end{bmatrix}.$	$\operatorname{adj}(\boldsymbol{A}) = \begin{bmatrix} 14 & 21 & 42 \\ -42 & -14 & 21 \\ 21 & -42 & 14 \end{bmatrix}.$
lyche-numerical-linear-algebra-042-EQ0066	042	■ 1.000 ●	$\operatorname{adj}(\boldsymbol{A})\boldsymbol{A}=\left[\begin{array}{ccc}343&0&0\\0&343&0\\0&0&343\end{array}\right]=\operatorname{det}(\boldsymbol{A})\boldsymbol{I}.$	$\operatorname{adj}(\boldsymbol{A}) \boldsymbol{A} = \begin{bmatrix} 343 & 0 & 0 \\ 0 & 343 & 0 \\ 0 & 0 & 343 \end{bmatrix} = \det(\boldsymbol{A}) \boldsymbol{I}.$	$\operatorname{adj}(\boldsymbol{A}) \boldsymbol{A} = \begin{bmatrix} 343 & 0 & 0 \\ 0 & 343 & 0 \\ 0 & 0 & 343 \end{bmatrix} = \det(\boldsymbol{A}) \boldsymbol{I}.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-042 EQ0067	042	■ 1.000 ● W +1	$\left \begin{array}{llll} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{array}\right = 0$	$\left \begin{array}{llll} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{array}\right = 0$	$\left \begin{array}{llll} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{array}\right = 0.$
lyche-numerical-linear-algebra-043 EQ0068	043	■ 1.000 ● W +0	$A(T)=\frac{1}{2}\left \begin{array}{ccc} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{array}\right .$	$A(T) = \frac{1}{2} \left \begin{array}{ccc} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{array} \right .$	$A(T) = \frac{1}{2} \left \begin{array}{ccc} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{array} \right .$
lyche-numerical-linear-algebra-043 EQ0069	043	■ 1.000 ● W +0	$\left \begin{array}{cccccc} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} & \\ \cdots & x_1^{n-1} & \cdots & 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} & \cdots & \\ x_2^2 & \cdots & x_2^{n-1} & \cdots & \vdots & \vdots & \vdots & \vdots & \cdots & \\ & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} & \end{array}\right = \prod_{i>j} (x_i - x_j),$	$\left \begin{array}{cccccc} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} & \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} & \\ \vdots & \vdots & \vdots & & \vdots & \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} & \end{array}\right = \prod_{i>j} (x_i - x_j),$	$\left \begin{array}{cccccc} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} & \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} & \\ \vdots & \vdots & \vdots & & \vdots & \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} & \end{array}\right = \prod_{i>j} (x_i - x_j),$
lyche-numerical-linear-algebra-043 EQ0070	043	■ 1.000 ● N +0	$\operatorname{det}(\boldsymbol{A})=P\ g(\boldsymbol{\alpha})\ g(\boldsymbol{\beta})$	$\det(\boldsymbol{A}) = P g(\boldsymbol{\alpha}) g(\boldsymbol{\beta})$	$\det(\boldsymbol{A}) = P g(\boldsymbol{\alpha}) g(\boldsymbol{\beta})$
lyche-numerical-linear-algebra-044 EQ0071	044	■ 1.000 ● N +0	$g(\boldsymbol{\gamma})=\prod_{i=2}^n\left(\gamma_i-\gamma_1\right)\left(\gamma_i-\gamma_2\right) \cdots\left(\gamma_i-\gamma_{i-1}\right)$	$g(\boldsymbol{\gamma}) = \prod_{i=2}^n (\gamma_i - \gamma_1) (\gamma_i - \gamma_2) \cdots (\gamma_i - \gamma_{i-1})$	$g(\boldsymbol{\gamma}) = \prod_{i=2}^n (\gamma_i - \gamma_1) (\gamma_i - \gamma_2) \cdots (\gamma_i - \gamma_{i-1})$
lyche-numerical-linear-algebra-044 EQ0072	044	■ 1.000 ● N +0	$q(\boldsymbol{\alpha}, \boldsymbol{\beta})=k\ g(\boldsymbol{\alpha})\ g(\boldsymbol{\beta})$	$q(\boldsymbol{\alpha}, \boldsymbol{\beta}) = k g(\boldsymbol{\alpha}) g(\boldsymbol{\beta})$	$q(\boldsymbol{\alpha}, \boldsymbol{\beta}) = k g(\boldsymbol{\alpha}) g(\boldsymbol{\beta})$
lyche-numerical-linear-algebra-044 EQ0074	044	■ 1.000 ● N +0	$A_{\{k\}}(x)=\prod_{s \neq k}\left(\frac{\alpha_{\{s\}}-x}{\alpha_{\{s\}}-\alpha_{\{k\}}}\right), \quad B_{\{k\}}(x)=\prod_{s \neq k}\left(\frac{\beta_{\{s\}}-x}{\beta_{\{s\}}-\beta_{\{k\}}}\right) .$	$A_k(x) = \prod_{s \neq k} \left(\frac{\alpha_s - x}{\alpha_s - \alpha_k} \right), \quad B_k(x) = \prod_{s \neq k} \left(\frac{\beta_s - x}{\beta_s - \beta_k} \right).$	$A_k(x) = \prod_{s \neq k} \left(\frac{\alpha_s - x}{\alpha_s - \alpha_k} \right), \quad B_k(x) = \prod_{s \neq k} \left(\frac{\beta_s - x}{\beta_s - \beta_k} \right).$
lyche-numerical-linear-algebra-044 EQ0075	044	■ 1.000 ● N +0	$t_{\{i, j\}^{\{n\}}}=\frac{f(i) \ f(j)}{i+j-1},$	$t_{i,j}^n = \frac{f(i)f(j)}{i+j-1},$	$t_{i,j}^n = \frac{f(i)f(j)}{i+j-1},$
lyche-numerical-linear-algebra-044 EQ0076	044	■ 1.000 ● N +0	$f(i+1)=\left(\frac{i^2-n^2}{i^2}\right) f(i), \quad i=1,2, \ldots, \quad f(1)=-n .$	$f(i+1) = \left(\frac{i^2 - n^2}{i^2} \right) f(i), \quad i = 1, 2, \dots, \quad f(1) = -n.$	$f(i+1) = \left(\frac{i^2 - n^2}{i^2} \right) f(i), \quad i = 1, 2, \dots, \quad f(1) = -n.$
lyche-numerical-linear-algebra-046 EQ0077 (2.1)	046	■ 1.000 ● N +0	$x_{\{i\}}=a+\frac{i-1}{n}(b-a), \quad i=1,2, \ldots, n+1$	$x_i = a + \frac{i-1}{n} (b-a), \quad i = 1, 2, \dots, n+1$	$x_i = a + \frac{i-1}{n} (b-a), \quad i = 1, 2, \dots, n+1$
lyche-numerical-linear-algebra-047 EQ0079 (2.3)	047	■ 1.000 ● N +0	$g(x)=y_1+\frac{y_2-y_1}{x_2-x_1}\left(x-x_1\right),$	$g(x) = y_1 + \frac{y_2 - y_1}{x_2 - x_1} (x - x_1),$	$g(x) = y_1 + \frac{y_2 - y_1}{x_2 - x_1} (x - x_1),$
lyche-numerical-linear-algebra-047 EQ0080	047	■ 1.000 ● N -2	$f(x)=\arctan \left(10\ x\right)+\pi / 2, \quad x \in[-1,1] .$	$f(x) = \arctan(10x) + \pi/2, \quad x \in [-1, 1].$	$f(x) = \arctan(10x) + \pi/2, \quad x \in [-1, 1].$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-alignatE00082 (2.5)	049	1.000	$g(x):=\begin{cases}p_1(x)=-\frac{1}{2}x+\frac{3}{2}x^3, & \text{if } 0\leq x<1, \\ p_2(x)=1+4(x-1)+\frac{9}{2}(x-1)^2+\frac{13}{2}(x-1)^3, & \text{if } 1\leq x\leq 2,\end{cases}$	$g(x):=\begin{cases}p_1(x)=-\frac{1}{2}x+\frac{3}{2}x^3, & \text{if } 0\leq x<1, \\ p_2(x)=1+4(x-1)+\frac{9}{2}(x-1)^2+\frac{13}{2}(x-1)^3, & \text{if } 1\leq x\leq 2,\end{cases}$	$g(x):=\begin{cases}p_1(x)=-\frac{1}{2}x+\frac{3}{2}x^3, & \text{if } 0\leq x<1, \\ p_2(x)=1+4(x-1)+\frac{9}{2}(x-1)^2+\frac{13}{2}(x-1)^3, & \text{if } 1\leq x\leq 2,\end{cases}$
lyche-numerical-linear-alignatE00083	049	1.000 N +0	$p_{i-1}^{(j)}(x_i)=p_i^{(j)}(x_i), \quad j=0,1,2, \quad i=2,\dots,n.$	$p_{i-1}^{(j)}(x_i)=p_i^{(j)}(x_i), \quad j=0,1,2, \quad i=2,\dots,n.$	$p_{i-1}^{(j)}(x_i)=p_i^{(j)}(x_i), \quad j=0,1,2, \quad i=2,\dots,n.$
lyche-numerical-linear-alignatE00085 (2.7)	050	1.000 N +0	$p_i(x)=c_{i,1}+c_{i,2}(x-x_i)+c_{i,3}(x-x_i)^2+c_{i,4}(x-x_i)^3 \quad i=1,\dots,n,$	$p_i(x)=c_{i,1}+c_{i,2}(x-x_i)+c_{i,3}(x-x_i)^2+c_{i,4}(x-x_i)^3 \quad i=1,\dots,n,$	$p_i(x)=c_{i,1}+c_{i,2}(x-x_i)+c_{i,3}(x-x_i)^2+c_{i,4}(x-x_i)^3 \quad i=1,\dots,n,$
lyche-numerical-linear-alignatE00086 (2.8)	050	1.000 W +2	$c_{i1}=y_i, \quad c_{i2}=\frac{y_{i+1}-y_i}{h}-\frac{h}{3}\mu_i-\frac{h}{6}\mu_{i+1}, \quad c_{i3}=\frac{\mu_i}{2}, \quad c_{i4}=\frac{\mu_{i+1}-\mu_i}{6h}.$	$c_{i1}=y_i, \quad c_{i2}=\frac{y_{i+1}-y_i}{h}-\frac{h}{3}\mu_i-\frac{h}{6}\mu_{i+1}, \quad c_{i3}=\frac{\mu_i}{2}, \quad c_{i4}=\frac{\mu_{i+1}-\mu_i}{6h}.$	$c_{i1}=y_i, \quad c_{i2}=\frac{y_{i+1}-y_i}{h}-\frac{h}{3}\mu_i-\frac{h}{6}\mu_{i+1}, \quad c_{i3}=\frac{\mu_i}{2}, \quad c_{i4}=\frac{\mu_{i+1}-\mu_i}{6h}.$
lyche-numerical-linear-alignatE00088 (2.10)	051	1.000 W +0	$\begin{aligned} & p_{i-1}^{(j)}(x_i) \\ & = c_{i-1,2} + 2h c_{i-1,3} + 3h^2 c_{i-1,4} \\ & = \frac{y_i - y_{i-1}}{h} - \frac{h}{3}\mu_{i-1} - \frac{h}{6}\mu_i + 2h\frac{\mu_{i-1}}{2} + 3h^2\frac{\mu_i - \mu_{i-1}}{6h} \\ & = \frac{y_i - y_{i-1}}{h} + \frac{h}{6}\mu_{i-1} + \frac{h}{3}\mu_i \\ & p_i'(x_i) = c_{i2} = \frac{y_{i+1} - y_i}{h} - \frac{h}{3}\mu_i - \frac{h}{6}\mu_{i+1}. \end{aligned}$	$\begin{aligned} & p_{i-1}'(x_i) = c_{i-1,2} + 2h c_{i-1,3} + 3h^2 c_{i-1,4} \\ & = \frac{y_i - y_{i-1}}{h} - \frac{h}{3}\mu_{i-1} - \frac{h}{6}\mu_i + 2h\frac{\mu_{i-1}}{2} + 3h^2\frac{\mu_i - \mu_{i-1}}{6h} \\ & = \frac{y_i - y_{i-1}}{h} + \frac{h}{6}\mu_{i-1} + \frac{h}{3}\mu_i \\ & p_i'(x_i) = c_{i2} = \frac{y_{i+1} - y_i}{h} - \frac{h}{3}\mu_i - \frac{h}{6}\mu_{i+1}. \end{aligned}$	$\begin{aligned} & p_{i-1}'(x_i) = c_{i-1,2} + 2h c_{i-1,3} + 3h^2 c_{i-1,4} \\ & = \frac{y_i - y_{i-1}}{h} - \frac{h}{3}\mu_{i-1} - \frac{h}{6}\mu_i + 2h\frac{\mu_{i-1}}{2} + 3h^2\frac{\mu_i - \mu_{i-1}}{6h} \\ & = \frac{y_i - y_{i-1}}{h} + \frac{h}{6}\mu_{i-1} + \frac{h}{3}\mu_i \\ & p_i'(x_i) = c_{i2} = \frac{y_{i+1} - y_i}{h} - \frac{h}{3}\mu_i - \frac{h}{6}\mu_{i+1}. \end{aligned}$
lyche-numerical-linear-alignatE00090 (2.12)	052	1.000 N +0	$ a_{ii} > \sum_{j \neq i} a_{ij} , \quad i=1,\dots,n.$	$ a_{ii} > \sum_{j \neq i} a_{ij} , \quad i=1,\dots,n.$	$ a_{ii} > \sum_{j \neq i} a_{ij} , \quad i=1,\dots,n.$
lyche-numerical-linear-alignatE00091 (2.13)	052	1.000 W +1	$\max_{1 \leq i \leq n} x_i \leq \max_{1 \leq i \leq n} \left(\frac{ b_i }{\sigma_i} \right), \quad \text{where } \sigma_i := a_{ii} - \sum_{j \neq i} a_{ij} .$	$\max_{1 \leq i \leq n} x_i \leq \max_{1 \leq i \leq n} \left(\frac{ b_i }{\sigma_i} \right), \quad \text{where } \sigma_i := a_{ii} - \sum_{j \neq i} a_{ij} .$	$\max_{1 \leq i \leq n} x_i \leq \max_{1 \leq i \leq n} \left(\frac{ b_i }{\sigma_i} \right), \quad \text{where } \sigma_i := a_{ii} - \sum_{j \neq i} a_{ij} .$
lyche-numerical-linear-alignatE00092	052	1.000 W +0	$ b_k = \left a_{kk}x_k + \sum_{j \neq k} a_{kj}x_j \right \geq a_{kk} x_k - \sum_{j \neq k} a_{kj} x_j \geq x_k \left(a_{kk} - \sum_{j \neq k} a_{kj} \right),$	$ b_k = \left a_{kk}x_k + \sum_{j \neq k} a_{kj}x_j \right \geq a_{kk} x_k - \sum_{j \neq k} a_{kj} x_j \geq x_k \left(a_{kk} - \sum_{j \neq k} a_{kj} \right),$	$ b_k = a_{kk}x_k + \sum_{j \neq k} a_{kj}x_j \geq a_{kk} x_k - \sum_{j \neq k} a_{kj} x_j \geq x_k (a_{kk} - \sum_{j \neq k} a_{kj}),$
lyche-numerical-linear-alignatE00093	052	1.000 N +0	$\begin{bmatrix} 4 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 4 \end{bmatrix} \begin{bmatrix} \mu_2 \\ \mu_3 \\ \mu_4 \end{bmatrix} = \begin{bmatrix} 2 \\ -6 \\ 2 \end{bmatrix}.$	$\begin{bmatrix} 4 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 4 \end{bmatrix} \begin{bmatrix} \mu_2 \\ \mu_3 \\ \mu_4 \end{bmatrix} = \begin{bmatrix} 2 \\ -6 \\ 2 \end{bmatrix}.$	$\begin{bmatrix} 4 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 4 \end{bmatrix} \begin{bmatrix} \mu_2 \\ \mu_3 \\ \mu_4 \end{bmatrix} = \begin{bmatrix} 2 \\ -6 \\ 2 \end{bmatrix}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-053 EQ0094 (2.14)	053	1.000	$g(x) := \begin{cases} p_1(x) = \frac{1}{6}x^3, & \text{if } 0 \leq x < 1, \\ p_2(x) = \frac{1}{6} + \frac{1}{2}(x-1) + \frac{1}{2}(x-1)^2 - \frac{1}{2}(x-1)^3, & \text{if } 1 \leq x < 2, \\ p_3(x) = \frac{2}{3} - (x-2)^2 + \frac{1}{2}(x-2)^3, & \text{if } 2 \leq x < 3, \\ p_4(x) = \frac{1}{6} - \frac{1}{2}(x-3) + \frac{1}{2}(x-3)^2 - \frac{1}{6}(x-3)^3, & \text{if } 3 \leq x \leq 4, \end{cases}$		
lyche-numerical-linear-algebra-054 EQ0095 (2.15)	054	1.000 W +2	$\begin{bmatrix} d_1 & c_1 & & & \\ a_1 & d_2 & c_2 & & \\ & \ddots & \ddots & \ddots & \\ & & a_{n-2} & d_{n-1} & c_{n-1} \\ & & & a_{n-1} & d_n \end{bmatrix} = \begin{bmatrix} 1 & & & & \\ l_1 & 1 & & & \\ & \ddots & \ddots & & \\ & & l_{n-1} & 1 & \end{bmatrix} \begin{bmatrix} u_1 & c_1 & & & \\ & \ddots & \ddots & & \\ & & u_{n-1} & c_{n-1} & \\ & & & u_n & \end{bmatrix}.$		$\begin{bmatrix} d_1 & c_1 & & & \\ a_1 & d_2 & c_2 & & \\ & \ddots & \ddots & \ddots & \\ & & a_{n-2} & d_{n-1} & c_{n-1} \\ & & & a_{n-1} & d_n \end{bmatrix} = \begin{bmatrix} 1 & & & & \\ l_1 & 1 & & & \\ & \ddots & \ddots & & \\ & & l_{n-1} & 1 & \end{bmatrix} \begin{bmatrix} u_1 & c_1 & & & \\ & \ddots & \ddots & & \\ & & u_{n-1} & c_{n-1} & \\ & & & u_n & \end{bmatrix}.$
lyche-numerical-linear-algebra-055 EQ0100 (2.17)	055	1.000 N +0	$\left u_k\right \geq \sigma_k+\left c_k\right , \quad \text { where, } \sigma_k:=\left d_k\right -\left a_{k-1}\right -\left c_k\right >0, k=1, \ldots, n,$	$\left u_k\right \geq \sigma_k+\left c_k\right , \quad \text { where, } \sigma_k:=\left d_k\right -\left a_{k-1}\right -\left c_k\right >0, k=1, \ldots, n,$	$\left u_k\right \geq \sigma_k+\left c_k\right , \quad \text { where, } \sigma_k:=\left d_k\right -\left a_{k-1}\right -\left c_k\right >0, k=1, \ldots, n,$
lyche-numerical-linear-algebra-056 EQ0104 (2.20)	056	1.000 N +0	$-u''(x)=f(x), \quad x \in[0,1], \quad u(0)=0, u(1)=0,$	$-u''(x)=f(x), \quad x \in[0,1], \quad u(0)=0, u(1)=0,$	$-u''(x)=f(x), \quad x \in[0,1], \quad u(0)=0, u(1)=0,$
lyche-numerical-linear-algebra-056 EQ0105	056	1.000 W +0	$g'(x)=\lim _{h \rightarrow 0} \frac{g\left(x+\frac{h}{2}\right)-g\left(x-\frac{h}{2}\right)}{h}$	$g'(x)=\lim _{h \rightarrow 0} \frac{g\left(x+\frac{h}{2}\right)-g\left(x-\frac{h}{2}\right)}{h}$	$g'(x)=\lim _{h \rightarrow 0} \frac{g\left(x+\frac{h}{2}\right)-g\left(x-\frac{h}{2}\right)}{h}$
lyche-numerical-linear-algebra-057 EQ0107	057	1.000 N +0	$\frac{-v_{j-1}+2 v_j-v_{j+1}}{h^2}=f(j h), \quad j=1, \ldots, m, \quad v_0=v_{m+1}=0 .$	$\frac{-v_{j-1}+2 v_j-v_{j+1}}{h^2}=f(j h), \quad j=1, \ldots, m, \quad v_0=v_{m+1}=0 .$	$\frac{-v_{j-1}+2 v_j-v_{j+1}}{h^2}=f(j h), \quad j=1, \ldots, m, \quad v_0=v_{m+1}=0 .$
lyche-numerical-linear-algebra-057 EQ0109 (2.22)	057	1.000 N +0	$\left a_{i i}\right \geq \sum_{j \neq i}\left a_{i j}\right , i=1, \ldots, n .$	$\left a_{i i}\right \geq \sum_{j \neq i}\left a_{i j}\right , i=1, \ldots, n .$	$\left a_{i i}\right \geq \sum_{j \neq i}\left a_{i j}\right , i=1, \ldots, n .$
lyche-numerical-linear-algebra-058 EQ0111 (2.23)	058	1.000 C +3	$\left u_{k+1}\right =\left d_{k+1}-l_k c_k\right =\left d_{k+1}-\frac{a_k c_k}{u_k}\right \geq\left d_{k+1}\right -\frac{\left a_k\right \left c_k\right }{\left u_k\right }>\left d_{k+1}\right -\left a_k\right .$	$\left u_{k+1}\right =\left d_{k+1}-l_k c_k\right =\left d_{k+1}-\frac{a_k c_k}{u_k}\right \geq\left d_{k+1}\right -\frac{\left a_k\right \left c_k\right }{\left u_k\right }>\left d_{k+1}\right -\left a_k\right .$	$\left u_{k+1}\right =\left d_{k+1}-l_k c_k\right =\left d_{k+1}-\frac{a_k c_k}{u_k}\right \geq\left d_{k+1}\right -\frac{\left a_k\right \left c_k\right }{\left u_k\right }>\left d_{k+1}\right -\left a_k\right .$
lyche-numerical-linear-algebra-059 EQ0112 (2.24)	059	1.000 N +0	$R y''(x)=-F y(x), \quad y(0)=y(L)=0,$	$R y''(x)=-F y(x), \quad y(0)=y(L)=0,$	$R y''(x)=-F y(x), \quad y(0)=y(L)=0,$
lyche-numerical-linear-algebra-059 EQ0113 (2.25)	059	1.000 N +0	$u''(t)=-K u(t), \quad u(0)=u(1)=0, \quad K:=\frac{F L^2}{R} .$	$u''(t)=-K u(t), \quad u(0)=u(1)=0, \quad K:=\frac{F L^2}{R} .$	$u''(t)=-K u(t), \quad u(0)=u(1)=0, \quad K:=\frac{F L^2}{R} .$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-060 EQ0115	059	■ 1.000 ● N +0	$\frac{-v_{j-1}+2v_j-v_{j+1}}{h^2}=Kv_j, \quad j=1, \ldots, m, h=\frac{1}{m+1}, \quad v_0=v_{m+1}=0,$	$\frac{-v_{j-1}+2v_j-v_{j+1}}{h^2}=Kv_j, \quad j=1, \ldots, m, h=\frac{1}{m+1}, \quad v_0=v_{m+1}=0,$	$\frac{-v_{j-1}+2v_j-v_{j+1}}{h^2}=Kv_j, \quad j=1, \ldots, m, h=\frac{1}{m+1}, \quad v_0=v_{m+1}=0,$
lyche-numerical-linear-algebra-060 EQ0116 (2.26)	060	■ 1.000 ●	$\boldsymbol{T} \boldsymbol{v}=\lambda \boldsymbol{v}, \quad \text { with } \boldsymbol{v}=\left[v_1, \ldots, v_m\right]^T,$	$\boldsymbol{T} \boldsymbol{v}=\lambda \boldsymbol{v}, \text { with } \boldsymbol{v}=\left[v_1, \ldots, v_m\right]^T,$	$\boldsymbol{T} \boldsymbol{v}=\lambda \boldsymbol{v}, \text { with } \boldsymbol{v}=\left[v_1, \ldots, v_m\right]^T,$
lyche-numerical-linear-algebra-060 EQ0118	060	■ 1.000 ● N +0	$F=\frac{4 \sin ^{2}(\pi h / 2) R}{h^2 L^2} .$	$F=\frac{4 \sin ^2(\pi h / 2) R}{h^2 L^2} .$	$F=\frac{4 \sin ^2(\pi h / 2) R}{h^2 L^2} .$
lyche-numerical-linear-algebra-060 EQ0119 (2.28)	060	■ 1.000 ● N -1	$\boldsymbol{T}_1:=\operatorname{tridiag}(a, d, a)$	$\boldsymbol{T}_1:=\operatorname{tridiag}(a, d, a)$	$\boldsymbol{T}_1:=\operatorname{tridiag}(a, d, a)$
lyche-numerical-linear-algebra-060 EQ0120 (2.29)	060	■ 1.000 ● N +0	$\boldsymbol{S}=\left[\sin \frac{j k \pi}{m+1}\right]_{j, k=1}^m \in \mathbb{R}^{m \times m} .$	$\boldsymbol{S}=\left[\sin \frac{j k \pi}{m+1}\right]_{j, k=1}^m \in \mathbb{R}^{m \times m} .$	$\boldsymbol{S}=\left[\sin \frac{j k \pi}{m+1}\right]_{j, k=1}^m \in \mathbb{R}^{m \times m} .$
lyche-numerical-linear-algebra-060 EQ0126	062	■ 1.000 ● W +0	$\sum_{k=0}^m \cos (2 k j \pi h)+i \sum_{k=0}^m \sin (2 k j \pi h)=\sum_{k=0}^m e^{2 i k j \pi h}=\frac{e^{2 i(m+1) j \pi h}-1}{e^{2 i j \pi h}-1}=0,$	$\sum_{k=0}^m \cos (2 k j \pi h)+i \sum_{k=0}^m \sin (2 k j \pi h)=\sum_{k=0}^m e^{2 i k j \pi h}=\frac{e^{2 i(m+1) j \pi h}-1}{e^{2 i j \pi h}-1}=0,$	$\sum_{k=0}^m \cos (2 k j \pi h)+i \sum_{k=0}^m \sin (2 k j \pi h)=\sum_{k=0}^m e^{2 i k j \pi h}=\frac{e^{2 i(m+1) j \pi h}-1}{e^{2 i j \pi h}-1}=0,$
lyche-numerical-linear-algebra-060 EQ0130	063	■ 1.000 ● N +1	$\boldsymbol{A} \boldsymbol{B}=\left[\boldsymbol{A} \boldsymbol{b}_{: 1}, \boldsymbol{A} \boldsymbol{b}_{: 2}, \ldots, \boldsymbol{A} \boldsymbol{b}_{: n}\right] .$	$\boldsymbol{A} \boldsymbol{B}=\left[\boldsymbol{A} \boldsymbol{b}_{: 1}, \boldsymbol{A} \boldsymbol{b}_{: 2}, \ldots, \boldsymbol{A} \boldsymbol{b}_{: n}\right] .$	$\boldsymbol{A} \boldsymbol{B}=\left[\boldsymbol{A} \boldsymbol{b}_{: 1}, \boldsymbol{A} \boldsymbol{b}_{: 2}, \ldots, \boldsymbol{A} \boldsymbol{b}_{: n}\right] .$
lyche-numerical-linear-algebra-060 EQ0131	063	■ 1.000 ● K +0	$\boldsymbol{A}=\boldsymbol{A} \boldsymbol{I}=\boldsymbol{A}\left[\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_p\right]=\left[\boldsymbol{A} \boldsymbol{e}_1, \boldsymbol{A} \boldsymbol{e}_2, \ldots, \boldsymbol{A} \boldsymbol{e}_p\right]$	$\boldsymbol{A}=\boldsymbol{A} \boldsymbol{I}=\boldsymbol{A}\left[\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_p\right]=\left[\boldsymbol{A} \boldsymbol{e}_1, \boldsymbol{A} \boldsymbol{e}_2, \ldots, \boldsymbol{A} \boldsymbol{e}_p\right]$	$\boldsymbol{A}=\boldsymbol{A} \boldsymbol{I}=\boldsymbol{A}\left[\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_p\right]=\left[\boldsymbol{A} \boldsymbol{e}_1, \boldsymbol{A} \boldsymbol{e}_2, \ldots, \boldsymbol{A} \boldsymbol{e}_p\right]$
lyche-numerical-linear-algebra-060 EQ0133	063	■ 1.000 ● N +0	$\boldsymbol{A} \boldsymbol{x}=x_1 \boldsymbol{a}_{: 1}+x_2 \boldsymbol{a}_{: 2}+\cdots+x_p \boldsymbol{a}_{: p} .$	$\boldsymbol{A} \boldsymbol{x}=x_1 \boldsymbol{a}_{: 1}+x_2 \boldsymbol{a}_{: 2}+\cdots+x_p \boldsymbol{a}_{: p} .$	$\boldsymbol{A} \boldsymbol{x}=x_1 \boldsymbol{a}_{: 1}+x_2 \boldsymbol{a}_{: 2}+\cdots+x_p \boldsymbol{a}_{: p} .$
lyche-numerical-linear-algebra-060 EQ0134	064	■ 1.000 ● K +0	$\boldsymbol{A}\left[\boldsymbol{B}_1, \boldsymbol{B}_2\right]=\left[\boldsymbol{A} \boldsymbol{B}_1, \boldsymbol{A} \boldsymbol{B}_2\right] .$	$\boldsymbol{A}\left[\boldsymbol{B}_1, \boldsymbol{B}_2\right]=\left[\boldsymbol{A} \boldsymbol{B}_1, \boldsymbol{A} \boldsymbol{B}_2\right] .$	$\boldsymbol{A}\left[\boldsymbol{B}_1, \boldsymbol{B}_2\right]=\left[\boldsymbol{A} \boldsymbol{B}_1, \boldsymbol{A} \boldsymbol{B}_2\right] .$
lyche-numerical-linear-algebra-060 EQ0136	064	■ 1.000 ● K +0	$\left[\boldsymbol{A}_1, \boldsymbol{A}_2\right]\left[\begin{array}{l} \boldsymbol{B}_1 \\ \boldsymbol{B}_2 \end{array}\right]=\left[\boldsymbol{A}_1 \boldsymbol{B}_1+\boldsymbol{A}_2 \boldsymbol{B}_2\right] .$	$\left[\boldsymbol{A}_1, \boldsymbol{A}_2\right]\left[\begin{array}{l} \boldsymbol{B}_1 \\ \boldsymbol{B}_2 \end{array}\right]=\left[\boldsymbol{A}_1 \boldsymbol{B}_1+\boldsymbol{A}_2 \boldsymbol{B}_2\right] .$	$\left[\boldsymbol{A}_1, \boldsymbol{A}_2\right]\left[\begin{array}{l} \boldsymbol{B}_1 \\ \boldsymbol{B}_2 \end{array}\right]=\left[\boldsymbol{A}_1 \boldsymbol{B}_1+\boldsymbol{A}_2 \boldsymbol{B}_2\right] .$
lyche-numerical-linear-algebra-060 EQ0140	065	■ 1.000 ● N +0	$\boldsymbol{C}_{i j}=\sum_{k=1}^s \boldsymbol{A}_{i k} \boldsymbol{B}_{k j}, \quad i=1, \ldots, p, j=1, \ldots, q$	$\boldsymbol{C}_{i j}=\sum_{k=1}^s \boldsymbol{A}_{i k} \boldsymbol{B}_{k j}, \quad i=1, \ldots, p, j=1, \ldots, q$	$\boldsymbol{C}_{i j}=\sum_{k=1}^s \boldsymbol{A}_{i k} \boldsymbol{B}_{k j}, \quad i=1, \ldots, p, j=1, \ldots, q$

Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-065 EQ0141	065	1.000	$\left[\begin{array}{ccc} \boldsymbol{A}_{11} & \cdots & \boldsymbol{A}_{1s} \\ \vdots & & \vdots \\ \boldsymbol{A}_{p1} & \cdots & \boldsymbol{A}_{ps} \end{array}\right] \left[\begin{array}{ccc} \boldsymbol{B}_{11} & \cdots & \boldsymbol{B}_{1q} \\ \vdots & & \vdots \\ \boldsymbol{B}_{s1} & \cdots & \boldsymbol{B}_{sq} \end{array}\right] = \left[\begin{array}{ccc} \boldsymbol{C}_{11} & \cdots & \boldsymbol{C}_{1q} \\ \vdots & & \vdots \\ \boldsymbol{C}_{p1} & \cdots & \boldsymbol{C}_{pq} \end{array}\right].$	$\begin{bmatrix} \boldsymbol{A}_{11} & \cdots & \boldsymbol{A}_{1s} \\ \vdots & & \vdots \\ \boldsymbol{A}_{p1} & \cdots & \boldsymbol{A}_{ps} \end{bmatrix} \begin{bmatrix} \boldsymbol{B}_{11} & \cdots & \boldsymbol{B}_{1q} \\ \vdots & & \vdots \\ \boldsymbol{B}_{s1} & \cdots & \boldsymbol{B}_{sq} \end{bmatrix} = \begin{bmatrix} \boldsymbol{C}_{11} & \cdots & \boldsymbol{C}_{1q} \\ \vdots & & \vdots \\ \boldsymbol{C}_{p1} & \cdots & \boldsymbol{C}_{pq} \end{bmatrix}.$	
lyche-numerical-linear-algebra-065 EQ0143 (2.33)	065	1.000 K +0	$\boldsymbol{A}^{\{-1\}}=\left[\begin{array}{cc} \boldsymbol{A}_{11}^{\{-1\}} & \boldsymbol{C} \\ \boldsymbol{0} & \boldsymbol{A}_{22}^{\{-1\}} \end{array}\right],$	$\boldsymbol{A}^{-1}=\left[\begin{array}{cc} \boldsymbol{A}_{11}^{-1} & \boldsymbol{C} \\ \boldsymbol{0} & \boldsymbol{A}_{22}^{-1} \end{array}\right],$	$\boldsymbol{A}^{-1}=\left[\begin{array}{cc} \boldsymbol{A}_{11}^{-1} & \boldsymbol{C} \\ \boldsymbol{0} & \boldsymbol{A}_{22}^{-1} \end{array}\right],$
lyche-numerical-linear-algebra-066 EQ0146	066	1.000 K +0	$\boldsymbol{B}_{12}=\boldsymbol{C}=-\boldsymbol{A}_{11}^{\{-1\}}\boldsymbol{A}_{12}\boldsymbol{A}_{22}^{\{-1\}}.$	$\boldsymbol{B}_{12}=\boldsymbol{C}=-\boldsymbol{A}_{11}^{-1}\boldsymbol{A}_{12}\boldsymbol{A}_{22}^{-1}.$	$\boldsymbol{B}_{12}=\boldsymbol{C}=-\boldsymbol{A}_{11}^{-1}\boldsymbol{A}_{12}\boldsymbol{A}_{22}^{-1}.$
lyche-numerical-linear-algebra-066 EQ0148	066	1.000 N +0	$\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_k & \boldsymbol{a}_k \\ \boldsymbol{0} & \boldsymbol{a}_{k+1,k+1} \end{array}\right]$	$\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_k & \boldsymbol{a}_k \\ \boldsymbol{0} & \boldsymbol{a}_{k+1,k+1} \end{array}\right]$	$\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_k & \boldsymbol{a}_k \\ \boldsymbol{0} & \boldsymbol{a}_{k+1,k+1} \end{array}\right]$
lyche-numerical-linear-algebra-066 EQ0149	066	1.000 N +0	$\boldsymbol{A}^{\{-1\}}=\left[\begin{array}{cc} \boldsymbol{A}_k^{\{-1\}} & \boldsymbol{c} \\ \boldsymbol{0} & \boldsymbol{a}_{k+1,k+1}^{\{-1\}} \end{array}\right],$	$\boldsymbol{A}^{-1}=\left[\begin{array}{cc} \boldsymbol{A}_k^{-1} & \boldsymbol{c} \\ \boldsymbol{0} & \boldsymbol{a}_{k+1,k+1}^{-1} \end{array}\right],$	$\boldsymbol{A}^{-1}=\left[\begin{array}{cc} \boldsymbol{A}_k^{-1} & \boldsymbol{c} \\ \boldsymbol{0} & \boldsymbol{a}_{k+1,k+1}^{-1} \end{array}\right],$
lyche-numerical-linear-algebra-067 EQ0150	067	1.000 N +0	$\max_{2\leq j\leq n} \mu_j \leq\frac{3}{h^2}\max_{2\leq i\leq n} y_{i+1}-2y_i+y_{i-1} .$	$\max_{2\leq j\leq n} \mu_j \leq\frac{3}{h^2}\max_{2\leq i\leq n} y_{i+1}-2y_i+y_{i-1} .$	$\max_{2\leq j\leq n} \mu_j \leq\frac{3}{h^2}\max_{2\leq i\leq n} y_{i+1}-2y_i+y_{i-1} .$
lyche-numerical-linear-algebra-067 EQ0151 (2.34)	067	1.000 C -8	$\left[\begin{array}{ccccc} 2&1&&&\\ 1&4&1&&\\ &\ddots&\ddots&\ddots&\\ &&1&4&1\\ &&&1&2 \end{array}\right]\left[\begin{array}{c} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \\ \mu_{n+1} \end{array}\right]=\frac{6}{h^2}\left[\begin{array}{c} y_2-y_1-hs_1 \\ \delta^2y_2 \\ \delta^2y_3 \\ \vdots \\ \delta^2y_{n-1} \\ \delta^2y_n \\ hs_{n+1}-y_{n+1}+y_n \end{array}\right],$	$\left[\begin{array}{ccc} 2&1&\\ 1&4&1\\ &\ddots&\ddots&\ddots\\ &&1&4&1\\ &&&1&2 \end{array}\right]\left[\begin{array}{c} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \\ \mu_{n+1} \end{array}\right]=\frac{6}{h^2}\left[\begin{array}{c} y_2-y_1-hs_1 \\ \delta^2y_2 \\ \delta^2y_3 \\ \vdots \\ \delta^2y_{n-1} \\ \delta^2y_n \\ hs_{n+1}-y_{n+1}+y_n \end{array}\right],$	$\left[\begin{array}{ccc} 2&1&\\ 1&4&1\\ &\ddots&\ddots&\ddots\\ &&1&4&1\\ &&&1&2 \end{array}\right]\left[\begin{array}{c} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \\ \mu_{n+1} \end{array}\right]=\frac{6}{h^2}\left[\begin{array}{c} y_2-y_1-hs_1 \\ \delta^2y_2 \\ \delta^2y_3 \\ \vdots \\ \delta^2y_{n-1} \\ \delta^2y_n \\ hs_{n+1}-y_{n+1}+y_n \end{array}\right],$
lyche-numerical-linear-algebra-068 EQ0152	068	1.000 N +0	$\int_a^b(g''(x))^2\,dx\leq\int_a^b(h''(x))^2\,dx$	$\int_a^b(g''(x))^2\,dx\leq\int_a^b(h''(x))^2\,dx$	$\int_a^b(g''(x))^2\,dx\leq\int_a^b(h''(x))^2\,dx$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-068 EQ0153 (2.35)	068	■ 1.000 ● N +0	$\int_a^b g^{\prime\prime\prime} e^{\prime\prime} = 0, \quad e := h - g.$	$\int_a^b g^{\prime\prime\prime} e^{\prime\prime} = 0, \quad e := h - g.$	$\int_a^b g^{\prime\prime} e^{\prime\prime} = 0, \quad e := h - g.$
lyche-numerical-linear-algebra-068 EQ0154	068	■ 1.000 ●	$\int_a^b g^{\prime\prime\prime} e^{\prime\prime} = \sum_{i=1}^n \int_{x_i}^{x_{i+1}} g^{\prime\prime\prime} e^{\prime\prime} = \sum_{i=1}^n v_i \int_{x_i}^{x_{i+1}} e^{\prime\prime} = \sum_{i=1}^n v_i (e(x_{i+1}) - e(x_i)) = 0.$	$\int_a^b g^{\prime\prime\prime} e^{\prime\prime} = \sum_{i=1}^n \int_{x_i}^{x_{i+1}} g^{\prime\prime\prime} e^{\prime\prime} = \sum_{i=1}^n v_i \int_{x_i}^{x_{i+1}} e^{\prime\prime} = \sum_{i=1}^n v_i (e(x_{i+1}) - e(x_i)) = 0.$	$\int_a^b g^{\prime\prime\prime} e^{\prime\prime} = \sum_{i=1}^n \int_{x_i}^{x_{i+1}} g^{\prime\prime\prime} e^{\prime\prime} = \sum_{i=1}^n v_i \int_{x_i}^{x_{i+1}} e^{\prime\prime} = \sum_{i=1}^n v_i (e(x_{i+1}) - e(x_i)) = 0.$
lyche-numerical-linear-algebra-068 EQ0155	068	■ 1.000 ● W +0	$\begin{aligned} \int_a^b (h^{\prime\prime})^2 &= \int_a^b (g^{\prime\prime} + e^{\prime\prime})^2 \\ &= \int_a^b (g^{\prime\prime})^2 + \int_a^b (e^{\prime\prime})^2 + 2 \int_a^b g^{\prime\prime} e^{\prime\prime} \\ &= \int_a^b (g^{\prime\prime})^2 + \int_a^b (e^{\prime\prime})^2 \geq \int_a^b (g^{\prime\prime})^2 \end{aligned}$	$\begin{aligned} \int_a^b (h^{\prime\prime})^2 &= \int_a^b (g^{\prime\prime} + e^{\prime\prime})^2 \\ &= \int_a^b (g^{\prime\prime})^2 + \int_a^b (e^{\prime\prime})^2 + 2 \int_a^b g^{\prime\prime} e^{\prime\prime} \\ &= \int_a^b (g^{\prime\prime})^2 + \int_a^b (e^{\prime\prime})^2 \geq \int_a^b (g^{\prime\prime})^2 \end{aligned}$	$\begin{aligned} \int_a^b (h^{\prime\prime})^2 &= \int_a^b (g^{\prime\prime} + e^{\prime\prime})^2 \\ &= \int_a^b (g^{\prime\prime})^2 + \int_a^b (e^{\prime\prime})^2 + 2 \int_a^b g^{\prime\prime} e^{\prime\prime} \\ &= \int_a^b (g^{\prime\prime})^2 + \int_a^b (e^{\prime\prime})^2 \geq \int_a^b (g^{\prime\prime})^2 \end{aligned}$
lyche-numerical-linear-algebra-071 EQ0157	071	■ 1.000 ● N +0	$\delta^2 f(x) = f^{\prime\prime}(\eta_2), \quad x - h < \eta_2 < x + h.$	$\delta^2 f(x) = f^{\prime\prime}(\eta_2), \quad x - h < \eta_2 < x + h.$	$\delta^2 f(x) = f^{\prime\prime}(\eta_2), \quad x - h < \eta_2 < x + h.$
lyche-numerical-linear-algebra-071 EQ0158	071	■ 1.000 ● N +0	$\delta^2 f(x) = f^{\prime\prime}(x) + \frac{h^2}{12} f^{(4)}(\eta_4), \quad x - h < \eta_4 < x + h.$	$\delta^2 f(x) = f^{\prime\prime}(x) + \frac{h^2}{12} f^{(4)}(\eta_4), \quad x - h < \eta_4 < x + h.$	$\delta^2 f(x) = f^{\prime\prime}(x) + \frac{h^2}{12} f^{(4)}(\eta_4), \quad x - h < \eta_4 < x + h.$
lyche-numerical-linear-algebra-071 EQ0159 (2.36)	071	■ 1.000 ● S +0	$-u^{\prime\prime}(x) + r(x)u'(x) + q(x)u(x) = f(x), \text{ for } x \in [a, b],$ $u(a) = g_0, \quad u(b) = g_1.$	$-u^{\prime\prime}(x) + r(x)u'(x) + q(x)u(x) = f(x), \text{ for } x \in [a, b],$ $u(a) = g_0, \quad u(b) = g_1.$	$-u^{\prime\prime}(x) + r(x)u'(x) + q(x)u(x) = f(x), \text{ for } x \in [a, b],$ $u(a) = g_0, \quad u(b) = g_1.$
lyche-numerical-linear-algebra-071 EQ0160 (2.37)	071	■ 1.000 ● N +0	$\frac{-v_{j-1} + 2v_j - v_{j+1}}{h^2} + r(x_j) \frac{v_{j+1} - v_{j-1}}{2h} + q(x_j)v_j = f(x_j), \quad j = 1, \dots, m,$	$\frac{-v_{j-1} + 2v_j - v_{j+1}}{h^2} + r(x_j) \frac{v_{j+1} - v_{j-1}}{2h} + q(x_j)v_j = f(x_j), \quad j = 1, \dots, m,$	$\frac{-v_{j-1} + 2v_j - v_{j+1}}{h^2} + r(x_j) \frac{v_{j+1} - v_{j-1}}{2h} + q(x_j)v_j = f(x_j), \quad j = 1, \dots, m,$
lyche-numerical-linear-algebra-072 EQ0162	072	■ 1.000 ● W +0	$b_j = \begin{cases} h^2 f(x_1) - a_1 g_0, & \text{if } j = 1, \\ h^2 f(x_j), & \text{if } 2 \leq j \leq m - 1, \\ h^2 f(x_m) - c_m g_1, & \text{if } j = m. \end{cases}$	$b_j = \begin{cases} h^2 f(x_1) - a_1 g_0, & \text{if } j = 1, \\ h^2 f(x_j), & \text{if } 2 \leq j \leq m - 1, \\ h^2 f(x_m) - c_m g_1, & \text{if } j = m. \end{cases}$	$b_j = \begin{cases} h^2 f(x_1) - a_1 g_0, & \text{if } j = 1, \\ h^2 f(x_j), & \text{if } 2 \leq j \leq m - 1, \\ h^2 f(x_m) - c_m g_1, & \text{if } j = m. \end{cases}$
lyche-numerical-linear-algebra-072 EQ0163	072	■ 1.000 ● N +0	$F = \frac{4 \sin^2(\pi h/2) R}{h^2 L^2} = \frac{\pi^2 R}{L^2} + O(h^2).$	$F = \frac{4 \sin^2(\pi h/2) R}{h^2 L^2} = \frac{\pi^2 R}{L^2} + O(h^2).$	$F = \frac{4 \sin^2(\pi h/2) R}{h^2 L^2} = \frac{\pi^2 R}{L^2} + O(h^2).$
lyche-numerical-linear-algebra-073 EQ0165 (2.39)	073	■ 1.000 ● N +0	$s_{i,j} = s_{j,i} = \frac{1}{m+1} j(m+1-i), \quad 1 \leq j \leq i \leq m.$	$s_{i,j} = s_{j,i} = \frac{1}{m+1} j(m+1-i), \quad 1 \leq j \leq i \leq m.$	$s_{i,j} = s_{j,i} = \frac{1}{m+1} j(m+1-i), \quad 1 \leq j \leq i \leq m.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0167	073	1.000 N +0	$\boldsymbol{A} \boldsymbol{X} = \boldsymbol{B}$ $\quad \Longleftrightarrow \quad \boldsymbol{A} \boldsymbol{x}_{:j} = \boldsymbol{b}_{:j}, j = 1, \dots, p$		$\boldsymbol{A} \boldsymbol{X} = \boldsymbol{B} \quad \Longleftrightarrow \quad \boldsymbol{A} \boldsymbol{x}_{:j} = \boldsymbol{b}_{:j}, j = 1, \dots, p.$
lyche-numerical-linear-algebra EQ0173	076	1.000	$\begin{aligned} & x_3 = b_3^{(3)} / a_{33}^{(3)} \\ & \& x_2 = \left(b_2^{(2)} - a_{23}^{(2)} x_3 \right) / a_{22}^{(2)} \\ & \& x_1 = \left(b_1^{(1)} - a_{12}^{(1)} x_2 - a_{13}^{(1)} x_3 \right) / a_{11}^{(1)}. \end{aligned}$	$x_3 = b_3^{(3)} / a_{33}^{(3)}$ $x_2 = \left(b_2^{(2)} - a_{23}^{(2)} x_3 \right) / a_{22}^{(2)}$ $x_1 = \left(b_1^{(1)} - a_{12}^{(1)} x_2 - a_{13}^{(1)} x_3 \right) / a_{11}^{(1)}.$	$x_3 = b_3^{(3)} / a_{33}^{(3)}$ $x_2 = (b_2^{(2)} - a_{23}^{(2)} x_3) / a_{22}^{(2)}$ $x_1 = (b_1^{(1)} - a_{12}^{(1)} x_2 - a_{13}^{(1)} x_3) / a_{11}^{(1)}.$
lyche-numerical-linear-algebra EQ0177 (3.2)	077	1.000 K +0	$\text{for } j = k : n$	$\text{for } j = k : n$	$\text{for } j = k : n$
lyche-numerical-linear-algebra EQ0178	077	1.000 W +0	$a_{ij}^{(k+1)} = a_{ij}^{(k)} - l_{ik}^{(k)} a_{kj}^{(k)}$	$a_{ij}^{(k+1)} = a_{ij}^{(k)} - l_{ik}^{(k)} a_{kj}^{(k)}$	$a_{ij}^{(k+1)} = a_{ij}^{(k)} - l_{ik}^{(k)} a_{kj}^{(k)}$
lyche-numerical-linear-algebra EQ0179	078	1.000 N +0	$\boldsymbol{A}_{[k]} := \boldsymbol{A}(1 : k, 1 : k) = \begin{bmatrix} a_{11} & \cdots & a_{k1} \\ \vdots & & \vdots \\ a_{k1} & \cdots & a_{kk} \end{bmatrix}$	$\boldsymbol{A}_{[k]} := \boldsymbol{A}(1 : k, 1 : k) = \begin{bmatrix} a_{11} & \cdots & a_{k1} \\ \vdots & & \vdots \\ a_{k1} & \cdots & a_{kk} \end{bmatrix}$	$\boldsymbol{A}_{[k]} := \boldsymbol{A}(1 : k, 1 : k) = \begin{bmatrix} a_{11} & \cdots & a_{k1} \\ \vdots & & \vdots \\ a_{k1} & \cdots & a_{kk} \end{bmatrix}$
lyche-numerical-linear-algebra EQ0180	078	1.000 K +0	$b_{i,j} = a_{r_i, r_j}, \quad i, j = 1, \dots, k.$	$b_{i,j} = a_{r_i, r_j}, \quad i, j = 1, \dots, k.$	$b_{i,j} = a_{r_i, r_j}, \quad i, j = 1, \dots, k.$
lyche-numerical-linear-algebra EQ0182	078	1.000 K +0	$[1], \left[\begin{array}{cc} 1 & 2 \\ 4 & 5 \end{array} \right], \boldsymbol{A}.$	$[1], \left[\begin{array}{cc} 1 & 2 \\ 4 & 5 \end{array} \right], \boldsymbol{A}.$	$[1], \left[\begin{array}{cc} 1 & 2 \\ 4 & 5 \end{array} \right], \boldsymbol{A}.$
lyche-numerical-linear-algebra EQ0183 (3.3)	078	1.000 W +0	$\operatorname{det}(\boldsymbol{A}_{[k]}) = a_{11}^{(1)} a_{22}^{(2)} \cdots a_{kk}^{(k)}, \quad k = 1, \dots, n.$	$\det(\boldsymbol{A}_{[k]}) = a_{11}^{(1)} a_{22}^{(2)} \cdots a_{kk}^{(k)}, \quad k = 1, \dots, n.$	$\det(\boldsymbol{A}_{[k]}) = a_{11}^{(1)} a_{22}^{(2)} \cdots a_{kk}^{(k)}, \quad k = 1, \dots, n.$
lyche-numerical-linear-algebra EQ0187 (3.6)	079	1.000 W +0	$(\boldsymbol{L} \boldsymbol{U})_{ij} = \sum_{k=1}^{j-1} l_{ik}^{(k)} a_{kj}^{(k)} + l_{ij} a_{jj}^{(j)} = \sum_{k=1}^{j-1} \left(a_{ij}^{(k)} - a_{ij}^{(k+1)} \right) + a_{ij}^{(j)} = a_{ij}.$	$(\boldsymbol{L} \boldsymbol{U})_{ij} = \sum_{k=1}^{j-1} l_{ik}^{(k)} a_{kj}^{(k)} + l_{ij} a_{jj}^{(j)} = \sum_{k=1}^{j-1} \left(a_{ij}^{(k)} - a_{ij}^{(k+1)} \right) + a_{ij}^{(j)} = a_{ij}.$	$(\boldsymbol{L} \boldsymbol{U})_{ij} = \sum_{k=1}^{j-1} l_{ik}^{(k)} a_{kj}^{(k)} + l_{ij} a_{jj}^{(j)} = \sum_{k=1}^{j-1} \left(a_{ij}^{(k)} - a_{ij}^{(k+1)} \right) + a_{ij}^{(j)} = a_{ij}.$
lyche-numerical-linear-algebra EQ0188	080	1.000 W +0	$\left[\begin{array}{ccc} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{array} \right] \left[\begin{array}{c} x_1 \\ x_2 \\ x_3 \end{array} \right] = \left[\begin{array}{c} b_1 \\ b_2 \\ b_3 \end{array} \right].$	$\left[\begin{array}{ccc} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{array} \right] \left[\begin{array}{c} x_1 \\ x_2 \\ x_3 \end{array} \right] = \left[\begin{array}{c} b_1 \\ b_2 \\ b_3 \end{array} \right].$	$\left[\begin{array}{ccc} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{array} \right] \left[\begin{array}{c} x_1 \\ x_2 \\ x_3 \end{array} \right] = \left[\begin{array}{c} b_1 \\ b_2 \\ b_3 \end{array} \right].$

Conf. MathPix confidence:

≥0.9

0.5–0.9

<0.5

— no value

LaTeX source MathPix’s reading, unchanged;
Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-080 EQ0191	080	1.000 W +0	$\left[\begin{array}{ccccc} a_{11} & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 \\ 0 & a_{32} & a_{33} & 0 & 0 \\ 0 & 0 & a_{43} & a_{44} & 0 \\ 0 & 0 & 0 & a_{54} & a_{55} \end{array}\right], \quad \left[\begin{array}{ccccc} a_{11} & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 & 0 \\ 0 & a_{42} & a_{43} & a_{44} & 0 \\ 0 & 0 & a_{53} & a_{54} & a_{55} \end{array}\right]$	$\begin{bmatrix} a_{11} & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 \\ 0 & a_{32} & a_{33} & 0 & 0 \\ 0 & 0 & a_{43} & a_{44} & 0 \\ 0 & 0 & 0 & a_{54} & a_{55} \end{bmatrix}, \quad \begin{bmatrix} a_{11} & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 & 0 \\ 0 & a_{42} & a_{43} & a_{44} & 0 \\ 0 & 0 & a_{53} & a_{54} & a_{55} \end{bmatrix}$	
lyche-numerical-linear-algebra-081 EQ0192	081	1.000	$\left[\begin{array}{ccccc} a_{k,k} & 0 & \cdots & 0 \\ a_{k+1,k} & a_{k+1,k+1} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ a_{n,k} & & \cdots & a_{n\times n} \end{array}\right] \begin{bmatrix} x_k \\ x_{k+1} \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_k \\ b_{k+1} \\ \vdots \\ b_n \end{bmatrix}.$	$\begin{bmatrix} a_{k,k} & 0 & \cdots & 0 \\ a_{k+1,k} & a_{k+1,k+1} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ a_{n,k} & & \cdots & a_{n\times n} \end{bmatrix} \begin{bmatrix} x_k \\ x_{k+1} \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_k \\ b_{k+1} \\ \vdots \\ b_n \end{bmatrix}.$	
lyche-numerical-linear-algebra-082 EQ0194	082	1.000 N +0	$b((k+1):n)=b((k+1):n)-x(k)*A((k+1):n,k).$	$b((k+1):n)=b((k+1):n)-x(k)*A((k+1):n,k).$	$b((k+1):n)=b((k+1):n)-x(k)*A((k+1):n,k).$
lyche-numerical-linear-algebra-082 EQ0195 (3.9)	082	1.000 C +10	$\begin{array}{l} \text{\texttt{\textbackslashbegin{figure} \captionsetup{labelformat=empty} \\ \caption{Listing 3.3 cforwardsolve} \\ \text{\texttt{\textbackslash[N_{L U}:= \\ \text{\texttt{\textbackslashfrac{2}{3} n^3}-\text{\texttt{\textbackslashfrac{1}{2} n^2}-\text{\texttt{\textbackslashfrac{1}{6} n \\ \text{\texttt{\textbackslashend{figure}}}}}} \end{array}$	(not rendered)	$N_{LU}:=\frac{2}{3}n^3-\frac{1}{2}n^2-\frac{1}{6}n$
lyche-numerical-linear-algebra-083 EQ0196	083	1.000 N +0	$M+D+A+S=\frac{2}{3}n(n-1)\left(n-\frac{1}{2}\right)+\frac{1}{2}n(n-1)=N_{LU}$	$M+D+A+S=\frac{2}{3}n(n-1)\left(n-\frac{1}{2}\right)+\frac{1}{2}n(n-1)=N_{LU}$	$M+D+A+S=\frac{2}{3}n(n-1)\left(n-\frac{1}{2}\right)+\frac{1}{2}n(n-1)=N_{LU}$
lyche-numerical-linear-algebra-083 EQ0197	083	1.000 N +0	$M+S=2\sum_{k=1}^{n-1}(n-k)^2\approx 2\int_1^{n-1}(n-k)^2dk\approx 2\int_0^n(n-k)^2dk=\frac{2}{3}n^3$	$M+S=2\sum_{k=1}^{n-1}(n-k)^2\approx 2\int_1^{n-1}(n-k)^2dk\approx 2\int_0^n(n-k)^2dk=\frac{2}{3}n^3$	$M+S=2\sum_{k=1}^{n-1}(n-k)^2\approx 2\int_1^{n-1}(n-k)^2dk\approx 2\int_0^n(n-k)^2dk=\frac{2}{3}n^3$
lyche-numerical-linear-algebra-083 EQ0198	083	1.000 N +1	$N_S=2\sum_{k=1}^n(2k-1)\approx 2\int_1^n(2k-1)dk\approx 4\int_0^n kdk=2n^2$	$N_S=2\sum_{k=1}^n(2k-1)\approx 2\int_1^n(2k-1)dk\approx 4\int_0^n kdk=2n^2$	$N_S=2\sum_{k=1}^n(2k-1)\approx 2\int_1^n(2k-1)dk\approx 4\int_0^n kdk=2n^2.$
lyche-numerical-linear-algebra-084 EQ0199	084	1.000 N +0	$\boldsymbol{P}=\boldsymbol{I}(:,\boldsymbol{p})=\left[\boldsymbol{e}_{i_1},\boldsymbol{e}_{i_2},\ldots,\boldsymbol{e}_{i_n}\right]\in\mathbb{R}^{n\times n},$	$\boldsymbol{P}=\boldsymbol{I}(:,\boldsymbol{p})=\left[\boldsymbol{e}_{i_1},\boldsymbol{e}_{i_2},\ldots,\boldsymbol{e}_{i_n}\right]\in\mathbb{R}^{n\times n},$	$\boldsymbol{P}=\boldsymbol{I}(:,\boldsymbol{p})=\left[\boldsymbol{e}_{i_1},\boldsymbol{e}_{i_2},\ldots,\boldsymbol{e}_{i_n}\right]\in\mathbb{R}^{n\times n},$
lyche-numerical-linear-algebra-084 EQ0200 (3.10)	084	1.000 K +0	$\boldsymbol{A}\boldsymbol{P}=\boldsymbol{A}(:,\boldsymbol{p}),\quad \boldsymbol{P}^T\boldsymbol{A}=\boldsymbol{A}(\boldsymbol{p},:).$	$\boldsymbol{A}\boldsymbol{P}=\boldsymbol{A}(:,\boldsymbol{p}),\quad \boldsymbol{P}^T\boldsymbol{A}=\boldsymbol{A}(\boldsymbol{p},:).$	$\boldsymbol{A}\boldsymbol{P}=\boldsymbol{A}(:,\boldsymbol{p}),\quad \boldsymbol{P}^T\boldsymbol{A}=\boldsymbol{A}(\boldsymbol{p},:).$
lyche-numerical-linear-algebra-086 EQ0206	086	1.000 W -1	$a_{p_i,k}^{(k)}a_{p_k,j}^{(k)}=a_{p_i,j}^{(k)}-a_{p_i,j}^{(k+1)}\text{ for }k<\min(i,j),\text{ and }a_{p_i,j}^{(k)}a_{p_j,j}^{(j)}=a_{p_i,j}^{(j)}\text{ for }i>j.$	$a_{p_i,k}^{(k)}a_{p_k,j}^{(k)}=a_{p_i,j}^{(k)}-a_{p_i,j}^{(k+1)}\text{ for }k<\min(i,j),\text{ and }a_{p_i,j}^{(k)}a_{p_j,j}^{(j)}=a_{p_i,j}^{(j)}\text{ for }i>j.$	$a_{p_i,k}^{(k)}a_{p_k,j}^{(k)}=a_{p_i,j}^{(k)}-a_{p_i,j}^{(k+1)}\text{ for }k<\min(i,j),\text{ and }a_{p_i,j}^{(k)}a_{p_j,j}^{(j)}=a_{p_i,j}^{(j)}\text{ for }i>j.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-087 EQ0209	087	■ 1.000 ● N +0	$\left a_{r_k,k}, k\right ^{\left(k\right)}:=\max \left\{\left a_{i,k}, k\right ^{\left(k\right)}: k \leq i \leq n\right\}$	$\left a_{r_k,k}\right :=\max \left\{\left a_{i,k}\right : k \leq i \leq n\right\}$	$\left a_{r_k,k}^{(k)}\right :=\max \left\{\left a_{i,k}^{(k)}\right : k \leq i \leq n\right\}$
lyche-numerical-linear-algebra-087 EQ0211	087	■ 1.000 ● N +0	$\begin{aligned} 10^{-4} x_1+2 x_2 &=4 \\ \left(1-2 \times 10^4\right) x_2 &=3-4 \times 10^4 \end{aligned}$	$10^{-4} x_1+2 x_2=4$ $\left(1-2 \times 10^4\right) x_2=3-4 \times 10^4$	$10^{-4} x_1+2 x_2=4$ $\left(1-2 \times 10^4\right) x_2=3-4 \times 10^4$
lyche-numerical-linear-algebra-087 EQ0212	087	■ 1.000 ● N +1	$x_2=\frac{-39997}{-19999} \approx 2, \quad x_1=\frac{4-2 x_2}{10^{-4}}=\frac{20000}{19999} \approx 1 .$	$x_2=\frac{-39997}{-19999} \approx 2, \quad x_1=\frac{4-2 x_2}{10^{-4}}=\frac{20000}{19999} \approx 1 .$	$x_2=\frac{-39997}{-19999} \approx 2, \quad x_1=\frac{4-2 x_2}{10^{-4}}=\frac{20000}{19999} \approx 1 .$
lyche-numerical-linear-algebra-087 EQ0214	087	■ 1.000 ● K +0	$\begin{aligned} x_1+x_2 &=3 \\ 10^{-4} x_1+2 x_2 &=4 \end{aligned}$	$x_1+x_2=3$ $10^{-4} x_1+2 x_2=4$	$x_1+x_2=3$ $10^{-4} x_1+2 x_2=4$
lyche-numerical-linear-algebra-087 EQ0215	087	■ 1.000 ●	$\begin{aligned} x_1+x_2 &=3 \\ \left(2-10^{-4}\right) x_2 &=4-3 \times 10^{-4} \end{aligned}$	$x_1+x_2=3$ $\left(2-10^{-4}\right) x_2=4-3 \times 10^{-4}$	$x_1+x_2=3$ $\left(2-10^{-4}\right) x_2=4-3 \times 10^{-4}$
lyche-numerical-linear-algebra-088 EQ0217	088	■ 1.000 ● W +0	$\frac{\left a_{r_k,k}, k\right ^{\left(k\right)}}{s_k}:=\max \left\{\frac{\left a_{i,k}, k\right ^{\left(k\right)}}{s_k}: k \leq i \leq n\right\}, \quad s_k:=\max _{1 \leq j \leq n}\left a_{k j}\right .$	$\frac{\left a_{r_k,k}^{(k)}\right }{s_k}:=\max \left\{\frac{\left a_{i,k}^{(k)}\right }{s_k}: k \leq i \leq n\right\}, \quad s_k:=\max _{1 \leq j \leq n}\left a_{k j}\right .$	$\frac{\left a_{r_k,k}^{(k)}\right }{s_k}:=\max \left\{\frac{\left a_{i,k}^{(k)}\right }{s_k}: k \leq i \leq n\right\}, \quad s_k:=\max _{1 \leq j \leq n}\left a_{k j}\right .$
lyche-numerical-linear-algebra-088 EQ0218	088	■ 1.000 ● N -1	$a_{r_k,k}, s_k\right ^{\left(k\right)}:=\max \left\{\left a_{i,j}, k\right ^{\left(k\right)}: k \leq i, j \leq n\right\}$	$a_{r_k,s_k}^{(k)}:=\max \left\{\left a_{i,j}^{(k)}\right : k \leq i, j \leq n\right\}$	$a_{r_k,s_k}^{(k)}:=\max \left\{\left a_{i,j}^{(k)}\right : k \leq i, j \leq n\right\}$
lyche-numerical-linear-algebra-088 EQ0219	088	■ 1.000 ● S +0	$\boldsymbol{L}=\left[\begin{array}{ccc} l_{1,1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ l_{n,1} & \cdots & l_{n,n} \end{array}\right], \quad \boldsymbol{U}=\left[\begin{array}{ccc} u_{1,1} & \cdots & u_{1,n} \\ \vdots & \ddots & \vdots \\ 0 & \cdots & u_{n,n} \end{array}\right] .$	$\boldsymbol{L}=\left[\begin{array}{ccc} l_{1,1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ l_{n,1} & \cdots & l_{n,n} \end{array}\right], \quad \boldsymbol{U}=\left[\begin{array}{ccc} u_{1,1} & \cdots & u_{1,n} \\ \vdots & \ddots & \vdots \\ 0 & \cdots & u_{n,n} \end{array}\right] .$	$\boldsymbol{L}=\left[\begin{array}{ccc} l_{1,1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ l_{n,1} & \cdots & l_{n,n} \end{array}\right], \quad \boldsymbol{U}=\left[\begin{array}{ccc} u_{1,1} & \cdots & u_{1,n} \\ \vdots & \ddots & \vdots \\ 0 & \cdots & u_{n,n} \end{array}\right] .$
lyche-numerical-linear-algebra-089 EQ0221 (3.15)	089	■ 1.000 ● N +0	$u_1=a, \quad u_2=b, \quad a l_1=c, \quad b l_1+u_3=d .$	$u_1=a, \quad u_2=b, \quad a l_1=c, \quad b l_1+u_3=d .$	$u_1=a, \quad u_2=b, \quad a l_1=c, \quad b l_1+u_3=d .$
lyche-numerical-linear-algebra-089 EQ0222	089	■ 1.000 ● N +0	$\boldsymbol{A}_1:=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right], \quad \boldsymbol{A}_2:=\left[\begin{array}{cc} 0 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{A}_3:=\left[\begin{array}{cc} 0 & 1 \\ 0 & 2 \end{array}\right], \quad \boldsymbol{A}_4:=\left[\begin{array}{cc} 1 & 1 \\ 1 & 1 \end{array}\right] .$	$\boldsymbol{A}_1:=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right], \quad \boldsymbol{A}_2:=\left[\begin{array}{cc} 0 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{A}_3:=\left[\begin{array}{cc} 0 & 1 \\ 0 & 2 \end{array}\right], \quad \boldsymbol{A}_4:=\left[\begin{array}{cc} 1 & 1 \\ 1 & 1 \end{array}\right] .$	$\boldsymbol{A}_1:=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right], \quad \boldsymbol{A}_2:=\left[\begin{array}{cc} 0 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{A}_3:=\left[\begin{array}{cc} 0 & 1 \\ 0 & 2 \end{array}\right], \quad \boldsymbol{A}_4:=\left[\begin{array}{cc} 1 & 1 \\ 1 & 1 \end{array}\right] .$
lyche-numerical-linear-algebra-089 EQ0223	089	■ 1.000 ● K +0	$\boldsymbol{A}_{[k]}:=\left[\begin{array}{ccc} a_{11} & \cdots & a_{k 1} \\ \vdots & & \vdots \\ a_{k 1} & \cdots & a_{k k} \end{array}\right]$	$\boldsymbol{A}_{[k]}:=\left[\begin{array}{ccc} a_{11} & \cdots & a_{k 1} \\ \vdots & & \vdots \\ a_{k 1} & \cdots & a_{k k} \end{array}\right]$	$\boldsymbol{A}_{[k]}:=\left[\begin{array}{ccc} a_{11} & \cdots & a_{k 1} \\ \vdots & & \vdots \\ a_{k 1} & \cdots & a_{k k} \end{array}\right]$

Conf. MathPix confidence:

■ ≥0.9

■ 0.5–0.9

■ <0.5

 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

● C component

● W weak

● S stable

● N noise

● K clean

● not measured

[refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-E00225 (3.17)	090	■ 1.000 ● S +0	$\begin{array}{cc} \boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{[n-1]} & \boldsymbol{c}_n \\ \boldsymbol{r}_n^T & a_{nn} \end{array}\right] & \boldsymbol{A}_{[n-1]} & \boldsymbol{c}_n \\ \boldsymbol{r}_n^T & a_{nn} \end{array} \right] = \left[\begin{array}{cc} \boldsymbol{L}_{n-1} & \boldsymbol{0} \\ \boldsymbol{l}_n^T & 1 \end{array}\right] \left[\begin{array}{cc} \boldsymbol{U}_{n-1} & \boldsymbol{u}_n \\ 0 & u_{nn} \end{array}\right] = \left[\begin{array}{cc} \boldsymbol{L}_{n-1}\boldsymbol{U}_{n-1} & \boldsymbol{L}_{n-1}\boldsymbol{u}_n \\ \boldsymbol{l}_n^T\boldsymbol{U}_{n-1} & \boldsymbol{l}_n^T\boldsymbol{u}_n + u_{nn} \end{array}\right], \end{array}$	$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_{[n-1]} & \boldsymbol{c}_n \\ \boldsymbol{r}_n^T & a_{nn} \end{bmatrix} = \begin{bmatrix} \boldsymbol{L}_{n-1} & \boldsymbol{0} \\ \boldsymbol{l}_n^T & 1 \end{bmatrix} \begin{bmatrix} \boldsymbol{U}_{n-1} & \boldsymbol{u}_n \\ 0 & u_{nn} \end{bmatrix} = \begin{bmatrix} \boldsymbol{L}_{n-1}\boldsymbol{U}_{n-1} & \boldsymbol{L}_{n-1}\boldsymbol{u}_n \\ \boldsymbol{l}_n^T\boldsymbol{U}_{n-1} & \boldsymbol{l}_n^T\boldsymbol{u}_n + u_{nn} \end{bmatrix},$	
lyche-numerical-linear-algebra-E00226 (3.18)	090	■ 1.000 ● S +0	$\begin{array}{cc} \boldsymbol{U}_{n-1}^T \boldsymbol{l}_n = \boldsymbol{r}_n, & \boldsymbol{L}_{n-1} \boldsymbol{u}_n = \boldsymbol{c}_n, & u_{nn} = a_{nn} - \boldsymbol{l}_n^T \boldsymbol{u}_n. \end{array}$	$\boldsymbol{U}_{n-1}^T \boldsymbol{l}_n = \boldsymbol{r}_n, \quad \boldsymbol{L}_{n-1} \boldsymbol{u}_n = \boldsymbol{c}_n, \quad u_{nn} = a_{nn} - \boldsymbol{l}_n^T \boldsymbol{u}_n.$	
lyche-numerical-linear-algebra-E00227	091	■ 1.000 ●	$\begin{array}{cc} \boldsymbol{U}_{k-1}^T \boldsymbol{l}_k = \boldsymbol{r}_k, & \left[\begin{array}{cc} \boldsymbol{L}_{k-1} & \boldsymbol{0} \\ \boldsymbol{l}_k^T & 1 \end{array}\right] \left[\begin{array}{c} \boldsymbol{u}_k \\ u_{kk} \end{array}\right] = \left[\begin{array}{c} \boldsymbol{c}_k \\ a_{kk} \end{array}\right]. \end{array}$	$\boldsymbol{U}_{k-1}^T \boldsymbol{l}_k = \boldsymbol{r}_k, \quad \left[\begin{array}{cc} \boldsymbol{L}_{k-1} & \boldsymbol{0} \\ \boldsymbol{l}_k^T & 1 \end{array}\right] \begin{bmatrix} \boldsymbol{u}_k \\ u_{kk} \end{bmatrix} = \begin{bmatrix} \boldsymbol{c}_k \\ a_{kk} \end{bmatrix}.$	
lyche-numerical-linear-algebra-E00230	092	■ 1.000 ● N +0	$\boldsymbol{A}_{\{k\}} := \left[\begin{array}{ccc} \boldsymbol{A}_{11} & \cdots & \boldsymbol{A}_{1k} \\ \vdots & & \vdots \\ \boldsymbol{A}_{k1} & \cdots & \boldsymbol{A}_{kk} \end{array}\right]$	$\boldsymbol{A}_{\{k\}} := \begin{bmatrix} \boldsymbol{A}_{11} & \cdots & \boldsymbol{A}_{1k} \\ \vdots & & \vdots \\ \boldsymbol{A}_{k1} & \cdots & \boldsymbol{A}_{kk} \end{bmatrix}$	
lyche-numerical-linear-algebra-E00232 (3.22)	093	■ 1.000 ● C +54	$\begin{array}{cc} \begin{array}{c} \boldsymbol{A}((k+1):n, (k+1):n) \boldsymbol{b}_k((k+1):n) = -\boldsymbol{A}((k+1):n, k) \boldsymbol{b}_k(k), \\ \text{for } k=1, \dots, n-1. \end{array} \end{array}$	$\boldsymbol{A}((k+1):n, (k+1):n) \boldsymbol{b}_k((k+1):n) = -\boldsymbol{A}((k+1):n, k) \boldsymbol{b}_k(k), \quad k = 1, \dots, n-1.$	
lyche-numerical-linear-algebra-E00233 (3.23)	094	■ 1.000 ● K +0	$\boldsymbol{A}(1:k, 1:k) \boldsymbol{b}_k(1:k) = \boldsymbol{I}(1:k, k), \quad k = n, n-1, \dots, 1,$	$\boldsymbol{A}(1:k, 1:k) \boldsymbol{b}_k(1:k) = \boldsymbol{I}(1:k, k), \quad k = n, n-1, \dots, 1,$	
lyche-numerical-linear-algebra-E00235 (3.28)	095	■ 1.000 ● K +0	$\boldsymbol{C} := \boldsymbol{U} + \boldsymbol{v} \boldsymbol{e}_1^T,$	$\boldsymbol{C} := \boldsymbol{U} + \boldsymbol{v} \boldsymbol{e}_1^T,$	
lyche-numerical-linear-algebra-E00236 (3.29)	095	■ 1.000 ● N +0	$\boldsymbol{P} := \boldsymbol{I}_{1,2} \boldsymbol{I}_{2,3} \cdots \boldsymbol{I}_{n-1,n},$	$\boldsymbol{P} := \boldsymbol{I}_{1,2} \boldsymbol{I}_{2,3} \cdots \boldsymbol{I}_{n-1,n},$	
lyche-numerical-linear-algebra-E00237 (3.30)	096	■ 1.000 ● K +0	$\boldsymbol{B} := \boldsymbol{A} + \boldsymbol{w} \boldsymbol{e}_1^T.$	$\boldsymbol{B} := \boldsymbol{A} + \boldsymbol{w} \boldsymbol{e}_1^T.$	
lyche-numerical-linear-algebra-E00238	096	■ 1.000 ● N +0	$\det(\boldsymbol{A}_{[k]}) = u_{11} u_{22} \cdots u_{kk} \text{ for } k = 1, \dots, n.$	$\det(\boldsymbol{A}_{[k]}) = u_{11} u_{22} \cdots u_{kk} \text{ for } k = 1, \dots, n.$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix's reading, unchanged, Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-097 EQ0239 (3.31)	097	1.000 N +0	$u_{11}=a_{11}, \quad u_{kk}=\frac{\operatorname{det}\left(\boldsymbol{A}_{\left[k\right]}\right)}{\operatorname{det}\left(\boldsymbol{A}_{\left[k-1\right]}\right)}, \quad \text { for } k=2, \ldots, n .$	$u_{11}=a_{11}, \quad u_{kk}=\frac{\det\left(\boldsymbol{A}_{[k]}\right)}{\det\left(\boldsymbol{A}_{[k-1]}\right)}, \quad \text { for } k=2, \ldots, n .$	$u_{11}=a_{11}, \quad u_{kk}=\frac{\det(\boldsymbol{A}_{[k]})}{\det(\boldsymbol{A}_{[k-1]})}, \quad \text { for } k=2, \ldots, n .$
lyche-numerical-linear-algebra-098 EQ0242	098	1.000 N +0	$\boldsymbol{B}:=\left[\boldsymbol{a}_1, \ldots, \boldsymbol{a}_{p-1}, \boldsymbol{a}_{p+1}, \ldots, \boldsymbol{a}_n, \boldsymbol{b}\right] \in \mathbb{R}^{n \times n} .$	$\boldsymbol{B}:=\left[\boldsymbol{a}_1, \ldots, \boldsymbol{a}_{p-1}, \boldsymbol{a}_{p+1}, \ldots, \boldsymbol{a}_n, \boldsymbol{b}\right] \in \mathbb{R}^{n \times n} .$	$\boldsymbol{B}:=\left[\boldsymbol{a}_1, \ldots, \boldsymbol{a}_{p-1}, \boldsymbol{a}_{p+1}, \ldots, \boldsymbol{a}_n, \boldsymbol{b}\right] \in \mathbb{R}^{n \times n} .$
lyche-numerical-linear-algebra-098 EQ0243	098	1.000 N +0	$A:=\left[\begin{array}{rr} -3 & -2 \\ 4 & 2 \end{array}\right] .$	$A:=\left[\begin{array}{rr} -3 & -2 \\ 4 & 2 \end{array}\right] .$	$\boldsymbol{A}:=\left[\begin{array}{rr} -3 & -2 \\ 4 & 2 \end{array}\right] .$
lyche-numerical-linear-algebra-098 EQ0244	098	1.000 N +0	$\boldsymbol{P}:=\left[\boldsymbol{e}_n, \boldsymbol{e}_{n-1}, \ldots, \boldsymbol{e}_1\right] .$	$\boldsymbol{P}:=\left[\boldsymbol{e}_n, \boldsymbol{e}_{n-1}, \ldots, \boldsymbol{e}_1\right] .$	$\boldsymbol{P}:=\left[\boldsymbol{e}_n, \boldsymbol{e}_{n-1}, \ldots, \boldsymbol{e}_1\right] .$
lyche-numerical-linear-algebra-101 EQ0246 (4.1)	101	1.000 N +0	$d_1=a . \quad a l_1=b, \quad d_2=d-a\left l_1\right ^2 .$	$d_1=a . \quad a l_1=b, \quad d_2=d-a\left l_1\right ^2 .$	$d_1=a . \quad a l_1=b, \quad d_2=d-a\left l_1\right ^2 .$
lyche-numerical-linear-algebra-102 EQ0251	102	1.000 N +0	$f(\boldsymbol{x})=\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}=\sum_{i=1}^n \sum_{j=1}^n a_{i j} \bar{x}_i x_j$	$f(\boldsymbol{x})=\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}=\sum_{i=1}^n \sum_{j=1}^n a_{i j} \bar{x}_i x_j$	$f(\boldsymbol{x})=\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}=\sum_{i=1}^n \sum_{j=1}^n a_{i j} \bar{x}_i x_j$
lyche-numerical-linear-algebra-104 EQ0255 (4.5)	104	1.000	$\boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r})=\boldsymbol{X}^* \boldsymbol{A} \boldsymbol{X}, \quad \boldsymbol{X}:=\left[\boldsymbol{e}_{r_1}, \ldots, \boldsymbol{e}_{r_k}\right] \in \mathbb{C}^{n \times k} .$	$\boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r})=\boldsymbol{X}^* \boldsymbol{A} \boldsymbol{X}, \quad \boldsymbol{X}:=\left[\boldsymbol{e}_{r_1}, \ldots, \boldsymbol{e}_{r_k}\right] \in \mathbb{C}^{n \times k} .$	$\boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r})=\boldsymbol{X}^* \boldsymbol{A} \boldsymbol{X}, \quad \boldsymbol{X}:=\left[\boldsymbol{e}_{r_1}, \ldots, \boldsymbol{e}_{r_k}\right] \in \mathbb{C}^{n \times k} .$
lyche-numerical-linear-algebra-104 EQ0256 (4.6)	104	1.000 W +1	$\boldsymbol{y}^* \boldsymbol{B} \boldsymbol{y}=\boldsymbol{y}^* \boldsymbol{X}^* \boldsymbol{A} \boldsymbol{X} \boldsymbol{y}=\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} \geq 0, \quad \boldsymbol{y} \in \mathbb{C}^k, \quad \boldsymbol{x}:=\boldsymbol{X} \boldsymbol{y} .$	$\boldsymbol{y}^* \boldsymbol{B} \boldsymbol{y}=\boldsymbol{y}^* \boldsymbol{X}^* \boldsymbol{A} \boldsymbol{X} \boldsymbol{y}=\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} \geq 0, \quad \boldsymbol{y} \in \mathbb{C}^k, \quad \boldsymbol{x}:=\boldsymbol{X} \boldsymbol{y} .$	$\boldsymbol{y}^* \boldsymbol{B} \boldsymbol{y}=\boldsymbol{y}^* \boldsymbol{X}^* \boldsymbol{A} \boldsymbol{X} \boldsymbol{y}=\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} \geq 0, \quad \boldsymbol{y} \in \mathbb{C}^k, \quad \boldsymbol{x}:=\boldsymbol{X} \boldsymbol{y} .$
lyche-numerical-linear-algebra-105 EQ0257 (0)	105	1.000 W -1	$\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right]=\left[\begin{array}{cc} 1 & 0 \\ 0 & \frac{3}{2} \end{array}\right]\left[\begin{array}{cc} 1 & -\frac{1}{2} \\ 0 & 1 \end{array}\right]=\left[\begin{array}{cc} \sqrt{2} & 0 \\ -1 / \sqrt{2} & \sqrt{3} / 2 \end{array}\right]\left[\begin{array}{cc} \sqrt{2} & -1 / \sqrt{2} \\ 0 & \sqrt{3} / 2 \end{array}\right] .$	$\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right]=\left[\begin{array}{cc} 1 & 0 \\ -\frac{1}{2} & 1 \end{array}\right]\left[\begin{array}{cc} 2 & 0 \\ 0 & \frac{3}{2} \end{array}\right]\left[\begin{array}{cc} 1 & -\frac{1}{2} \\ 0 & 1 \end{array}\right]=\left[\begin{array}{cc} \sqrt{2} & 0 \\ -1 / \sqrt{2} & \sqrt{3} / 2 \end{array}\right]\left[\begin{array}{cc} \sqrt{2} & -1 / \sqrt{2} \\ 0 & \sqrt{3} / 2 \end{array}\right] .$	$\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right]=\left[\begin{array}{cc} 1 & 0 \\ -\frac{1}{2} & 1 \end{array}\right]\left[\begin{array}{cc} 2 & 0 \\ 0 & \frac{3}{2} \end{array}\right]\left[\begin{array}{cc} 1 & -\frac{1}{2} \\ 0 & 1 \end{array}\right]=\left[\begin{array}{cc} \sqrt{2} & 0 \\ -1 / \sqrt{2} & \sqrt{3} / 2 \end{array}\right]\left[\begin{array}{cc} \sqrt{2} & -1 / \sqrt{2} \\ 0 & \sqrt{3} / 2 \end{array}\right] .$
lyche-numerical-linear-algebra-106 EQ0258 (4.7)	106	1.000 C +54	$\begin{array}{l} \text {\captionsetup{labelformat=empty}} \\ \text {\caption{Listing 4.2 bandcholesky}} \\ \text {\[} \theta<(\backslash\leq)\left(\alpha \boldsymbol{e}_i+\beta \boldsymbol{e}_j\right)^* \boldsymbol{A} \backslash\left(\alpha \boldsymbol{e}_i+\beta \boldsymbol{e}_j\right) \\ \text {\right)=} \alpha ^2 a_{i i}+ \beta ^2 a_{j j}+2 \operatorname{Re}\left(\bar{\alpha} \beta a_{i j}\right) . \end{array}$	<i>(not rendered)</i>	$0 < (\leq)(\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j)^* \boldsymbol{A} (\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j) = \alpha ^2 a_{ii} + \beta ^2 a_{jj} + 2 \operatorname{Re}(\bar{\alpha} \beta a_{ij}).$
lyche-numerical-linear-algebra-106 EQ0259	106	1.000 N +0	$0<\left a_{i j}\right ^2 a_{i i}+a_{i i}^2 a_{j j}-2\left a_{i j}\right ^2 a_{i i}=a_{i i}\left(a_{i i} a_{j j}-\left a_{i j}\right ^2\right) .$	$0 < \left a_{i j}\right ^2 a_{i i}+a_{i i}^2 a_{j j}-2\left a_{i j}\right ^2 a_{i i}=a_{i i}\left(a_{i i} a_{j j}-\left a_{i j}\right ^2\right) .$	$0 < \left a_{i j}\right ^2 a_{i i}+a_{i i}^2 a_{j j}-2\left a_{i j}\right ^2 a_{i i}=a_{i i}\left(a_{i i} a_{j j}-\left a_{i j}\right ^2\right) .$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-106 EQ0260	106	1.000 N +0	$\left a_{ij}\right =\left b_{ij}\right <\sqrt{b_{ii}b_{jj}}$ $b_{ij}=\sqrt{\left(a_{ii}+\varepsilon\right)\left(a_{jj}+\varepsilon\right)}$, $\quad i\neq j$.	$ a_{ij} = b_{ij} < \sqrt{b_{ii}b_{jj}} = \sqrt{(a_{ii} + \varepsilon)(a_{jj} + \varepsilon)}, \quad i \neq j.$	$ a_{ij} = b_{ij} < \sqrt{b_{ii}b_{jj}} = \sqrt{(a_{ii} + \varepsilon)(a_{jj} + \varepsilon)}, \quad i \neq j.$
lyche-numerical-linear-algebra-107 EQ0261	107	1.000 N +0	$\boldsymbol{A}_1=\left[\begin{array}{ll} 0 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{A}_2=\left[\begin{array}{ll} 1 & 2 \\ 2 & 2 \end{array}\right], \quad \boldsymbol{A}_3=\left[\begin{array}{ll} -2 & 1 \\ 1 & 2 \end{array}\right].$	$\boldsymbol{A}_1 = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad \boldsymbol{A}_2 = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}, \quad \boldsymbol{A}_3 = \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix}.$	$\boldsymbol{A}_1 = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad \boldsymbol{A}_2 = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}, \quad \boldsymbol{A}_3 = \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix}.$
lyche-numerical-linear-algebra-109 EQ0266 (4.10)	109	1.000 K +0	$\boldsymbol{L}^*:=\left[\begin{array}{cc} \beta & \boldsymbol{v}^*/\beta \\ \boldsymbol{0} & \boldsymbol{L}_1^* \end{array}\right], \quad \beta:=\sqrt{\alpha},$	$\boldsymbol{L}^* := \begin{bmatrix} \beta & \boldsymbol{v}^*/\beta \\ \boldsymbol{0} & \boldsymbol{L}_1^* \end{bmatrix}, \quad \beta := \sqrt{\alpha},$	$\boldsymbol{L}^* := \begin{bmatrix} \beta & \boldsymbol{v}^*/\beta \\ \boldsymbol{0} & \boldsymbol{L}_1^* \end{bmatrix}, \quad \beta := \sqrt{\alpha},$
lyche-numerical-linear-algebra-113 EQ0271	113	1.000 N +0	$\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x}=2x_1^2+(2-a)x_1x_2+ax_2x_1+x_2^2=x_1^2+(x_1+x_2)^2>0.$	$\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x} = 2x_1^2 + (2 - a)x_1x_2 + ax_2x_1 + x_2^2 = x_1^2 + (x_1 + x_2)^2 > 0.$	$\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x} = 2x_1^2 + (2 - a)x_1x_2 + ax_2x_1 + x_2^2 = x_1^2 + (x_1 + x_2)^2 > 0.$
lyche-numerical-linear-algebra-114 EQ0273	114	1.000 K +0	$\boldsymbol{A}=\boldsymbol{L}\left(\boldsymbol{I}+\boldsymbol{v}\boldsymbol{v}^T\right)\boldsymbol{L}^T,$	$\boldsymbol{A} = \boldsymbol{L} \left(\boldsymbol{I} + \boldsymbol{v} \boldsymbol{v}^T \right) \boldsymbol{L}^T,$	$\boldsymbol{A} = \boldsymbol{L} (\boldsymbol{I} + \boldsymbol{v} \boldsymbol{v}^T) \boldsymbol{L}^T,$
lyche-numerical-linear-algebra-114 EQ0275	114	1.000	$\boldsymbol{L}_+=\left[\begin{array}{cc} \boldsymbol{L} & \boldsymbol{0} \\ \boldsymbol{y}^T & \lambda \end{array}\right],$	$\boldsymbol{L}_+ = \begin{bmatrix} \boldsymbol{L} & \boldsymbol{0} \\ \boldsymbol{y}^T & \lambda \end{bmatrix},$	$\boldsymbol{L}_+ = \begin{bmatrix} \boldsymbol{L} & \boldsymbol{0} \\ \boldsymbol{y}^T & \lambda \end{bmatrix},$
lyche-numerical-linear-algebra-117 EQ0276	117	1.000 N +0	$\langle \boldsymbol{x}, \boldsymbol{a}\boldsymbol{y} + \boldsymbol{b}\boldsymbol{z} \rangle = \overline{\langle \boldsymbol{a}\boldsymbol{y} + \boldsymbol{b}\boldsymbol{z}, \boldsymbol{x} \rangle} = \overline{\langle \boldsymbol{y}, \boldsymbol{x} \rangle} + \overline{\langle \boldsymbol{z}, \boldsymbol{x} \rangle} = \overline{\langle \boldsymbol{y}, \boldsymbol{x} \rangle} + \overline{\langle \boldsymbol{z}, \boldsymbol{x} \rangle}$	$\langle \boldsymbol{x}, \boldsymbol{a}\boldsymbol{y} + \boldsymbol{b}\boldsymbol{z} \rangle = \overline{\langle \boldsymbol{a}\boldsymbol{y} + \boldsymbol{b}\boldsymbol{z}, \boldsymbol{x} \rangle} = \overline{\langle \boldsymbol{y}, \boldsymbol{x} \rangle} + \overline{\langle \boldsymbol{z}, \boldsymbol{x} \rangle} = \overline{\langle \boldsymbol{y}, \boldsymbol{x} \rangle} + \overline{\langle \boldsymbol{z}, \boldsymbol{x} \rangle}$	$\langle \boldsymbol{x}, \boldsymbol{a}\boldsymbol{y} + \boldsymbol{b}\boldsymbol{z} \rangle = \overline{\langle \boldsymbol{a}\boldsymbol{y} + \boldsymbol{b}\boldsymbol{z}, \boldsymbol{x} \rangle} = \overline{\langle \boldsymbol{y}, \boldsymbol{x} \rangle} + \overline{\langle \boldsymbol{z}, \boldsymbol{x} \rangle} = \overline{\langle \boldsymbol{y}, \boldsymbol{x} \rangle} + \overline{\langle \boldsymbol{z}, \boldsymbol{x} \rangle}$
lyche-numerical-linear-algebra-117 EQ0277 (5.1)	117	1.000 N +1	$\langle \boldsymbol{x}, \boldsymbol{a}\boldsymbol{y} + \boldsymbol{b}\boldsymbol{z} \rangle = \overline{\langle \boldsymbol{a}\boldsymbol{y} + \boldsymbol{b}\boldsymbol{z}, \boldsymbol{x} \rangle} = \overline{\langle \boldsymbol{y}, \boldsymbol{x} \rangle} + \overline{\langle \boldsymbol{z}, \boldsymbol{x} \rangle} = \overline{\langle \boldsymbol{y}, \boldsymbol{x} \rangle} + \overline{\langle \boldsymbol{z}, \boldsymbol{x} \rangle}$	$\langle \boldsymbol{x}, \boldsymbol{a}\boldsymbol{y} + \boldsymbol{b}\boldsymbol{z} \rangle = \overline{\langle \boldsymbol{a}\boldsymbol{y} + \boldsymbol{b}\boldsymbol{z}, \boldsymbol{x} \rangle} = \overline{\langle \boldsymbol{y}, \boldsymbol{x} \rangle} + \overline{\langle \boldsymbol{z}, \boldsymbol{x} \rangle} = \overline{\langle \boldsymbol{y}, \boldsymbol{x} \rangle} + \overline{\langle \boldsymbol{z}, \boldsymbol{x} \rangle}$	$\langle \boldsymbol{x}, \boldsymbol{a}\boldsymbol{y} + \boldsymbol{b}\boldsymbol{z} \rangle = \overline{\langle \boldsymbol{a}\boldsymbol{y} + \boldsymbol{b}\boldsymbol{z}, \boldsymbol{x} \rangle} = \overline{\langle \boldsymbol{y}, \boldsymbol{x} \rangle} + \overline{\langle \boldsymbol{z}, \boldsymbol{x} \rangle} = \overline{\langle \boldsymbol{y}, \boldsymbol{x} \rangle} + \overline{\langle \boldsymbol{z}, \boldsymbol{x} \rangle}$
lyche-numerical-linear-algebra-117 EQ0280 (5.2)	117	1.000 N +0	$\ \cdot\ :\mathcal{V}\rightarrow\mathbb{R}, \quad \boldsymbol{x}\mapsto\ \boldsymbol{x}\ :=\sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle}$	$\ \cdot\ : \mathcal{V} \rightarrow \mathbb{R}, \quad \boldsymbol{x} \mapsto \ \boldsymbol{x}\ := \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle}$	$\ \cdot\ : \mathcal{V} \rightarrow \mathbb{R}, \quad \boldsymbol{x} \mapsto \ \boldsymbol{x}\ := \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle}$
lyche-numerical-linear-algebra-118 EQ0281 (5.3)	118	1.000 N +0	$ \langle \boldsymbol{x}, \boldsymbol{y} \rangle \leq \ \boldsymbol{x}\ \ \boldsymbol{y}\ ,$	$ \langle \boldsymbol{x}, \boldsymbol{y} \rangle \leq \ \boldsymbol{x}\ \ \boldsymbol{y}\ ,$	$ \langle \boldsymbol{x}, \boldsymbol{y} \rangle \leq \ \boldsymbol{x}\ \ \boldsymbol{y}\ ,$
lyche-numerical-linear-algebra-119 EQ0285 (5.5)	119	1.000 W +0	$\ \boldsymbol{x} + \boldsymbol{a}\boldsymbol{y}\ ^2 = \ \boldsymbol{x}\ ^2 + \langle \boldsymbol{y}, \boldsymbol{x} \rangle + \overline{\langle \boldsymbol{x}, \boldsymbol{y} \rangle} + \boldsymbol{a} ^2 \ \boldsymbol{y}\ ^2, \quad \boldsymbol{a} \in \mathbb{C}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}.$	$\ \boldsymbol{x} + \boldsymbol{a}\boldsymbol{y}\ ^2 = \ \boldsymbol{x}\ ^2 + \langle \boldsymbol{y}, \boldsymbol{x} \rangle + \overline{\langle \boldsymbol{x}, \boldsymbol{y} \rangle} + \boldsymbol{a} ^2 \ \boldsymbol{y}\ ^2, \quad \boldsymbol{a} \in \mathbb{C}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}.$	$\ \boldsymbol{x} + \boldsymbol{a}\boldsymbol{y}\ ^2 = \ \boldsymbol{x}\ ^2 + \langle \boldsymbol{y}, \boldsymbol{x} \rangle + \overline{\langle \boldsymbol{x}, \boldsymbol{y} \rangle} + \boldsymbol{a} ^2 \ \boldsymbol{y}\ ^2, \quad \boldsymbol{a} \in \mathbb{C}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0287 (5.6)	119	■ 1.000 ● N +0	$\cos \theta = \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\ \boldsymbol{x}\ \ \boldsymbol{y}\ }.$	$\cos \theta = \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\ \boldsymbol{x}\ \ \boldsymbol{y}\ }.$	$\cos \theta = \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\ \boldsymbol{x}\ \ \boldsymbol{y}\ }.$
lyche-numerical-linear-algebra- EQ0288 (5.7)	119	■ 1.000 ● N +0	$\ \boldsymbol{x} + \boldsymbol{y}\ ^2 = \ \boldsymbol{x}\ ^2 + \ \boldsymbol{y}\ ^2, \quad \text{if } \boldsymbol{x} \perp \boldsymbol{y}.$	$\ \boldsymbol{x} + \boldsymbol{y}\ ^2 = \ \boldsymbol{x}\ ^2 + \ \boldsymbol{y}\ ^2, \quad \text{if } \boldsymbol{x} \perp \boldsymbol{y}.$	$\ \boldsymbol{x} + \boldsymbol{y}\ ^2 = \ \boldsymbol{x}\ ^2 + \ \boldsymbol{y}\ ^2, \quad \text{if } \boldsymbol{x} \perp \boldsymbol{y}.$
lyche-numerical-linear-algebra- EQ0290	120	■ 1.000 ● N +0	$\left\{ \frac{\boldsymbol{v}_1}{\ \boldsymbol{v}_1\ }, \dots, \frac{\boldsymbol{v}_k}{\ \boldsymbol{v}_k\ } \right\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_k\} := \left\{ \frac{\boldsymbol{v}_1}{\ \boldsymbol{v}_1\ }, \dots, \frac{\boldsymbol{v}_k}{\ \boldsymbol{v}_k\ } \right\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_k\} := \left\{ \frac{\boldsymbol{v}_1}{\ \boldsymbol{v}_1\ }, \dots, \frac{\boldsymbol{v}_k}{\ \boldsymbol{v}_k\ } \right\}$
lyche-numerical-linear-algebra- EQ0291	120	■ 1.000 ● C +3	$\left\langle \boldsymbol{v}_j, \boldsymbol{v}_l \right\rangle = \left\langle \boldsymbol{s}_j, \boldsymbol{v}_l \right\rangle - \sum_{i=1}^{j-1} \frac{\langle \boldsymbol{s}_j, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle} \langle \boldsymbol{v}_i, \boldsymbol{v}_l \rangle = \left\langle \boldsymbol{s}_j, \boldsymbol{v}_l \right\rangle - \frac{\langle \boldsymbol{s}_j, \boldsymbol{v}_l \rangle}{\langle \boldsymbol{v}_l, \boldsymbol{v}_l \rangle} \langle \boldsymbol{v}_l, \boldsymbol{v}_l \rangle = 0$	$\langle \boldsymbol{v}_j, \boldsymbol{v}_l \rangle = \langle \boldsymbol{s}_j, \boldsymbol{v}_l \rangle - \sum_{i=1}^{j-1} \frac{\langle \boldsymbol{s}_j, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle} \langle \boldsymbol{v}_i, \boldsymbol{v}_l \rangle = \langle \boldsymbol{s}_j, \boldsymbol{v}_l \rangle - \frac{\langle \boldsymbol{s}_j, \boldsymbol{v}_l \rangle}{\langle \boldsymbol{v}_l, \boldsymbol{v}_l \rangle} \langle \boldsymbol{v}_l, \boldsymbol{v}_l \rangle = 0$	$\langle \boldsymbol{v}_j, \boldsymbol{v}_l \rangle = \langle \boldsymbol{s}_j, \boldsymbol{v}_l \rangle - \sum_{i=1}^{j-1} \frac{\langle \boldsymbol{s}_j, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle} \langle \boldsymbol{v}_i, \boldsymbol{v}_l \rangle = \langle \boldsymbol{s}_j, \boldsymbol{v}_l \rangle - \frac{\langle \boldsymbol{s}_j, \boldsymbol{v}_l \rangle}{\langle \boldsymbol{v}_l, \boldsymbol{v}_l \rangle} \langle \boldsymbol{v}_l, \boldsymbol{v}_l \rangle = 0$
lyche-numerical-linear-algebra- EQ0292	121	■ 1.000 ●	$\dim(\mathcal{S} \oplus \mathcal{T}) = \dim(\mathcal{S}) + \dim(\mathcal{T}).$	$\dim(\mathcal{S} \oplus \mathcal{T}) = \dim(\mathcal{S}) + \dim(\mathcal{T}).$	$\dim(\mathcal{S} \oplus \mathcal{T}) = \dim(\mathcal{S}) + \dim(\mathcal{T}).$
lyche-numerical-linear-algebra- EQ0295	122	■ 1.000 ● W +0	$\left\langle \boldsymbol{s}_0, \boldsymbol{v}_j \right\rangle = \left\langle \sum_{i=1}^k \frac{\langle \boldsymbol{v}, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle} \boldsymbol{v}_i, \boldsymbol{v}_j \right\rangle = \sum_{i=1}^k \frac{\langle \boldsymbol{v}, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle} \langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = \langle \boldsymbol{v}, \boldsymbol{v}_j \rangle.$	$\langle \boldsymbol{s}_0, \boldsymbol{v}_j \rangle = \left\langle \sum_{i=1}^k \frac{\langle \boldsymbol{v}, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle} \boldsymbol{v}_i, \boldsymbol{v}_j \right\rangle = \sum_{i=1}^k \frac{\langle \boldsymbol{v}, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle} \langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = \langle \boldsymbol{v}, \boldsymbol{v}_j \rangle.$	$\langle \boldsymbol{s}_0, \boldsymbol{v}_j \rangle = \left\langle \sum_{i=1}^k \frac{\langle \boldsymbol{v}, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle} \boldsymbol{v}_i, \boldsymbol{v}_j \right\rangle = \sum_{i=1}^k \frac{\langle \boldsymbol{v}, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle} \langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = \langle \boldsymbol{v}, \boldsymbol{v}_j \rangle.$
lyche-numerical-linear-algebra- EQ0297	123	■ 1.000 ● N +0	$\ \boldsymbol{v} - \boldsymbol{s}\ ^2 = \ \boldsymbol{v} - \boldsymbol{s}_0 + \boldsymbol{u}\ ^2 = \ \boldsymbol{v} - \boldsymbol{s}_0\ ^2 + \ \boldsymbol{u}\ ^2 > \ \boldsymbol{v} - \boldsymbol{s}_0\ ^2.$	$\ \boldsymbol{v} - \boldsymbol{s}\ ^2 = \ \boldsymbol{v} - \boldsymbol{s}_0 + \boldsymbol{u}\ ^2 = \ \boldsymbol{v} - \boldsymbol{s}_0\ ^2 + \ \boldsymbol{u}\ ^2 > \ \boldsymbol{v} - \boldsymbol{s}_0\ ^2.$	$\ \boldsymbol{v} - \boldsymbol{s}\ ^2 = \ \boldsymbol{v} - \boldsymbol{s}_0 + \boldsymbol{u}\ ^2 = \ \boldsymbol{v} - \boldsymbol{s}_0\ ^2 + \ \boldsymbol{u}\ ^2 > \ \boldsymbol{v} - \boldsymbol{s}_0\ ^2.$
lyche-numerical-linear-algebra- EQ0299	123	■ 1.000 ● K +0	$\boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_j = \langle \boldsymbol{A} \boldsymbol{e}_j, \boldsymbol{e}_i \rangle = \langle \boldsymbol{e}_j, \overline{\boldsymbol{A} \boldsymbol{e}_i} \rangle = \boldsymbol{e}_i^T \boldsymbol{A}^T \boldsymbol{e}_j.$	$\boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_j = \langle \boldsymbol{A} \boldsymbol{e}_j, \boldsymbol{e}_i \rangle = \langle \boldsymbol{e}_j, \overline{\boldsymbol{A} \boldsymbol{e}_i} \rangle = \boldsymbol{e}_i^T \boldsymbol{A}^T \boldsymbol{e}_j.$	$\boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_j = \langle \boldsymbol{A} \boldsymbol{e}_j, \boldsymbol{e}_i \rangle = \langle \boldsymbol{e}_j, \overline{\boldsymbol{A} \boldsymbol{e}_i} \rangle = \boldsymbol{e}_i^T \boldsymbol{A}^T \boldsymbol{e}_j.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0302	124	■1.000 ● N -1	$\boldsymbol{H}:=\boldsymbol{I}-\boldsymbol{u}\boldsymbol{u}^*, \text{ where } \boldsymbol{u} \in \mathbb{C}^n \text{ and } \boldsymbol{u}^*\boldsymbol{u} = 2$	$\boldsymbol{H} := \boldsymbol{I} - \boldsymbol{u}\boldsymbol{u}^*, \text{ where } \boldsymbol{u} \in \mathbb{C}^n \text{ and } \boldsymbol{u}^*\boldsymbol{u} = 2$	$\boldsymbol{H} := \boldsymbol{I} - \boldsymbol{u}\boldsymbol{u}^*, \text{ where } \boldsymbol{u} \in \mathbb{C}^n \text{ and } \boldsymbol{u}^*\boldsymbol{u} = 2$
lyche-numerical-linear-algebra-EQ0304	124	■1.000 ● N +0	$\boldsymbol{H}^*\boldsymbol{H}=\boldsymbol{H}^2=\left(\boldsymbol{I}-\boldsymbol{u}\boldsymbol{u}^*\right)\left(\boldsymbol{I}-\boldsymbol{u}\boldsymbol{u}^*\right)=\boldsymbol{I}-2\boldsymbol{u}\boldsymbol{u}^*+\boldsymbol{u}\left(\boldsymbol{u}^*\boldsymbol{u}\right)\boldsymbol{u}^*=\boldsymbol{I}.$	$\boldsymbol{H}^*\boldsymbol{H} = \boldsymbol{H}^2 = (\boldsymbol{I} - \boldsymbol{u}\boldsymbol{u}^*)(\boldsymbol{I} - \boldsymbol{u}\boldsymbol{u}^*) = \boldsymbol{I} - 2\boldsymbol{u}\boldsymbol{u}^* + \boldsymbol{u}(\boldsymbol{u}^*\boldsymbol{u})\boldsymbol{u}^* = \boldsymbol{I}.$	$\boldsymbol{H}^*\boldsymbol{H} = \boldsymbol{H}^2 = (\boldsymbol{I} - \boldsymbol{u}\boldsymbol{u}^*)(\boldsymbol{I} - \boldsymbol{u}\boldsymbol{u}^*) = \boldsymbol{I} - 2\boldsymbol{u}\boldsymbol{u}^* + \boldsymbol{u}(\boldsymbol{u}^*\boldsymbol{u})\boldsymbol{u}^* = \boldsymbol{I}.$
lyche-numerical-linear-algebra-EQ0305 (5.12)	124	■1.000 ● K +0	$\boldsymbol{H}:=\boldsymbol{I}-2\frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}}$	$\boldsymbol{H} := \boldsymbol{I} - 2\frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}}$	$\boldsymbol{H} := \boldsymbol{I} - 2\frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}}$
lyche-numerical-linear-algebra-EQ0307	125	■1.000 ● K +0	$\boldsymbol{H}=\boldsymbol{I}-\frac{2}{\boldsymbol{v}^*\boldsymbol{v}}\boldsymbol{v}\boldsymbol{v}^*=\boldsymbol{I}-\frac{2}{\boldsymbol{v}^*\boldsymbol{v}}\boldsymbol{P}, \text{ where } \boldsymbol{P}:=\boldsymbol{I}-\frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}},$	$\boldsymbol{H} = \boldsymbol{I} - \frac{2\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}} = \boldsymbol{P} - \frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}}, \text{ where } \boldsymbol{P} := \boldsymbol{I} - \frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}},$	$\boldsymbol{H} = \boldsymbol{I} - \frac{2\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}} = \boldsymbol{P} - \frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}}, \text{ where } \boldsymbol{P} := \boldsymbol{I} - \frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}},$
lyche-numerical-linear-algebra-EQ0315	128	■1.000 ● K +0	$\left[\begin{array}{cccc} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right], \quad \left[\begin{array}{cccc} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right], \quad \left[\begin{array}{ccc} x & x & x \\ 0 & x & x \\ 0 & 0 & x \end{array}\right].$	$\left[\begin{array}{cccc} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right], \quad \left[\begin{array}{cccc} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right], \quad \left[\begin{array}{ccc} x & x & x \\ 0 & x & x \\ 0 & 0 & x \end{array}\right].$	$\left[\begin{array}{cccc} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right], \quad \left[\begin{array}{cccc} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right], \quad \left[\begin{array}{ccc} x & x & x \\ 0 & x & x \\ 0 & 0 & x \end{array}\right].$
lyche-numerical-linear-algebra-EQ0316	128	■1.000 ●	$\boldsymbol{A}_{n+1}:=\boldsymbol{H}_{n+1}\boldsymbol{A}_n=\boldsymbol{H}_{n+1}\boldsymbol{H}_n\cdots\boldsymbol{H}_1\boldsymbol{A}=\left[\begin{array}{c} \boldsymbol{R}_1 \\ \boldsymbol{0} \end{array}\right]=\boldsymbol{R},$	$\boldsymbol{A}_{n+1} := \boldsymbol{H}_n\boldsymbol{H}_{n-1}\cdots\boldsymbol{H}_1\boldsymbol{A} = \left[\begin{array}{c} \boldsymbol{R}_1 \\ \boldsymbol{0} \end{array}\right] = \boldsymbol{R},$	$\boldsymbol{A}_{n+1} := \boldsymbol{H}_n\boldsymbol{H}_{n-1}\cdots\boldsymbol{H}_1\boldsymbol{A} = \left[\begin{array}{c} \boldsymbol{R}_1 \\ \boldsymbol{0} \end{array}\right] = \boldsymbol{R},$
lyche-numerical-linear-algebra-EQ0317	128	■1.000 ● N +0	$\boldsymbol{A}_{k+1}:=\boldsymbol{H}_{k+1}\boldsymbol{A}_k=\boldsymbol{H}_{k+1}\boldsymbol{H}_k\cdots\boldsymbol{H}_1\boldsymbol{A}=\left[\begin{array}{c} \boldsymbol{R}_1 \\ \boldsymbol{0} \end{array}\right]=\boldsymbol{R},$	$\boldsymbol{A}_1 := \boldsymbol{A}, \quad \boldsymbol{A}_{k+1} = \boldsymbol{H}_k\boldsymbol{A}_k, \quad k = 1, 2, \dots, n.$	$\boldsymbol{A}_1 := \boldsymbol{A}, \quad \boldsymbol{A}_{k+1} = \boldsymbol{H}_k\boldsymbol{A}_k, \quad k = 1, 2, \dots, n.$
lyche-numerical-linear-algebra-EQ0319	129	■1.000 ● N +0	$\boldsymbol{A}_{k+1}:=\boldsymbol{H}_{k+1}\boldsymbol{A}_k=\boldsymbol{H}_{k+1}\boldsymbol{H}_k\cdots\boldsymbol{H}_1\boldsymbol{A}=\left[\begin{array}{c} \boldsymbol{R}_1 \\ \boldsymbol{0} \end{array}\right]=\boldsymbol{R},$	$\boldsymbol{A}_{k+1} := \boldsymbol{H}_k\boldsymbol{A}_k = \left[\begin{array}{cc} \boldsymbol{B}_k & \boldsymbol{C}_k \\ \boldsymbol{0} & \hat{\boldsymbol{H}}_k\boldsymbol{D}_k \end{array}\right] = \left[\begin{array}{cc} \boldsymbol{B}_{k+1} & \boldsymbol{C}_{k+1} \\ \boldsymbol{0} & \boldsymbol{D}_{k+1} \end{array}\right],$	$\boldsymbol{A}_{k+1} := \boldsymbol{H}_k\boldsymbol{A}_k = \left[\begin{array}{cc} \boldsymbol{B}_k & \boldsymbol{C}_k \\ \boldsymbol{0} & \hat{\boldsymbol{H}}_k\boldsymbol{D}_k \end{array}\right] = \left[\begin{array}{cc} \boldsymbol{B}_{k+1} & \boldsymbol{C}_{k+1} \\ \boldsymbol{0} & \boldsymbol{D}_{k+1} \end{array}\right],$
lyche-numerical-linear-algebra-EQ0322	131	■1.000 ● N +0	$\boldsymbol{R}=\left[\begin{array}{c} \boldsymbol{R}_1 \\ \boldsymbol{0}_{m-n,n} \end{array}\right]$	$\boldsymbol{R} = \left[\begin{array}{c} \boldsymbol{R}_1 \\ \boldsymbol{0}_{m-n,n} \end{array}\right]$	$\boldsymbol{R} = \left[\begin{array}{c} \boldsymbol{R}_1 \\ \boldsymbol{0}_{m-n,n} \end{array}\right]$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value

LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: **basis**] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0325 (5.17)	133	■ 1.000 ● W +0	$\begin{aligned} \boldsymbol{v}_{\{1\}} &= \boldsymbol{a}_{\{1\}}, \quad \boldsymbol{v}_{\{j\}} = \boldsymbol{a}_{\{j\}} - \sum_{i=1}^{j-1} \frac{\boldsymbol{a}_{\{j\}}^T \boldsymbol{v}_{\{i\}}}{\boldsymbol{v}_{\{i\}}^T \boldsymbol{v}_{\{i\}}} \boldsymbol{v}_{\{i\}}, \quad \text{for } j=2, \ldots, n. \end{aligned}$	$\boldsymbol{v}_1 = \boldsymbol{a}_1, \quad \boldsymbol{v}_j = \boldsymbol{a}_j - \sum_{i=1}^{j-1} \frac{\boldsymbol{a}_j^T \boldsymbol{v}_i}{\boldsymbol{v}_i^T \boldsymbol{v}_i} \boldsymbol{v}_i, \quad \text{for } j = 2, \ldots, n.$	$\boldsymbol{v}_1 = \boldsymbol{a}_1, \quad \boldsymbol{v}_j = \boldsymbol{a}_j - \sum_{i=1}^{j-1} \frac{\boldsymbol{a}_j^T \boldsymbol{v}_i}{\boldsymbol{v}_i^T \boldsymbol{v}_i} \boldsymbol{v}_i, \quad \text{for } j = 2, \ldots, n.$
lyche-numerical-linear-algebra EQ0327	133	■ 1.000 ● N +0	$\begin{aligned} \boldsymbol{a}_{\{j\}} &= \boldsymbol{v}_{\{j\}} + \sum_{i=1}^{j-1} \frac{\boldsymbol{a}_{\{j\}}^T \boldsymbol{v}_{\{i\}}}{\boldsymbol{v}_{\{i\}}^T \boldsymbol{v}_{\{i\}}} \boldsymbol{v}_{\{i\}} \\ &= r_{jj} \boldsymbol{q}_j + \sum_{i=1}^{j-1} \boldsymbol{q}_i r_{ij} = \boldsymbol{Q}_1 \boldsymbol{R}_1 \boldsymbol{e}_j, \quad j = 1, \ldots, n. \end{aligned}$	$\boldsymbol{a}_j = \boldsymbol{v}_j + \sum_{i=1}^{j-1} \frac{\boldsymbol{a}_j^T \boldsymbol{v}_i}{\boldsymbol{v}_i^T \boldsymbol{v}_i} \boldsymbol{v}_i = r_{jj} \boldsymbol{q}_j + \sum_{i=1}^{j-1} \boldsymbol{q}_i r_{ij} = \boldsymbol{Q}_1 \boldsymbol{R}_1 \boldsymbol{e}_j, j = 1, \ldots, n.$	$\boldsymbol{a}_j = \boldsymbol{v}_j + \sum_{i=1}^{j-1} \frac{\boldsymbol{a}_j^T \boldsymbol{v}_i}{\boldsymbol{v}_i^T \boldsymbol{v}_i} \boldsymbol{v}_i = r_{jj} \boldsymbol{q}_j + \sum_{i=1}^{j-1} \boldsymbol{q}_i r_{ij} = \boldsymbol{Q}_1 \boldsymbol{R}_1 \boldsymbol{e}_j, \quad j = 1, \ldots, n.$
lyche-numerical-linear-algebra EQ0333	135	■ 1.000 ● N +0	$\begin{bmatrix} p_{ii} & p_{ij} \\ p_{ji} & p_{jj} \end{bmatrix} = \begin{bmatrix} c & s \\ -s & c \end{bmatrix}, \quad \text{where } c^2 + s^2 = 1.$	$\begin{bmatrix} p_{ii} & p_{ij} \\ p_{ji} & p_{jj} \end{bmatrix} = \begin{bmatrix} c & s \\ -s & c \end{bmatrix}, \quad \text{where } c^2 + s^2 = 1.$	$\begin{bmatrix} p_{ii} & p_{ij} \\ p_{ji} & p_{jj} \end{bmatrix} = \begin{bmatrix} c & s \\ -s & c \end{bmatrix}, \quad \text{where } c^2 + s^2 = 1.$
lyche-numerical-linear-algebra EQ0334	135	■ 1.000 ● C -6	$\begin{aligned} \boldsymbol{P}_{\{12\}} &= \begin{bmatrix} c & s & 0 & 0 \\ -s & c & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{P}_{\{13\}} = \begin{bmatrix} c & 0 & s & 0 \\ 0 & 1 & 0 & 0 \\ -s & 0 & c & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{P}_{\{23\}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & s & c & 0 \\ 0 & -s & c & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \end{aligned}$	$\boldsymbol{P}_{12} = \begin{bmatrix} c & s & 0 & 0 \\ -s & c & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{P}_{13} = \begin{bmatrix} c & 0 & s & 0 \\ 0 & 1 & 0 & 0 \\ -s & 0 & c & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{P}_{23} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & s & c & 0 \\ 0 & -s & c & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$	$\boldsymbol{P}_{12} = \begin{bmatrix} c & s & 0 & 0 \\ -s & c & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{P}_{13} = \begin{bmatrix} c & 0 & s & 0 \\ 0 & 1 & 0 & 0 \\ -s & 0 & c & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{P}_{23} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & s & c & 0 \\ 0 & -s & c & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$
lyche-numerical-linear-algebra EQ0337 (5.20)	136	■ 1.000 ● N +0	$\cos \theta = \frac{ \langle \boldsymbol{x}, \boldsymbol{y} \rangle }{\ \boldsymbol{x}\ \ \boldsymbol{y}\ }.$	$\cos \theta = \frac{ \langle \boldsymbol{x}, \boldsymbol{y} \rangle }{\ \boldsymbol{x}\ \ \boldsymbol{y}\ }.$	$\cos \theta = \frac{ \langle \boldsymbol{x}, \boldsymbol{y} \rangle }{\ \boldsymbol{x}\ \ \boldsymbol{y}\ }.$
lyche-numerical-linear-algebra EQ0339	137	■ 1.000 ●	$\boldsymbol{H} = \begin{bmatrix} -\cos \phi & \sin \phi \\ \sin \phi & \cos \phi \end{bmatrix}.$	$\boldsymbol{H} = \begin{bmatrix} -\cos \phi & \sin \phi \\ \sin \phi & \cos \phi \end{bmatrix}.$	$\boldsymbol{H} = \begin{bmatrix} -\cos \phi & \sin \phi \\ \sin \phi & \cos \phi \end{bmatrix}.$
lyche-numerical-linear-algebra EQ0340	137	■ 1.000 ● N +1	$\boldsymbol{H} \boldsymbol{x} = \boldsymbol{y}, \quad \text{where } \boldsymbol{H} := \boldsymbol{I} - 2 \frac{\boldsymbol{v} \boldsymbol{v}^T}{\boldsymbol{v}^T \boldsymbol{v}}.$	$\boldsymbol{H} \boldsymbol{x} = \boldsymbol{y}, \quad \text{where } \boldsymbol{H} := \boldsymbol{I} - 2 \frac{\boldsymbol{v} \boldsymbol{v}^T}{\boldsymbol{v}^T \boldsymbol{v}}.$	$\boldsymbol{H} \boldsymbol{x} = \boldsymbol{y}, \quad \text{where } \boldsymbol{H} := \boldsymbol{I} - 2 \frac{\boldsymbol{v} \boldsymbol{v}^T}{\boldsymbol{v}^T \boldsymbol{v}}.$
lyche-numerical-linear-algebra EQ0343	137	■ 1.000 ● K +0	$\boldsymbol{A} := \begin{bmatrix} 1 & 0 & 1 \\ -2 & -1 & 0 \\ 2 & 2 & 1 \end{bmatrix}.$	$\boldsymbol{A} := \begin{bmatrix} 1 & 0 & 1 \\ -2 & -1 & 0 \\ 2 & 2 & 1 \end{bmatrix}.$	$\boldsymbol{A} := \begin{bmatrix} 1 & 0 & 1 \\ -2 & -1 & 0 \\ 2 & 2 & 1 \end{bmatrix}.$
lyche-numerical-linear-algebra EQ0344 (5.21)	138	■ 1.000 ● N +0	$ \operatorname{det}(\boldsymbol{A}) \leq \prod_{j=1}^n \ \boldsymbol{a}_j\ _2.$	$ \det(\boldsymbol{A}) \leq \prod_{j=1}^n \ \boldsymbol{a}_j\ _2.$	$ \det(\boldsymbol{A}) \leq \prod_{j=1}^n \ \boldsymbol{a}_j\ _2.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0346	139	1.000 K +0	$A:=\left[\begin{array}{rr} 2 & 1 \\ 2 & -3 \\ -2 & -1 \\ -2 & 3 \end{array}\right]$	$A:=\left[\begin{array}{cc} 2 & 1 \\ 2 & -3 \\ -2 & -1 \\ -2 & 3 \end{array}\right]$	$A:=\left[\begin{array}{cc} 2 & 1 \\ 2 & -3 \\ -2 & -1 \\ -2 & 3 \end{array}\right]$
lyche-numerical-linear-algebra- EQ0347	140	1.000 K +0	$\boldsymbol{G}_3\boldsymbol{G}_2\boldsymbol{G}_1\boldsymbol{H}=\boldsymbol{R}$	$\boldsymbol{G}_3\boldsymbol{G}_2\boldsymbol{G}_1\boldsymbol{H}=\boldsymbol{R}$	$\boldsymbol{G}_3\boldsymbol{G}_2\boldsymbol{G}_1\boldsymbol{H}=\boldsymbol{R}$
lyche-numerical-linear-algebra- EQ0349	141	1.000 N +0	$\boldsymbol{P}=\left[\begin{array}{cc} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{array}\right]$	$\boldsymbol{P}=\left[\begin{array}{cc} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{array}\right]$	$\boldsymbol{P}=\left[\begin{array}{cc} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{array}\right]$
lyche-numerical-linear-algebra- EQ0351	141	1.000 K +0	$\left(a_{1,1}, a_{2,2}, \ldots, a_{n,n}\right)$	$\left(a_{1,1}, a_{2,2}, \ldots, a_{n,n}\right)$	$\left(a_{1,1}, a_{2,2}, \ldots, a_{n,n}\right)$
lyche-numerical-linear-algebra- EQ0352	141	1.000 N +0	$\left(a_{2,1}, a_{3,2}, \ldots, a_{n,n-1}\right)$.	$\left(a_{2,1}, a_{3,2}, \ldots, a_{n,n-1}\right)$.	$\left(a_{2,1}, a_{3,2}, \ldots, a_{n,n-1}\right)$.
lyche-numerical-linear-algebra- EQ0353	142	1.000 N +0	$\boldsymbol{P}_{m-1, m} \boldsymbol{P}_{m-2, m-1} \cdots \boldsymbol{P}_{2,3} \boldsymbol{P}_{1,2} \boldsymbol{H}$	$\boldsymbol{P}_{m-1, m} \boldsymbol{P}_{m-2, m-1} \cdots \boldsymbol{P}_{2,3} \boldsymbol{P}_{1,2} \boldsymbol{H}$	$\boldsymbol{P}_{m-1, m} \boldsymbol{P}_{m-2, m-1} \cdots \boldsymbol{P}_{2,3} \boldsymbol{P}_{1,2} \boldsymbol{H}$
lyche-numerical-linear-algebra- EQ0356	145	1.000 K +0	$\pi_{\boldsymbol{A}}(\lambda)=\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})$.	$\pi_{\boldsymbol{A}}(\lambda)=\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})$.	$\pi_{\boldsymbol{A}}(\lambda)=\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})$.
lyche-numerical-linear-algebra- EQ0357	146	1.000 N +0	$\sum_{j=1}^m c_j \boldsymbol{x}_j=\boldsymbol{0} \Rightarrow \sum_{j=1}^m c_j \boldsymbol{A} \boldsymbol{x}_j=\sum_{j=1}^m c_j \lambda_j \boldsymbol{x}_j=\boldsymbol{0}$.	$\sum_{j=1}^m c_j \boldsymbol{x}_j=\boldsymbol{0} \Rightarrow \sum_{j=1}^m c_j \boldsymbol{A} \boldsymbol{x}_j=\sum_{j=1}^m c_j \lambda_j \boldsymbol{x}_j=\boldsymbol{0}$.	$\sum_{j=1}^m c_j \boldsymbol{x}_j=\boldsymbol{0} \Rightarrow \sum_{j=1}^m c_j \boldsymbol{A} \boldsymbol{x}_j=\sum_{j=1}^m c_j \lambda_j \boldsymbol{x}_j=\boldsymbol{0}$.
lyche-numerical-linear-algebra- EQ0358	146	1.000 K +0	$\boldsymbol{I}:=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right], \quad \boldsymbol{J}:=\left[\begin{array}{ll} 1 & 1 \\ 0 & 1 \end{array}\right]$.	$\boldsymbol{I}:=\left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array}\right], \quad \boldsymbol{J}:=\left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \end{array}\right]$.	$\boldsymbol{I}:=\left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array}\right], \quad \boldsymbol{J}:=\left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \end{array}\right]$.
lyche-numerical-linear-algebra- EQ0359	146	1.000 N +0	$\boldsymbol{x}=\sum_{j=1}^n c_j \boldsymbol{x}_j$ for some scalars $c_1, \ldots, c_n$	$\boldsymbol{x}=\sum_{j=1}^n c_j \boldsymbol{x}_j$ for some scalars $c_1, \ldots, c_n$	$\boldsymbol{x}=\sum_{j=1}^n c_j \boldsymbol{x}_j$ for some scalars $c_1, \ldots, c_n$.
lyche-numerical-linear-algebra- EQ0360	146	1.000 K +0	$\boldsymbol{x}=\frac{x_1+x_2}{2}\left[\begin{array}{l} 1 \\ 1 \end{array}\right]+\frac{x_1-x_2}{2}\left[\begin{array}{c} 1 \\ -1 \end{array}\right]$.	$\boldsymbol{x}=\frac{x_1+x_2}{2}\left[\begin{array}{l} 1 \\ 1 \end{array}\right]+\frac{x_1-x_2}{2}\left[\begin{array}{l} 1 \\ -1 \end{array}\right]$.	$\boldsymbol{x}=\frac{x_1+x_2}{2}\left[\begin{array}{l} 1 \\ 1 \end{array}\right]+\frac{x_1-x_2}{2}\left[\begin{array}{l} 1 \\ -1 \end{array}\right]$.
lyche-numerical-linear-algebra- EQ0365 (6.3)	148	1.000	$\mathcal{N}(\boldsymbol{A}-\lambda \boldsymbol{I}):=\left\{\boldsymbol{x} \in \mathbb{C}^n: \left(\boldsymbol{A}-\lambda \boldsymbol{I}\right) \boldsymbol{x}=\boldsymbol{0}\right\}$	$\mathcal{N}(\boldsymbol{A}-\lambda \boldsymbol{I}):=\left\{\boldsymbol{x} \in \mathbb{C}^n: \left(\boldsymbol{A}-\lambda \boldsymbol{I}\right) \boldsymbol{x}=\boldsymbol{0}\right\}$	$\mathcal{N}(\boldsymbol{A}-\lambda \boldsymbol{I}):=\left\{\boldsymbol{x} \in \mathbb{C}^n: \left(\boldsymbol{A}-\lambda \boldsymbol{I}\right) \boldsymbol{x}=\boldsymbol{0}\right\}$
lyche-numerical-linear-algebra- EQ0368 (6.6)	150	1.000 W +0	$\boldsymbol{U}_i=\operatorname{diag}\left(\boldsymbol{J}_{m_i, 1}\left(\lambda_i\right), \ldots, \boldsymbol{J}_{m_i, g_i}\left(\lambda_i\right)\right)$.	$\boldsymbol{U}_i=\operatorname{diag}\left(\boldsymbol{J}_{m_i, 1}\left(\lambda_i\right), \ldots, \boldsymbol{J}_{m_i, g_i}\left(\lambda_i\right)\right)$.	$\boldsymbol{U}_i=\operatorname{diag}\left(\boldsymbol{J}_{m_i, 1}\left(\lambda_i\right), \ldots, \boldsymbol{J}_{m_i, g_i}\left(\lambda_i\right)\right)$.
lyche-numerical-linear-algebra- EQ0372	152	1.000 N +0	$\boldsymbol{V}^* \boldsymbol{A} \boldsymbol{V} \boldsymbol{e}_1=\boldsymbol{V}^* \boldsymbol{A} \boldsymbol{v}_1=\lambda_1 \boldsymbol{V}^* \boldsymbol{v}_1=\lambda_1 \boldsymbol{e}_1$.	$\boldsymbol{V}^* \boldsymbol{A} \boldsymbol{V} \boldsymbol{e}_1=\boldsymbol{V}^* \boldsymbol{A} \boldsymbol{v}_1=\lambda_1 \boldsymbol{V}^* \boldsymbol{v}_1=\lambda_1 \boldsymbol{e}_1$.	$\boldsymbol{V}^* \boldsymbol{A} \boldsymbol{V} \boldsymbol{e}_1=\boldsymbol{V}^* \boldsymbol{A} \boldsymbol{v}_1=\lambda_1 \boldsymbol{V}^* \boldsymbol{v}_1=\lambda_1 \boldsymbol{e}_1$.
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-alignatE00376	153	■1.000 ● N +0	$\boldsymbol{V}=\left[\begin{array}{ccc} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{array}\right]$.	$\boldsymbol{V} = \begin{bmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}.$	
lyche-numerical-linear-alignatE00377	153	■1.000 ● N +0	$\boldsymbol{M}=\left[\begin{array}{cc} 2 & -\sqrt{2} \\ -\sqrt{2} & 2 \end{array}\right]$.	$\boldsymbol{M} = \begin{bmatrix} 2 & -\sqrt{2} \\ -\sqrt{2} & 2 \end{bmatrix}.$	
lyche-numerical-linear-alignatE00378	153	■1.000 ● N +0	$\boldsymbol{A}^*\boldsymbol{A}=\operatorname{diag}(\overline{d_1},\ldots,\overline{d_n})=\operatorname{diag}(d_1 ^2,\ldots, d_n ^2)=\boldsymbol{A}\boldsymbol{A}^*$,	$\boldsymbol{A}^*\boldsymbol{A}=\operatorname{diag}(\overline{d_1}d_1,\ldots,\overline{d_n}d_n)=\operatorname{diag}(d_1 ^2,\ldots, d_n ^2)=\boldsymbol{A}\boldsymbol{A}^*,$	
lyche-numerical-linear-alignatE00382	154	■1.000 ● U +0	$e_{ii}=\sum_{k=1}^i b_{ki} ^2= b_{ii} ^2=\sum_{k=i}^n b_{ik} ^2=f_{ii}$	$e_{ii} = \sum_{k=1}^i b_{ki} ^2 = b_{ii} ^2 = \sum_{k=i}^n b_{ik} ^2 = f_{ii}$	
lyche-numerical-linear-alignatE00385 (6.10)	155	■1.000 ● K +0	$\boldsymbol{M}=\left[\begin{array}{cc} \mu & v \\ -v & \mu \end{array}\right]$	$\boldsymbol{M} = \begin{bmatrix} \mu & v \\ -v & \mu \end{bmatrix}$	
lyche-numerical-linear-alignatE00388	156	■1.000 ● K +0	$\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U}=\operatorname{diag}(\lambda_1,\ldots,\lambda_n)$.	$\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U}=\operatorname{diag}(\lambda_1,\ldots,\lambda_n).$	
lyche-numerical-linear-alignatE00389	156	■1.000 ● N +0	$\boldsymbol{U}^T\boldsymbol{A}\boldsymbol{U}=\operatorname{diag}(\lambda_1,\lambda_2,\ldots,\lambda_n)$.	$\boldsymbol{U}^T\boldsymbol{A}\boldsymbol{U}=\operatorname{diag}(\lambda_1,\lambda_2,\ldots,\lambda_n).$	
lyche-numerical-linear-alignatE00391	157	■1.000 ● N +0	$\operatorname{dim}(S\cap S')=\operatorname{dim}(S)+\operatorname{dim}(S')-\operatorname{dim}(S+S')\geq(n-k+1)+k-n=1$.	$\operatorname{dim}(S\cap S')=\operatorname{dim}(S)+\operatorname{dim}(S')-\operatorname{dim}(S+S')\geq(n-k+1)+k-n=1.$	
lyche-numerical-linear-alignatE00395 (6.14)	158	■1.000 ● N +0	$\alpha_k+\varepsilon_n\leq\beta_k\leq\alpha_k+\varepsilon_1$, for $k=1,\ldots,n$,	$\alpha_k+\varepsilon_n\leq\beta_k\leq\alpha_k+\varepsilon_1, \text{ for } k=1,\ldots,n,$	
lyche-numerical-linear-alignatE00398 (6.16)	159	■1.000 ● N +1	$\boldsymbol{v}=\sum_{j=1}^n\frac{\boldsymbol{y}_j^*\boldsymbol{v}}{\boldsymbol{y}_j^*\boldsymbol{x}_j}\boldsymbol{x}_j=\sum_{k=1}^n\frac{\boldsymbol{x}_k^*\boldsymbol{v}}{\boldsymbol{y}_k^*\boldsymbol{x}_k}\boldsymbol{y}_k$, $\boldsymbol{v}\in\mathbb{C}^n$	$\boldsymbol{v} = \sum_{j=1}^n \frac{\boldsymbol{y}_j^*\boldsymbol{v}}{\boldsymbol{y}_j^*\boldsymbol{x}_j} \boldsymbol{x}_j = \sum_{k=1}^n \frac{\boldsymbol{x}_k^*\boldsymbol{v}}{\boldsymbol{y}_k^*\boldsymbol{x}_k} \boldsymbol{y}_k, \quad \boldsymbol{v} \in \mathbb{C}^n.$	
lyche-numerical-linear-alignatE00399	160	■1.000 ●	$\boldsymbol{X}:=\left[\boldsymbol{x}_1,\ldots,\boldsymbol{x}_n\right]$, $\boldsymbol{Y}:=\left[\boldsymbol{y}_1,\ldots,\boldsymbol{y}_n\right]$, $\boldsymbol{D}:=\operatorname{diag}(\lambda_1,\ldots,\lambda_n)$.	$\boldsymbol{X} := [\boldsymbol{x}_1,\ldots,\boldsymbol{x}_n], \quad \boldsymbol{Y} := [\boldsymbol{y}_1,\ldots,\boldsymbol{y}_n], \quad \boldsymbol{D} := \operatorname{diag}(\lambda_1,\ldots,\lambda_n).$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-160 EQ0400 (6.17)	160	■ 1.000 ● N +0	$\boldsymbol{v}=\sum_{j=1}^n\frac{\boldsymbol{x}_j^*\boldsymbol{v}}{\boldsymbol{x}_j^*\boldsymbol{x}_j}\boldsymbol{x}_j$	$\boldsymbol{v}=\sum_{j=1}^n\frac{\boldsymbol{x}_j^*\boldsymbol{v}}{\boldsymbol{x}_j^*\boldsymbol{x}_j}\boldsymbol{x}_j.$	$\boldsymbol{v}=\sum_{j=1}^n\frac{\boldsymbol{x}_j^*\boldsymbol{v}}{\boldsymbol{x}_j^*\boldsymbol{x}_j}\boldsymbol{x}_j.$
lyche-numerical-linear-algebra-161 EQ0402	161	■ 1.000 ● N +0	$\left(\boldsymbol{V}^*\boldsymbol{A}^*\boldsymbol{V}\right)\boldsymbol{u}=\boldsymbol{V}^*\boldsymbol{A}^*\boldsymbol{y}=\bar{\lambda}\boldsymbol{V}^*\boldsymbol{y}=\bar{\lambda}\boldsymbol{u},$	$\left(\boldsymbol{V}^*\boldsymbol{A}^*\boldsymbol{V}\right)\boldsymbol{u}=\boldsymbol{V}^*\boldsymbol{A}^*\boldsymbol{y}=\bar{\lambda}\boldsymbol{V}^*\boldsymbol{y}=\bar{\lambda}\boldsymbol{u},$	$\left(\boldsymbol{V}^*\boldsymbol{A}^*\boldsymbol{V}\right)\boldsymbol{u}=\boldsymbol{V}^*\boldsymbol{A}^*\boldsymbol{y}=\bar{\lambda}\boldsymbol{V}^*\boldsymbol{y}=\bar{\lambda}\boldsymbol{u},$
lyche-numerical-linear-algebra-162 EQ0405 (6.18)	162	■ 1.000 ● N +0	$\boldsymbol{A}^k\boldsymbol{x}=\sum_{j=1}^nc_j\lambda_j^k\boldsymbol{x}_j$ for some scalars $c_1,\ldots,c_n$	$\boldsymbol{A}^k\boldsymbol{x}=\sum_{j=1}^nc_j\lambda_j^k\boldsymbol{x}_j$ for some scalars $c_1,\ldots,c_n$	$\boldsymbol{A}^k\boldsymbol{x}=\sum_{j=1}^nc_j\lambda_j^k\boldsymbol{x}_j$ for some scalars $c_1,\ldots,c_n$.
lyche-numerical-linear-algebra-164 EQ0411	164	■ 1.000 ● N +0	$p(\boldsymbol{A}):=\sum_{j=0}^rb_j\boldsymbol{A}^j,$	$p(\boldsymbol{A}):=\sum_{j=0}^rb_j\boldsymbol{A}^j,$	$p(\boldsymbol{A}):=\sum_{j=0}^rb_j\boldsymbol{A}^j,$
lyche-numerical-linear-algebra-168 EQ0417 (7.1)	168	■ 1.000 ● K +0	$\boldsymbol{A}=\boldsymbol{U}\boldsymbol{D}\boldsymbol{U}^*,$ where $\boldsymbol{U}^*\boldsymbol{U}=\boldsymbol{I}.$	$\boldsymbol{A}=\boldsymbol{U}\boldsymbol{D}\boldsymbol{U}^*,$ where $\boldsymbol{U}^*\boldsymbol{U}=\boldsymbol{I}.$	$\boldsymbol{A}=\boldsymbol{U}\boldsymbol{D}\boldsymbol{U}^*,$ where $\boldsymbol{U}^*\boldsymbol{U}=\boldsymbol{I}.$
lyche-numerical-linear-algebra-169 EQ0419	169	■ 1.000 ● N +0	$\lambda_1\geq\cdots\geq\lambda_r>0=\lambda_{r+1}=\cdots=\lambda_n.$	$\lambda_1\geq\cdots\geq\lambda_r>0=\lambda_{r+1}=\cdots=\lambda_n.$	$\lambda_1\geq\cdots\geq\lambda_r>0=\lambda_{r+1}=\cdots=\lambda_n.$
lyche-numerical-linear-algebra-170 EQ0421 (7.3)	170	■ 1.000 ● N +0	$\lambda=\frac{\boldsymbol{v}^*\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{v}}{\boldsymbol{v}^*\boldsymbol{v}}=\frac{\ \boldsymbol{A}\boldsymbol{v}\ _2^2}{\ \boldsymbol{v}\ _2^2}\geq0.$	$\lambda=\frac{\boldsymbol{v}^*\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{v}}{\boldsymbol{v}^*\boldsymbol{v}}=\frac{\ \boldsymbol{A}\boldsymbol{v}\ _2^2}{\ \boldsymbol{v}\ _2^2}\geq0.$	$\lambda=\frac{\boldsymbol{v}^*\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{v}}{\boldsymbol{v}^*\boldsymbol{v}}=\frac{\ \boldsymbol{A}\boldsymbol{v}\ _2^2}{\ \boldsymbol{v}\ _2^2}\geq0.$
lyche-numerical-linear-algebra-171 EQ0423	171	■ 1.000 ● K +0	$\boldsymbol{B}:=\boldsymbol{A}^T\boldsymbol{A}=\frac{1}{25}\left[\begin{array}{ll}52&36\\36&73\end{array}\right]$	$\boldsymbol{B}:=\boldsymbol{A}^T\boldsymbol{A}=\frac{1}{25}\left[\begin{array}{ll}52&36\\36&73\end{array}\right]$	$\boldsymbol{B}:=\boldsymbol{A}^T\boldsymbol{A}=\frac{1}{25}\left[\begin{array}{ll}52&36\\36&73\end{array}\right]$
lyche-numerical-linear-algebra-171 EQ0424	171	■ 1.000 ● N +0	$\boldsymbol{B}\left[\begin{array}{l}3\\4\end{array}\right]=4\left[\begin{array}{l}3\\4\end{array}\right],\quad\boldsymbol{B}\left[\begin{array}{l}4\\-3\end{array}\right]=1\left[\begin{array}{l}4\\-3\end{array}\right].$	$\boldsymbol{B}\left[\begin{array}{l}3\\4\end{array}\right]=4\left[\begin{array}{l}3\\4\end{array}\right],\quad\boldsymbol{B}\left[\begin{array}{l}4\\-3\end{array}\right]=1\left[\begin{array}{l}4\\-3\end{array}\right].$	$\boldsymbol{B}\left[\begin{array}{l}3\\4\end{array}\right]=4\left[\begin{array}{l}3\\4\end{array}\right],\quad\boldsymbol{B}\left[\begin{array}{l}4\\-3\end{array}\right]=1\left[\begin{array}{l}4\\-3\end{array}\right].$
lyche-numerical-linear-algebra-172 EQ0428 (7.6)	172	■ 1.000 ● N +0	$\boldsymbol{A}=\sum_{j=1}^r\sigma_j\boldsymbol{u}_j\boldsymbol{v}_j^*.$	$\boldsymbol{A}=\sum_{j=1}^r\sigma_j\boldsymbol{u}_j\boldsymbol{v}_j^*.$	$\boldsymbol{A}=\sum_{j=1}^r\sigma_j\boldsymbol{u}_j\boldsymbol{v}_j^*.$
lyche-numerical-linear-algebra-172 EQ0430	172	■ 1.000 ● K +0	$\boldsymbol{B}:=\boldsymbol{A}^T\boldsymbol{A}=\left[\begin{array}{ll}2&2\\2&2\end{array}\right]$	$\boldsymbol{B}:=\boldsymbol{A}^T\boldsymbol{A}=\left[\begin{array}{ll}2&2\\2&2\end{array}\right]$	$\boldsymbol{B}:=\boldsymbol{A}^T\boldsymbol{A}=\left[\begin{array}{ll}2&2\\2&2\end{array}\right]$
lyche-numerical-linear-algebra-173 EQ0433	173	■ 1.000 ● K +0	$\boldsymbol{A}=\frac{1}{\sqrt{2}}\left[\begin{array}{l}1\\1\\0\end{array}\right][2]\frac{1}{\sqrt{2}}\left[\begin{array}{ll}1&1\end{array}\right].$	$\boldsymbol{A}=\frac{1}{\sqrt{2}}\left[\begin{array}{l}1\\1\\0\end{array}\right][2]\frac{1}{\sqrt{2}}\left[\begin{array}{ll}1&1\end{array}\right].$	$\boldsymbol{A}=\frac{1}{\sqrt{2}}\left[\begin{array}{l}1\\1\\0\end{array}\right][2]\frac{1}{\sqrt{2}}\left[\begin{array}{ll}1&1\end{array}\right].$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0434	173	■ 1.000	$\boldsymbol{s}_1=\left[\begin{array}{l} 1 \\ 1 \\ 0 \end{array}\right], \quad \boldsymbol{s}_2=\left[\begin{array}{l} 0 \\ 1 \\ 0 \end{array}\right], \quad \boldsymbol{s}_3=\left[\begin{array}{l} 0 \\ 0 \\ 1 \end{array}\right],$	$s_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad s_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad s_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$	
lyche-numerical-linear-algebra-EQ0437	174	■ 1.000 ● W +0	$\mathcal{E}=\boldsymbol{U}_1\tilde{\mathcal{E}} \text{ where } \tilde{\mathcal{E}}:=\left\{\boldsymbol{y}=\left[y_1,\ldots,y_n\right]^T\in\mathbb{R}^n:\frac{y_1^2}{\sigma_1^2}+\cdots+\frac{y_n^2}{\sigma_n^2}=1\right\}.$	$\mathcal{E} = \boldsymbol{U}_1 \tilde{\mathcal{E}} \text{ where } \tilde{\mathcal{E}} := \left\{ \boldsymbol{y} = [y_1, \dots, y_n]^T \in \mathbb{R}^n : \frac{y_1^2}{\sigma_1^2} + \dots + \frac{y_n^2}{\sigma_n^2} = 1 \right\}.$	
lyche-numerical-linear-algebra-EQ0441 (7.9)	176	■ 1.000 ● K +0	$\ \boldsymbol{A}\ _F:=\left(\sum_{i=1}^m\sum_{j=1}^n\left a_{ij}\right ^2\right)^{1/2}.$	$\ A\ _F := \left(\sum_{i=1}^m \sum_{j=1}^n a_{ij} ^2 \right)^{1/2}.$	
lyche-numerical-linear-algebra-EQ0444	177	■ 1.000 ● N +0	$\left\ \boldsymbol{A}-\boldsymbol{A}'\right\ _F=\left\ \boldsymbol{U}\left[\begin{array}{c} \boldsymbol{D}-\boldsymbol{D}' \\ \boldsymbol{0} \end{array}\right]\boldsymbol{V}^*\right\ _F=\left\ \left[\begin{array}{c} \boldsymbol{D}-\boldsymbol{D}' \\ \boldsymbol{0} \end{array}\right]\right\ _F=\sqrt{\sigma_{r+1}^2+\cdots+\sigma_n^2}<\epsilon.$	$\ A-A'\ _F = \ U \begin{bmatrix} \boldsymbol{D}-\boldsymbol{D}' \\ \boldsymbol{0} \end{bmatrix} V^*\ _F = \left\ \begin{bmatrix} \boldsymbol{D}-\boldsymbol{D}' \\ \boldsymbol{0} \end{bmatrix} \right\ _F = \sqrt{\sigma_{r+1}^2 + \dots + \sigma_n^2} < \epsilon.$	
lyche-numerical-linear-algebra-EQ0449	180	■ 1.000 ● U +0	$\boldsymbol{C}:=\left[\begin{array}{cc} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \mathbf{0} \end{array}\right] \in \mathbb{R}^{(m+n) \times(m+n)}$	$\boldsymbol{C}:=\left[\begin{array}{cc} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \mathbf{0} \end{array}\right] \in \mathbb{R}^{(m+n) \times(m+n)}$	
lyche-numerical-linear-algebra-EQ0450	180	■ 1.000 ● N +0	$\left\{\left(\sigma_1, \boldsymbol{p}_1\right), \ldots,\left(\sigma_n, \boldsymbol{p}_n\right),\left(-\sigma_1, \boldsymbol{q}_1\right), \ldots,\left(-\sigma_n, \boldsymbol{q}_n\right),\left(0, \boldsymbol{r}_{n+1}\right), \ldots,\left(0, \boldsymbol{r}_m\right)\right\},$	$\{(\sigma_1, \boldsymbol{p}_1), \dots, (\sigma_n, \boldsymbol{p}_n), (-\sigma_1, \boldsymbol{q}_1), \dots, (-\sigma_n, \boldsymbol{q}_n), (0, \boldsymbol{r}_{n+1}), \dots, (0, \boldsymbol{r}_m)\},$	
lyche-numerical-linear-algebra-EQ0452 (7.12)	180	■ 1.000 ● N +0	$\boldsymbol{Q}:=\boldsymbol{U} \boldsymbol{V}^T, \quad \boldsymbol{P}:=\boldsymbol{V} \boldsymbol{\Sigma} \boldsymbol{V}^T$	$\boldsymbol{Q}:=\boldsymbol{U} \boldsymbol{V}^T, \quad \boldsymbol{P}:=\boldsymbol{V} \boldsymbol{\Sigma} \boldsymbol{V}^T$	
lyche-numerical-linear-algebra-EQ0453 (7.13)	180	■ 1.000 ● N +0	$\boldsymbol{A}=\boldsymbol{Q} \boldsymbol{P}$	$\boldsymbol{A}=\boldsymbol{Q} \boldsymbol{P}$	
lyche-numerical-linear-algebra-EQ0455 (7.15)	181	■ 1.000 ● N +0	$\boldsymbol{X}_{k+1}-\boldsymbol{Q}=\frac{1}{2} \boldsymbol{X}_k^{-T}\left(\boldsymbol{X}_k^T-\boldsymbol{Q}^T\right)\left(\boldsymbol{X}_k-\boldsymbol{Q}\right)$	$\boldsymbol{X}_{k+1}-\boldsymbol{Q}=\frac{1}{2} \boldsymbol{X}_k^{-T}\left(\boldsymbol{X}_k^T-\boldsymbol{Q}^T\right)\left(\boldsymbol{X}_k-\boldsymbol{Q}\right)$	
lyche-numerical-linear-algebra-EQ0456 (7.16)	181	■ 1.000 ● N +0	$\left\ \boldsymbol{X}_{k+1}-\boldsymbol{Q}\right\ _F \leq \frac{1}{2}\left\ \boldsymbol{X}_k^{-1}\right\ _F\left\ \boldsymbol{X}_k-\boldsymbol{Q}\right\ _F^2.$	$\ \boldsymbol{X}_{k+1}-\boldsymbol{Q}\ _F \leq \frac{1}{2}\ \boldsymbol{X}_k^{-1}\ _F\ \boldsymbol{X}_k-\boldsymbol{Q}\ _F^2.$	
lyche-numerical-linear-algebra-EQ0458	181	■ 1.000 ● K +0	$\boldsymbol{B}=\left[\begin{array}{ccc} 1 & 0 & -1 \\ 1 & 1 & 1 \end{array}\right].$	$\boldsymbol{B}=\left[\begin{array}{ccc} 1 & 0 & -1 \\ 1 & 1 & 1 \end{array}\right].$	

Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-alignatEQ0461	182	1.000	$\boldsymbol{A}=\left[\begin{array}{rrrr} 1 & -1 & -1 & -1 \\ \theta & 1 & -1 & -1 \\ \theta & \theta & 1 & -1 \\ \theta & \theta & \theta & 1 \end{array}\right]$.	$\boldsymbol{A} = \begin{bmatrix} 1 & -1 & -1 & -1 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$	$\boldsymbol{A} = \begin{bmatrix} 1 & -1 & -1 & -1 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$
lyche-numerical-linear-alignatEQ0464	183	1.000 N +0	$\left \boldsymbol{A}-\boldsymbol{A}'\right _{F}^{2}=\sigma_{r+1}^{2}+\cdots+\sigma_{n}^{2}$.	$\ \boldsymbol{A}-\boldsymbol{A}'\ _F^2 = \sigma_{r+1}^2 + \cdots + \sigma_n^2.$	$\ \boldsymbol{A}-\boldsymbol{A}'\ _F^2 = \sigma_{r+1}^2 + \cdots + \sigma_n^2.$
lyche-numerical-linear-alignatEQ0472	187	1.000 N +0	$\mathcal{S}:=\left\{\boldsymbol{y} \in \mathcal{V}: \boldsymbol{y}'=1\right\}$.	$\mathcal{S}:=\{\boldsymbol{y} \in \mathcal{V}: \ \boldsymbol{y}\ '=1\}.$	$\mathcal{S}:=\{\boldsymbol{y} \in \mathcal{V}: \ \boldsymbol{y}\ '=1\}.$
lyche-numerical-linear-alignatEQ0473 (8.8)	187	1.000 N +0	$m \leq \boldsymbol{y} \leq M, \quad \boldsymbol{y} \in \mathcal{S}.$	$m \leq \ \boldsymbol{y}\ \leq M, \quad \boldsymbol{y} \in \mathcal{S}.$	$m \leq \ \boldsymbol{y}\ \leq M, \quad \boldsymbol{y} \in \mathcal{S}.$
lyche-numerical-linear-alignatEQ0474	188	1.000 N +0	$\left \boldsymbol{A}\right _{F}:=\left(\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ^2\right)^{1 / 2}$	$\ \boldsymbol{A}\ _F := \left(\sum_{i=1}^m \sum_{j=1}^n a_{ij} ^2\right)^{1/2}$	$\ \boldsymbol{A}\ _F := \left(\sum_{i=1}^m \sum_{j=1}^n a_{ij} ^2\right)^{1/2}$
lyche-numerical-linear-alignatEQ0475	188	1.000 K +0	$\mu\ \boldsymbol{A}\ \leq\left\ \boldsymbol{A}'\right\ \leq M\ \boldsymbol{A}\ $	$\mu\ \boldsymbol{A}\ \leq\left\ \boldsymbol{A}'\right\ \leq M\ \boldsymbol{A}\ $	$\mu\ \boldsymbol{A}\ \leq\left\ \boldsymbol{A}'\right\ \leq M\ \boldsymbol{A}\ $
lyche-numerical-linear-alignatEQ0476 (8.9)	188	1.000 S -1	$\left \boldsymbol{A}\right _{S}:=\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right , \quad\left \boldsymbol{A}\right _{F}:=\left(\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ^2\right)^{1 / 2}, \quad\left \boldsymbol{A}\right _{M}:=\max _{i, j}\left a_{i j}\right .$	$\ \boldsymbol{A}\ _S := \sum_{i=1}^m \sum_{j=1}^n a_{ij} , \quad \ \boldsymbol{A}\ _F := \left(\sum_{i=1}^m \sum_{j=1}^n a_{ij} ^2\right)^{1/2}, \quad \ \boldsymbol{A}\ _M := \max_{i,j} a_{ij} .$	$\ \boldsymbol{A}\ _S := \sum_{i=1}^m \sum_{j=1}^n a_{ij} , \quad \ \boldsymbol{A}\ _F := \left(\sum_{i=1}^m \sum_{j=1}^n a_{ij} ^2\right)^{1/2}, \quad \ \boldsymbol{A}\ _M := \max_{i,j} a_{ij} .$
lyche-numerical-linear-alignatEQ0478 (8.11)	189	1.000 K +0	$\left\ \boldsymbol{A} \boldsymbol{x}\right\ \leq\left\ \boldsymbol{A}\right\ \ \boldsymbol{x}\ ,$	$\ \boldsymbol{A} \boldsymbol{x}\ \leq\left\ \boldsymbol{A}\right\ \ \boldsymbol{x}\ ,$	$\ \boldsymbol{A} \boldsymbol{x}\ \leq\left\ \boldsymbol{A}\right\ \ \boldsymbol{x}\ ,$
lyche-numerical-linear-alignatEQ0479 (8.12)	190	1.000 N +0	$\left\ \boldsymbol{A}\right\ :=\max _{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\left\ \boldsymbol{A} \boldsymbol{x}\right\ }{\ \boldsymbol{x}\ }.$	$\ \boldsymbol{A}\ := \max_{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ }{\ \boldsymbol{x}\ }.$	$\ \boldsymbol{A}\ := \max_{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ }{\ \boldsymbol{x}\ }.$
lyche-numerical-linear-alignatEQ0480 (8.13)	190	1.000 U +0	$\left\ \boldsymbol{A}\right\ =\max _{\ \boldsymbol{x}\ =1}\left\ \boldsymbol{A} \boldsymbol{x}\right\ .$	$\ \boldsymbol{A}\ = \max_{\ \boldsymbol{x}\ =1} \ \boldsymbol{A} \boldsymbol{x}\ .$	$\ \boldsymbol{A}\ = \max_{\ \boldsymbol{x}\ =1} \ \boldsymbol{A} \boldsymbol{x}\ .$
lyche-numerical-linear-alignatEQ0483 (8.15)	190	1.000 N +0	$\left\ \boldsymbol{A} \boldsymbol{x}\right\ \leq\left\ \boldsymbol{A}\right\ \ \boldsymbol{x}\ $ for all $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ and $\boldsymbol{x} \in \mathbb{C}^n$.	$\ \boldsymbol{A} \boldsymbol{x}\ \leq\left\ \boldsymbol{A}\right\ \ \boldsymbol{x}\ $ for all $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ and $\boldsymbol{x} \in \mathbb{C}^n$.	$\ \boldsymbol{A} \boldsymbol{x}\ \leq\left\ \boldsymbol{A}\right\ \ \boldsymbol{x}\ $ for all $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ and $\boldsymbol{x} \in \mathbb{C}^n$.
lyche-numerical-linear-alignatEQ0489	192	1.000 N +0	$\left \boldsymbol{A} \boldsymbol{x}\right _1=\sum_{k=1}^m\left \sum_{j=1}^n a_{k j} x_j\right \leq \sum_{k=1}^m \sum_{j=1}^n\left a_{k j}\right \left x_j\right =\sum_{j=1}^n\left(\sum_{k=1}^m\left a_{k j}\right \right)\left x_j\right \leq K_1.$	$\ \boldsymbol{A} \boldsymbol{x}\ _1 = \sum_{k=1}^m \left \sum_{j=1}^n a_{kj} x_j \right \leq \sum_{k=1}^m \sum_{j=1}^n a_{kj} x_j = \sum_{j=1}^n \left(\sum_{k=1}^m a_{kj} \right) x_j \leq K_1.$	$\ \boldsymbol{A} \boldsymbol{x}\ _1 = \sum_{k=1}^m \left \sum_{j=1}^n a_{kj} x_j \right \leq \sum_{k=1}^m \sum_{j=1}^n a_{kj} x_j = \sum_{j=1}^n \left(\sum_{k=1}^m a_{kj} \right) x_j \leq K_1.$
lyche-numerical-linear-alignatEQ0491	192	1.000 N +0	$\left\ \boldsymbol{A}\right _1=\frac{29}{15}, \quad\left\ \boldsymbol{A}\right _2=2, \quad\left\ \boldsymbol{A}\right _{\infty}=\frac{37}{15}, \quad\left\ \boldsymbol{A}\right _F=\sqrt{5}.$	$\ \boldsymbol{A}\ _1 = \frac{29}{15}, \quad \ \boldsymbol{A}\ _2 = 2, \quad \ \boldsymbol{A}\ _{\infty} = \frac{37}{15}, \quad \ \boldsymbol{A}\ _F = \sqrt{5}.$	$\ \boldsymbol{A}\ _1 = \frac{29}{15}, \quad \ \boldsymbol{A}\ _2 = 2, \quad \ \boldsymbol{A}\ _{\infty} = \frac{37}{15}, \quad \ \boldsymbol{A}\ _F = \sqrt{5}.$
lyche-numerical-linear-alignatEQ0495	194	1.000 S -1	$\left\ \boldsymbol{U} \boldsymbol{A}\right\ _2=\max _{\ \boldsymbol{x}\ _2=1}\left\ \boldsymbol{U} \boldsymbol{A} \boldsymbol{x}\right\ _2=\max _{\ \boldsymbol{x}\ _2=1}\left\ \boldsymbol{A} \boldsymbol{x}\right\ _2=\left\ \boldsymbol{A}\right\ _2.$	$\ \boldsymbol{U} \boldsymbol{A}\ _2 = \max_{\ \boldsymbol{x}\ _2=1} \ \boldsymbol{U} \boldsymbol{A} \boldsymbol{x}\ _2 = \max_{\ \boldsymbol{x}\ _2=1} \ \boldsymbol{A} \boldsymbol{x}\ _2 = \ \boldsymbol{A}\ _2.$	$\ \boldsymbol{U} \boldsymbol{A}\ _2 = \max_{\ \boldsymbol{x}\ _2=1} \ \boldsymbol{U} \boldsymbol{A} \boldsymbol{x}\ _2 = \max_{\ \boldsymbol{x}\ _2=1} \ \boldsymbol{A} \boldsymbol{x}\ _2 = \ \boldsymbol{A}\ _2.$
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-alignat-E00496	194	■ 1.000 ● K +0	$\ \boldsymbol{A} \boldsymbol{\boldsymbol{V}}\ _2 = \ (\boldsymbol{A} \boldsymbol{\boldsymbol{V}})^*\ _2 = \ \boldsymbol{V}^* \boldsymbol{A}^*\ _2 = \ \boldsymbol{A}^*\ _2 = \ \boldsymbol{A}\ _2.$	$\ \boldsymbol{A} \boldsymbol{V}\ _2 = \ (\boldsymbol{A} \boldsymbol{V})^*\ _2 = \ \boldsymbol{V}^* \boldsymbol{A}^*\ _2 = \ \boldsymbol{A}^*\ _2 = \ \boldsymbol{A}\ _2.$	
lyche-numerical-linear-alignat-E00501 (8.22)	195	■ 1.000 ●	$\frac{1}{K(\boldsymbol{A})} \frac{\ \boldsymbol{e}\ }{\ \boldsymbol{b}\ } \leq \frac{\ \boldsymbol{y} - \boldsymbol{x}\ }{\ \boldsymbol{x}\ } \leq K(\boldsymbol{A}) \frac{\ \boldsymbol{e}\ }{\ \boldsymbol{b}\ },$	$\frac{1}{K(\boldsymbol{A})} \frac{\ \boldsymbol{e}\ }{\ \boldsymbol{b}\ } \leq \frac{\ \boldsymbol{y} - \boldsymbol{x}\ }{\ \boldsymbol{x}\ } \leq K(\boldsymbol{A}) \frac{\ \boldsymbol{e}\ }{\ \boldsymbol{b}\ },$	
lyche-numerical-linear-alignat-E00503 (8.24)	196	■ 1.000 ● W +0	$\begin{aligned} & \frac{\ \boldsymbol{y} - \boldsymbol{x}\ }{\ \boldsymbol{y}\ } \leq r \leq K(\boldsymbol{A}) \frac{\ \boldsymbol{E}\ }{\ \boldsymbol{A}\ }, \\ & \frac{\ \boldsymbol{y} - \boldsymbol{x}\ }{\ \boldsymbol{x}\ } \leq \frac{r}{1-r} \leq \frac{K(\boldsymbol{A}) \ \boldsymbol{E}\ }{1-r \ \boldsymbol{A}\ }. \end{aligned}$	$\frac{\ \boldsymbol{y} - \boldsymbol{x}\ }{\ \boldsymbol{y}\ } \leq r \leq K(\boldsymbol{A}) \frac{\ \boldsymbol{E}\ }{\ \boldsymbol{A}\ },$ $\frac{\ \boldsymbol{y} - \boldsymbol{x}\ }{\ \boldsymbol{x}\ } \leq \frac{r}{1-r} \leq \frac{K(\boldsymbol{A}) \ \boldsymbol{E}\ }{1-r \ \boldsymbol{A}\ }.$	
lyche-numerical-linear-alignat-E00506 (8.28)	198	■ 1.000 ● N +0	$\frac{1}{1+r} \leq \frac{\ \boldsymbol{B}^{-1}\ }{\ \boldsymbol{A}^{-1}\ } \leq \frac{1}{1-r}.$	$\frac{1}{1+r} \leq \frac{\ \boldsymbol{B}^{-1}\ }{\ \boldsymbol{A}^{-1}\ } \leq \frac{1}{1-r}.$	
lyche-numerical-linear-alignat-E00507 (8.29)	198	■ 1.000 ● N +0	$\frac{\ \boldsymbol{B}^{-1} - \boldsymbol{A}^{-1}\ }{\ \boldsymbol{A}^{-1}\ } \leq \frac{r}{1-r} \leq \frac{K(\boldsymbol{A}) \ \boldsymbol{E}\ }{1-r \ \boldsymbol{A}\ }.$	$\frac{\ \boldsymbol{B}^{-1} - \boldsymbol{A}^{-1}\ }{\ \boldsymbol{A}^{-1}\ } \leq \frac{r}{1-r} \leq \frac{K(\boldsymbol{A}) \ \boldsymbol{E}\ }{1-r \ \boldsymbol{A}\ }.$	
lyche-numerical-linear-alignat-E00508 (8.30)	198	■ 1.000 ● K +0	$\boldsymbol{B}^{-1} - \boldsymbol{A}^{-1} = -\boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{B}^{-1} = -\boldsymbol{B}^{-1} \boldsymbol{E} \boldsymbol{A}^{-1}.$	$\boldsymbol{B}^{-1} - \boldsymbol{A}^{-1} = -\boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{B}^{-1} = -\boldsymbol{B}^{-1} \boldsymbol{E} \boldsymbol{A}^{-1}.$	
lyche-numerical-linear-alignat-E00512 (8.31)	199	■ 1.000 ● K +0	$f\left((1-\lambda)x_1 + \lambda x_2\right) \leq (1-\lambda)f(x_1) + \lambda f(x_2)$	$f\left((1-\lambda)x_1 + \lambda x_2\right) \leq (1-\lambda)f(x_1) + \lambda f(x_2)$	
lyche-numerical-linear-alignat-E00514	200	■ 1.000 ● K +0	$x = \frac{x_2 - x}{x_2 - x_1} x_1 + \frac{x - x_1}{x_2 - x_1} x_2 = (1-\lambda)x_1 + \lambda x_2$	$x = \frac{x_2 - x}{x_2 - x_1} x_1 + \frac{x - x_1}{x_2 - x_1} x_2 = (1-\lambda)x_1 + \lambda x_2$	
lyche-numerical-linear-alignat-E00515	200	■ 1.000 ● N +0	$f\left(\sum_{j=1}^n \lambda_j z_j\right) \leq \sum_{j=1}^n \lambda_j f(z_j).$	$f\left(\sum_{j=1}^n \lambda_j z_j\right) \leq \sum_{j=1}^n \lambda_j f(z_j).$	
lyche-numerical-linear-alignat-E00516	200	■ 1.000 ● N +0	$f\left(\sum_{j=1}^n \lambda_j z_j\right) = f(\lambda_1 z_1 + (1-\lambda_1)u) \leq \lambda_1 f(z_1) + (1-\lambda_1)f(u) \leq \sum_{j=1}^n \lambda_j f(z_j)$	$f\left(\sum_{j=1}^n \lambda_j z_j\right) = f(\lambda_1 z_1 + (1-\lambda_1)u) \leq \lambda_1 f(z_1) + (1-\lambda_1)f(u) \leq \sum_{j=1}^n \lambda_j f(z_j)$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value					
LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					
[refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-201-EQ0518	201	1.000 S + 0	$-\log \left(\sum_{j=1}^n \lambda_j a_j \right) \leq -\sum_{j=1}^n \lambda_j \log(a_j) = -\log \left(a_1^{\lambda_1} \cdots a_n^{\lambda_n} \right)$	$-\log \left(\sum_{j=1}^n \lambda_j a_j \right) \leq -\sum_{j=1}^n \lambda_j \log(a_j) = -\log \left(a_1^{\lambda_1} \cdots a_n^{\lambda_n} \right)$	
lyche-numerical-linear-algebra-201-EQ0519 (8.33)	201	1.000 N + 0	$\left(a_1 a_2 \cdots a_n \right)^{\frac{1}{n}} \leq \frac{1}{n} \sum_{j=1}^n a_j.$	$(a_1 a_2 \cdots a_n)^{\frac{1}{n}} \leq \frac{1}{n} \sum_{j=1}^n a_j.$	
lyche-numerical-linear-algebra-201-EQ0520	201	1.000	$\sum_{j=1}^n x_j y_j \leq \ \mathbf{x}\ _p \ \mathbf{y}\ _q, \text{ where } \frac{1}{p} + \frac{1}{q} = 1$	$\sum_{j=1}^n x_j y_j \leq \ \mathbf{x}\ _p \ \mathbf{y}\ _q, \text{ where } \frac{1}{p} + \frac{1}{q} = 1$	$\sum_{j=1}^n x_j y_j \leq \ \mathbf{x}\ _p \ \mathbf{y}\ _q, \text{ where } \frac{1}{p} + \frac{1}{q} = 1.$
lyche-numerical-linear-algebra-201-EQ0522	201	1.000 W + 0	$\frac{1}{\ \mathbf{x}\ _p \ \mathbf{y}\ _q} \sum_{j=1}^n x_j y_j = \sum_{j=1}^n \left(\frac{ x_j ^p}{\ \mathbf{x}\ _p^p} \right)^{\frac{1}{p}} \left(\frac{ y_j ^q}{\ \mathbf{y}\ _q^q} \right)^{\frac{1}{q}} \leq \sum_{j=1}^n \left(\frac{1}{p} \frac{ x_j ^p}{\ \mathbf{x}\ _p^p} + \frac{1}{q} \frac{ y_j ^q}{\ \mathbf{y}\ _q^q} \right) = 1$	$\frac{1}{\ \mathbf{x}\ _p \ \mathbf{y}\ _q} \sum_{j=1}^n x_j y_j = \sum_{j=1}^n \left(\frac{ x_j ^p}{\ \mathbf{x}\ _p^p} \right)^{\frac{1}{p}} \left(\frac{ y_j ^q}{\ \mathbf{y}\ _q^q} \right)^{\frac{1}{q}} \leq \sum_{j=1}^n \left(\frac{1}{p} \frac{ x_j ^p}{\ \mathbf{x}\ _p^p} + \frac{1}{q} \frac{ y_j ^q}{\ \mathbf{y}\ _q^q} \right) = 1$	
lyche-numerical-linear-algebra-202-EQ0523	202	1.000 N + 0	$\ \mathbf{x} + \mathbf{y}\ _p \leq \ \mathbf{x}\ _p + \ \mathbf{y}\ _p.$	$\ \mathbf{x} + \mathbf{y}\ _p \leq \ \mathbf{x}\ _p + \ \mathbf{y}\ _p.$	
lyche-numerical-linear-algebra-202-EQ0524	202	1.000 N + 0	$\ \mathbf{x} + \mathbf{y}\ _p^p = \sum_{j=1}^n x_j + y_j ^p \leq \sum_{j=1}^n x_j x_j + y_j ^{p-1} + \sum_{j=1}^n y_j x_j + y_j ^{p-1}.$	$\ \mathbf{x} + \mathbf{y}\ _p^p = \sum_{j=1}^n x_j + y_j ^p \leq \sum_{j=1}^n x_j x_j + y_j ^{p-1} + \sum_{j=1}^n y_j x_j + y_j ^{p-1}.$	
lyche-numerical-linear-algebra-202-EQ0525	202	1.000 N + 0	$\ \mathbf{x} + \mathbf{y}\ _p^p \leq \ \mathbf{x}\ _p \ \mathbf{x} + \mathbf{y}\ _p^{p/q} + \ \mathbf{y}\ _p \ \mathbf{x} + \mathbf{y}\ _p^{p/q} = (\ \mathbf{x}\ _p + \ \mathbf{y}\ _p) \ \mathbf{x} + \mathbf{y}\ _p^{p-1},$	$\ \mathbf{x} + \mathbf{y}\ _p^p \leq \ \mathbf{x}\ _p \ \mathbf{x} + \mathbf{y}\ _p^{p/q} + \ \mathbf{y}\ _p \ \mathbf{x} + \mathbf{y}\ _p^{p/q} = (\ \mathbf{x}\ _p + \ \mathbf{y}\ _p) \ \mathbf{x} + \mathbf{y}\ _p^{p-1},$	
lyche-numerical-linear-algebra-202-EQ0526 (8.35)	202	1.000 N + 0	$\ \mathbf{x} + \mathbf{y}\ ^2 + \ \mathbf{x} - \mathbf{y}\ ^2 = 2\ \mathbf{x}\ ^2 + 2\ \mathbf{y}\ ^2,$	$\ \mathbf{x} + \mathbf{y}\ ^2 + \ \mathbf{x} - \mathbf{y}\ ^2 = 2\ \mathbf{x}\ ^2 + 2\ \mathbf{y}\ ^2,$	
lyche-numerical-linear-algebra-203-EQ0534	203	1.000 N + 0	$\left\langle \frac{n}{m} \mathbf{x}, \mathbf{y} \right\rangle = \frac{n}{m} \langle \mathbf{x}, \mathbf{y} \rangle.$	$\left\langle \frac{n}{m} \mathbf{x}, \mathbf{y} \right\rangle = \frac{n}{m} \langle \mathbf{x}, \mathbf{y} \rangle.$	
lyche-numerical-linear-algebra-204-EQ0539	204	1.000 N + 0	$\tilde{\mathbf{b}}_k := \mathbf{b}_k - \mathbf{B}_{k-1} \left(\mathbf{B}_{k-1}^T \mathbf{A} \mathbf{B}_{k-1} \right)^{-1} \mathbf{B}_{k-1}^T \mathbf{A} \mathbf{b}_k, \quad k = 2, \dots, n.$	$\tilde{\mathbf{b}}_k := \mathbf{b}_k - \mathbf{B}_{k-1} \left(\mathbf{B}_{k-1}^T \mathbf{A} \mathbf{B}_{k-1} \right)^{-1} \mathbf{B}_{k-1}^T \mathbf{A} \mathbf{b}_k, \quad k = 2, \dots, n.$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-205 E08540	205	■ 1.000 ● N +0	$\left\ \boldsymbol{b}_1\right\ _{\boldsymbol{A}}\left\ \boldsymbol{b}_2\right\ _{\boldsymbol{A}} \cdots\left\ \boldsymbol{b}_n\right\ _{\boldsymbol{A}} \leq 2^{n(n-1) / 4} \sqrt{\operatorname{det}(\boldsymbol{A})}.$	$\ \boldsymbol{b}_1\ _{\boldsymbol{A}}\ \boldsymbol{b}_2\ _{\boldsymbol{A}} \cdots\ \boldsymbol{b}_n\ _{\boldsymbol{A}} \leq 2^{n(n-1) / 4} \sqrt{\det(\boldsymbol{A})}.$	$\ \boldsymbol{b}_1\ _{\boldsymbol{A}}\ \boldsymbol{b}_2\ _{\boldsymbol{A}} \cdots\ \boldsymbol{b}_n\ _{\boldsymbol{A}} \leq 2^{n(n-1) / 4} \sqrt{\det(\boldsymbol{A})}.$
lyche-numerical-linear-algebra-205 E08541	205	■ 1.000 ● N +0	$\ \boldsymbol{A}\ :=\sqrt{m n}\ \boldsymbol{A}\ _M, \quad \boldsymbol{A} \in \mathbb{C}^{m \times n}$	$\ \boldsymbol{A}\ :=\sqrt{m n}\ \boldsymbol{A}\ _M, \quad \boldsymbol{A} \in \mathbb{C}^{m \times n}$	$\ \boldsymbol{A}\ :=\sqrt{m n}\ \boldsymbol{A}\ _M, \quad \boldsymbol{A} \in \mathbb{C}^{m \times n}$
lyche-numerical-linear-algebra-205 E08542	205	■ 1.000 ● N -1	$\ \boldsymbol{A}\ _2=\max _{\ \boldsymbol{y}\ _2=1}\left\ \boldsymbol{A} \boldsymbol{y}\right\ _2=\left\ \boldsymbol{A}\right\ _2$	$\ \boldsymbol{A}\ _2=\max _{\ \boldsymbol{x}\ _2=\ \boldsymbol{y}\ _2=1}\left \boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x}\right .$	$\ \boldsymbol{A}\ _2=\max _{\ \boldsymbol{x}\ _2=\ \boldsymbol{y}\ _2=1}\left \boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x}\right .$
lyche-numerical-linear-algebra-207 E08545	207	■ 1.000 ●	$\boldsymbol{x}_{k+1}=\boldsymbol{x}_k+\frac{1}{\lambda_{k+1}}\left\{\boldsymbol{r}_k,\right. \text { where } \left.\boldsymbol{r}_k=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_k\right.$	$\boldsymbol{x}_{k+1}=\boldsymbol{x}_k+\frac{1}{\lambda_{k+1}} \boldsymbol{r}_k, \text { where } \boldsymbol{r}_k=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_k.$	$\boldsymbol{x}_{k+1}=\boldsymbol{x}_k+\frac{1}{\lambda_{k+1}} \boldsymbol{r}_k, \text { where } \boldsymbol{r}_k=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_k.$
lyche-numerical-linear-algebra-207 E08546	207	■ 1.000 ● N +0	$\boldsymbol{r}_k=\sum_{i=1}^n c_{i k} \boldsymbol{u}_i,$	$\boldsymbol{r}_k=\sum_{i=1}^n c_{i k} \boldsymbol{u}_i,$	$\boldsymbol{r}_k=\sum_{i=1}^n c_{i k} \boldsymbol{u}_i,$
lyche-numerical-linear-algebra-207 E08548	207	■ 1.000 ● N +0	$\lambda_j=d+2 c \cos \left(\frac{j \pi}{n+1}\right), \quad j=1, \ldots, n .$	$\lambda_j=d+2 c \cos \left(\frac{j \pi}{n+1}\right), \quad j=1, \ldots, n .$	$\lambda_j=d+2 c \cos \left(\frac{j \pi}{n+1}\right), \quad j=1, \ldots, n .$
lyche-numerical-linear-algebra-207 E08549 (8.40)	207	■ 1.000 ● K +0	$\boldsymbol{T} \boldsymbol{x}_{k+1}=\boldsymbol{b}-\boldsymbol{B} \boldsymbol{x}_k .$	$\boldsymbol{T} \boldsymbol{x}_{k+1}=\boldsymbol{b}-\boldsymbol{B} \boldsymbol{x}_k .$	$\boldsymbol{T} \boldsymbol{x}_{k+1}=\boldsymbol{b}-\boldsymbol{B} \boldsymbol{x}_k .$
lyche-numerical-linear-algebra-207 E08550	207	■ 1.000 ● W -2	$\min \left\{\max _i \sum_{j=1}^n\left b_{i j}\right ,\max _j \sum_{i=1}^n\left b_{i j}\right \right\}<d-2 c .$	$\min \left\{\max _i \sum_{j=1}^n\left b_{i j}\right ,\max _j \sum_{i=1}^n\left b_{i j}\right \right\}<d-2 c .$	$\min \left\{\max _i \sum_{j=1}^n\left b_{i j}\right ,\max _j \sum_{i=1}^n\left b_{i j}\right \right\}<d-2 c .$
lyche-numerical-linear-algebra-208 E08552 (8.41)	208	■ 1.000 ● N +0	$\frac{4}{\pi^2}(m+1)^2-2 / 3<\operatorname{cond}_p(\boldsymbol{T}) \leq \frac{1}{2}(m+1)^2, \quad \text {quad } p=1,2, \infty,$	$\frac{4}{\pi^2}(m+1)^2-2 / 3<\operatorname{cond}_p(\boldsymbol{T}) \leq \frac{1}{2}(m+1)^2, \quad p=1,2, \infty,$	$\frac{4}{\pi^2}(m+1)^2-2 / 3<\operatorname{cond}_p(\boldsymbol{T}) \leq \frac{1}{2}(m+1)^2, \quad p=1,2, \infty,$
lyche-numerical-linear-algebra-208 E08554	208	■ 1.000 ● N -1	$\operatorname{cond}_2(\boldsymbol{T})=\cot ^2\left(\frac{\pi h}{2}\right)=1 / \tan ^2\left(\frac{\pi h}{2}\right) .$	$\operatorname{cond}_2(\boldsymbol{T})=\cot ^2\left(\frac{\pi h}{2}\right)=1 / \tan ^2\left(\frac{\pi h}{2}\right) .$	$\operatorname{cond}_2(\boldsymbol{T})=\cot ^2\left(\frac{\pi h}{2}\right)=1 / \tan ^2\left(\frac{\pi h}{2}\right) .$
lyche-numerical-linear-algebra-209 E08555 (8.43)	209	■ 1.000 ● N +0	$\frac{4}{\pi^2} h^{-2}-\frac{2}{3}<\operatorname{cond}_2(\boldsymbol{T})<\frac{4}{\pi^2} h^{-2} .$	$\frac{4}{\pi^2} h^{-2}-\frac{2}{3}<\operatorname{cond}_2(\boldsymbol{T})<\frac{4}{\pi^2} h^{-2} .$	$\frac{4}{\pi^2} h^{-2}-\frac{2}{3}<\operatorname{cond}_2(\boldsymbol{T})<\frac{4}{\pi^2} h^{-2} .$
lyche-numerical-linear-algebra-209 E08556	209	■ 1.000 ● N +0	$\frac{\left\ (\boldsymbol{I}-\boldsymbol{E})^{-1}-\boldsymbol{I}\right\ }{\left\ (\boldsymbol{I}-\boldsymbol{E})^{-1}\right\ } \leq\ \boldsymbol{E}\ $	$\frac{\ (\boldsymbol{I}-\boldsymbol{E})^{-1}-\boldsymbol{I}\ }{\ (\boldsymbol{I}-\boldsymbol{E})^{-1}\ } \leq\ \boldsymbol{E}\ $	$\frac{\ (\boldsymbol{I}-\boldsymbol{E})^{-1}-\boldsymbol{I}\ }{\ (\boldsymbol{I}-\boldsymbol{E})^{-1}\ } \leq\ \boldsymbol{E}\ $
lyche-numerical-linear-algebra-209 E08557	209	■ 1.000 ● K +0	$\frac{1}{1+\ \boldsymbol{E}\ } \leq\left\ (\boldsymbol{I}-\boldsymbol{E})^{-1}\right\ \leq \frac{1}{1-\ \boldsymbol{E}\ }$	$\frac{1}{1+\ \boldsymbol{E}\ } \leq\left\ (\boldsymbol{I}-\boldsymbol{E})^{-1}\right\ \leq \frac{1}{1-\ \boldsymbol{E}\ }$	$\frac{1}{1+\ \boldsymbol{E}\ } \leq\left\ (\boldsymbol{I}-\boldsymbol{E})^{-1}\right\ \leq \frac{1}{1-\ \boldsymbol{E}\ }$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-209 E09558	209	■ 1.000 ● N +0	$\left \left(\boldsymbol{I}-\boldsymbol{E}\right)^{-1}-\boldsymbol{I}\right \leq\frac{\left\ \boldsymbol{E}\right\ }{1-\left\ \boldsymbol{E}\right\ }.$	$\left\ \left(\boldsymbol{I}-\boldsymbol{E}\right)^{-1}-\boldsymbol{I}\right\ \leq\frac{\left\ \boldsymbol{E}\right\ }{1-\left\ \boldsymbol{E}\right\ }.$	$\left\ \left(\boldsymbol{I}-\boldsymbol{E}\right)^{-1}-\boldsymbol{I}\right\ \leq\frac{\left\ \boldsymbol{E}\right\ }{1-\left\ \boldsymbol{E}\right\ }.$
lyche-numerical-linear-algebra-209 E09560	209	■ 1.000 ● N +0	$\frac{K\left(\boldsymbol{B}\right)^{-1}}{1+r}\frac{\left\ \boldsymbol{E}\right\ }{\left\ \boldsymbol{A}\right\ }\leq\frac{\left\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\ }{\left\ \boldsymbol{A}^{-1}\right\ }.$	$\frac{K\left(\boldsymbol{B}\right)^{-1}}{1+r}\frac{\left\ \boldsymbol{E}\right\ }{\left\ \boldsymbol{A}\right\ }\leq\frac{\left\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\ }{\left\ \boldsymbol{A}^{-1}\right\ }.$	$\frac{K\left(\boldsymbol{B}\right)^{-1}}{1+r}\frac{\left\ \boldsymbol{E}\right\ }{\left\ \boldsymbol{A}\right\ }\leq\frac{\left\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\ }{\left\ \boldsymbol{A}^{-1}\right\ }.$
lyche-numerical-linear-algebra-209 E09561	209	■ 1.000 ● K +0	$a=x_0<x_1<\cdots <x_n=b,$	$a=x_0<x_1<\cdots <x_n=b,$	$a=x_0<x_1<\cdots <x_n=b,$
lyche-numerical-linear-algebra-210 E09562	210	■ 1.000 ● W +0	$\begin{array}{ll} g\left(x_i\right)=y_i, & i=0,1,\ldots,n, \\ \ldots, n, & \\ g^{\prime}(a)=g^{\prime}(b), & g^{\prime \prime}(a)=g^{\prime \prime}(b). \end{array}$	$g\left(x_i\right)=y_i, \quad i=0,1,\ldots,n, \\ g^{\prime}(a)=g^{\prime}(b), \quad g^{\prime \prime}(a)=g^{\prime \prime}(b).$	$g\left(x_i\right)=y_i, \quad i=0,1,\ldots,n, \\ g^{\prime}(a)=g^{\prime}(b), \quad g^{\prime \prime}(a)=g^{\prime \prime}(b).$
lyche-numerical-linear-algebra-210 E09565	210	■ 1.000 ●	$\lambda _i:=\frac{h_i}{h_{i-1}+h_i}, \quad \mu _i:=\frac{h_{i-1}}{h_{i-1}+h_i}, \quad i=1,\ldots,n, \\ \frac{h_i}{h_{i-1}+h_i}, \quad \mu _i:=\frac{h_{i-1}}{h_{i-1}+h_i}, \quad i=1,\ldots,n,$	$\lambda _i:=\frac{h_i}{h_{i-1}+h_i}, \quad \mu _i:=\frac{h_{i-1}}{h_{i-1}+h_i}, \quad i=1,\ldots,n,$	$\lambda _i:=\frac{h_i}{h_{i-1}+h_i}, \quad \mu _i:=\frac{h_{i-1}}{h_{i-1}+h_i}, \quad i=1,\ldots,n,$
lyche-numerical-linear-algebra-210 E09566	210	■ 1.000 ● N -1	$h_i=x_{i+1}-x_i, \quad i=0,\ldots,n-1, \text{ and } h_n=h_0.$	$h_i=x_{i+1}-x_i, \quad i=0,\ldots,n-1, \text{ and } h_n=h_0.$	$h_i=x_{i+1}-x_i, \quad i=0,\ldots,n-1, \text{ and } h_n=h_0.$
lyche-numerical-linear-algebra-210 E09567	210	■ 1.000 ● N +0	$\frac{1}{2}\leq\frac{h_i}{h_{i-1}}\leq2, \quad i=1,\ldots,n.$	$\frac{1}{2}\leq\frac{h_i}{h_{i-1}}\leq2, \quad i=1,\ldots,n.$	$\frac{1}{2}\leq\frac{h_i}{h_{i-1}}\leq2, \quad i=1,\ldots,n.$
lyche-numerical-linear-algebra-211 E09568	211	■ 1.000 ● N -1	$\left\ \boldsymbol{A}\right\ _{\infty}=3 \quad \text{and that} \quad \left\ \boldsymbol{A}\right\ _1\leq\frac{10}{3}.$	$\left\ \boldsymbol{A}\right\ _{\infty}=3 \quad \text{and that} \quad \left\ \boldsymbol{A}\right\ _1\leq\frac{10}{3}.$	$\left\ \boldsymbol{A}\right\ _{\infty}=3 \quad \text{and that} \quad \left\ \boldsymbol{A}\right\ _1\leq\frac{10}{3}.$
lyche-numerical-linear-algebra-211 E09571 (8.45)	211	■ 1.000 ● N +0	$\langle\boldsymbol{x},\boldsymbol{y}\rangle:=\frac{1}{4}\left(\left\ \boldsymbol{x}+\boldsymbol{y}\right\ ^2-\left\ \boldsymbol{x}-\boldsymbol{y}\right\ ^2+i\left\ \boldsymbol{x}+i\boldsymbol{y}\right\ ^2-i\left\ \boldsymbol{x}-i\boldsymbol{y}\right\ ^2\right),$	$\langle\boldsymbol{x},\boldsymbol{y}\rangle:=\frac{1}{4}\left(\left\ \boldsymbol{x}+\boldsymbol{y}\right\ ^2-\left\ \boldsymbol{x}-\boldsymbol{y}\right\ ^2+i\left\ \boldsymbol{x}+i\boldsymbol{y}\right\ ^2-i\left\ \boldsymbol{x}-i\boldsymbol{y}\right\ ^2\right),$	$\langle\boldsymbol{x},\boldsymbol{y}\rangle:=\frac{1}{4}\left(\left\ \boldsymbol{x}+\boldsymbol{y}\right\ ^2-\left\ \boldsymbol{x}-\boldsymbol{y}\right\ ^2+i\left\ \boldsymbol{x}+i\boldsymbol{y}\right\ ^2-i\left\ \boldsymbol{x}-i\boldsymbol{y}\right\ ^2\right),$
lyche-numerical-linear-algebra-213 E09574	213	■ 1.000 ● K +0	$E\left(\boldsymbol{x}\right):=\left\ \boldsymbol{A}\boldsymbol{x}-\boldsymbol{b}\right\ _2^2,$	$E\left(\boldsymbol{x}\right):=\left\ \boldsymbol{A}\boldsymbol{x}-\boldsymbol{b}\right\ _2^2,$	$E\left(\boldsymbol{x}\right):=\left\ \boldsymbol{A}\boldsymbol{x}-\boldsymbol{b}\right\ _2^2,$
lyche-numerical-linear-algebra-214 E09576	214	■ 1.000 ● N +1	$\left\ \boldsymbol{A}\boldsymbol{x}-\boldsymbol{b}\right\ _2^2=\left(x_1-1\right)^2+\left(x_1-1\right)^2+\left(x_1-2\right)^2=3x_1^2-8x_1+6.$	$\left\ \boldsymbol{A}\boldsymbol{x}-\boldsymbol{b}\right\ _2^2=\left(x_1-1\right)^2+\left(x_1-1\right)^2+\left(x_1-2\right)^2=3x_1^2-8x_1+6.$	$\left\ \boldsymbol{A}\boldsymbol{x}-\boldsymbol{b}\right\ _2^2=\left(x_1-1\right)^2+\left(x_1-1\right)^2+\left(x_1-2\right)^2=3x_1^2-8x_1+6.$
lyche-numerical-linear-algebra-215 E09581	215	■ 1.000 ● N +0	$y=\boldsymbol{u}^*\boldsymbol{x}=\sum_{i=1}^n u_i x_i,$	$y=\boldsymbol{u}^*\boldsymbol{x}=\sum_{i=1}^n u_i x_i,$	$y=\boldsymbol{u}^*\boldsymbol{x}=\sum_{i=1}^n u_i x_i,$
lyche-numerical-linear-algebra-216 E09585 (9.2)	216	■ 1.000 ● N +0	$E\left(\boldsymbol{x}\right):=\left\ \boldsymbol{A}\boldsymbol{x}-\boldsymbol{b}\right\ _2^2=\sum_{k=1}^m\left(\sum_{j=1}^n x_j \phi_j\left(t_k\right)-y_k\right)^2.$	$E\left(\boldsymbol{x}\right):=\left\ \boldsymbol{A}\boldsymbol{x}-\boldsymbol{b}\right\ _2^2=\sum_{k=1}^m\left(\sum_{j=1}^n x_j \phi_j\left(t_k\right)-y_k\right)^2.$	$E\left(\boldsymbol{x}\right):=\left\ \boldsymbol{A}\boldsymbol{x}-\boldsymbol{b}\right\ _2^2=\sum_{k=1}^m\left(\sum_{j=1}^n x_j \phi_j\left(t_k\right)-y_k\right)^2.$
lyche-numerical-linear-algebra-217 E09586	217	■ 1.000 ● N +0	$E\left(\boldsymbol{x}\right):=\sum_{k=1}^m w_k\left(\sum_{j=1}^n x_j \phi_j\left(t_k\right)-y_k\right)^2.$	$E\left(\boldsymbol{x}\right):=\sum_{k=1}^m w_k\left(\sum_{j=1}^n x_j \phi_j\left(t_k\right)-y_k\right)^2.$	$E\left(\boldsymbol{x}\right):=\sum_{k=1}^m w_k\left(\sum_{j=1}^n x_j \phi_j\left(t_k\right)-y_k\right)^2.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-228 E00625	228	■ 1.000 ● W +0	$\left \left(\boldsymbol{A}+\boldsymbol{E}\right)^{\dagger}\right _{-2}=\frac{1}{\beta_r}\leq\frac{1}{\alpha_r-\epsilon_1}=\frac{1/\alpha_r}{1-\epsilon_1/\alpha_r}=\frac{\left\ \boldsymbol{A}^{\dagger}\right\ _2}{1-\left\ \boldsymbol{A}^{\dagger}\right\ _2\left\ \boldsymbol{E}\right\ _2}.$	$\left\ \left(\boldsymbol{A}+\boldsymbol{E}\right)^{\dagger}\right\ _2=\frac{1}{\beta_r}\leq\frac{1}{\alpha_r-\epsilon_1}=\frac{1/\alpha_r}{1-\epsilon_1/\alpha_r}=\frac{\left\ \boldsymbol{A}^{\dagger}\right\ _2}{1-\left\ \boldsymbol{A}^{\dagger}\right\ _2\left\ \boldsymbol{E}\right\ _2}.$	$\left\ \left(\boldsymbol{A}+\boldsymbol{E}\right)^{\dagger}\right\ _2=\frac{1}{\beta_r}\leq\frac{1}{\alpha_r-\epsilon_1}=\frac{1/\alpha_r}{1-\epsilon_1/\alpha_r}=\frac{\left\ \boldsymbol{A}^{\dagger}\right\ _2}{1-\left\ \boldsymbol{A}^{\dagger}\right\ _2\left\ \boldsymbol{E}\right\ _2}.$
lyche-numerical-linear-algebra-228 E00626 (9.19)	228	■ 1.000 ● N +0	$\sum_{j=1}^n\left \mu_j-\lambda_j\right ^2\leq\left\ \boldsymbol{A}-\boldsymbol{B}\right\ _F^2:=\sum_{i=1}^n\sum_{j=1}^n\left a_{ij}-b_{ij}\right ^2,$	$\sum_{j=1}^n\left \mu_j-\lambda_j\right ^2\leq\left\ \boldsymbol{A}-\boldsymbol{B}\right\ _F^2:=\sum_{i=1}^n\sum_{j=1}^n\left a_{ij}-b_{ij}\right ^2,$	$\sum_{j=1}^n\left \mu_j-\lambda_j\right ^2\leq\left\ \boldsymbol{A}-\boldsymbol{B}\right\ _F^2:=\sum_{i=1}^n\sum_{j=1}^n\left a_{ij}-b_{ij}\right ^2,$
lyche-numerical-linear-algebra-229 E00627 (9.20)	229	■ 1.000 ● N +0	$\sum_{j=1}^n\left \beta_j-\alpha_j\right ^2\leq\left\ \boldsymbol{A}-\boldsymbol{B}\right\ _F^2.$	$\sum_{j=1}^n\left \beta_j-\alpha_j\right ^2\leq\left\ \boldsymbol{A}-\boldsymbol{B}\right\ _F^2.$	$\sum_{j=1}^n\left \beta_j-\alpha_j\right ^2\leq\left\ \boldsymbol{A}-\boldsymbol{B}\right\ _F^2.$
lyche-numerical-linear-algebra-229 E00629 (9.21)	229	■ 1.000 ● N +0	$\sum_{j=1}^{m+n}\left \lambda_j-\mu_j\right ^2\leq\left\ \boldsymbol{C}-\boldsymbol{D}\right\ _F^2.$	$\sum_{j=1}^{m+n}\left \lambda_j-\mu_j\right ^2\leq\left\ \boldsymbol{C}-\boldsymbol{D}\right\ _F^2.$	$\sum_{j=1}^{m+n}\left \lambda_j-\mu_j\right ^2\leq\left\ \boldsymbol{C}-\boldsymbol{D}\right\ _F^2.$
lyche-numerical-linear-algebra-229 E00631	229	■ 1.000 ● N +0	$t:=\max\left(r,s\right).$	$t:=\max(r,s).$	$t:=\max(r,s).$
lyche-numerical-linear-algebra-230 E00633	230	■ 1.000 ●	$\sum_{j=1}^{m+n}\left \lambda_j-\mu_j\right ^2=\sum_{i=1}^t\left \alpha_i-\beta_i\right ^2+\sum_{i=1}^t\left -\alpha_i+\beta_i\right ^2=2\sum_{i=1}^t\left \alpha_i-\beta_i\right ^2$	$\sum_{j=1}^{m+n}\left \lambda_j-\mu_j\right ^2=\sum_{i=1}^t\left \alpha_i-\beta_i\right ^2+\sum_{i=1}^t\left -\alpha_i+\beta_i\right ^2=2\sum_{i=1}^t\left \alpha_i-\beta_i\right ^2$	$\sum_{j=1}^{m+n}\left \lambda_j-\mu_j\right ^2=\sum_{i=1}^t\left \alpha_i-\beta_i\right ^2+\sum_{i=1}^t\left -\alpha_i+\beta_i\right ^2=2\sum_{i=1}^t\left \alpha_i-\beta_i\right ^2$
lyche-numerical-linear-algebra-230 E00635 (9.22)	230	■ 1.000 ● N +0	$\left(t_i-c_1\right)^2+\left(y_i-c_2\right)^2=r^2,i=1,\ldots,m,$	$\left(t_i-c_1\right)^2+\left(y_i-c_2\right)^2=r^2,i=1,\ldots,m,$	$\left(t_i-c_1\right)^2+\left(y_i-c_2\right)^2=r^2,i=1,\ldots,m,$
lyche-numerical-linear-algebra-230 E00636 (9.23)	230	■ 1.000 ● W +1	$t_ix_1+y_ix_2+x_3=t_i^2+y_i^2,i=1,\ldots,m,$	$t_ix_1+y_ix_2+x_3=t_i^2+y_i^2,i=1,\ldots,m,$	$t_ix_1+y_ix_2+x_3=t_i^2+y_i^2,i=1,\ldots,m,$
lyche-numerical-linear-algebra-231 E00637	231	■ 1.000 ● N +0	$\sum_{i=1}^3\left\ p\left(x_i\right)-y_i\right\ ^2,$	$\sum_{i=1}^3\left\ p\left(x_i\right)-y_i\right\ ^2,$	$\sum_{i=1}^3\left\ p\left(x_i\right)-y_i\right\ ^2,$
lyche-numerical-linear-algebra-231 E00638	231	■ 1.000 ● N +0	$F\left(\boldsymbol{x}\right):=\left\ \boldsymbol{Ax}-\boldsymbol{b}\right\ _2^2+\left\ \boldsymbol{x}\right\ _2^2.$	$F(\boldsymbol{x}):=\left\ \boldsymbol{Ax}-\boldsymbol{b}\right\ _2^2+\left\ \boldsymbol{x}\right\ _2^2.$	$F(\boldsymbol{x}):=\left\ \boldsymbol{Ax}-\boldsymbol{b}\right\ _2^2+\left\ \boldsymbol{x}\right\ _2^2.$
lyche-numerical-linear-algebra-231 E00639	231	■ 1.000 ● N +1	$F\left(\boldsymbol{x}\right)=\boldsymbol{x}^T\boldsymbol{Bx}-2\boldsymbol{c}^T\boldsymbol{x}+\boldsymbol{b}^T\boldsymbol{b},\quad\text{where}\quad\boldsymbol{c}=\boldsymbol{A}^T\boldsymbol{b}.$	$F(\boldsymbol{x})=\boldsymbol{x}^T\boldsymbol{Bx}-2\boldsymbol{c}^T\boldsymbol{x}+\boldsymbol{b}^T\boldsymbol{b},\quad\text{where}\quad\boldsymbol{c}=\boldsymbol{A}^T\boldsymbol{b}.$	$F(\boldsymbol{x})=\boldsymbol{x}^T\boldsymbol{Bx}-2\boldsymbol{c}^T\boldsymbol{x}+\boldsymbol{b}^T\boldsymbol{b},\quad\text{where}\quad\boldsymbol{c}=\boldsymbol{A}^T\boldsymbol{b}.$
lyche-numerical-linear-algebra-231 E00640 (9.24)	231	■ 1.000 ● N +1	$\left \boldsymbol{r}\left(\boldsymbol{x}\right)\right _D^2:=\sum_{i=1}^mr_i\left(x\right)^2d_i,\quad\boldsymbol{x}\in\mathbb{R}^n,$	$\left\ \boldsymbol{r}(\boldsymbol{x})\right\ _D^2:=\sum_{i=1}^mr_i(x)^2d_i,\quad\boldsymbol{x}\in\mathbb{R}^n,$	$\left\ \boldsymbol{r}(\boldsymbol{x})\right\ _D^2:=\sum_{i=1}^mr_i(x)^2d_i,\quad\boldsymbol{x}\in\mathbb{R}^n,$
lyche-numerical-linear-algebra-232 E00642	232	■ 1.000 ● K +0	$\boldsymbol{A}^T\boldsymbol{DAx}=\boldsymbol{A}^T\boldsymbol{Db}.$	$\boldsymbol{A}^T\boldsymbol{DAx}=\boldsymbol{A}^T\boldsymbol{Db}.$	$\boldsymbol{A}^T\boldsymbol{DAx}=\boldsymbol{A}^T\boldsymbol{Db}.$
lyche-numerical-linear-algebra-232 E00643	232	■ 1.000 ● N +0	$K_2\left(\boldsymbol{A}^T\boldsymbol{DA}\right)\leq K_2\left(\boldsymbol{A}^T\boldsymbol{A}\right)K_2\left(\boldsymbol{D}\right),$	$K_2\left(\boldsymbol{A}^T\boldsymbol{DA}\right)\leq K_2\left(\boldsymbol{A}^T\boldsymbol{A}\right)K_2\left(\boldsymbol{D}\right),$	$K_2\left(\boldsymbol{A}^T\boldsymbol{DA}\right)\leq K_2\left(\boldsymbol{A}^T\boldsymbol{A}\right)K_2\left(\boldsymbol{D}\right),$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-233 EQ0646 (9.25)	233	■ 1.000 ● N +0	$\begin{aligned} & \text{\boldsymbol{P}}_{\mathcal{R}(A)} = \text{\boldsymbol{A}}^{\dagger} \text{\boldsymbol{A}} = \text{\boldsymbol{V}}_1 \text{\boldsymbol{V}}_1^* = \sum_{j=1}^r \text{\boldsymbol{v}}_j \text{\boldsymbol{v}}_j^* \in \mathbb{C}^{n \times n} \\ & \text{\boldsymbol{P}}_{\mathcal{N}(A)} = \text{\boldsymbol{I}} - \text{\boldsymbol{A}}^{\dagger} \text{\boldsymbol{A}} = \text{\boldsymbol{V}}_2 \text{\boldsymbol{V}}_2^* = \sum_{j=r+1}^n \text{\boldsymbol{v}}_j \text{\boldsymbol{v}}_j^* \in \mathbb{C}^{n \times n}. \end{aligned}$	$\begin{aligned} \text{\boldsymbol{P}}_{\mathcal{R}(A^*)} &= \text{\boldsymbol{A}}^{\dagger} \text{\boldsymbol{A}} = \text{\boldsymbol{V}}_1 \text{\boldsymbol{V}}_1^* = \sum_{j=1}^r \text{\boldsymbol{v}}_j \text{\boldsymbol{v}}_j^* \in \mathbb{C}^{n \times n} \\ \text{\boldsymbol{P}}_{\mathcal{N}(A)} &= \text{\boldsymbol{I}} - \text{\boldsymbol{A}}^{\dagger} \text{\boldsymbol{A}} = \text{\boldsymbol{V}}_2 \text{\boldsymbol{V}}_2^* = \sum_{j=r+1}^n \text{\boldsymbol{v}}_j \text{\boldsymbol{v}}_j^* \in \mathbb{C}^{n \times n}. \end{aligned}$	$\text{\boldsymbol{P}}_{\mathcal{R}(A^*)} = \text{\boldsymbol{A}}^{\dagger} \text{\boldsymbol{A}} = \text{\boldsymbol{V}}_1 \text{\boldsymbol{V}}_1^* = \sum_{j=1}^r \text{\boldsymbol{v}}_j \text{\boldsymbol{v}}_j^* \in \mathbb{C}^{n \times n}$ $\text{\boldsymbol{P}}_{\mathcal{N}(A)} = \text{\boldsymbol{I}} - \text{\boldsymbol{A}}^{\dagger} \text{\boldsymbol{A}} = \text{\boldsymbol{V}}_2 \text{\boldsymbol{V}}_2^* = \sum_{j=r+1}^n \text{\boldsymbol{v}}_j \text{\boldsymbol{v}}_j^* \in \mathbb{C}^{n \times n}.$
lyche-numerical-linear-algebra-234 EQ0652	234	■ 1.000 ● K +0	A=\left[\begin{array}{l} B \\ C \end{array}\right]	$A = \left[\begin{array}{c} B \\ C \end{array} \right]$	$A = \begin{bmatrix} B \\ C \end{bmatrix}$
lyche-numerical-linear-algebra-235 EQ0653	235	■ 1.000 ● N +0	$\text{\boldsymbol{A}} = \left[\begin{array}{ll} 1 & 2 \\ 1 & 1 \\ 1 & 1 \end{array} \right], \quad \text{\boldsymbol{b}} = \left[\begin{array}{c} b_1 \\ b_2 \\ b_3 \end{array} \right].$	$\text{\boldsymbol{A}} = \left[\begin{array}{cc} 1 & 2 \\ 1 & 1 \\ 1 & 1 \end{array} \right], \quad \text{\boldsymbol{b}} = \left[\begin{array}{c} b_1 \\ b_2 \\ b_3 \end{array} \right].$	$\text{\boldsymbol{A}} = \left[\begin{array}{cc} 1 & 2 \\ 1 & 1 \\ 1 & 1 \end{array} \right], \quad \text{\boldsymbol{b}} = \left[\begin{array}{c} b_1 \\ b_2 \\ b_3 \end{array} \right].$
lyche-numerical-linear-algebra-235 EQ0655	235	■ 1.000 ●	$ \sigma_i(\epsilon) - \sigma_i(0) \leq \epsilon , \quad i = 1, \dots, n,$	$ \sigma_i(\epsilon) - \sigma_i(0) \leq \epsilon , \quad i = 1, \dots, n,$	$ \sigma_i(\epsilon) - \sigma_i(0) \leq \epsilon , \quad i = 1, \dots, n,$
lyche-numerical-linear-algebra-238 EQ0657	238	■ 1.000 ● N +0	$\bar{\Omega}_h := \{(jh, kh) : j, k = 0, 1, \dots, m+1\}, \quad \text{where } h = 1/(m+1).$	$\bar{\Omega}_h := \{(jh, kh) : j, k = 0, 1, \dots, m+1\}, \quad \text{where } h = 1/(m+1).$	$\bar{\Omega}_h := \{(jh, kh) : j, k = 0, 1, \dots, m+1\}, \quad \text{where } h = 1/(m+1).$
lyche-numerical-linear-algebra-239 EQ0658 (10.2)	239	■ 1.000 ● N +0	$\frac{\partial^2 u(jh, kh)}{\partial x^2} \approx \frac{v_{j-1,k} - 2v_{j,k} + v_{j+1,k}}{h^2}, \quad \frac{\partial^2 u(jh, kh)}{\partial y^2} \approx \frac{v_{j,k-1} - 2v_{j,k} + v_{j,k+1}}{h^2}.$	$\frac{\partial^2 u(jh, kh)}{\partial x^2} \approx \frac{v_{j-1,k} - 2v_{j,k} + v_{j+1,k}}{h^2}, \quad \frac{\partial^2 u(jh, kh)}{\partial y^2} \approx \frac{v_{j,k-1} - 2v_{j,k} + v_{j,k+1}}{h^2}.$	$\frac{\partial^2 u(jh, kh)}{\partial x^2} \approx \frac{v_{j-1,k} - 2v_{j,k} + v_{j+1,k}}{h^2}, \quad \frac{\partial^2 u(jh, kh)}{\partial y^2} \approx \frac{v_{j,k-1} - 2v_{j,k} + v_{j,k+1}}{h^2}.$
lyche-numerical-linear-algebra-239 EQ0660 (10.4)	239	■ 1.000 ● N +0	$-\Delta_h v_{j,k} := \frac{-v_{j-1,k} + 2v_{j,k} - v_{j+1,k}}{h^2} + \frac{-v_{j,k-1} + 2v_{j,k} - v_{j,k+1}}{h^2}.$	$-\Delta_h v_{j,k} := \frac{-v_{j-1,k} + 2v_{j,k} - v_{j+1,k}}{h^2} + \frac{-v_{j,k-1} + 2v_{j,k} - v_{j,k+1}}{h^2}.$	$-\Delta_h v_{j,k} := \frac{-v_{j-1,k} + 2v_{j,k} - v_{j+1,k}}{h^2} + \frac{-v_{j,k-1} + 2v_{j,k} - v_{j,k+1}}{h^2}.$
lyche-numerical-linear-algebra-239 EQ0662 (10.6)	239	■ 1.000 ● C -4	$\text{\boldsymbol{V}} := \left[\begin{array}{ccc} v_{1,1} & \cdots & v_{1,m} \\ \vdots & & \vdots \\ v_{m,1} & \cdots & v_{m,m} \end{array} \right] \in \mathbb{R}^{m \times m}, \quad \text{\boldsymbol{F}} := \left[\begin{array}{ccc} f_{1,1} & \cdots & f_{1,m} \\ \vdots & & \vdots \\ f_{m,1} & \cdots & f_{m,m} \end{array} \right] \in \mathbb{R}^{m \times m}.$	$\text{\boldsymbol{V}} := \left[\begin{array}{ccc} v_{1,1} & \cdots & v_{1,m} \\ \vdots & & \vdots \\ v_{m,1} & \cdots & v_{m,m} \end{array} \right] \in \mathbb{R}^{m \times m}, \quad \text{\boldsymbol{F}} := \left[\begin{array}{ccc} f_{1,1} & \cdots & f_{1,m} \\ \vdots & & \vdots \\ f_{m,1} & \cdots & f_{m,m} \end{array} \right] \in \mathbb{R}^{m \times m}.$	$\text{\boldsymbol{V}} := \left[\begin{array}{ccc} v_{1,1} & \cdots & v_{1,m} \\ \vdots & & \vdots \\ v_{m,1} & \cdots & v_{m,m} \end{array} \right] \in \mathbb{R}^{m \times m}, \quad \text{\boldsymbol{F}} := \left[\begin{array}{ccc} f_{1,1} & \cdots & f_{1,m} \\ \vdots & & \vdots \\ f_{m,1} & \cdots & f_{m,m} \end{array} \right] \in \mathbb{R}^{m \times m}.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-239 E09663	239	1.000 W +0	$\sum_{i=1}^m \boldsymbol{T}_{j,i} v_{i,k} + \sum_{i=1}^m v_{j,i} \boldsymbol{T}_{i,k} = -v_{j-1,k} + 2v_{j,k} - v_{j+1,k} - v_{j,k-1} + 2v_{j,k} - v_{j,k+1},$	$\sum_{i=1}^m \boldsymbol{T}_{j,i} v_{i,k} + \sum_{i=1}^m v_{j,i} \boldsymbol{T}_{i,k} = -v_{j-1,k} + 2v_{j,k} - v_{j+1,k} - v_{j,k-1} + 2v_{j,k} - v_{j,k+1},$	$\sum_{i=1}^m \boldsymbol{T}_{j,i} v_{i,k} + \sum_{i=1}^m v_{j,i} \boldsymbol{T}_{i,k} = -v_{j-1,k} + 2v_{j,k} - v_{j+1,k} - v_{j,k-1} + 2v_{j,k} - v_{j,k+1},$
lyche-numerical-linear-algebra-239 E09664	239	1.000 S +0	$\operatorname{vec}(\boldsymbol{B}) := \left[b_{11}, \ldots, b_{m1}, b_{12}, \ldots, b_{m2}, \ldots, b_{1n}, \ldots, b_{mn} \right]^T \in \mathbb{R}^{mn}$	$\operatorname{vec}(\boldsymbol{B}) := [b_{11}, \ldots, b_{m1}, b_{12}, \ldots, b_{m2}, \ldots, b_{1n}, \ldots, b_{mn}]^T \in \mathbb{R}^{mn}$	$\operatorname{vec}(\boldsymbol{B}) := [b_{11}, \ldots, b_{m1}, b_{12}, \ldots, b_{m2}, \ldots, b_{1n}, \ldots, b_{mn}]^T \in \mathbb{R}^{mn}$
lyche-numerical-linear-algebra-240 E09665	240	1.000 N +0	$4x_i - x_{i-1} - x_{i+1} - x_{i-m} - x_{i+m} = b_i,$	$4x_i - x_{i-1} - x_{i+1} - x_{i-m} - x_{i+m} = b_i,$	$4x_i - x_{i-1} - x_{i+1} - x_{i-m} - x_{i+m} = b_i,$
lyche-numerical-linear-algebra-241 E09669 (10.9)	241	1.000 N -2	$\boldsymbol{T}_{\{1\}} := \operatorname{tridiag}(a, d, a) .$	$\boldsymbol{T}_1 := \operatorname{tridiag}(a, d, a).$	$\boldsymbol{T}_1 := \operatorname{tridiag}(a, d, a).$
lyche-numerical-linear-algebra-242 E09672	242	1.000 N +0	$\boldsymbol{C} = \left[\begin{array}{cccc} \boldsymbol{A} b_{1,1} & \boldsymbol{A} b_{1,2} & \cdots & \boldsymbol{A} b_{1,s} \\ \boldsymbol{A} b_{2,1} & \boldsymbol{A} b_{2,2} & \cdots & \boldsymbol{A} b_{2,s} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{A} b_{r,1} & \boldsymbol{A} b_{r,2} & \cdots & \boldsymbol{A} b_{r,s} \end{array} \right] .$	$\boldsymbol{C} = \left[\begin{array}{cccc} \boldsymbol{A} b_{1,1} & \boldsymbol{A} b_{1,2} & \cdots & \boldsymbol{A} b_{1,s} \\ \boldsymbol{A} b_{2,1} & \boldsymbol{A} b_{2,2} & \cdots & \boldsymbol{A} b_{2,s} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{A} b_{r,1} & \boldsymbol{A} b_{r,2} & \cdots & \boldsymbol{A} b_{r,s} \end{array} \right] .$	$\boldsymbol{C} = \left[\begin{array}{cccc} \boldsymbol{A} b_{1,1} & \boldsymbol{A} b_{1,2} & \cdots & \boldsymbol{A} b_{1,s} \\ \boldsymbol{A} b_{2,1} & \boldsymbol{A} b_{2,2} & \cdots & \boldsymbol{A} b_{2,s} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{A} b_{r,1} & \boldsymbol{A} b_{r,2} & \cdots & \boldsymbol{A} b_{r,s} \end{array} \right] .$
lyche-numerical-linear-algebra-242 E09673	242	1.000 N +1	$\boldsymbol{T}_{\{1\}} = \left[\begin{array}{ccc} d & a & 0 \\ a & d & a \\ 0 & a & d \end{array} \right], \quad \boldsymbol{I} = \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]$	$\boldsymbol{T}_1 = \left[\begin{array}{ccc} d & a & 0 \\ a & d & a \\ 0 & a & d \end{array} \right], \quad \boldsymbol{I} = \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]$	$\boldsymbol{T}_1 = \left[\begin{array}{ccc} d & a & 0 \\ a & d & a \\ 0 & a & d \end{array} \right], \quad \boldsymbol{I} = \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]$
lyche-numerical-linear-algebra-243 E09675 (10.12)	243	1.000	$\boldsymbol{T}_{\{2\}} = \boldsymbol{T}_{\{1\}} \otimes \boldsymbol{T}_{\{1\}} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}_{\{1\}} \otimes \boldsymbol{T}_{\{1\}},$ $\quad \boldsymbol{I} \in \mathbb{R}^{m \times m}, \quad \boldsymbol{T}_2 \in \mathbb{R}^{(m^2) \times (m^2)}.$	$\boldsymbol{T}_2 = \boldsymbol{T}_1 \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}_1, \quad \boldsymbol{T}_1, \boldsymbol{I} \in \mathbb{R}^{m \times m}, \quad \boldsymbol{T}_2 \in \mathbb{R}^{(m^2) \times (m^2)}.$	$\boldsymbol{T}_2 = \boldsymbol{T}_1 \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}_1, \quad \boldsymbol{T}_1, \boldsymbol{I} \in \mathbb{R}^{m \times m}, \quad \boldsymbol{T}_2 \in \mathbb{R}^{(m^2) \times (m^2)}.$
lyche-numerical-linear-algebra-243 E09678	243	1.000 N +0	$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D}) = \left[\begin{array}{ccc} \boldsymbol{A} b_{1,1} & \cdots & \boldsymbol{A} b_{1,t} \\ \vdots & & \vdots \\ \boldsymbol{A} b_{r,1} & \cdots & \boldsymbol{A} b_{r,t} \end{array} \right] \left[\begin{array}{ccc} \boldsymbol{C} d_{1,1} & \cdots & \boldsymbol{C} d_{1,s} \\ \vdots & & \vdots \\ \boldsymbol{C} d_{t,1} & \cdots & \boldsymbol{C} d_{t,s} \end{array} \right] .$	$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D}) = \left[\begin{array}{ccc} \boldsymbol{A} b_{1,1} & \cdots & \boldsymbol{A} b_{1,t} \\ \vdots & & \vdots \\ \boldsymbol{A} b_{r,1} & \cdots & \boldsymbol{A} b_{r,t} \end{array} \right] \left[\begin{array}{ccc} \boldsymbol{C} d_{1,1} & \cdots & \boldsymbol{C} d_{1,s} \\ \vdots & & \vdots \\ \boldsymbol{C} d_{t,1} & \cdots & \boldsymbol{C} d_{t,s} \end{array} \right] .$	$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D}) = \left[\begin{array}{ccc} \boldsymbol{A} b_{1,1} & \cdots & \boldsymbol{A} b_{1,t} \\ \vdots & & \vdots \\ \boldsymbol{A} b_{r,1} & \cdots & \boldsymbol{A} b_{r,t} \end{array} \right] \left[\begin{array}{ccc} \boldsymbol{C} d_{1,1} & \cdots & \boldsymbol{C} d_{1,s} \\ \vdots & & \vdots \\ \boldsymbol{C} d_{t,1} & \cdots & \boldsymbol{C} d_{t,s} \end{array} \right] .$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-245 EQ0681	245	■1.000 ●W +0	$\begin{aligned} & \left(\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B} \right) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F}) \\ & \Leftrightarrow \quad \left(\boldsymbol{A} \boldsymbol{V} \boldsymbol{I}_s^T + \boldsymbol{I}_r \boldsymbol{V} \boldsymbol{B}^T \right) = \boldsymbol{F} \quad \Leftrightarrow \quad \boldsymbol{A} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{B}^T = \boldsymbol{F}. \end{aligned}$	$\begin{aligned} & (\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F}) \\ & \Leftrightarrow \quad \left(\boldsymbol{A} \boldsymbol{V} \boldsymbol{I}_s^T + \boldsymbol{I}_r \boldsymbol{V} \boldsymbol{B}^T \right) = \boldsymbol{F} \quad \Leftrightarrow \quad \boldsymbol{A} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{B}^T = \boldsymbol{F}. \end{aligned}$	$\begin{aligned} & (\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F}) \\ & \Leftrightarrow \quad \left(\boldsymbol{A} \boldsymbol{V} \boldsymbol{I}_s^T + \boldsymbol{I}_r \boldsymbol{V} \boldsymbol{B}^T \right) = \boldsymbol{F} \quad \Leftrightarrow \quad \boldsymbol{A} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{B}^T = \boldsymbol{F}. \end{aligned}$
lyche-numerical-linear-algebra-246 EQ0683 (10.16)	246	■1.000 ●N +0	$\boldsymbol{s}_j = [\sin(j\pi h), \sin(2j\pi h), \dots, \sin(mj\pi h)]^T.$	$\boldsymbol{s}_j = [\sin(j\pi h), \sin(2j\pi h), \dots, \sin(mj\pi h)]^T.$	$\boldsymbol{s}_j = [\sin(j\pi h), \sin(2j\pi h), \dots, \sin(mj\pi h)]^T.$
lyche-numerical-linear-algebra-246 EQ0685 (10.18)	246	■1.000 ●N +0	$\boldsymbol{T}_2(\boldsymbol{s}_j \otimes \boldsymbol{s}_k) = (\lambda_j + \lambda_k)(\boldsymbol{s}_j \otimes \boldsymbol{s}_k) \quad j, k = 1, \dots, m,$	$\boldsymbol{T}_2(\boldsymbol{s}_j \otimes \boldsymbol{s}_k) = (\lambda_j + \lambda_k)(\boldsymbol{s}_j \otimes \boldsymbol{s}_k) \quad j, k = 1, \dots, m,$	$\boldsymbol{T}_2(\boldsymbol{s}_j \otimes \boldsymbol{s}_k) = (\lambda_j + \lambda_k)(\boldsymbol{s}_j \otimes \boldsymbol{s}_k) \quad j, k = 1, \dots, m,$
lyche-numerical-linear-algebra-246 EQ0688 (10.20)	246	■1.000 ●N +1	$\ \boldsymbol{A}\ _2 \ \boldsymbol{A}^{-1}\ _2 = \frac{\cos^2 w}{\sin^2 w}, \quad w := \frac{\pi}{2(m+1)}$	$\ \boldsymbol{A}\ _2 \ \boldsymbol{A}^{-1}\ _2 = \frac{\cos^2 w}{\sin^2 w}, \quad w := \frac{\pi}{2(m+1)}$	$\ \boldsymbol{A}\ _2 \ \boldsymbol{A}^{-1}\ _2 = \frac{\cos^2 w}{\sin^2 w}, \quad w := \frac{\pi}{2(m+1)}.$
lyche-numerical-linear-algebra-246 EQ0689	246	■1.000 ●N +0	$\lambda_{j,k} = 4 - 2\cos(2jw) - 2\cos(2kw) = 4\sin^2(jw) + 4\sin^2(kw), \quad j, k = 1, \dots, m.$	$\lambda_{j,k} = 4 - 2\cos(2jw) - 2\cos(2kw) = 4\sin^2(jw) + 4\sin^2(kw), \quad j, k = 1, \dots, m.$	$\lambda_{j,k} = 4 - 2\cos(2jw) - 2\cos(2kw) = 4\sin^2(jw) + 4\sin^2(kw), \quad j, k = 1, \dots, m.$
lyche-numerical-linear-algebra-246 EQ0690	246	■1.000 ●N +0	$\lambda_{\max} = 8\cos^2 w, \quad \lambda_{\min} = 8\sin^2 w.$	$\lambda_{\max} = 8\cos^2 w, \quad \lambda_{\min} = 8\sin^2 w.$	$\lambda_{\max} = 8\cos^2 w, \quad \lambda_{\min} = 8\sin^2 w.$
lyche-numerical-linear-algebra-249 EQ0696	249	■1.000 ●N +0	$(\boldsymbol{T} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}) \operatorname{vec}(\boldsymbol{V}) = h^2 \operatorname{vec}(\boldsymbol{F}), \quad (\boldsymbol{T} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}) \operatorname{vec}(\boldsymbol{U}) = h^2 \operatorname{vec}(\boldsymbol{V}).$	$(\boldsymbol{T} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}) \operatorname{vec}(\boldsymbol{V}) = h^2 \operatorname{vec}(\boldsymbol{F}), \quad (\boldsymbol{T} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}) \operatorname{vec}(\boldsymbol{U}) = h^2 \operatorname{vec}(\boldsymbol{V}).$	$(\boldsymbol{T} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}) \operatorname{vec}(\boldsymbol{V}) = h^2 \operatorname{vec}(\boldsymbol{F}), \quad (\boldsymbol{T} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}) \operatorname{vec}(\boldsymbol{U}) = h^2 \operatorname{vec}(\boldsymbol{V}).$
lyche-numerical-linear-algebra-249 EQ0697 (10.25)	249	■1.000 ●N -1	$\boldsymbol{T}^2 \boldsymbol{U} + 2\boldsymbol{T} \boldsymbol{U} \boldsymbol{T} + \boldsymbol{U} \boldsymbol{T}^2 = h^4 \boldsymbol{F} \quad \text{and} \quad \boldsymbol{A} \boldsymbol{x} = \boldsymbol{b},$	$\boldsymbol{T}^2 \boldsymbol{U} + 2\boldsymbol{T} \boldsymbol{U} \boldsymbol{T} + \boldsymbol{U} \boldsymbol{T}^2 = h^4 \boldsymbol{F} \quad \text{and} \quad \boldsymbol{A} \boldsymbol{x} = \boldsymbol{b},$	$\boldsymbol{T}^2 \boldsymbol{U} + 2\boldsymbol{T} \boldsymbol{U} \boldsymbol{T} + \boldsymbol{U} \boldsymbol{T}^2 = h^4 \boldsymbol{F} \quad \text{and} \quad \boldsymbol{A} \boldsymbol{x} = \boldsymbol{b},$
lyche-numerical-linear-algebra-252 EQ0700 (11.2)	252	■1.000 ●	$\boldsymbol{U}_1 = \boldsymbol{D}_1, \quad \boldsymbol{L}_k = \boldsymbol{A}_k \boldsymbol{U}_k^{-1}, \quad \boldsymbol{U}_{k+1} = \boldsymbol{D}_{k+1} - \boldsymbol{L}_k \boldsymbol{C}_k, \quad k = 1, 2, \dots, m-1.$	$\boldsymbol{U}_1 = \boldsymbol{D}_1, \quad \boldsymbol{L}_k = \boldsymbol{A}_k \boldsymbol{U}_k^{-1}, \quad \boldsymbol{U}_{k+1} = \boldsymbol{D}_{k+1} - \boldsymbol{L}_k \boldsymbol{C}_k, \quad k = 1, 2, \dots, m-1.$	$\boldsymbol{U}_1 = \boldsymbol{D}_1, \quad \boldsymbol{L}_k = \boldsymbol{A}_k \boldsymbol{U}_k^{-1}, \quad \boldsymbol{U}_{k+1} = \boldsymbol{D}_{k+1} - \boldsymbol{L}_k \boldsymbol{C}_k, \quad k = 1, 2, \dots, m-1.$
lyche-numerical-linear-algebra-253 EQ0704 (11.4)	253	■1.000 ●N +0	$\boldsymbol{S} := [\boldsymbol{s}_1, \dots, \boldsymbol{s}_m] = [\sin(jk\pi h)]_{j,k=1}^m \in \mathbb{R}^{m \times m}, \quad \boldsymbol{D} = \operatorname{diag}(\lambda_1, \dots, \lambda_m).$	$\boldsymbol{S} := [\boldsymbol{s}_1, \dots, \boldsymbol{s}_m] = [\sin(jk\pi h)]_{j,k=1}^m \in \mathbb{R}^{m \times m}, \quad \boldsymbol{D} = \operatorname{diag}(\lambda_1, \dots, \lambda_m).$	$\boldsymbol{S} := [\boldsymbol{s}_1, \dots, \boldsymbol{s}_m] = [\sin(jk\pi h)]_{j,k=1}^m \in \mathbb{R}^{m \times m}, \quad \boldsymbol{D} = \operatorname{diag}(\lambda_1, \dots, \lambda_m).$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-253 EQ0705 (2h)	253	■ 1.000 ● N +0	$\boldsymbol{T} \boldsymbol{S} = \left[\boldsymbol{T} \boldsymbol{s}_1, \dots, \boldsymbol{T} \boldsymbol{s}_m \right] = \left[\lambda_1 \boldsymbol{s}_1, \dots, \lambda_m \boldsymbol{s}_m \right] = \boldsymbol{S} \boldsymbol{D}, \quad \boldsymbol{S}^2 = \boldsymbol{S}^T \boldsymbol{S} = \frac{1}{2h} \boldsymbol{I}.$	$\boldsymbol{T} \boldsymbol{S} = [\boldsymbol{T} \boldsymbol{s}_1, \dots, \boldsymbol{T} \boldsymbol{s}_m] = [\lambda_1 \boldsymbol{s}_1, \dots, \lambda_m \boldsymbol{s}_m] = \boldsymbol{S} \boldsymbol{D}, \quad \boldsymbol{S}^2 = \boldsymbol{S}^T \boldsymbol{S} = \frac{1}{2h} \boldsymbol{I}.$	
lyche-numerical-linear-algebra-253 EQ0708	253	■ 1.000 ● N +0	$x_{j,k} = 4h^4 g_{j,k} / (\lambda_j + \lambda_k) = h^4 g_{j,k} / (\sigma_j + \sigma_k), \quad \sigma_j := \lambda_j / 4 = \sin^2(j\pi h / 2).$	$x_{j,k} = 4h^4 g_{j,k} / (\lambda_j + \lambda_k) = h^4 g_{j,k} / (\sigma_j + \sigma_k), \quad \sigma_j := \lambda_j / 4 = \sin^2(j\pi h / 2).$	
lyche-numerical-linear-algebra-255 EQ0710	255	■ 1.000 ● N +0	$w_j = \sum_{k=1}^m \sin\left(\frac{jk\pi}{m+1}\right) v_k, \quad j = 1, \dots, m$	$w_j = \sum_{k=1}^m \sin\left(\frac{jk\pi}{m+1}\right) v_k, \quad j = 1, \dots, m$	
lyche-numerical-linear-algebra-255 EQ0711 (11.5)	255	■ 1.000 ● K +0	$\omega_N = \exp^{-2\pi i / N} = \cos(2\pi / N) - i \sin(2\pi / N),$	$\omega_N = \exp^{-2\pi i / N} = \cos(2\pi / N) - i \sin(2\pi / N),$	
lyche-numerical-linear-algebra-256 EQ0712	256	■ 1.000 ● N +0	$z_{j+1} = \sum_{k=0}^{N-1} \omega_N^{jk} y_{k+1}, \quad j = 0, \dots, N-1,$	$z_{j+1} = \sum_{k=0}^{N-1} \omega_N^{jk} y_{k+1}, \quad j = 0, \dots, N-1,$	
lyche-numerical-linear-algebra-256 EQ0713	256	■ 1.000 ● K +0	$\omega_4 = \exp^{-2\pi i / 4} = \cos(\pi / 2) - i \sin(\pi / 2) = -i$	$\omega_4 = \exp^{-2\pi i / 4} = \cos(\pi / 2) - i \sin(\pi / 2) = -i$	
lyche-numerical-linear-algebra-256 EQ0715	256	■ 1.000 ● K +0	$\sin w = \frac{e^{iw} - e^{-iw}}{2i}.$	$\sin w = \frac{e^{iw} - e^{-iw}}{2i}.$	
lyche-numerical-linear-algebra-256 EQ0716	256	■ 1.000 ● N +0	$\boldsymbol{z}^T = [0, \boldsymbol{x}^T, 0, -\boldsymbol{x}_B^T] \in \mathbb{R}^{2m+2}, \quad \boldsymbol{x}_B^T := [x_m, \dots, x_2, x_1].$	$\boldsymbol{z}^T = [0, \boldsymbol{x}^T, 0, -\boldsymbol{x}_B^T] \in \mathbb{R}^{2m+2}, \quad \boldsymbol{x}_B^T := [x_m, \dots, x_2, x_1].$	
lyche-numerical-linear-algebra-256 EQ0718	256	■ 1.000 ● N +0	$\omega^{jk} = e^{-\pi i j k / (m+1)}, \quad \omega^{(2m+2-j)k} = e^{-2\pi i} e^{\pi i j k / (m+1)} = e^{\pi i j k / (m+1)}.$	$\omega^{jk} = e^{-\pi i j k / (m+1)}, \quad \omega^{(2m+2-j)k} = e^{-2\pi i} e^{\pi i j k / (m+1)} = e^{\pi i j k / (m+1)}.$	
lyche-numerical-linear-algebra-257 EQ0720	257	■ 1.000 ● K +0	$\boldsymbol{P}_N = [\boldsymbol{e}_1, \boldsymbol{e}_3, \dots, \boldsymbol{e}_{N-1}, \boldsymbol{e}_2, \boldsymbol{e}_4, \dots, \boldsymbol{e}_N],$	$\boldsymbol{P}_N = [\boldsymbol{e}_1, \boldsymbol{e}_3, \dots, \boldsymbol{e}_{N-1}, \boldsymbol{e}_2, \boldsymbol{e}_4, \dots, \boldsymbol{e}_N],$	
lyche-numerical-linear-algebra-258 EQ0724	258	■ 1.000 ● K +0	$\boldsymbol{F}_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad \boldsymbol{D}_2 \boldsymbol{F}_2 = \begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix},$	$\boldsymbol{F}_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad \boldsymbol{D}_2 \boldsymbol{F}_2 = \begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix},$	
lyche-numerical-linear-algebra-258 EQ0727 (11.8)	258	■ 1.000 ●	$\boldsymbol{D}_m = \text{diag}(1, \omega_N, \omega_N^2, \dots, \omega_N^{m-1}).$	$\boldsymbol{D}_m = \text{diag}(1, \omega_N, \omega_N^2, \dots, \omega_N^{m-1}).$	
lyche-numerical-linear-algebra-258 EQ0728	258	■ 1.000 ● W +0	$\omega_m^m = 1, \omega_{2m}^{2k} = \omega_m^k, \omega_{2m}^m = -1, (\boldsymbol{F}_m)_{p,q} = \omega_m^{jk}, (\boldsymbol{D}_m \boldsymbol{F}_m)_{p,q} = \omega_{2m}^j \omega_m^{jk},$	$\omega_m^m = 1, \omega_{2m}^{2k} = \omega_m^k, \omega_{2m}^m = -1, (\boldsymbol{F}_m)_{p,q} = \omega_m^{jk}, (\boldsymbol{D}_m \boldsymbol{F}_m)_{p,q} = \omega_{2m}^j \omega_m^{jk},$	
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-258 EQ0730	258	1.000 N +0	$\boldsymbol{w}_{1}^T=\left[y_{1}, y_{3}, \ldots, y_{2 m-1}, \ldots, y_{2 m-1}\right], \quad \boldsymbol{w}_{2}^T=\left[y_{2}, y_{4}, \ldots, y_{2 m}\right] .$	$\boldsymbol{w}_1^T = [y_1, y_3, \dots, y_{2m-1}], \quad \boldsymbol{w}_2^T = [y_2, y_4, \dots, y_{2m}].$	
lyche-numerical-linear-algebra-259 EQ0732	259	1.000 N +0	$\boldsymbol{q}_{1}=\boldsymbol{F}_{m} \boldsymbol{w}_{1}, \quad \text { and } \quad \boldsymbol{q}_{2}=\boldsymbol{D}_{m}\left(\boldsymbol{F}_{m} \boldsymbol{w}_{2}\right) .$	$\boldsymbol{q}_1 = \boldsymbol{F}_m \boldsymbol{w}_1, \quad \text{and} \quad \boldsymbol{q}_2 = \boldsymbol{D}_m(\boldsymbol{F}_m \boldsymbol{w}_2).$	
lyche-numerical-linear-algebra-260 EQ0733	260	1.000 N +0	$\frac{8 N^2}{5 N \log _2 N} \approx 84000 .$	$\frac{8 N^2}{5 N \log _2 N} \approx 84000.$	
lyche-numerical-linear-algebra-260 EQ0734	260	1.000 N +0	$8 \gamma m(m+1) \log _2(2 m+2) \approx 8 \gamma m^2 \log _2 m=4 \gamma n \log _2 n,$	$8 \gamma m(m+1) \log _2(2 m+2) \approx 8 \gamma m^2 \log _2 m=4 \gamma n \log _2 n,$	
lyche-numerical-linear-algebra-261 EQ0735	261	1.000 W +1	$v_{i j}=\frac{1}{(m+1)^4} \sum_{p=1}^m \sum_{r=1}^m \sum_{k=1}^m \sum_{l=1}^m \frac{\sin \left(\frac{i p \pi}{m+1}\right) \sin \left(\frac{j r \pi}{m+1}\right) \sin \left(\frac{k p \pi}{m+1}\right) \sin \left(\frac{l r \pi}{m+1}\right)}{\left[\sin \left(\frac{p \pi}{2(m+1)}\right)\right]^2+\left[\sin \left(\frac{r \pi}{2(m+1)}\right)\right]^2} f_{k, l} .$	$v_{ij} = \frac{1}{(m+1)^4} \sum_{p=1}^m \sum_{r=1}^m \sum_{k=1}^m \sum_{l=1}^m \frac{\sin\left(\frac{ip\pi}{m+1}\right) \sin\left(\frac{jr\pi}{m+1}\right) \sin\left(\frac{kp\pi}{m+1}\right) \sin\left(\frac{lr\pi}{m+1}\right)}{\left[\sin\left(\frac{p\pi}{2(m+1)}\right)\right]^2 + \left[\sin\left(\frac{r\pi}{2(m+1)}\right)\right]^2} f_{k,l}.$	
lyche-numerical-linear-algebra-261 EQ0737 (11.9)	261	1.000 N +0	$\left(\boldsymbol{T}+\lambda_j \boldsymbol{I}\right) \boldsymbol{x}_j=\boldsymbol{c}_j \quad j=1, \ldots, m,$	$(\boldsymbol{T}+\lambda_j \boldsymbol{I}) \boldsymbol{x}_j = \boldsymbol{c}_j \quad j = 1, \dots, m,$	
lyche-numerical-linear-algebra-261 EQ0738	261	1.000 N +0	$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}-\frac{1}{6} \boldsymbol{T} \boldsymbol{V} \boldsymbol{T}=h^2 \mu \boldsymbol{F},$	$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}-\frac{1}{6} \boldsymbol{T} \boldsymbol{V} \boldsymbol{T}=h^2 \mu \boldsymbol{F},$	
lyche-numerical-linear-algebra-262 EQ0741	262	1.000 K +0	$\boldsymbol{T}^2 \boldsymbol{U}+2 \boldsymbol{T} \boldsymbol{U} \boldsymbol{T}+\boldsymbol{U} \boldsymbol{T}^2=h^4 \boldsymbol{F} .$	$\boldsymbol{T}^2 \boldsymbol{U}+2 \boldsymbol{T} \boldsymbol{U} \boldsymbol{T}+\boldsymbol{U} \boldsymbol{T}^2=h^4 \boldsymbol{F}.$	
lyche-numerical-linear-algebra-262 EQ0742	262	1.000 N +0	$\boldsymbol{D}^2 \boldsymbol{X}+2 \boldsymbol{D} \boldsymbol{X} \boldsymbol{D}+\boldsymbol{X} \boldsymbol{D}^2=4 h^6 \boldsymbol{G}, \text { where } \boldsymbol{G}=\boldsymbol{S} \boldsymbol{F} \boldsymbol{S},$	$\boldsymbol{D}^2 \boldsymbol{X}+2 \boldsymbol{D} \boldsymbol{X} \boldsymbol{D}+\boldsymbol{X} \boldsymbol{D}^2=4 h^6 \boldsymbol{G}, \text{ where } \boldsymbol{G} = \boldsymbol{S} \boldsymbol{F} \boldsymbol{S},$	
lyche-numerical-linear-algebra-262 EQ0743	262	1.000 W +0	$x_{j, k}=\frac{h^6 g_{j, k}}{4\left(\sigma_j+\sigma_k\right)^2}, \text { where } \sigma_j=\sin ^2((j \pi h) / 2) \text { for } j, k=1,2, \ldots, m .$	$x_{j,k} = \frac{h^6 g_{j,k}}{4(\sigma_j + \sigma_k)^2}, \text{ where } \sigma_j = \sin^2((j\pi h)/2) \text{ for } j, k = 1, 2, \dots, m.$	
lyche-numerical-linear-algebra-266 EQ0744 (12.1)	266	1.000 W +0	$\boldsymbol{x}^{(i)}=\left(-\sum_{j=1}^{i-1} a_{i j} \boldsymbol{x}(j)-\sum_{j=i+1}^n a_{i j} \boldsymbol{x}(j)+b_i\right) / a_{i i}, \quad i=1,2, \ldots, n .$	$\boldsymbol{x}^{(i)} = \left(-\sum_{j=1}^{i-1} a_{ij} \boldsymbol{x}(j) - \sum_{j=i+1}^n a_{ij} \boldsymbol{x}(j) + b_i\right) / a_{ii}, \quad i = 1, 2, \dots, n.$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-E00745 (12.2)	266	1.000	$\boldsymbol{x}_{k+1}(i)=\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_k(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii},\text{ for }i=1,2,\ldots,n.$	$\boldsymbol{x}_{k+1}(i)=\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_k(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii},\text{ for }i=1,2,\ldots,n.$	$\boldsymbol{x}_{k+1}(i)=\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_k(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii},\text{ for }i=1,2,\ldots,n.$
lyche-numerical-linear-algebra-E00746 (12.3)	266	1.000 W +0	$\boldsymbol{x}_{k+1}(i)=\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii},\text{ for }i=1,2,\ldots,n.$	$\boldsymbol{x}_{k+1}(i)=\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii},\text{ for }i=1,2,\ldots,n.$	$\boldsymbol{x}_{k+1}(i)=\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii},\text{ for }i=1,2,\ldots,n.$
lyche-numerical-linear-algebra-E00747 (12.4)	266	1.000 N +0	$\boldsymbol{x}_{k+1}(i)=\omega\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii}+(1-\omega)\boldsymbol{x}_k(i).$	$\boldsymbol{x}_{k+1}(i)=\omega\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii}+(1-\omega)\boldsymbol{x}_k(i).$	$\boldsymbol{x}_{k+1}(i)=\omega\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii}+(1-\omega)\boldsymbol{x}_k(i).$
lyche-numerical-linear-algebra-E00748 (12.5)	267	1.000 N +0	$\boldsymbol{x}_{k+1}(i)=\omega\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1/2}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_{k+1}(j)+b_i\right)/a_{ii}+(1-\omega)\boldsymbol{x}_{k+1/2}(i)$	$\boldsymbol{x}_{k+1}(i)=\omega\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1/2}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_{k+1}(j)+b_i\right)/a_{ii}+(1-\omega)\boldsymbol{x}_{k+1/2}(i)$	$\boldsymbol{x}_{k+1}(i)=\omega\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1/2}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_{k+1}(j)+b_i\right)/a_{ii}+(1-\omega)\boldsymbol{x}_{k+1/2}(i)$
lyche-numerical-linear-algebra-E00749	267	1.000 N +0	$4\boldsymbol{v}(i,j)-\boldsymbol{v}(i-1,j)-\boldsymbol{v}(i+1,j)-\boldsymbol{v}(i,j-1)-\boldsymbol{v}(i,j+1)=h^2f_{i,j},\quad i,j=1,\ldots,m,$	$4\boldsymbol{v}(i,j)-\boldsymbol{v}(i-1,j)-\boldsymbol{v}(i+1,j)-\boldsymbol{v}(i,j-1)-\boldsymbol{v}(i,j+1)=h^2f_{i,j},\quad i,j=1,\ldots,m,$	$4\boldsymbol{v}(i,j)-\boldsymbol{v}(i-1,j)-\boldsymbol{v}(i+1,j)-\boldsymbol{v}(i,j-1)-\boldsymbol{v}(i,j+1)=h^2f_{i,j},\quad i,j=1,\ldots,m,$
lyche-numerical-linear-algebra-E00750 (12.6)	267	1.000 N +0	$\boldsymbol{v}(i,j)=\left(\boldsymbol{v}(i-1,j)+\boldsymbol{v}(i+1,j)+\boldsymbol{v}(i,j-1)+\boldsymbol{v}(i,j+1)+e_{i,j}\right)/4,$	$\boldsymbol{v}(i,j)=\left(\boldsymbol{v}(i-1,j)+\boldsymbol{v}(i+1,j)+\boldsymbol{v}(i,j-1)+\boldsymbol{v}(i,j+1)+e_{i,j}\right)/4,$	$\boldsymbol{v}(i,j)=\left(\boldsymbol{v}(i-1,j)+\boldsymbol{v}(i+1,j)+\boldsymbol{v}(i,j-1)+\boldsymbol{v}(i,j+1)+e_{i,j}\right)/4,$
lyche-numerical-linear-algebra-E00755 (12.10)	270	1.000 N +0	$\boldsymbol{x}_{k+1}:=\boldsymbol{G}\boldsymbol{x}_k+\boldsymbol{c},\quad \boldsymbol{G}=\boldsymbol{I}-\boldsymbol{M}^{-1}\boldsymbol{A},\quad \boldsymbol{c}=\boldsymbol{M}^{-1}\boldsymbol{b}.$	$\boldsymbol{x}_{k+1}:=\boldsymbol{G}\boldsymbol{x}_k+\boldsymbol{c},\quad \boldsymbol{G}=\boldsymbol{I}-\boldsymbol{M}^{-1}\boldsymbol{A},\quad \boldsymbol{c}=\boldsymbol{M}^{-1}\boldsymbol{b}.$	$\boldsymbol{x}_{k+1}:=\boldsymbol{G}\boldsymbol{x}_k+\boldsymbol{c},\quad \boldsymbol{G}=\boldsymbol{I}-\boldsymbol{M}^{-1}\boldsymbol{A},\quad \boldsymbol{c}=\boldsymbol{M}^{-1}\boldsymbol{b}.$
lyche-numerical-linear-algebra-E00756	270	1.000 W +0	$\boldsymbol{x}=\lim_{k\rightarrow\infty}\boldsymbol{x}_k=\lim_{k\rightarrow\infty}(\boldsymbol{G}\boldsymbol{x}_k+\boldsymbol{c})=\boldsymbol{G}\lim_{k\rightarrow\infty}\boldsymbol{x}_k+\boldsymbol{c}=\boldsymbol{G}\boldsymbol{x}+\boldsymbol{c}.$	$\boldsymbol{x}=\lim_{k\rightarrow\infty}\boldsymbol{x}_{k+1}=\lim_{k\rightarrow\infty}(\boldsymbol{G}\boldsymbol{x}_k+\boldsymbol{c})=\boldsymbol{G}\lim_{k\rightarrow\infty}\boldsymbol{x}_k+\boldsymbol{c}=\boldsymbol{G}\boldsymbol{x}+\boldsymbol{c}.$	$\boldsymbol{x}=\lim_{k\rightarrow\infty}\boldsymbol{x}_{k+1}=\lim_{k\rightarrow\infty}(\boldsymbol{G}\boldsymbol{x}_k+\boldsymbol{c})=\boldsymbol{G}\lim_{k\rightarrow\infty}\boldsymbol{x}_k+\boldsymbol{c}=\boldsymbol{G}\boldsymbol{x}+\boldsymbol{c}.$
lyche-numerical-linear-algebra-E00758 (12.12)	271	1.000 N +0	$\boldsymbol{M}_J=\boldsymbol{D},\quad \boldsymbol{M}_1=\boldsymbol{D}-\boldsymbol{A}_L,\quad \boldsymbol{M}_\omega=\omega^{-1}\boldsymbol{D}-\boldsymbol{A}_L.$	$\boldsymbol{M}_J=\boldsymbol{D},\quad \boldsymbol{M}_1=\boldsymbol{D}-\boldsymbol{A}_L,\quad \boldsymbol{M}_\omega=\omega^{-1}\boldsymbol{D}-\boldsymbol{A}_L.$	$\boldsymbol{M}_J=\boldsymbol{D},\quad \boldsymbol{M}_1=\boldsymbol{D}-\boldsymbol{A}_L,\quad \boldsymbol{M}_\omega=\omega^{-1}\boldsymbol{D}-\boldsymbol{A}_L.$
lyche-numerical-linear-algebra-E00759 (12.13)	271	1.000 N +0	$\boldsymbol{D}\boldsymbol{x}_{k+1}=\omega(\boldsymbol{A}_L\boldsymbol{x}_{k+1}+\boldsymbol{A}_R\boldsymbol{x}_k+\boldsymbol{b})+(1-\omega)\boldsymbol{D}\boldsymbol{x}_k.$	$\boldsymbol{D}\boldsymbol{x}_{k+1}=\omega(\boldsymbol{A}_L\boldsymbol{x}_{k+1}+\boldsymbol{A}_R\boldsymbol{x}_k+\boldsymbol{b})+(1-\omega)\boldsymbol{D}\boldsymbol{x}_k.$	$\boldsymbol{D}\boldsymbol{x}_{k+1}=\omega(\boldsymbol{A}_L\boldsymbol{x}_{k+1}+\boldsymbol{A}_R\boldsymbol{x}_k+\boldsymbol{b})+(1-\omega)\boldsymbol{D}\boldsymbol{x}_k.$
lyche-numerical-linear-algebra-E00760	271	1.000 N +0	$\begin{bmatrix}2&-1\\-1&2\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix}=\begin{bmatrix}1\\1\end{bmatrix}$	$\begin{bmatrix}2&-1\\-1&2\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix}=\begin{bmatrix}1\\1\end{bmatrix}$	$\begin{bmatrix}2&-1\\-1&2\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix}=\begin{bmatrix}1\\1\end{bmatrix}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0761	271	■ 1.000 ●	$\boldsymbol{A}_{\boldsymbol{L}}=\left[\begin{array}{ll} 0 & 0 \\ 1 & 0 \end{array}\right], \quad \boldsymbol{D}=\left[\begin{array}{ll} 2 & 0 \\ 0 & 2 \end{array}\right], \quad \boldsymbol{A}_{\boldsymbol{R}}=\left[\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array}\right],$	$\boldsymbol{A}_L = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad \boldsymbol{D} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad \boldsymbol{A}_R = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix},$	$\boldsymbol{A}_L = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad \boldsymbol{D} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad \boldsymbol{A}_R = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix},$
lyche-numerical-linear-algebra EQ0762	271	■ 1.000 ● K +0	$\boldsymbol{M}_{\boldsymbol{J}}=\boldsymbol{D}=\left[\begin{array}{ll} 2 & 0 \\ 0 & 2 \end{array}\right], \quad \boldsymbol{M}_{\boldsymbol{\omega}}=\omega^{-1}\boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}}=\left[\begin{array}{cc} 2\omega^{-1} & 0 \\ -1 & 2\omega^{-1} \end{array}\right].$	$\boldsymbol{M}_J = \boldsymbol{D} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad \boldsymbol{M}_\omega = \omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L = \begin{bmatrix} 2\omega^{-1} & 0 \\ -1 & 2\omega^{-1} \end{bmatrix}.$	$\boldsymbol{M}_J = \boldsymbol{D} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad \boldsymbol{M}_\omega = \omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L = \begin{bmatrix} 2\omega^{-1} & 0 \\ -1 & 2\omega^{-1} \end{bmatrix}.$
lyche-numerical-linear-algebra EQ0764 (12.15)	271	■ 1.000 ● K +0	$\boldsymbol{G}_{\boldsymbol{J}}=\boldsymbol{I}-\boldsymbol{D}^{-1}\boldsymbol{A}=\left[\begin{array}{cc} 0 & 1/2 \\ 1/2 & 0 \end{array}\right], \quad \boldsymbol{G}_1=\left[\begin{array}{cc} 0 & 1/2 \\ 0 & 1/4 \end{array}\right].$	$\boldsymbol{G}_J = \boldsymbol{I} - \boldsymbol{D}^{-1}\boldsymbol{A} = \begin{bmatrix} 0 & 1/2 \\ 1/2 & 0 \end{bmatrix}, \quad \boldsymbol{G}_1 = \begin{bmatrix} 0 & 1/2 \\ 0 & 1/4 \end{bmatrix}.$	$\boldsymbol{G}_J = \boldsymbol{I} - \boldsymbol{D}^{-1}\boldsymbol{A} = \begin{bmatrix} 0 & 1/2 \\ 1/2 & 0 \end{bmatrix}, \quad \boldsymbol{G}_1 = \begin{bmatrix} 0 & 1/2 \\ 0 & 1/4 \end{bmatrix}.$
lyche-numerical-linear-algebra EQ0771 (12.18)	273	■ 1.000 ● N +0	$\boldsymbol{x}_{k+1}=\boldsymbol{x}_{\boldsymbol{k}}+\alpha\left(\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_{\boldsymbol{k}}\right).$	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha(\boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}_k).$	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha(\boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}_k).$
lyche-numerical-linear-algebra EQ0773	274	■ 1.000 ● N -1	$\rho_{\alpha}:=\rho(\boldsymbol{I}-\alpha\boldsymbol{A}):=\max _j\left 1-\alpha \lambda_j\right =\left\{\begin{array}{ll} 1-\alpha \lambda_1, & \text { if } \alpha \leq 0 \\ 1-\alpha \lambda_n, & \text { if } 0<\alpha \leq \alpha_o \\ \alpha \lambda_1-1, & \text { if } \alpha>\alpha_o, \end{array}\right.$	$\rho_\alpha := \rho(\boldsymbol{I} - \alpha\boldsymbol{A}) := \max_j 1 - \alpha\lambda_j = \begin{cases} 1 - \alpha\lambda_1, & \text{if } \alpha \leq 0 \\ 1 - \alpha\lambda_n, & \text{if } 0 < \alpha \leq \alpha_o \\ \alpha\lambda_1 - 1, & \text{if } \alpha > \alpha_o, \end{cases}$	$\rho_\alpha := \rho(\boldsymbol{I} - \alpha\boldsymbol{A}) := \max_j 1 - \alpha\lambda_j = \begin{cases} 1 - \alpha\lambda_1, & \text{if } \alpha \leq 0 \\ 1 - \alpha\lambda_n, & \text{if } 0 < \alpha \leq \alpha_o \\ \alpha\lambda_1 - 1, & \text{if } \alpha > \alpha_o, \end{cases}$
lyche-numerical-linear-algebra EQ0774	274	■ 1.000 ● N +1	$\rho_{\alpha_o}=\alpha_o \lambda_1-1=\frac{\lambda_1-\lambda_n}{\lambda_1+\lambda_n}=\frac{\kappa-1}{\kappa+1}<1.$	$\rho_{\alpha_o} = \alpha_o\lambda_1 - 1 = \frac{\lambda_1 - \lambda_n}{\lambda_1 + \lambda_n} = \frac{\kappa - 1}{\kappa + 1} < 1.$	$\rho_{\alpha_o} = \alpha_o\lambda_1 - 1 = \frac{\lambda_1 - \lambda_n}{\lambda_1 + \lambda_n} = \frac{\kappa - 1}{\kappa + 1} < 1.$
lyche-numerical-linear-algebra EQ0776 (12.21)	275	■ 1.000 ● K +0	$\boldsymbol{G}_{\boldsymbol{\omega}}=(\boldsymbol{I}-\omega\boldsymbol{L})^{-1}(\omega\boldsymbol{R}+(1-\omega)\boldsymbol{I}).$	$\boldsymbol{G}_\omega = (\boldsymbol{I} - \omega\boldsymbol{L})^{-1}(\omega\boldsymbol{R} + (1 - \omega)\boldsymbol{I}).$	$\boldsymbol{G}_\omega = (\boldsymbol{I} - \omega\boldsymbol{L})^{-1}(\omega\boldsymbol{R} + (1 - \omega)\boldsymbol{I}).$
lyche-numerical-linear-algebra EQ0777 (12.22)	275	■ 1.000 ● K +0	$\omega^{-1}(2-\omega)\left 1-\lambda\right ^2\boldsymbol{x}^* \boldsymbol{D} \boldsymbol{x}=\left(1-\left \lambda\right ^2\right) \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x},$	$\omega^{-1}(2 - \omega) 1 - \lambda ^2\boldsymbol{x}^*\boldsymbol{D}\boldsymbol{x} = (1 - \lambda ^2)\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x},$	$\omega^{-1}(2 - \omega) 1 - \lambda ^2\boldsymbol{x}^*\boldsymbol{D}\boldsymbol{x} = (1 - \lambda ^2)\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x},$
lyche-numerical-linear-algebra EQ0778	275	■ 1.000 ● N +0	$\boldsymbol{G}_{\omega} \boldsymbol{x}=(\boldsymbol{I}-\boldsymbol{M}_{\omega}^{-1} \boldsymbol{A}) \boldsymbol{x}=\boldsymbol{x}-\left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}}\right)^{-1} \boldsymbol{A} \boldsymbol{x}$	$\boldsymbol{G}_\omega\boldsymbol{x} = (\boldsymbol{I} - \boldsymbol{M}_\omega^{-1}\boldsymbol{A})\boldsymbol{x} = \boldsymbol{x} - (\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L)^{-1}\boldsymbol{A}\boldsymbol{x}$	$\boldsymbol{G}_\omega\boldsymbol{x} = (\boldsymbol{I} - \boldsymbol{M}_\omega^{-1}\boldsymbol{A})\boldsymbol{x} = \boldsymbol{x} - (\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L)^{-1}\boldsymbol{A}\boldsymbol{x}$
lyche-numerical-linear-algebra EQ0779 (12.23)	276	■ 1.000 ● N +0	$\boldsymbol{A} \boldsymbol{x}=\left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}}\right) \boldsymbol{y}, \quad \boldsymbol{y}:=(1-\lambda) \boldsymbol{x}.$	$\boldsymbol{A}\boldsymbol{x} = (\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L)\boldsymbol{y}, \quad \boldsymbol{y} := (1 - \lambda)\boldsymbol{x}.$	$\boldsymbol{A}\boldsymbol{x} = (\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L)\boldsymbol{y}, \quad \boldsymbol{y} := (1 - \lambda)\boldsymbol{x}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-276 EQ0780 (12.24)	276	■ 1.000 ● N +0	$\boldsymbol{E} \boldsymbol{y}=\lambda \boldsymbol{A} \boldsymbol{x}, \quad \boldsymbol{E}:=\omega^{-1} \boldsymbol{D}+\boldsymbol{A}_R-\boldsymbol{D}=\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_L-\boldsymbol{A}.$	$\boldsymbol{E} \boldsymbol{y}=\lambda \boldsymbol{A} \boldsymbol{x}, \quad \boldsymbol{E}:=\omega^{-1} \boldsymbol{D}+\boldsymbol{A}_R-\boldsymbol{D}=\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_L-\boldsymbol{A}.$	
lyche-numerical-linear-algebra-276 EQ0781	276	■ 1.000 ●	$\boldsymbol{E} \boldsymbol{y}=\left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_L-\boldsymbol{A}\right) \boldsymbol{y}=\boldsymbol{A} \boldsymbol{x}-\boldsymbol{A} \boldsymbol{y}=\boldsymbol{A} \boldsymbol{x}-\left(1-\lambda\right) \boldsymbol{A} \boldsymbol{x}=\lambda \boldsymbol{A} \boldsymbol{x}.$	$\boldsymbol{E} \boldsymbol{y}=\left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_L-\boldsymbol{A}\right) \boldsymbol{y}=\boldsymbol{A} \boldsymbol{x}-\boldsymbol{A} \boldsymbol{y}=\boldsymbol{A} \boldsymbol{x}-\left(1-\lambda\right) \boldsymbol{A} \boldsymbol{x}=\lambda \boldsymbol{A} \boldsymbol{x}.$	
lyche-numerical-linear-algebra-276 EQ0784	276	■ 1.000 ● N +0	$\lambda_{j, k}=4-2 \cos (j \pi h)-2 \cos (k \pi h), \quad j, k=1, \ldots, m, h=1 /(m+1) .$	$\lambda_{j, k}=4-2 \cos (j \pi h)-2 \cos (k \pi h), \quad j, k=1, \ldots, m, h=1 /(m+1) .$	
lyche-numerical-linear-algebra-276 EQ0785 (12.25)	276	■ 1.000 ● N +0	$\mu_{j, k}=1-\frac{1}{4} \lambda_{j, k}=\frac{1}{2} \cos (j \pi h)+\frac{1}{2} \cos (k \pi h), \quad j, k=1, \ldots, m .$	$\mu_{j, k}=1-\frac{1}{4} \lambda_{j, k}=\frac{1}{2} \cos (j \pi h)+\frac{1}{2} \cos (k \pi h), \quad j, k=1, \ldots, m .$	
lyche-numerical-linear-algebra-277 EQ0786 (12.26)	277	■ 1.000 ● N +0	$\left\ \boldsymbol{x}_k-\boldsymbol{x}\right\ _2 \leq\left\ \boldsymbol{G}_J\right\ _2^k\left\ \boldsymbol{x}_0-\boldsymbol{x}\right\ _2=\cos ^k(\pi h)\left\ \boldsymbol{x}_0-\boldsymbol{x}\right\ _2, \quad k=0,1,2, \ldots$	$\left\ \boldsymbol{x}_k-\boldsymbol{x}\right\ _2 \leq\left\ \boldsymbol{G}_J\right\ _2^k\left\ \boldsymbol{x}_0-\boldsymbol{x}\right\ _2=\cos ^k(\pi h)\left\ \boldsymbol{x}_0-\boldsymbol{x}\right\ _2, \quad k=0,1,2, \ldots$	
lyche-numerical-linear-algebra-277 EQ0788 (12.28)	277	■ 1.000 ● K +0	$\omega^{*}:=\frac{2}{1+\sqrt{1-\beta^2}} >1 .$	$\omega^{*}:=\frac{2}{1+\sqrt{1-\beta^2}} >1 .$	
lyche-numerical-linear-algebra-277 EQ0790 (12.30)	277	■ 1.000 ● W +2	$\omega^{*}=\frac{2}{1+\sin (\pi h)}, \quad \rho\left(\boldsymbol{G}_{\omega^{*}}\right)=\omega^{*}-1=\frac{1-\sin (\pi h)}{1+\sin (\pi h)}, \quad h=\frac{1}{m+1} .$	$\omega^{*}=\frac{2}{1+\sin (\pi h)}, \quad \rho\left(\boldsymbol{G}_{\omega^{*}}\right)=\omega^{*}-1=\frac{1-\sin (\pi h)}{1+\sin (\pi h)}, \quad h=\frac{1}{m+1} .$	
lyche-numerical-linear-algebra-278 EQ0792 (12.31)	278	■ 1.000 ● K +0	$\tilde{k}:=\frac{s \log (10)}{\eta}$	$\tilde{k}:=\frac{s \log (10)}{\eta}$	
lyche-numerical-linear-algebra-279 EQ0794	279	■ 1.000 ● N +0	$\tilde{k}_n=\frac{2 \log (10) s}{\pi^2} n+O\left(n^{-1}\right)=O(n) .$	$\tilde{k}_n=\frac{2 \log (10) s}{\pi^2} n+O\left(n^{-1}\right)=O(n) .$	
lyche-numerical-linear-algebra-279 EQ0795	279	■ 1.000 ● N +0	$\tilde{k}_n=\frac{\log (10) s}{\pi^2} n+O\left(n^{-1}\right)=O(n) .$	$\tilde{k}_n=\frac{\log (10) s}{\pi^2} n+O\left(n^{-1}\right)=O(n) .$	
lyche-numerical-linear-algebra-279 EQ0796	279	■ 1.000 ● N +0	$\tilde{k}_n=\frac{\log (10) s}{2 \pi} \sqrt{n}+O\left(n^{-1 / 2}\right)=O(\sqrt{n}) .$	$\tilde{k}_n=\frac{\log (10) s}{2 \pi} \sqrt{n}+O\left(n^{-1 / 2}\right)=O(\sqrt{n}) .$	
lyche-numerical-linear-algebra-280 EQ0797 (12.32)	280	■ 1.000 ● K +0	$\left\ \boldsymbol{x}_k-\boldsymbol{x}_{k-1}\right\ \geq \frac{1-\left\ \boldsymbol{G}\right\ }{\left\ \boldsymbol{G}\right\ }\left\ \boldsymbol{x}_k-\boldsymbol{x}\right\ , \quad k \geq 1 .$	$\left\ \boldsymbol{x}_k-\boldsymbol{x}_{k-1}\right\ \geq \frac{1-\left\ \boldsymbol{G}\right\ }{\left\ \boldsymbol{G}\right\ }\left\ \boldsymbol{x}_k-\boldsymbol{x}\right\ , \quad k \geq 1 .$	
lyche-numerical-linear-algebra-280 EQ0799 (12.33)	280	■ 1.000 ● N +0	$\boldsymbol{x}_{k+1}=\boldsymbol{x}_k+\boldsymbol{M}^{-1} \boldsymbol{r}_k, \quad \boldsymbol{r}_k=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_k .$	$\boldsymbol{x}_{k+1}=\boldsymbol{x}_k+\boldsymbol{M}^{-1} \boldsymbol{r}_k, \quad \boldsymbol{r}_k=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_k .$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-281 EQ8888	281	■ 1.000 ● N +0	$\lim_{k\rightarrow\infty}\boldsymbol{A}^k=\mathbf{0}$ $\Longleftarrow\longrightarrow\rho(\boldsymbol{A})<1$,	$\lim_{k\rightarrow\infty}\boldsymbol{A}^k=\mathbf{0}\iff\rho(\boldsymbol{A})<1$,	$\lim_{k\rightarrow\infty}\boldsymbol{A}^k=\mathbf{0}\iff\rho(\boldsymbol{A})<1$,
lyche-numerical-linear-algebra-281 EQ8881	281	■ 1.000 ● N +0	$\boldsymbol{D}_t\boldsymbol{R}\boldsymbol{D}_t^{-1}=\left[\begin{array}{ccc}\lambda_1&t^{-1}r_{12}&t^{-2}r_{13}\\0&\lambda_2&t^{-1}r_{23}\\0&0&\lambda_3\end{array}\right]$.	$\boldsymbol{D}_t\boldsymbol{R}\boldsymbol{D}_t^{-1}=\left[\begin{array}{ccc}\lambda_1&t^{-1}r_{12}&t^{-2}r_{13}\\0&\lambda_2&t^{-1}r_{23}\\0&0&\lambda_3\end{array}\right]$.	$\boldsymbol{D}_t\boldsymbol{R}\boldsymbol{D}_t^{-1}=\left[\begin{array}{ccc}\lambda_1&t^{-1}r_{12}&t^{-2}r_{13}\\0&\lambda_2&t^{-1}r_{23}\\0&0&\lambda_3\end{array}\right]$.
lyche-numerical-linear-algebra-282 EQ8883 (12.34)	282	■ 1.000 ●	$\lim_{k\rightarrow\infty}\boldsymbol{A}^k\left \right =\rho(\boldsymbol{A})$.	$\lim_{k\rightarrow\infty}\ \boldsymbol{A}^k\ ^{1/k}=\rho(\boldsymbol{A})$.	$\lim_{k\rightarrow\infty}\ \boldsymbol{A}^k\ ^{1/k}=\rho(\boldsymbol{A})$.
lyche-numerical-linear-algebra-282 EQ8884	282	■ 1.000 ● N +0	$\left \right \boldsymbol{A}^k\right =\left \right (\rho(\boldsymbol{A})+\epsilon)^k\boldsymbol{B}^k\right =(\rho(\boldsymbol{A})+\epsilon)^k\left \right \boldsymbol{B}^k\right <(\rho(\boldsymbol{A})+\epsilon)^k$.	$\ \boldsymbol{A}^k\ =\ (\rho(\boldsymbol{A})+\epsilon)^k\boldsymbol{B}^k\ =(\rho(\boldsymbol{A})+\epsilon)^k\ \boldsymbol{B}^k\ <(\rho(\boldsymbol{A})+\epsilon)^k$.	$\ \boldsymbol{A}^k\ =\ (\rho(\boldsymbol{A})+\epsilon)^k\boldsymbol{B}^k\ =(\rho(\boldsymbol{A})+\epsilon)^k\ \boldsymbol{B}^k\ <(\rho(\boldsymbol{A})+\epsilon)^k$.
lyche-numerical-linear-algebra-282 EQ8885	282	■ 1.000 ● N +0	$\sum_{k=0}^{\infty}\boldsymbol{B}^k$	$\sum_{k=0}^{\infty}\boldsymbol{B}^k$	$\sum_{k=0}^{\infty}\boldsymbol{B}^k$
lyche-numerical-linear-algebra-283 EQ8887	283	■ 1.000 ● S +0	$\left \right \sum_{k=m+1}^l\boldsymbol{B}^k\right \leq\sum_{k=m+1}^l\left \right \boldsymbol{B}^k\right \leq\left \right \boldsymbol{B}^{m+1}\sum_{k=0}^{\infty}\boldsymbol{B}^k\right =\frac{\left \right \boldsymbol{B}^{m+1}\right }{1-\left \right \boldsymbol{B}\right }$.	$\ S_l-S_m\ =\left\ \sum_{k=m+1}^l\boldsymbol{B}^k\right\ \leq\sum_{k=m+1}^l\ \boldsymbol{B}\ ^k\leq\ \boldsymbol{B}\ ^{m+1}\sum_{k=0}^{\infty}\ \boldsymbol{B}\ ^k=\frac{\ \boldsymbol{B}\ ^{m+1}}{1-\ \boldsymbol{B}\ }$.	$\ S_l-S_m\ =\left\ \sum_{k=m+1}^l\boldsymbol{B}^k\right\ \leq\sum_{k=m+1}^l\ \boldsymbol{B}\ ^k\leq\ \boldsymbol{B}\ ^{m+1}\sum_{k=0}^{\infty}\ \boldsymbol{B}\ ^k=\frac{\ \boldsymbol{B}\ ^{m+1}}{1-\ \boldsymbol{B}\ }$.
lyche-numerical-linear-algebra-283 EQ8888 (12.36)	283	■ 1.000 ● N +0	$\left(\sum_{k=0}^m\boldsymbol{B}^k\right)(\boldsymbol{I}-\boldsymbol{B})=\boldsymbol{I}+\boldsymbol{B}+\cdots+\boldsymbol{B}^m-(\boldsymbol{B}+\cdots+\boldsymbol{B}^{m+1})=\boldsymbol{I}-\boldsymbol{B}^{m+1}$.	$\left(\sum_{k=0}^m\boldsymbol{B}^k\right)(\boldsymbol{I}-\boldsymbol{B})=\boldsymbol{I}+\boldsymbol{B}+\cdots+\boldsymbol{B}^m-(\boldsymbol{B}+\cdots+\boldsymbol{B}^{m+1})=\boldsymbol{I}-\boldsymbol{B}^{m+1}$.	$(\sum_{k=0}^m\boldsymbol{B}^k)(\boldsymbol{I}-\boldsymbol{B})=\boldsymbol{I}+\boldsymbol{B}+\cdots+\boldsymbol{B}^m-(\boldsymbol{B}+\cdots+\boldsymbol{B}^{m+1})=\boldsymbol{I}-\boldsymbol{B}^{m+1}$.
lyche-numerical-linear-algebra-283 EQ8889 (12.37)	283	■ 1.000 ● N +0	$\frac{1}{4}(v_{i-1,j}+v_{i,j-1}+v_{i+1,j}+v_{i,j+1})=\mu v_{i,j}$, $i,j=1,\dots,m$,	$\frac{1}{4}(v_{i-1,j}+v_{i,j-1}+v_{i+1,j}+v_{i,j+1})=\mu v_{i,j}$, $i,j=1,\dots,m$,	$\frac{1}{4}(v_{i-1,j}+v_{i,j-1}+v_{i+1,j}+v_{i,j+1})=\mu v_{i,j}$, $i,j=1,\dots,m$,
lyche-numerical-linear-algebra-283 EQ8810 (12.38)	283	■ 1.000 ● K +0	$(\omega\boldsymbol{R}+\lambda\omega\boldsymbol{L})\boldsymbol{w}=(\lambda+\omega-1)\boldsymbol{w}$.	$(\omega\boldsymbol{R}+\lambda\omega\boldsymbol{L})\boldsymbol{w}=(\lambda+\omega-1)\boldsymbol{w}$.	$(\omega\boldsymbol{R}+\lambda\omega\boldsymbol{L})\boldsymbol{w}=(\lambda+\omega-1)\boldsymbol{w}$.
lyche-numerical-linear-algebra-284 EQ8811 (12.39)	284	■ 1.000 ● N +1	$\frac{\omega}{4}(\lambda w_{i-1,j}+\lambda w_{i,j-1}+w_{i+1,j}+w_{i,j+1})=(\lambda+\omega-1)w_{i,j}$	$\frac{\omega}{4}(\lambda w_{i-1,j}+\lambda w_{i,j-1}+w_{i+1,j}+w_{i,j+1})=(\lambda+\omega-1)w_{i,j}$	$\frac{\omega}{4}(\lambda w_{i-1,j}+\lambda w_{i,j-1}+w_{i+1,j}+w_{i,j+1})=(\lambda+\omega-1)w_{i,j}$
lyche-numerical-linear-algebra-284 EQ8812 (12.40)	284	■ 1.000 ● K +0	$\mu:=\frac{\lambda+\omega-1}{\omega\lambda^{1/2}}$	$\mu:=\frac{\lambda+\omega-1}{\omega\lambda^{1/2}}$	$\mu:=\frac{\lambda+\omega-1}{\omega\lambda^{1/2}}$
lyche-numerical-linear-algebra-284 EQ8813 (12.41)	284	■ 1.000 ● K +0	$\mu\omega\lambda^{1/2}=\lambda+\omega-1$	$\mu\omega\lambda^{1/2}=\lambda+\omega-1$	$\mu\omega\lambda^{1/2}=\lambda+\omega-1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0814	284	■ 1.000 ● K +0	$\frac{\omega}{4}\left(v_{i-1,j}+v_{i,j-1}+v_{i+1,j}+v_{i,j+1}\right)=\lambda^{\text{-1 / 2}}(\lambda+\omega-1)v_{i,j}$.	$\frac{\omega}{4}\left(v_{i-1,j}+v_{i,j-1}+v_{i+1,j}+v_{i,j+1}\right)=\lambda^{-1/2}(\lambda+\omega-1)v_{i,j}.$	$\frac{\omega}{4}\left(v_{i-1,j}+v_{i,j-1}+v_{i+1,j}+v_{i,j+1}\right)=\lambda^{-1/2}(\lambda+\omega-1)v_{i,j}.$
lyche-numerical-linear-algebra- EQ0817	284	■ 1.000 ● N +1	$\frac{\omega}{4}\left(\lambda w_{i-1,j}+\lambda w_{i,j-1}+w_{i+1,j}+w_{i,j+1}\right)=\omega \mu \lambda^{1 / 2} w_{i,j}$	$\frac{\omega}{4}\left(\lambda w_{i-1,j}+\lambda w_{i,j-1}+w_{i+1,j}+w_{i,j+1}\right)=\omega \mu \lambda^{1/2} w_{i,j}$	$\frac{\omega}{4}\left(\lambda w_{i-1,j}+\lambda w_{i,j-1}+w_{i+1,j}+w_{i,j+1}\right)=\omega \mu \lambda^{1/2} w_{i,j},$
lyche-numerical-linear-algebra- EQ0818 (12.42)	285	■ 1.000 ● K +0	$\lambda(\mu):=\frac{1}{4}\left(\omega \mu \pm \sqrt{\left(\omega \mu\right)^2-4(\omega-1)}\right)^2.$	$\lambda(\mu):=\frac{1}{4}\left(\omega \mu \pm \sqrt{\left(\omega \mu\right)^2-4(\omega-1)}\right)^2.$	$\lambda(\mu):=\frac{1}{4}\left(\omega \mu \pm \sqrt{\left(\omega \mu\right)^2-4(\omega-1)}\right)^2.$
lyche-numerical-linear-algebra- EQ0821 (12.43)	285	■ 1.000 ● N +0	$\omega=\tilde{\omega}(\mu):=2 \frac{1-\sqrt{1-\mu^2}}{\mu^2}=\frac{2}{1+\sqrt{1-\mu^2}}.$	$\omega=\tilde{\omega}(\mu):=2 \frac{1-\sqrt{1-\mu^2}}{\mu^2}=\frac{2}{1+\sqrt{1-\mu^2}}.$	$\omega=\tilde{\omega}(\mu):=2 \frac{1-\sqrt{1-\mu^2}}{\mu^2}=\frac{2}{1+\sqrt{1-\mu^2}}.$
lyche-numerical-linear-algebra- EQ0822	285	■ 1.000 ●	$ \lambda(\mu) =\frac{1}{4}\left((\omega \mu)^2+4(\omega-1)-(\omega \mu)^2\right)=\omega-1, \quad \tilde{\omega}(\mu)<\omega<2.$	$ \lambda(\mu) =\frac{1}{4}\left((\omega \mu)^2+4(\omega-1)-(\omega \mu)^2\right)=\omega-1, \quad \tilde{\omega}(\mu)<\omega<2.$	$ \lambda(\mu) =\frac{1}{4}\left((\omega \mu)^2+4(\omega-1)-(\omega \mu)^2\right)=\omega-1, \quad \tilde{\omega}(\mu)<\omega<2.$
lyche-numerical-linear-algebra- EQ0823 (12.44)	285	■ 1.000 ● K +0	$\lambda(\mu)=\frac{1}{4}\left(\omega \mu+\sqrt{\left(\omega \mu\right)^2-4(\omega-1)}\right)^2, \quad 0<\omega \leq \tilde{\omega}(\mu).$	$\lambda(\mu)=\frac{1}{4}\left(\omega \mu+\sqrt{\left(\omega \mu\right)^2-4(\omega-1)}\right)^2, \quad 0<\omega \leq \tilde{\omega}(\mu).$	$\lambda(\mu)=\frac{1}{4}\left(\omega \mu+\sqrt{\left(\omega \mu\right)^2-4(\omega-1)}\right)^2, \quad 0<\omega \leq \tilde{\omega}(\mu).$
lyche-numerical-linear-algebra- EQ0824	285	■ 1.000 ● N +0	$\frac{d}{d \omega}\left(\omega \beta+\sqrt{\left(\omega \beta\right)^2-4(\omega-1)}\right)=\frac{\beta \sqrt{\left(\omega \beta\right)^2-4(\omega-1)}+\omega \beta^2-2}{\sqrt{\left(\omega \beta\right)^2-4(\omega-1)}}.$	$\frac{d}{d \omega}\left(\omega \beta+\sqrt{\left(\omega \beta\right)^2-4(\omega-1)}\right)=\frac{\beta \sqrt{\left(\omega \beta\right)^2-4(\omega-1)}+\omega \beta^2-2}{\sqrt{\left(\omega \beta\right)^2-4(\omega-1)}}.$	$\frac{d}{d \omega}\left(\omega \beta+\sqrt{\left(\omega \beta\right)^2-4(\omega-1)}\right)=\frac{\beta \sqrt{\left(\omega \beta\right)^2-4(\omega-1)}+\omega \beta^2-2}{\sqrt{\left(\omega \beta\right)^2-4(\omega-1)}}.$
lyche-numerical-linear-algebra- EQ0826	286	■ 1.000 ● N +0	$\rho\left(\boldsymbol{G}_1\right)=\left a_{21} a_{12} / a_{11} a_{22}\right .$	$\rho\left(\boldsymbol{G}_1\right)=\left a_{21} a_{12} / a_{11} a_{22}\right .$	$\rho\left(\boldsymbol{G}_1\right)=\left a_{21} a_{12} / a_{11} a_{22}\right .$
lyche-numerical-linear-algebra- EQ0829 (12.45)	287	■ 1.000 ● N +0	$\boldsymbol{M} \boldsymbol{x}_{k+1}=\boldsymbol{N} \boldsymbol{x}_k+\boldsymbol{b},$	$\boldsymbol{M} \boldsymbol{x}_{k+1}=\boldsymbol{N} \boldsymbol{x}_k+\boldsymbol{b},$	$\boldsymbol{M} \boldsymbol{x}_{k+1}=\boldsymbol{N} \boldsymbol{x}_k+\boldsymbol{b},$
lyche-numerical-linear-algebra- EQ0832 (12.48)	288	■ 1.000 ● N +0	$x(i)=b(i)-\sum_{j \neq i} a(i, j) x(j)$	$x(i)=b(i)-\sum_{j \neq i} a(i, j) x(j)$	$x(i)=b(i)-\sum_{j \neq i} a(i, j) x(j)$
lyche-numerical-linear-algebra- EQ0838 (13.4)	291	■ 1.000 ● W +0	$Q(x, y):=\frac{1}{2}\left[\begin{array}{ll} x & y \end{array}\right]\left[\begin{array}{rr} 2 & -1 \\ -1 & 2 \end{array}\right]\left[\begin{array}{l} x \\ y \end{array}\right]=x^2-x y+y^2$	$Q(x, y):=\frac{1}{2}\left[\begin{array}{ll} x & y \end{array}\right]\left[\begin{array}{rr} 2 & -1 \\ -1 & 2 \end{array}\right]\left[\begin{array}{l} x \\ y \end{array}\right]=x^2-x y+y^2$	$Q(x, y):=\frac{1}{2}\left[\begin{array}{ll} x & y \end{array}\right]\left[\begin{array}{rr} 2 & -1 \\ -1 & 2 \end{array}\right]\left[\begin{array}{l} x \\ y \end{array}\right]=x^2-x y+y^2$
lyche-numerical-linear-algebra- EQ0839 (13.5)	291	■ 1.000 ● N +1	$Q\left(\boldsymbol{y}+\varepsilon \boldsymbol{h}\right)=Q\left(\boldsymbol{y}\right)-\varepsilon \boldsymbol{h}^T \boldsymbol{r}(\boldsymbol{y})+\frac{1}{2} \varepsilon^2 \boldsymbol{h}^T \boldsymbol{A} \boldsymbol{h}, \text { where } \boldsymbol{r}(\boldsymbol{y}):=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{y}.$	$Q\left(\boldsymbol{y}+\varepsilon \boldsymbol{h}\right)=Q\left(\boldsymbol{y}\right)-\varepsilon \boldsymbol{h}^T \boldsymbol{r}(\boldsymbol{y})+\frac{1}{2} \varepsilon^2 \boldsymbol{h}^T \boldsymbol{A} \boldsymbol{h}, \text { where } \boldsymbol{r}(\boldsymbol{y}):=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{y}.$	$Q\left(\boldsymbol{y}+\varepsilon \boldsymbol{h}\right)=Q\left(\boldsymbol{y}\right)-\varepsilon \boldsymbol{h}^T \boldsymbol{r}(\boldsymbol{y})+\frac{1}{2} \varepsilon^2 \boldsymbol{h}^T \boldsymbol{A} \boldsymbol{h}, \text { where } \boldsymbol{r}(\boldsymbol{y}):=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{y}.$
lyche-numerical-linear-algebra- EQ0841	293	■ 1.000 ● N +0	$Q\left(\boldsymbol{x}_{k+1}\right)=Q\left(\boldsymbol{x}_k+\alpha \boldsymbol{p}_k\right)=\min _{\alpha \in \mathbb{R}} Q\left(\boldsymbol{x}_k+\alpha \boldsymbol{p}_k\right).$	$Q\left(\boldsymbol{x}_{k+1}\right)=Q\left(\boldsymbol{x}_k+\alpha \boldsymbol{p}_k\right)=\min _{\alpha \in \mathbb{R}} Q\left(\boldsymbol{x}_k+\alpha \boldsymbol{p}_k\right).$	$Q\left(\boldsymbol{x}_{k+1}\right)=Q\left(\boldsymbol{x}_k+\alpha \boldsymbol{p}_k\right)=\min _{\alpha \in \mathbb{R}} Q\left(\boldsymbol{x}_k+\alpha \boldsymbol{p}_k\right).$

Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra E08842 (13.7)	293	■ 1.000 ● W +0	$\alpha_{\mathbf{k}} := \frac{\boldsymbol{p}_{\mathbf{k}}}{\boldsymbol{r}_{\mathbf{k}}^T \boldsymbol{p}_{\mathbf{k}}} \boldsymbol{r}_{\mathbf{k}}$	$\alpha_k := \frac{\boldsymbol{p}_k^T \boldsymbol{r}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k}.$	$\alpha_k := \frac{\boldsymbol{p}_k^T \boldsymbol{r}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k}.$
lyche-numerical-linear-algebra E08843 (13.8)	293	■ 1.000 ● N +0	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_{\mathbf{k}} + \left(\frac{\boldsymbol{r}_{\mathbf{k}}^T \boldsymbol{r}_{\mathbf{k}}}{\boldsymbol{r}_{\mathbf{k}}^T \boldsymbol{A} \boldsymbol{r}_{\mathbf{k}}} \right) \boldsymbol{r}_{\mathbf{k}}$	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \left(\frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k} \right) \boldsymbol{r}_k.$	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \left(\frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k} \right) \boldsymbol{r}_k.$
lyche-numerical-linear-algebra E08844 (13.9)	293	■ 1.000 ●	$\begin{aligned} \boldsymbol{p}_{\mathbf{k}} &= \boldsymbol{r}_{\mathbf{k}}, \boldsymbol{t}_{\mathbf{k}} = \boldsymbol{A} \boldsymbol{p}_{\mathbf{k}}, \\ \alpha_{\mathbf{k}} &= (\boldsymbol{p}_{\mathbf{k}}^T \boldsymbol{r}_{\mathbf{k}}) / (\boldsymbol{p}_{\mathbf{k}}^T \boldsymbol{t}_{\mathbf{k}}), \\ \boldsymbol{x}_{k+1} &= \boldsymbol{x}_{\mathbf{k}} + \alpha_{\mathbf{k}} \boldsymbol{p}_{\mathbf{k}}, \\ \boldsymbol{r}_{k+1} &= \boldsymbol{r}_{\mathbf{k}} - \alpha_{\mathbf{k}} \boldsymbol{t}_{\mathbf{k}}. \end{aligned}$	$\boldsymbol{p}_k = \boldsymbol{r}_k, \boldsymbol{t}_k = \boldsymbol{A} \boldsymbol{p}_k,$ $\alpha_k = (\boldsymbol{p}_k^T \boldsymbol{r}_k) / (\boldsymbol{p}_k^T \boldsymbol{t}_k),$ $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k,$ $\boldsymbol{r}_{k+1} = \boldsymbol{r}_k - \alpha_k \boldsymbol{t}_k.$	$\boldsymbol{p}_k = \boldsymbol{r}_k, \boldsymbol{t}_k = \boldsymbol{A} \boldsymbol{p}_k,$ $\alpha_k = (\boldsymbol{p}_k^T \boldsymbol{r}_k) / (\boldsymbol{p}_k^T \boldsymbol{t}_k),$ $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k,$ $\boldsymbol{r}_{k+1} = \boldsymbol{r}_k - \alpha_k \boldsymbol{t}_k.$
lyche-numerical-linear-algebra E08845 (13.10)	293	■ 1.000 ● N +0	$\boldsymbol{r}_{k+1} = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_{k+1} = \boldsymbol{b} - \boldsymbol{A} (\boldsymbol{x}_{\mathbf{k}} + \alpha_{\mathbf{k}} \boldsymbol{p}_{\mathbf{k}}) = \boldsymbol{r}_{\mathbf{k}} - \alpha_{\mathbf{k}} \boldsymbol{A} \boldsymbol{p}_{\mathbf{k}}$	$\boldsymbol{r}_{k+1} = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_{k+1} = \boldsymbol{b} - \boldsymbol{A} (\boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k) = \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k.$	$\boldsymbol{r}_{k+1} = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_{k+1} = \boldsymbol{b} - \boldsymbol{A} (\boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k) = \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k.$
lyche-numerical-linear-algebra E08853 (13.17)	296	■ 1.000 ● W +0	$\boldsymbol{p}_{k+1} := \boldsymbol{r}_{k+1} + \beta_{\mathbf{k}} \boldsymbol{p}_{\mathbf{k}}, \quad \beta_{\mathbf{k}} := \frac{\boldsymbol{r}_{k+1}^T \boldsymbol{r}_{k+1}}{\boldsymbol{r}_{\mathbf{k}}^T \boldsymbol{r}_{\mathbf{k}}}$	$\boldsymbol{p}_{k+1} := \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k, \quad \beta_k := \frac{\boldsymbol{r}_{k+1}^T \boldsymbol{r}_{k+1}}{\boldsymbol{r}_k^T \boldsymbol{r}_k}.$	$\boldsymbol{p}_{k+1} := \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k, \quad \beta_k := \frac{\boldsymbol{r}_{k+1}^T \boldsymbol{r}_{k+1}}{\boldsymbol{r}_k^T \boldsymbol{r}_k}.$
lyche-numerical-linear-algebra E08856	298	■ 1.000 ● S -2	$\boldsymbol{A} = \text{tridiag}_{\mathbf{m}}(\boldsymbol{a}, \boldsymbol{d}, \boldsymbol{a}) \otimes \boldsymbol{I}_{\mathbf{m}} + \boldsymbol{I}_{\mathbf{m}} \otimes \text{tridiag}_{\mathbf{m}}(\boldsymbol{a}, \boldsymbol{d}, \boldsymbol{a}) \in \mathbb{R}^{(m^2) \times (m^2)}$	$\boldsymbol{A} = \text{tridiag}_m(\boldsymbol{a}, \boldsymbol{d}, \boldsymbol{a}) \otimes \boldsymbol{I}_m + \boldsymbol{I}_m \otimes \text{tridiag}_m(\boldsymbol{a}, \boldsymbol{d}, \boldsymbol{a}) \in \mathbb{R}^{(m^2) \times (m^2)}$	$\boldsymbol{A} = \text{tridiag}_m(\boldsymbol{a}, \boldsymbol{d}, \boldsymbol{a}) \otimes \boldsymbol{I}_m + \boldsymbol{I}_m \otimes \text{tridiag}_m(\boldsymbol{a}, \boldsymbol{d}, \boldsymbol{a}) \in \mathbb{R}^{(m^2) \times (m^2)}.$
lyche-numerical-linear-algebra E08859	300	■ 1.000 ● C -3	$\boldsymbol{T}_2 := \text{tridiag}_{\mathbf{m}}(\boldsymbol{a}, \boldsymbol{d}, \boldsymbol{a}) \otimes \boldsymbol{I}_{\mathbf{m}} + \boldsymbol{I}_{\mathbf{m}} \otimes \text{tridiag}_{\mathbf{m}}(\boldsymbol{a}, \boldsymbol{d}, \boldsymbol{a}) \in \mathbb{R}^{(m^2) \times (m^2)}$	$\boldsymbol{T}_2 := \text{tridiag}_m(\boldsymbol{a}, \boldsymbol{d}, \boldsymbol{a}) \otimes \boldsymbol{I}_m + \boldsymbol{I}_m \otimes \text{tridiag}_m(\boldsymbol{a}, \boldsymbol{d}, \boldsymbol{a}) \in \mathbb{R}^{(m^2) \times (m^2)}.$	$\boldsymbol{T}_2 := \text{tridiag}_m(\boldsymbol{a}, \boldsymbol{d}, \boldsymbol{a}) \otimes \boldsymbol{I}_m + \boldsymbol{I}_m \otimes \text{tridiag}_m(\boldsymbol{a}, \boldsymbol{d}, \boldsymbol{a}) \in \mathbb{R}^{(m^2) \times (m^2)}.$
lyche-numerical-linear-algebra E08860 (13.21)	300	■ 1.000 ● N +0	$\lambda_{j, \mathbf{k}} = 2d + 2a \cos(j \pi h) + 2a \cos(k \pi h), \quad j, \mathbf{k} = 1, \dots, m$	$\lambda_{j,k} = 2d + 2a \cos(j \pi h) + 2a \cos(k \pi h), \quad j, k = 1, \dots, m.$	$\lambda_{j,k} = 2d + 2a \cos(j \pi h) + 2a \cos(k \pi h), \quad j, k = 1, \dots, m.$
lyche-numerical-linear-algebra E08864	301	■ 1.000 ● S +0	$\mathbb{W}_{\mathbf{0}} \subset \mathbb{W}_{\mathbf{1}} \subset \mathbb{W}_{\mathbf{2}} \subset \dots \subset \mathbb{W}_{\mathbf{n}} \subset \mathbb{R}^n$	$\mathbb{W}_0 \subset \mathbb{W}_1 \subset \mathbb{W}_2 \subset \dots \subset \mathbb{W}_n \subset \mathbb{R}^n$	$\mathbb{W}_0 \subset \mathbb{W}_1 \subset \mathbb{W}_2 \subset \dots \subset \mathbb{W}_n \subset \mathbb{R}^n$
lyche-numerical-linear-algebra E08865 (13.22)	301	■ 1.000 ● N +0	$\boldsymbol{x}_{\mathbf{k}} - \boldsymbol{x}_{\mathbf{0}} \in \mathbb{W}_{\mathbf{k}}, \quad \boldsymbol{r}_{\mathbf{k}}, \boldsymbol{p}_{\mathbf{k}} \in \mathbb{W}_{k+1}, \quad k = 0, 1, \dots,$	$\boldsymbol{x}_k - \boldsymbol{x}_0 \in \mathbb{W}_k, \quad \boldsymbol{r}_k, \boldsymbol{p}_k \in \mathbb{W}_{k+1}, \quad k = 0, 1, \dots,$	$\boldsymbol{x}_k - \boldsymbol{x}_0 \in \mathbb{W}_k, \quad \boldsymbol{r}_k, \boldsymbol{p}_k \in \mathbb{W}_{k+1}, \quad k = 0, 1, \dots,$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-alignatE09867 (13.24)	301	1.000 N -1	$\left \left \boldsymbol{x}-\boldsymbol{x}_{k}\right \right _{\left\{\boldsymbol{A}\right\}=\min _{\boldsymbol{w} \in \mathbb{W}_{k}}\left \left \boldsymbol{x}-\boldsymbol{x}_{0}-\boldsymbol{w}\right \right _{\left\{\boldsymbol{A}\right\}}\right $.	$\left\ \boldsymbol{x}-\boldsymbol{x}_k\right\ _{\boldsymbol{A}}=\min _{\boldsymbol{w} \in \mathbb{W}_k}\left\ \boldsymbol{x}-\boldsymbol{x}_0-\boldsymbol{w}\right\ _{\boldsymbol{A}}.$	$\left\ \boldsymbol{x}-\boldsymbol{x}_k\right\ _{\boldsymbol{A}}=\min _{\boldsymbol{w} \in \mathbb{W}_k}\left\ \boldsymbol{x}-\boldsymbol{x}_0-\boldsymbol{w}\right\ _{\boldsymbol{A}}.$
lyche-numerical-linear-alignatE09869 (13.25)	302	1.000 N +0	$\left \left \boldsymbol{x}-\boldsymbol{x}_{0}-\boldsymbol{w}\right \right _{\left\{\boldsymbol{A}\right\}^2=\sum_{j=1}^n\left\{\frac{\sigma_{\left\{j\right\}}^2}{\lambda_{\left\{j\right\}}} Q\left(\lambda_{\left\{j\right\}}\right)^2\right\} Q\left(\lambda_{\left\{j\right\}}\right)^2}, \quad Q(t):=1-t P(t) .$	$\left\ \boldsymbol{x}-\boldsymbol{x}_0-\boldsymbol{w}\right\ _{\boldsymbol{A}}^2=\sum_{j=1}^n \frac{\sigma_j^2}{\lambda_j} Q\left(\lambda_j\right)^2, \quad Q(t):=1-t P(t) .$	$\left\ \boldsymbol{x}-\boldsymbol{x}_0-\boldsymbol{w}\right\ _{\boldsymbol{A}}^2=\sum_{j=1}^n \frac{\sigma_j^2}{\lambda_j} Q\left(\lambda_j\right)^2, \quad Q(t):=1-t P(t) .$
lyche-numerical-linear-alignatE09870 (13.26)	302	1.000 N +0	$\left \left \boldsymbol{x}-\boldsymbol{x}_{0}-P(\boldsymbol{A}) \boldsymbol{r}_0\right \right _{\boldsymbol{A}}=\boldsymbol{c}^T \boldsymbol{A}^{-1} \boldsymbol{c}$ where $\boldsymbol{c}=(\boldsymbol{I}-\boldsymbol{A} P(\boldsymbol{A})) \boldsymbol{r}_0=Q(\boldsymbol{A}) \boldsymbol{r}_0$.	$\left\ \boldsymbol{x}-\boldsymbol{x}_0-P(\boldsymbol{A}) \boldsymbol{r}_0\right\ _{\boldsymbol{A}}^2=\boldsymbol{c}^T \boldsymbol{A}^{-1} \boldsymbol{c}$ where $\boldsymbol{c}=(\boldsymbol{I}-\boldsymbol{A} P(\boldsymbol{A})) \boldsymbol{r}_0=Q(\boldsymbol{A}) \boldsymbol{r}_0$.	$\left\ \boldsymbol{x}-\boldsymbol{x}_0-P(\boldsymbol{A}) \boldsymbol{r}_0\right\ _{\boldsymbol{A}}^2=\boldsymbol{c}^T \boldsymbol{A}^{-1} \boldsymbol{c}$ where $\boldsymbol{c}=(\boldsymbol{I}-\boldsymbol{A} P(\boldsymbol{A})) \boldsymbol{r}_0=Q(\boldsymbol{A}) \boldsymbol{r}_0$.
lyche-numerical-linear-alignatE09871 (13.27)	302	1.000	$\boldsymbol{c}=\sum_{j=1}^n \sigma_j Q\left(\lambda_j\right) \boldsymbol{u}_j, \quad \boldsymbol{A}^{-1} \boldsymbol{c}=\sum_{i=1}^n \sigma_i \frac{Q\left(\lambda_i\right)}{\lambda_i} \boldsymbol{u}_i .$	$\boldsymbol{c}=\sum_{j=1}^n \sigma_j Q\left(\lambda_j\right) \boldsymbol{u}_j, \quad \boldsymbol{A}^{-1} \boldsymbol{c}=\sum_{i=1}^n \sigma_i \frac{Q\left(\lambda_i\right)}{\lambda_i} \boldsymbol{u}_i .$	$\boldsymbol{c}=\sum_{j=1}^n \sigma_j Q\left(\lambda_j\right) \boldsymbol{u}_j, \quad \boldsymbol{A}^{-1} \boldsymbol{c}=\sum_{i=1}^n \sigma_i \frac{Q\left(\lambda_i\right)}{\lambda_i} \boldsymbol{u}_i .$
lyche-numerical-linear-alignatE09876	303	1.000 N +0	$e^{\left\{2 / x\right\}}=e^{\left\{2 y\right\}}=\sum_{k=0}^{\infty} \frac{\left(2 y\right)^k}{k!}<-1+2 \sum_{k=0}^{\infty} y^k=\frac{1+y}{1-y}=\frac{x+1}{x-1}$	$e^{2 / x}=e^{2 y}=\sum_{k=0}^{\infty} \frac{\left(2 y\right)^k}{k!}<-1+2 \sum_{k=0}^{\infty} y^k=\frac{1+y}{1-y}=\frac{x+1}{x-1}$	$e^{2 / x}=e^{2 y}=\sum_{k=0}^{\infty} \frac{\left(2 y\right)^k}{k!}<-1+2 \sum_{k=0}^{\infty} y^k=\frac{1+y}{1-y}=\frac{x+1}{x-1}$
lyche-numerical-linear-alignatE09877	304	1.000 N +0	$\left f\right _{\left\{\infty\right\}}=\max _{a \leq x \leq b} f(x) $.	$\ f\ _{\infty}=\max _{a \leq x \leq b} f(x) .$	$\ f\ _{\infty}=\max _{a \leq x \leq b} f(x) .$
lyche-numerical-linear-alignatE09878	304	1.000 N -1	$\left Q^*\right _{\left\{\infty\right\}}=\min _{Q \in \mathcal{S}_k}\ Q\ _{\infty} .$	$\left\ Q^*\right\ _{\infty}=\min _{Q \in \mathcal{S}_k}\ Q\ _{\infty} .$	$\left\ Q^*\right\ _{\infty}=\min _{Q \in \mathcal{S}_k}\ Q\ _{\infty} .$
lyche-numerical-linear-alignatE09879	304	1.000 N +0	$T_{n+1}(t)=2 t T_n(t)-T_{n-1}(t), \quad n \geq 1, \quad t \in \mathbb{R},$	$T_{n+1}(t)=2 t T_n(t)-T_{n-1}(t), \quad n \geq 1, \quad t \in \mathbb{R},$	$T_{n+1}(t)=2 t T_n(t)-T_{n-1}(t), \quad n \geq 1, \quad t \in \mathbb{R},$
lyche-numerical-linear-alignatE09880	304	1.000 N +0	$P_{n+1}(t)+P_{n-1}(t)=\cos (n+1) \phi+\cos (n-1) \phi=2 \cos \phi \cos n \phi=2 t P_n(t),$	$P_{n+1}(t)+P_{n-1}(t)=\cos (n+1) \phi+\cos (n-1) \phi=2 \cos \phi \cos n \phi=2 t P_n(t),$	$P_{n+1}(t)+P_{n-1}(t)=\cos (n+1) \phi+\cos (n-1) \phi=2 \cos \phi \cos n \phi=2 t P_n(t),$
lyche-numerical-linear-alignatE09882	305	1.000 K +0	$z_{\left\{1\right\}}=t+\sqrt{t^2-1}, \quad z_{\left\{2\right\}}=t-\sqrt{t^2-1}=\left(t+\sqrt{t^2-1}\right)^{-1} .$	$z_1=t+\sqrt{t^2-1}, \quad z_2=t-\sqrt{t^2-1}=\left(t+\sqrt{t^2-1}\right)^{-1} .$	$z_1=t+\sqrt{t^2-1}, \quad z_2=t-\sqrt{t^2-1}=\left(t+\sqrt{t^2-1}\right)^{-1} .$
lyche-numerical-linear-alignatE09883 (13.32)	305	1.000 N +0	$Q^{\left\{* \right\}}(x)=\frac{T_{\left\{k\right\}}(u(x))}{T_{\left\{k\right\}}(u(c))}, \quad u(x)=\frac{b+a-2 x}{b-a} .$	$Q^*(x)=\frac{T_k(u(x))}{T_k(u(c))}, \quad u(x)=\frac{b+a-2 x}{b-a} .$	$Q^*(x)=\frac{T_k(u(x))}{T_k(u(c))}, \quad u(x)=\frac{b+a-2 x}{b-a} .$
lyche-numerical-linear-alignatE09884	305	1.000 K +0	$\sigma_{\left\{j\right\}}=f\left(\mu_{\left\{j-1\right\}}\right) f\left(\mu_{\left\{j\right\}}\right) .$	$\sigma_j=f\left(\mu_{j-1}\right) f\left(\mu_j\right) .$	$\sigma_j=f\left(\mu_{j-1}\right) f\left(\mu_j\right) .$
lyche-numerical-linear-alignatE09885	305	1.000 N +0	$Q\left(\mu_{\left\{j\right\}}\right) \leq\ Q\ _{\infty} \leq\left\ Q^*\right\ _{\infty}=Q^*\left(\mu_{\left\{j\right\}}\right)$	$Q\left(\mu_j\right) \leq\ Q\ _{\infty} \leq\left\ Q^*\right\ _{\infty}=Q^*\left(\mu_j\right)$	$Q\left(\mu_j\right) \leq\ Q\ _{\infty} \leq\left\ Q^*\right\ _{\infty}=Q^*\left(\mu_j\right)$
lyche-numerical-linear-alignatE09886	305	1.000 N +0	$-Q\left(\mu_{\left\{j-1\right\}}\right) \leq\ Q\ _{\infty} \leq\left\ Q^*\right\ _{\infty}=-Q^*\left(\mu_{\left\{j-1\right\}}\right) .$	$-Q\left(\mu_{j-1}\right) \leq\ Q\ _{\infty} \leq\left\ Q^*\right\ _{\infty}=-Q^*\left(\mu_{j-1}\right) .$	$-Q\left(\mu_{j-1}\right) \leq\ Q\ _{\infty} \leq\left\ Q^*\right\ _{\infty}=-Q^*\left(\mu_{j-1}\right) .$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0887 (13.33)	306	1.000 K +0	$Q^{*}(x)=T_k\left(\frac{b+a-2x}{b-a}\right)/T_k\left(\frac{b+a}{b-a}\right).$	$Q^{*}(x)=T_k\left(\frac{b+a-2x}{b-a}\right)/T_k\left(\frac{b+a}{b-a}\right).$	$Q^{*}(x)=T_k\left(\frac{b+a-2x}{b-a}\right)/T_k\left(\frac{b+a}{b-a}\right).$
lyche-numerical-linear-algebra- EQ0888 (13.34)	306	1.000 N -1	$\max_{a\leq x\leq b}\left T_k\left(\frac{b+a-2x}{b-a}\right)\right =\max_{-1\leq t\leq 1}\left T_k(t)\right =1.$	$\max_{a\leq x\leq b}\left T_k\left(\frac{b+a-2x}{b-a}\right)\right =\max_{-1\leq t\leq 1} T_k(t) =1.$	$\max_{a\leq x\leq b}\left T_k\left(\frac{b+a-2x}{b-a}\right)\right =\max_{-1\leq t\leq 1} T_k(t) =1.$
lyche-numerical-linear-algebra- EQ0889	306	1.000 N +1	$t+\sqrt{t^2-1}=\frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1},\quad \kappa=b/a.$	$t+\sqrt{t^2-1}=\frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1},\quad \kappa=b/a.$	$t+\sqrt{t^2-1}=\frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1},\quad \kappa=b/a.$
lyche-numerical-linear-algebra- EQ0890 (13.35)	306	1.000 W +0	$T_k\left(\frac{b+a}{b-a}\right)=T_k\left(\frac{\kappa+1}{\kappa-1}\right)=\frac{1}{2}\left[\left(\frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1}\right)^k+\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k\right]$	$T_k\left(\frac{b+a}{b-a}\right)=T_k\left(\frac{\kappa+1}{\kappa-1}\right)=\frac{1}{2}\left[\left(\frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1}\right)^k+\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k\right]$	$T_k\left(\frac{b+a}{b-a}\right)=T_k\left(\frac{\kappa+1}{\kappa-1}\right)=\frac{1}{2}\left[\left(\frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1}\right)^k+\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k\right]$
lyche-numerical-linear-algebra- EQ0893 (13.38)	307	1.000 N +0	$t_i=\frac{c_i^2}{\sum_{j=1}^nc_j^2}\geq 0,\quad i=1,\dots,n\text{ and }\sum_{i=1}^nt_i=1.$	$t_i=\frac{c_i^2}{\sum_{j=1}^nc_j^2}\geq 0,\quad i=1,\dots,n\text{ and }\sum_{i=1}^nt_i=1.$	$t_i=\frac{c_i^2}{\sum_{j=1}^nc_j^2}\geq 0,\quad i=1,\dots,n\text{ and }\sum_{i=1}^nt_i=1.$
lyche-numerical-linear-algebra- EQ0894	307	1.000	$\sqrt{a\,b}=\sqrt{(a\,c)(b\,/\,c)}\leq (a\,c+b\,/\,c)\,/\,2=\frac{1}{2}\sum_{i=1}^nt_i\left(\frac{\lambda_i}{\lambda_i}\right)\leq \frac{1}{2}\sum_{i=1}^nt_i\left(\lambda_i+\frac{1}{\lambda_i}\right)=\frac{1}{2}\sum_{i=1}^nt_i\left(\lambda_i+\frac{1}{\lambda_i}\right).$	$\sqrt{ab}=\sqrt{(ac)(b/c)}\leq (ac+b/c)/2=\frac{1}{2}\sum_{i=1}^nt_i(\lambda_ic+1/(\lambda_ic))=\frac{1}{2}\sum_{i=1}^nt_if(\lambda_ic),$	$\sqrt{ab}=\sqrt{(ac)(b/c)}\leq (ac+b/c)/2=\frac{1}{2}\sum_{i=1}^nt_i(\lambda_ic+1/(\lambda_ic))=\frac{1}{2}\sum_{i=1}^nt_if(\lambda_ic),$
lyche-numerical-linear-algebra- EQ0895	307	1.000 N +0	$\sqrt{a\,b}\leq \frac{1}{2}\max_{m\,c\leq x\leq M\,c}f(x).$	$\sqrt{ab}\leq \frac{1}{2}\max_{mc\leq x\leq Mc}f(x).$	$\sqrt{ab}\leq \frac{1}{2}\max_{mc\leq x\leq Mc}f(x).$
lyche-numerical-linear-algebra- EQ0896 (13.39)	307	1.000 N -1	$\sqrt{a\,b}\leq \frac{1}{2}\max\{f(mc),f(Mc)\}.$	$\sqrt{ab}\leq \frac{1}{2}\max\{f(mc),f(Mc)\}.$	$\sqrt{ab}\leq \frac{1}{2}\max\{f(mc),f(Mc)\}.$
lyche-numerical-linear-algebra- EQ0898	308	1.000 S +0	$1=\left(\sum_{i=1}^nt_i\right)^2=\left(\sum_{i=1}^n(t_i\lambda_i)^{1/2}(t_i/\lambda_i)^{1/2}\right)^2\leq \left(\sum_{i=1}^nt_i\lambda_i\right)\left(\sum_{i=1}^nt_i/\lambda_i\right)=ab.$	$1=\left(\sum_{i=1}^nt_i\right)^2=\left(\sum_{i=1}^n(t_i\lambda_i)^{1/2}(t_i/\lambda_i)^{1/2}\right)^2\leq \left(\sum_{i=1}^nt_i\lambda_i\right)\left(\sum_{i=1}^nt_i/\lambda_i\right)=ab.$	$1=\left(\sum_{i=1}^nt_i\right)^2=\left(\sum_{i=1}^n(t_i\lambda_i)^{1/2}(t_i/\lambda_i)^{1/2}\right)^2\leq \left(\sum_{i=1}^nt_i\lambda_i\right)\left(\sum_{i=1}^nt_i/\lambda_i\right)=ab.$
lyche-numerical-linear-algebra- EQ0900	308	1.000 N +0	$\epsilon_{k+1}=\epsilon_k-\alpha_k\mathbf{r}_k,\quad \alpha_k:=\frac{\mathbf{r}_k^T\mathbf{r}_k}{\mathbf{r}_k^T\mathbf{A}\mathbf{r}_k}.$	$\epsilon_{k+1}=\epsilon_k-\alpha_k\mathbf{r}_k,\quad \alpha_k:=\frac{\mathbf{r}_k^T\mathbf{r}_k}{\mathbf{r}_k^T\mathbf{A}\mathbf{r}_k}.$	$\epsilon_{k+1}=\epsilon_k-\alpha_k\mathbf{r}_k,\quad \alpha_k:=\frac{\mathbf{r}_k^T\mathbf{r}_k}{\mathbf{r}_k^T\mathbf{A}\mathbf{r}_k}.$

Conf. MathPix confidence:

≥0.9

0.5–0.9

<0.5

— no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ
Residual render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ9982	308	■ 1.000 ● W +0	$\frac{\left \left \boldsymbol{\epsilon}_{k+1}\right \right }{\left \left \boldsymbol{\epsilon}_k\right \right _{\boldsymbol{A}}^2} = 1 - \frac{\left(\boldsymbol{r}_k^T \boldsymbol{r}_k\right)^2}{\left(\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k\right)\left(\boldsymbol{r}_k^T \boldsymbol{A}^{-1} \boldsymbol{r}_k\right)} \leq 1 - \frac{4 \lambda_{\min } \lambda_{\max }}{\left(\lambda_{\min } + \lambda_{\max }\right)^2} = \left(\frac{\kappa - 1}{\kappa + 1}\right)^2$	$\frac{\left\ \boldsymbol{\epsilon}_{k+1}\right\ _{\boldsymbol{A}}^2}{\left\ \boldsymbol{\epsilon}_k\right\ _{\boldsymbol{A}}^2} = 1 - \frac{\left(\boldsymbol{r}_k^T \boldsymbol{r}_k\right)^2}{\left(\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k\right)\left(\boldsymbol{r}_k^T \boldsymbol{A}^{-1} \boldsymbol{r}_k\right)} \leq 1 - \frac{4 \lambda_{\min } \lambda_{\max }}{\left(\lambda_{\min } + \lambda_{\max }\right)^2} = \left(\frac{\kappa - 1}{\kappa + 1}\right)^2$	$\frac{\left\ \boldsymbol{\epsilon}_{k+1}\right\ _{\boldsymbol{A}}^2}{\left\ \boldsymbol{\epsilon}_k\right\ _{\boldsymbol{A}}^2} = 1 - \frac{\left(\boldsymbol{r}_k^T \boldsymbol{r}_k\right)^2}{\left(\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k\right)\left(\boldsymbol{r}_k^T \boldsymbol{A}^{-1} \boldsymbol{r}_k\right)} \leq 1 - \frac{4 \lambda_{\min } \lambda_{\max }}{\left(\lambda_{\min } + \lambda_{\max }\right)^2} = \left(\frac{\kappa - 1}{\kappa + 1}\right)^2$
lyche-numerical-linear-algebra-EQ9985 (13.41)	309	■ 1.000 ● W +0	$\boldsymbol{\epsilon}_{\{j\}} = \sum_{i=j}^m \alpha_{\{i\}} \boldsymbol{p}_{\{i\}}, \quad \alpha_{\{i\}} = \frac{\boldsymbol{r}_{\{i\}}^T \boldsymbol{r}_{\{i\}}}{\boldsymbol{p}_{\{i\}}^T \boldsymbol{A} \boldsymbol{p}_{\{i\}}}$	$\boldsymbol{\epsilon}_j = \sum_{i=j}^m \alpha_i \boldsymbol{p}_i, \quad \alpha_i = \frac{\boldsymbol{r}_i^T \boldsymbol{r}_i}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i}.$	$\boldsymbol{\epsilon}_j = \sum_{i=j}^m \alpha_i \boldsymbol{p}_i, \quad \alpha_i = \frac{\boldsymbol{r}_i^T \boldsymbol{r}_i}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i}.$
lyche-numerical-linear-algebra-EQ9986 (13.42)	309	■ 1.000 ● W +0	$\left \left \boldsymbol{\epsilon}_{\{j\}}\right \right _{\boldsymbol{A}}^2 = \boldsymbol{\epsilon}_{\{j\}}^T \boldsymbol{\epsilon}_{\{j\}} = \sum_{i=j}^m \alpha_{\{i\}}^2 \boldsymbol{p}_{\{i\}}^T \boldsymbol{A} \boldsymbol{p}_{\{i\}} = \sum_{i=j}^m \frac{\left(\boldsymbol{r}_{\{i\}}^T \boldsymbol{r}_{\{i\}}\right)^2}{\boldsymbol{p}_{\{i\}}^T \boldsymbol{A} \boldsymbol{p}_{\{i\}}}$	$\left\ \boldsymbol{\epsilon}_j\right\ _{\boldsymbol{A}}^2 = \boldsymbol{\epsilon}_j^T \boldsymbol{A} \boldsymbol{\epsilon}_j = \sum_{i=j}^m \alpha_i^2 \boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i = \sum_{i=j}^m \frac{\left(\boldsymbol{r}_i^T \boldsymbol{r}_i\right)^2}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i}.$	$\left\ \boldsymbol{\epsilon}_j\right\ _{\boldsymbol{A}}^2 = \boldsymbol{\epsilon}_j^T \boldsymbol{A} \boldsymbol{\epsilon}_j = \sum_{i=j}^m \alpha_i^2 \boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i = \sum_{i=j}^m \frac{\left(\boldsymbol{r}_i^T \boldsymbol{r}_i\right)^2}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i}.$
lyche-numerical-linear-algebra-EQ9987	309	■ 1.000 ●	$\boldsymbol{p}_{\{i\}}^T \boldsymbol{p}_{\{k\}} = \left(\boldsymbol{r}_{\{i\}} + \beta_{\{i-1\}} \boldsymbol{p}_{\{i-1\}}\right)^T \boldsymbol{p}_{\{k\}} = \beta_{i-1} \boldsymbol{p}_{i-1}^T \boldsymbol{p}_k = \cdots = \beta_{i-1} \cdots \beta_k \left(\boldsymbol{p}_k^T \boldsymbol{p}_k\right),$	$\boldsymbol{p}_i^T \boldsymbol{p}_k = \left(\boldsymbol{r}_i + \beta_{i-1} \boldsymbol{p}_{i-1}\right)^T \boldsymbol{p}_k = \beta_{i-1} \boldsymbol{p}_{i-1}^T \boldsymbol{p}_k = \cdots = \beta_{i-1} \cdots \beta_k \left(\boldsymbol{p}_k^T \boldsymbol{p}_k\right),$	$\boldsymbol{p}_i^T \boldsymbol{p}_k = \left(\boldsymbol{r}_i + \beta_{i-1} \boldsymbol{p}_{i-1}\right)^T \boldsymbol{p}_k = \beta_{i-1} \boldsymbol{p}_{i-1}^T \boldsymbol{p}_k = \cdots = \beta_{i-1} \cdots \beta_k \left(\boldsymbol{p}_k^T \boldsymbol{p}_k\right),$
lyche-numerical-linear-algebra-EQ9988 (13.43)	309	■ 1.000 ● N +1	$\boldsymbol{p}_{\{i\}}^T \boldsymbol{p}_{\{k\}} = \frac{\boldsymbol{r}_{\{i\}}^T \boldsymbol{r}_{\{i\}}}{\boldsymbol{r}_{\{k\}}^T \boldsymbol{r}_{\{k\}}} \boldsymbol{p}_{\{k\}}^T \boldsymbol{p}_{\{k\}}, \quad i \geq k$	$\boldsymbol{p}_i^T \boldsymbol{p}_k = \frac{\boldsymbol{r}_i^T \boldsymbol{r}_i}{\boldsymbol{r}_k^T \boldsymbol{r}_k} \boldsymbol{p}_k^T \boldsymbol{p}_k, \quad i \geq k$	$\boldsymbol{p}_i^T \boldsymbol{p}_k = \frac{\boldsymbol{r}_i^T \boldsymbol{r}_i}{\boldsymbol{r}_k^T \boldsymbol{r}_k} \boldsymbol{p}_k^T \boldsymbol{p}_k, \quad i \geq k.$
lyche-numerical-linear-algebra-EQ9989	309	■ 1.000 ● K +0	$\left \left \boldsymbol{\epsilon}_{\{k\}}\right \right _{\boldsymbol{A}}^2 = \left \left \boldsymbol{\epsilon}_{\{k+1\}} + \boldsymbol{x}_{\{k+1\}} - \boldsymbol{x}_{\{k+1\}}\right \right _{\boldsymbol{A}}^2 = \left \left \boldsymbol{\epsilon}_{\{k+1\}} + \alpha_{\{k\}} \boldsymbol{p}_{\{k\}}\right \right _{\boldsymbol{A}}^2,$	$\left\ \boldsymbol{\epsilon}_k\right\ _2^2 = \left\ \boldsymbol{\epsilon}_{k+1} + \boldsymbol{x}_{k+1} - \boldsymbol{x}_k\right\ _2^2 = \left\ \boldsymbol{\epsilon}_{k+1} + \alpha_k \boldsymbol{p}_k\right\ _2^2,$	$\left\ \boldsymbol{\epsilon}_k\right\ _2^2 = \left\ \boldsymbol{\epsilon}_{k+1} + \boldsymbol{x}_{k+1} - \boldsymbol{x}_k\right\ _2^2 = \left\ \boldsymbol{\epsilon}_{k+1} + \alpha_k \boldsymbol{p}_k\right\ _2^2,$
lyche-numerical-linear-algebra-EQ9913 (13.49)	311	■ 1.000 ● K +0	$\boldsymbol{C}^T \left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T\right) \boldsymbol{C}^{-T} = \boldsymbol{B} \boldsymbol{A}.$	$\boldsymbol{C}^T \left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T\right) \boldsymbol{C}^{-T} = \boldsymbol{B} \boldsymbol{A}.$	$\boldsymbol{C}^T \left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T\right) \boldsymbol{C}^{-T} = \boldsymbol{B} \boldsymbol{A}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0916 (13.50)	311	1.000 W +0	$\begin{aligned} & \boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k, \\ & \boldsymbol{r}_{k+1} = \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k, \\ & \boldsymbol{s}_{k+1} = \boldsymbol{s}_k - \alpha_k \boldsymbol{B} \boldsymbol{A} \boldsymbol{p}_k, \\ & \boldsymbol{p}_{k+1} = \boldsymbol{s}_{k+1} + \beta_k \boldsymbol{p}_k, \quad \beta_k = \frac{\boldsymbol{s}_{k+1}^T \boldsymbol{r}_{k+1}}{\boldsymbol{s}_k^T \boldsymbol{r}_k}. \end{aligned}$	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k, \quad \alpha_k = \frac{\boldsymbol{s}_k^T \boldsymbol{r}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k},$ $\boldsymbol{r}_{k+1} = \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k,$ $\boldsymbol{s}_{k+1} = \boldsymbol{s}_k - \alpha_k \boldsymbol{B} \boldsymbol{A} \boldsymbol{p}_k,$ $\boldsymbol{p}_{k+1} = \boldsymbol{s}_{k+1} + \beta_k \boldsymbol{p}_k, \quad \beta_k = \frac{\boldsymbol{s}_{k+1}^T \boldsymbol{r}_{k+1}}{\boldsymbol{s}_k^T \boldsymbol{r}_k}.$	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k, \quad \alpha_k = \frac{\boldsymbol{s}_k^T \boldsymbol{r}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k},$ $\boldsymbol{r}_{k+1} = \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k,$ $\boldsymbol{s}_{k+1} = \boldsymbol{s}_k - \alpha_k \boldsymbol{B} \boldsymbol{A} \boldsymbol{p}_k,$ $\boldsymbol{p}_{k+1} = \boldsymbol{s}_{k+1} + \beta_k \boldsymbol{p}_k, \quad \beta_k = \frac{\boldsymbol{s}_{k+1}^T \boldsymbol{r}_{k+1}}{\boldsymbol{s}_k^T \boldsymbol{r}_k}.$
lyche-numerical-linear-algebra EQ0921	313	1.000 S +0	$\begin{aligned} & I_m := \{(j, k) : 1 \leq j, k \leq m\}, \\ & \bar{I}_m := \{(j, k) : 0 \leq j, k \leq m + 1\}, \\ & \partial I_m := \bar{I}_m \setminus I_m. \end{aligned}$	$I_m := \{(j, k) : 1 \leq j, k \leq m\},$ $\bar{I}_m := \{(j, k) : 0 \leq j, k \leq m + 1\},$ $\partial I_m := \bar{I}_m \setminus I_m.$	$I_m := \{(j, k) : 1 \leq j, k \leq m\},$ $\bar{I}_m := \{(j, k) : 0 \leq j, k \leq m + 1\},$ $\partial I_m := \bar{I}_m \setminus I_m.$
lyche-numerical-linear-algebra EQ0922	313	1.000 N +0	$\left\{ \left(\left(x_j, y_k \right) = (j \, h, \, k \, h) : (j, \, k) \in \bar{I}_m \right) \right\}$	$\{(x_j, y_k) = (jh, kh) : (j, k) \in \bar{I}_m\}$	$\{(x_j, y_k) = (jh, kh) : (j, k) \in \bar{I}_m\}$
lyche-numerical-linear-algebra EQ0923 (13.54)	314	1.000	$\begin{aligned} & \frac{d}{dt} \left(f(t) \frac{d}{dt} g(t) \right) \approx \left(f \left(t + \frac{h}{2} \right) \frac{d}{dt} g(t + h/2) - f \left(t - \frac{h}{2} \right) \frac{d}{dt} g \left(t - \frac{h}{2} \right) \right) / h \\ & \approx \left(f \left(t + \frac{h}{2} \right) (g(t + h) - g(t)) - f \left(t - \frac{h}{2} \right) (g(t) - g(t - h)) \right) / h^2 \end{aligned}$	$\frac{d}{dt} \left(f(t) \frac{d}{dt} g(t) \right) \approx \left(f \left(t + \frac{h}{2} \right) \frac{d}{dt} g(t + h/2) - f \left(t - \frac{h}{2} \right) \frac{d}{dt} g \left(t - \frac{h}{2} \right) \right) / h$ $\approx \left(f \left(t + \frac{h}{2} \right) (g(t + h) - g(t)) - f \left(t - \frac{h}{2} \right) (g(t) - g(t - h)) \right) / h^2$	$\frac{d}{dt} \left(f(t) \frac{d}{dt} g(t) \right) \approx \left(f \left(t + \frac{h}{2} \right) \frac{d}{dt} g(t + h/2) - f \left(t - \frac{h}{2} \right) \frac{d}{dt} g \left(t - \frac{h}{2} \right) \right) / h$ $\approx \left(f \left(t + \frac{h}{2} \right) (g(t + h) - g(t)) - f \left(t - \frac{h}{2} \right) (g(t) - g(t - h)) \right) / h^2$
lyche-numerical-linear-algebra EQ0927	314	1.000 N +0	$\boldsymbol{V} := \left[\begin{array}{ccc} v_{1,1} & \dots & v_{1,m} \\ \vdots & & \vdots \\ v_{m,1} & \dots & v_{m,m} \end{array} \right]$	$\boldsymbol{V} := \begin{bmatrix} v_{1,1} & \dots & v_{1,m} \\ \vdots & & \vdots \\ v_{m,1} & \dots & v_{m,m} \end{bmatrix}$	$\boldsymbol{V} := \begin{bmatrix} v_{1,1} & \dots & v_{1,m} \\ \vdots & & \vdots \\ v_{m,1} & \dots & v_{m,m} \end{bmatrix}$
lyche-numerical-linear-algebra EQ0928 (13.57)	314	1.000 N +0	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{A} \operatorname{vec}(\boldsymbol{V}) := h^2 \operatorname{vec}(L_h \boldsymbol{v}).$	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{A} \operatorname{vec}(\boldsymbol{V}) := h^2 \operatorname{vec}(L_h \boldsymbol{v}).$	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{A} \operatorname{vec}(\boldsymbol{V}) := h^2 \operatorname{vec}(L_h \boldsymbol{v}).$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-315 EQ0931 (13.60)	315	■ 1.000 ● W +0	$\sum_{i=1}^m \left(a_{i-1} \left(b_i - b_{i-1} \right) - a_i \left(b_{i+1} - b_i \right) \right) c_i = \sum_{i=0}^m a_i \left(b_{i+1} - b_i \right) \left(c_{i+1} - c_i \right).$	$\sum_{i=1}^m \left(a_{i-1} \left(b_i - b_{i-1} \right) - a_i \left(b_{i+1} - b_i \right) \right) c_i = \sum_{i=0}^m a_i \left(b_{i+1} - b_i \right) \left(c_{i+1} - c_i \right).$	$\sum_{i=1}^m \left(a_{i-1} \left(b_i - b_{i-1} \right) - a_i \left(b_{i+1} - b_i \right) \right) c_i = \sum_{i=0}^m a_i \left(b_{i+1} - b_i \right) \left(c_{i+1} - c_i \right).$
lyche-numerical-linear-algebra-315 EQ0932	315	■ 1.000 ● S +0	$\sum_{i=0}^m a_i \left(b_{i+1} - b_i \right) c_{i+1} - \sum_{i=0}^m a_i \left(b_{i+1} - b_i \right) c_i,$	$\sum_{i=0}^m a_i \left(b_{i+1} - b_i \right) c_{i+1} - \sum_{i=0}^m a_i \left(b_{i+1} - b_i \right) c_i,$	$\sum_{i=0}^m a_i \left(b_{i+1} - b_i \right) c_{i+1} - \sum_{i=0}^m a_i \left(b_{i+1} - b_i \right) c_i,$
lyche-numerical-linear-algebra-316 EQ0933	316	■ 1.000 ● N +0	$c(x, y) = e^{-x+y} \text{ and } f(x, y) = 1.$	$c(x, y) = e^{-x+y} \text{ and } f(x, y) = 1.$	$c(x, y) = e^{-x+y} \text{ and } f(x, y) = 1.$
lyche-numerical-linear-algebra-317 EQ0934	317	■ 1.000 ● N +0	$\kappa = \frac{\lambda_{\max}}{\lambda_{\min}} \leq \frac{c_1}{c_0}.$	$\kappa = \frac{\lambda_{\max}}{\lambda_{\min}} \leq \frac{c_1}{c_0}.$	$\kappa = \frac{\lambda_{\max}}{\lambda_{\min}} \leq \frac{c_1}{c_0}.$
lyche-numerical-linear-algebra-317 EQ0936	317	■ 1.000 ● N +0	$\sum_{j=0}^m \left(v_{i,j+1} - v_{i,j} \right)^2 + \sum_{j=1}^m \left(v_{i+1,j} - v_{i,j} \right)^2.$	$\sum_{j=0}^m \left(v_{i,j+1} - v_{i,j} \right)^2 + \sum_{j=1}^m \left(v_{i+1,j} - v_{i,j} \right)^2.$	$\sum_{j=0}^m \left(v_{i,j+1} - v_{i,j} \right)^2 + \sum_{j=1}^m \left(v_{i+1,j} - v_{i,j} \right)^2.$
lyche-numerical-linear-algebra-318 EQ0938	318	■ 1.000 ● N +0	$Q(y) = \frac{1}{2} \sum_{j=1}^n \lambda_j v_j^2 - \sum_{j=1}^n c_j v_j.$	$Q(y) = \frac{1}{2} \sum_{j=1}^n \lambda_j v_j^2 - \sum_{j=1}^n c_j v_j.$	$Q(y) = \frac{1}{2} \sum_{j=1}^n \lambda_j v_j^2 - \sum_{j=1}^n c_j v_j.$
lyche-numerical-linear-algebra-318 EQ0939	318	■ 1.000 ● K +0	$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$	$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$	$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$
lyche-numerical-linear-algebra-319 EQ0940	319	■ 1.000 ● N -1	$\hat{\boldsymbol{x}} = \boldsymbol{w} \text{ for all } \boldsymbol{w} \in \mathbb{W},$	$\boldsymbol{w}^T \boldsymbol{A}^T \boldsymbol{A} \hat{\boldsymbol{x}} = \boldsymbol{w}^T \boldsymbol{A}^T \boldsymbol{b}, \text{ for all } \boldsymbol{w} \in \mathbb{W},$	$\boldsymbol{w}^T \boldsymbol{A}^T \boldsymbol{A} \hat{\boldsymbol{x}} = \boldsymbol{w}^T \boldsymbol{A}^T \boldsymbol{b}, \text{ for all } \boldsymbol{w} \in \mathbb{W},$
lyche-numerical-linear-algebra-319 EQ0941	319	■ 1.000 ● N -1	$\ \boldsymbol{b} - \boldsymbol{A} \hat{\boldsymbol{x}}\ _2 \leq \ \boldsymbol{b} - \boldsymbol{A} \boldsymbol{w}\ _2, \text{ for all } \boldsymbol{w} \in \mathbb{W}.$	$\ \boldsymbol{b} - \boldsymbol{A} \hat{\boldsymbol{x}}\ _2 \leq \ \boldsymbol{b} - \boldsymbol{A} \boldsymbol{w}\ _2, \text{ for all } \boldsymbol{w} \in \mathbb{W}.$	$\ \boldsymbol{b} - \boldsymbol{A} \hat{\boldsymbol{x}}\ _2 \leq \ \boldsymbol{b} - \boldsymbol{A} \boldsymbol{w}\ _2, \text{ for all } \boldsymbol{w} \in \mathbb{W}.$
lyche-numerical-linear-algebra-319 EQ0943	319	■ 1.000 ●	$\langle \boldsymbol{v}, \boldsymbol{w} \rangle_{\boldsymbol{A}} := \boldsymbol{v}^T \boldsymbol{A}^T \boldsymbol{A} \boldsymbol{w}, \quad \boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^n.$	$\langle \boldsymbol{v}, \boldsymbol{w} \rangle_{\boldsymbol{A}} := \boldsymbol{v}^T \boldsymbol{A}^T \boldsymbol{A} \boldsymbol{w}, \quad \boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^n.$	$\langle \boldsymbol{v}, \boldsymbol{w} \rangle_{\boldsymbol{A}} := \boldsymbol{v}^T \boldsymbol{A}^T \boldsymbol{A} \boldsymbol{w}, \quad \boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^n.$
lyche-numerical-linear-algebra-320 EQ0944	320	■ 1.000 ● K +0	$\langle \boldsymbol{p}_k, \boldsymbol{w} \rangle_{\boldsymbol{A}} = 0, \text{ for all } \boldsymbol{w} \in \mathbb{W}_k.$	$\langle \boldsymbol{p}_k, \boldsymbol{w} \rangle_{\boldsymbol{A}} = 0, \text{ for all } \boldsymbol{w} \in \mathbb{W}_k.$	$\langle \boldsymbol{p}_k, \boldsymbol{w} \rangle_{\boldsymbol{A}} = 0, \text{ for all } \boldsymbol{w} \in \mathbb{W}_k.$
lyche-numerical-linear-algebra-321 EQ0952 (13.65)	321	■ 1.000 ● K +0	$\langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{w} \rangle = \langle \boldsymbol{b}, \boldsymbol{w} \rangle \text{ for all } \boldsymbol{w} \in \mathcal{W},$	$\langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{w} \rangle = \langle \boldsymbol{b}, \boldsymbol{w} \rangle \text{ for all } \boldsymbol{w} \in \mathcal{W},$	$\langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{w} \rangle = \langle \boldsymbol{b}, \boldsymbol{w} \rangle \text{ for all } \boldsymbol{w} \in \mathcal{W},$
lyche-numerical-linear-algebra-322 EQ0953 (13.66)	322	■ 1.000 ● N +1	$\sum_{j=1}^k x_j \langle \boldsymbol{A} \boldsymbol{w}_j, \boldsymbol{w}_i \rangle = \langle \boldsymbol{b}, \boldsymbol{w}_i \rangle, \quad i = 1, \dots, k$	$\sum_{j=1}^k x_j \langle \boldsymbol{A} \boldsymbol{w}_j, \boldsymbol{w}_i \rangle = \langle \boldsymbol{b}, \boldsymbol{w}_i \rangle, \quad i = 1, \dots, k$	$\sum_{j=1}^k x_j \langle \boldsymbol{A} \boldsymbol{w}_j, \boldsymbol{w}_i \rangle = \langle \boldsymbol{b}, \boldsymbol{w}_i \rangle, \quad i = 1, \dots, k.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-322 EQ0955	322	1.000 N -1	$\mathcal{W}_k := \operatorname{span}(\boldsymbol{b}, \boldsymbol{Bb}, \dots, \boldsymbol{B}^{k-1}\boldsymbol{b})$.	$\mathcal{W}_k := \operatorname{span}(\boldsymbol{b}, \boldsymbol{Bb}, \dots, \boldsymbol{B}^{k-1}\boldsymbol{b})$.	
lyche-numerical-linear-algebra-322 EQ0956	322	1.000 N +0	$\left\langle \boldsymbol{Ax}_k, \boldsymbol{w} \right\rangle = \left\langle \boldsymbol{b}, \boldsymbol{w} \right\rangle$, for all $\boldsymbol{w} \in \mathcal{W}_k$.	$\langle \boldsymbol{Ax}_k, \boldsymbol{w} \rangle = \langle \boldsymbol{b}, \boldsymbol{w} \rangle$, for all $\boldsymbol{w} \in \mathcal{W}_k$.	
lyche-numerical-linear-algebra-322 EQ0958	322	1.000 N +0	$\rho_k \neq 0$, $k=0, \dots, m$.	$\rho_k \neq 0$, $k=0, \dots, m$.	
lyche-numerical-linear-algebra-323 EQ0962 (13.72)	323	1.000 N +0	$\left\langle \boldsymbol{r}_{k+1}, \boldsymbol{A}^{-1}\boldsymbol{r}_{k+1} \right\rangle = \left\langle \boldsymbol{r}_{k+1}, \boldsymbol{A}^{-1}\boldsymbol{r}_j \right\rangle$, $j=0, 1, \dots, k$.	$\langle \boldsymbol{r}_{k+1}, \boldsymbol{A}^{-1}\boldsymbol{r}_{k+1} \rangle = \langle \boldsymbol{r}_{k+1}, \boldsymbol{A}^{-1}\boldsymbol{r}_j \rangle$, $j=0, 1, \dots, k$.	
lyche-numerical-linear-algebra-323 EQ0963 (13.70)	323	1.000 N -1	$T_n(t) = \cosh(n \operatorname{arccosh} t)$ for $t \geq 1$,	$T_n(t) = \cosh(n \operatorname{arccosh} t)$ for $t \geq 1$,	
lyche-numerical-linear-algebra-324 EQ0964	324	1.000 C -6	$\boldsymbol{A} \boldsymbol{x} = \begin{bmatrix} a_{1,1} & -c_{\frac{3}{2},1} & -c_{1,\frac{3}{2}} & 0 \\ -c_{\frac{3}{2},1} & a_{2,2} & 0 & -c_{2,\frac{3}{2}} \\ -c_{1,\frac{3}{2}} & 0 & a_{3,3} & -c_{\frac{3}{2},2} \\ 0 & -c_{2,\frac{3}{2}} & -c_{\frac{3}{2},2} & a_{4,4} \end{bmatrix} \begin{bmatrix} v_{1,1} \\ v_{2,1} \\ v_{1,2} \\ v_{2,2} \end{bmatrix} = \begin{bmatrix} (dv)_{1,1} \\ (dv)_{2,1} \\ (dv)_{1,2} \\ (dv)_{2,2} \end{bmatrix}$,	$\boldsymbol{Ax} = \begin{bmatrix} a_{1,1} & -c_{\frac{3}{2},1} & -c_{1,\frac{3}{2}} & 0 \\ -c_{\frac{3}{2},1} & a_{2,2} & 0 & -c_{2,\frac{3}{2}} \\ -c_{1,\frac{3}{2}} & 0 & a_{3,3} & -c_{\frac{3}{2},2} \\ 0 & -c_{2,\frac{3}{2}} & -c_{\frac{3}{2},2} & a_{4,4} \end{bmatrix} \begin{bmatrix} v_{1,1} \\ v_{2,1} \\ v_{1,2} \\ v_{2,2} \end{bmatrix} = \begin{bmatrix} (dv)_{1,1} \\ (dv)_{2,1} \\ (dv)_{1,2} \\ (dv)_{2,2} \end{bmatrix}$,	
lyche-numerical-linear-algebra-324 EQ0965	324	1.000	$\left[\begin{array}{l} a_{1,1} \\ a_{2,2} \\ a_{3,3} \\ a_{4,4} \end{array} \right] = \left[\begin{array}{l} c_{\frac{1}{2},1} + c_{1,\frac{1}{2}} + c_{1,\frac{3}{2}} + c_{\frac{3}{2},1} \\ c_{\frac{3}{2},1} + c_{2,\frac{1}{2}} + c_{2,\frac{3}{2}} + c_{\frac{5}{2},1} \\ c_{\frac{1}{2},2} + c_{1,\frac{3}{2}} + c_{1,\frac{5}{2}} + c_{\frac{3}{2},2} \\ c_{\frac{3}{2},2} + c_{2,\frac{3}{2}} + c_{2,\frac{5}{2}} + c_{\frac{5}{2},2} \end{array} \right]$.	$\begin{bmatrix} a_{1,1} \\ a_{2,2} \\ a_{3,3} \\ a_{4,4} \end{bmatrix} = \begin{bmatrix} c_{\frac{1}{2},1} + c_{1,\frac{1}{2}} + c_{1,\frac{3}{2}} + c_{\frac{3}{2},1} \\ c_{\frac{3}{2},1} + c_{2,\frac{1}{2}} + c_{2,\frac{3}{2}} + c_{\frac{5}{2},1} \\ c_{\frac{1}{2},2} + c_{1,\frac{3}{2}} + c_{1,\frac{5}{2}} + c_{\frac{3}{2},2} \\ c_{\frac{3}{2},2} + c_{2,\frac{3}{2}} + c_{2,\frac{5}{2}} + c_{\frac{5}{2},2} \end{bmatrix}$.	
lyche-numerical-linear-algebra-326 EQ0966	326	1.000 N -1	$\boldsymbol{A} = \operatorname{diag}(\boldsymbol{A}_1, \boldsymbol{A}_2, \dots, \boldsymbol{A}_r)$, $\boldsymbol{A}_i \in \mathbb{C}^{m_i \times m_i}$.	$\boldsymbol{A} = \operatorname{diag}(\boldsymbol{A}_1, \boldsymbol{A}_2, \dots, \boldsymbol{A}_r)$, $\boldsymbol{A}_i \in \mathbb{C}^{m_i \times m_i}$.	
lyche-numerical-linear-algebra-327 EQ0968	327	1.000 N +0	$\operatorname{det}(\boldsymbol{A} - \lambda \boldsymbol{I}) = \prod_{i=1}^r \operatorname{det}(\boldsymbol{A}_{ii} - \lambda \boldsymbol{I})$	$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = \prod_{i=1}^r \det(\boldsymbol{A}_{ii} - \lambda \boldsymbol{I})$	

Conf. MathPix confidence:

1.000

0.5-0.9

<0.5

— no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

not measured

[refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-328 EQ0970	328	■ 1.000 ● N +0	$\left \lambda-a_{ii}\right =\left \sum_{j \neq i} a_{ij} x_j / x_i\right \leq \sum_{j \neq i}\left a_{ij}\right \left x_j / x_i\right \leq r_i$	$ \lambda-a_{ii} =\left \sum_{j \neq i} a_{ij} x_j / x_i\right \leq \sum_{j \neq i}\left a_{ij}\right \left x_j / x_i\right \leq r_i$	$ \lambda-a_{ii} =\left \sum_{j \neq i} a_{ij} x_j / x_i\right \leq \sum_{j \neq i}\left a_{ij}\right \left x_j / x_i\right \leq r_i$
lyche-numerical-linear-algebra-328 EQ0971	328	■ 1.000 ● N +0	$R_i=\{z \in \mathbb{R}: z-2 \leq 2\}, \quad i=2,3, \ldots, m-1 .$	$R_i=\{z \in \mathbb{R}: z-2 \leq 2\}, \quad i=2,3, \ldots, m-1 .$	$R_i=\{z \in \mathbb{R}: z-2 \leq 2\}, \quad i=2,3, \ldots, m-1 .$
lyche-numerical-linear-algebra-328 EQ0972	328	■ 1.000 ● N +0	$\lambda_j=4\left[\sin \frac{j \pi}{2(m+1)}\right]^2, \quad j=1,2, \ldots, m .$	$\lambda_j=4\left[\sin \frac{j \pi}{2(m+1)}\right]^2, \quad j=1,2, \ldots, m .$	$\lambda_j=4\left[\sin \frac{j \pi}{2(m+1)}\right]^2, \quad j=1,2, \ldots, m .$
lyche-numerical-linear-algebra-329 EQ0973	329	■ 1.000 ● K +0	$\boldsymbol{A}(t):=\boldsymbol{D}+t(\boldsymbol{A}-\boldsymbol{D}), \quad \boldsymbol{D}:=\operatorname{diag}\left(a_{11}, \ldots, a_{nn}\right), \quad t \in[0,1] .$	$\boldsymbol{A}(t):=\boldsymbol{D}+t(\boldsymbol{A}-\boldsymbol{D}), \quad \boldsymbol{D}:=\operatorname{diag}\left(a_{11}, \ldots, a_{nn}\right), \quad t \in[0,1] .$	$\boldsymbol{A}(t):=\boldsymbol{D}+t(\boldsymbol{A}-\boldsymbol{D}), \quad \boldsymbol{D}:=\operatorname{diag}\left(a_{11}, \ldots, a_{nn}\right), \quad t \in[0,1] .$
lyche-numerical-linear-algebra-329 EQ0974	329	■ 1.000 ● N +0	$R_i(t)=\left\{z \in \mathbb{C}: \left z-a_{ii}\right \leq t r_i\right\}, \quad r_i:=\sum_{j=1 \atop j \neq i}^n\left a_{ij}\right .$	$R_i(t)=\{z \in \mathbb{C}: z-a_{ii} \leq t r_i\}, \quad r_i:=\sum_{j=1 \atop j \neq i}^n\left a_{ij}\right .$	$R_i(t)=\{z \in \mathbb{C}: z-a_{ii} \leq t r_i\}, \quad r_i:=\sum_{j=1 \atop j \neq i}^n\left a_{ij}\right .$
lyche-numerical-linear-algebra-330 EQ0978	330	■ 1.000 ● N +1	$\begin{aligned} & \left \mu-\lambda_1\right ^n \leq \prod_{j=1}^n\left \mu-\lambda_j\right =\left \operatorname{det}(\mu \boldsymbol{I}-\boldsymbol{A})\right =\left \operatorname{det}(\mu \boldsymbol{I}-\boldsymbol{A})\right \leq \prod_{j=1}^n\left \mu-\lambda_j\right \\ & \leq\left\ (\mu \boldsymbol{I}-\boldsymbol{A}) \boldsymbol{u}_1\right\ _2 \prod_{j=2}^n\left\ (\mu \boldsymbol{I}-\boldsymbol{A}) \boldsymbol{u}_j\right\ _2 \leq\left\ \boldsymbol{E}\right\ _2\left(\left\ (\boldsymbol{A}+\boldsymbol{E})\right\ _2+\left\ \boldsymbol{A}\right\ _2\right)^{n-1} . \end{aligned}$	$\left \mu-\lambda_1\right ^n \leq \prod_{j=1}^n\left \mu-\lambda_j\right =\left \operatorname{det}(\mu \boldsymbol{I}-\boldsymbol{A})\right =\left \operatorname{det}((\mu \boldsymbol{I}-\boldsymbol{A})\left[\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n\right])\right $ $\leq\left\ (\mu \boldsymbol{I}-\boldsymbol{A}) \boldsymbol{u}_1\right\ _2 \prod_{j=2}^n\left\ (\mu \boldsymbol{I}-\boldsymbol{A}) \boldsymbol{u}_j\right\ _2 \leq\left\ \boldsymbol{E}\right\ _2\left(\left\ (\boldsymbol{A}+\boldsymbol{E})\right\ _2+\left\ \boldsymbol{A}\right\ _2\right)^{n-1} .$	$\left \mu-\lambda_1\right ^n \leq \prod_{j=1}^n\left \mu-\lambda_j\right =\left \operatorname{det}(\mu \boldsymbol{I}-\boldsymbol{A})\right =\left \operatorname{det}((\mu \boldsymbol{I}-\boldsymbol{A})\left[\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n\right])\right $ $\leq\left\ (\mu \boldsymbol{I}-\boldsymbol{A}) \boldsymbol{u}_1\right\ _2 \prod_{j=2}^n\left\ (\mu \boldsymbol{I}-\boldsymbol{A}) \boldsymbol{u}_j\right\ _2 \leq\left\ \boldsymbol{E}\right\ _2\left(\left\ (\boldsymbol{A}+\boldsymbol{E})\right\ _2+\left\ \boldsymbol{A}\right\ _2\right)^{n-1} .$
lyche-numerical-linear-algebra-331 EQ0979 (14.2)	331	■ 1.000 ●	$ \lambda-\mu \leq K_p(\boldsymbol{X})\ \boldsymbol{r}\ _p, \quad 1 \leq p \leq \infty,$	$ \lambda-\mu \leq K_p(\boldsymbol{X})\ \boldsymbol{r}\ _p, \quad 1 \leq p \leq \infty,$	$ \lambda-\mu \leq K_p(\boldsymbol{X})\ \boldsymbol{r}\ _p, \quad 1 \leq p \leq \infty,$
lyche-numerical-linear-algebra-331 EQ0980 (14.3)	331	■ 1.000 ● N +1	$ \lambda-\mu \leq K_p(\boldsymbol{X})\ \boldsymbol{E}\ _p, \quad 1 \leq p \leq \infty,$	$ \lambda-\mu \leq K_p(\boldsymbol{X})\ \boldsymbol{E}\ _p, \quad 1 \leq p \leq \infty,$	$ \lambda-\mu \leq K_p(\boldsymbol{X})\ \boldsymbol{E}\ _p, \quad 1 \leq p \leq \infty,$
lyche-numerical-linear-algebra-332 EQ0983	332	■ 1.000 ● N +0	$\left \lambda-\mu\right \leq K_p(\boldsymbol{X}) K_p(\boldsymbol{A}) \frac{\left\ \boldsymbol{r}\right\ _p}{\left\ \boldsymbol{A}\right\ _p}, \quad 1 \leq p \leq \infty,$	$\left\ \boldsymbol{A} \boldsymbol{x}\right\ _p=\left\ \left[\lambda_1 x_1, \ldots, \lambda_n x_n\right]^T\right\ _p=\left(\sum_{j=1}^n\left \lambda_j\right ^p\left x_j\right ^p\right)^{1 / p} \leq \rho(\boldsymbol{A})\left\ \boldsymbol{x}\right\ _p .$	$\left\ \boldsymbol{A} \boldsymbol{x}\right\ _p=\left\ \left[\lambda_1 x_1, \ldots, \lambda_n x_n\right]^T\right\ _p=\left(\sum_{j=1}^n\left \lambda_j\right ^p\left x_j\right ^p\right)^{1 / p} \leq \rho(\boldsymbol{A})\left\ \boldsymbol{x}\right\ _p .$
lyche-numerical-linear-algebra-332 EQ0984 (14.4)	332	■ 1.000 ● N +0	$\frac{\left \lambda-\mu\right }{ \lambda } \leq K_p(\boldsymbol{X}) K_p(\boldsymbol{A}) \frac{\left\ \boldsymbol{r}\right\ _p}{\left\ \boldsymbol{A}\right\ _p}, \quad 1 \leq p \leq \infty,$	$\frac{ \lambda-\mu }{ \lambda } \leq K_p(\boldsymbol{X}) K_p(\boldsymbol{A}) \frac{\ \boldsymbol{r}\ _p}{\ \boldsymbol{A}\ _p}, \quad 1 \leq p \leq \infty,$	$\frac{ \lambda-\mu }{ \lambda } \leq K_p(\boldsymbol{X}) K_p(\boldsymbol{A}) \frac{\ \boldsymbol{r}\ _p}{\ \boldsymbol{A}\ _p}, \quad 1 \leq p \leq \infty,$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ09985 (14.5)	332	1.000 N +0	$\frac{ \lambda - \mu }{ \lambda } \leq K_p(\boldsymbol{X}) \left\ \boldsymbol{A}^{-1} \boldsymbol{E} \right\ _p \leq K_p(\boldsymbol{X}) K_p(\boldsymbol{A}) \frac{\ \boldsymbol{E}\ _p}{\ \boldsymbol{A}\ _p}, \quad 1 \leq p \leq \infty,$	$\frac{ \lambda - \mu }{ \lambda } \leq K_p(\boldsymbol{X}) \left\ \boldsymbol{A}^{-1} \boldsymbol{E} \right\ _p \leq K_p(\boldsymbol{X}) K_p(\boldsymbol{A}) \frac{\ \boldsymbol{E}\ _p}{\ \boldsymbol{A}\ _p}, \quad 1 \leq p \leq \infty,$	
lyche-numerical-linear-algebra- EQ09987	333	1.000 K +0	$\begin{aligned} (\boldsymbol{B} + \boldsymbol{F} - \boldsymbol{I}) \boldsymbol{x} &= (\mu \boldsymbol{A}^{-1} - \boldsymbol{A}^{-1} \boldsymbol{E} - \boldsymbol{I}) \boldsymbol{x} = \boldsymbol{A}^{-1} (\mu \boldsymbol{I} - (\boldsymbol{E} + \boldsymbol{A})) \boldsymbol{x} = \mathbf{0} \\ \boldsymbol{x} &= \boldsymbol{A} (\boldsymbol{B} + \boldsymbol{F} - \boldsymbol{I})^{-1} (\mu \boldsymbol{I} - (\boldsymbol{E} + \boldsymbol{A})) \boldsymbol{x} = \mathbf{0} \end{aligned}$	$(\boldsymbol{B} + \boldsymbol{F} - \boldsymbol{I}) \boldsymbol{x} = (\mu \boldsymbol{A}^{-1} - \boldsymbol{A}^{-1} \boldsymbol{E} - \boldsymbol{I}) \boldsymbol{x} = \boldsymbol{A}^{-1} (\mu \boldsymbol{I} - (\boldsymbol{E} + \boldsymbol{A})) \boldsymbol{x} = \mathbf{0}.$	$(\boldsymbol{B} + \boldsymbol{F} - \boldsymbol{I}) \boldsymbol{x} = (\mu \boldsymbol{A}^{-1} - \boldsymbol{A}^{-1} \boldsymbol{E} - \boldsymbol{I}) \boldsymbol{x} = \boldsymbol{A}^{-1} (\mu \boldsymbol{I} - (\boldsymbol{E} + \boldsymbol{A})) \boldsymbol{x} = \mathbf{0}.$
lyche-numerical-linear-algebra- EQ09988	333	1.000 N +0	$\boldsymbol{A}_{k+1}^* = (\boldsymbol{H}_k \boldsymbol{A}_k \boldsymbol{H}_k)^* = \boldsymbol{H}_k \boldsymbol{A}_k^* \boldsymbol{H}_k = \boldsymbol{A}_{k+1}.$	$\boldsymbol{A}_{k+1}^* = (\boldsymbol{H}_k \boldsymbol{A}_k \boldsymbol{H}_k)^* = \boldsymbol{H}_k \boldsymbol{A}_k^* \boldsymbol{H}_k = \boldsymbol{A}_{k+1}.$	
lyche-numerical-linear-algebra- EQ09990	334	1.000 N +0	$\boldsymbol{H}_k = \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{V}_k \end{bmatrix} \in \mathbb{C}^{n \times n}.$	$\boldsymbol{H}_k = \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{V}_k \end{bmatrix} \in \mathbb{C}^{n \times n}.$	
lyche-numerical-linear-algebra- EQ09993	335	1.000 W +0	$\boldsymbol{Q}_k = \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I} - \boldsymbol{v}_k \boldsymbol{v}_k^T \end{bmatrix} * \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{U}_k \end{bmatrix} = \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{U}_k - \boldsymbol{v}_k (\boldsymbol{v}_k^T \boldsymbol{U}_k) \end{bmatrix}.$	$\boldsymbol{Q}_k = \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I} - \boldsymbol{v}_k \boldsymbol{v}_k^T \end{bmatrix} * \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{U}_k \end{bmatrix} = \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{U}_k - \boldsymbol{v}_k (\boldsymbol{v}_k^T \boldsymbol{U}_k) \end{bmatrix}.$	
lyche-numerical-linear-algebra- EQ09996 (14.7)	336	1.000 N +0	$p_{k+1}(x) = (x - d_{k+1}) p_k(x) - c_k^2 p_{k-1}(x).$	$p_{k+1}(x) = (x - d_{k+1}) p_k(x) - c_k^2 p_{k-1}(x).$	$p_{k+1}(x) = (x - d_{k+1}) p_k(x) - c_k^2 p_{k-1}(x).$
lyche-numerical-linear-algebra- EQ1001	337	1.000 K +0	$p_{k+1}(y_j) = (-1)^{k+1-j} p_{k+1}(y_j) \neq 0, \text{ for } j = 0, 1, \dots, k+1.$	$p_{k+1}(y_j) = (-1)^{k+1-j} p_{k+1}(y_j) \neq 0, \text{ for } j = 0, 1, \dots, k+1.$	$p_{k+1}(y_j) = (-1)^{k+1-j} p_{k+1}(y_j) \neq 0, \text{ for } j = 0, 1, \dots, k+1.$
lyche-numerical-linear-algebra- EQ1002	337	1.000	$p_{k+1}(y_j) = -c_k^2 p_{k-1}(y_j) = -c_k^2 (y_j - z_1) \cdots (y_j - z_{k-1}).$	$p_{k+1}(y_j) = -c_k^2 p_{k-1}(y_j) = -c_k^2 (y_j - z_1) \cdots (y_j - z_{k-1}).$	$p_{k+1}(y_j) = -c_k^2 p_{k-1}(y_j) = -c_k^2 (y_j - z_1) \cdots (y_j - z_{k-1}).$
lyche-numerical-linear-algebra- EQ1003	337	1.000 N +0	$\boldsymbol{y}^* \boldsymbol{A} \boldsymbol{y} = \sum_{i=1}^n \lambda_i y_i ^2 = \sum_{i=1}^k \lambda_i x_i ^2 > 0.$	$\boldsymbol{y}^* \boldsymbol{A} \boldsymbol{y} = \sum_{i=1}^n \lambda_i y_i ^2 = \sum_{i=1}^k \lambda_i x_i ^2 > 0.$	$\boldsymbol{y}^* \boldsymbol{A} \boldsymbol{y} = \sum_{i=1}^n \lambda_i y_i ^2 = \sum_{i=1}^k \lambda_i x_i ^2 > 0.$
lyche-numerical-linear-algebra- EQ1005 (14.8)	338	1.000 N +0	$d_1(\alpha) = d_1 - \alpha, d_k(\alpha) = d_k - \alpha - c_{k-1}^2 / d_{k-1}(\alpha), k = 2, 3, \dots, n.$	$d_1(\alpha) = d_1 - \alpha, d_k(\alpha) = d_k - \alpha - c_{k-1}^2 / d_{k-1}(\alpha), k = 2, 3, \dots, n.$	$d_1(\alpha) = d_1 - \alpha, d_k(\alpha) = d_k - \alpha - c_{k-1}^2 / d_{k-1}(\alpha), k = 2, 3, \dots, n.$
lyche-numerical-linear-algebra- EQ1006	338	1.000 K +0	$\rho(x) := \# \{k : d_k(x) > 0 \text{ for } k = 1, \dots, n\}$	$\rho(x) := \# \{k : d_k(x) > 0 \text{ for } k = 1, \dots, n\}$	$\rho(x) := \# \{k : d_k(x) > 0 \text{ for } k = 1, \dots, n\}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ1007	340	■ 1.000 ● K +0	$\boldsymbol{A}=\left(\begin{array}{llll} 4 & 1 & 0 & 0 \\ 1 & 4 & 1 & 0 \\ 0 & 1 & 4 & 1 \\ 0 & 0 & 1 & 4 \end{array}\right)$.	$\boldsymbol{A} = \begin{pmatrix} 4 & 1 & 0 & 0 \\ 1 & 4 & 1 & 0 \\ 0 & 1 & 4 & 1 \\ 0 & 0 & 1 & 4 \end{pmatrix}.$	$\boldsymbol{A} = \begin{pmatrix} 4 & 1 & 0 & 0 \\ 1 & 4 & 1 & 0 \\ 0 & 1 & 4 & 1 \\ 0 & 0 & 1 & 4 \end{pmatrix}.$
lyche-numerical-linear-algebra-EQ1008	340	■ 1.000 ● N -1	$\boldsymbol{A}(t):=\boldsymbol{D}+t(\boldsymbol{A}-\boldsymbol{D}), \quad \boldsymbol{D}:=\text{diag}(a_{11},\dots,a_{nn}), \quad t\in\mathbb{R}.$ $\boldsymbol{D}:=\operatorname{diag}(\boldsymbol{a}_{11},\,\ldots,\,\boldsymbol{a}_{nn}), \quad t\in\mathbb{R}.$	$\boldsymbol{A}(t) := \boldsymbol{D} + t(\boldsymbol{A} - \boldsymbol{D}), \quad \boldsymbol{D} := \text{diag}(a_{11}, \dots, a_{nn}), \quad t \in \mathbb{R}.$	$\boldsymbol{A}(t) := \boldsymbol{D} + t(\boldsymbol{A} - \boldsymbol{D}), \quad \boldsymbol{D} := \text{diag}(a_{11}, \dots, a_{nn}), \quad t \in \mathbb{R}.$
lyche-numerical-linear-algebra-EQ1009	340	■ 1.000 ● N +0	$ \lambda-\mu \leq C\left(t_2-t_1\right)^{1 / n}, \quad \text { where } C \leq 2\left(\ \boldsymbol{D}\ _2+\ \boldsymbol{A}-\boldsymbol{D}\ _2\right).$	$ \lambda-\mu \leq C\left(t_2-t_1\right)^{1 / n}, \quad \text { where } C \leq 2\left(\ \boldsymbol{D}\ _2+\ \boldsymbol{A}-\boldsymbol{D}\ _2\right).$	$ \lambda-\mu \leq C\left(t_2-t_1\right)^{1 / n}, \quad \text { where } C \leq 2\left(\ \boldsymbol{D}\ _2+\ \boldsymbol{A}-\boldsymbol{D}\ _2\right).$
lyche-numerical-linear-algebra-EQ1013	342	■ 1.000 ● N +0	$\boldsymbol{D}^{-1 / 2}:=\operatorname{diag}\left(d_1^{-1 / 2}, \ldots, d_n^{-1 / 2}\right).$	$\boldsymbol{D}^{-1 / 2} := \text{diag}\left(d_1^{-1 / 2}, \dots, d_n^{-1 / 2}\right).$	$\boldsymbol{D}^{-1 / 2} := \text{diag}\left(d_1^{-1 / 2}, \dots, d_n^{-1 / 2}\right).$
lyche-numerical-linear-algebra-EQ1027	347	■ 1.000 ● N +0	$\rho(\lambda)^2=\left \boldsymbol{u}^* \boldsymbol{A} \boldsymbol{u}-\lambda\right ^2+\left\ \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{u}\right\ _2^2,$	$\rho(\lambda)^2 = \left \boldsymbol{u}^* \boldsymbol{A} \boldsymbol{u} - \lambda\right ^2 + \left\ \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{u}\right\ _2^2,$	$\rho(\lambda)^2 = \left \boldsymbol{u}^* \boldsymbol{A} \boldsymbol{u} - \lambda\right ^2 + \left\ \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{u}\right\ _2^2,$
lyche-numerical-linear-algebra-EQ1028	347	■ 1.000 ● N +0	$\frac{ \lambda-\mu }{ \lambda } \leq K_2(\boldsymbol{X}) K_2(\boldsymbol{A}) \frac{\left\ \boldsymbol{A} \boldsymbol{u}-\mu \boldsymbol{u}\right\ _2}{\ \boldsymbol{A}\ _2}.$	$\frac{ \lambda-\mu }{ \lambda } \leq K_2(\boldsymbol{X}) K_2(\boldsymbol{A}) \frac{\left\ \boldsymbol{A} \boldsymbol{u}-\mu \boldsymbol{u}\right\ _2}{\ \boldsymbol{A}\ _2}.$	$\frac{ \lambda-\mu }{ \lambda } \leq K_2(\boldsymbol{X}) K_2(\boldsymbol{A}) \frac{\left\ \boldsymbol{A} \boldsymbol{u}-\mu \boldsymbol{u}\right\ _2}{\ \boldsymbol{A}\ _2}.$
lyche-numerical-linear-algebra-EQ1033	349	■ 1.000 ● N +0	$\rho_k:=\frac{\boldsymbol{y}_k^* \boldsymbol{x}_{k-1}}{\boldsymbol{y}_k^* \boldsymbol{y}_k}, \quad \boldsymbol{w}_k:=\frac{\boldsymbol{x}_{k-1}}{\left\ \boldsymbol{y}_k\right\ _2}.$	$\rho_k := \frac{\boldsymbol{y}_k^* \boldsymbol{x}_{k-1}}{\boldsymbol{y}_k^* \boldsymbol{y}_k}, \quad \boldsymbol{w}_k := \frac{\boldsymbol{x}_{k-1}}{\left\ \boldsymbol{y}_k\right\ _2}.$	$\rho_k := \frac{\boldsymbol{y}_k^* \boldsymbol{x}_{k-1}}{\boldsymbol{y}_k^* \boldsymbol{y}_k}, \quad \boldsymbol{w}_k := \frac{\boldsymbol{x}_{k-1}}{\left\ \boldsymbol{y}_k\right\ _2}.$
lyche-numerical-linear-algebra-EQ1036 (15.9)	351	■ 1.000 ● N +0	$\boldsymbol{A}_{k+1}=\boldsymbol{R}_k \boldsymbol{Q}_k=\boldsymbol{Q}_k^* \boldsymbol{A}_k \boldsymbol{Q}_k,$	$\boldsymbol{A}_{k+1} = \boldsymbol{R}_k \boldsymbol{Q}_k = \boldsymbol{Q}_k^* \boldsymbol{A}_k \boldsymbol{Q}_k,$	$\boldsymbol{A}_{k+1} = \boldsymbol{R}_k \boldsymbol{Q}_k = \boldsymbol{Q}_k^* \boldsymbol{A}_k \boldsymbol{Q}_k,$
lyche-numerical-linear-algebra-EQ1037	351	■ 1.000 ● N +0	$\boldsymbol{A}_1=\boldsymbol{A}=\left(\begin{array}{ll} 2 & 1 \\ 1 & 2 \end{array}\right)=\left(\frac{1}{\sqrt{5}}\left[\begin{array}{rr} -2 & -1 \\ -1 & 2 \end{array}\right]\right)*\left(\frac{1}{\sqrt{5}}\left[\begin{array}{rr} -5 & -4 \\ 0 & 3 \end{array}\right]\right)=\boldsymbol{Q}_1 \boldsymbol{R}_1$	$\boldsymbol{A}_1 = \boldsymbol{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = \left(\frac{1}{\sqrt{5}} \begin{bmatrix} -2 & -1 \\ -1 & 2 \end{bmatrix}\right) * \left(\frac{1}{\sqrt{5}} \begin{bmatrix} -5 & -4 \\ 0 & 3 \end{bmatrix}\right) = \boldsymbol{Q}_1 \boldsymbol{R}_1$	$\boldsymbol{A}_1 = \boldsymbol{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = \left(\frac{1}{\sqrt{5}} \begin{bmatrix} -2 & -1 \\ -1 & 2 \end{bmatrix}\right) * \left(\frac{1}{\sqrt{5}} \begin{bmatrix} -5 & -4 \\ 0 & 3 \end{bmatrix}\right) = \boldsymbol{Q}_1 \boldsymbol{R}_1$
lyche-numerical-linear-algebra-EQ1038	351	■ 1.000 ●	$\boldsymbol{A}_2=\boldsymbol{R}_1 \boldsymbol{Q}_1=\frac{1}{5}\left[\begin{array}{rr} -5 & -4 \\ 0 & 3 \end{array}\right]*\left[\begin{array}{rr} -2 & -1 \\ -1 & 2 \end{array}\right]=\frac{1}{5}\left[\begin{array}{rr} 14 & -3 \\ -3 & 6 \end{array}\right]=\left[\begin{array}{rr} 2.8 & -0.6 \\ -0.6 & 1.2 \end{array}\right].$	$\boldsymbol{A}_2 = \boldsymbol{R}_1 \boldsymbol{Q}_1 = \frac{1}{5} \begin{bmatrix} -5 & -4 \\ 0 & 3 \end{bmatrix} * \begin{bmatrix} -2 & -1 \\ -1 & 2 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 14 & -3 \\ -3 & 6 \end{bmatrix} = \begin{bmatrix} 2.8 & -0.6 \\ -0.6 & 1.2 \end{bmatrix}.$	$\boldsymbol{A}_2 = \boldsymbol{R}_1 \boldsymbol{Q}_1 = \frac{1}{5} \begin{bmatrix} -5 & -4 \\ 0 & 3 \end{bmatrix} * \begin{bmatrix} -2 & -1 \\ -1 & 2 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 14 & -3 \\ -3 & 6 \end{bmatrix} = \begin{bmatrix} 2.8 & -0.6 \\ -0.6 & 1.2 \end{bmatrix}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-alignat-EQ1039	351	■ 1.000 ● K +0	$\boldsymbol{A}_{\{4\}} \approx \left[\begin{array}{cc} 2.997 & -0.074 \\ -0.074 & 1.0027 \end{array} \right], \quad \boldsymbol{A}_{\{10\}} \approx \left[\begin{array}{cc} 3.0000 & -0.0001 \\ -0.0001 & 1.0000 \end{array} \right]$	$\boldsymbol{A}_4 \approx \left[\begin{array}{cc} 2.997 & -0.074 \\ -0.074 & 1.0027 \end{array} \right], \quad \boldsymbol{A}_{10} \approx \left[\begin{array}{cc} 3.0000 & -0.0001 \\ -0.0001 & 1.0000 \end{array} \right]$	$\boldsymbol{A}_4 \approx \left[\begin{array}{cc} 2.997 & -0.074 \\ -0.074 & 1.0027 \end{array} \right], \quad \boldsymbol{A}_{10} \approx \left[\begin{array}{cc} 3.0000 & -0.0001 \\ -0.0001 & 1.0000 \end{array} \right]$
lyche-numerical-linear-alignat-EQ1040	352	■ 1.000 ● N +0	$\boldsymbol{A}_{\{1\}} = \left[\begin{array}{cccc} 0.9501 & 0.8913 & 0.8214 & 0.9218 \\ 0.8214 & 0.9218 & 0.2311 & 0.7621 \\ 0.2311 & 0.7621 & 0.4447 & 0.7382 \\ 0.6068 & 0.4565 & 0.6154 & 0.1763 \end{array} \right]$	$\boldsymbol{A}_1 = \boldsymbol{A} = \left[\begin{array}{cccc} 0.9501 & 0.8913 & 0.8214 & 0.9218 \\ 0.2311 & 0.7621 & 0.4447 & 0.7382 \\ 0.6068 & 0.4565 & 0.6154 & 0.1763 \\ 0.4860 & 0.0185 & 0.7919 & 0.4057 \end{array} \right]$	$\boldsymbol{A}_1 = \boldsymbol{A} = \left[\begin{array}{cccc} 0.9501 & 0.8913 & 0.8214 & 0.9218 \\ 0.2311 & 0.7621 & 0.4447 & 0.7382 \\ 0.6068 & 0.4565 & 0.6154 & 0.1763 \\ 0.4860 & 0.0185 & 0.7919 & 0.4057 \end{array} \right]$
lyche-numerical-linear-alignat-EQ1042 (15.10)	352	■ 1.000 ● K +0	$\tilde{\boldsymbol{Q}}_{\{k\}} := \boldsymbol{Q}_{\{1\}} \cdots \boldsymbol{Q}_{\{k\}} \text{ and } \tilde{\boldsymbol{R}}_{\{k\}} := \boldsymbol{R}_{\{1\}} \cdots \boldsymbol{R}_{\{k\}}$	$\tilde{\boldsymbol{Q}}_k := \boldsymbol{Q}_1 \cdots \boldsymbol{Q}_k \text{ and } \tilde{\boldsymbol{R}}_k := \boldsymbol{R}_k \cdots \boldsymbol{R}_1,$	$\tilde{\boldsymbol{Q}}_k := \boldsymbol{Q}_1 \cdots \boldsymbol{Q}_k \text{ and } \tilde{\boldsymbol{R}}_k := \boldsymbol{R}_k \cdots \boldsymbol{R}_1,$
lyche-numerical-linear-alignat-EQ1044 (15.11)	353	■ 1.000 ● N +0	$\boldsymbol{A}_{\{k\}} = \boldsymbol{A}_{\{k-1\}}^* \boldsymbol{A}_{\{k-1\}} \boldsymbol{Q}_{k-1} = \boldsymbol{Q}_{k-1}^* \boldsymbol{Q}_{k-2}^* \boldsymbol{A}_{k-2} \boldsymbol{Q}_{k-2} \boldsymbol{Q}_{k-1} = \cdots = \tilde{\boldsymbol{Q}}_{k-1}^* \boldsymbol{A} \tilde{\boldsymbol{Q}}_{k-1}.$	$\boldsymbol{A}_k = \boldsymbol{Q}_{k-1}^* \boldsymbol{A}_{k-1} \boldsymbol{Q}_{k-1} = \boldsymbol{Q}_{k-1}^* \boldsymbol{Q}_{k-2}^* \boldsymbol{A}_{k-2} \boldsymbol{Q}_{k-2} \boldsymbol{Q}_{k-1} = \cdots = \tilde{\boldsymbol{Q}}_{k-1}^* \boldsymbol{A} \tilde{\boldsymbol{Q}}_{k-1}.$	$\boldsymbol{A}_k = \boldsymbol{Q}_{k-1}^* \boldsymbol{A}_{k-1} \boldsymbol{Q}_{k-1} = \boldsymbol{Q}_{k-1}^* \boldsymbol{Q}_{k-2}^* \boldsymbol{A}_{k-2} \boldsymbol{Q}_{k-2} \boldsymbol{Q}_{k-1} = \cdots = \tilde{\boldsymbol{Q}}_{k-1}^* \boldsymbol{A} \tilde{\boldsymbol{Q}}_{k-1}.$
lyche-numerical-linear-alignat-EQ1046	353	■ 1.000 ● N +0	$\boldsymbol{A}^k \boldsymbol{e}_1 = \tilde{\boldsymbol{Q}}_k \tilde{\boldsymbol{R}}_k \boldsymbol{e}_1 = \tilde{r}_{11}^{(k)} \tilde{\boldsymbol{Q}}_k \boldsymbol{e}_1 \text{ or } \tilde{q}_1^{(k)} := \tilde{\boldsymbol{Q}}_k \boldsymbol{e}_1 = \frac{1}{\tilde{r}_{11}^{(k)}} \boldsymbol{A}^k \boldsymbol{e}_1.$	$\boldsymbol{A}^k \boldsymbol{e}_1 = \tilde{\boldsymbol{Q}}_k \tilde{\boldsymbol{R}}_k \boldsymbol{e}_1 = \tilde{r}_{11}^{(k)} \tilde{\boldsymbol{Q}}_k \boldsymbol{e}_1 \text{ or } \tilde{q}_1^{(k)} := \tilde{\boldsymbol{Q}}_k \boldsymbol{e}_1 = \frac{1}{\tilde{r}_{11}^{(k)}} \boldsymbol{A}^k \boldsymbol{e}_1.$	$\boldsymbol{A}^k \boldsymbol{e}_1 = \tilde{\boldsymbol{Q}}_k \tilde{\boldsymbol{R}}_k \boldsymbol{e}_1 = \tilde{r}_{11}^{(k)} \tilde{\boldsymbol{Q}}_k \boldsymbol{e}_1 \text{ or } \tilde{q}_1^{(k)} := \tilde{\boldsymbol{Q}}_k \boldsymbol{e}_1 = \frac{1}{\tilde{r}_{11}^{(k)}} \boldsymbol{A}^k \boldsymbol{e}_1.$
lyche-numerical-linear-alignat-EQ1048	355	■ 1.000 ● N +0	$\boldsymbol{A}_{\{k+1\}} = \boldsymbol{A}_{\{k\}}^* \boldsymbol{A}_{\{k\}} \boldsymbol{Q}_k + s_k \boldsymbol{I} = \boldsymbol{Q}_k^* \boldsymbol{A}_k \boldsymbol{Q}_k$	$\boldsymbol{A}_{k+1} = \boldsymbol{Q}_k^* (\boldsymbol{A}_k - s_k \boldsymbol{I}) \boldsymbol{Q}_k + s_k \boldsymbol{I} = \boldsymbol{Q}_k^* \boldsymbol{A}_k \boldsymbol{Q}_k$	$\boldsymbol{A}_{k+1} = \boldsymbol{Q}_k^* (\boldsymbol{A}_k - s_k \boldsymbol{I}) \boldsymbol{Q}_k + s_k \boldsymbol{I} = \boldsymbol{Q}_k^* \boldsymbol{A}_k \boldsymbol{Q}_k$
lyche-numerical-linear-alignat-EQ1050 (16.1)	358	■ 1.000 ● N +0	$\nabla \boldsymbol{f} := \left[\begin{array}{c} D_1 \boldsymbol{f} \\ \vdots \\ D_n \boldsymbol{f} \end{array} \right], \quad \boldsymbol{H} \boldsymbol{f} := \nabla \nabla^T \boldsymbol{f} := \left[\begin{array}{ccc} D_1 D_1 \boldsymbol{f} & \cdots & D_1 D_n \boldsymbol{f} \\ \vdots & & \vdots \\ D_n D_1 & \cdots & D_n D_n \boldsymbol{f} \end{array} \right],$	$\nabla \boldsymbol{f} := \left[\begin{array}{c} D_1 \boldsymbol{f} \\ \vdots \\ D_n \boldsymbol{f} \end{array} \right], \quad \boldsymbol{H} \boldsymbol{f} := \nabla \nabla^T \boldsymbol{f} := \left[\begin{array}{ccc} D_1 D_1 \boldsymbol{f} & \cdots & D_1 D_n \boldsymbol{f} \\ \vdots & & \vdots \\ D_n D_1 & \cdots & D_n D_n \boldsymbol{f} \end{array} \right],$	$\nabla \boldsymbol{f} := \left[\begin{array}{c} D_1 \boldsymbol{f} \\ \vdots \\ D_n \boldsymbol{f} \end{array} \right], \quad \boldsymbol{H} \boldsymbol{f} := \nabla \nabla^T \boldsymbol{f} := \left[\begin{array}{ccc} D_1 D_1 \boldsymbol{f} & \cdots & D_1 D_n \boldsymbol{f} \\ \vdots & & \vdots \\ D_n D_1 & \cdots & D_n D_n \boldsymbol{f} \end{array} \right],$
lyche-numerical-linear-alignat-EQ1051	359	■ 1.000 ●	$\nabla^T \boldsymbol{f} := \left[\begin{array}{ccc} D_1 \boldsymbol{f}_1 & \cdots & D_n \boldsymbol{f}_1 \\ \vdots & & \vdots \\ D_1 \boldsymbol{f}_m & \cdots & D_n \boldsymbol{f}_m \end{array} \right].$	$\nabla^T \boldsymbol{f} := \left[\begin{array}{ccc} D_1 \boldsymbol{f}_1 & \cdots & D_n \boldsymbol{f}_1 \\ \vdots & & \vdots \\ D_1 \boldsymbol{f}_m & \cdots & D_n \boldsymbol{f}_m \end{array} \right].$	$\nabla^T \boldsymbol{f} := \left[\begin{array}{ccc} D_1 \boldsymbol{f}_1 & \cdots & D_n \boldsymbol{f}_1 \\ \vdots & & \vdots \\ D_1 \boldsymbol{f}_m & \cdots & D_n \boldsymbol{f}_m \end{array} \right].$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean					
[refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ1053 (16.2)	359	■ 1.000 ● N +0	$f(\boldsymbol{x}+\boldsymbol{h})=f(\boldsymbol{x})+\boldsymbol{h}^T\nabla f(\boldsymbol{x})+\frac{1}{2}\boldsymbol{h}^T\nabla\nabla^T f(\boldsymbol{c})\boldsymbol{h},\text{ for some }\boldsymbol{c}\in L.$	$f(\boldsymbol{x}+\boldsymbol{h})=f(\boldsymbol{x})+\boldsymbol{h}^T\nabla f(\boldsymbol{x})+\frac{1}{2}\boldsymbol{h}^T\nabla\nabla^T f(\boldsymbol{c})\boldsymbol{h},\text{ for some }\boldsymbol{c}\in L.$	$f(\boldsymbol{x}+\boldsymbol{h})=f(\boldsymbol{x})+\boldsymbol{h}^T\nabla f(\boldsymbol{x})+\frac{1}{2}\boldsymbol{h}^T\nabla\nabla^T f(\boldsymbol{c})\boldsymbol{h},\text{ for some }\boldsymbol{c}\in L.$
lyche-numerical-linear-algebra EQ1055	359	■ 1.000 ● N +0	$g(1)=g(0)+g^{\prime}(0)+\frac{1}{2}g^{\prime\prime}(u),\text{ for some }u\in(0,1),$	$g(1)=g(0)+g'(0)+\frac{1}{2}g''(u),\text{ for some }u\in(0,1),$	$g(1)=g(0)+g'(0)+\frac{1}{2}g''(u),\text{ for some }u\in(0,1),$
lyche-numerical-linear-algebra EQ1056	360	■ 1.000 ● S +1	$\begin{aligned} D_i\left(\boldsymbol{x}^T\boldsymbol{B}\boldsymbol{x}\right) &= \lim_{h\rightarrow 0}\frac{1}{h}\left(\left(\boldsymbol{x}+h\boldsymbol{e}_i\right)^T\boldsymbol{B}\left(\boldsymbol{x}+h\boldsymbol{e}_i\right)-\boldsymbol{x}^T\boldsymbol{B}\boldsymbol{x}\right) \\ &= \lim_{h\rightarrow 0}\left(\boldsymbol{e}_i^T\boldsymbol{B}\boldsymbol{x}+\boldsymbol{x}^T\boldsymbol{B}\boldsymbol{e}_i+h\boldsymbol{e}_i^T\boldsymbol{e}_i\right)=\boldsymbol{e}_i^T\left(\boldsymbol{B}+\boldsymbol{B}^T\right)\boldsymbol{x}, \end{aligned}$	$D_i\left(\boldsymbol{x}^T\boldsymbol{B}\boldsymbol{x}\right) = \lim_{h\rightarrow 0}\frac{1}{h}\left(\left(\boldsymbol{x}+h\boldsymbol{e}_i\right)^T\boldsymbol{B}\left(\boldsymbol{x}+h\boldsymbol{e}_i\right)-\boldsymbol{x}^T\boldsymbol{B}\boldsymbol{x}\right) \\ = \lim_{h\rightarrow 0}\left(\boldsymbol{e}_i^T\boldsymbol{B}\boldsymbol{x}+\boldsymbol{x}^T\boldsymbol{B}\boldsymbol{e}_i+h\boldsymbol{e}_i^T\boldsymbol{e}_i\right)=\boldsymbol{e}_i^T\left(\boldsymbol{B}+\boldsymbol{B}^T\right)\boldsymbol{x},$	$D_i(\boldsymbol{x}^T\boldsymbol{B}\boldsymbol{x}) = \lim_{h\rightarrow 0}\frac{1}{h}\left((\boldsymbol{x}+h\boldsymbol{e}_i)^T\boldsymbol{B}(\boldsymbol{x}+h\boldsymbol{e}_i)-\boldsymbol{x}^T\boldsymbol{B}\boldsymbol{x}\right) \\ = \lim_{h\rightarrow 0}\left(\boldsymbol{e}_i^T\boldsymbol{B}\boldsymbol{x}+\boldsymbol{x}^T\boldsymbol{B}\boldsymbol{e}_i+h\boldsymbol{e}_i^T\boldsymbol{e}_i\right)=\boldsymbol{e}_i^T(\boldsymbol{B}+\boldsymbol{B}^T)\boldsymbol{x},$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

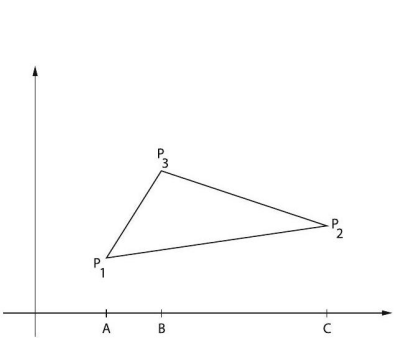
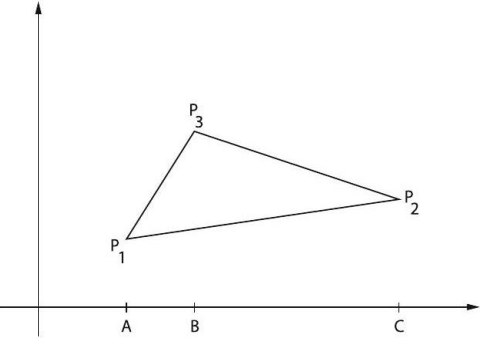
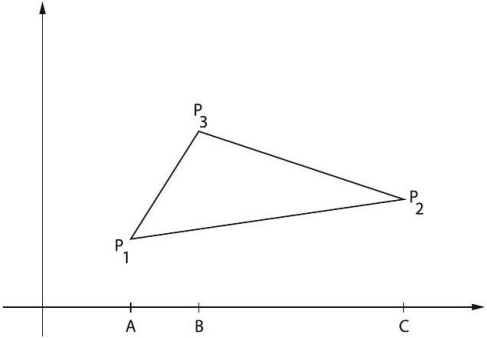
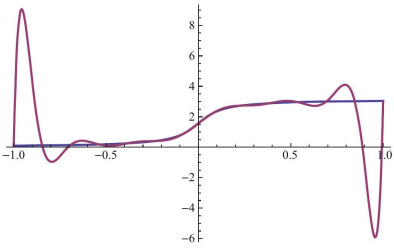
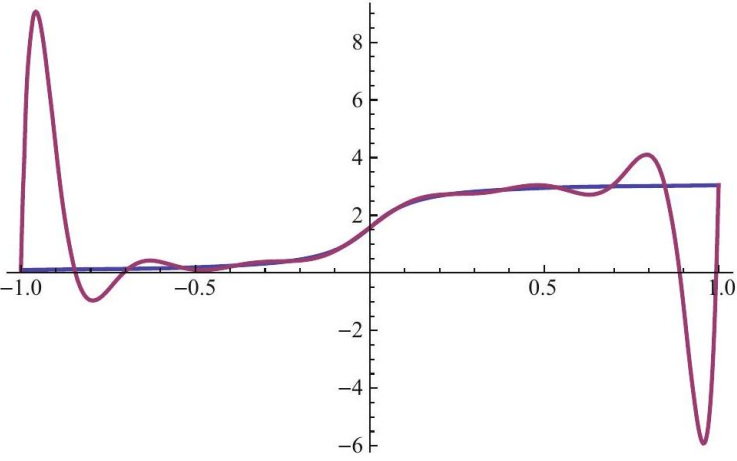
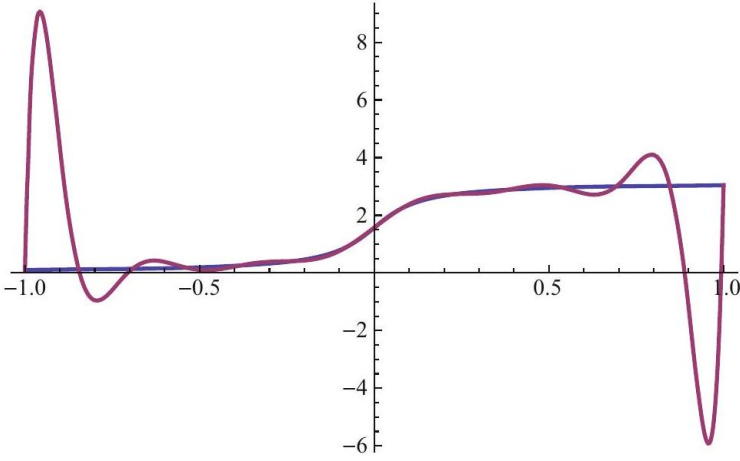
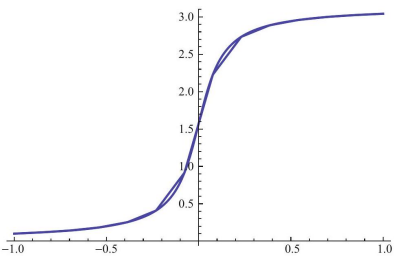
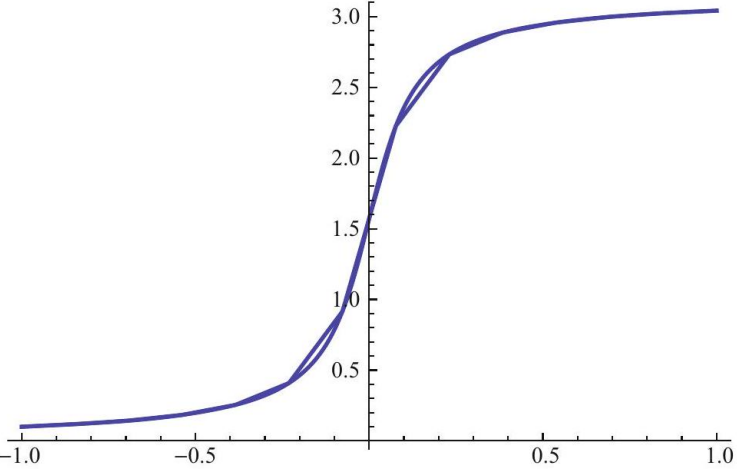
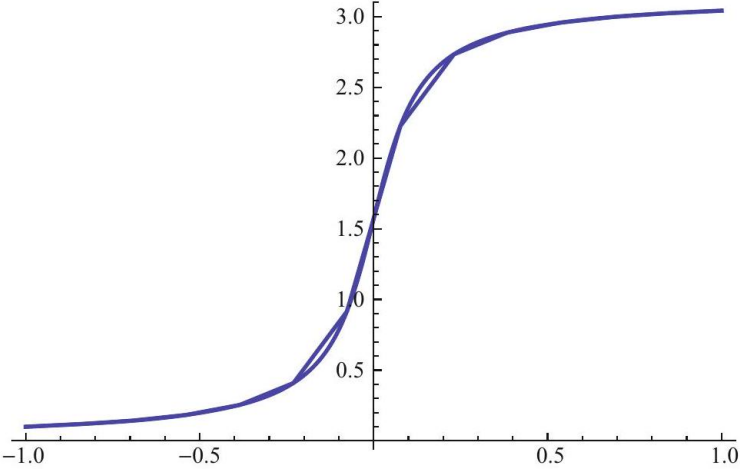
Tables

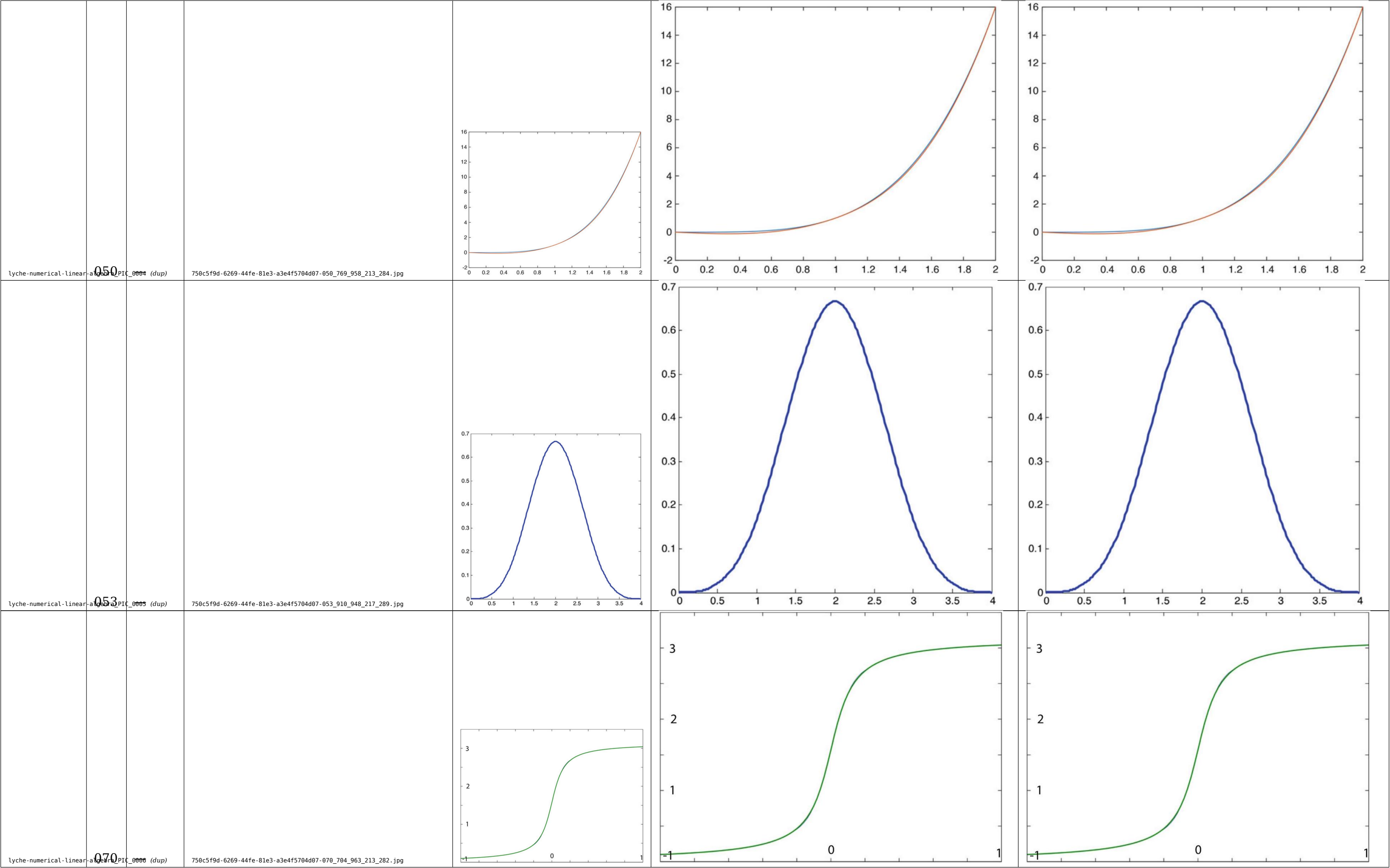
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																
lyche-numerical-linear-algebra-008	008		$\begin{tabular}{ t l } \hline \text{LU and QR factorizations} \\ \& 2.4, 2.5, 3.2, 3.3, 3.5, 4.1, 4.2, 5.1-5.4, 5.6 \\ \hline \text{SVD, norms and LSQ} \\ \& 6.1, 6.3, 7.1-7.4, 8.1-8.3, 9.1-9.3, 9.4.1 \\ \hline \text{Kronecker products} \\ \& 10.1, 10.2, 10.3, 11.1, 11.2, 11.3 \\ \hline \text{Iterative methods} \\ \& 12.1-12.4, 13.1-13.3, 13.5 \\ \hline \text{Eigenpairs} \\ \& 14.1-14.5, 15.1-15.3 \\ \hline \end{tabular}$	<table><tr><td>LU and QR factorizations</td><td>2.4, 2.5, 3.2, 3.3, 3.5, 4.1, 4.2, 5.1-5.4, 5.6</td></tr><tr><td>SVD, norms and LSQ</td><td>6.1, 6.3, 7.1-7.4, 8.1-8.3, 9.1-9.3, 9.4.1</td></tr><tr><td>Kronecker products</td><td>10.1, 10.2, 10.3, 11.1, 11.2, 11.3</td></tr><tr><td>Iterative methods</td><td>12.1-12.4, 13.1-13.3, 13.5</td></tr><tr><td>Eigenpairs</td><td>14.1-14.5, 15.1-15.3</td></tr></table>	LU and QR factorizations	2.4, 2.5, 3.2, 3.3, 3.5, 4.1, 4.2, 5.1-5.4, 5.6	SVD, norms and LSQ	6.1, 6.3, 7.1-7.4, 8.1-8.3, 9.1-9.3, 9.4.1	Kronecker products	10.1, 10.2, 10.3, 11.1, 11.2, 11.3	Iterative methods	12.1-12.4, 13.1-13.3, 13.5	Eigenpairs	14.1-14.5, 15.1-15.3	LU and QR factorizations 2.4, 2.5, 3.2, 3.3, 3.5, 4.1, 4.2, 5.1 – 5.4, 5.6 SVD, norms and LSQ 6.1, 6.3, 7.1 – 7.4, 8.1 – 8.3, 9.1 – 9.3, 9.4.1 Kronecker products 10.1, 10.2, 10.3, 11.1, 11.2, 11.3 Iterative methods 12.1 – 12.4, 13.1 – 13.3, 13.5 Eigenpairs 14.1 – 14.5, 15.1 – 15.3																																						
LU and QR factorizations	2.4, 2.5, 3.2, 3.3, 3.5, 4.1, 4.2, 5.1-5.4, 5.6																																																				
SVD, norms and LSQ	6.1, 6.3, 7.1-7.4, 8.1-8.3, 9.1-9.3, 9.4.1																																																				
Kronecker products	10.1, 10.2, 10.3, 11.1, 11.2, 11.3																																																				
Iterative methods	12.1-12.4, 13.1-13.3, 13.5																																																				
Eigenpairs	14.1-14.5, 15.1-15.3																																																				
lyche-numerical-linear-algebra-268	268		$\begin{tabular}{ t l l l l l } \hline \& \text{(k_{100})} \\ \& \text{(k_{2500})} \\ \& \text{(k_{10} 000)} \\ \& \text{(k_{40000})} \\ \& \text{(k_{160} 000)} \\ \hline \text{J} \\ \& 385 \\ \& 8386 \\ \hline \text{GS} \\ \& 194 \\ \& 4194 \\ \hline \text{SOR} \\ \& 35 \\ \& 164 \\ \& 324 \\ \& 645 \\ \& 1286 \\ \hline \end{tabular}$	<table><tr><td></td><td>k_{100}</td><td>k_{2500}</td><td>$k_{10\,000}$</td><td>$k_{40\,000}$</td><td>$k_{160\,000}$</td></tr><tr><td>J</td><td>385</td><td>8386</td><td></td><td></td><td></td></tr><tr><td>GS</td><td>194</td><td>4194</td><td></td><td></td><td></td></tr><tr><td>SOR</td><td>35</td><td>164</td><td>324</td><td>645</td><td>1286</td></tr></table>		k_{100}	k_{2500}	$k_{10\,000}$	$k_{40\,000}$	$k_{160\,000}$	J	385	8386				GS	194	4194				SOR	35	164	324	645	1286	<table><tr><td></td><td>k_{100}</td><td>k_{2500}</td><td>$k_{10\,000}$</td><td>$k_{40\,000}$</td><td>$k_{160\,000}$</td></tr><tr><td>J</td><td>385</td><td>8386</td><td></td><td></td><td></td></tr><tr><td>GS</td><td>194</td><td>4194</td><td></td><td></td><td></td></tr><tr><td>SOR</td><td>35</td><td>164</td><td>324</td><td>645</td><td>1286</td></tr></table>		k_{100}	k_{2500}	$k_{10\,000}$	$k_{40\,000}$	$k_{160\,000}$	J	385	8386				GS	194	4194				SOR	35	164	324	645	1286
	k_{100}	k_{2500}	$k_{10\,000}$	$k_{40\,000}$	$k_{160\,000}$																																																
J	385	8386																																																			
GS	194	4194																																																			
SOR	35	164	324	645	1286																																																
	k_{100}	k_{2500}	$k_{10\,000}$	$k_{40\,000}$	$k_{160\,000}$																																																
J	385	8386																																																			
GS	194	4194																																																			
SOR	35	164	324	645	1286																																																
lyche-numerical-linear-algebra-278	278		$\begin{tabular}{ t l l l l l } \hline \& \text{(n=100)} \\ \& \text{(n=2500)} \\ \& \text{(k_{100})} \\ \& \text{(k_{2500})} \\ \hline \text{J} \\ \& 0.959493 \\ \& 0.998103 \\ \& 446 \\ \& 9703 \\ \hline \text{GS} \\ \& 0.920627 \\ \& 0.99621 \\ \& 223 \\ \& 4852 \\ \hline \text{SOR} \\ \& 0.56039 \\ \& 0.88402 \\ \& 32 \\ \& 150 \\ \hline \end{tabular}$	<table><tr><td></td><td>n = 100</td><td>n = 2500</td><td>k_{100}</td><td>k_{2500}</td></tr><tr><td>J</td><td>0.959493</td><td>0.998103</td><td>446</td><td>9703</td></tr><tr><td>GS</td><td>0.920627</td><td>0.99621</td><td>223</td><td>4852</td></tr><tr><td>SOR</td><td>0.56039</td><td>0.88402</td><td>32</td><td>150</td></tr></table>		n = 100	n = 2500	k_{100}	k_{2500}	J	0.959493	0.998103	446	9703	GS	0.920627	0.99621	223	4852	SOR	0.56039	0.88402	32	150	<table><tr><td></td><td>n=100</td><td>n = 2500</td><td>k_{100}</td><td>k_{2500}</td></tr><tr><td>J</td><td>0.959493</td><td>0.998103</td><td>446</td><td>9703</td></tr><tr><td>GS</td><td>0.920627</td><td>0.99621</td><td>223</td><td>4852</td></tr><tr><td>SOR</td><td>0.56039</td><td>0.88402</td><td>32</td><td>150</td></tr></table>		n=100	n = 2500	k_{100}	k_{2500}	J	0.959493	0.998103	446	9703	GS	0.920627	0.99621	223	4852	SOR	0.56039	0.88402	32	150								
	n = 100	n = 2500	k_{100}	k_{2500}																																																	
J	0.959493	0.998103	446	9703																																																	
GS	0.920627	0.99621	223	4852																																																	
SOR	0.56039	0.88402	32	150																																																	
	n=100	n = 2500	k_{100}	k_{2500}																																																	
J	0.959493	0.998103	446	9703																																																	
GS	0.920627	0.99621	223	4852																																																	
SOR	0.56039	0.88402	32	150																																																	
lyche-numerical-linear-algebra-298	298		$\begin{tabular}{ t l l l l l } \hline \text{(n)} \\ \& 2500 \\ \& 10000 \\ \& 40000 \\ \& 100000 \\ \& 400000 \\ \hline \text{(K)} \\ \& 19 \\ \& 18 \\ \& 16 \\ \& 15 \\ \hline \end{tabular}$	<table><tr><td>n</td><td>2500</td><td>10000</td><td>40000</td><td>1000000</td><td>4000000</td></tr><tr><td>K</td><td>19</td><td>18</td><td>18</td><td>16</td><td>15</td></tr></table>	n	2500	10000	40000	1000000	4000000	K	19	18	18	16	15	<table><tr><td>n</td><td>2500</td><td>10000</td><td>40000</td><td>1000000</td><td>4000000</td></tr><tr><td>K</td><td>19</td><td>18</td><td>18</td><td>16</td><td>15</td></tr></table>	n	2500	10000	40000	1000000	4000000	K	19	18	18	16	15																								
n	2500	10000	40000	1000000	4000000																																																
K	19	18	18	16	15																																																
n	2500	10000	40000	1000000	4000000																																																
K	19	18	18	16	15																																																
lyche-numerical-linear-algebra-298	298		$\begin{tabular}{ t l l l l l } \hline \text{(n)} \\ \& 2500 \\ \& 10000 \\ \& 40000 \\ \& 160000 \\ \hline \text{(K)} \\ \& 94 \\ \& 188 \\ \& 370 \\ \& 735 \\ \hline \text{(K / \sqrt{n})} \\ \& 1.88 \\ \& 1.88 \\ \& 1.85 \\ \& 1.84 \\ \hline \end{tabular}$	<table><tr><td>n</td><td>2500</td><td>10000</td><td>40000</td><td>160000</td></tr><tr><td>K</td><td>94</td><td>188</td><td>370</td><td>735</td></tr><tr><td>$K/\sqrt{n}$</td><td>1.88</td><td>1.88</td><td>1.85</td><td>1.84</td></tr></table>	n	2500	10000	40000	160000	K	94	188	370	735	$K/\sqrt{n}$	1.88	1.88	1.85	1.84	<table><tr><td>n</td><td>2500</td><td>10000</td><td>40000</td><td>160000</td></tr><tr><td>K</td><td>94</td><td>188</td><td>370</td><td>735</td></tr><tr><td>$K/\sqrt{n}$</td><td>1.88</td><td>1.88</td><td>1.85</td><td>1.84</td></tr></table>	n	2500	10000	40000	160000	K	94	188	370	735	$K/\sqrt{n}$	1.88	1.88	1.85	1.84																		
n	2500	10000	40000	160000																																																	
K	94	188	370	735																																																	
$K/\sqrt{n}$	1.88	1.88	1.85	1.84																																																	
n	2500	10000	40000	160000																																																	
K	94	188	370	735																																																	
$K/\sqrt{n}$	1.88	1.88	1.85	1.84																																																	
lyche-numerical-linear-algebra-316	316		$\begin{tabular}{ t l l l l l } \hline \text{(n)} \\ \& 2500 \\ \& 10000 \\ \& 22500 \\ \& 40000 \\ \& 62500 \\ \hline \text{(K)} \\ \& 222 \\ \& 472 \\ \& 728 \\ \& 986 \\ \& 1246 \\ \hline \text{(K / \sqrt{n})} \\ \& 4.44 \\ \& 4.72 \\ \& 4.85 \\ \& 4.93 \\ \& 4.98 \\ \hline \text{(K_{\text{pre}})} \\ \& 22 \\ \& 23 \\ \& 23 \\ \& 23 \\ \& 23 \\ \hline \end{tabular}$	<table><tr><td>n</td><td>2500</td><td>10000</td><td>22500</td><td>40000</td><td>62500</td></tr><tr><td>K</td><td>222</td><td>472</td><td>728</td><td>986</td><td>1246</td></tr><tr><td>$K/\sqrt{n}$</td><td>4.44</td><td>4.72</td><td>4.85</td><td>4.93</td><td>4.98</td></tr><tr><td>K_{pre}</td><td>22</td><td>23</td><td>23</td><td>23</td><td>23</td></tr></table>	n	2500	10000	22500	40000	62500	K	222	472	728	986	1246	$K/\sqrt{n}$	4.44	4.72	4.85	4.93	4.98	K_{pre}	22	23	23	23	23	<table><tr><td>n</td><td>2500</td><td>10000</td><td>22500</td><td>40000</td><td>62500</td></tr><tr><td>K</td><td>222</td><td>472</td><td>728</td><td>986</td><td>1246</td></tr><tr><td>$K/\sqrt{n}$</td><td>4.44</td><td>4.72</td><td>4.85</td><td>4.93</td><td>4.98</td></tr><tr><td>K_{pre}</td><td>22</td><td>23</td><td>23</td><td>23</td><td>23</td></tr></table>	n	2500	10000	22500	40000	62500	K	222	472	728	986	1246	$K/\sqrt{n}$	4.44	4.72	4.85	4.93	4.98	K_{pre}	22	23	23	23	23
n	2500	10000	22500	40000	62500																																																
K	222	472	728	986	1246																																																
$K/\sqrt{n}$	4.44	4.72	4.85	4.93	4.98																																																
K_{pre}	22	23	23	23	23																																																
n	2500	10000	22500	40000	62500																																																
K	222	472	728	986	1246																																																
$K/\sqrt{n}$	4.44	4.72	4.85	4.93	4.98																																																
K_{pre}	22	23	23	23	23																																																
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.																																																					

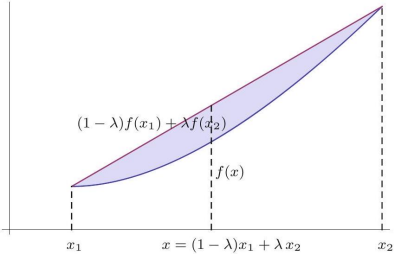
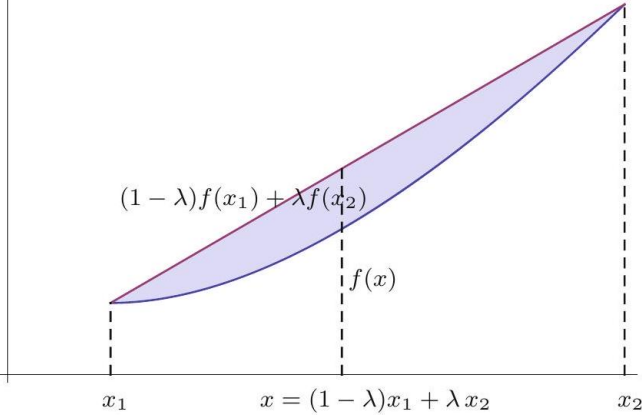
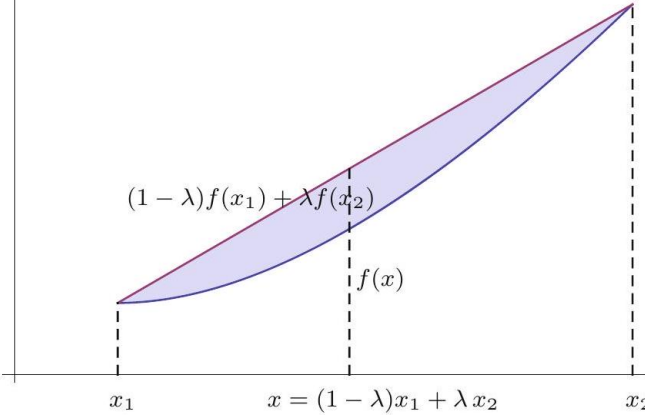
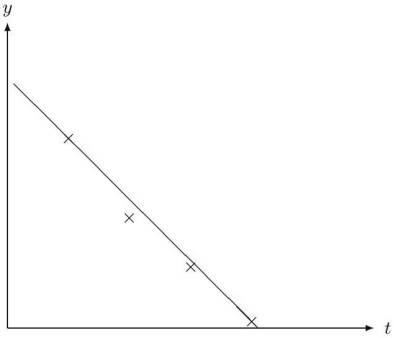
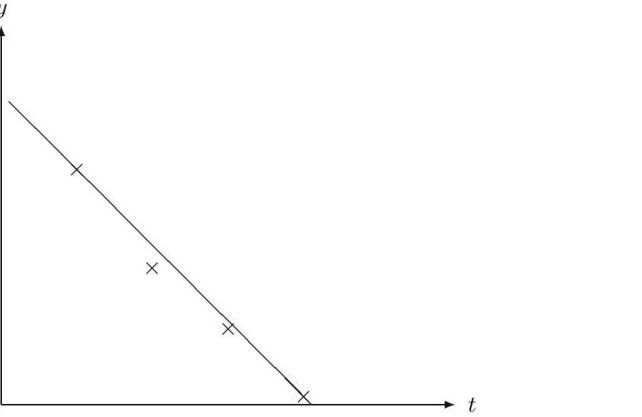
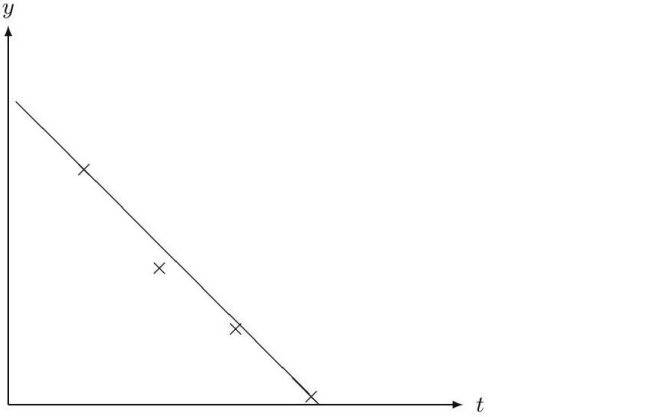
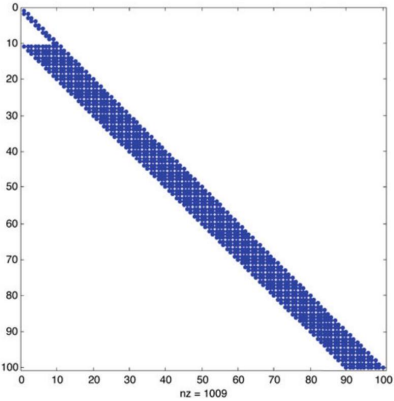
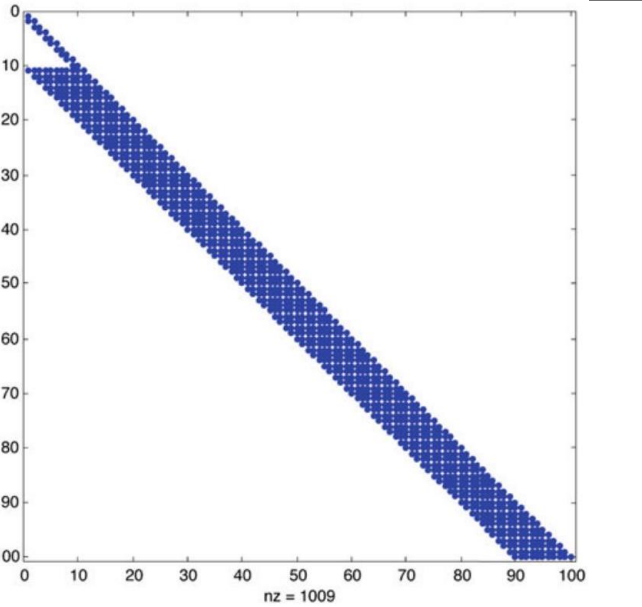
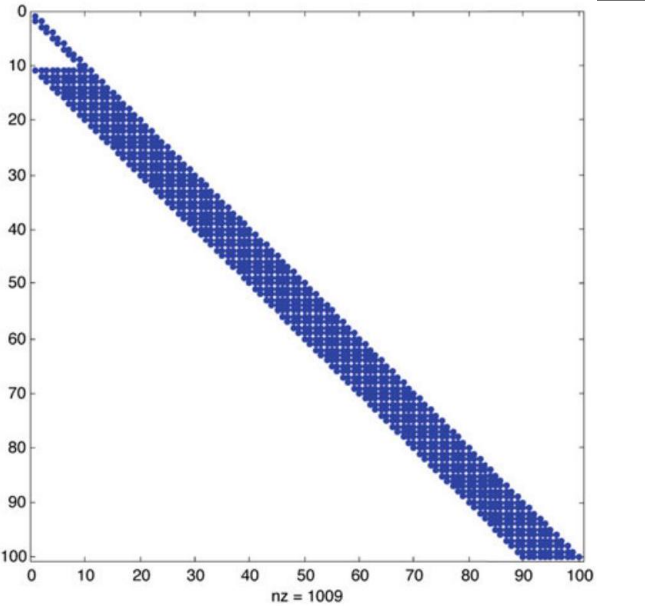
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																				
lyche-numerical-linear-algebra	350	0.7	<pre>\begin{tabular}{t}{ l l l l l l } \hline \ (k\) & 1 & 2 & 3 & 4 & 5 \\ \hline \ (\boldsymbol{r}\ _{2}\) & \ (1.0 \mathrm{e}+000\) & 7.7e-002 & 1.6e-004 & 8.2e-010 & 2.0e-020 \\ \hline \ (\left s-\lambda_{1}\right \) & 3.7e-001 & -1.2e-002 & -2.9e-005 & -1.4e-010 & -2.2e-016 \\ \hline \end{tabular}</pre>	<table><tr><td>k</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>$\ \boldsymbol{r}\ _2$</td><td>1.0e + 000</td><td>7.7e-002</td><td>1.6e-004</td><td>8.2e-010</td><td>2.0e-020</td></tr><tr><td>$s - \lambda_1$</td><td>3.7e-001</td><td>-1.2e-002</td><td>-2.9e-005</td><td>-1.4e-010</td><td>-2.2e-016</td></tr></table>	k	1	2	3	4	5	$\ \boldsymbol{r}\ _2$	1.0e + 000	7.7e-002	1.6e-004	8.2e-010	2.0e-020	$ s - \lambda_1 $	3.7e-001	-1.2e-002	-2.9e-005	-1.4e-010	-2.2e-016	<table><tr><td>k</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>$\ \boldsymbol{r}\ _2$</td><td>1.0e+000</td><td>7.7e−002</td><td>1.6e−004</td><td>8.2e−010</td><td>2.0e−020</td></tr><tr><td>$s - \lambda_1$</td><td>3.7e−001</td><td>−1.2e−002</td><td>−2.9e−005</td><td>−1.4e−010</td><td>−2.2e−016</td></tr></table>	k	1	2	3	4	5	$\ \boldsymbol{r}\ _2$	1.0e+000	7.7e−002	1.6e−004	8.2e−010	2.0e−020	$ s - \lambda_1 $	3.7e−001	−1.2e−002	−2.9e−005	−1.4e−010	−2.2e−016
k	1	2	3	4	5																																				
$\ \boldsymbol{r}\ _2$	1.0e + 000	7.7e-002	1.6e-004	8.2e-010	2.0e-020																																				
$ s - \lambda_1 $	3.7e-001	-1.2e-002	-2.9e-005	-1.4e-010	-2.2e-016																																				
k	1	2	3	4	5																																				
$\ \boldsymbol{r}\ _2$	1.0e+000	7.7e−002	1.6e−004	8.2e−010	2.0e−020																																				
$ s - \lambda_1 $	3.7e−001	−1.2e−002	−2.9e−005	−1.4e−010	−2.2e−016																																				
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.																																									

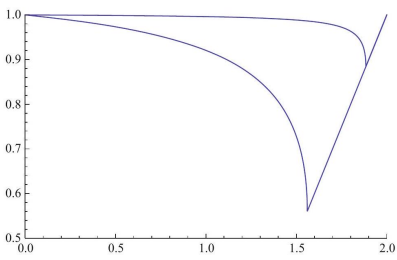
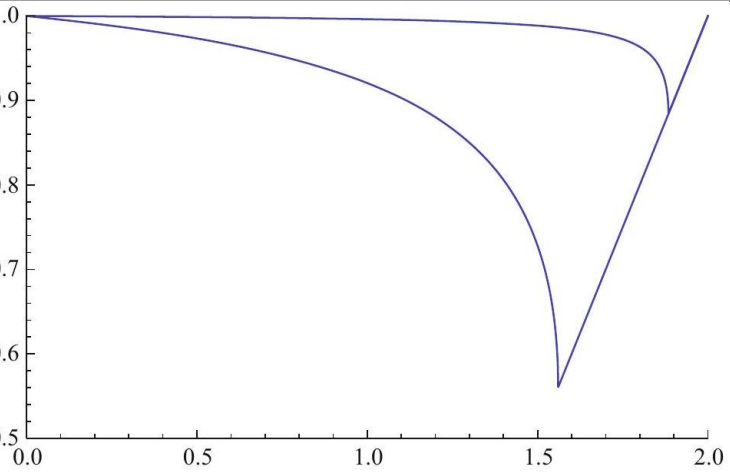
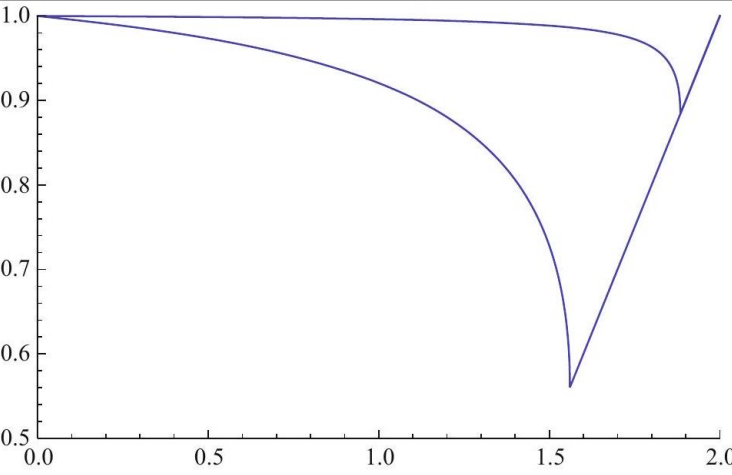
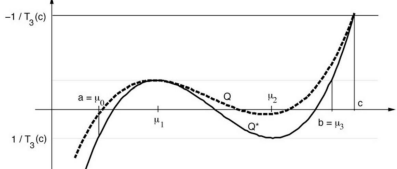
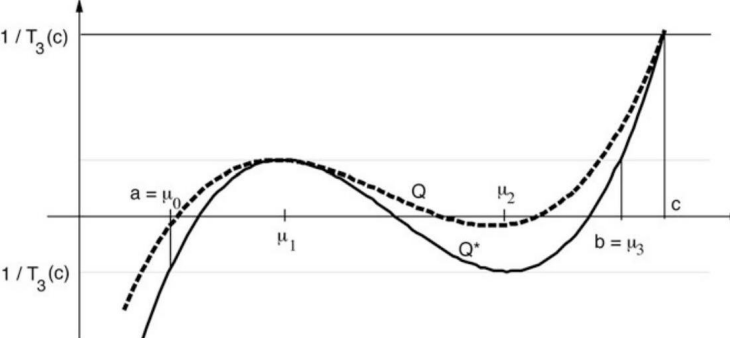
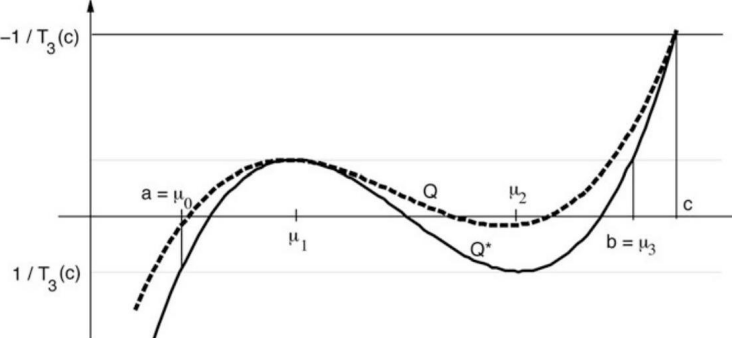
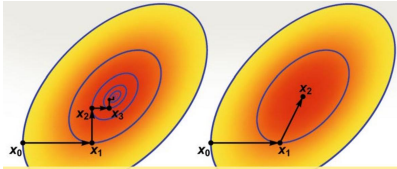
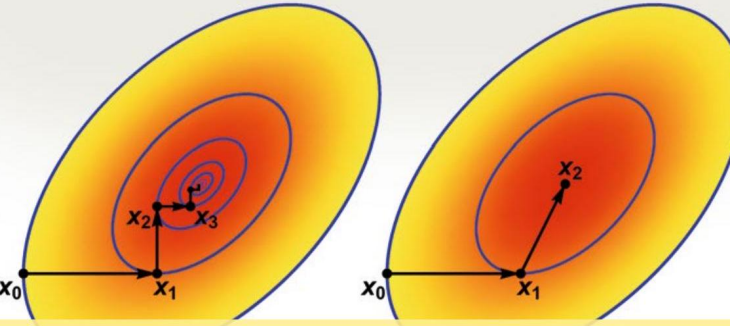
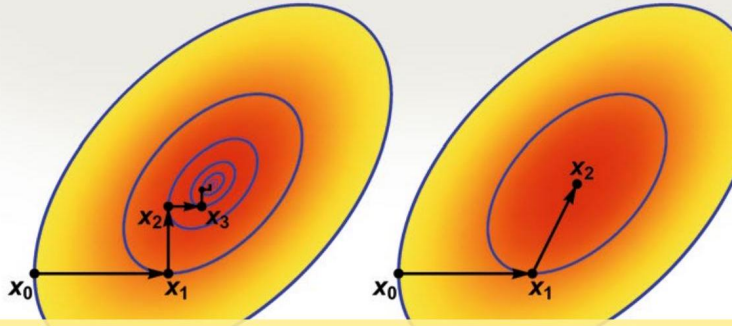
Image regions — rendered against scan

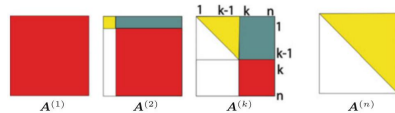
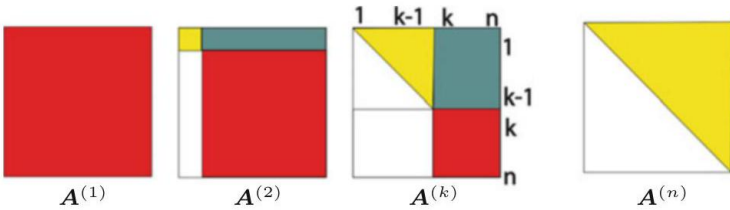
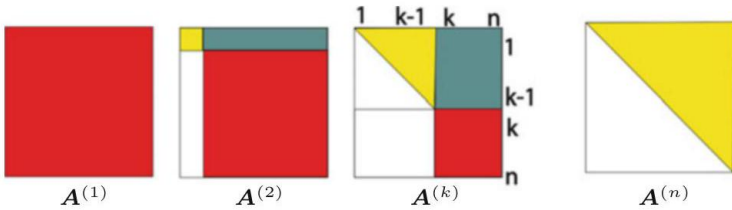
Two comparable cells per row. **Rendered:** the region’s own LaTeX compiled as its OWN document, or — where no LaTeX exists — the scan again, so every row has two cells and the ink difference reads as a floor rather than as missing-versus-present. A duplicated row is marked *(dup)* and its distance is a SELF-comparison, not agreement between two sources. **Scan:** the tex.zip image whose filename region 5-tuple matches this row, else the CDN crop. The Class column carries the residual class beside MathPix’s confidence, as the equation rows do. Setting a region’s LaTeX does NOT say it renders to what is on the page — that is what the two columns are for, and 281 is the open question they exist to let a reader answer. A region with no LaTeX can be reconstructed with pdfdrill vision; verify any LLM result against the real ink with inkdrill.

Identifier	Page	Class	Source	Author source	Rendered	Scan
lyche-numerical-linear-algebra-PIC_0001	043	(dup)	750c5f9d-6269-44fe-81e3-a3e4f5704d07-043_441_635_213_705.jpg			
lyche-numerical-linear-algebra-PIC_0002	048	(dup)	750c5f9d-6269-44fe-81e3-a3e4f5704d07-048_598_971_213_276.jpg			
lyche-numerical-linear-algebra-PIC_0003	048	(dup)	750c5f9d-6269-44fe-81e3-a3e4f5704d07-048_615_969_946_278.jpg			

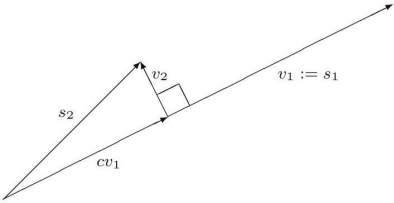
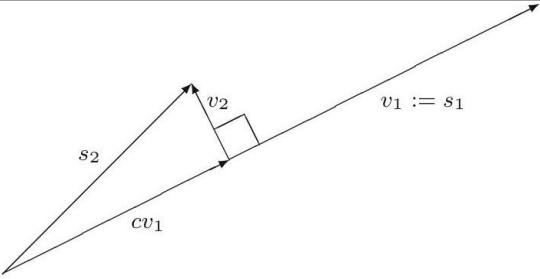
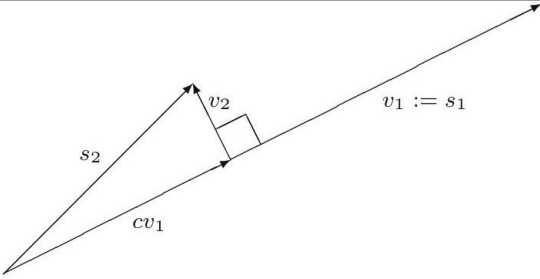
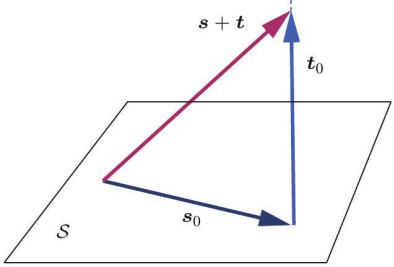
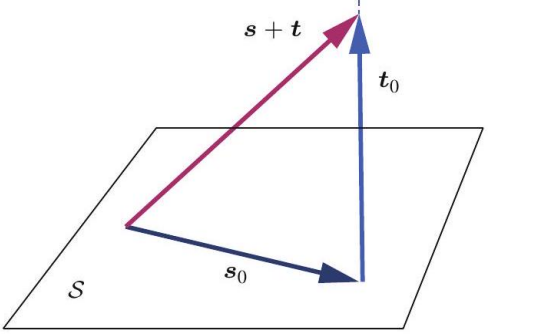
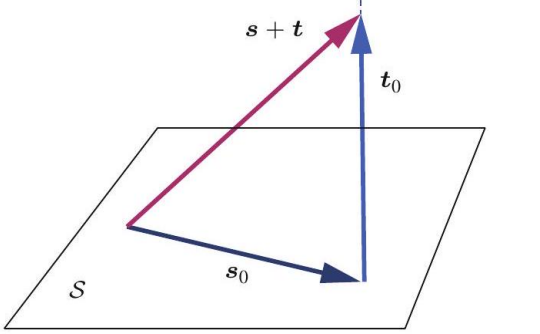
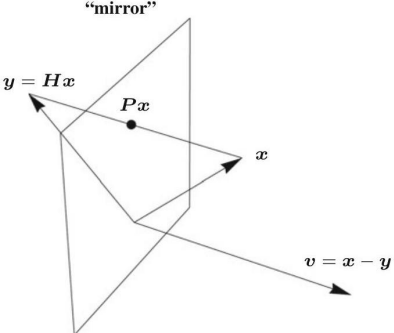
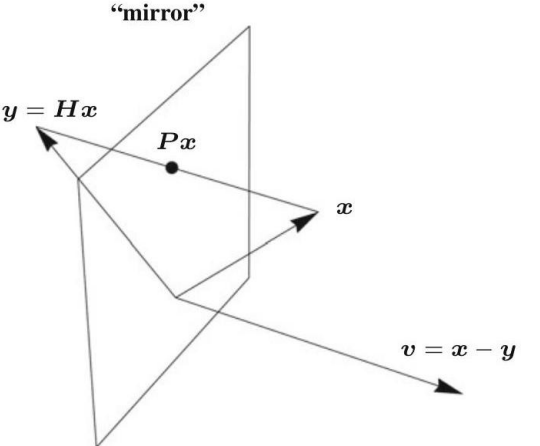
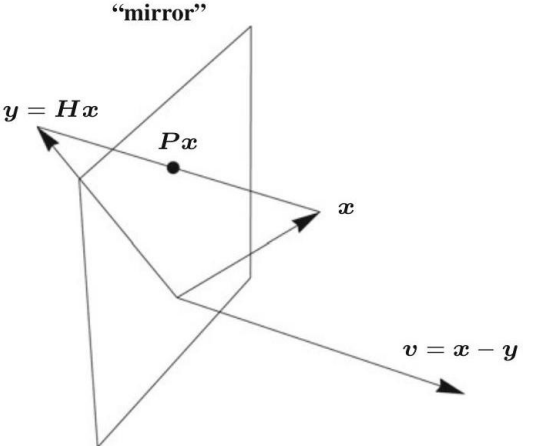


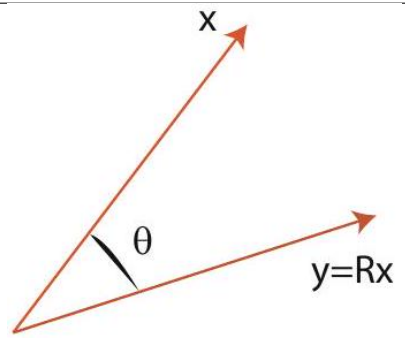
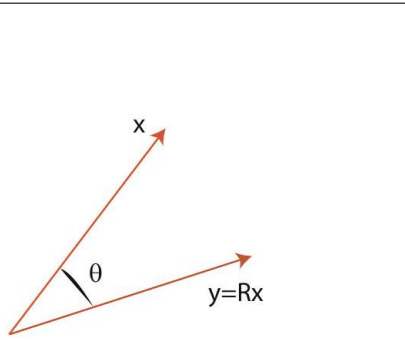
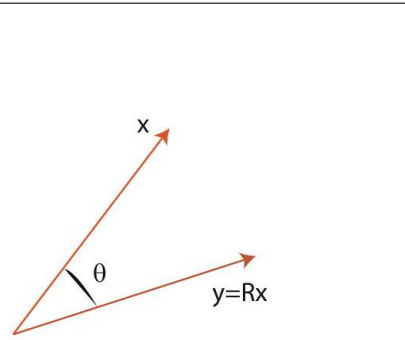
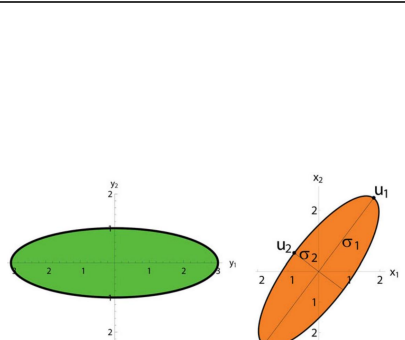
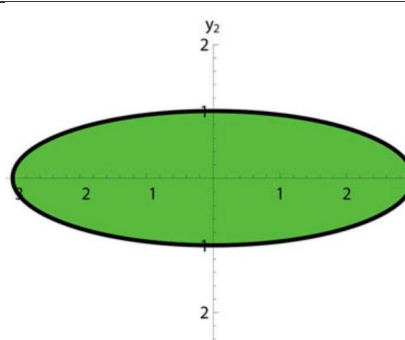
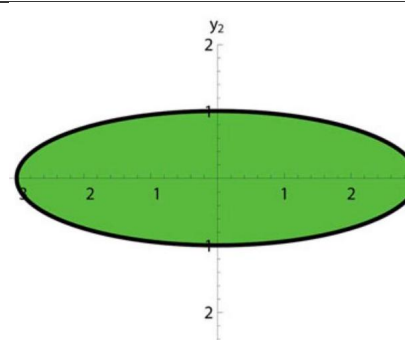
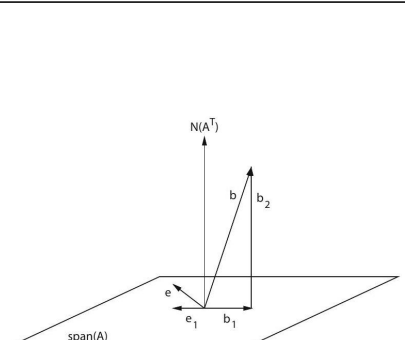
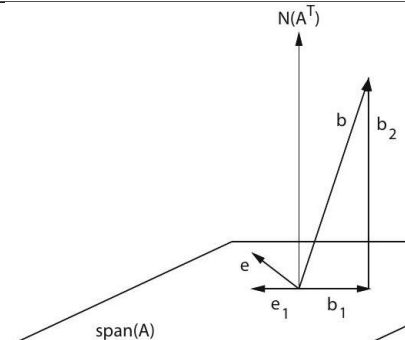
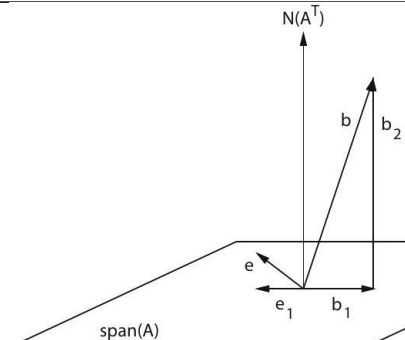
lyche-numerical-linear-algebra-PIE_0007 (dup)	199	750c5f9d-6269-44fe-81e3-a3e4f5704d07-199_550_851_1359_337.jpg			
lyche-numerical-linear-algebra-PIE_0008 (dup)	215	750c5f9d-6269-44fe-81e3-a3e4f5704d07-215_543_636_1546_706.jpg			
lyche-numerical-linear-algebra-PIE_0009 (dup)	251	750c5f9d-6269-44fe-81e3-a3e4f5704d07-251_788_777_1215_376.jpg			

lyche-numerical-linear-algebra-DIA_0010 (dup)	278	750c5f9d-6269-44fe-81e3-a3e4f5704d07-278_609_969_215_278.jpg			
lyche-numerical-linear-algebra-DIA_0011 (dup)	306	750c5f9d-6269-44fe-81e3-a3e4f5704d07-306_527_1156_1449_185.jpg			
lyche-numerical-linear-algebra-DIA_0012 (dup)	339	750c5f9d-6269-44fe-81e3-a3e4f5704d07-339_40_320_596_624.jpg	$\lambda = (a + b)/2$; return	$\lambda = (a + b)/2$; return	$\lambda = (a + b)/2$; return
lyche-numerical-linear-algebra-DIA_0013 (dup)	354	750c5f9d-6269-44fe-81e3-a3e4f5704d07-354_42_316_1949_278.jpg	$A_{k+1} = R_k Q_k + s_k I.$	$A_{k+1} = R_k Q_k + s_k I.$	$A_{k+1} = R_k Q_k + s_k I.$
lyche-numerical-linear-algebra-DIA_0001 (dup)	001	750c5f9d-6269-44fe-81e3-a3e4f5704d07-001_500_1166_59_176.jpg			
lyche-numerical-linear-algebra-DIA_0002 (dup)	055	no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre>function [l,u]=trifactor(a,d,c) % [l,u]=trifactor(a,d,c) u=d; l=a; for k =1:length(a) l(k)=a(k)/u(k); u(k+1)=d(k+1)-l(k)*c(k); end</pre> Listing 2.1 trifactor	<pre>function [l,u]=trifactor(a,d,c) % [l,u]=trifactor(a,d,c) u=d; l=a; for k =1:length(a) l(k)=a(k)/u(k); u(k+1)=d(k+1)-l(k)*c(k); end</pre> Listing 2.1 trifactor
lyche-numerical-linear-algebra-DIA_0003 (dup)	055	no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre>function x = trisolve (l,u,c,b) % x = trisolve (l,u,c,b) x=b; n= size(b,1); for k =2:n x(k,:)=b(k,:)-l(k-1)*x(k-1,:); end x(n,:)=x(n,:)/u(n); for k=(n-1):-1:1 x(k,:)=(x(k,:)-c(k)*x(k+1,:))/u(k); end</pre> Listing 2.2 trisolve	<pre>function x = trisolve (l,u,c,b) % x = trisolve (l,u,c,b) x=b; n= size(b,1); for k =2:n x(k,:)=b(k,:)-l(k-1)*x(k-1,:); end x(n,:)=x(n,:)/u(n); for k=(n-1):-1:1 x(k,:)=(x(k,:)-c(k)*x(k+1,:))/u(k); end</pre> Listing 2.2 trisolve

lyche-numerical-linear-algebra-DIA_0694 (dup)	069		no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre>function [x,C]=splineint(a,b,y,mu1,munp1) % [x,C]=splineint(a,b,y,mu1,munp1) y=y(:); n=length(y)-1; h=(b-a)/n; x=a:h:b-h; c=ones(n-2,1); [l,u]= trifactor(c,4*ones(n-1,1),c); b1=6/h^2*(y(3:n+1)-2*y(2:n)+y(1:n-1)); b1(1)=b1(1)-mu1; b1(n-1)=b1(n-1)-munp1; mu= [mu1;trisolve(l,u,c,b1);munp1]; C=zeros(4*n,1); C(1:4:4*n-3)=y(1:n); C(2:4:4*n-2)=(y(2:n+1)-y(1:n))/h... -h*mu(1:n)/3-h*mu(2:n+1)/6; C(3:4:4*n-1)=mu(1:n)/2; C(4:4:4*n)=(mu(2:n+1)-mu(1:n))/(6*h); C=reshape(C,4,n)'; end</pre> Listing 2.3 splineint	<pre>function [x,C]=splineint(a,b,y,mu1,munp1) % [x,C]=splineint(a,b,y,mu1,munp1) y=y(:); n=length(y)-1; h=(b-a)/n; x=a:h:b-h; c=ones(n-2,1); [l,u]= trifactor(c,4*ones(n-1,1),c); b1=6/h^2*(y(3:n+1)-2*y(2:n)+y(1:n-1)); b1(1)=b1(1)-mu1; b1(n-1)=b1(n-1)-munp1; mu= [mu1;trisolve(l,u,c,b1);munp1]; C=zeros(4*n,1); C(1:4:4*n-3)=y(1:n); C(2:4:4*n-2)=(y(2:n+1)-y(1:n))/h... -h*mu(1:n)/3-h*mu(2:n+1)/6; C(3:4:4*n-1)=mu(1:n)/2; C(4:4:4*n)=(mu(2:n+1)-mu(1:n))/(6*h); C=reshape(C,4,n)'; end</pre> Listing 2.3 splineint
lyche-numerical-linear-algebra-DIA_0695 (dup)	069		no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre>function i = findsubintervals(t,x) %i = findsubintervals(t,x) k=length(t); m=length(x); if k<2 i=ones(m,1); else t(1)=min(x(1),t(1))-1; [~,j]=sort([t(:)',x(:)']); i=(find(j>k)-(1:m))'; end</pre> Listing 2.4 findsubintervals	<pre>function i = findsubintervals(t,x) %i = findsubintervals(t,x) k=length(t); m=length(x); if k<2 i=ones(m,1); else t(1)=min(x(1),t(1))-1; [~,j]=sort([t(:)',x(:)']); i=(find(j>k)-(1:m))'; end</pre> Listing 2.4 findsubintervals
lyche-numerical-linear-algebra-DIA_0700 (dup)	070		no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre>function [X,G]=splineval(x,C,X) % [X,G]=splineval(x,C,X) m=length(X); i=findsubintervals(x,X); G=zeros(m,1); for j=1:m k=i(j); t=X(j)-x(k); G(j)=[1,t,t^2,t^3]*C(k,:)'; end</pre> Listing 2.5 splineval	<pre>function [X,G]=splineval(x,C,X) % [X,G]=splineval(x,C,X) m=length(X); i=findsubintervals(x,X); G=zeros(m,1); for j=1:m k=i(j); t=X(j)-x(k); G(j)=[1,t,t^2,t^3]*C(k,:)'; end</pre> Listing 2.5 splineval
lyche-numerical-linear-algebra-DIA_0777 (dup)	077	750c5f9d-6269-44fe-81e3-a3e4f5704d07-077_274_965_1748_280.jpg				
lyche-numerical-linear-algebra-DIA_0810 (dup)	081		no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre>function x=rforwardsolve(A,b,d) % x=rforwardsolve(A,b,d) n=length(b); x=b; x(1)=b(1)/A(1,1); for k=2:n lk=max(1,k-d); x(k)=(b(k)-A(k,lk:(k-1))*x(lk:(k-1)))/A(k,k); end</pre> Listing 3.1 rforwardsolve	<pre>function x=rforwardsolve(A,b,d) % x=rforwardsolve(A,b,d) n=length(b); x=b; x(1)=b(1)/A(1,1); for k=2:n lk=max(1,k-d); x(k)=(b(k)-A(k,lk:(k-1))*x(lk:(k-1)))/A(k,k); end</pre> Listing 3.1 rforwardsolve
lyche-numerical-linear-algebra-DIA_0811 (dup)	081		no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre>function x=rbacksolve(A,b,d) % x=rbacksolve(A,b,d) n=length(b); x=b; x(n)=b(n)/A(n,n); for k=n-1:-1:1 uk=min(n,k+d); x(k)=(b(k)-A(k,(k+1):uk)*x((k+1):uk))/A(k,k); end</pre> Listing 3.2 rbacksolve	<pre>function x=rbacksolve(A,b,d) % x=rbacksolve(A,b,d) n=length(b); x=b; x(n)=b(n)/A(n,n); for k=n-1:-1:1 uk=min(n,k+d); x(k)=(b(k)-A(k,(k+1):uk)*x((k+1):uk))/A(k,k); end</pre> Listing 3.2 rbacksolve

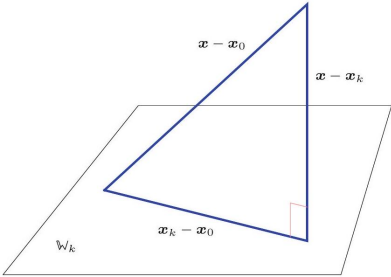
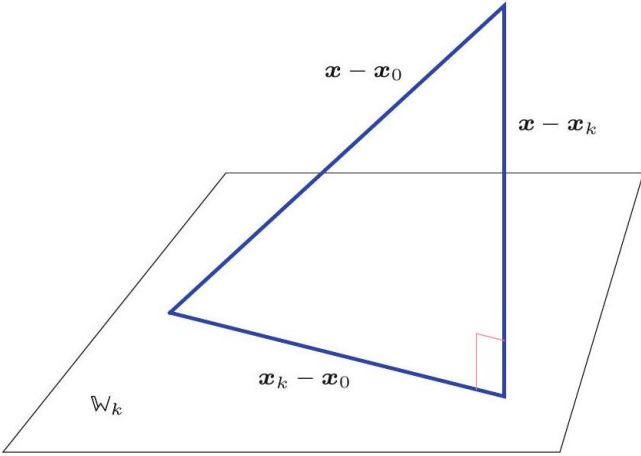
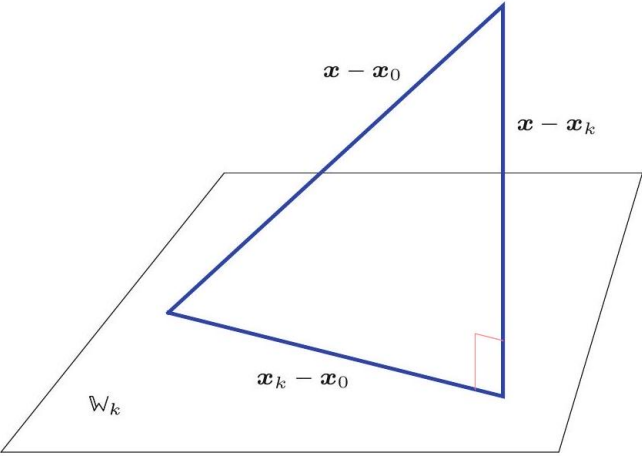
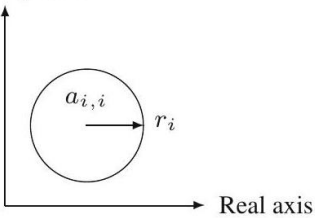
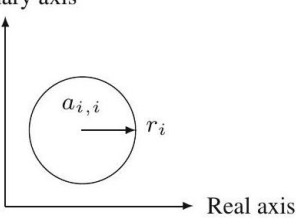
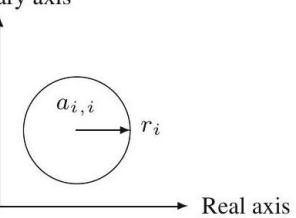
lyche-numerical-linear-algebra_DIA_0010 (dup)	082		no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre>function x=cforwardsolve(A,b,d) %x=cforwardsolve(A,b,d) x=b; n=length(b); for k=1:n-1 x(k)=b(k)/A(k,k); uk=min(n,k+d); b((k+1):uk)=b((k+1):uk)-A((k+1):uk,k)*x(k); end x(n)=b(n)/A(n,n); end</pre> Listing 3.3 cforwardsolve	<pre>function x=cforwardsolve(A,b,d) %x=cforwardsolve(A,b,d) x=b; n=length(b); for k=1:n-1 x(k)=b(k)/A(k,k); uk=min(n,k+d); b((k+1):uk)=b((k+1):uk)-A((k+1):uk,k)*x(k); end x(n)=b(n)/A(n,n); end</pre> Listing 3.3 cforwardsolve
lyche-numerical-linear-algebra_DIA_0011 (dup)	091		no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre>function [L,U]=L1U(A,d) % [L,U]=L1U(A,d) n=length(A); L=eye(n,n); U=zeros(n,n);U(1,1)=A(1,1); for k=2:n km=max(1,k-d); L(k,km:(k-1))=rforwardsolve(U(km:(k-1) ... ,km:(k-1))',A(k,km:(k-1))',d)'; U(km:k,k)=rforwardsolve(L(km:k,km:k),A(km:k,k),d); end</pre> Listing 3.4 L1U	<pre>function [L,U]=L1U(A,d) % [L,U]=L1U(A,d) n=length(A); L=eye(n,n); U=zeros(n,n);U(1,1)=A(1,1); for k=2:n km=max(1,k-d); L(k,km:(k-1))=rforwardsolve(U(km:(k-1) ... ,km:(k-1))',A(k,km:(k-1))',d)'; U(km:k,k)=rforwardsolve(L(km:k,km:k),A(km:k,k),d); end</pre> Listing 3.4 L1U
lyche-numerical-linear-algebra_DIA_0012 (dup)	093		no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre>function x=cbacksolve(A,b,d) % x=cbacksolve(A,b,d) x=b; n=length(b); for k=n:-1:2 x(k)=b(k)/A(k,k); lk=max(1,k-d); b(lk:(k-1))=b(lk:(k-1))-A(lk:(k-1),k)*x(k); end x(1)=b(1)/A(1,1); end</pre> Listing 3.5 cbacksolve	<pre>function x=cbacksolve(A,b,d) % x=cbacksolve(A,b,d) x=b; n=length(b); for k=n:-1:2 x(k)=b(k)/A(k,k); lk=max(1,k-d); b(lk:(k-1))=b(lk:(k-1))-A(lk:(k-1),k)*x(k); end x(1)=b(1)/A(1,1); end</pre> Listing 3.5 cbacksolve
lyche-numerical-linear-algebra_DIA_0013 (dup)	102		no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre>function [L,dg]=LDLs(A,d) % [L,dg]=LDLs(A,d) n=length(A); L=eye(n,n); dg=zeros(n,1);dg(1)=A(1,1); for k=2:n m=rforwardsolve(L(1:k-1,1:k-1),A(1:k-1,k),d); L(k,1:k-1)=m./dg(1:k-1); dg(k)=A(k,k)-L(k,1:k-1)*m; end</pre> Listing 4.1 LDLs	<pre>function [L,dg]=LDLs(A,d) % [L,dg]=LDLs(A,d) n=length(A); L=eye(n,n); dg=zeros(n,1);dg(1)=A(1,1); for k=2:n m=rforwardsolve(L(1:k-1,1:k-1),A(1:k-1,k),d); L(k,1:k-1)=m./dg(1:k-1); dg(k)=A(k,k)-L(k,1:k-1)*m; end</pre> Listing 4.1 LDLs
lyche-numerical-linear-algebra_DIA_0014 (dup)	106		no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre>function L=bandcholesky(A,d) %L=bandcholesky(A,d) [L,dg]=LDL(A,d); L=L*diag(sqrt(dg)); end</pre> Listing 4.2 bandcholesky	<pre>function L=bandcholesky(A,d) %L=bandcholesky(A,d) [L,dg]=LDL(A,d); L=L*diag(sqrt(dg)); end</pre> Listing 4.2 bandcholesky
lyche-numerical-linear-algebra_DIA_0015 (dup)	111		no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre>function L=bandsemicholeskyL(A,d) %L=bandsemicholeskyL(A,d) n=length(A); for k=1:n kp=min(n,k+d); if A(k,k)>0 A(k,k)=sqrt(A(k,k)); A((k+1):kp,k)=A((k+1):kp,k)/A(k,k); for j=k+1:kp A(j:kp,j)=A(j:kp,j)-A(j,k)*A(j:kp,k); end else A(k:kp,k)=zeros(kp-k+1,1); end end L=tril(A); end</pre> Listing 4.3 bandsemicholeskyL	<pre>function L=bandsemicholeskyL(A,d) %L=bandsemicholeskyL(A,d) n=length(A); for k=1:n kp=min(n,k+d); if A(k,k)>0 A(k,k)=sqrt(A(k,k)); A((k+1):kp,k)=A((k+1):kp,k)/A(k,k); for j=k+1:kp A(j:kp,j)=A(j:kp,j)-A(j,k)*A(j:kp,k); end else A(k:kp,k)=zeros(kp-k+1,1); end end L=tril(A); end</pre> Listing 4.3 bandsemicholeskyL

lyche-numerical-linear-algebra-DIA_0016 (dup)	120	750c5f9d-6269-44fe-81e3-a3e4f5704d07-120_360_708_227_632.jpg			
lyche-numerical-linear-algebra-DIA_0017 (dup)	122	750c5f9d-6269-44fe-81e3-a3e4f5704d07-122_438_638_213_704.jpg			
lyche-numerical-linear-algebra-DIA_0018 (dup)	125	750c5f9d-6269-44fe-81e3-a3e4f5704d07-125_586_680_1348_664.jpg			
lyche-numerical-linear-algebra-DIA_0019 (dup)	127	no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre>function [u,a]=housegen(x) % [u,a]=housegen(x) a=norm(x); if a==0 u=x; u(1)=sqrt(2); return; end if x(1)== 0 r=1; else r=x(1)/abs(x(1)); end u=conj(r)*x/a; u(1)=u(1)+1; u=u/sqrt(u(1)); a=-r*a; end</pre> Listing 5.1 housegen	<pre>function [u,a]=housegen(x) % [u,a]=housegen(x) a=norm(x); if a==0 u=x; u(1)=sqrt(2); return; end if x(1)== 0 r=1; else r=x(1)/abs(x(1)); end u=conj(r)*x/a; u(1)=u(1)+1; u=u/sqrt(u(1)); a=-r*a; end</pre> Listing 5.1 housegen
lyche-numerical-linear-algebra-DIA_0020 (dup)	129	no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre>function [R,C] = housetriang(A,B) % [R,C] = housetriang(A,B) [m,n]=size(A); r=size(B,2); A=[A,B]; for k=1:min(n,m-1) [v,A(k,k)]=housegen(A(k:m,k)); C=A(k:m,k+1:n+r); A(k:m,k+1:n+r)=C-v*(v'*C); end R=triu(A(:,1:n)); C=A(:,n+1:n+r); end</pre> Listing 5.2 housetriang	<pre>function [R,C] = housetriang(A,B) % [R,C] = housetriang(A,B) [m,n]=size(A); r=size(B,2); A=[A,B]; for k=1:min(n,m-1) [v,A(k,k)]=housegen(A(k:m,k)); C=A(k:m,k+1:n+r); A(k:m,k+1:n+r)=C-v*(v'*C); end R=triu(A(:,1:n)); C=A(:,n+1:n+r); end</pre> Listing 5.2 housetriang

lyche-numerical-linear-algebra-DIA_0021 (dup)	134	750c5f9d-6269-44fe-81e3-a3e4f5704d07-134_291_343_1660_1001.jpg																														
lyche-numerical-linear-algebra-DIA_0022 (dup)	140	no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre>function x=rothesstri(A,b) % x=rothesstri(A,b) n=length(A); A=[A b]; for k=1:n-1 r=norm([A(k,k),A(k+1,k)]); if r>0 c=A(k,k)/r; s=A(k+1,k)/r; A([k k+1],k+1:n+1) ... =[c s;-s c]*A([k k+1],k+1:n+1); end A(k,k)=r; A(k+1,k)=0; end x=rbacksolve(A(:,1:n),A(:,n+1),n); end</pre> Listing 5.3 rothesstri	<pre>function x=rothesstri(A,b) % x=rothesstri(A,b) n=length(A); A=[A b]; for k=1:n-1 r=norm([A(k,k),A(k+1,k)]); if r>0 c=A(k,k)/r; s=A(k+1,k)/r; A([k k+1],k+1:n+1) ... =[c s;-s c]*A([k k+1],k+1:n+1); end A(k,k)=r; A(k+1,k)=0; end x=rbacksolve(A(:,1:n),A(:,n+1),n); end</pre> Listing 5.3 rothesstri																											
lyche-numerical-linear-algebra-DIA_0023 (dup)	175	750c5f9d-6269-44fe-81e3-a3e4f5704d07-175_524_1142_1371_196.jpg																														
lyche-numerical-linear-algebra-DIA_0024 (dup)	225	750c5f9d-6269-44fe-81e3-a3e4f5704d07-225_449_765_1559_575.jpg																														
lyche-numerical-linear-algebra-DIA_0025 (dup)	240	750c5f9d-6269-44fe-81e3-a3e4f5704d07-240_385_935_211_295.jpg	<table><tr><td>1,1 1,2 1,3</td><td>1,3 2,3 3,3</td><td>7 8 9</td></tr><tr><td>2,1 2,2 2,3</td><td>1,2 2,2 3,2</td><td>4 5 6</td></tr><tr><td>3,1 3,2 3,3</td><td>1,1 2,1 3,1</td><td>1 2 3</td></tr></table> $v_{j,k}$ in V - matrix $v_{j,k}$ in grid x_i in grid	1,1 1,2 1,3	1,3 2,3 3,3	7 8 9	2,1 2,2 2,3	1,2 2,2 3,2	4 5 6	3,1 3,2 3,3	1,1 2,1 3,1	1 2 3	<table><tr><td>1,1 1,2 1,3</td><td>1,3 2,3 3,3</td><td>7 8 9</td></tr><tr><td>2,1 2,2 2,3</td><td>1,2 2,2 3,2</td><td>4 5 6</td></tr><tr><td>3,1 3,2 3,3</td><td>1,1 2,1 3,1</td><td>1 2 3</td></tr></table> $v_{j,k}$ in V - matrix $v_{j,k}$ in grid x_i in grid	1,1 1,2 1,3	1,3 2,3 3,3	7 8 9	2,1 2,2 2,3	1,2 2,2 3,2	4 5 6	3,1 3,2 3,3	1,1 2,1 3,1	1 2 3	<table><tr><td>1,1 1,2 1,3</td><td>1,3 2,3 3,3</td><td>7 8 9</td></tr><tr><td>2,1 2,2 2,3</td><td>1,2 2,2 3,2</td><td>4 5 6</td></tr><tr><td>3,1 3,2 3,3</td><td>1,1 2,1 3,1</td><td>1 2 3</td></tr></table> $v_{j,k}$ in V - matrix $v_{j,k}$ in grid x_i in grid	1,1 1,2 1,3	1,3 2,3 3,3	7 8 9	2,1 2,2 2,3	1,2 2,2 3,2	4 5 6	3,1 3,2 3,3	1,1 2,1 3,1	1 2 3
1,1 1,2 1,3	1,3 2,3 3,3	7 8 9																														
2,1 2,2 2,3	1,2 2,2 3,2	4 5 6																														
3,1 3,2 3,3	1,1 2,1 3,1	1 2 3																														
1,1 1,2 1,3	1,3 2,3 3,3	7 8 9																														
2,1 2,2 2,3	1,2 2,2 3,2	4 5 6																														
3,1 3,2 3,3	1,1 2,1 3,1	1 2 3																														
1,1 1,2 1,3	1,3 2,3 3,3	7 8 9																														
2,1 2,2 2,3	1,2 2,2 3,2	4 5 6																														
3,1 3,2 3,3	1,1 2,1 3,1	1 2 3																														

lyche-numerical-linear-algebra-DIA_0026 (dup)	240		750c5f9d-6269-44fe-81e3-a3e4f5704d07-240_316_958_704_284.jpg		
lyche-numerical-linear-algebra-DIA_0027 (dup)	241		750c5f9d-6269-44fe-81e3-a3e4f5704d07-241_398_1079_209_230.jpg		
lyche-numerical-linear-algebra-DIA_0028 (dup)	254	no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre>function V=fastpoisson(F) %function V=fastpoisson(F) m=length(F); h=1/(m+1); hv=pi*h*(1:m)'; sigma=sin(hv/2).^2; S=sin(hv*(1:m)); G=S*F*S; X=h^4*G./(sigma*ones(1,m)+ones(m,1)*sigma'); V=zeros(m+2,m+2); V(2:m+1,2:m+1)=S*X*S; end</pre> Listing 11.1 fastpoisson	<pre>function V=fastpoisson(F) %function V=fastpoisson(F) m=length(F); h=1/(m+1); hv=pi*h*(1:m)'; sigma=sin(hv/2).^2; S=sin(hv*(1:m)); G=S*F*S; X=h^4*G./(sigma*ones(1,m)+ones(m,1)*sigma'); V=zeros(m+2,m+2); V(2:m+1,2:m+1)=S*X*S; end</pre> Listing 11.1 fastpoisson
lyche-numerical-linear-algebra-DIA_0029 (dup)	259	no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre>function z=fftrez(y) %function z=fftrez(y) y=y(:); n=length(y); if n==1 z=y; else q1=fftrez(y(1:2:n-1)); q2=exp(-2*pi*1i/n).^(0:n/2-1)'.*fftrez(y(2:2:n)); z=[q1+q2; q1-q2]; end</pre> Listing 11.2 fftrec	<pre>function z=fftrez(y) %function z=fftrez(y) y=y(:); n=length(y); if n==1 z=y; else q1=fftrez(y(1:2:n-1)); q2=exp(-2*pi*1i/n).^(0:n/2-1)'.*fftrez(y(2:2:n)); z=[q1+q2; q1-q2]; end</pre> Listing 11.2 fftrec
lyche-numerical-linear-algebra-DIA_0030 (dup)	262	no image for this region (28 in zip)	(no tex.zip crop for this row)	function U=simplefastbiharmonic(F)	function U=simplefastbiharmonic(F)
lyche-numerical-linear-algebra-DIA_0031 (dup)	268	no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre>function k=jdp(F,K,tol) % k=jdp(F,K,tol) m=length(F); U=fastpoisson(F); V=zeros(m+2,m+2); E=F/(m+1)^2; for k=1:K V(2:m+1,2:m+1)=(V(1:m,2:m+1)+V(3:m+2,2:m+1)... +V(2:m+1,1:m)+V(2:m+1,3:m+2)+E)/4; if max(max(abs(V-U)))<tol, return end end k=K+1; end</pre> Listing 12.1 jdp	<pre>function k=jdp(F,K,tol) % k=jdp(F,K,tol) m=length(F); U=fastpoisson(F); V=zeros(m+2,m+2); E=F/(m+1)^2; for k=1:K V(2:m+1,2:m+1)=(V(1:m,2:m+1)+V(3:m+2,2:m+1)... +V(2:m+1,1:m)+V(2:m+1,3:m+2)+E)/4; if max(max(abs(V-U)))<tol, return end end k=K+1; end</pre> Listing 12.1 jdp

lyche-numerical-linear-algebra-DIA_0032 (dup)	269		no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre>function k=sordp(F,K,w,tol) % k=sordp(F,K,w,tol) m=length(F); U=fastpoisson(F); V=zeros(m+2,m+2); E=F/(m+1)^2; for k=1:K for j=2:m+1 for i=2:m+1 V(i,j)=w*(V(i-1,j)+V(i+1,j)+V(i,j-1)... +V(i,j+1)+E(i-1,j-1))/4+(1-w)*V(i,j); end end if max(max(abs(V-U)))<tol, return end end k=K+1; end</pre> Listing 12.2 sordp	<pre>function k=sordp(F,K,w,tol) % k=sordp(F,K,w,tol) m=length(F); U=fastpoisson(F); V=zeros(m+2,m+2); E=F/(m+1)^2; for k=1:K for j=2:m+1 for i=2:m+1 V(i,j)=w*(V(i-1,j)+V(i+1,j)+V(i,j-1)... +V(i,j+1)+E(i-1,j-1))/4+(1-w)*V(i,j); end end if max(max(abs(V-U)))<tol, return end end k=K+1; end</pre> Listing 12.2 sordp
lyche-numerical-linear-algebra-DIA_0033 (dup)	274	750c5f9d-6269-44fe-81e3-a3e4f5704d07-274_459_956_209_286.jpg				
lyche-numerical-linear-algebra-DIA_0034 (dup)	292	750c5f9d-6269-44fe-81e3-a3e4f5704d07-292_586_847_211_339.jpg				
lyche-numerical-linear-algebra-DIA_0035 (dup)	297	no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre>function [x,K]=cg(A,b,x,tol,itmax) % [x,K]=cg(A,b,x,tol,itmax) r=b-A*x; p=r; rho0=b'*b; rho=r'*r; for k=0:itmax if sqrt(rho/rho0)<= tol K=k; return end t=A*p; a=rho/(p'*t); x=x+a*p; r=r-a*t; rhos=rho; rho=r'*r; p=r+(rho/rhos)*p; end K=itmax+1; end</pre> Listing 13.1 cg	<pre>function [x,K]=cg(A,b,x,tol,itmax) % [x,K]=cg(A,b,x,tol,itmax) r=b-A*x; p=r; rho0=b'*b; rho=r'*r; for k=0:itmax if sqrt(rho/rho0)<= tol K=k; return end t=A*p; a=rho/(p'*t); x=x+a*p; r=r-a*t; rhos=rho; rho=r'*r; p=r+(rho/rhos)*p; end K=itmax+1; end</pre> Listing 13.1 cg	

lyche-numerical-linear-algebra-DIA_0036 (dup)	298	no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre> function [V,K]=cgtest(m,a,d,tol,itmax) % [V,K]=cgtest(m,a,d,tol,itmax) R=ones(m)/(m+1)^2; rho=sum(sum(R.*R)); rho0=rho; P=R; V=zeros(m,m); Tl=sparse(tridiagonal(a,d,a,m)); for k=1:itmax if sqrt(rho/rho0)<= tol K=k; return end T=Tl*P+P*Tl; a=rho/sum(sum(P.*T)); V=V+a*P; R=R-a*T; rhos=rho; rho=sum(sum(R.*R)); P=R+(rho/rhos)*P; end K=itmax+1; end </pre> Listing 13.2 cgtest	<pre> function [V,K]=cgtest(m,a,d,tol,itmax) % [V,K]=cgtest(m,a,d,tol,itmax) R=ones(m)/(m+1)^2; rho=sum(sum(R.*R)); rho0=rho; P=R; V=zeros(m,m); Tl=sparse(tridiagonal(a,d,a,m)); for k=1:itmax if sqrt(rho/rho0)<= tol K=k; return end T=Tl*P+P*Tl; a=rho/sum(sum(P.*T)); V=V+a*P; R=R-a*T; rhos=rho; rho=sum(sum(R.*R)); P=R+(rho/rhos)*P; end K=itmax+1; end </pre> Listing 13.2 cgtest
lyche-numerical-linear-algebra-DIA_0037 (dup)	302	750c5f9d-6269-44fe-81e3-a3e4f5704d07-302_596_847_211_339.jpg			
lyche-numerical-linear-algebra-DIA_0038 (dup)	312	no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre> function [x,K]=pcg(A,B,b,x,tol,itmax) % [x,K]=pcg(A,B,b,x,tol,itmax) r=b-A*x; p=B*r; s=p; rho=s'*r; rho0=b'*b; for k=0:itmax if sqrt(rho/rho0)<= tol K=k; return end t=A*p; a=rho/(p'*t); x=x+a*p; r=r-a*t; w=B*t; s=s-a*w; rhos=rho; rho=s'*r; p=r+(rho/rhos)*p; end K=itmax+1; end </pre> Listing 13.3 pcg	<pre> function [x,K]=pcg(A,B,b,x,tol,itmax) % [x,K]=pcg(A,B,b,x,tol,itmax) r=b-A*x; p=B*r; s=p; rho=s'*r; rho0=b'*b; for k=0:itmax if sqrt(rho/rho0)<= tol K=k; return end t=A*p; a=rho/(p'*t); x=x+a*p; r=r-a*t; w=B*t; s=s-a*w; rhos=rho; rho=s'*r; p=r+(rho/rhos)*p; end K=itmax+1; end </pre> Listing 13.3 pcg
lyche-numerical-linear-algebra-DIA_0039 (dup)	328	750c5f9d-6269-44fe-81e3-a3e4f5704d07-328_308_484_213_853.jpg	Imaginary axis  Real axis	Imaginary axis  Real axis	Imaginary axis  Real axis
lyche-numerical-linear-algebra-DIA_0040 (dup)	334	no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre> function [L,B] = hesshousegen(A) n=length(A); L=zeros(n,n); B=A; for k=1:n-2 [v,B(k+1,k)]=housegen(B(k+1:n,k)); L(k+1:n,k)=v; B(k+2:n,k)=zeros(n-k-1,1); C=B(k+1:n,(k+1):n); B(k+1:n,(k+1):n)=C-v*(v'*C); C=B(1:n,(k+1):n); B(1:n,(k+1):n)=C-(C*v)*v'; end end </pre> Listing 14.1 hesshousegen	<pre> function [L,B] = hesshousegen(A) n=length(A); L=zeros(n,n); B=A; for k=1:n-2 [v,B(k+1,k)]=housegen(B(k+1:n,k)); L(k+1:n,k)=v; B(k+2:n,k)=zeros(n-k-1,1); C=B(k+1:n,(k+1):n); B(k+1:n,(k+1):n)=C-v*(v'*C); C=B(1:n,(k+1):n); B(1:n,(k+1):n)=C-(C*v)*v'; end end </pre> Listing 14.1 hesshousegen

lyche-numerical-linear-algebra-DIA_0041 (dup)	335		no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre>function Q = accumulateQ(L) n=length(L); Q=eye(n); for k=n-2:-1:1 v=L((k+1):n,k); C=Q((k+1):n,(k+1):n); Q((k+1):n,(k+1):n)=C-v*(v'*C); end</pre> Listing 14.2 accumulateQ	<pre>function Q = accumulateQ(L) n=length(L); Q=eye(n); for k=n-2:-1:1 v=L((k+1):n,k); C=Q((k+1):n,(k+1):n); Q((k+1):n,(k+1):n)=C-v*(v'*C); end</pre> Listing 14.2 accumulateQ
lyche-numerical-linear-algebra-DIA_0042 (dup)	339		no image for this region (28 in zip)	(no tex.zip crop for this row)	$h = b - a;$ $\text{for } j = 1 : itmax$ $c = (a + b)/2;$ $\text{if } b - a < eps * h$ $\lambda = (a + b)/2; \text{ return}$ end $k = \rho(c);$ $\text{if } k \geq m \text{ } a = c \text{ else } b = c;$ end	$h = b - a;$ $\text{for } j = 1 : itmax$ $c = (a + b)/2;$ $\text{if } b - a < eps * h$ $\lambda = (a + b)/2; \text{ return}$ end $k = \rho(c);$ $\text{if } k \geq m \text{ } a = c \text{ else } b = c;$ end
lyche-numerical-linear-algebra-DIA_0043 (dup)	347		no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre>function [l,x,it]=powerit(A,z,K,tol) % [l,x,it]=powerit(A,z,K,tol) af=norm(A,'fro'); x=z/norm(z); for k=1:K y=A*x; l=x'*y; if norm(y-l*x)/af<tol it=k; x=y/norm(y); return end x=y/norm(y); end it=K+1; end</pre> Listing 15.1 powerit	<pre>function [l,x,it]=powerit(A,z,K,tol) % [l,x,it]=powerit(A,z,K,tol) af=norm(A,'fro'); x=z/norm(z); for k=1:K y=A*x; l=x'*y; if norm(y-l*x)/af<tol it=k; x=y/norm(y); return end x=y/norm(y); end it=K+1; end</pre> Listing 15.1 powerit
lyche-numerical-linear-algebra-DIA_0044 (dup)	350		no image for this region (28 in zip)	(no tex.zip crop for this row)	<pre>function [x,s,nr]=rayleighit(A,x,s) % [x,s,nr]=rayleighit(A,x,s) n=length(x); y=(A-s*eye(n,n))\x; yn=norm(y); w=x/yn; x=y/yn; rho=x'*w; s=s+rho; nr=norm(w-rho*x); end</pre> Listing 15.2 rayleighit	<pre>function [x,s,nr]=rayleighit(A,x,s) % [x,s,nr]=rayleighit(A,x,s) n=length(x); y=(A-s*eye(n,n))\x; yn=norm(y); w=x/yn; x=y/yn; rho=x'*w; s=s+rho; nr=norm(w-rho*x); end</pre> Listing 15.2 rayleighit
lyche-numerical-linear-algebra-DIA_0045 (dup)	354		no image for this region (28 in zip)	(no tex.zip crop for this row)	$A_1 = A$ $\text{for } k = 1, 2, \dots$ Choose a shift s_k $Q_k R_k = A_k - s_k I \quad (\text{QR factorization of } A_k - s I)$ $A_{k+1} = R_k Q_k + s_k I.$ end	$A_1 = A$ $\text{for } k = 1, 2, \dots$ Choose a shift s_k $Q_k R_k = A_k - s_k I \quad (\text{QR factorization of } A_k - s I)$ $A_{k+1} = R_k Q_k + s_k I.$ end